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Guidelines for computing crop water requirements

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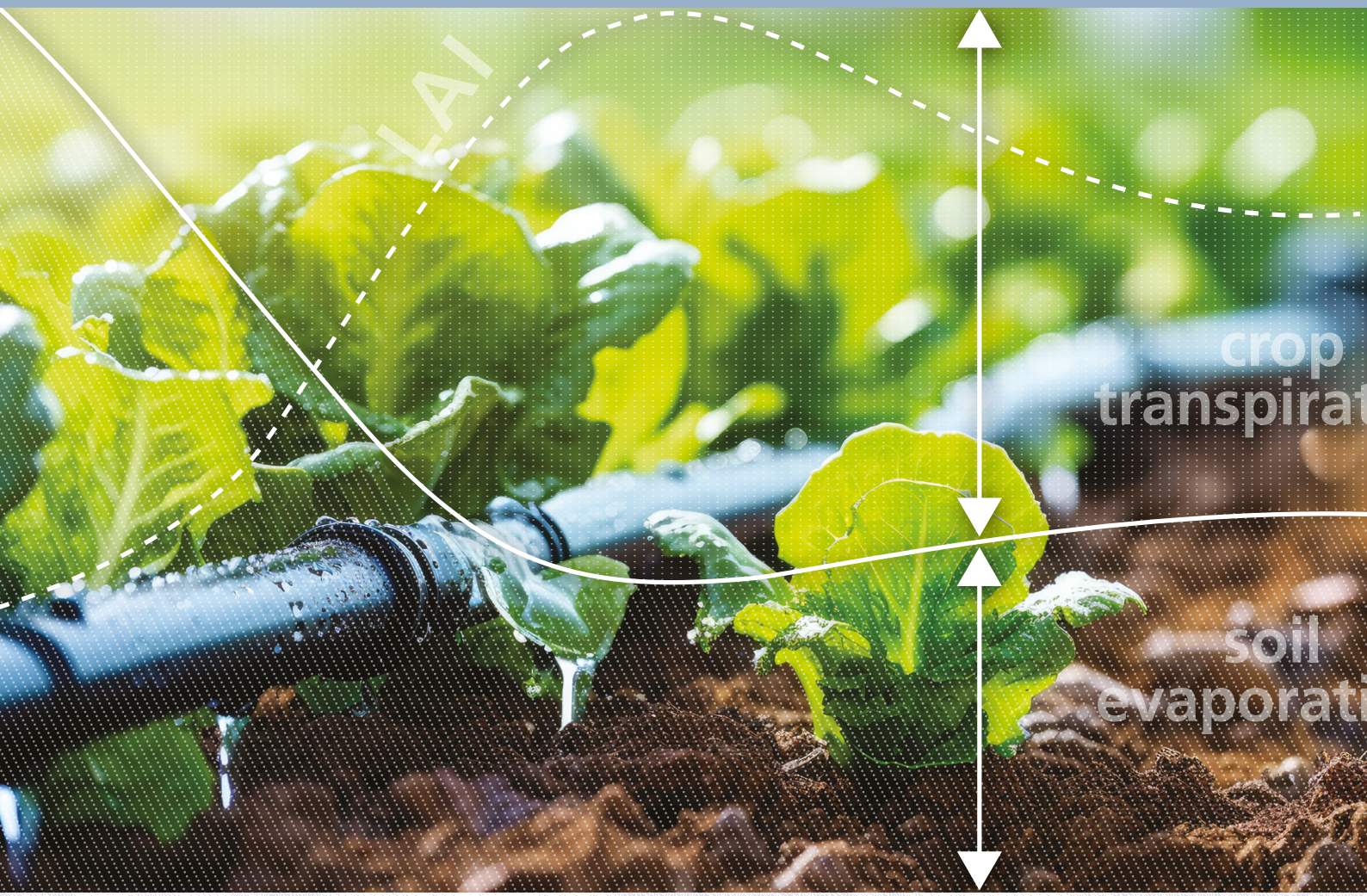

Crop evapotranspiration

Guidelines for computing crop water requirements

2025 revised edition

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IRRIGATION
AND DRAINAGE
PAPER
56
Rev.1

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Foreword

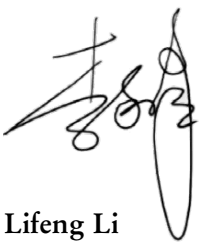
Water management in agriculture is increasingly critical as the world faces escalating challenges from climate change, water scarcity, and the demands of an increasing global population. The FAO Irrigation and Drainage Paper No. 56, "*Crop Evapotranspiration—Guidelines for Computing Crop Water Requirements*", has been a cornerstone of agricultural water management since its first edition in 1998.

The standardization of methodologies and the adoption of the Penman-Monteith ET_o equation for computing the reference evapotranspiration and then estimating crop evapotranspiration enabled agronomists, researchers, policymakers, and farmers to effectively manage water resources, optimize irrigation practices, and promote sustainable agricultural systems. Its influence is globally reflected in thousands of citations and numerous studies validating its overall accuracy.

In response to significant progress in both science and technology over the past two decades, FAO, in collaboration with key partners, has revised these guidelines. The revamped version not only reaffirms the value of the original guidelines but also integrates cutting-edge techniques such as remote sensing, the Internet of Things (IoT), and improved energy balance methods. These tools enhance the accuracy and applicability of the methods included in the guidelines, addressing the dynamic and complex needs of today's agricultural systems. Moreover, the new version includes new approaches and datasets, making it adaptable to diverse regional conditions, especially in arid and semi-arid areas.

The revised guidelines are also key for addressing the pressing challenges posed by climate change. Rising temperatures and shifting precipitation patterns reinforce the need for more precise information on crop water requirements and more efficient water use in agriculture. This revamped version introduces refined methodologies for optimizing water allocation, incorporating water-saving strategies, and advancing crop coefficient estimation and irrigation scheduling. These enhancements enable stakeholders to tailor their irrigation practices to local conditions, preserve water, and support food security while minimizing environmental impacts.

By integrating the latest scientific knowledge with practical tools, the revised guidelines remain an indispensable resource for agriculture to effectively respond to changing climate and water availability conditions, and to become more efficient, inclusive, resilient and sustainable in the future.



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Chapter 1
Introduction to evapotranspiration

This chapter introduces the evapotranspiration process and explains the concepts of and the differences between reference crop evapotranspiration (ET_o) and crop evapotranspiration under standard conditions (ET_c) and various management and environmental conditions ($ET_{c\ act}$). It examines the factors that affect evapotranspiration, the units in which it is normally expressed and the way in which it can be determined. It also describes the field methods used to determine the $ET_{c\ act}$, as well as the main remote sensing ET methods.

1.1 EVAPOTRANSPIRATION PROCESS

The combination of two separate processes whereby water is transferred to the atmosphere, on the one hand from the soil surface by evaporation and, on the other hand, from the crop by transpiration, is referred to as evapotranspiration (ET).

1.1.1 Evaporation

Evaporation is the process whereby liquid water is converted to water vapour (vaporization) and removed from the evaporating surface (vapour removal). Water evaporates from a variety of surfaces, such as lakes, rivers, pavements, soils and vegetation.

Energy is required to change the state of the molecules of water from liquid to vapour. Direct solar radiation and, to a lesser extent, the ambient temperature of the air provide this energy. The driving force to remove water vapour from the evaporating surface is the difference between the water vapour pressure at the evaporating surface and that of the surrounding atmosphere. As evaporation proceeds, the surrounding air becomes gradually saturated and the process will slow down, and might stop if the wet air is not transferred to the atmosphere. The replacement of the saturated air with drier air depends greatly on wind speed. Hence, solar radiation, air temperature, air humidity and wind speed are the essential climatological parameters to consider when assessing the evaporation process.

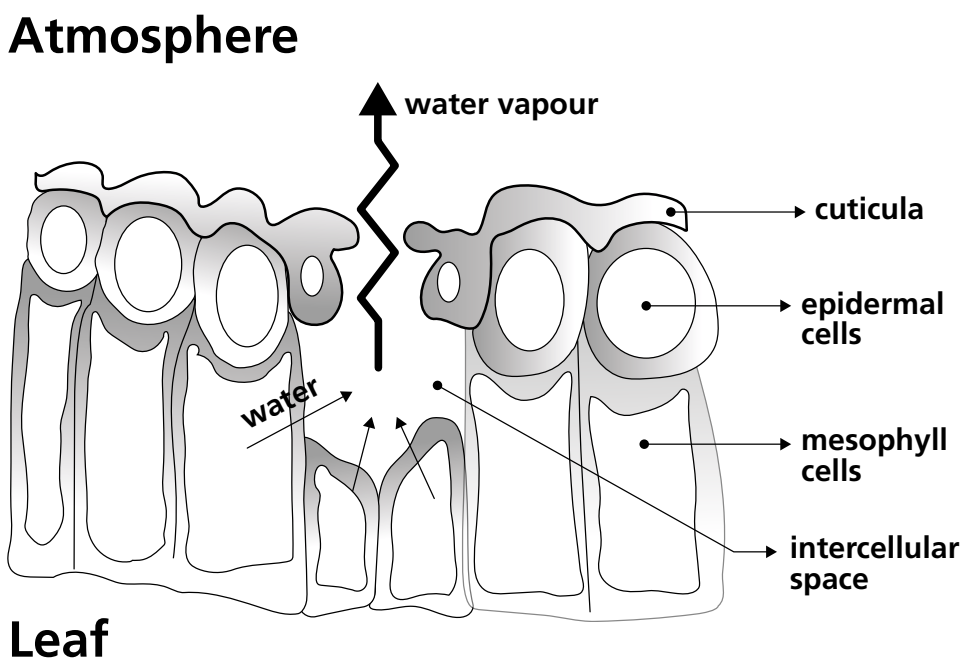
Where the evaporating surface is the soil surface, the degree of shading by the crop canopy and the amount of water available at the evaporating surface are other factors that affect the evaporation process. Frequent rains, irrigation and water transported upwards in a soil from a shallow water table wet the soil surface. Where the soil is able to supply water fast enough to satisfy the evaporation demand, the evaporation from the soil is determined only by the meteorological conditions. However, where the interval between rains and irrigation becomes large and the ability of the soil to conduct moisture to near the surface is small, the water content in the topsoil drops and the soil surface dries out. Under these circumstances, the limited availability of water exerts a controlling influence on soil evaporation. In the absence of any supply of water to the soil surface, evaporation decreases rapidly and may cease almost completely within a few days.

1.1.2 Transpiration

Transpiration consists of the vaporization of liquid water contained in plant tissues and the vapour removal to the atmosphere. Crops predominately lose their water through stomata. These are small openings on the plant leaf through which gases and

water vapour pass (Figure 1.1). The water, together with some nutrients, is taken up by the roots and transported through the plant. The vaporization occurs within the leaf, namely in the intercellular spaces, and the vapour exchange with the atmosphere is controlled by the stomatal aperture. Nearly all water taken up is lost by transpiration and only a tiny fraction is used within the plant.

FIGURE 1.1
Schematic representation of a stoma



Source: FAO56, 1998

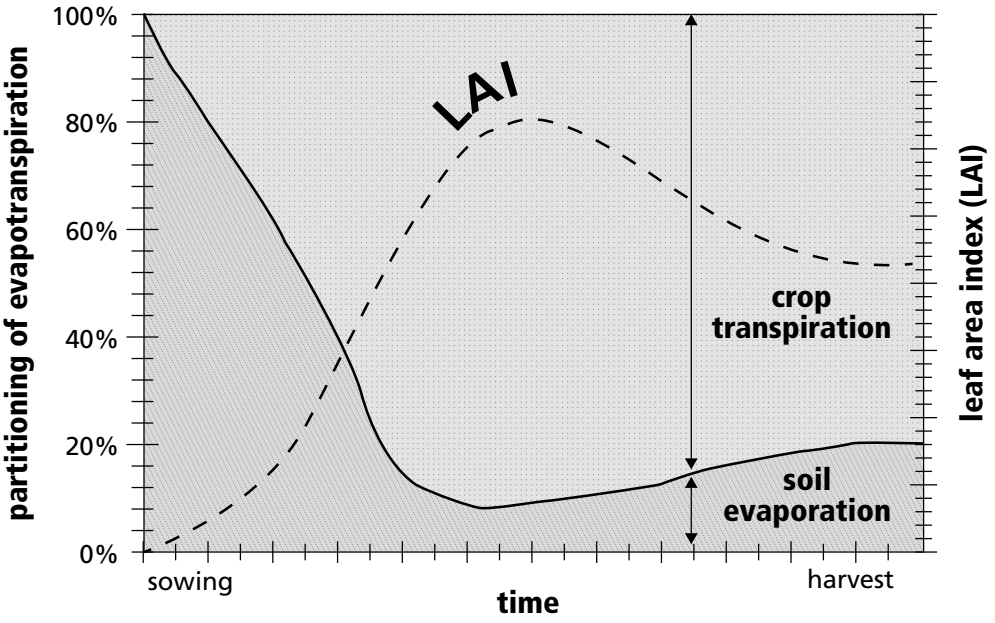
Transpiration, like direct evaporation, depends on the energy supply, vapour pressure gradient and wind. Hence, radiation, air temperature, air humidity and wind terms should be considered when assessing transpiration. The soil water content and the ability of the soil to conduct water to the roots also determine the transpiration rate, as do waterlogging and soil water salinity. The transpiration rate is also influenced by crop characteristics, environmental aspects and cultivation practices. Different kinds of plants may have different transpiration rates. Not only the type of crop, but also the crop development stage, environment and management should be considered when assessing transpiration.

1.1.3 Evapotranspiration (ET)

Evaporation and transpiration occur simultaneously and there is no easy way of distinguishing between the two processes unless specific instruments are used. Apart from the water availability in the topsoil, the evaporation from a cropped soil is mainly determined by the fraction of the solar radiation reaching the soil surface. This fraction decreases over the growing period as the crop develops and the crop canopy shades more and more of the ground area. When the crop is small, water is predominately lost by soil evaporation, but once the crop is well developed and completely covers the soil, transpiration becomes the main process. In Figure 1.2,

the partitioning of evapotranspiration into evaporation and transpiration is plotted in correspondence to leaf area per unit surface of soil below it. At sowing, nearly 100 percent of ET comes from evaporation; while at full crop cover, more than 90 percent of ET comes from transpiration.

FIGURE 1.2
The partitioning of evapotranspiration into evaporation and transpiration over the growing period for an annual field crop



Source: FAO56, 1998

1.2 UNITS

The evapotranspiration rate is normally expressed in millimetres (mm) per unit time. The rate expresses the amount of water lost from a cropped surface in units of water depth. The time unit can be an hour or fraction, a day, decade, month or even an entire growing period or year.

As one hectare has a surface of 10 000 m^2 and 1 mm is equal to 0.001 m, a loss of 1 mm of water corresponds to a loss of 10 m^3 of water per hectare. In other words, 1 mm day^{-1} is equivalent to 10 m^3 ha^{-1} day^{-1} (or 1 litre per square meter).

Water depths can also be expressed in terms of energy received per unit area, i.e. the energy or heat required to vaporize free water. This energy, known as the latent heat of vaporization (λ), is a function of the water temperature. For example, at 20 °C, λ is about 2.45 MJ kg^{-1}. In other words, 2.45 MJ is needed to vaporize 1 kg or 0.001 m^3 of water. Hence, an energy input of 2.45 MJ per m^2 is able to vaporize 0.001 m or 1 mm of water; therefore, 1 mm of water is equivalent to 2.45 MJ m^{-2} (Example 1.1). The evapotranspiration rate when expressed in units of MJ m^{-2} day^{-1} is represented by λET, the latent heat flux.

Table 1.1. summarizes the units used to express the evapotranspiration rate and the conversion factors.

TABLE 1.1
Conversion factors for evapotranspiration

	depth	volume per unit area		energy per unit area*
	mm day^{-1}	m^3 ha^{-1} day^{-1}	l s^{-1} ha^{-1}	MJ m^{-2} day^{-1}
1 mm day^{-1}	1	10	0.116	2.45
1 m^3 ha^{-1} day^{-1}	0.1	1	0.012	0.245
1 l s^{-1} ha^{-1}	8.640	86.40	1	21.17
1 MJ m^{-2} day^{-1}	0.408	4.082	0.047	1

*For water with a density of 1 000 kg m^{-3} and at 20 °C

EXAMPLE 1.1
Converting evaporation from one unit to another

On a summer day, net solar energy received at a lake reaches 15 MJ per square meter per day. If 80 percent of the energy is used to vaporize water, how large could the depth of evaporation be?

From Table 1.1:	1 MJ m^{-2} day^{-1} =	0.408	mm day^{-1}
Therefore:	0.8 x 15 MJ m^{-2} day^{-1} = 0.8 x 15 x 0.408		
	mm day^{-1} =	4.9	mm day^{-1}

The evaporation rate could be 4.9 mm day^{-1}

1.3 FACTORS AFFECTING EVAPOTRANSPIRATION

Weather parameters, crop characteristics, management and environmental aspects are factors affecting evaporation and transpiration. The related ET concepts presented in Figure 1.3 are discussed in the section on evapotranspiration concepts.

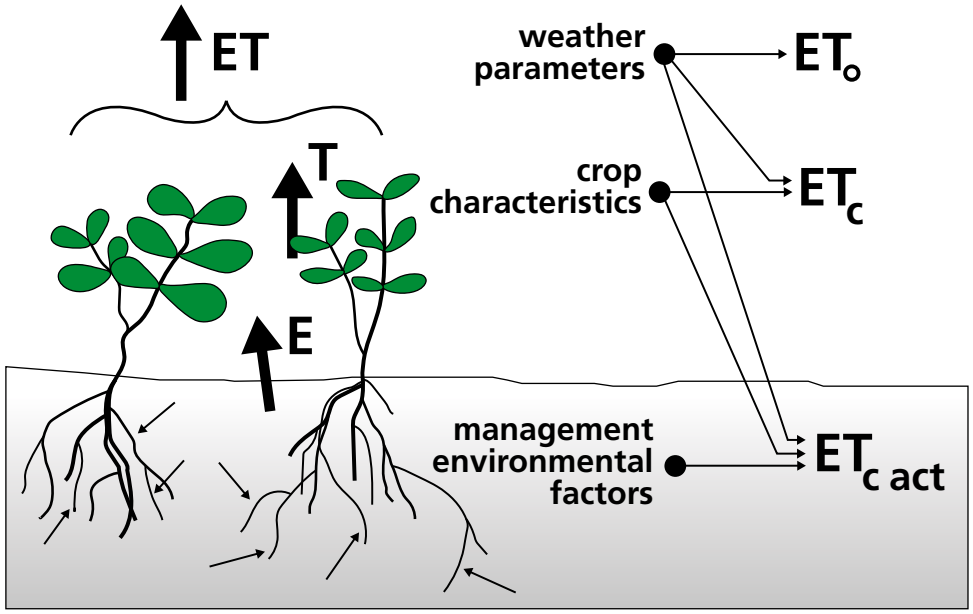
1.3.1 Weather parameters

The principal weather parameters affecting evapotranspiration are radiation, air temperature, humidity and wind speed. Several procedures have been developed to assess the evaporation rate from these parameters. The evaporation power of the atmosphere is expressed by the reference crop evapotranspiration (ET_o). The reference crop evapotranspiration represents the evapotranspiration from a standardized vegetated surface. The ET_o is described in detail later in this Chapter and in Chapters 2 and 4.

1.3.2 Influencing crop factors

The crop type, variety and development stage should be considered when assessing the evapotranspiration from crops grown in large, well-managed fields. Differences in resistance to transpiration, crop height, crop roughness, reflection, ground cover and crop rooting characteristics result in different ET levels in different types of crops under identical environmental conditions. Crop evapotranspiration under standard conditions (ET_c) refers to the evaporating demand from crops that are grown in large fields under optimum soil water, excellent management and environmental conditions, and achieve full production under the given climatic conditions. The ET_c is detailed in Chapters 6 and 7.

FIGURE 1.3
Factors affecting evapotranspiration with indication of related ET concepts



Source: FAO56, 1998

1.3.3 Management and environmental conditions

Factors such as soil salinity, poor land fertility, limited application of fertilizers, the presence of hard or impenetrable soil horizons, the absence of control of diseases and pests and poor soil management may limit crop development and reduce evapotranspiration. Other factors to be considered when assessing ET are ground cover, plant density and the soil water content. The effect of soil water content on ET is conditioned primarily by the magnitude of the water deficit and the type of soil. On the other hand, too much water will result in waterlogging which might damage the root and limit root water uptake by inhibiting respiration.

When assessing the ET rate, additional consideration should be given to the range of management practices that act on the climatic and crop factors affecting the ET process. Cultivation practices and the type of irrigation methods can alter the microclimate, affect the crop characteristics or affect the wetting of the soil and crop surface. A windbreak reduces wind velocities and decreases the ET rate of the field directly downwind of the barrier. The effect can be significant especially in windy, warm and dry conditions although evapotranspiration from the trees themselves may offset any reduction in the field. Soil evaporation in a young orchard, where trees are widely spaced, can be reduced by using a well-designed drip or trickle irrigation system. The drippers apply water directly to the soil near trees, thereby leaving the major part of the soil surface dry, and limiting the evaporation losses. The use of mulches, especially when the crop is small, is another way of substantially reducing soil evaporation. Anti-transpirants, such as stomata-closing, film-forming or reflecting material, reduce the water losses from the crop and hence the transpiration rate. $ET_{c\ act}$ is discussed in Chapters 8, 9 and 10.

Where field conditions differ from the standard conditions, correction factors are required to adjust ET_c to actual conditions, i.e. $ET_{c\ act}$. The adjustment reflects the effect on crop evapotranspiration by the environmental and management conditions of the field.

1.4 EVAPOTRANSPIRATION CONCEPTS

Distinctions are made (Figure 1.4) between reference crop evapotranspiration (ET_o), crop evapotranspiration under standard conditions (ET_c) and crop evapotranspiration under non-standard conditions, i.e. actual conditions ($ET_{c\ act}$). ET_o is a climatic parameter expressing the evaporation power of the atmosphere. ET_c refers to the evapotranspiration from excellently managed and disease-free, large, well-watered fields that achieve full production under the given climatic conditions. Due to sub-optimal crop management and environmental constraints that affect crop growth and yield and any presence of water stress, evapotranspiration may be limited, and therefore the actual ET_c under non-standard conditions, $ET_{c\ act}$, requires a correction.

1.4.1 Reference crop evapotranspiration (ET_o)

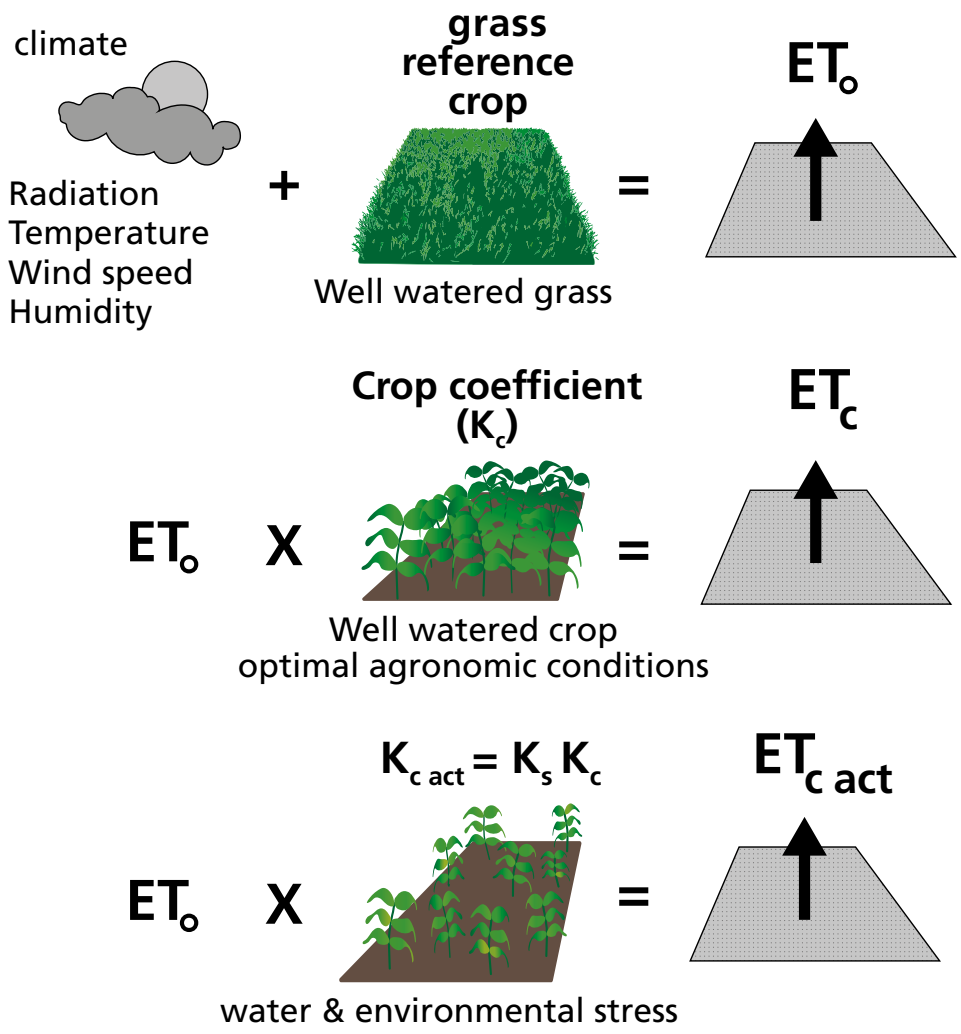
The evapotranspiration rate from a crop reference surface, not short of water, is called the reference crop evapotranspiration or reference evapotranspiration and is denoted as ET_o. The reference surface is a hypothetical grass reference crop with specific characteristics. The use of other denominations and terms such as potential ET is strongly discouraged due to ambiguities in their definitions.

The concept of reference evapotranspiration was introduced to study the evaporative demand of the atmosphere independent of crop type, crop development and management practices. As water is abundantly available at the reference evapotranspiring surface, soil factors do not affect ET_o. Relating ET_o to a specific surface provides a reference to which ET from other surfaces can be related. It obviates the need to define a separate ET equation or level for each crop and stage of growth. ET_o values measured or calculated at different locations or in different seasons are comparable as they refer to the ET from the same reference surface.

The only factors affecting ET_o are climatic parameters. Consequently, ET_o is a climatic parameter and must be computed from weather data. ET_o expresses the evaporating power of the atmosphere at a specific location and time of the year and does not consider crop characteristics and soil factors. Given the specific characteristics of the standardized hypothetical grass reference crop, that are not possible to consistently and accurately reproduce in a lysimeter, ET_o, in general, cannot be easily estimated in an operational manner from lysimetric observations. Thus, the FAO Penman-Monteith method (FAO-PM ET_o) is recommended as the sole method for determining ET_o. The method has been selected because it closely approximates grass ET_o at the locations evaluated, is physically based, and explicitly incorporates both physiological and aerodynamic parameters. Moreover, procedures have been developed for estimating missing climatic parameters.

Typical ranges for ET_o values for different agroclimatic regions are given in Table 1.2. These values are intended to familiarize users with typical ranges that may be compared with locally computed values and are not intended for direct application. The calculation of the reference crop evapotranspiration is discussed in Part A of these guidelines (Box 1.1).

FIGURE 1.4
Reference evapotranspiration (ET_o), standard crop evapotranspiration (ET_c) and actual (non-standard) crop evapotranspiration ($ET_{c\ act}$)



Source: FAO56, 1998

TABLE 1.2
Average ET_o, in mm day^{-1}, for different agroclimatic regions

Regions	Mean daily temperature (°C)		
	Cool ~ 10 °C	Moderate ~ 20 °C	Warm > 30 °C
Tropics and subtropics			
• humid and sub-humid	2 - 3	3 - 5	5 - 7
• arid and semi-arid	2 - 4	4 - 6	6 - 8
Temperate region			
• humid and sub-humid	1 - 2	2 - 4	4 - 7
• arid and semi-arid	1 - 3	4 - 7	6 - 9

BOX 1.1
Chapters concerning the calculation of the reference crop evapotranspiration (ET_o)

PART A

Chapter 2 - FAO Penman-Monteith ET_o equation:

The Chapter introduces the user to the need to standardize one method to compute ET_o from meteorological data. The FAO-PM method is recommended as the method for determining reference ET_o. The method, its derivation, the required meteorological data; the corresponding definition of the reference surface are described in this chapter. Also included is the correction of data collected over non-reference weather sites and application of QAQC to measured meteorological data.

Chapter 3 - Meteorological variables and parameters:

The FAO-PM ET_o method requires radiation, air temperature, air humidity, and wind speed data. Calculation procedures to derive climatic parameters from the meteorological data are presented. Procedures to estimate missing meteorological variables required for calculating ET_o are outlined. This Chapter discussed the source, measurement and computation of all data required for the calculation of the reference evapotranspiration (ET_o) by means of the FAO-PM method under circumstances, even in the case of missing climatic data or, for example, having only air temperature data. Different examples illustrate the various calculation procedures.

Chapter 4 - Determination of ET_o:

The calculation of ET_o by means of the FAO-PM ET_o equation, with different time steps, from the principal weather parameters and with missing data is described. The determination of ET_o from pan evaporation is also presented.

1.4.2 Crop evapotranspiration under standard conditions (ET_c)

The crop evapotranspiration under standard conditions, denoted as ET_c, is the evapotranspiration from disease-free, well-fertilized crops, grown in large fields, under optimum soil water conditions, and achieving full production under the given climatic conditions. It must be noted that the target is achieving full production and not necessarily attaining maximum evapotranspiration. In addition, any reductions to be made to ET_c to exercise limitations specified by water use policy must be applied outside of the ET_c calculation process.

The amount of water required to satisfy the evapotranspiration from the cropped field is defined as the crop water requirement. Although the values for crop evapotranspiration and crop water requirement are identical, crop water requirement refers to the amount of water that needs to be supplied naturally or through irrigation, while crop evapotranspiration refers to the amount of water that is transferred to the atmosphere through evapotranspiration. The irrigation water requirement basically represents the difference between the crop water requirement and effective precipitation. The irrigation water requirement also includes additional water for the leaching of salts and to compensate for the non-uniformity of water application.

Crop evapotranspiration can be calculated from climatic data and by integrating directly the crop resistance, albedo and air resistance factors for the standardized reference crop into the Penman-Monteith combination equation. As there is still a considerable lack of information for different crops, the Penman-Monteith method is used for the estimation of the standard reference crop to determine its evapotranspiration rate, i.e.

ET_o. Experimentally, determined ratios of ET_c/ET_o, called crop coefficients (K_c), are used to relate ET_c to ET_o as $ET_c = K_c\ ET_o$.

Differences in leaf anatomy, stomatal characteristics, aerodynamic properties and even albedo cause the crop evapotranspiration to differ from the reference crop evapotranspiration under the same climatic conditions. Due to variations in the crop characteristics throughout its growing season, K_c for a given crop changes from sowing to harvest. The calculation of crop evapotranspiration under standard conditions (ET_c) is discussed in Part B of this guidelines (Box 1.2).

BOX 1.2
Chapters concerning the calculation of crop evapotranspiration under standard conditions (ET_c)

PART B

Chapter 5 - Introduction to crop evapotranspiration (ET_c) and its measurement:

This chapter introduces the user to the "$K_c\ ET_o$" approach for calculating crop evapotranspiration. The effects of characteristics that distinguish field crops from the reference grass crop are integrated into the crop coefficient K_c. Differences in evaporation and transpiration between field crops and the reference grass surface can be integrated into a single crop coefficient (K_c) or separated into two coefficients: a basal crop (K_{cb}) and a soil evaporation coefficient (K_e), i.e. $K_c = K_{cb} + K_e$. The approach to follow should be selected as a function of the purpose of the calculation, the accuracy required and the data available. A review of ET_c measurement using ground and remote observations and sensors is added.

Chapter 6 - ET_c and single crop coefficients (K_c):

This chapter deals with the calculation of crop evapotranspiration (ET_c) under standard, pristine, non-stressed cropping conditions. ET_c is determined by the single crop coefficient approach whereby the effects of the various weather conditions are incorporated into ET_o and the crop characteristics into the standard K_c coefficient. This chapter presents the tabulated K_c values of numerous annual and perennial crops as well as the main types of natural vegetation. Guidelines for the use and computation of K_c are presented.

Chapter 7 - ET_c and dual crop coefficient ($K_c = K_{cb} + K_e$):

This chapter presents the dual calculation method for crop evapotranspiration. K_c is split into two separate coefficients, one for crop transpiration (i.e. the basal crop coefficient K_{cb}) and one for soil evaporation (K_e). Guidelines for the daily computation are given and tabulated values for standard K_{cb} of annual and perennial crops are provided.

1.4.3 Crop evapotranspiration under non-standard conditions ($ET_{c\ act}$)

The crop evapotranspiration under non-standard conditions, i.e. actual crop conditions ($ET_{c\ act}$) is the evapotranspiration from crops grown under management and environmental conditions that differ from the standard conditions. When cultivating crops in fields, the real crop evapotranspiration may fall below ET_c due to non-optimal conditions such as the presence of pests and diseases, soil salinity and/or low soil fertility that impact plant size or vigour, and by water shortage or waterlogging, as well as non-optimal planting density and other crop management issues. This may result in scanty plant growth, low canopy density and may reduce the evapotranspiration rate below ET_c.

The crop evapotranspiration under non-standard conditions is calculated by using a water stress coefficient K_s and/or by adjusting K_c for all kinds of other stresses and environmental constraints on crop evapotranspiration. The adjustment of ET_c for water stress, management and environmental constraints is discussed in Part C of these guidelines (Box 1.3).

BOX 1.3
Chapters concerning the calculation of crop evapotranspiration under non-standard conditions ($ET_{c\ act}$)

PART C

Chapter 8 - ET_c under soil water and salinity stress conditions:

This chapter discusses the reduction in transpiration induced by soil moisture stress or soil and water salinity. The resulting evapotranspiration will decrease compared to the crop evapotranspiration under standard conditions. The evapotranspiration is computed by using a water stress coefficient, K_s, describing the effect of water stress on crop transpiration. Various parameters related to the water and salinity stress are defined and tabulated for numerous crops and particular attention is paid to the soil water conditions as affected by stresses.

Chapter 9 - Actual crop ET ($ET_{c\ act}$) by agricultural and natural vegetation:

Procedures that can be used to make adjustments to the K_c to account for less than perfect growing conditions or stand characteristics are discussed. The procedures can also be used to determine K_c for agricultural crops not listed in the tables of Part B. The estimation of crop coefficients from ground cover and height is discussed in detail, including tabulated computational parameters.

Chapter 10 - ET_c under various crop management practices:

This chapter discusses various types of management practices that may cause the values for K_c and ET_c to deviate from the standard conditions described in Part B. Adjustment procedures for K_c to account for surface mulches, intercropping, small areas of vegetation, management induced stress and non-growing periods under various types of surface conditions are presented. In addition, strategies, methods and technologies to achieve water savings are discussed with particular attention to irrigation scheduling.

1.5 DETERMINING EVAPOTRANSPIRATION

1.5.1 ET measurement and estimation

Evapotranspiration is not easy to measure. Specific devices and accurate measurements of various physical parameters include the soil water balance in lysimeters or the field, as well as the energy balance performed on the ground or through remote sensing, and are required to determine evapotranspiration. The methods are commonly expensive, demanding in terms of accuracy of measurements and can only be fully exploited by well-trained research personnel. Although the methods are inappropriate for routine measurements, they remain important for the evaluation of ET estimates obtained by more indirect methods.

This section describes the main methods for measuring actual ET_c or indirectly deriving ET_c estimates, and emphasizes the need for accuracy in measurements, describing some essential aspects that affect measurement accuracy.

When used correctly, the accuracy of the described measurement systems can be very high. However, in many instances, understanding the underlying physics of turbulence and heat, water and energy transfer that govern the measurement is crucial to avoid subtle biases and mistakes that will compromise the accuracy of the data. For the purpose of producing representative and reliable data, the deployment of equilibrium air-boundary layer systems, such as eddy covariance, the Bowen ratio, and scintillometers, must abide by fetch requirements and minimum equipment heights.

Energy balance methods like the Bowen ratio and scintillometers must incorporate representative measurements of net radiation and soil heat flux density, which typically require multiple locations for sensors, especially when used in spatially non-uniform systems such as riparian and other non-agricultural systems. Measurements of eddy covariance and scintillometry require physically based "corrections", for example, to obtain the so-called energy balance closure where the sum of λET, sensible heat flux and ground heat flux equals the net radiation as introduced in the next section. Many practitioners may lack the necessary training or experience to use ET_c measurement methods, which can lead to significant measurement biases.

Lysimeters and soil water balance methods are older techniques that can potentially provide reliable measurements, but only if certain fundamental criteria for the representativeness of vegetation and environmental conditions are met. Sap flow sensors rely on empirical correction factors derived from the physiology and anatomy of species under consideration, and on the accuracy of the scaling techniques used to go from a branch or tree to a group of plants and larger area estimates of ET, including evaporation from soil.

In order to achieve high integrity data, the following sections provide a brief overview of ET measurement and estimation systems, as well as different problems and requirements related to each one of these.

1.5.2 Energy balance and micro-climatological methods

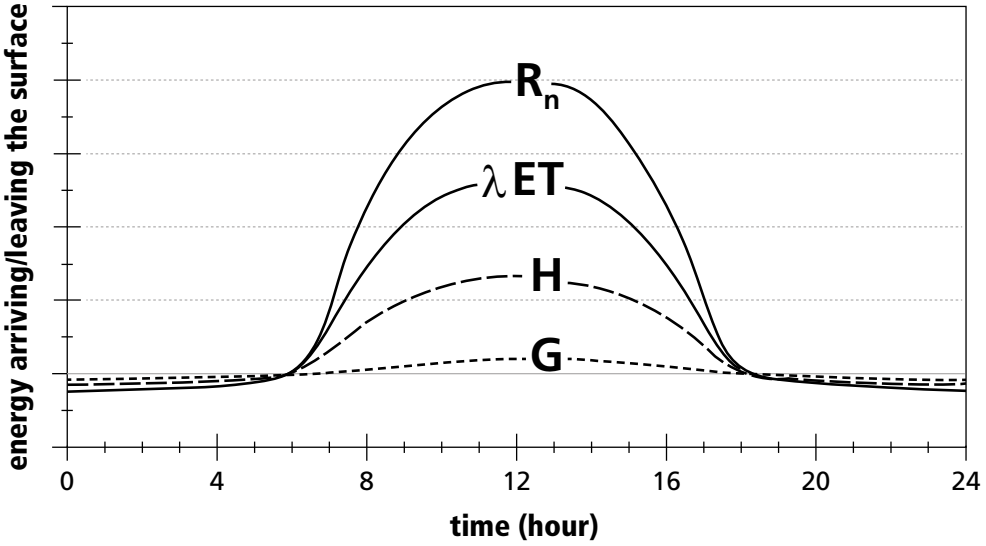
Evaporation of water requires relatively large amounts of energy, either in the form of sensible heat or radiant energy. Therefore, the evapotranspiration process is governed by energy exchange at the vegetation surface and is limited by the amount of energy available. Because of this limitation, it is possible to estimate the evapotranspiration rate by applying the principle of energy conservation. The energy arriving at the surface must equal the energy leaving the surface for the same period. The energy leaving the surface includes the energy consumed by the λET process. The λ variable represents the latent heat of vaporization, described earlier, that is required per unit of ET. The use of the word "latent" means that this energy is "hidden" in the vapour that is produced by the ET.

All primary fluxes of energy should be considered when deriving an energy balance equation. The equation for an evaporating surface can be written as:

$$R_n - G - \lambda ET - H = 0 \quad (1.1)$$

where R_n is the net radiation, H is the sensible heat flux, G is the soil heat flux and λET is the latent heat flux. The various terms can be either positive or negative. Positive R_n supplies energy to the surface and positive G, λET and H remove energy from the surface (Figure 1.5).

FIGURE 1.5
Schematic representation of the diurnal variation of the components of the energy balance above a well-watered transpiring surface on a cloudless day



Source: FAO56, 1998

In Equation 1.1. only vertical fluxes are considered and any energy that is being transferred horizontally, by local advection of dry air or heat near the edge of a field that is surrounded by dryness, is ignored. Therefore, the equation is to be applied for large, extensive surfaces of homogeneous vegetation only. In addition, the methodology, especially the estimation of ET_o, does not apply to greenhouses and other protected cultivation facilities where aerodynamic conditions and energy exchange can be much different than those experienced for open agricultural fields. The equation is restricted to the four components: R_n, λET, H and G. Other energy terms, such as heat stored or released in the plant, or the energy used in metabolic activities and photosynthesis, are not considered. These terms account for only a small fraction of the daily net radiation and can be considered negligible when compared with the other four components.

The latent heat flux (λET) representing the evapotranspiration fraction can be derived from the energy balance equation if all other components are known. Net radiation (R_n) and soil heat fluxes (G) can be measured or estimated from climatic parameters. Measurements of the sensible heat (H) are however complex and cannot be easily obtained. H requires accurate measurement of temperature gradients above the surface.

Another method for estimating evapotranspiration is using a mass transfer method. This approach considers the vertical movement of small parcels of air (eddies) above a large homogeneous surface. The eddies transport material (water vapour) and energy (heat and momentum) from and towards the evaporating surface. By assuming steady state conditions and that the eddy transfer coefficients for water vapour are proportional to those for heat and momentum, the evapotranspiration rate can be computed from the vertical gradients of air temperature and water vapour above the surface via the **Bowen Ratio Energy Balance** (BREB). The Bowen ratio BR is defined as the ratio of H to λET (BR = H/λET) and λET is determined as λET = (R_n–G)/(1+BR).

Other direct estimation methods use gradients of wind speed and water vapour to directly estimate the vapour flux. These methods and other methods, such as Eddy Covariance (EC), require accurate measurement of vapour pressure, and air temperature or wind speed at different levels above the surface. Therefore, their application is restricted to primarily research situations.

The **BREB technique** performs best when ET is not constrained by soil water so that λET tends to be larger than H and relative error is reduced. The Bowen ratio rises and the relative error in ET increases as water becomes less readily available.

Requirements of the BREB system include:

- sufficient distance of fetch to establish an equilibrium boundary layer (EBL) deeper than the instrument height (this requirement is frequently violated);
- sufficient elevation of the instrument above the canopy to avoid the roughness sublayer where gradients of temperature and vapour pressure are difficult to establish and may not be in proportion to the true BR;
- representative and accurate measurement of R_n and G; and
- multiple net radiometers and soil heat flux plates, required when measurements are performed over heterogeneous or sparse areas.

Eddy covariance (EC) systems are often used for crop ET monitoring due to their relative simplicity of setup, moderate sensor prices and, depending on the equipment settings, the ability to simultaneously measure H, λET and CO_2 fluxes. The concept of eddy covariance (also known as eddy correlation) refers to the statistical covariance between vertical fluxes of vapour or sensible heat within the upward and downward legs of turbulent eddies. Quick response sensors must be used to make high speed measurements of air temperature (T), vertical wind velocity (w), and mean vapour pressure (e) or mean specific humidity (q), normally at frequencies of 5-20 Hz (5-20 times per second), typically at 10 Hz. Additionally, ET can be calculated as a residual from the energy balance equation as $\lambda ET = R_n - G - H$, which has the advantage of eliminating the need to measure q, although it does require accurate measurements of R_n and G.

A large number of corrections are required with the EC procedure, and there is a variety of software available for EC flux processing. Many studies have reported a lack of energy balance closure for EC data since the sum of the measured turbulent fluxes ($H + \lambda ET$) usually underestimates the measured available energy ($R_n - G$). This underestimation can often be as large as 30 percent. Potential causes for the lack of closure may be horizontal advection, storage of heat in the canopies, energy used by photosynthesis, change in storage of heat in the developing boundary layer below the instrumentation (causing flux divergence), errors in the measurements of turbulent fluxes, error in measurement of the vertical wind velocity by the sonic anemometer, and inadequate or too much separation between the anemometer and humidity sensor. Additionally, an error can occur in R_n or G measurements due to miscalibration and improper installation or nonrepresentative placement of sensors over nonuniform vegetation such as with rows of trees.

The literature is not conclusive about the method to be followed for resolving the energy imbalance, although the most important issue is that the EC measurements are consistent with the conservation of energy. Generally, users force the energy balance closure using the residual technique by scaling H and λET in the same proportion until the sum equals $R_n - G$, or by assuming correct H measurement and

underestimation of λET. Days associated with vertical regional advection of warm, dry air may show a poor energy balance closure as compared to the rest of the days. Therefore, energy balance closure cannot be considered as a data quality assessment, since closure error can still be present even if data acquired by the EC system are of good quality and vice versa.

The EC system requirements are as follows:

- a uniform fetch with sufficient distance to create an equilibrium boundary layer (EBL) deeper than the instrument height (this is frequently violated);
- sufficient elevation of instruments above the canopy to reduce roughness sublayer impacts and to increase mean eddy size to match the sensor path lengths;
- accomplishing the necessary data "corrections";
- accurate, high frequency instrumentation and a good knowledge and understanding of the physics of turbulence; and
- utilization of net radiation and soil heat flux measurements if energy balance closure error is to be identified and corrected.

A **scintillometer** is an optical device that measures small fluctuations in the refractive index of air caused by temperature, humidity, and pressure induced variations in density. The most common scintillometers measure sensible heat flux, H, by relating the structure parameter to a temperature parameter and the Monin-Obukhov stability parameters. A scintillometer system for measuring sensible heat flux consists of an optical transmitter and a receiver at the ends of an atmospheric propagation path. The receiver detects and evaluates the intensity of fluctuations of the transmitted signal, called scintillations. λET is estimated as a residual of the surface energy balance. To obtain λET, measurements of R_n and G are needed, which can be problematic and biased in heterogeneous and sparse vegetation systems.

Scintillometers are relatively simple to operate and maintain. Their cost can equal or exceed those of BREB and EC systems. A distinct advantage of scintillometry is the ability to derive sensible heat flux that is integrated over a long transect, up to several km in length. This integration is especially useful for measuring H over complex natural vegetation where spatial variation can have scales of hundreds of meters. However, insufficient representativeness of R_n and G measurement over a transect can greatly reduce the accuracy of ET determined from the scintillometer-based integration of H. R_n measurements may need augmentation by satellite-based images of albedo and surface temperature. Soil heat flux densities can vary in the same manner along a mixed transect.

1.5.3 Soil water balance

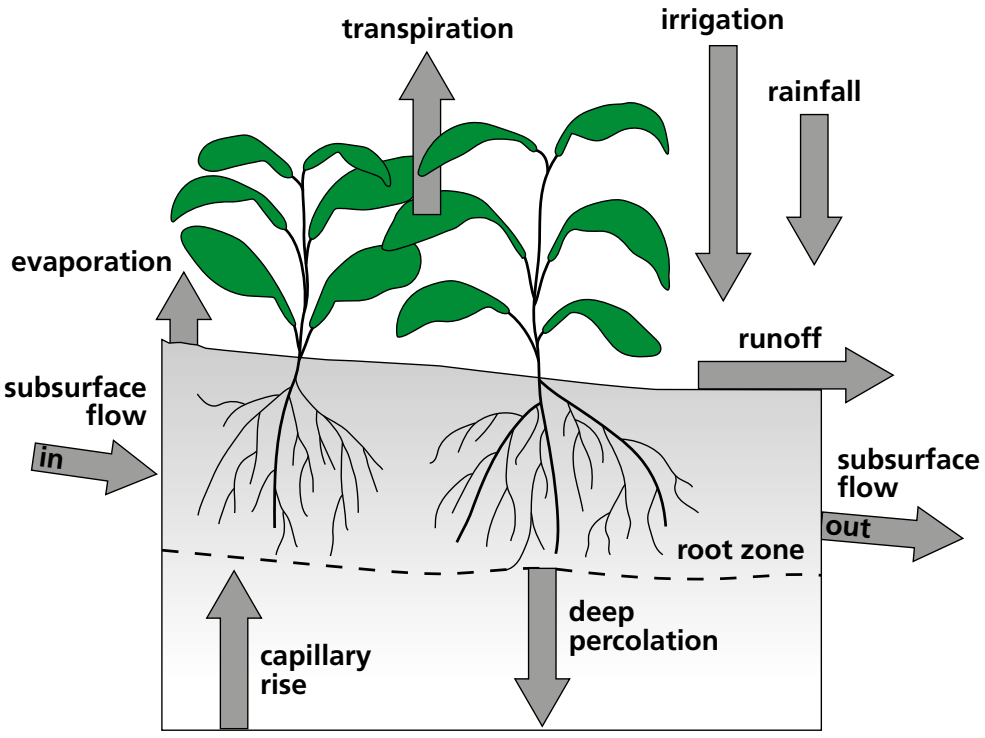
Evapotranspiration can also be determined by measuring the various components of the soil water (SW) balance. The method consists of assessing the incoming and outgoing water flux into the crop root zone over some time period (Figure 1.6). Irrigation (I) and rainfall (P) add water to the root zone. Part of I and P might be lost by surface runoff (RO) and by deep percolation (DP) that will eventually recharge the water table. Water might also be transported upward by capillary rise (CR) from a shallow water table towards the root zone or even be transferred horizontally by subsurface flow in (SF_{in}) or out (SF_{out}) of the root zone. In many situations, however, except under conditions with large slopes, SF_{in} and SF_{out} are minor and can be ignored. Soil evaporation and crop transpiration deplete water

from the root zone. If all fluxes other than evapotranspiration (ET) can be assessed, the evapotranspiration can be deduced from the change in soil water content (ΔSW) over the selected time period:

$$ET = I + P - RO - DP + CR - \Delta SF - \Delta SW \quad (1.2)$$

A positive ΔSW indicates an increase in total soil water content. Some fluxes such as subsurface flow, deep percolation and capillary rise from a water table are difficult to assess when very short time periods are considered. The soil water balance method can generally give ET estimates over long time periods on the order of week-long or ten-day periods but can only provide accurate daily ET estimates when appropriate and highly accurate soil water content sensors are adopted and appropriate modelling software is used. It may be difficult to capture the increase in evaporation following a large irrigation event when soils are such that one day or more is required for water within the root zone to drain.

FIGURE 1.6
Soil water balance of the root zone



Source: FAO56, 1998

Evapotranspiration from mass balance over large areas, also referred to as the inflow-outflow or mass balance method, is a water balance technique that can be applied over large, integrated areas of land and water to develop watershed scale estimates of ET. These estimates may be useful for calibration of or comparison to watershed and remote-sensing based models. Examples of ET by water balance are applications to large areas such as valleys where the inflow and outflow are determined from stream flow and precipitation measurements, and where the basin is confined to eliminate other significant sources of inflow or outflow such as groundwater underflow. Such applications generally provide only gross seasonal estimates of the average ET from

agricultural and non-agricultural areas within the watershed. The results represent ET from a combination of vegetation types and are generally applicable only to climatic, vegetation and water availability conditions similar to those existing in the study area.

Mass balances require precipitation as the primary input and consequently, the ET estimate can only be as accurate as the precipitation estimate. Fine resolution grids of precipitation on the scale of a km or less may be required in areas having substantial precipitation from convective storms and areas having substantial relief with orographic influence on precipitation amounts. As with soil water balance applications, precipitation data from standard gauges may need to be corrected for under-catch. Accurate measurement of basin or irrigation project outflow is necessary as is locating the measurement area boundaries to minimize groundwater inflows and outflows.

1.5.4 Lysimeters

By isolating the crop root zone from its environment and controlling the processes that are difficult to measure, the different terms in the soil water balance equation can be determined with greater accuracy. This is done in lysimeters where the crop grows in isolated tanks filled with either disturbed or undisturbed soil. In *precision weighing lysimeters*, where the water loss is directly measured by measuring the change of mass of the complete system, evapotranspiration can be obtained with an accuracy of a few hundredths of a millimetre, and small time periods such as an hour or even less can be considered. In *non-weighing drainage lysimeters*, the evapotranspiration for a given time period is determined by deducting the drainage water, collected at the bottom of the lysimeters, from the total water input. Non-weighing drainage lysimeters can generally only provide ET information at a weekly or longer time scale. On the contrary, *static water table lysimeters* can be very accurate and provide for daily time step or longer measurements since ET corresponds to the amount of water transferred during the considered time period from the groundwater. These lysimeters are of particular interest for studies referring to the contribution of the groundwater to the crop under study. They are commonly used to define the target depth of the water table, or when the groundwater is saline.

Many studies have used lysimeter ET measurements for evaluation, calibration and validation of different ET approaches. However, lysimeter measurements are very sensitive to environmental aspects, which are often not well interpreted or even taken into account. Moreover, there tends to be a lack of necessary information reported for associated meteorological and environmental conditions, and processing of lysimeter data in published scientific papers, which calls into question the validity and representativeness of certain reported data sets.

A requirement of lysimeters is that the vegetation both inside and immediately outside of the lysimeter be perfectly matched (same height and leaf area index) to that of the surrounding field. This requirement has historically not been closely adhered to in a majority of lysimeter studies and has resulted in severely erroneous and unrepresentative ET_c and K_c data sets. Typical positive measurement biases include a clothesline effect where lysimeter vegetation is taller than that outside and situations where vegetation from inside the lysimeter creeps outside the lysimeter due to lower vegetation density outside the lysimeter. Lower vegetation density immediately outside a lysimeter can occur due to foot traffic or soil compaction or a thick lysimeter boundary. The outward extension of vegetation from the lysimeter can thereby increase the amount of solar radiation captured by lysimeter vegetation and converted to ET. This causes an over-measurement and overstatement of true field-wide ET.

In a similar manner, one serious problem in lysimeter data is produced by a so-called "bloom effect" on small lysimeters, which means that the area of exposure of plant canopy to solar radiation is higher than the considered effective area of the lysimeter. Finally, radiation emission from warmed sides of a lysimeter into the inner vegetation can increase lysimeter ET to rates higher than the surrounding field.

Recent studies have established guidelines for the requirements of lysimeter-based ET measurements, including a minimum fetch of 50 m around the lysimeter, while plant population, crop height, leaf area, ground coverage, irrigation water management and soil conditions inside the lysimeter must be identical to those in the rest of the field. In conclusion, because a lysimeter represents a small sample of ET for an entire field condition, it is essential that the lysimeter represents an exact reproduction of the field condition. As lysimeters are difficult and expensive to construct and as their operation and maintenance require special care, their use is limited to specific and careful research purposes.

1.5.5 Sap flow systems

Sap flow methods have been used to estimate transpiration from individual plants. The systems introduce a small source of heat into the trunk or branch of a plant and estimate the flow of water in the xylem by either measuring the velocity of a heat pulse carried away from the heat source in the transpiration stream or by the dissipation of heat energy in the stem due to convection in the transpiration stream. These methods do not measure the direct evaporation component of ET, but only the plant transpiration component. This can be an advantage if the goal is to measure plant water use, but is a disadvantage if the goal is to estimate total ET over an extended heterogeneous surface. Three main methods can be employed. Literature suggests that no one set of theory and instrumentation is applicable to all sizes or species of trees.

Heat pulse velocity measurements can determine transpiration in forests and orchards, where the sensors can be applied to any woody stem larger than about 40 mm in diameter. The system is inexpensive, easy to install, and suited to automated data collection. The most commonly used correction method is the compensation heat pulse method. The method may require a knowledge of the wood anatomy of the species being measured to determine sapwood area.

The *Granier heat dissipation* method is suited to transpiration measurements in trees, and it involves inserting a heating needle into the sapwood layer of a large trunk or branch. Heat is applied at a constant rate rather than as a pulse. Different calibration coefficients may be needed for each tree species and the method assumes that sap flow is radially uniform around the trunk.

The *tissue heat balance method* is the most direct sap flux method with the fewest assumptions. It involves wrapping a heating wire around an entire branch segment and measuring conductive heat losses upstream, downstream and radially away from the heat source by thermocouples referenced to the temperature at the heating wire. The energy balance is conducted on the entire stem segment so that no knowledge of the xylem anatomy or area of conducting sapwood is needed. Unfortunately, this is the most difficult method to implement and can only be applied to branches under 50 mm in diameter, because the heat source must be able to uniformly heat the entire segment of the branch around which it is wrapped. Whereas heat pulse and heat dissipation sensors placed in a tree trunk can measure transpiration through the entire tree, numerous tissue heat balance sensors, placed on individual branches would be needed to sample sap flux on a single tree, and a scaling method would then be needed to extrapolate from the branches to the whole tree.

The heat pulse and thermal dissipation methods must be calibrated for each new application, ideally through the use of weighing lysimeters or by pulling water through stem sections of the plants of interest while measuring velocity and heat dissipation parameters with sap flux. Scaling ET from individual limbs to full plants and from full plants to large areas is uncertain due to differences among tree structure, radiation interception per tree, and water availability. Shaded limbs tend to have lower rates of transpiration than sunlit limbs, and limbs lower in a canopy tend to have less aerodynamic exchange of energy and vapor. Besides scaling of transpiration from specific species of trees, additional uncertainty in areal ET is caused by stands of multiple species, the presence of understory vegetation, and evaporation from bare soil.

1.5.6 Remote Sensing

Remote sensing from vegetation indices. The estimation of ET using a vegetation index (VI) is possible due to the generally close correspondence between vegetation amount and transpiration, where, as vegetation cover increases, leaf area increases, and the relative rate of transpiration increases. Generally, the estimation of ET is made using the $K_{c\ act}$ ET_o approach where the value for $K_{c\ act}$ is estimated from the VI. The VI is usually updated over time using satellite or drone imagery. Because the $K_{c\ act}$ from VI represents the transpiration component, challenges with VI-based methods include estimating evaporation from bare soil associated with precipitation and irrigation events. Also challenging is the estimation of reduced ET associated with soil-water shortage, because these processes are not generally shown by the VI. Under conditions of long periods of water stress, vegetation growth is expected to be less and therefore the VI may be less. However, during conditions of short periods of water stress, the VI may appear to be at unstressed levels even though the ET may decline due to partial or full closure of stomates. Therefore, VI-based methods are often best used to estimate ET under adequate water supplies. More discussion on estimating K_c from VI is given in Chapter 5 and Chapter 9.

Remote sensing from energy balance. Remote sensing surface energy balance (RSSEB) methods take advantage of the physical fact that conversion of liquid water to the vapour form requires very substantial amounts of energy. This conversion of liquid to vapour during the evaporation process involves the same physical requirements whether the process is the transpiration of water from inside a plant leaf or the evaporation of water from a wet soil surface. Approximately 2.4 million Joules (2.4 megajoules) of energy are required to evaporate one kilogram of water at room temperature. The energy source can be in the form of heat energy from the surrounding air (this is termed "sensible heat" since it can be sensed with a thermometer), and the energy source also includes the radiant energy from the sun and atmosphere in the form of photons contacting the evaporating surface. Energy from photons is converted into freeing water molecules from a liquid to vapour state.

The surface energy balance determination of $ET_{c\ act}$ is usually conducted using Equation 1.1. RSSEB is useful in that it produces the combined ET including both transpiration from vegetation and evaporation of water from wet soil surfaces. Thermal sensor-based RSSEB models are capable of estimating actual crop ET, $ET_{c\ act}$. RSSEB-based estimation of $ET_{c\ act}$ is useful in situations where $ET_{c\ act}$ is lower than the potential crop ET, ET_c, during times of water shortage or due to poor water management. The more dependable and universal satellite-based models employ a complete surface energy balance (SEB), introduced in Equation 1.1, where ET is computed as a residual of surface energy. This determination requires a thermal imager onboard the satellite. The focus of RSSEB is often on the production of field-scale resolution maps of actual ET, meaning those produced from moderately high resolution thermal sensor-equipped

satellites such as Landsat that have 30-m pixel size, a thermal sensor, and a relatively frequent revisit time (8 days for Landsat). RSSEB can help evaluate variation in $ET_{c\ act}$ across large populations of fields having the same crop type but differing cropping dates, crop varieties or water management.

RSSEB provides a valuable complement to the $K_c\ ET_o$ approach. RSSEB can be used to confirm or calibrate crop coefficients for new crops or in new areas. In turn, the $K_c\ ET_o$ approach is useful in checking or confirming $ET_{c\ act}$ estimates made by RSSEB models. Some RSSEB models, for example, the METRIC model, utilize reference ET to establish upper limits on the estimated $ET_{c\ act}$.

1.5.7 ET computed from meteorological data

Owing to the difficulty of obtaining accurate field measurements, ET is commonly computed from weather data. A large number of empirical or semi-empirical equations have been developed for assessing crop or reference crop evapotranspiration from meteorological data. Some of the methods are only valid under specific climatic and agronomic conditions and cannot be applied under conditions different from those under which they were originally developed.

Numerous researchers have analyzed the performance of the various calculation methods for different locations. As a result of an Expert Consultation held in May 1990, the FAO Penman-Monteith ET_o method is recommended as the standard method for the definition and computation of the grass reference evapotranspiration (ET_o). The ET from crop surfaces under standard conditions is determined by crop coefficients (K_c) that relate ET_c to ET_o. The ET from crop surfaces under non-standard conditions is adjusted by a water stress coefficient (K_s) and/or by modifying the crop coefficient ($ET_{c\ act} = K_s\ K_c\ ET_o = K_{c\ act}\ ET_o$).

1.5.8 ET estimated from pan evaporation

Evaporation from an open water surface provides an index of the integrated effects of radiation, air temperature, air humidity and wind on evapotranspiration. However, differences in the water and cropped surface produce significant differences in the water evaporation from an open water surface and the crop canopy. The pan has proved its practical value and has been used successfully to estimate reference evapotranspiration by observing the evaporation loss from a water surface and applying empirical coefficients to relate pan evaporation to ET_o. The procedure is outlined in Chapter 4.

Chapter 2
FAO Penman-Monteith ET_o equation

This chapter introduces the user to the need to standardize one method to compute reference evapotranspiration (ET_o) from meteorological data. The FAO Penman-Monteith method is recommended as the sole standard ET_o method for determining reference evapotranspiration. The method, its derivation, the required meteorological data and reference weather site conditions, and the corresponding definition of the reference surface are described in this chapter. Also included is the correction of data collected over non-reference weather sites and the application of Quality Assurance and Quality Control (QAQC) to measured meteorological data.

2.1 NEED FOR A STANDARD ET_o METHOD

A large number of more or less empirical methods have been developed over the last 80 years by numerous scientists and specialists worldwide to estimate evapotranspiration using different climatic variables. Relationships were often subject to rigorous local calibrations and have proved to have limited global validity. Testing the accuracy of ET methods under a new set of conditions is laborious, time-consuming and costly, and yet evapotranspiration data are frequently needed at short notice for project planning or irrigation scheduling design. To meet this need, guidelines were developed and published in the FAO Irrigation and Drainage Paper No. 24 "Crop water requirements". To accommodate users with different data availability, four methods were presented to calculate the reference crop evapotranspiration (ET_o): the Blaney-Criddle, radiation, modified Penman and pan evaporation methods. The modified Penman method was considered to offer the best results with a minimum possible error in relation to a living grass reference crop. It was expected that the pan method would give acceptable estimates, depending on the location of the pan. The radiation method was suggested in FAO Paper No. 24 for areas where available climatic data include measured air temperature and sunshine, cloudiness or radiation, but not measured wind speed and air humidity. Finally, the publication proposed the use of the Blaney-Criddle method for areas where available climatic data cover air temperature data only.

These climatic methods to calculate ET_o were all calibrated for ten-day or monthly calculations, not for daily or hourly calculations. The Blaney-Criddle method was recommended for periods of one month or longer. For the pan method, it was suggested that calculations should be done for periods of ten days or longer. Users have not always respected these conditions and calculations have often been done on daily time steps.

Advances in research and the more accurate assessment of crop water use have revealed weaknesses in the methodologies. Numerous researchers have analyzed the performance of the four methods for different locations. Although the results of such analyses could have been influenced by site or measurement conditions, or by bias in weather data collection, it became evident that the proposed methods do not behave the same way in different locations around the world. Deviations from computed to observed values were often found to exceed ranges indicated by FAO. The modified Penman was frequently found to overestimate ET_o, even by up to 20 percent for low evaporative conditions. The other FAO recommended equations showed variable adherence to the reference crop evapotranspiration standard of clipped, cool-season grass.

To evaluate the performance of these and other estimation procedures under different climatological conditions, a major study was undertaken under the auspices of the Committee on Irrigation Water Requirements of the American Society of Civil Engineers (ASCE). The ASCE study analyzed the performance of 20 different methods, using detailed procedures to assess the validity of the methods compared to a set of carefully screened lysimeter data from 11 international locations with a broad range of climatic conditions. The study proved very revealing and showed the widely varying performance of the methods under different climatic conditions. In a parallel study commissioned by the European Community, a consortium of European research institutes evaluated the performance of various evapotranspiration methods using data from different lysimeter studies in Europe.

The studies confirmed the overestimation of the modified Penman introduced in FAO Irrigation and Drainage Paper No. 24, and the variable performance of the different methods depending on their adaptation to local conditions. The comparative studies may be summarized as follows:

- The semi-empirical Penman methods may require local calibration of the wind function to achieve satisfactory results.
- The radiation methods show good results in humid climates where the aerodynamic term is relatively small, but performance in arid conditions is erratic and tends to underestimate evapotranspiration.
- Temperature methods remain empirical and require local calibration in order to achieve satisfactory results. A possible exception is the 1985 Hargreaves-Samani method, which has shown reasonable ET_o results with a global validity. However, it requires validated parameterization.
- Pan evapotranspiration methods clearly reflect the shortcomings of predicting crop evapotranspiration from open water evaporation collected from a shallow pan. The methods are susceptible to the microclimatic conditions under which the pans are operating and the rigour of station maintenance. Their performance proves erratic.
- The relatively accurate and consistent performance of the FAO Penman-Monteith approach (FAO-PM ET_o) in both arid and humid climates has been indicated in both the ASCE and European studies.

The analysis of the performance of the various calculation methods reveals the need for formulating a standard method for the computation of ET_o. The FAO-PM method is recommended as the sole standard method. It is a method with a strong likelihood of correctly predicting ET_o in a wide range of locations and climates and has provision for application in data-short situations. The use of older FAO or other reference ET methods is no longer encouraged. As a result, the FAO-PM method was utilized in the 1998 edition of FAO Irrigation and Drainage Paper No. 56 as the sole standard ET_o method. Its adoption has been widely accepted by the international community as attested by citations in more than 47 000 publications and a large number of tests on overall accuracy. The FAO-PM ET_o method continues to be the ET_o method utilized in this 2025 edition of FAO Irrigation and Drainage Paper No. 56.

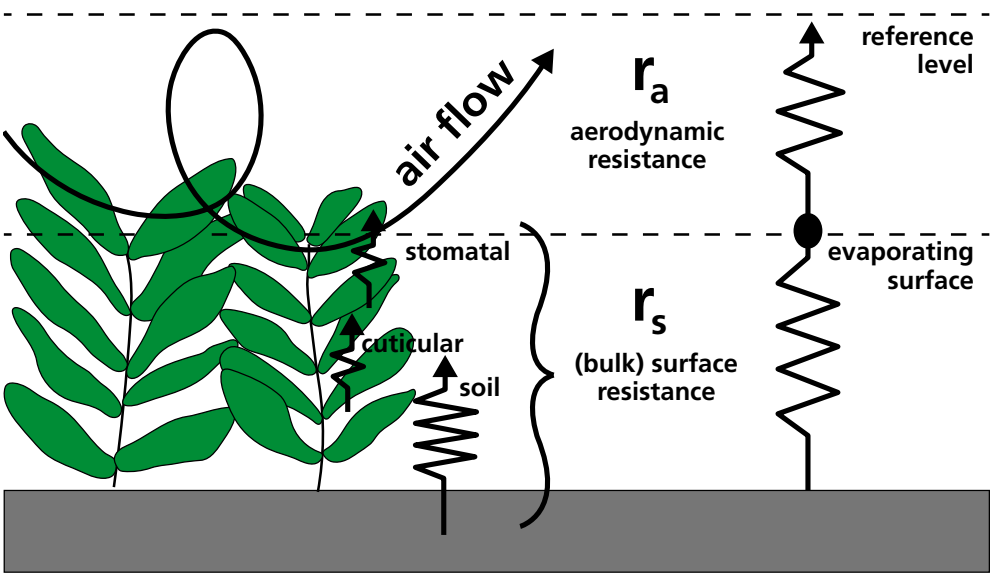
2.2 FORMULATION OF THE PENMAN-MONTEITH EQUATION

2.2.1 Penman-Monteith combination equation

In 1948, Penman combined the energy balance with the mass transfer method and derived an equation to compute the evaporation from an open water surface from standard climatological records of sunshine, temperature, air humidity and wind speed. This so-called combination method was further developed by many researchers, including Monteith in 1965, and was extended to cropped surfaces by introducing resistance factors.

The resistance nomenclature distinguishes between aerodynamic resistance and surface resistance factors (Figure 2.1). The surface resistance parameters are often combined into one parameter, the "bulk" surface resistance parameter, which operates in series with the aerodynamic resistance. The surface resistance, r_s, describes the resistance of vapour flow through stomata openings, total leaf area and soil surface. The aerodynamic resistance, r_a, describes the resistance to vapour and heat flow from the vegetation upward and involves friction from air flowing over vegetative surfaces (Figure 2.1). Although the exchange process in a vegetation layer is too complex to be fully described by the two resistance factors, good correlations can be obtained between measured and calculated evapotranspiration rates, especially for a uniform grass reference surface.

FIGURE 2.1
Simplified representation of the (bulk) surface and aerodynamic resistances for water vapour flow



Source: FAO56, 1998

The derivation and formulation of the Penman and Penman-Monteith combination equations do require a degree of exactness in the representation of temperature and vapour pressure as measured above the surface to that which occurs over a fully transpiring surface. This is because vapour transport from the surface increases vapour pressure a few meters above the surface and cools the air temperature. The Penman-Monteith (PM) combination equation expects these behaviours of the near-surface

boundary layer to occur and incorporates these expectations into the ET estimate. Applying the combination equation with high temperature and low vapour pressure measured over a non-transpiring surface instructs the equation to estimate ET that is biased high. This occurs because the equation, through its formulation, is being instructed to estimate the ET that explains the non-representative high temperature and low vapour pressure being assumed to exist above a fully transpiring surface. Weather data collected above a non- or low-transpiring surface is not synchronized with the assumptions involved in the formulation of the combination equation.

The Penman-Monteith (PM) combination equation (Figure 2.1) is:

$$\lambda ET = \frac{\Delta (R_n - G) + \rho_a c_p \dfrac{(e_s - e_a)}{r_a}}{\Delta + \gamma \left(1 + \dfrac{r_s}{r_a}\right)} \qquad (2.1)$$

where λET is the latent heat flux representing the energy consumed by the ET process,

R_n is the net radiation,

G is the soil heat flux,

$(e_s - e_a)$ represents the vapour pressure deficit of the air,

ρ_a is the mean air density at constant pressure,

c_p is the specific heat of the air,

Δ represents the slope of the saturation vapour pressure-temperature relationship,

γ is the psychrometric constant,

r_s and r_a are the (bulk) surface and aerodynamic resistances respectively.

The parameters of the equation are defined in Chapter 3.

The Penman-Monteith combination equation (2.1) as formulated above includes all parameters that govern energy exchange and corresponding latent heat flux (evapotranspiration) from uniform expanses of transpiring vegetation. Most of the parameters are measured or can be readily calculated from weather data. This is due to the elimination of surface temperature and vapour pressure by combining the energy balance with the mass transfer method in the combination equation. The equation can be utilized for the direct calculation of any crop evapotranspiration as the surface and aerodynamic resistances are crop specific.

2.2.2 Aerodynamic resistance (r_a)

The transfer of heat and water vapour from the evaporating surface into the air above the canopy is determined by the aerodynamic resistance:

$$r_a = \frac{\ln\left[\dfrac{z_m - d}{z_{om}}\right] \ln\left[\dfrac{z_h - d}{z_{oh}}\right]}{k^2 u_z} \qquad (2.2)$$

where r_a aerodynamic resistance [s m^{-1}],

z_m height of wind measurements [m],

z_h height of humidity measurements [m],

d	zero plane displacement height [m],
z_{om}	roughness length governing momentum transfer [m],
z_{oh}	roughness length governing the transfer of heat and vapour [m],
k	von Karman's constant, 0.41 [-],
u_z	wind speed at height z [m s^{-1}].

The equation is restricted to neutral or nearly neutral stability conditions, i.e. where temperature, atmospheric pressure, and wind velocity distributions follow nearly adiabatic conditions (only small amounts of heat exchange) and where vertical transport due to buoyancy effects is small. The application of the equation for short time periods (hourly or less) may require the inclusion of corrections for stability. However, when predicting ET_o from the defined well-watered reference surface, heat exchanged is small, and therefore stability correction is normally not required.

Many studies have explored the nature of the wind regime in plant canopies. Zero displacement heights and roughness lengths have to be considered when the surface is covered by vegetation. The factors depend upon the crop height and architecture. Several empirical equations for the estimate of d, z_{om} and z_{oh} have been developed. The derivation of the aerodynamic resistance for the grass reference surface is presented in Box 2.1.

BOX 2.1
The aerodynamic resistance for a grass reference surface

For a wide range of crops, the zero plane displacement height, d [m], and the roughness length governing momentum transfer, z_{om} [m], can be estimated from the crop height h [m] by the following equations:

$$d = 2/3\ h$$

$$z_{om} = 0.123\ h$$

The roughness length governing the transfer of heat and vapour, z_{oh} [m], can be approximated by:

$$z_{oh} = 0.1\ z_{om}$$

Assuming a constant crop height of 0.12 m and a standardized height for wind speed, air temperature and humidity at 2 m ($z_m = z_h = 2$ m), the aerodynamic resistance r_a [s m^{-1}] for the standardized 0.12 m grass reference surface becomes (Equation 2.2):

$$r_a = \frac{\ln\left[\frac{2-2/3(0.12)}{0.123(0.12)}\right]\ln\left[\frac{2-2/3(0.12)}{(0.1)0.123(0.12)}\right]}{(0.41)^2\ u_2} = \frac{208}{u_2}$$

where u_2 is the wind speed [m s^{-1}] at 2 m.

2.2.3 (Bulk) surface resistance (r_s)

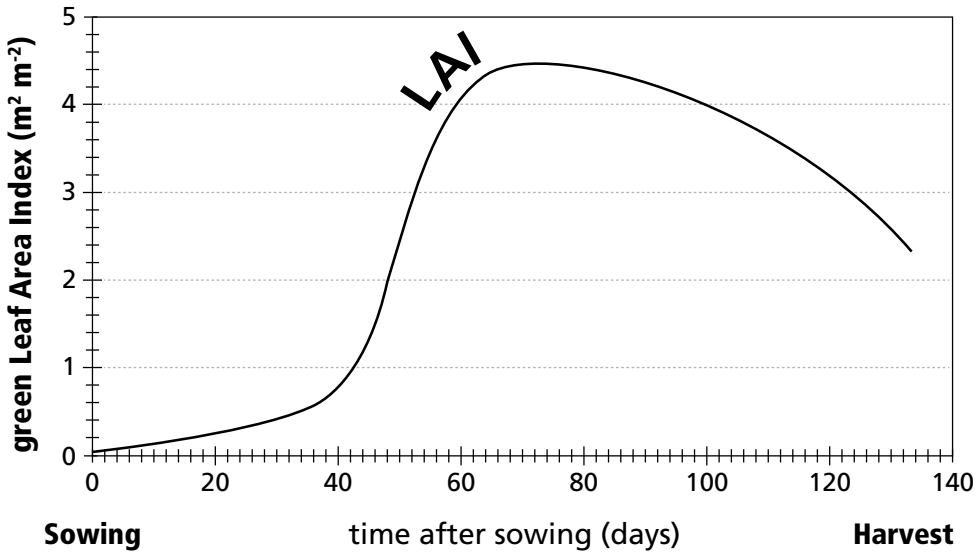
The "bulk" surface resistance describes the resistance to vapour flow through the transpiring crop and evaporating soil surface. Where the vegetation does not completely cover the soil, the resistance factor should indeed include the effects of the evaporation from the soil surface. If the crop is not transpiring at a potential rate, the resistance depends also on the water status of the vegetation. An acceptable approximation to a much more complex relation of the surface resistance of dense full cover vegetation is:

$$r_S = \frac{r_l}{LAI_{active}} \tag{2.3}$$

where r_s (bulk) surface resistance [s m^{-1}],
r_l bulk stomatal resistance of a well-illuminated leaf [s m^{-1}],
LAI_{active} active (sunlit) leaf area index [m^2 (leaf area) m^{-2} (soil surface)].

The Leaf Area Index (LAI), a dimensionless quantity, is the leaf area (upper side only) per unit area of soil below it. It is expressed as m^2 leaf area per m^2 ground area. The active LAI is the index of the leaf area that actively contributes to surface heat and vapour transfer. It is generally the upper, sunlit portion of a dense canopy. The LAI values for various crops differ widely but values of 3 to 5 are common for many mature crops. For a given crop, green LAI changes throughout the season and normally reaches its maximum before or at flowering (Figure 2.2). LAI further depends on the plant density and the crop variety.

FIGURE 2.2
Typical presentation of the variation in the active (green) Leaf Area Index over the growing season for a maize crop



Source: FAO56, 1998

The bulk stomatal resistance, r_l, is the average resistance of an individual sunlit leaf. This resistance is crop specific and differs among crop varieties and crop management. It usually increases as the crop ages and begins to ripen. There is, however, a lack of consolidated information on changes in r_l over time for the different crops. The information available in the literature on stomatal conductance or resistance is often oriented toward physiological or ecophysiological studies. The derivation of the surface resistance for the grass reference surface is presented in Box 2.2.

BOX 2.2
The (bulk) surface resistance for a grass reference crop

A general equation for LAI_{active} is:

$$LAI_{active} = 0.5\ LAI$$

This takes into consideration the fact that generally only the upper half of dense clipped grass is actively contributing to the surface heat and vapour transfer. For clipped grass:

$$LAI = 24\ h$$

where h is the crop height [m].

The stomatal resistance, r_l, of a single leaf has a value of about 100 s m^{-1} under well-watered conditions when averaged over a day and when 24-h time steps are used for weather data and calculations of ET. By assuming a crop height of 0.12 m, the surface resistance, r_s [s m^{-1}], for the grass reference surface becomes, for 24-h time steps (Eq. 2.3):

$$r_{s\ 24\text{-}h} = \frac{100}{0.5\ (24)\ (0.12)} \approx 70\ s\ m^{-1}$$

For hourly and shorter calculation time steps, the value for r_l for a single leaf tends to have a lower value of about 60 to 80 s m^{-1} under well-watered conditions and during daylight hours. By assuming a crop height of 0.12 m, the surface resistance, r_s [s m^{-1}], for the grass reference surface becomes, for hourly or shorter calculation time steps (Eq. 2.3):

$$r_{s\ hrly} = \frac{70}{0.5\ (24)\ (0.12)} \approx 50\ s\ m^{-1}$$

The stomatal resistance, r_l, is influenced by climate and by water availability. However, influences vary from one crop to another, and different varieties can be affected differently. The resistance increases when the crop is water stressed and the soil water availability limits crop evapotranspiration. Some studies indicate that stomatal resistance is influenced to some extent by radiation intensity, temperature, and vapour pressure deficit.

2.3 REFERENCE SURFACE

To obviate the need to define unique evaporation parameters for each crop and stage of growth, the concept of a reference surface was introduced. Evapotranspiration rates of the various crops are related to the evapotranspiration rate from the reference surface (ET_o) by means of crop coefficients.

In the past, an open water surface has been proposed as a reference surface. However, the differences in aerodynamic, vegetation control and radiation characteristics present a strong challenge in relating ET to measurements of free water evaporation. Relating ET_o to a specific crop has the advantage of incorporating the biological and physical processes involved in ET from cropped surfaces.

Grass, together with alfalfa, is a well-studied crop regarding its aerodynamic and surface characteristics and is accepted worldwide as a reference surface. Because the resistance to diffusion of vapour strongly depends on crop height, ground cover, LAI and soil moisture conditions, the characteristics of the reference crop should be well

defined and fixed. Changes in crop height result in variations in the roughness and LAI. Consequently, the associated canopy and aerodynamic resistances will vary appreciably with time. Moreover, water stress and the degree of ground cover have an effect on the resistances and also on the albedo.

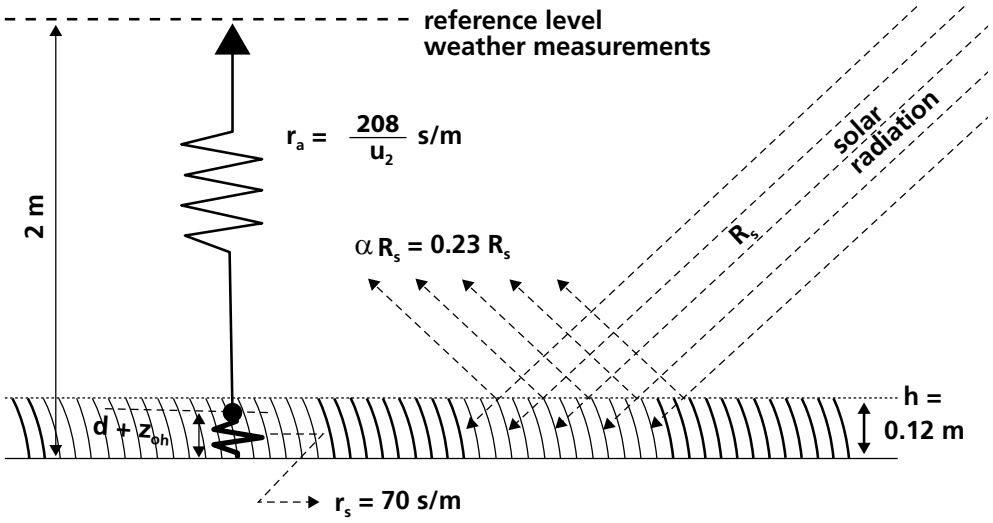
To avoid problems of local calibration which would require demanding and expensive studies, a hypothetical grass reference has been selected with well-defined properties. Difficulties with a living grass reference result from the fact that the grass variety and morphology can significantly affect the evapotranspiration rate, especially during peak water use. Large differences may exist between warm-season and cool-season grass types. Cool-season grasses have a lower degree of stomatal control and hence higher rates of evapotranspiration. However, it may be difficult to grow cool-season grasses in some arid, tropical climates.

The 1990 FAO Expert Consultation on Revision of FAO Methodologies for Crop Water Requirements accepted the following unambiguous definition for the reference surface when applied using 24-hour calculation timesteps (Figure 2.3):

> *A hypothetical reference crop with an assumed crop height of 0.12 m, a fixed surface resistance of 70 s m^{-1} and an albedo of 0.23.*

The reference surface closely resembles an extensive surface of cool-season green grass of uniform height, actively growing, completely shading the ground and with adequate water. The requirements that the grass surface should be extensive and uniform result from the assumption that all fluxes are one-dimensional upwards and that the values for air temperature, vapour pressure and wind speed two to three meters above the surface are in equilibrium with the ET_o rate of the surface. An ideal example of an extensive surface is an irrigated field of more than 4 ha in size.

FIGURE 2.3
Characteristics of the hypotetical reference crop for 24-hour time steps



Source: FAO56, 1998

The FAO Penman-Monteith method is selected as the method by which the evapotranspiration of this reference surface (ET_o) can be unambiguously determined, and as the method which provides consistent ET_o values in all regions and climates.

2.4 FAO PENMAN-MONTEITH EQUATION

2.4.1 Equation

The consultation of experts and researchers was organized by FAO in May 1990, in collaboration with the International Commission for Irrigation and Drainage and with the World Meteorological Organization, to review the FAO methodologies on crop water requirements and to advise on the revision and update of procedures. The panel of experts recommended the adoption of the Penman-Monteith combination equation as a new standard for reference evapotranspiration and advised on procedures for the calculation of the various parameters. By *defining the reference crop as a hypothetical crop with an assumed height of 0.12 m having a surface resistance of 70 s m^{-1} for 24-hour time steps and an albedo of 0.23, closely resembling the evaporation of an extensive surface of cool-season green grass of uniform height, actively growing and adequately watered*, the FAO Penman-Monteith ET_o equation was developed (Figure 2.3). Research and experience demonstrated that the method provides ET_o values consistent with actual crop water use data worldwide.

From the original Penman-Monteith combination equation (Equation 2.1) and the equations of the aerodynamic (Equation 2.2) and surface resistance (Equation 2.3), the FAO Penman-Monteith method to estimate ET_o can be derived (Box 2.3) *for 24-hour calculation time steps:*

$$ET_o = \frac{0.408\,\Delta\,(R_n - G) + \gamma \dfrac{900}{T + 273}\,u_2\,(e_s - e_a)}{\Delta + \gamma\,(1 + 0.34\,u_2)} \qquad (2.4)$$

where
- ET_o reference evapotranspiration [mm day^{-1}],
- R_n net radiation at the crop surface [MJ m^{-2} day^{-1}],
- G soil heat flux density [MJ m^{-2} day^{-1}],
- T mean daily air temperature at 2 m height [°C],
- u_2 wind speed at 2 m height [m s^{-1}],
- e_s saturation vapour pressure [kPa],
- e_a actual vapour pressure [kPa],
- $e_s - e_a$ saturation vapour pressure deficit [kPa],
- Δ slope vapour pressure curve [kPa °C^{-1}],
- γ psychrometric constant [kPa °C^{-1}].

The FAO-PM method to estimate ET_o (Box 2.3) for *hourly or shorter calculation time steps* has been recommended to use r_s = 50 s m^{-1} for daytime and r_s = 200 s m^{-1} for night time periods. This use also provides, on average, good agreement with computations made on a 24-hour time step basis. Hence, the following equations are recommended for hourly calculation time steps during daytime ($R_n \geq 0$) and for hourly time steps during night time ($R_n < 0$).

- **hourly calculation time steps during daytime (when $R_n \geq 0$):**

$$ET_o = \frac{0.408\,\Delta\,(R_n - G) + \gamma\,\frac{37}{T + 273}\,u_2\,(e_s - e_a)}{\Delta + \gamma\,(1 + 0.24\,u_2)} \qquad (2.5a)$$

where ET_o reference evapotranspiration [mm h^{-1}],
R_n net radiation at the crop surface [MJ m^{-2} h^{-1}],
G soil heat flux density [MJ m^{-2} h^{-1}],
T mean hourly air temperature at 2 m height [°C],
u_2 wind speed at 2 m height [m s^{-1}],
e_s saturation vapour pressure [kPa],
e_a actual vapour pressure [kPa],
$e_s - e_a$ saturation vapour pressure deficit [kPa],
Δ slope vapour pressure curve [kPa °C^{-1}],
γ psychrometric constant [kPa °C^{-1}].

- **hourly calculation time steps during nighttime (when $R_n < 0$):**

$$ET_o = \frac{0.408\,\Delta\,(R_n - G) + \gamma\,\frac{37}{T + 273}\,u_2\,(e_s - e_a)}{\Delta + \gamma\,(1 + 0.96\,u_2)} \qquad (2.5b)$$

In Equation 2.5a for hourly time steps, the coefficient in the numerator is 37 rather than 900 that is used for 24-hour time steps, and the coefficient in the denominator is 0.24 rather than the 0.34 that is used for 24-hour. During nighttime (when $R_n < 0$) the 0.24 coefficient in the denominator of Equation 2.5 (hourly) is changed to 0.96 as shown in Equation 2.5b.

BOX 2.3

Derivation of the FAO-PM ET_o equation for the grass reference crop

With standardized height for wind speed, temperature, and humidity measurements at 2 m ($z_m = z_h = 2$ m) and the crop height h = 0.12 m, the aerodynamic (r_a) and surface (r_s) resistances become (Boxes 2.1, 2.2):

$$r_a = 208/u_2, \text{ s m}^{-1} \text{ (with } u_2 \text{ wind speed at 2 m height)}$$

for 24-hour calculation time steps:

$$r_s = 70 \text{ s m}^{-1}$$

$$(1 + r_s/r_a) = (1 + 0.34\ u_2)$$

for hourly or shorter calculation time steps during daytime ($R_n \geq 0$):

$$r_s = 50 \text{ s m}^{-1}$$

$$(1 + r_s/r_a) = (1 + 0.24\ u_2)$$

for hourly or shorter calculation timesteps during nighttime ($R_n < 0$):

$$r_s = 200 \text{ s m}^{-1}$$

$$(1 + r_s/r_a) = (1 + 0.96\ u_2)$$

R_n and G is energy available per unit area and expressed in MJ m^{-2} day^{-1}. To convert the energy units for radiation to equivalent water depths (mm) the latent heat of vaporization, λ, is used as a conversion factor (Chapter 1).

The conversion from energy values to equivalent depths of water or vice versa is given by the equation of radiation:

$$\text{Radiation [mm day}^{-1}] \approx \frac{\text{Radiation [MJ day}^{-1}]}{2.45} = 0.408 \text{ Radiation [MJ day}^{-1}]$$

By substituting the specific heat at constant pressure (c_p) with a rearrangement of the c_p equation in Chapter 3:

$$c_p = \frac{\gamma\, \varepsilon\, \lambda}{P}$$

where γ psychrometric constant [kPa °C^{-1}],
ε ratio of molecular weight of water vapour/dry air = 0.622
λ latent heat of vaporization, 2.45 [MJ kg^{-1}],
P atmospheric pressure [kPa].

and considering the ideal gas law for the mean air density at constant pressure (ρ_a):

$$\rho_a = \frac{P}{T_{Kv}\, R}$$

where T_{Kv}, the virtual temperature in K, may be substituted by $T_{Kv} = 1.01(T + 273)$ results in:

$$\frac{c_p\, \rho_a}{r_a} = \frac{\gamma\, \varepsilon\, \lambda}{1.01(T + 273)\, R\, (208)}\, u_2 \quad [\text{MJ m}^{-2}\,°\text{C}^{-1}\,\text{s}^{-1}]$$

BOX 2.3 (CONT.)

where c_p specific heat at constant pressure [MJ kg^{-1} $°C^{-1}$],
ρ_a mean air density at constant pressure [kg m^{-3}],
r_a aerodynamic resistance [s m^{-1}],
γ psychrometric constant [kPa $°C^{-1}$],
ε ratio molecular weight of water vapour/dry air = 0.622,
λ latent heat of vaporization [MJ kg^{-1}],
u_2 wind speed at 2 m [m s^{-1}],
R specific gas constant = 0.287 kJ kg^{-1} K^{-1},
T air temperature [°C],
P atmospheric pressure [kPa],

Thus,

$$\frac{c_p \rho_a}{r_a} = 86400 \frac{\gamma\,(0.622)\,\lambda}{1.01(T + 273)\,(0.287)\,(208)} u_2 \quad [\text{MJ m}^{-2}\,°\text{C}^{-1}\,\text{day}^{-1}]$$

that, when divided by λ (λ = 2.45), for 24-h calculation time steps, results as:

$$\frac{c_p \rho_a}{r_a} \approx \gamma \frac{900}{T + 273} u_2 \quad [\text{mm}\,°\text{C}^{-1}\,\text{day}^{-1}]$$

For hourly or shorter calculation time steps is:

$$\frac{c_p \rho_a}{r_a} \approx \gamma \frac{37}{T + 273} u_2 \quad [\text{mm}\,°\text{C}^{-1}\,\text{hour}^{-1}]$$

The reference evapotranspiration, ET_o, provides, thus, a standard to which:

- evapotranspiration at different periods of the year or in other regions can be compared;
- evapotranspiration of other crops can be related.

The FAO-PM ET_o equation uses standard climatological records of solar radiation (sunshine), air temperature, humidity and wind speed. To ensure the integrity of computations, the weather measurements should be made at 2 m (or converted to that height) above an extensive surface of green grass, shading the ground and not short of water. Wind speed may often be measured at a height other than 2 m, for example, at 3 m or 10 m, to avoid influences by nearby tall crops. Measurements at those heights are transformed to an equivalent wind speed at 2 m height using a logarithmic transformation as shown later.

No weather-based evapotranspiration equation can be expected to predict evapotranspiration perfectly under every climatic situation due to simplification in formulation and errors in data measurement. It is probable that precision instruments under excellent environmental and biological management conditions will show the FAO-PM ET_o equation to deviate at times from true measurements of grass ET_o. However, the Expert Consultation agreed to use the hypothetical reference definition of the FAO-PM ET_o equation as the definition for grass ET_o when deriving and expressing crop coefficients.

During the last 25 years, the time between the publication of the first edition of FAO Paper 56 in 1998 and this second edition, a number of studies have been produced showing the relatively good accuracy, consistency and dependability of the FAO-PM ET_o equation and of the ET estimates computed as K_c ET_o. This indicates that the values assigned to the resistances of the FAO-PM ET_o equation and the standardized calculation of important parameters were properly selected.

It is important, *when comparing the FAO-PM ET_o equation to ET_o measurements, that the full Penman-Monteith equation (Equation 2.1) and associated equations for r_a and r_s (Equations 2.2 and 2.3) must be used to enable accounting for variation in ET due to variation in height of the grass measured.* Variations in measurement height can significantly change LAI, d and z_{om} and the corresponding ET_o measurement and predicted value. When evaluating results, it should be noted that local environmental and management factors, such as watering frequency, also affect ET_o observations.

The FAO-PM ET_o equation is a close, simple representation of the physical and physiological factors governing the evapotranspiration process. By using the FAO-PM ET_o equation, one may calculate crop coefficients at research sites by relating the measured crop evapotranspiration (ET_c) with the calculated ET_o, i.e. $K_c = ET_c/ET_o$. In the crop coefficient approach, differences in the crop canopy and aerodynamic resistance relative to the hypothetical reference crop are accounted for within the crop coefficient. The K_c factor serves as an aggregation of the physical and physiological differences between crops and the reference definition.

2.4.2 Impacts of advection

Many irrigated areas around the globe are located in semiarid and arid regions where irrigated agriculture is downwind of extensive regions of dryness. The upwind dry regions, through surface heating, increase air temperature and dryness to the overlying air mass. *Two types of advection can exist, local advection and regional advection, where advection is defined as the transport of warmer, drier air onto an irrigated surface.* When warmer, drier air is transported onto an irrigated surface, it increases the air temperature and decreases the humidity above the irrigated surface as compared to what would occur in the absence of the advection. The advection will tend to increase the corresponding ET rate.

Local advection

Local advection occurs when a nearby dry field generates warmer air directly upwind of an adjacent irrigated field. That warmer air can create an increase in ET at the edges of the downwind irrigated field and into the field for tens of meters. After the establishment of a new equilibrium boundary layer (EBL) some tens of meters into the irrigated field, the effects of the local advection are reduced.

The definition for ET_o given earlier in this chapter describes ET_o as the ET from an extensive surface of well-watered, clipped grass. The extensive surface is specified so that the impacts of local advection on the edges of a field are made small in proportion to the total field ET, and so that weather measurements are less influenced by local advection. The ET_o estimate is, thereby, intended to represent the basis for the ET produced from the majority of a medium to large, irrigated field. That ET is calculated as K_c ET_o. In other words, the ET_o estimate represents ET that occurs for the average well-watered field when surrounded by other agricultural fields that may have low or high amounts of vegetated cover at various times during the growing season. For this reason, *it is mandatory that weather stations are set up under reference conditions, i.e. over a well-watered grass field surrounded by other agricultural fields.*

Regional advection

Contrary to the impact of local advection, irrigated areas that are downwind of large areas of dryness, such as in semiarid and arid climates, are influenced by regional advection. Regional advection transports warmer and drier air from an arid planetary boundary layer toward an irrigated surface. The transport of warmer and drier air affects all fields of an irrigated area and is made possible by turbulent eddy transfer. The regional advection impacts the air temperature and humidity gradients above the irrigated surface, effectively increasing the air temperature and reducing the humidity at the 2 m height. These changes increase ET from the irrigated surface. The changes in gradients over the irrigated fields are assumed to be similar to those experienced at a reference weather station that has a well-watered, transpiring surface and is located in a similar environment. In those situations, the FAO Penman-Monteith equation responds to the impacted temperature and humidity gradients established by the weather data and produces higher ET_o. That higher ET_o, in turn, corresponds to the higher ET that occurs from well-watered fields under regional advection. This phenomenon is common to irrigated fields in arid and semiarid regions and is accurately treated by the FAO Penman-Monteith ET_o equation due to its use of solar radiation, air temperature, air humidity and wind speed measurements.

2.4.3 Data

Apart from the site location, the FAO-PM ET_o equation requires air temperature, humidity, radiation and wind speed data for hourly, daily, weekly, ten-day or monthly calculations. The computation of all data required for the calculation of the reference evapotranspiration is given in Chapter 3. Annex 3 presents methods for the completion of the weather data series and for analysis of the homogeneity of the data series. It is important to verify the units in which the weather data are reported. Factors to convert common units to standard unit are presented in Annex 1.

Location

The altitude above sea level (m) and latitude (degrees north or south) of the location should be specified. These data are needed to adjust some weather parameters for the local average value of atmospheric pressure (a function of the site elevation above mean sea level) and to compute extraterrestrial radiation (R_a) and, in some cases, daylight hours (N). In the calculation procedures for R_a and N, the latitude is expressed in radians (i.e. decimal degrees times $\pi/180$). A positive value is used for the northern hemisphere and a negative value for the southern hemisphere.

Temperature

The (average) daily maximum and minimum air temperatures in degrees Celsius (°C) are required for 24-hour calculation time steps. Where only (average) mean daily temperatures are available, the calculations can still be executed but some underestimation of ET_o will probably occur due to the non-linearity of the saturation vapour pressure-temperature relationship. Using mean air temperature instead of maximum and minimum air temperatures yields a lower saturation vapour pressure e_s, and hence a lower vapour pressure difference ($e_s - e_a$), and a lower reference evapotranspiration estimate.

Air humidity

The (average) daily actual vapour pressure, e_a, in kilopascals (kPa) is required. The actual vapour pressure, where not available, can be derived from maximum and minimum relative humidity (%), psychrometric data (dry and wet bulb temperatures in °C) or dewpoint temperature (°C) according to the procedures outlined in Chapter 3.

Solar radiation

The (average) daily net radiation expressed in megajoules per square meter per day (MJ m^{-2} day^{-1}) is required. These data are not commonly available but can be derived from the (average) shortwave radiation measured with a pyranometer or from the (average) daily actual duration of bright sunshine (hours per day) measured with a (Campbell-Stokes) sunshine recorder. The use of a sunshine recorder is becoming less common and a pyranometer is the preferred instrument. The calculation procedures are outlined in Chapter 3.

Wind speed

The (average) daily wind speed in meters per second (m s^{-1}) measured at 2 to 10 m above the ground level is required. It is important to verify the height at which wind speed is measured, as wind speeds measured at different heights above the ground surface differ. The calculation procedure to adjust wind speed to the standard height of 2 m, prior to entry into FAO-PM ET_o equation, is presented in Chapter 3.

2.4.4 Missing climatic data

Situations might occur where data for some weather variables are missing. The use of an alternative ET_o calculation procedure, requiring only limited meteorological parameters, should generally be avoided. It is recommended that one calculates ET_o using the standard FAO-PM ET_o method after resolving the specific problem of the missing data. Procedures for estimating missing climatic data are outlined in Chapter 3 with some material at the end of this Chapter. Differences between ET_o values obtained with the FAO-PM ET_o equation with, on the one hand, a limited data set and, on the other hand, a full data set, are expected, and have been observed to be smaller than or of similar magnitude to the differences resulting from the use of an alternative ET_o equation.

Even where the data set contains only maximum and minimum air temperature it is still possible to obtain reasonable estimates of daily, ten-day or monthly ET_o with the FAO-PM ET_o equation. As outlined in Chapter 3, radiation data can be derived from the air temperature difference, or, along with wind speed and humidity data, can be imported from a nearby weather station, or from reanalysis weather data, or from gridded stationary satellite radiation data. Humidity data can be estimated from daily minimum air temperature.

The procedures for estimating missing data should be validated at the regional level. This can be done for weather stations with full data sets by comparing ET_o calculated with full and with limited data sets. The ratio should be close to one. Where the ratio deviates significantly from one, the ratio can be used as a correction factor for estimates made with the limited data set. Where the standard error of estimate exceeds 20 percent of the mean ET_o, a sensitivity analysis should be performed to determine causes (and limits) for the method utilized to import the missing data. A validation should be completed for each month and variable for the monthly ET_o estimates as well as for the daily estimates.

2.4.5 Importance of utilizing weather data collected over transpiring (reference) surfaces

The development of the Penman and Penman-Monteith combination equations eliminated the need to measure or estimate surface temperature and vapour pressure at the surface by combining the energy balance with the mass transfer method. A requirement is that the equations should only be applied using standard weather measurements of air temperature, vapour pressure, solar radiation, and wind speed

collected over transpiring surfaces. The assumptions in the combination equation, including the calculation of the slope of the vapour pressure curve (Δ) using air temperature, require that the temperature of the surface be similar to the air temperature, or within a few degrees. This is usually the case for the reference ET_o because ET from a well-watered grassed surface consumes most of the energy available from R_n and G, so that sensible heat, H, is small. A small value for H means that differences between surface temperature and air temperature are also small.

Vapour transport from a transpiring surface increases the actual vapour pressure of the air for several meters above the surface and cools the air temperature. This is in contrast to a dry, non-transpiring surface, where heating of the surface adds substantial heat to the air above the surface, but no vapour. The FAO-PM ET_o equation requires that the near-surface boundary layer that resides for several meters above the surface behaves as that over a transpiring surface where the air temperature is cooled, and vapour pressure is increased. These expectations carry into the estimation of Δ using air temperature, the calculation of saturation and actual vapour pressure and net radiation. These calculations all utilize the measured air temperature and vapour pressure values.

Applying the combination equation using high temperature and low vapour pressure measured over a non-transpiring surface confuses the equation by instructing it to estimate the ET rate that would be required to occur in spite of high air temperature and low vapour pressure over a transpiring surface. This causes the combination equation to function under an unrealistic and perhaps even impossible situation. This impossible situation must be mitigated by limiting weather data sets to those that are collected over large, well-watered surfaces. Or, for non-conforming weather data sets that are influenced by the dryness of the measurement site, or by a weather grid-generating model, the data sets should be corrected by adjusting the near-surface air temperature and/or humidity data to levels expected and realistic for well-watered conditions.

In summary, an important requirement in applying the FAO-PM ET_o method to calculate ET_o for irrigated agriculture is the assumption that the surface underlying the weather measurements is evaporating or transpiring at nearly the same rate as the ET_o calculation produced by the FAO-PM ET_o equation. This condition is required so that the near-surface profiles for air temperature, humidity and wind speed are in equilibrium with a well-watered surface represented by the FAO-PM ET_o. In addition, the net radiation (R_n) estimate for the standardized FAO-PM ET_o equation assumes a relatively cooled surface having long-wave radiation emission that can be as much as 100 W m^{-2} less than the long-wave emission associated with a warmer non-evaporating or non-transpiring surface.

The next section describes relatively simple empirical, but effective, procedures to adjust air temperature and/or humidity (dewpoint temperature) data. *That adjustment or conditioning is done prior to using the weather data in the FAO-PM ET_o equation.*

2.5 ADJUSTMENT AND CORRECTION OF WEATHER DATA TO ACCOUNT FOR THE DRYNESS OF THE WEATHER STATION ENVIRONMENT

The concept on which the FAO-PM ET_o method for computing ET_o is based requires that weather data be measured under environmental conditions that correspond to, and are synchronized with, the definition of reference evapotranspiration. In other words, the weather data input to the FAO-PM ET_o equation are to be measured above a surface that corresponds to the reference definition: an extensive grass crop that is

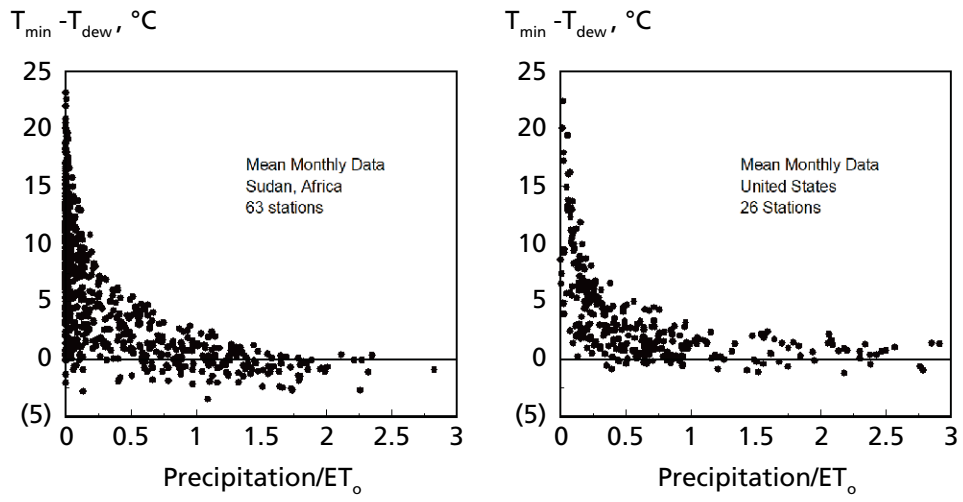
actively evapotranspiring, or at least in an environment having healthy transpiring vegetation that is not short of water. Computing ET_o using standardized estimates for $(R_n - G)$, r_a and r_s, ET_o will be overstated if calculated using air temperature and vapour pressure data collected in an arid, non-reference setting. A correction is therefore required to bring temperature and humidity data sets closer to those that would have been measured under reference conditions.

In an environment having healthy vegetation and adequate soil moisture (reference conditions), minimum air temperature T_{min} usually approaches dewpoint temperature, T_{dew}. This especially occurs if the wind speed dies down by early morning and when soil moisture is high. Air temperatures decrease during night-time by evaporative cooling when vapour pressure deficit (VPD) is positive and due to surface cooling caused by the emission of long-wave radiation. For reference conditions, simple relationships can be used to assess whether weather data adhere to reference ET conditions and to replace measured humidity data when the data have not been collected under reference ET conditions.

For non-reference weather sites and data sets, soil moisture and/or vegetation limitations cause the ET of the site to be substantially less than ET_o. As a result, T_{min} may remain well above the reported T_{dew}. One cause of this phenomenon is the large "reservoir" of sensible heat created during daytime in the atmosphere over an arid region that is transferred towards the dry surface during the night, reducing the effect of cooling by long wave radiation. This surface warming is sustained by the lack of evaporative cooling during nighttime due to the lack of soil moisture at a non-reference site. The result is the air temperature and T_{dew} measurements that are uncharacteristic of a transpiring reference site and that provide the FAO-PM equation with a wrong idea of the underlying ET_o. This phenomenon can be observed in Figure 2.4.

FIGURE 2.4

Comparison of differences between the monthly values for minimum and dewpoint temperature (T_{min} - T_{dew}) corresponding to monthly ratios of precipitation/ET_o within Sudan, Africa, and within United States



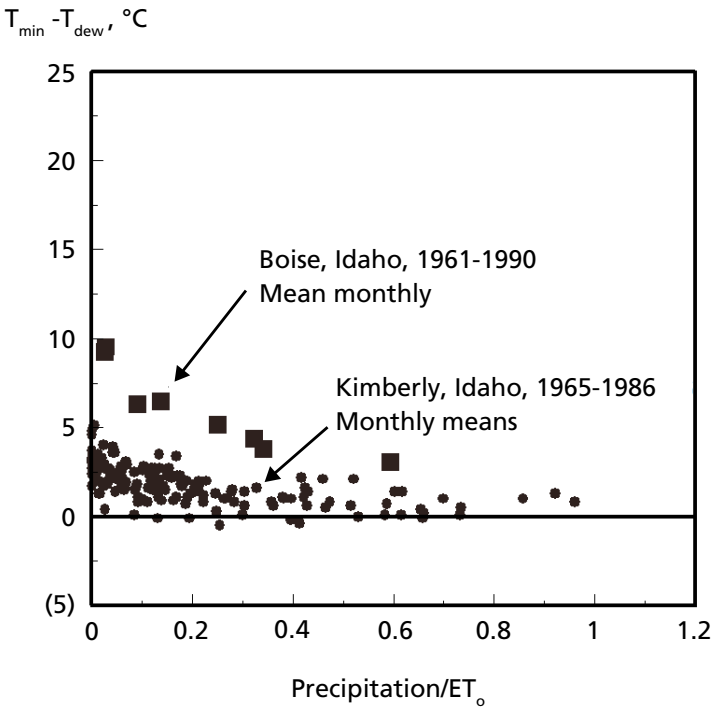
Source: Authors' own elaboration

In Figure 2.4, monthly means for $T_{min} - T_{dew}$ are plotted for weather stations operated by the national governments of two countries, Sudan and the United States. The data are plotted against the monthly ratios of precipitation to reference ET_o. The P/ET_o ratios indicate the availability of soil water to support reference (well-watered) conditions during any particular month in the absence of irrigation. As illustrated by the data, T_{min} approaches T_{dew} for nearly all stations when the ratio P/ET_o approaches and exceeds 1.0. This indicates that P is sufficient to supply potential ET rates and to cool and humidify near-surface air. When $P/ET_o < 1.0$, then the aridity of the area causes heating and reduces humidification, and T_{min} thereby may substantially exceed measured T_{dew}. The exception is for those weather stations that have $P/ET_o < 1.0$ but are irrigated or have adequate soil water reserves from a prior month. The similarity between data of Sudan and the United States indicates that this is a general phenomenon.

An additional comparison is given in Figure 2.5, where $T_{min} - T_{dew}$ are compared for two semiarid locations in Idaho, the United States, that are separated by 200 km. One location, Kimberly, is a reference site in the middle of a very large irrigated area. The other, Boise, is a non-reference site (annual P < 250 mm), located at an airport and surrounded by a mixture of irrigated and non-irrigated rangeland. It can be seen that T_{min} approaches T_{dew} frequently for the irrigated site at Kimberly, with only small differences between T_{min} and T_{dew} occurring during months where a dry climate occurs (low precipitation/ET_o ratio). On the contrary, T_{min} was as much as 10 ºC higher than T_{dew} for the non-reference Boise station. From this graphical comparison, one can conclude that data for the non-reference Boise site require appropriate correction before being utilized to compute ET_o using the FAO-PM method. This is necessary to avoid overestimation of ET_o due to overestimation of air temperature and VPD.

FIGURE 2.5

Comparison of differences between the monthly values for minimum and dewpoint temperature (T_{min} - T_{dew}) corresponding to monthly ratios of precipitation/ET_o for a reference site (Kimberly, Idaho, USA) and a non-reference site (Boise, Idaho, USA) in the same region



Source: FAO56, 1998

2.5.1 Adjustments to weather daily data collected over dry surfaces

Researchers, hydrologists and engineers have long recognized the impact of local aridity on near-surface weather measurements and the lack of synchronization and equilibrium of those measurements with that expected to occur over evaporating surfaces associated with irrigated agriculture. The lack of synchronization and equilibrium creates a consequent incongruence within ET_o equations such as the standardized FAO-PM ET_o as was described in a previous section.

Empirical methods have been developed to adjust arid air temperature downward or to adjust arid vapour pressure upward. Some of the more simple methods are described in this section. These empirical approaches generally employ linear adjustments based on observed differences between weather data collected from close-by irrigated and non-irrigated locations or by differences between daily minimum air temperature (T_{min}) and daily average dewpoint temperature (T_{dew}). The objective of the adjustments is to produce a corrected weather data set that best reflects weather conditions to be obtained in a well-watered, reference-type environment. These are the data expected by the standardized FAO-PM ET_o equation.

The following sections describe two procedures for recognizing and correcting non-reference weather data prior to its use to calculate ET_o. The second procedure adjusts both air temperature and dewpoint temperature whereas the first procedure is even simpler and adjusts dewpoint temperature data only. Both procedures can produce very helpful adjustments that account for the aridity of a weather station site and the data collected, and help to bring the data into synchronization with the reference surface.

A third type of procedure for adjusting air temperature, humidity and even wind speed data utilizes a more theoretical boundary layer profile theory. These procedures are still in their infancy, but may become more commonly used in the future due to their more physical basis and ability to automatically identify the extent of adjustments needed.

2.5.2 Adjustment to T_{dew} or e_a

A simple, yet relatively effective approach to correct weather data collected in a non-reference setting is to retain the measured air temperature data with no adjustment, but to abandon the use of all collected humidity data (T_{dew}, e_a or RH). Instead, replacement humidity data are synthesized to better describe conditions of a reference (well-watered) environment. The synthesized humidity data set works with the original reported air temperature data to produce a VPD, ($e_s - e_a$) that is expected to occur in a reference weather station environment.

An estimate of actual vapour pressure, e_a, can be obtained by assuming that the daily average dewpoint temperature (T_{dew}) is near the daily minimum temperature (T_{min}). This implicitly assumes that at sunrise, when the air temperature is close to T_{min}, that the near-surface air over a well-watered, transpiring surface has cooled, so that it is nearly saturated with water vapour, and the relative humidity is nearly 100 percent.

The relationship $T_{dew} \approx T_{min}$ holds for locations in humid climates and where the cover crop of the station is well watered. However, particularly for arid regions, the air might not be saturated when its temperature is at its minimum due to the entrainment of dry air from above the irrigated area. Hence, T_{min} might remain greater than T_{dew} and a further calibration is then recommended to estimate dewpoint temperature.

In these situations, T_{dew} may be estimated from T_{min} using subtractive correction factors, a_T, that vary with the general aridity of the surrounding region unless the weather station is cropped with irrigated green grass. Thus, a general approach is:

$$T_{dew} = T_{min} - a_T \qquad (2.6)$$

where a_T is a subtractive correction factor that is based on regional climate through the UNEP Aridity Index, AI, and assumes the following values:

a_T = 4-6 °C in Hyper-arid regions (AI<0.05),

a_T = 2-3 °C in Arid regions (0.05≤AI<0.20),

a_T = 1-2 °C in Semiarid regions (0.20≤AI<0.50),

a_T = 0-1 °C in Dry sub-humid regions (0.50≤AI<0.65),

a_T = 0 °C in Moist sub-humid and humid regions (0.65≤AI).

The upper limit 6 °C of the a_T values for hyper-arid regions is to be applied only under very stringent site aridity conditions.

The UNEP Aridity Index describes the general aridity of the region. The United Nations Environment Programme (UNEP) aridity index AI is defined by the ratio between the long term means of annual precipitation (P, mm) and the Thornthwaite (1948) annual climatic evaporation index (CEI_{TH}, mm) (Box 2.4 and Example 2.1). Both values are computed from 20 or more years of data whenever possible.

It is noted that the use of the $T_{dew} = T_{min} - a_T$ (Equation 2.6) adjustment may produce a slight upward or downward bias in computed net radiation (R_n). The adjustment to e_a when T_{min} is impacted by station aridity inflates e_a above levels expected for reference conditions. This error in e_a causes incoming atmospheric long-wave radiation to be overestimated, but the elevated T from using arid data causes outgoing long-wave radiation to be overestimated. These two impacts tend to cancel one another. One may expect up to 1 to 3% under- or overestimation in R_n.

BOX 2.4
Estimation of the UNEP Aridity index

The aridity index is the ratio between the mean annual precipitation (mm) and the Thornthwaite (1948) mean annual climatic evaporation index (CEI, mm) of a long-term data series of preferably 20-30 years of data or more. It is estimated as:

$$AI = P_t / CEI_{TH}$$

where P_t mean annual precipitation [mm],
CEI_{TH} mean annual Thornthwaite climatic evaporation index [mm].

The Thornthwaite (1948) mean annual climatic evaporation index (CEI_{TH}, mm) is estimated as:

$$CEI_{TH} = \sum_{i=1}^{12} 16 \left(\frac{10\, T_{mean,\,i}}{I} \right)^{\alpha}$$

where T_{mean} mean monthly temperature [°C],
I annual heat index (0 - 160),
α coefficient (0 - 4.25) which depends on the heat index.

The annual heat index (I) is estimated from the mean monthly temperature (T_{mean} in °C) as:

$$I = \sum_{i=1}^{12} \left(\frac{T_{mean,\,i}}{5} \right)^{1.514}$$

The coefficient α is estimated as:

$$\alpha = [(6.75)\,(10^{-7})\,I^3] - [(7.71)\,(10^{-5})\,I^2] + [(1.7912)\,(10^{-2})\,I] + 0.492$$

EXAMPLE 2.1
Determination of the UNEP Aridity index

Determine the Aridity Index from a station in Lerida, Spain (41°37' N, altitude 203 m). Using the long term CLIMWAT database the mean monthly temperatures of January, February, March, April, May, June, July, August, September, October, November, December was 5.2, 7.1, 11.6, 14.3, 18.3, 22.1, 24.95, 24.6, 21.5, 15.5, 9.45, 5.7 °C respectively. While the precipitation was 22, 18, 31, 46, 47, 43, 24, 34, 34, 30, 22, 29 mm, respectively.

From Eq. in Box 2.4	I	68.89	°C
	α	1.58	
	Annual CEI_{TH}	725.26	mm
	Annual P_T	380	mm
	AI= P_r/CEI_{TH}	0.52	[]
	Dry sub-humid climate		

The actual vapour pressure e_a can be directly calculated as (Example 2.2):

$$e_a = e^o(T_{min} - a_T) = 0.611 \exp\left[\frac{17.27\,(T_{min} - a_T)}{(T_{min} - a_T) + 237.3}\right] \quad (2.7)$$

where e_a estimated 24-hour actual vapour pressure of the near surface air (at 2 m) [kPa],

T_{min} daily minimum air temperature [°C];

with a_T ranging from 0 °C to 6 °C according to the climate type as for Equation 2.6.

EXAMPLE 2.2
Determination of actual vapour pressure from temperature data

Determine the monthly actual vapour pressure for August in the station at Lerida, Spain (41°37' N, altitude 203 m) in the presence and absence of RH observations. As per Example 2.1 the aridity index (AI) is 0.52, thus, indicating that the location has a dry sub-humid climate type. The T_{min} = 17.7 °C, and T_{max}= 31.5 °C

	$e^o(T_{max})$	4.62	kPa
	$e^o(T_{min})$	2.03	kPa
	$e_s = [e^o(T_{max}) + e^o(T_{min})]/2$ Eq.3.7	3.32	kPa
From Eq. in Box 2.4	$T_{dew} \approx T_{min} - 1$ =	16.7	°C
Then	e_a =	1.90	kPa

Equations 2.6 and 2.7 can not work directly when utilizing hourly or shorter time periods. Instead, one first applies the procedure with daily T_{min} determined from the hourly (or shorter) air temperature readings and then estimates a daily average T_{dew} or e_a from Equation 2.6 or 2.7. That estimate for T_{dew} or e_a is then applied to all periods of the day.

2.5.3 Customizing for a region

An alternative to selecting the a_T in Equation 2.6 based on AI is to develop a summary of a_T that is based on qualified weather stations in the region. The qualified stations should have high integrity humidity and air temperature data as well as be located in extensive well-watered environments. Despite being more laborious, the determination of a_T can be made on a monthly basis to capture any seasonality associated with the T_{dew} vs. T_{min} relationship. For example, in regions having a cold winter, values for a_T may even be negative in winter, e.g. Example 2.3, meaning daytime increases in vapour pressure increase the 24-hour average T_{dew} to above T_{min}.

An example of establishing monthly values for a_T is given in Example 2.3 where long-term average records of T_{dew} and T_{min} were derived from ten irrigated weather sites across southern Idaho, USA. The records were used to establish monthly average values for a_T that described the region and that varied by month. The a_T offsets averaged 1.5 °C during most of the growing season from April through September. However, negative values were found to occur during winter, which indicates that, during winter, T_{dew} may be higher than T_{min}.

EXAMPLE 2.3
Dewpoint temperature offsets (a_T) below T_{min} for reference weather stations in southern Idaho, USA

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
-2.0	-1.5	0.0	1.5	1.5	1.5	1.5	1.5	1.5	1.0	-1.0	-2.0

In applying this simple adjustment approach to only humidity data (T_{dew} or e_a), with no adjustment to air temperature, one recognizes that measured T_{min} collected in a non-reference weather station environment will remain higher than it would be in a reference setting. In addition, using the unadjusted minimum air temperature measurements from a non-reference setting to estimate dewpoint temperature using a_T as defined in Equation 2.6 will tend to overestimate the true T_{dew} and associated e_a that are expected to occur under reference conditions. However, a cancelling in biases occurs inside the ET_o calculation because saturation vapour pressure, e_s, in the FAO-PM ET_o equation is then estimated using the same T_{min} values used to estimate T_{dew}. Therefore the values for e_s and e_a will be equally "inflated" and cancelling of these effects will tend to occur. The VPD = $e_s - e_a$ difference in the standardized ET_o equation will, in general, thereby approximate the $e_s - e_a$ difference anticipated to occur over a reference surface. As a consequence, a more accurate estimate for ET_o should result than if the actual measurement of T_{dew} from the arid weather setting had been used with no correction. Thus, *when humidity data are adjusted using* $T_{dew} = T_{min} - a_T$, *no further adjustment is generally needed* to the air temperature data set to account for the effects of the aridity of the weather measurement site.

2.5.4 Adjustment of T_{max}, T_{min} and T_{dew}

The second empirical method for adjusting arid weather data provides correction to both air temperature (T_{max} and T_{min}) and dewpoint temperature in proportion to the difference ($T_{min} - T_{dew}$). The difference $T_{min} - T_{dew}$ functions as an indication of the aridity of the weather station environment. The methodology can be applied with or without a "trusted" reference station in the region.

When a "trusted" reference site is available in the region to compare against

If one is fortunate to have at least one weather station site in a region that is well-watered, in other words a station that is located within a sizeable irrigated area or a large wetland area, then one can use $T_{min} - T_{dew}$ relationships from that "trusted" reference station as a basis to develop adjustment parameters to apply to arid, non-reference weather station data. The adjustments can also be applied to arid, gridded weather data.

In this second procedure, adjustments are split between air temperature and dewpoint temperature. Monthly, daily or hourly weather data can be corrected using the following approach:

1. Compare $T_{min} - T_{dew}$ from a suspected non-reference site with that from a trusted reference site in the same region (if available) using a graphical procedure such as in Figure 2.4 or 2.5 and using monthly ratios of Precipitation/ET_o as the abscissa. Monthly data are utilized to compute $T_{min} - T_{dew}$ and for applying Equation 2.8.
2. For any particular month, when differences $T_{min} - T_{dew}$ for the non-reference site are systematically more than 2 °C to 3 °C higher than the $T_{min} - T_{dew}$ for the reference site, then one can conclude that the non-reference station is impacted by some degree of aridity and adjustments can be made to the data. One therefore computes the average difference between the $T_{min} - T_{dew}$ values:

$$\Delta T_{paired} = (T_{min} - T_{dew})_{n/ref} - (T_{min} - T_{dew})_{ref} \quad (2.8)$$

where ΔT_{paired} upward bias in $T_{min} - T_{dew}$ differences created by non-reference station aridity [°C],
n/ref subscript indicating the suspected non-reference weather station,
ref subscript indicating a "trusted" reference weather station.

3. Corrected daily maximum and minimum air temperatures for each month or day are calculated using a specified fraction of the ΔT_{paired} difference:

$$(T_{max})_{cor} = (T_{max})_{obs} - b_T \Delta T_{paired} \quad (2.9)$$

$$(T_{min})_{cor} = (T_{min})_{obs} - b_T \Delta T_{paired} \quad (2.10)$$

where b_T fraction of ΔT_{paired} correction to be applied to T_{max} and T_{min} [0 - 1.0],
subscripts *cor* and *obs* refer to corrected and observed values.

Corrections to T_{max} and T_{min} are assumed to be the same. These reductions to air temperature reduce observations by a b_T fraction of the distance of the aridity bias in $T_{min} - T_{dew}$ for the non-reference data. The other $(1 - b_T)$ of the bias is added to T_{dew} in the following step 4. The default value for b_T is 0.3, based on experience. The value 0.3 means that 30 percent of the $T_{min} - T_{dew}$ bias adjusts downward the high values for T_{max} and T_{min} and 70 percent of the $T_{min} - T_{dew}$ bias adjusts upward the low values for T_{dew}.

4. Correction to T_{dew}:

$$(T_{dew})_{cor} = (T_{dew})_{obs} + (1 - b_T)\,\Delta T_{paired} \quad (2.11)$$

The corrections by Equation 2.9 and 2.10 decrease the values for T_{max} and T_{min} and corrections by Equation 2.11 increase the value for T_{dew}.

When a "trusted" reference site is not available in the region to compare against

When there is no reference weather location and data in a region to use to establish expected relationships between T_{min} and T_{dew}, then the procedure of Equation 2.6 can be applied using a value for a_T from Equation 2.6. First the $T_{min} - T_{dew}$ departure is calculated for each month:

$$\Delta T = T_{min} - T_{dew} \quad (2.12)$$

Monthly values can be summed from hourly or daily data as needed.

1. Corrected temperatures for each day (or month) are then calculated as:

$$(T_{max})_{cor} = (T_{max})_{obs} - b_T\,(\Delta T - a_T) \quad (2.13)$$

$$(T_{min})_{cor} = (T_{min})_{obs} - b_T\,(\Delta T - a_T) \quad (2.14)$$

   for $\Delta T > a_T$, a_T is the expected difference estimated using the definitions for a_T following Equation 2.6. If $\Delta T \leq a_T$, then no correction is made to T_{max}, T_{min} or T_{dew}. The value b_T (~0.3) is the same as for Equations 2.9-2.11, where subscripts "cor" and "obs" refer to corrected and observed values, respectively.

2. Corrected T_{dew} for the same months or days is calculated as:

$$(T_{dew})_{cor} = (T_{dew})_{obs} + (1 - b_T)\,(\Delta T - a_T) \quad (2.15)$$

3. Following corrections, one recomputes ET_o using the corrected values for T_{max}, T_{min} and T_{dew}.

2.5.5 Application of aridity corrections to hourly data

Collection of weather data at hourly or shorter time steps is becoming more common using automated electronic weather stations. In some cases, the calculation of ET_o can be more accurate when using hourly weather data as compared to when using daily weather data. Therefore, the calculation of ET_o using hourly weather data is encouraged. However, the aridity correction procedures for air temperature and/or humidity data and QAQC should be equally applied to hourly data.

For correction of weather data for hourly or shorter time steps, the procedure begins by calculating the daily average T_{dew} data using the hourly data for each day. T_{min} and T_{max} are determined by selecting minimum and maximum hourly air temperature values during the day. The daily average T_{dew} as well as daily T_{max} and T_{min} are then averaged over each month to determine monthly values for ΔT_{paired} or ΔT using Equation 2.8 or Equation 2.12. The corrections to hourly air temperature become a single equation that is applied to each hourly or shorter (or longer) time step for the case of a trusted reference station:

$$(T_{hrly})_{cor} = (T_{hrly})_{obs} - b_T\,\Delta T_{paired} \quad (2.16)$$

When a trusted reference station does not exist:

$$(T_{hrly})_{cor} = (T_{hrly})_{obs} - b_T (\Delta T - a_T) \quad (2.17)$$

The "hrly" subscript refers to hourly or shorter timesteps (or even 3-hour time steps). ΔT_{paired} or ΔT are based on 24-hour periods, averaged over each month, even though corrections are made to hourly or shorter timestep data. Equations 2.15 – 2.17 require that $\Delta T_{paired} > 0$ and $\Delta T\text{-}a_T > 0$. Otherwise, no correction is made.

For hourly or shorter time step dewpoint correction, Equation 2.15 is applied to T_{dew} of each hourly or shorter time step:

$$(T_{dew\ hrly})_{cor} = (T_{dew\ hrly})_{obs} + (1 - b_T) (\Delta T - a_T) \quad (2.18)$$

In other words, the T_{air} and T_{dew} for each hourly period during a month are given the same adjustment.

2.5.6 Gridded weather data sets

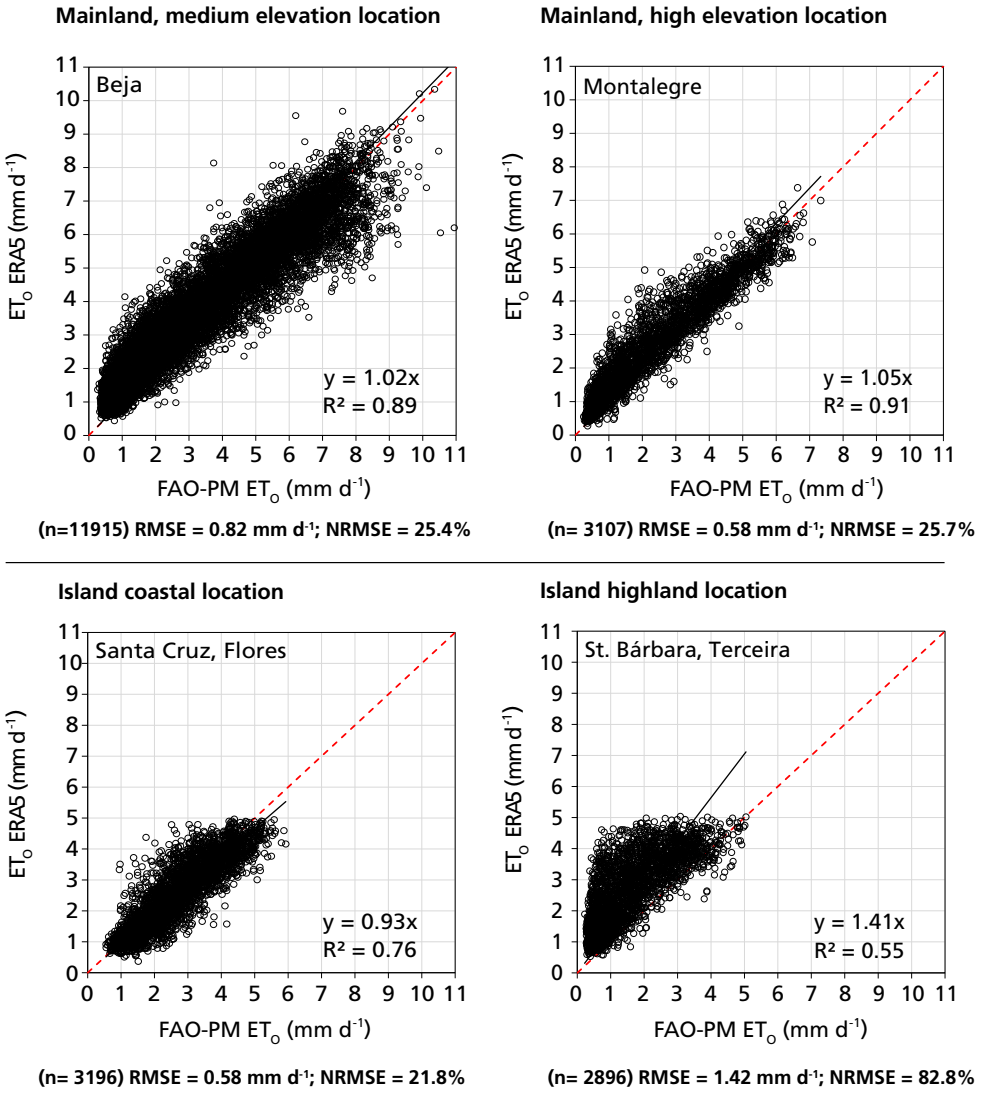
Since about the year 2000, the advent of gridded historical weather data derived from sophisticated land data assimilation system (LDAS) models has provided an alternative data source to the traditional use of point-based weather station data for estimating ET_o. *LDAS systems are principally operated for purposes of weather research and weather and climate forecasting.* LDAS systems are programmed to electronically assimilate weather data collected from select weather sites around the globe and to couple those data with land process (soil and vegetation water balance) models and with atmospheric models to extend the point data and to create gridded weather data sets that are generally traceable to the original point measurements.

Gridded weather data products are made available for the globe from a number of sources including the European Center for Medium-Range Weather Forecasts (ECMWF) and the European Satellite Applications Facility on Land Surface Analysis (LSA-SAF) framework. The LSA-SAF framework produces an ET_o estimate using a modified Makkink equation using solar radiation derived from the European Meteosat Second Generation (MSG) satellite system. In the USA, gridded weather data sets are produced by NOAA, NASA and NCEP. Many of the gridded weather data sets include 50-year or longer time periods and have spatial resolutions of 1/6 to 1/8 degree (~ 9 km x 9 km to 12 km x 12 km) but some gridded data sets have resolutions as large as 50 km. Time resolutions range from one hour to one day. Many gridded weather data sets contain daily precipitation, solar radiation, and near surface temperature and humidity (at ~2 or 3 m reference height), and wind speed often at ~10 m reference height. These data parameters enable the application of the FAO-PM ET_o equation.

Figure 2.6 shows four examples of computing the FAO-PM ET_o equation from ERA-5 reanalysis of the ECMWF relative to two continental weather station locations and two island weather station locations. For all cases, the regression coefficients are generally higher than 1.0, indicating a trend for over-estimation of calculated ET_o values, but the daily error of estimates is lower than 30 percent of the mean daily ET_o values. The exception is the island's location in altitude (Santa Bárbara, Terceira), which is likely related to the cloudy and windy climate typical of oceanic islands in altitude. Overall, the results of using ERA5 reanalysis suggest its appropriateness for use in practice for these locations in the absence of observed data.

FIGURE 2.6

FAO-PM ET_o equation computed with observed daily full weather data compared with ET_o estimated with gridded daily full weather data sets from ERA5 reanalysis for four locations in Portugal: Beja and Montalegre in mainland, Santa Cruz in the island of Flores and Santa Bárbara in the island of Terceira, Azores



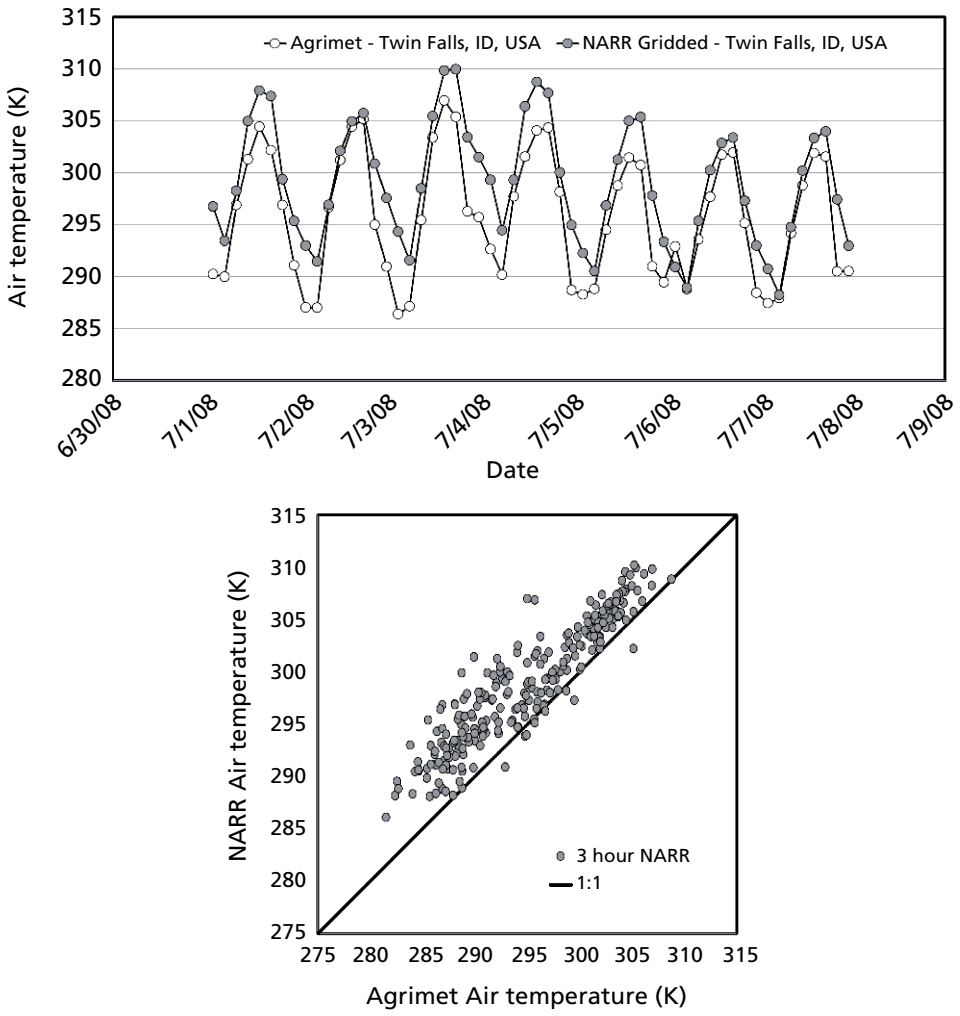
Source: Authors' own elaboration

Tendency for aridity artifacts in gridded weather data sets

When used to calculate ET_o, LDAS-derived and other assembled data sets should be reviewed to determine whether the weather data contain substantial "artifacts" of dryness from the original weather data sets used to drive the LSM models. These artifacts may produce weather data that are characteristic of regional weather systems, which in semiarid and arid climates, represent the general aridity of the region. However, for purposes of estimating ET_o for use in irrigation-related settings, the ambient gridded weather data may not exhibit the conditioning effects that irrigation has on near-surface air properties, as illustrated in Figure 2.7, where the air temperature is reduced and humidity content is increased by the ET supplied by irrigation (Agrimet – Twin Falls data set).

FIGURE 2.7

Three-hourly air temperature data from North American Regional Reanalysis (NARR) gridded data set for Twin Falls, Idaho, USA during July 2008, compared with measured air temperature collected at the irrigated USBR Agrimet weather station located near Twin Falls. The top figure shows the time series for 1 - 7 July and the bottom figure show NARR data plotted against the irrigated Agrimet data for the month of July. The NARR data were impacted by assimilation of ambient, non-irrigated data in the gridded model



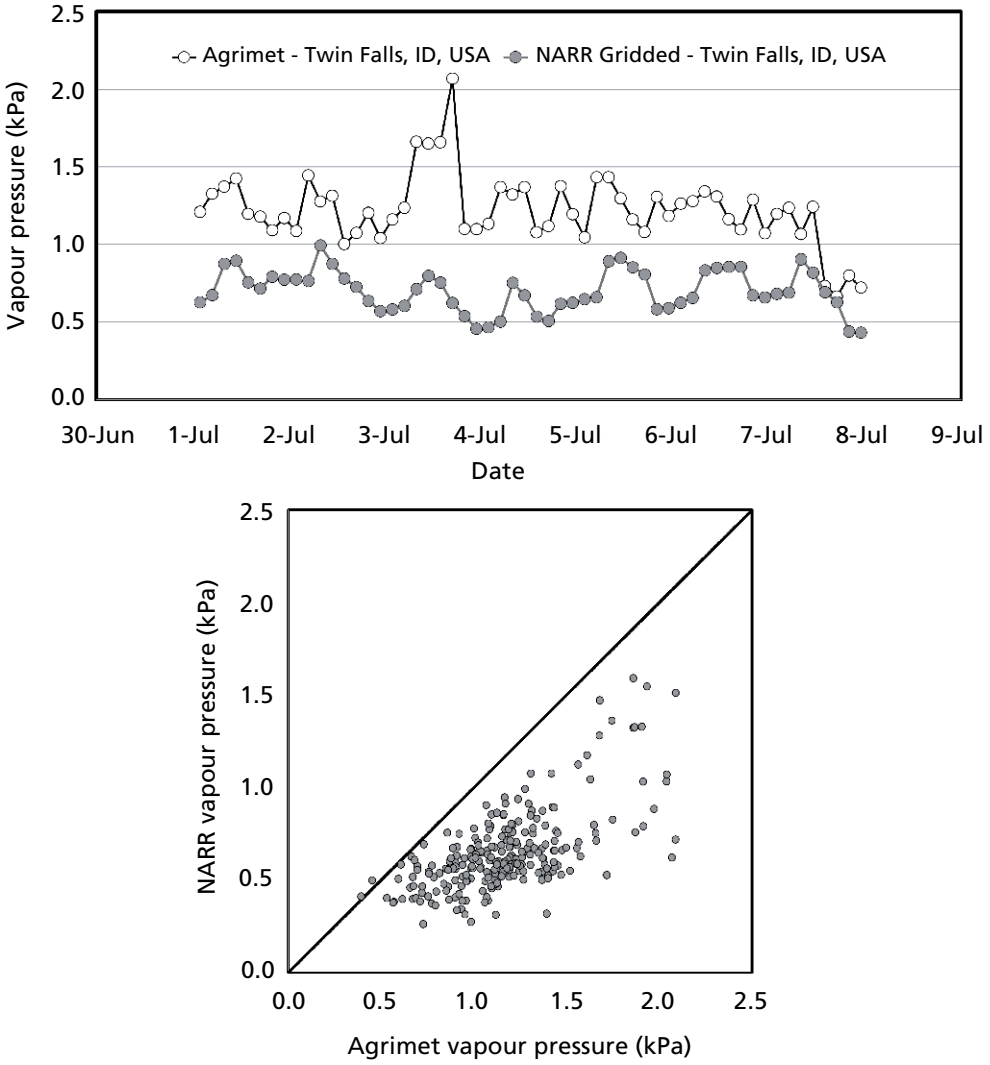
Source: Authors' own elaboration

Generally, the LDAS models creating the gridded data sets are not aware of, and do not include, irrigation in their daily soil water balances that are used to estimate ET, and the partitioning of available energy into sensible heat flux, H, and estimation of R_n. As a consequence, the ET in the model's energy balance is too low and the H is too high when developing the gridded weather data sets. As a result, the estimated gridded air temperature is too high and the vapour pressure is too low. In a 2020 USA study, summertime ET_o computed from major gridded weather data sets exceeded that computed from 100 agricultural reference weather stations by 10 to 30 percent.

As an illustration of aridity artifacts in gridded weather data sets and to demonstrate that aridity impacts all times of the day, Figures 2.7 and 2.8 show 3-hourly air temperature and 3-hourly vapour pressure from a North American Regional Reanalysis (NARR) gridded data set for a 32 km x 32 km grid cell located near Twin Falls, Idaho, USA, during July 2008.

FIGURE 2.8

Three-hourly vapour pressure from the NARR gridded data set for Twin Falls, Idaho, USA during July 2008, compared with measured at the irrigated USBR Agrimet weather station. Top figure shows the time series for 1 - 7 July and the bottom figure show NARR data plotted against the irrigated Agrimet data for the month of July. The Agrimet station is over irrigated grass and had nearly double the humidity than NARR data that were impacted by assimilation of ambient, non-irrigated data in the gridded model



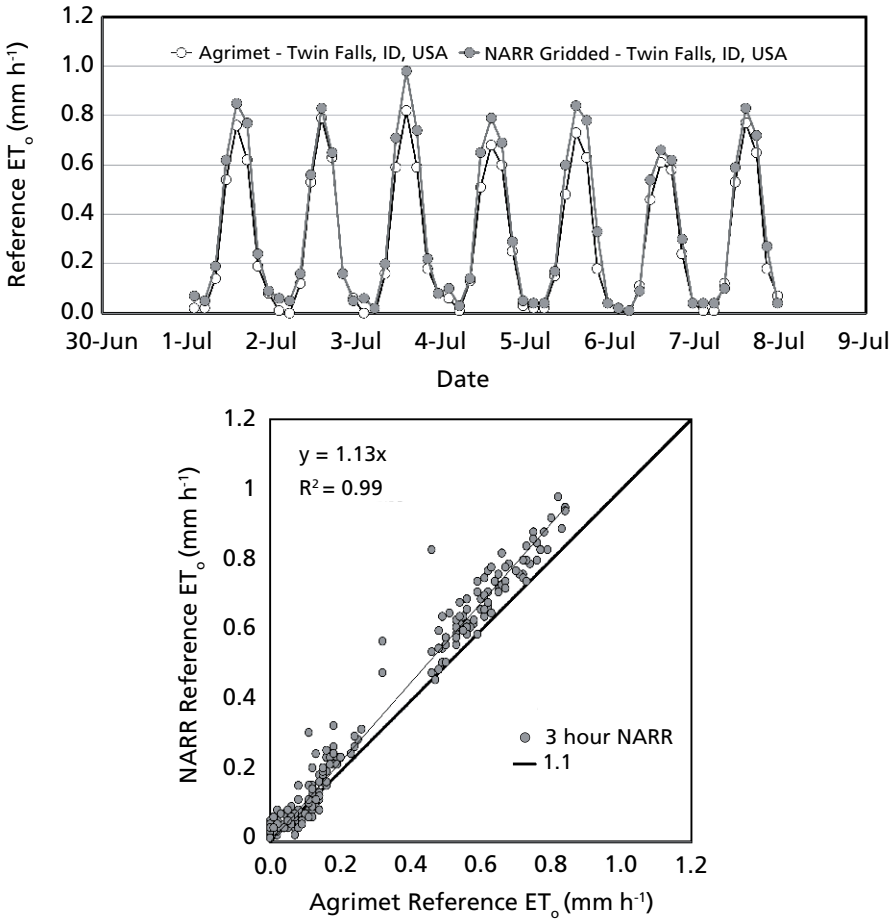
Source: Authors' own elaboration

The NARR gridded data are compared against measured air temperature and vapour pressure at an irrigated agricultural weather station (Agrimet) located in the same grid cell. The Agrimet station is an electronic, automated station located within a large irrigated agricultural area. While there is a close correspondence between the two data sets, which is an indication of correct assimilation and simulation of the gridded weather data at the regional scale, the NARR near surface air temperature overstated the Agrimet air temperature by up to 5 °C (Figure 2.7), and understated the near-surface vapour pressure by about one-half during July (Figure 2.8), which is a month having little or no rain in southern Idaho. Air temperature was overstated during peak periods of days and nighttime periods. The understatement of vapour pressure by 50 percent as compared to that measured in an irrigated environment is substantial and will cause an incorrect overstatement of ET_o by the PM equation if not corrected.

Figure 2.9 shows calculated ET_o for time periods three hours in length for the two Twin Falls data sets for the same July 1 – July 7, 2008 period. The 3-hour timestep matched the shortest timestep available with the NARR data set. The overestimation of calculated ET_o from the arid NARR data is subtle but significant, averaging about 17 percent higher in July than ET_o calculated from the irrigated Agrimet weather site. These examples are only one of many that show the serious overestimation of ET_o caused by the collection or assimilation of weather data from arid, non-transpiring local environments.

FIGURE 2.9

Three-hourly ET_o calculated with the FAO-PM ET_o equation from the NARR gridded data set over Twin Falls, Idaho, USA, during July 2008, compared with calculations using data from irrigated USBR Agrimet weather station. Top figure shows the time series for 1 - 7 July and the bottom figure shows ET_o calculated from NARR data plotted against ET_o calculated from irrigated Agrimet data for the month of July. The NARR data were impacted by assimilation of ambient, non-irrigated data in the gridded model



Source: Authors' own elaboration

Table 2.1 lists the total monthly ET_o calculated from the arid NARR gridded weather data set and irrigated Agrimet weather data set for Twin Falls, Idaho, USA during the growing season of 2008. The two data sets were in good agreement during the first two months of the growing season when ET rates were smaller and stored wintertime precipitation and precipitation during March and April tended to keep surfaces moist for both data sets so that air temperature and humidity levels were similar. The substantial separation between the two data sets began in May when desert grasses in areas outside irrigated areas dried due to lack of precipitation. During the May-

October period, irrigated crops in the region were often near full ground cover and were adequately watered to sustain near-maximum ET. Therefore, the measured air temperature was lower, the measured humidity was higher, and the calculated ET_o was lower for the irrigated region than for the arid gridded NARR data set.

TABLE 2.1
Total monthly ET_o calculated in mm month^{-1} from the arid NARR gridded data set and from the irrigated Agrimet weather data set for Twin Falls, Idaho, USA during the growing season of 2008

Month	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct
NARR-based ET_o (arid gridded data set)	58	109	159	197	232	204	147	88
Agrimet-based ET_o (irrigated station)	58	107	141	173	199	176	126	74
Percent error in NARR (overestimation)	0%	2%	13%	14%	17%	16%	17%	19%

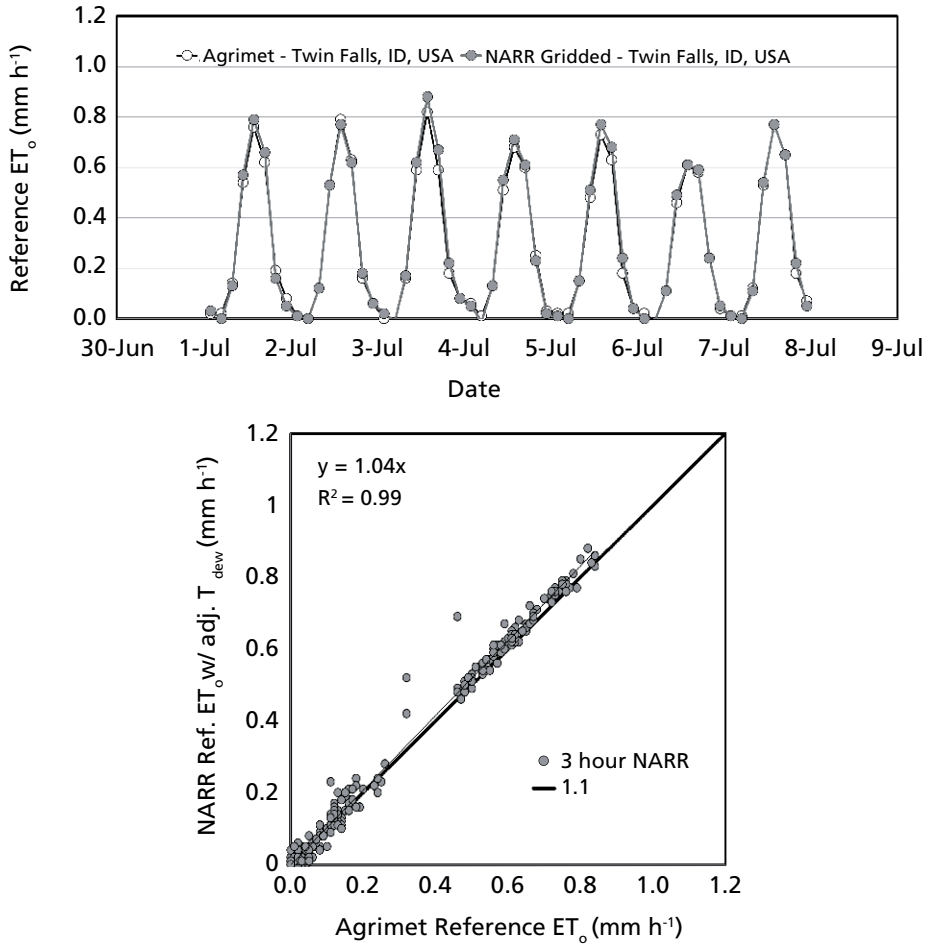
These types of outcomes and impacts on both measured and gridded weather data are common, and can cause overestimation of ET_o by as much as 20 to 30 percent when ignoring the influence of conditioning of the equilibrium boundary layer (EBL) by evaporative cooling that occurs over irrigated agricultural surfaces.

The results of the application of Equation 2.6 to the arid weather data from the NARR gridded weather data set shown earlier in Figures 2.7-2.9 are shown in Figure 2.10 where the application was made to the 3-hour data of NARR and compared to an equivalent 3-hour data set derived from the irrigated Agrimet weather data site. The NARR data were impacted by NARR's assimilation of ambient, non-irrigated data in the gridded data set and therefore, for Figure 2.10, T_{dew} was replaced by T_{dew} synthesized using T_{min}. In the application, the daily minimum air temperature was determined as the lowest 3-hourly air temperature for each day. That T_{min} was then used in Equation 2.6 to estimate a daily average T_{dew}. A value for a_T = 3 °C was used for the semiarid climate as recommended from Equation 2.6. The daily average T_{dew} was assigned to each 3-hour period during the day and then ET_o was computed.

The results of the replacement of the humidity data in the gridded NARR weather data set using synthesized T_{dew} from T_{min} was successful in producing ET_o calculations on a 3-hour time step that compared closely with ET_o calculated from the well-watered Agrimet weather station located in the interior of a large irrigated area (> 100 000 ha). Following the establishment of synthesized T_{dew} in the NARR gridded data set, the ET_o totals were within 4 percent of the irrigated data set for the month of July and within 1 percent for the March – October growing season. ET_o estimates for the March – October growing season were effectively reduced by 12 percent by the adjustment. Similar results would be expected for hourly or 24-hour time steps.

Similar results were obtained after correcting T_{max}, T_{min} and T_{dew} for the NARR data set using Equation 2.12, 2.15 and 2.17 with a_T = 3 °C, and b_T = 0.3. ET_o calculated using corrected NARR air temperature and dewpoint data was within 5 percent of ET_o calculated with the irrigated Agrimet data set. Similar results were also obtained after correcting T_{max}, T_{min} and T_{dew} for the NARR data set using the irrigated United States Bureau of Reclamation (USBR) Agrimet weather station as a "trusted" (paired) reference station based on Equation 2.8, 2.11 and 2.16 with a_T = 3 °C, and b_T = 0.3. ET_o calculated using corrected NARR air temperature and dewpoint data was within 2 percent of ET_o calculated with the irrigated Agrimet data set when the Agrimet data site was used as a trusted site.

FIGURE 2.10
Three-hourly ET_o calculated with the FAO-PM ET_o equation from the NARR gridded data set over Twin Falls, Idaho, USA, during July 2008, after estimating T_{dew} using T_{min} using Eq. 2.6 with a_T = 3° C, compared with calculations using data from the irrigated USBR Agrimet weather station. Top figure shows the time series for 1 - 7 July and the bottom figure shows ET_o, calculated from NARR data with estimeted T_{dew} plotted against ET_o calculated from irrigated Agrimet data for the month of July



Source: Authors' own elaboration

2.6 QUALITY ASSURANCE AND QUALITY CONTROL (QAQC) OF WEATHER DATA

Estimates of reference evapotranspiration (ET_o) are no better than weather data upon which they are based. Assessments of weather data integrity and quality need to be conducted before data are utilized in ET_o equations. Corrections to the data should be made to account for poor sensor calibration, poor maintenance or for a poor weather station environment. Some statistics-based corrections for completion of data sets and for for testing and correction of data sets for homogeneity are described in Annex 3.

A cautionary statement in data analysis and application is that "having no data is often better than having bad data." Bad data refers to data that are substantially biased or are faulty, or are not representative of a well-watered environment. When one has no data, one can look to regional weather or ET data summaries for information that might be useful to represent conditions within the local area. In the case of "bad" weather data,

the ET_o procedures of these guidelines will produce faulty estimates of ET_o and ET_c. The result is the application of evapotranspiration data or evapotranspiration calculations to irrigation water management, to water resources operations, or to irrigation and water resources systems design that can cause greater economic and hydrologic problems than if only reasonable estimates or even "textbook" values for ET_c had been used instead. Humanity can be worse off with erroneous data than with no data at all. This statement also applies to the measurements of evapotranspiration that have been used to develop or calibrate reference ET equations or to determine crop coefficients.

This section presents guidelines to be used to calculate both expected ranges for weather data measurements and also means to assess integrity of data that fall within the ranges.

2.6.1 Solar radiation using clear sky comparisons

Pyranometer operation and calibration accuracy should be evaluated for a particular weather location by plotting hourly or daily readings of solar radiation (R_s) against computed theoretical short-wave radiation that is expected to occur under clear sky conditions (R_{so}). R_{so} can be computed for any day or hour using Equation 3.35 of Chapter 3 along with calculated R_a. The theoretical R_{so} should be expected to plot as an upper envelope of measured R_s.

Pyranometers that measure solar radiation can be substantially impacted by a drift in the sensor calibration caused by aging or film or dust on the sensor. Pyranometers can also be affected by a lack of level for the sensor. Other error in R_s data can occur from the use of an incorrect calibration factor to convert millivolts of pyranometer signal into radiation units. In many situations, faulty R_s data can be corrected by multiplying by a derived coefficient that is determined using the R_{so} values. A simple linear multiplier is often adequate when the problem with sensor reporting is caused by calibration or dirtiness of the sensor.

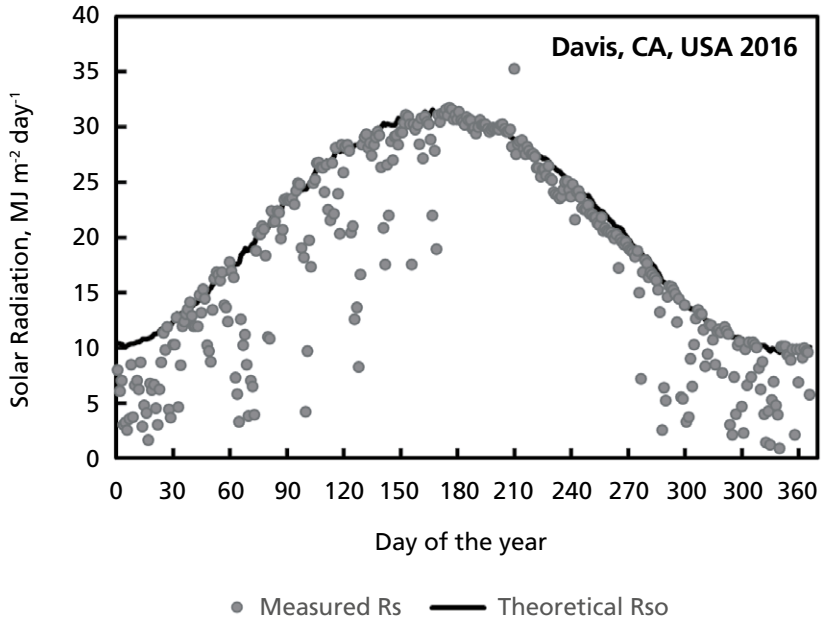
Daily R_s data

Figure 2.11 shows an example of plotting 24-hour measurements for R_s against the theoretical R_{so} calculated using Equation 3.35. The data are taken from the California Irrigation Management Information System (CIMIS) for a CIMIS station located near Davis, CA. In the figure, the highest observed values for R_s tend to correspond to the theoretical envelope of calculated R_{so}. Those observations represent those on cloud-free days so that pyranometer observations came close to values predicted using the theoretical R_{so} curve.

One can note that the R_{so} envelope in Figure 2.11 has day-to-day variations caused by variation in reported vapour pressure. Those variations impacted the estimation of precipitable water in the atmosphere. Observed R_s on cloud-free days tended to increase and decrease with the increases and decreases in R_{so}. This confirms that the actual R_s is, in fact, influenced by atmospheric water content.

Days 220 to 270 had measured R_s that were consistently close to the R_{so} curve, but about 3 to 5 percent below that curve. These appear to have been clear-sky days, but may have been impacted by some dust or aerosols in the air that increased turbidity of the atmosphere. It is notable in Figure 2.11 that days during the 180-270 period at Davis were nearly all clear. This is common for a Mediterranean climate such as exists for Davis. However, many other regions of the globe will tend to have more cloud-impacted days. Large numbers of cloud-impacted days may increase the difficulty in identifying enough clear-sky days on which to base an R_s data correction.

FIGURE 2.11
Daily (24-hour) solar radiation measured at Davis, CA, USA during year 2016 and the theoretical R_{so} curve calculated using Eq. 3.35



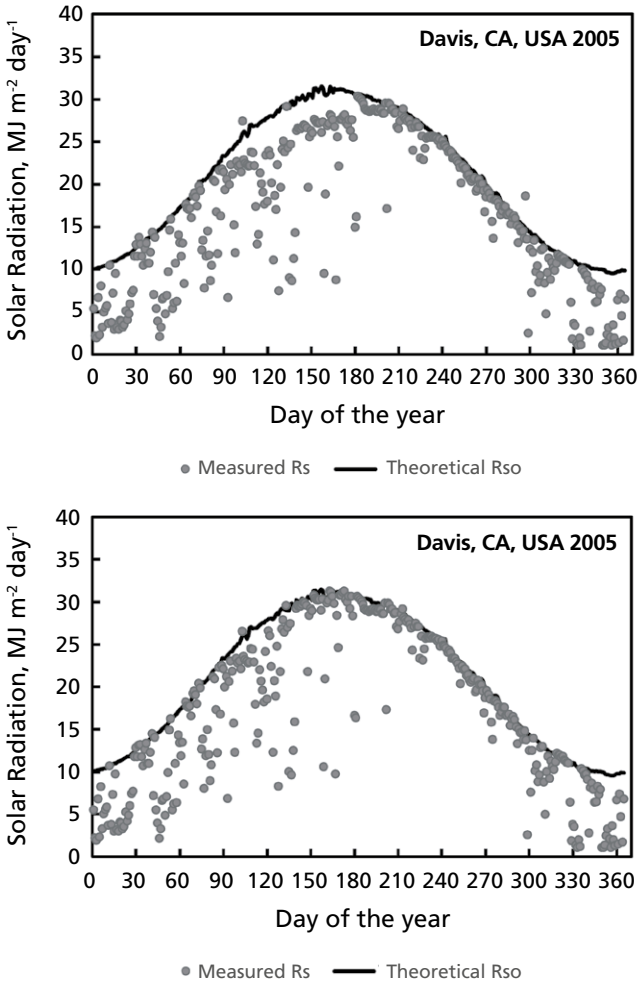
Source: Authors' own elaboration

Overall, the close agreement between measured R_s and theoretical R_{so} for the data shown in Figure 2.11 suggests that the pyranometer was well-calibrated and well-maintained during the year 2016 and that the R_s data can be used with confidence in calculating ET_o and with no correction.

Figure 2.12 provides an illustration of when R_s data do exhibit some faultiness and should be corrected. In this example, also for Davis, CA, USA, but for the year 2005, essentially all R_s measurements during days 90 through 200 fell as much as 10 percent lower than the R_{so} curve. This occurred on what appear to have been clear sky days, as indicated by the day-to-day consistency of observations during some extended periods. It appears that the pyranometer was then replaced or cleaned, or the calibration was updated on about day 200. Following day 200, the measurements on what appear to have been cloud-free days align well with the R_{so} envelope.

The bottom graph of Figure 2.12 shows the same data as for the top graph, but where R_s measurements were multiplied by 1.10 for days 120 through 180. The 1.10 multiplier was determined by identifying what were likely to have been clear days and then calculating the ratio of R_s to R_{so} for those specific days. This correction was made using the REF-ET software and associated QAQC routine available from the University of Idaho (Ref-ET Software | Kimberly Research and Extension Center (uidaho.edu)). That QAQC routine automatically calculates R_s/R_{so} based corrections for each 60 day period of a year. In this example, the 60-day interval that was automatically identified as requiring correction was for days 120 - 180.

FIGURE 2.12
Daily (24-hour) solar radiation measured at Davis, CA, USA during year 2005 and the theoretical R_{so} curve calculated using Eq. 3.35 (top) and corrected measured data (bottom)



Source: Authors' own elaboration

Manual corrections to R_s data can also be determined and applied for improved specificity. When making corrections, a constant multiplier is recommended to be applied to all data measurements during a specific period under the assumption that the pyranometer malfunction or improper calibration is relatively constant from day to day. That multiplier can be changed every 30 to 60 days or longer. An offset to the correction can be considered, in addition to a multiplier. However, justification for an offset is difficult to establish and its determination is difficult.

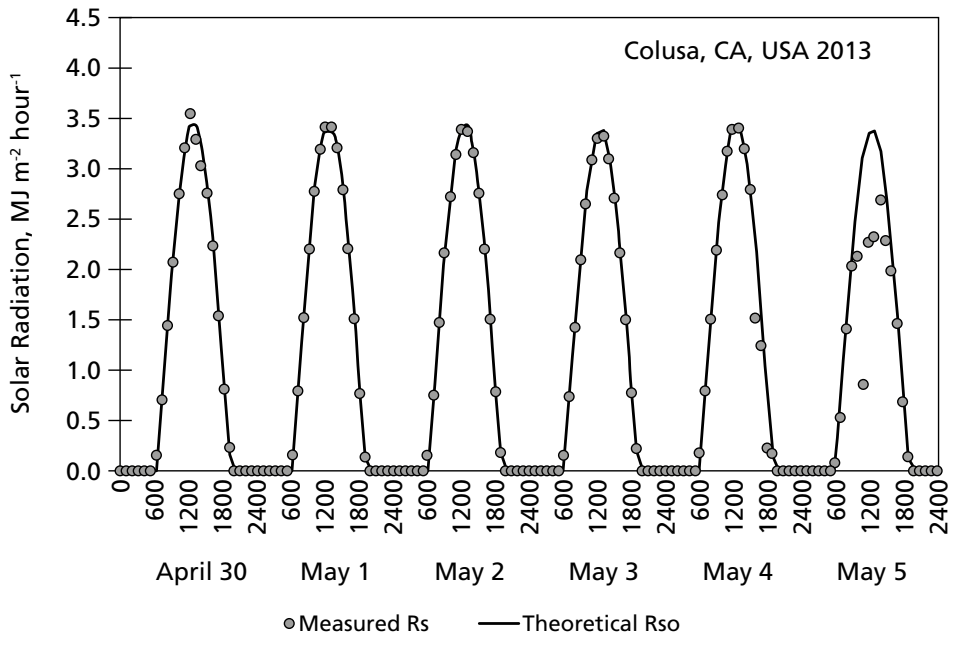
Following the correction of the daily R_s data in Figure 2.12 (bottom graph) for days 120-180, the resulting corrected data align well with the R_{so} envelope. The corrected data represent the measurements expected to have occurred if the pyranometer had been functioning and calibrated properly.

Hourly R_s data

A similar procedure for quality assessment and determination of a need to make corrections to R_s data can be applied for hourly or shorter time periods. The same equations are applied as for daily data. An incorrect shift in the time parameter will become obvious in the plots of R_s and R_{so} vs. time, where a shift in R_s causes the data to plot earlier or later relative to the R_{so} envelope. In that case, an adjustment to time should be made to the original data set and R_{so} (and ET_o) are recomputed. One can recall that, besides its use in QAQC of R_s data, R_{so} is used to estimate the impacts of cloud cover during the calculation of R_n. Therefore, it is important that the R_s/R_{so} ratio is correctly computed.

Figure 2.13 illustrates the plotting of hourly data from a CIMIS station near Colusa, CA, USA, for six days (April 30 – May 5). The R_{so} envelope plots as a solid line. The four days of May 1 to May 4 were apparently cloudless days and the R_s measurements plot on top of the R_{so} envelope. Therefore, one can conclude that the hourly R_s data are of good quality and do not require correction. The maximum height of the R_{so} envelope, reached at solar noon, varies among the four days due to differences in atmospheric water content. That same variation is noted in the peak R_s measurements.

FIGURE 2.13
Hourly solar radiation measured at Colusa, CA, USA during year 2013 and the theoretical R_{so} curve calculated using Eq. 3.35



Source: Authors' own elaboration

May 5 was a day with extensive cloudiness so that all R_s measurements during midday fell substantially below the R_{so} curve. April 30 exhibits an interesting, yet common, phenomenon where measured R_s exceeded the theoretical R_{so} at 1 200 hours. The high value for R_s was caused by the reflection of additional solar radiation from nearby clouds, but with the pyranometer apparently in direct sunlight. Therefore, the total flux of R_s was higher than theoretically expected for a cloud-free day. This occurrence is real and expected and does not require any adjustment or correction. One can note slight reductions in measured R_s during the 1 300-1 400 hourly periods on April 30 as small amounts of clouds passed over the pyranometer.

Obviously, the corrections based on the computed R_{so} envelopes presume that the envelopes are accurate. The accuracy of the R_{so} envelope may need to be confirmed for a region by using accurate radiation measurements obtained from a trusted, calibration-grade pyranometer that has a calibration coefficient that is traceable to the international standard. The calibration pyranometer should be used only for short term periods of 10 – 14 days, and then should be stored in darkness to extend its life and preserve the calibrated accuracy. Care should be exercised in the selection of the turbidity coefficient in Equation 3.35. Unfortunately, little generalized information is available on this factor.

2.6.2 Net radiation

The equations for estimating hourly and 24-hour average rates of net radiation (R_n) using R_s measurements are generally accurate under most conditions. Therefore, measured R_n data can be plotted against R_n data that have been estimated using the equations that are based on measured R_s, air temperature and vapour pressure. The value for albedo (α) used in the R_n estimating equation should represent conditions of the surface beneath the radiometer.

If measured values for R_n consistently deviate from estimated values by more than 3-5 percent, then the calibration or operation of the R_n device (net radiometer) should be scrutinized. This type of comparison between measured and calculated R_n can readily identify days or periods during which the radiometer device has malfunctioned due to effects of dust, bird droppings, moisture condensation inside plastic domes, deterioration of the domes, a lack of levelness of the instrument, an electronic malfunction, or a lack of green vegetation beneath the sensor. Of course, the R_s measurement used in the R_n equations must also be scrutinized as discussed in the previous section.

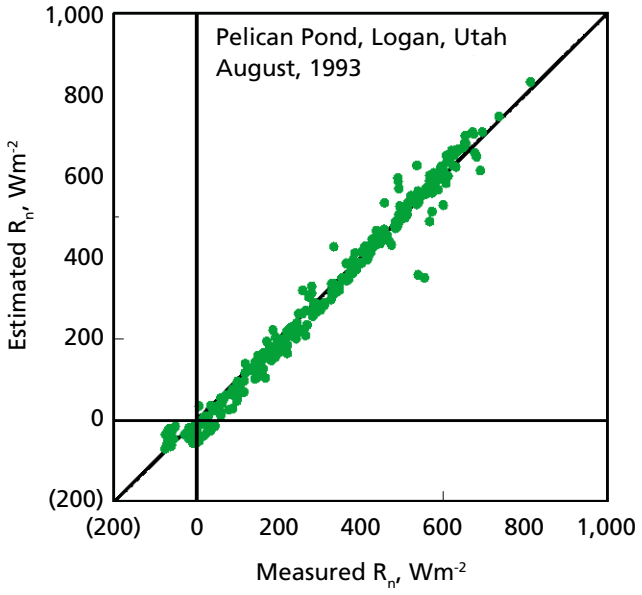
The user of net radiometer data must be aware that net radiometers manufactured by different companies may not yield the same measurements of radiation even when placed over the same surface. These differences are due to differences in sensitivities of various radiometers to long wave and short-wave radiation and variations among methods for calibrating sensors during manufacturing.

It is important that the type, density and height of vegetation beneath the net radiometer and the relative soil moisture content be monitored and reported with the data. Care should be exercised when positioning the radiometer to avoid shading the sensed vegetation with other instruments or structures and to ensure that the radiometer is not shaded by other instruments or structures at any time of the day or year.

If used to calculate ET_o, it is extremely important that measured R_n be taken from a setting where the net radiometer is placed over a well-watered, green vegetation that has albedo similar to the grass reference ($\alpha = 0.23$) and where the vegetation is cooled by the evapotranspiration process similar to that by the grass reference. Dry, non-transpiring vegetation or non-evaporating soil can have a surface temperature that is more than 30 ^{o}C hotter than that of transpiring vegetation. Entering the hotter temperature into the Stefan-Boltzmann equation shows that the R_{nl} for the surface can decrease by as much as 100 W m^{-2} (~5 MJ m^{-2} day^{-1} and ~0.4 MJ m^{-2} hour^{-1}). This is caused by the higher thermally induced long wave emission from the hot, dry surface of nearly 100 W m^{-2}. Use of these data would be in strict violation of the definition for ET_o and will cause the equation to underestimate. In those situations where R_n measurements are not collected in a reference or near-reference environment, it is highly recommended that one reverts to using calculated R_n that is based on measured R_s in the ET_o calculation.

Figure 2.14 shows measured and estimated R_n for a thick wetland cattail vegetation near Logan, UT, USA during 1993. The measurement and calculation time step was 20 minutes. The agreement between measurements and equation estimates was relatively good with an average relationship of $R_{n\,est} = -30 + 1.05\ R_{n\,meas}$ Wm^{-2}. Perfect agreement between the R_n measurements and R_n equations should not be expected, due to limitations of assumptions used in the equations (e.g. the value for albedo, the method for estimating the net long wave radiation component, etc.). One can conclude from the comparison in Figure 2.14 that the measurements appear to be valid and the calculated R_n appears to be valid.

FIGURE 2.14
Estimated vs measured R_n during 20 minute measurement periods over mature cattail vegetation near Logan, UT, USA during August, 1993



Source: FAO56, 1998

2.6.3 Soil heat flux

The Choudhury equation is commonly used for estimating soil heat flux density (G) under daylight conditions:

$$G = 0.4\ \exp^{(-0.5\ LAI)}\ R_n \qquad (2.19)$$

where LAI leaf area index,
exp() the natural number raised to the exponent,
G net radiation.

Equation 2.19 estimates $G = 0.1\ R_n$ for LAI = 2.8, which is typical for clipped grass during daylight. Soil heat flux under forage grass during nighttime hours was found to be about $0.5\ R_n$. Pruitt (1995, personal communication) observed $G = 0.3\ R_n$ during nighttime hours under clipped grass at Davis, CA. The user is reminded that the standardized FAO-PM ET_o equation uses G = 0 for 24-hour time steps, $G = 0.1\ R_n$ for hourly or shorter time steps during daytime and $G = 0.5\ R_n$ for hourly or shorter time steps during nighttime when R_n is negative.

Equation 2.19 can be used to check the functioning and relative accuracy of soil heat flux plates after correcting measurements for temperature change of soil above the plates. The relationship of Equation 2.19 does not hold for 24-hour data, as a positive 24-hour soil heat flux estimate would always result. The user must be aware that Equation 2.19 is only approximate and does not consider the effects of plant spacing, sun angle, soil colour, soil moisture, or soil texture, nor the sensible heat balance at the surface on the ratio of G to R_n. Generally, more than one soil heat flux plate is used due to spatial variation in soil, soil water content, and vegetation.

2.6.4 Wind speed

The accuracy of wind measurements is difficult to assess unless duplicate instruments are used. One should always scan wind records for the presence of consistently constant values occurring during periods of low wind speed. For electronic instruments, these recordings may represent a numerical "offset" in the anemometer calibration equation. The presence of these constant and consistent offsets in the data set indicates either the presence of exceptionally calm conditions (wind speeds less than about 0.5 m s^{-1} during the entire sampling period, which is rare) or malfunctioning of the wind speed sensor due to electrical shorting or perhaps due to fatigue of bearings. These problems may not be noticed by the station operator.

When possible, a second anemometer of the same design, but with fresh bearings, should be placed at the weather location for a three- or four-day period at least once each year, and recordings compared with the permanent instrument. Variations between recordings can signal a need to replace bearings, switches, or other parts. If a second data logger is used to record the temporary anemometer, one should be careful to synchronize data logger clocks. Also, one should be careful that anemometers do not interfere with one another's wind stream.

Comparisons of wind speed data can be made between two nearby stations using a double mass balance or other statistical method as described in Annex 3. One should expect differences between two locations that are time-related or that are caused by local relief, distance from the ocean and general location. When averaged over periods of a few weeks to a month, one should expect to see consistent ratios between two locations. Variation from those consistencies should give cause to look more closely at wind speed data.

As described earlier, anemometer heights should be several meters above surrounding vegetation. In the case of maize crops or orchards that can grow to more than 3 m, an anemometer that is located closer than about 20 to 50 m from a cropped field should be placed at 5 m height or higher.

2.6.5 Relative humidity and vapour pressure

Vapour pressure of air can be difficult to measure accurately. Some older electronic humidity sensors were commonly plagued by hysteresis, nonlinearity and calibration errors. Some of these errors are inherent in the sensor design and still plague some modern sensors. Other errors result from dust, moisture, insects, pollution, and age. As described earlier, most modern electronic RH sensors are well designed and are less prone to error. Nevertheless, those sensors can still succumb to age and electronic malfunction.

Replication of RH instruments

It is strongly recommended that duplicate RH and air temperature sensors be permanently employed in electronic weather stations, at least for some period each year. When duplicate RH and air temperature sensors yield similar measurements,

then it is likely that both sensors are functioning properly, provided proper calibration equations have been used. However, even though duplicate sensors are in agreement does not mean that the readings are free from calibration errors and biases due to basic sensor design, nonlinearity, etc.

Trends in computed dewpoint temperature with time

When air humidity is measured using RH sensors, the actual vapour pressure of the air (e_a) is calculated as:

$$e_a = \frac{RH}{100} e^o(T) \tag{2.20}$$

where e^o (T) is the saturation vapour pressure at air temperature T and RH is in %. It is essential that RH and T be taken for the same time period, preferably no longer than 1 hour.

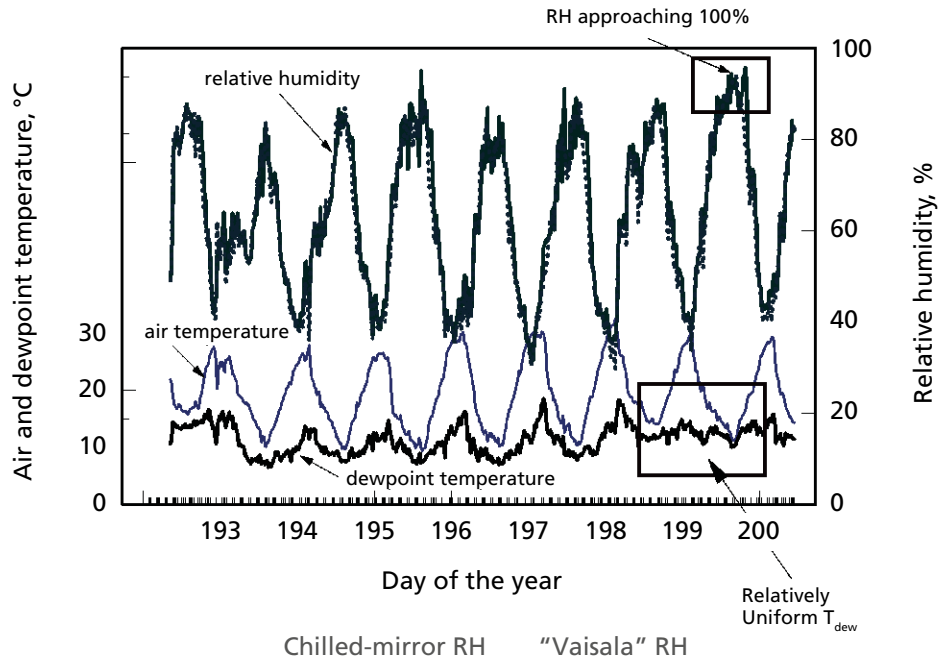
Hourly (or shorter period) measurements of RH, dewpoint temperature (T_{dew}) or vapour pressure (e_a) can be preliminarily assessed by plotting hourly measurements of computed T_{dew} or e_a with time. Relative humidity will vary significantly with time of day, and inversely with air temperature as shown in Figure 2.15. However, both T_{dew} and e_a, either measured directly, or computed using RH and T measurements, should remain somewhat constant throughout a 24-hour period when the air mass is stable and the advection of dry air from outside the area is small. During these stable periods, one should expect to see some rise in T_{dew} and e_a during daytime periods when ET fluxes humidify the equilibrium boundary layer. However, this increase is usually less than about 10 to 20 percent of the total vapour content. Variation in T_{dew} increases significantly when a weather front passes overhead. Because e_a is calculated as the product of RH and saturation vapour pressure at air temperature, any error in the RH calibration will cause false variation in T_{dew} and e_a with changing air temperature.

Figure 2.15 includes T_{dew} which was calculated from measurements of RH and air temperature at the weather station in the center of a wetland near Logan, UT (20-minute data). T_{dew} generally varied from hour to hour due to air mass instability and increased during most days of this period as evaporation from the local wetland vegetation added humidity to the air. The data sequence shows some periods of relatively constant measurement (calculation) of T_{dew} throughout a 24-hour period (for example day of year 199), even though air temperature varied substantially. This is a good indication that the RH sensor was probably functioning correctly and that the instrument calibrations were probably valid. Figure 2.15 also shows, for the same weather station, a comparison between RH measured using two different and independent relative humidity sensors.

The two sensors, one a "chilled-mirror" device that measures T_{dew} directly, with RH calculated using Equation 2.21 and 2.25, and the other, a device that measures RH directly, agreed very well with each other during the 8 days shown. The value of having "redundancy" in instrumentation is demonstrated in this example, where the two different devices measuring the same parameter (in this case RH) leave no question concerning the validity and accuracy of the RH measurements, due to the close agreement and complete independence between the two types of sensors. The use of only a single instrument would leave some questions as to accuracy.

One can notice in Figure 2.15 that the RH approached 100 percent on day 200, which is expected for a well-watered setting. The difference between minimum daily temperature and T_{dew} was generally 1 to 2 °C for many of the days. This small difference is expected in dry, advective environments when over an evaporating or transpiring surface.

FIGURE 2.15
T_{dew} and RH from measurements taken near Logan, Utah, USA during 1995 (20-minute data)



Source: FAO56, 1998

Observations during periods of dew or rainfall

In many climates, especially those where night-time dew occurs, T_{dew} during early morning hours before sunrise should coincide closely with recorded T_{min} and RH should approach 100 percent. For automatic recording weather stations where recording rain gauges are used, one should expect RH recordings during periods of rain or light drizzle to exceed 95 percent.

Relative humidity recordings that exceed 100 percent by more than 3-5 percent during early morning hours or during precipitation events indicate a need for recalibration and numerical adjustment of collected data.

Maximum daily relative humidity

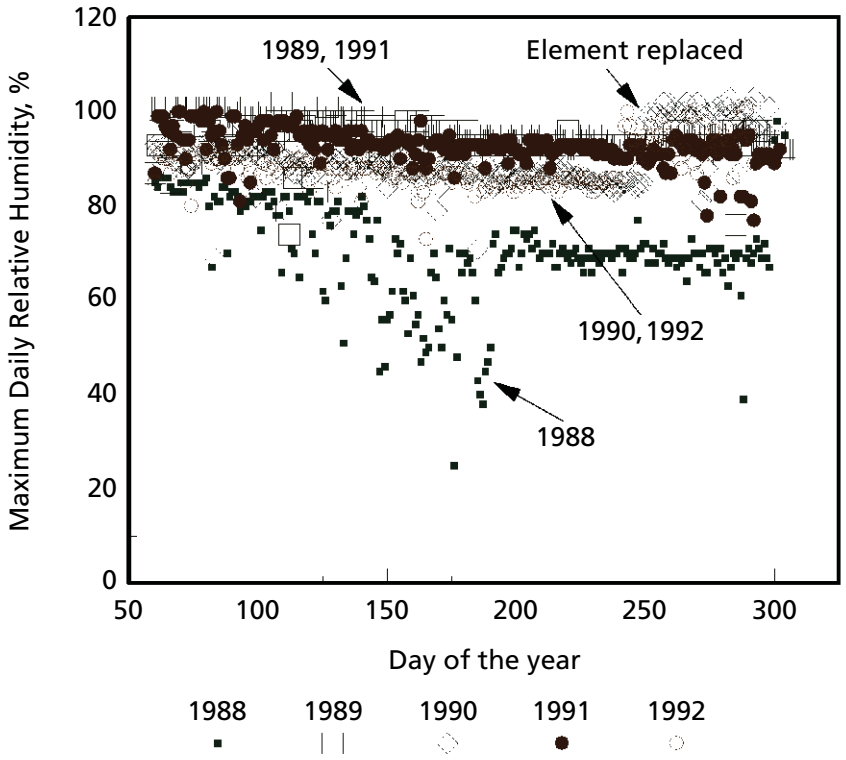
When humidity data are measured in a reference setting with extensive irrigated fetch, early morning RH will often approach 100 percent, even in semiarid areas if measurements are taken inside an irrigated region. Values of maximum relative humidity (RH_{max}) that consistently fall below 80 percent to 90 percent when in an irrigated or well-watered setting may indicate problems in RH sensor calibration or functioning or may indicate aridity of the measurement site and deviation from reference conditions. This may also be indicative of a very dry arid region surrounding the irrigated areas where dry air is continually advected over the irrigated areas.

Figure 2.16 shows daily measurements of RH_{max} from an electronic agricultural weather station located near North Baltimore, Ohio, USA over a five-year period. One would expect RH_{max} to approach 100 percent in this sub-humid setting. However, one can see clear evidence in Figure 2.16 that the RH sensor was undermeasuring RH_{max} for several years, with decreasing trends in RH_{max} visible during these years. This indicates that the RH sensor was functioning relatively well, electronically, except during 1988.

The calibration of the sensor element had seriously declined and was not valid for the year 1988, and had some decay in readings during 1990 and 1992. Sensor elements were typically replaced in September of each year. RH data for 1990 and 1992 could potentially be corrected by multiplying the RH measurements by a correction factor or by adding an offset.

The type of plotting and screening demonstrated in Figures 2.15 and 2.16 shows the simple types of integrity assessments that can be utilized in near-real-time or with historical data. These types of assessments can be applied to all weather data used in evapotranspiration estimation and should be adopted by operators of agricultural weather networks.

FIGURE 2.16
Daily values of measured RH_{max} at North Baltimore, Ohio (1988-92) showing inappropriate calibration of the sensor for years 1988, 1990 and 1992



Source: FAO56, 1998

Chapter 3

Meteorological variables and parameters

The methods for calculating evapotranspiration from meteorological data require climatological inputs and physical parameters. Some data are measured directly at weather stations. Other parameters are related to commonly measured data and can be derived with the help of a direct or empirical relationship. This chapter discusses the source, measurement and computation of all data required for the calculation of the reference evapotranspiration (ET_o) by means of the FAO Penman-Monteith method. Different examples illustrate the various calculation procedures.

Meteorological data can be expressed in several units. Conversion factors between various units and the standard International System of Units (S.I.) are given in Annex 1. Climatic parameters, calculated by means of the equations presented in this chapter are tabulated and displayed for different meteorological conditions in Annex 2. Only the standardized relationships are presented in this chapter. Annex 3 proposes procedures for completing data series and for statistical analysis of weather data.

3.1 METEOROLOGICAL FACTORS DETERMINING ET

The meteorological factors determining evapotranspiration are those weather parameters that provide energy for vaporization and that remove water vapour from the evaporating surface. The principal weather parameters to consider are presented below.

3.1.1 Solar radiation

Evapotranspiration involves the transformation of liquid water into vapour. That process requires large amounts of energy. As a result, the rate of the evapotranspiration process is determined by the amount of energy available to vaporize water. Solar radiation is the largest energy source and is able to change large quantities of liquid water into water vapour. The potential amount of radiation that can reach the evaporating surface is determined by the location and time of the year. Due to differences in the position of the sun, the potential radiation differs at various latitudes, time of day, and during different seasons. The actual solar radiation reaching the evaporating surface depends on the turbidity of the atmosphere and the presence of clouds that may reflect and absorb major parts of the radiation. When assessing the effect of solar radiation on evapotranspiration, one should bear in mind that not all available energy is used to vaporize water. Part of the solar energy is used to add warmth to the atmosphere and to the soil profile.

3.1.2 Air temperature

The solar radiation absorbed by the atmosphere and the heat emitted by the surface increase the air temperature. The sensible heat of the surrounding air can transfer energy to the crop, if the air is warmer than the surface, and exerts as such a controlling influence on the rate of evapotranspiration. In sunny, warm weather, the loss of water by evapotranspiration is greater than in cloudy and cool weather.

3.1.3 Air humidity

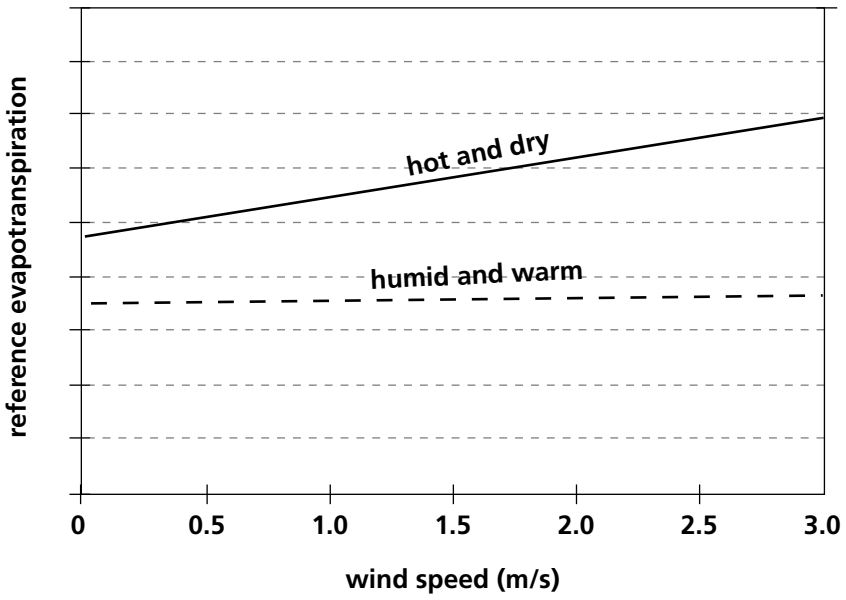
While the energy supply from the sun and the surrounding air is the main driving force for the vaporization of water, the difference between the water vapour pressure at the evapotranspiring surface and the surrounding air is the determining factor for the vapour removal. Well-watered fields in hot, dry arid regions consume large amounts of water due to the abundance of energy and the desiccating power of the dry atmosphere. In humid tropical regions, notwithstanding the high energy input, the high humidity of the air will reduce the evapotranspiration demand. In such an environment, the air may already be close to saturation, so that less additional water can be stored in the air and hence the evapotranspiration rate is lower than in arid regions.

3.1.4 Wind speed

The process of vapour removal depends to a large extent on wind and air turbulence which transfers large quantities of air over the evaporating surface. When vaporizing water, the air above the evaporating surface becomes gradually more saturated with water vapour. If this air is not continuously replaced with drier air by turbulence and air movement, the driving force for water vapour removal and the evapotranspiration rate decrease.

The combined effect of climatic factors affecting evapotranspiration is illustrated in Figure 3.1 for two different climatic conditions.

FIGURE 3.1
Illustration of the effect of wind speed on evapotranspiration in hot-dry and humid-warm weather conditions



Source: FAO56, 1998

The evapotranspiration demand is high in hot dry weather due to the dryness of the air and generally large amounts of energy available as direct solar radiation and sensible heat. Under these circumstances, much water vapour can be stored in the air while wind may promote the transport of water allowing more water vapour to be taken up. On the other hand, under humid weather conditions, the high humidity of the air and the presence of clouds cause the evapotranspiration rate to be lower. The effect on

evapotranspiration of increasing wind speeds for the two different climatic conditions is illustrated by the slope of the curves in Figure 3.1. The drier the atmosphere, the larger the effect of wind speed on ET and the greater the slope of the curve. For humid conditions, the wind can only replace saturated air at the plant surface with slightly less saturated air and remove heat energy. Consequently, the wind speed affects the evapotranspiration rate to a lesser extent than under arid conditions where small variations in wind speed may result in larger variations in the evapotranspiration rate.

3.2 ATMOSPHERIC PARAMETERS

Several relationships are available to express climatic parameters. The effect of the principal weather parameters on evapotranspiration can be assessed with the help of these equations. Some of the relationships require parameters that express a specific characteristic of the atmosphere. Before studying the four principal weather parameters, some atmospheric parameters will be discussed.

3.2.1 Atmospheric pressure (P)

The atmospheric pressure, P, is the pressure exerted by the weight of the earth's atmosphere. Evaporation at high altitudes is promoted due to low atmospheric pressure as expressed in the psychrometric constant. The sensitivity of ET_o to P is, however, for small variations in P, and in the calculation procedures, the average value for a location is sufficient. A simplification of the ideal gas law, assuming 20 °C for a standard atmosphere, can be employed to calculate P as a function of elevation:

$$P = 101.3\left(\frac{293 - 0.0065\ z}{293}\right)^{5.26} \qquad (3.1)$$

where P atmospheric pressure [kPa],
z elevation above sea level [m].

Values for atmospheric pressure as a function of altitude are given in Annex 2 (Table A2.1).

3.2.2 Latent heat of vaporization (λ)

The latent heat of vaporization, λ, expresses the energy required to transform a unit mass of water from liquid to water vapour in a constant pressure and constant temperature process. The value of the latent heat varies as a function of temperature (T). At a high temperature, less energy will be required than at lower temperatures.

As λ varies only slightly over normal temperature ranges, a single value of 2.45 MJ kg^{-1} is taken in the simplification of the FAO Penman-Monteith ET_o (FAO-PM ET_o) equation. This is the latent heat for an air temperature of ~ 20 °C. A full equation for λ is:

$$\lambda = 2.501 - (2.361 \times 10^{-3})\,T \qquad (3.2)$$

3.2.3 Psychrometric constant (γ)

The psychrometric constant, γ, is given by:

$$\gamma = \frac{c_p\,P}{\varepsilon\,\lambda} = 0.665 \times 10^{-3}\,P \qquad (3.3)$$

where γ psychrometric constant [kPa °C^{-1}],
P atmospheric pressure [kPa],
λ latent heat of vaporization, 2.45 [MJ kg^{-1}],
c_p specific heat at constant pressure, 1.013 10^{-3} [MJ kg^{-1} °C^{-1}],
ε ratio of molecular weight of water vapour/dry air = 0.622.

The specific heat at constant pressure is the amount of energy required to increase the temperature of a unit mass of air by one degree at constant pressure. Its value depends on the composition of the air, i.e. on its humidity. For average atmospheric conditions, a value c_p = 1.013 10^{-3} MJ kg^{-1} °C^{-1} can be used. As an average atmospheric pressure is used for each location (Equation 3.1), the psychrometric constant is kept constant for each location. Values for the psychrometric constant as a function of altitude are given in Annex 2 (Table A2.2). Example 3.1 refers to the determination of the psychrometric constant, γ.

EXAMPLE 3.1
Determination of atmospheric parameters

Determine the atmospheric pressure and the psychrometric constant at an elevation of 1 800 m.

With:	z =	1 800	m
From Eq. 3.1:	P = 101.3 [(293 – (0.0065) 1 800)/293]$^{5.26}$ =	81.8	kPa
From Eq. 3.3:	γ = 0.665 10^{-3} (81.8) =	0.054	kPa °C^{-1}

The average value for the atmospheric pressure is 81.8 kPa.
The psychrometric constant, γ, is 0.054 kPa °C^{-1}.

3.2.4 Air temperature

Agrometeorology is concerned with the air temperature near the level of the crop canopy. In traditional and modern automatic weather stations the air temperature is measured inside shelters (Stevenson screens, or ventilated, or naturally aspirated radiation shields) placed in accordance with World Meteorological Organization (WMO) standards at 2 m above the ground. The shelters are designed to protect the instruments from direct exposure to solar heating. The louvered construction promotes free air movement around the instruments.

Air temperature is measured with thermometers, thermistors or thermocouples mounted in the shelter. Minimum and maximum thermometers record the minimum and maximum air temperature over a 24-hour period and use mercury and alcohol. Older measurement systems have used bimetallic thermographs as mechanical temperature recorders. These were easy to read and maintain. However, mechanical thermographs require verification and adjustment of the position of the pen recorder to plot the instantaneous temperature on paper over a day or week. These instruments were usually installed in shelters that were naturally ventilated.

Modern temperature sensors are generally the thermistor or the thermocouple. These sensors provide accurate analogue measurements and are normally utilized in automated weather stations. Thermistors provide direct measurements of air (or soil) temperature, whereas thermocouples require an additional base temperature reading at

the datalogger terminal that is normally provided by a thermistor. Electronic weather stations often sample air temperature each second or minute and report hourly averages in addition to 24-hour maximum and minimum values.

Due to the non-linearity of humidity data required in the FAO-PM ET_o equation, the vapour pressure for a certain period should be computed as the average of the vapour pressure at the daily maximum and minimum air temperatures during that period. The daily maximum air temperature (T_{max}) and daily minimum air temperature (T_{min}) are, respectively, the maximum and minimum air temperature observed during the 24-hour period, beginning at midnight. T_{max} and T_{min} for longer periods such as weeks, 10 days or months are obtained by dividing the sum of the respective daily values by the number of days in the period. The mean daily air temperature (T_{mean}) is only employed in the 24-hour FAO-PM ET_o equation to calculate the slope of the saturation vapour pressure curve (Δ) and the impact of mean air density (ρ_a) as the effect of temperature variations on the value of the climatic parameter is small in these cases. For standardization, the T_{mean} for 24-hour periods is defined as the average of the daily maximum (T_{max}) and minimum temperatures (T_{min}) rather than as the average of hourly temperature measurements.

$$T_{mean} = \frac{T_{max} + T_{min}}{2} \tag{3.4}$$

The temperature is given in degrees Celsius (°C) or Fahrenheit (°F). The conversion table is given in Annex 1. In some calculation procedures, the temperature is required in Kelvin (K), which can be obtained by adding 273.15 to the temperature expressed in degrees Celsius (in practice K = °C + 273.15). The Kelvin and Celsius scales have the same scale interval.

3.2.5 Air humidity

Concepts

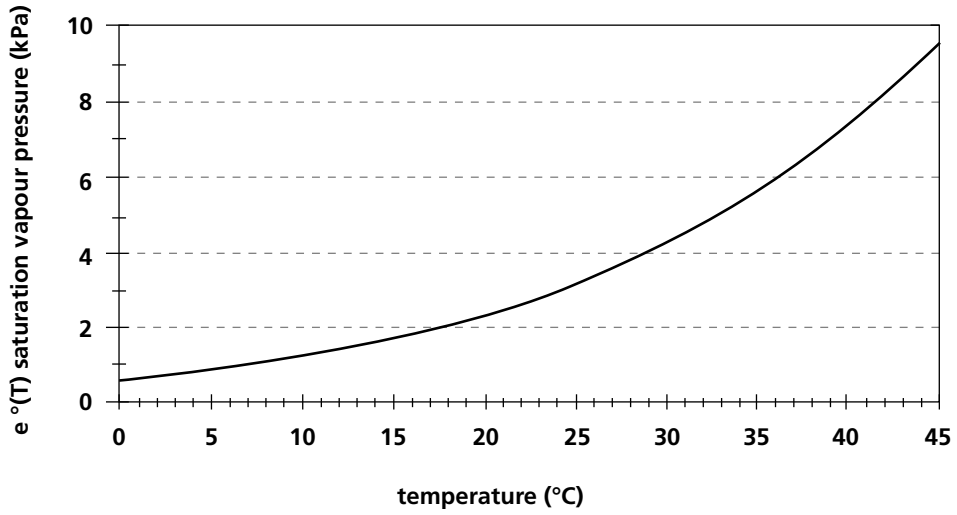
The water content of the air can be expressed in several ways. In agrometeorology, vapour pressure, dewpoint temperature and relative humidity are common expressions to indicate air humidity.

Vapour pressure

Water vapour is a gas, and its pressure contributes to the total atmospheric pressure. The amount of water in the air is related directly to the partial pressure exerted by the water vapour in the air and is, therefore, a direct measure of the air water content.

When air is enclosed above an evaporating water surface, an equilibrium is reached between the water molecules escaping and returning to the water reservoir. At that moment, the air is said to be saturated since it cannot store any extra water molecules. The corresponding pressure is called the saturation vapour pressure ($e^o(T)$). The number of water molecules that can be stored in the air depends on the temperature (T). The higher the air temperature, the higher the storage capacity, and the higher its saturation vapour pressure (Figure 3.2).

FIGURE 3.2
Saturation vapour pressure shown as a function of temperature: $e^o(T)$ curve



Source: FAO56, 1998

As can be seen from Figure 3.2, the slope of the curve changes exponentially with temperature. At low temperatures, the slope is small and varies only slightly as the temperature rises. At high temperatures, the slope is large and small changes in T result in large changes in slope. The slope of the saturation vapour pressure curve, Δ, is an important parameter in describing vaporization and is required in the equations for calculating ET_o from climatic data.

The actual vapour pressure (e_a) is the vapour pressure exerted by the actual quantity of water in the air. When the air is not saturated, the actual vapour pressure will be lower than the saturation vapour pressure. The difference between the saturation and actual vapour pressure is called the vapour pressure deficit or saturation deficit and is an accurate indicator of the actual evaporative capacity of the air.

> In standard S.I. units, pressure is no longer expressed in centimetre of water, millimetre of mercury, bars, atmosphere, etc., but in pascals (Pa). Conversion factors between various units and Pa are given in Annex 1. As a pascal refers to a relatively small force (1 newton) applied on a relatively large surface (1 m^2), multiples of the basic unit are often used. Vapour pressure is expressed in kilopascals (1 kPa = 1 000 Pa).

Dewpoint temperature

The dewpoint temperature is the temperature to which the air needs to be cooled to make the air saturated. The actual vapour pressure of the air is the saturation vapour pressure at the dewpoint temperature. The drier the air, the larger the difference between the air temperature and dewpoint temperature.

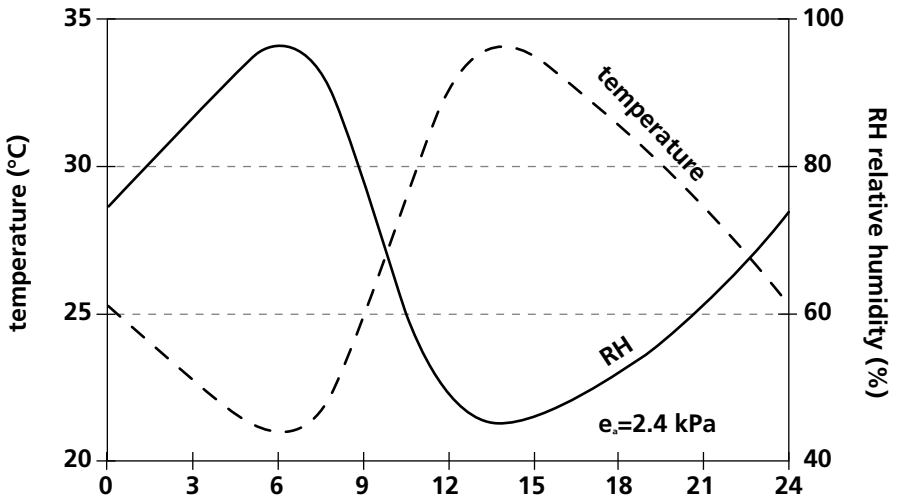
Relative humidity

The relative humidity (RH) expresses the degree of saturation of the air as a ratio of the actual (e_a) to the saturation ($e^o(T)$) vapour pressure at the same temperature (T):

$$RH = 100 \frac{e_a}{e^o(T)} \quad (3.5)$$

Relative humidity is the ratio between the amount of water the ambient air actually holds and the amount it could hold at the same temperature. It is dimensionless and is commonly given as a percentage. Although the actual vapour pressure might be relatively constant throughout the day, the relative humidity fluctuates between a maximum near sunrise and a minimum around early afternoon (Figure 3.3). The variation of the relative humidity is the result of the fact that the saturation vapour pressure is determined by the air temperature. As the temperature changes during the day, the relative humidity also changes substantially.

FIGURE 3.3
Variation of the relative humidity over 24 hours for a constant actual vapour pressure of 2.4 kPa



Source: FAO56, 1998

3.2.6 Measurement

It is not possible to directly measure the actual vapour pressure. The vapour pressure is commonly derived from relative humidity or dewpoint temperature. Relative humidity is measured directly with hygrometers. Older mechanical systems in mechanically-based field stations used hair hygrometers where the hair changes its length in response to changes in air humidity. These systems were normally operated as mechanical hygrographs. Measurements with hair hygrometers tend to lose accuracy with dust and aging of the hair. Modern electronic hygrometers use films from a dielectric polymer that changes its dielectric constant with changes in surface moisture, thus inducing a variation in the capacity of a condensator using that dielectric. These instruments are normally called dielectric polymer capacitive hygrometers. Accuracy can be higher than for hair hygrometers, the systems can be fully automated, and modern electronic sensors require little maintenance.

Vapour pressure can be measured indirectly with psychrometers that measure the temperature difference between two thermometers, the so-called dry and wet bulb thermometers. The dry bulb thermometer measures the temperature of the air. The bulb of the wet bulb thermometer is covered with a wick that is kept constantly saturated with distilled water. Evaporation of water from the wick, requiring energy, lowers the temperature of the thermometer. The drier the air, the larger the evaporative cooling and the larger the temperature drop. The difference between the dry and wet

bulb temperatures is called the wet bulb depression and is a measure of the air humidity. Usually, the dry and wet bulb thermometers are swung in the air by hand to produce sufficient air flow past the bulbs. The Assmann psychrometer has a forced ventilation of the wet bulb and dry bulb thermometers. When wet and dry bulb instruments are placed inside a naturally ventilated inside a shelter, problems can arise if air flow is not sufficient to maintain an appropriate evaporation rate and associated cooling. In more modern systems, the dry and wet bulb temperatures are measured by thermocouples or by thermistors, the so-called thermocouple psychrometers and thermo-sound psychrometers. These psychrometers are used in automatic weather stations and, when properly maintained and operated, tend to provide accurate measurements.

The dewpoint temperature is measured with dewpoint meters. The underlying principle of some types of apparatus is the cooling of the ambient air until dew formation occurs. The corresponding temperature is the dewpoint temperature. Measurement systems can use a mirror like metallic surface that is artificially cooled until vapour condenses on it and blocks the reading of reflected light. Other dew sensor systems use chemical or electric properties of certain materials that are altered when absorbing water vapour. Instruments for measuring dew point temperature require careful operation and maintenance and are seldom available in weather stations. The accuracy of estimation of the actual vapour pressure from T_{dew} is generally good under good maintenance protocol.

Relative humidity and dewpoint temperature data can be notoriously plagued by measurement errors. Measurement error is common for both older hygro-thermograph types of instruments and older electronic instruments prior to about the year 2000. The more modern electronic instruments since about the year 2000 tend to have improved dependability, consistency, and accuracy. However, the data collected still must be reviewed and verified.

Great care should be taken to assess the accuracy and integrity of RH and dewpoint data. The user is encouraged to always compare computed dewpoint temperatures to daily minimum air temperatures, as described near the end of Chapter 2. Frequently, it is better to utilize a dewpoint temperature that is estimated from daily minimum air temperature, rather than to use unreliable relative humidity measurements. The user is encouraged to utilize good judgement in this area.

3.2.7 Calculation procedures

Mean saturation vapour pressure (e_s)

As saturation vapour pressure is related to air temperature, it can be calculated from the air temperature. The relationship is expressed by:

$$e^o(T) = 0.6108 \exp\left[\frac{17.27\,T}{T + 237.3}\right] \tag{3.6}$$

where $e^o(T)$ saturation vapour pressure at the air temperature T [kPa],
T air temperature [°C],
exp[..] 2.7183 (base of natural logarithm) raised to the power [..].

Values for saturation vapour pressure as a function of air temperature are given in Annex 2 (Table A2.3). Due to the non-linearity of the above equation, the mean saturation vapour pressure for a day, week, decade or month (e_s) should be computed

as the mean between the saturation vapour pressure at the mean daily maximum and minimum air temperatures for that period:

$$e_S = \frac{e^o(T_{max}) + e^o(T_{min})}{2} \quad (3.7)$$

Example 3.2 refers to the determination of mean saturation vapour pressure, e_s. Using mean air temperature instead of daily minimum and maximum temperatures results in lower and less correct estimates for the mean saturation vapour pressure. The corresponding vapour pressure deficit (a parameter expressing the evaporating power of the atmosphere) will also be smaller and the result will be some underestimation of the reference crop evapotranspiration. Therefore, the mean saturation vapour pressure, e_s, should be calculated as the mean between the saturation vapour pressure at both the daily maximum and minimum air temperature. For hourly or shorter time periods, e_s can be calculated using the mean hourly air temperature since differences between T_{max} and T_{min} over the hourly period commonly are relatively small.

EXAMPLE 3.2
Determination of mean saturation vapour pressure

The daily maximum and minimum air temperature are respectively 24.5 and 15 °C. Determine the saturation vapour pressure for that day.

From Eq. 3.6	$e^o(T_{max}) = 0.6108 \exp[17.27(24.5)/(24.5+237.3)]$	3.075	kPa
From Eq. 3.6	$e^o(T_{min}) = 0.6108 \exp[17.27(15)/(15+237.3)]$	1.705	kPa
From Eq. 3.7	$e_s = (3.075 + 1.705)/2$	2.39	kPa
	Note that for temperature 19.75 °C (which is T_{mean}), $e^o(T)$ =	2.30	kPa

The mean saturation vapour pressure is 2.39 kPa.

Slope of saturation vapour pressure curve (Δ)

For the calculation of evapotranspiration, the slope of the relationship between saturation vapour pressure and temperature, Δ, is required. The slope of the curve (Figure 3.2) at a given temperature is given by:

$$\Delta = \frac{4098\left[0.6108 \exp\left(\frac{17.27\,T}{T+237.3}\right)\right]}{(T+237.3)^2} \quad (3.8)$$

where Δ slope of saturation vapour pressure curve at air temperature T [kPa °C^{-1}],
T air temperature [°C],
exp(…) 2.7183 (base of natural logarithm) raised to the power [..].

Values of slope Δ for different air temperatures are given in Annex 2 (Table A2.4). In the FAO-PM ET_o equation, where Δ occurs in the numerator and denominator, the slope of the vapour pressure curve is calculated using mean air temperature (Equation 3.4).

Actual vapour pressure (e_a) derived from dewpoint temperature

As the dewpoint temperature is the temperature to which the air needs to be cooled to make the air saturated, the actual vapour pressure (e_a) [kPa] is the saturation vapour pressure at the dewpoint temperature (T_{dew}) [ºC], or:

$$e_a = e^o(T_{dew}) = 0.6108 \exp\left[\frac{17.27\ T_{dew}}{T_{dew} + 237.3}\right] \quad (3.9)$$

For a day, week, decade or month, e_a should be computed using the mean dewpoint temperature for that period. For hourly time periods, e_a is calculated using the hourly dewpoint temperature.

Equation 3.9 can be inverted to solve for T_{dew} given e_a:

$$T_{dew} = \frac{116.91 + 237.3 \ln(e_a)}{16.78 - \ln(e_a)} \quad (3.10)$$

Actual vapour pressure (e_a) derived from psychrometric data

The actual vapour pressure can be determined from the difference between the dry and wet bulb temperatures, the so-called wet bulb depression. The relationship is expressed (Example 3.3) by the following equation:

$$e_a = e^o(T_{wet}) - \gamma_{psy}(T_{dry} - T_{wet}) \quad (3.11)$$

where e_a actual vapour pressure [kPa],

$e^\circ(T_{wet})$ saturation vapour pressure at wet bulb temperature [kPa],

γ_{psy} psychrometric constant of the instrument [kPa °C^{-1}],

$T_{dry} - T_{wet}$ wet bulb depression, with T_{dry} the dry bulb and T_{wet} the wet bulb temperature [°C].

The psychrometric constant of the instrument is given by:

$$\gamma_{psy} = a_{psy}\ P \quad (3.12)$$

where a_{psy} is a coefficient depending on the type of ventilation of the wet bulb [°C^{-1}], and P is the atmospheric pressure [kPa]. The coefficient a_{psy} depends mainly on the design of the psychrometer and the rate of ventilation around the wet bulb. The following values are used for a_{psy}:

0.000662 for ventilated (Assmann type) psychrometers, with an air movement of some 5 m s^{-1} (this is the most common value used),

0.000800 for naturally ventilated psychrometers (ventilation of about 1 m s^{-1}),

0.001200 for non-ventilated psychrometers installed indoors.

EXAMPLE 3.3
Determination of actual vapour pressure from psychrometric readings

Determine the vapour pressure from the readings of an aspirated psychrometer in a location at an elevation of 1 200 m. The temperatures measured by the dry and wet bulb thermometers are 25.6 and 19.5 °C respectively.

From Eq. 3.1 (Table A2.1), at:	z=	1 200	m
Then:	P =	87.9	kPa
From Eq. 3.6 (Table A2.3), for	T_{wet} =	19.5	°C
Then:	$e°(T_{wet})$ =	2.267	kPa
Ventilated psychrometer	a_{psy} =	0.000662	$°C^{-1}$
From Eq. 3.11:	$e_a = 2.267 - 0.000662\ (87.9)\ (25.6 - 19.5)$ =	1.91	kPa

The actual vapour pressure is 1.91 kPa.

Actual vapour pressure (e_a) derived from relative humidity data

The actual vapour pressure is routinely calculated from the relative humidity (RH). Depending on the availability of the humidity data, different equations should be used.

- For RH_{max} and RH_{min}:

$$e_a = \frac{e^o(T_{min})\frac{RH_{max}}{100} + e^o(T_{max})\frac{RH_{min}}{100}}{2} \quad (3.13)$$

where e_a actual vapour pressure [kPa],
$e°(T_{min})$ saturation vapour pressure at daily minimum temperature [kPa],
$e°(T_{max})$ saturation vapour pressure at daily maximum temperature [kPa],
RH_{max} maximum relative humidity [%],
RH_{min} minimum relative humidity [%].

For periods of a week, ten days, or a month, RH_{max} and RH_{min} are obtained by dividing the sum of the daily values by the number of days in that period. For hourly or shorter time periods, either Equation 3.13 or Equation 3.15 can be used with little difference (Example 3.4).

- For RH_{max}:

When using equipment where errors in estimating RH_{min} can be large, or when RH data integrity is in doubt, then one can use only RH_{max}:

$$e_a = e^o(T_{min})\ \frac{RH_{max}}{100} \quad (3.14)$$

RH_{max} tends to be more dependable and accurate than RH_{min}, since RH_{max} may often approach 100 percent during early morning hours and can thereby be more readily QAQC'ed.

- For RH_{mean}:

In the absence of RH_{max} and RH_{min}, equations based on RH_{mean} can be used to estimate e_a:

$$e_a = \frac{RH_{mean}}{100} e^o(T_{mean}) \quad (3.15)$$

and

$$e_a = \frac{RH_{mean}}{100} \left[\frac{e^o(T_{max}) + e^o(T_{min})}{2} \right] \quad (3.16)$$

where RH_{mean} the mean relative humidity, defined as the average between RH_{max} and RH_{min},

T_{mean} daily mean air temperature.

EXAMPLE 3.4
Determination of actual vapour pressure from relative humidity

Given the following daily minimum and maximum air temperature and the corresponding relative humidity data:

T_{min} = 18° C and RH_{max} = 82%

T_{max} = 25° C and RH_{min} = 54%

Determine the actual vapour pressure.

From Eq. 3.6 (Table A2.3), at:	T_{min} =	18	°C
Then:	$e^o(T_{min})$ =	2.064	kPa
From Eq. 3.6 (Table A2.3), at:	T_{max} =	25	°C
Then:	$e^o(T_{max})$ =	3.168	kPa
From Eq. 3.13:	e_a = [2.064 (82/100) + 3.168 (54/100)] / 2 =	1.70	$°C^{-1}$
Note that when using Eq. 3.15:	T_{mean} = (25 + 18) / 2 =	21.5	°C
	$e^o(T_{mean})$ =	2.564	kPa
	e_a = [(82 + 54) / 2] [2.564] =	1.74	kPa
and that when using Eq. 3.16:	e_a = [(82 + 54) / 2] [2.064 + 3.168] / 2 =	1.78	kPa

When reported RH data are known to have been derived from measured or calculated vapour pressure or dewpoint temperature, then the same calculations that were originally used to derive the RH data should be followed to calculate e_a. In other words, if RH_{mean} had been derived from a daily average vapour pressure and daily average air temperature, then Equation 3.15 should be utilized. If RH_{mean} data had been derived as an average of daily RH_{max} and RH_{min}, then Equation 3.16 should be utilized.

Studies reported by the American Society of Civil Engineers in 2016 indicated that Equation 3.15 may produce more representative estimates of daily average e_a than Equation 3.16. However, Equations 3.15 and 3.16 are usually less desirable than Equations 3.13 or 3.14. Estimating e_a from RH_{min} and T_{max} alone is discouraged because of tendencies for large uncertainty in measured RH_{min} that can be magnified by large values for $e^o(T_{max})$. Estimating e_a from daily average dewpoint temperature or hourly average dewpoint temperature is the most desirable approach. For hourly or shorter time periods, Equation 3.15 is valid, where T is the hourly average air temperature.

Vapour pressure deficit ($e_s - e_a$)

The vapour pressure deficit is the difference between the saturation (e_s) and actual vapour pressure (e_a) for a given time period. For time periods such as a week, ten days or a month e_s is computed from Equation 3.7 using the T_{max} and T_{min} averaged over the time period. Similarly, the e_a is computed with one of the Equations 3.9 to 3.16, using average measurements over the period (Example 3.5). When desired, e_s and e_a for long time periods can also be calculated as averages of values computed for each day of the period. The calculation sheet for the vapour pressure deficit is given in Box 3.1.

Using mean air temperature and not T_{max} and T_{min} in Equation 3.7 for daily calculations results in a lower estimate of e_s, thus in a lower vapour pressure deficit and hence an underestimation of the ET_o (see Box 3.1). When desired, e_s and e_a for long time periods can also be calculated as averages of values computed for each day of the period.

BOX 3.1
Calculation sheet for vapour pressure deficit ($e_s - e_a$)

Saturation vapour pressure: e_s (Eq. 3.6 or Table A2.3)

T_{max}		°C	$e^o(T_{max}) = 0.6108 \exp\left[\frac{17.27\, T_{max}}{T_{max}+237.3}\right]$		kPa
T_{min}		°C	$e^o(T_{min}) = 0.6108 \exp\left[\frac{17.27\, T_{min}}{T_{min}+237.3}\right]$		kPa
Saturation vapour pressure $e_s = [e^o(T_{max})+e^o(T_{min})]/2$ (Eq. 3.7)					kPa

Actual vapour pressure: e_a

1. e_a derived from dewpoint temperature (Eq. 3.9 or Table A2.3)

T_{dew}		°C	$e_a = 0.6108 \exp\left[\frac{17.27\, T_{dew}}{T_{dew}+237.3}\right]$		kPa

OR 2. e_a derived from maximum and minimum relative humidity

RH_{max}		%	$e^o(T_{min})\ \frac{RH_{max}}{100}$		kPa
RH_{min}		%	$e^o(T_{max})\ \frac{RH_{min}}{100}$		kPa
Thus: $e_a = [e^o(T_{min})\ RH_{max}/100 + e^o(T_{max})\ RH_{min}/100] / 2$ (Eq. 3.13)					kPa

OR 3. e_a derived from maximum relative humidity (when there are errors in RH_{min})

RH_{max}		%	$e_a = e^o(T_{min})\ RH_{max}/100$ (Eq. 3.14)		kPa

OR 4. e_a derived from mean relative humidity (when only RH_{mean} is available)

RH_{mean}		%	$e_a = e_s\ (RH_{mean})/100$ (Eq. 3.15 or 3.16)		kPa
Vapour pressure deficit: (e_s - e_a)					kPa

EXAMPLE 3.5
Determination of vapour pressure deficit

Determine the vapour pressure deficit with the data of the previous Example 3.4 for a daily timestep.

From Example 3.4:	$e°(T_{min})$ =	2.064	kPa
	$e°(T_{max})$ =	3.168	kPa
	e_a =	1.70	kPa
	$e_s - e_a = (2.064 + 3.168)/2 - 1.70$ =	0.91	kPa

The vapour pressure deficit is 0.91 kPa.

3.3 RADIATION

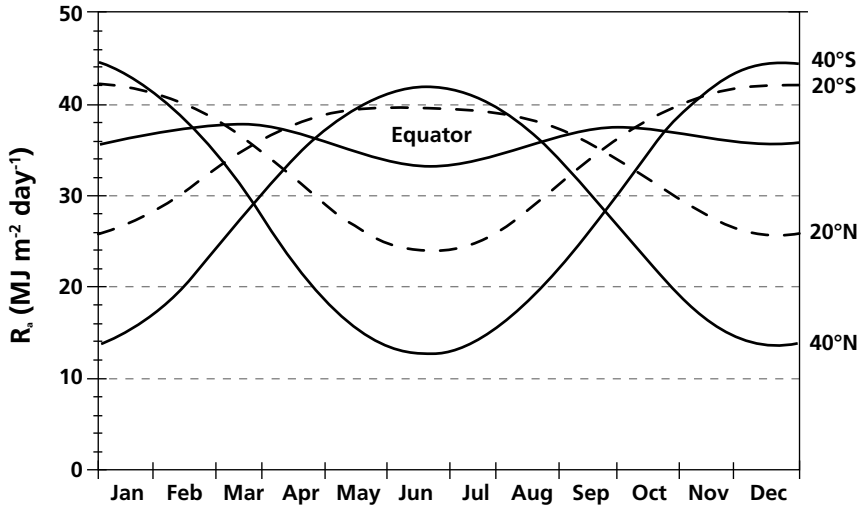
3.3.1 Concepts

Extraterrestrial radiation (R_a)

The radiation striking a surface perpendicular to the sun's rays at the top of the earth's atmosphere, called the solar constant, is about 0.082 MJ m^{-2} min^{-1}. The local intensity of radiation is, however, determined by the angle between the direction of the sun's rays and the normal to the surface of the atmosphere. This angle will change during the day and will be different at different latitudes and during different seasons. The solar radiation received at the top of the earth's atmosphere on a horizontal surface is called the extraterrestrial (solar) or exo-atmospheric radiation, R_a.

If the sun is directly overhead, the angle of incidence is zero and the extraterrestrial radiation is 0.0820 MJ m^{-2} min^{-1}. As seasons change, the position of the sun, the length of the day and, hence, R_a change as well. Extraterrestrial radiation is thus a function of latitude, date and time of day. Daily values of R_a throughout the year for different latitudes are plotted in Figure 3.4.

FIGURE 3.4
Annual variation in extraterrestrial radiation (R_a) at the Equator, 20°and 40° north and south



Source: FAO56, 1998

Solar or shortwave radiation (R_s)

As the radiation penetrates the atmosphere, some of the radiation is scattered, reflected or absorbed by the atmospheric gases, clouds and dust. The amount of radiation reaching a horizontal plane is known as the solar radiation, R_s. Because the sun emits energy by means of electromagnetic waves characterized by short wavelengths, solar radiation is also referred to as shortwave radiation.

For a cloudless day, R_s is roughly 75 percent of extra-terrestrial radiation due to about 25 percent of R_a absorbed or scattered in the atmosphere. On a cloudy day, additional radiation is scattered or absorbed by clouds, but even with extremely dense cloud cover, about 25 percent of the extraterrestrial radiation may still reach the earth's surface mainly as diffuse sky radiation. *Solar radiation is also known as global radiation, meaning that it is the sum of direct shortwave radiation from the sun and diffuse sky radiation from all upward angles.*

Relative shortwave radiation (R_s/R_{so})

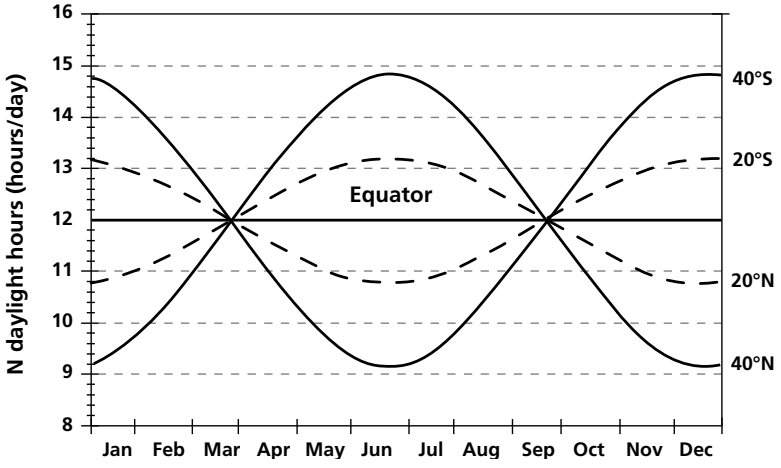
The relative shortwave radiation is the ratio of the solar radiation (R_s) to the clear-sky solar radiation (R_{so}). R_s is the solar radiation that actually reaches the earth's surface in a given period, while R_{so} is the solar radiation that would reach the same surface during the same period but under cloudless conditions.

The relative shortwave radiation is a way to express the cloudiness of the atmosphere; the cloudier the sky the smaller the ratio. The ratio varies between about 0.33 (dense cloud cover) and 1 (clear sky). In the absence of a direct measurement of R_n, the relative shortwave radiation is used in the computation of the net longwave radiation.

Relative sunshine duration (n/N)

The relative sunshine duration is another ratio that expresses the cloudiness of the atmosphere. It is the ratio of the actual duration of sunshine, n, to the maximum possible duration of sunshine or daylight hours N. In the absence of any clouds, the actual duration of sunshine is equal to the daylight hours (n = N) and the ratio is one, while on completely cloudy days n and, consequently, the ratio may be zero. In the absence of a direct measurement of R_s, the relative sunshine duration, n/N, can be used to derive solar radiation from extraterrestrial radiation. As with extraterrestrial radiation, the daylength N depends on the position of the sun and is hence a function of latitude and date. Daily values of N throughout the year for different latitudes are plotted in Figure 3.5.

FIGURE 3.5
Annual variation of the daylight hours (N) at the Equator, 20° and 40° north and south



Source: FAO56, 1998

Albedo (α) and net solar radiation (R_{ns})

A considerable amount of solar radiation reaching the earth's surface is reflected. The fraction, α, of the solar radiation reflected by the surface is known as the albedo. The albedo is highly variable for different surfaces and for the angle of incidence or slope of the ground surface. It may be as large as 0.95 for freshly fallen snow and as small as 0.05 for wet bare soil or open water. A green vegetation cover has an albedo of about 0.20-0.25. For the green grass reference crop, α is assumed to have a value of 0.23. The net solar radiation, R_{ns}, is the fraction of the solar radiation R_s that is not reflected from the surface. Its value is $(1 - \alpha)R_s$.

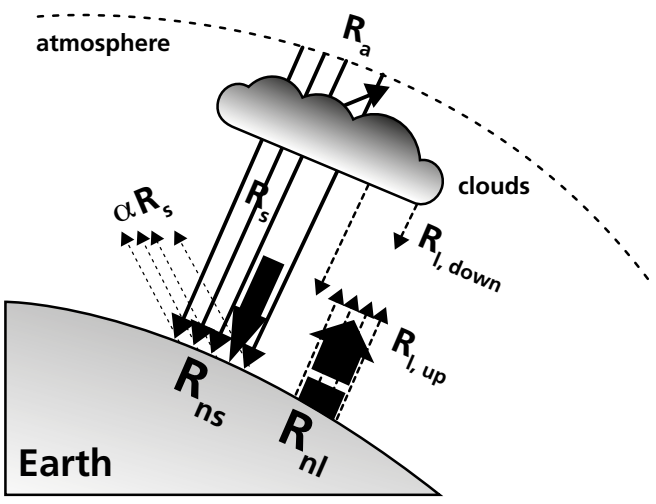
Net longwave radiation (R_{nl})

The solar radiation absorbed by the earth is converted to heat energy. By several processes, including the emission of radiation, the earth loses this energy. The earth, which is at a much lower temperature than the sun, emits radiative energy with wavelengths longer than those from the sun. Therefore, the terrestrial radiation is referred to as longwave radiation. The emitted longwave radiation ($R_{l,\,up}$) is absorbed by the atmosphere or is lost into space. As a consequence of the atmosphere being above absolute zero K, the atmosphere radiates energy of its own, as illustrated in Figure 3.6. Part of the radiation finds its way back to the earth's surface. The longwave radiation received at the surface from the atmosphere ($R_{l,\,down}$) helps to sustain surface temperature. Consequently, the earth's surface both emits and receives longwave radiation. The difference between outgoing and incoming longwave radiation is called the net longwave radiation, R_{nl}. As the outgoing longwave radiation is almost always greater than the incoming longwave radiation, a positive R_{nl} represents an energy loss from the surface.

Net radiation (R_n)

The net radiation, R_n, is the difference between incoming and outgoing radiation of both short and long wavelengths. It is the balance between the energy absorbed, reflected, and emitted by the earth's surface or the difference between the incoming net shortwave (R_{ns}) and the net outgoing longwave (R_{nl}) radiation (Figure 3.6). R_n is normally positive during the daytime and negative during the nighttime. The total daily value for R_n is almost always positive over a period of 24 hours, except in extreme conditions at high latitudes.

FIGURE 3.6
Various components of radiation



Source: FAO56, 1998

Soil heat flux (G)

In making estimates of evapotranspiration, all terms of the energy balance (Equation 1.1) should be considered. The soil heat flux, G, is the energy that is utilized in heating the soil. G is positive when the soil is warming and negative when the soil is cooling. Although the soil heat flux is small compared to R_n and may often be ignored, the amount of energy gained or lost by the soil in this process should theoretically be subtracted or added to R_n when estimating evapotranspiration.

3.3.2 Units

The standard unit used in this guideline to express energy received on a unit surface per unit time is megajoules per square meter per day (MJ m^{-2} day^{-1}) or megajoules per square meter per hour (MJ m^{-2} hr^{-1}). For hourly or shorter time periods, radiation is expressed in (MJ m^{-2} hr^{-1}) for use in the FAO-PM ET_o equation. It may also be expressed as watts per square meter (W m^{-2}). In some meteorological bulletins, other units might be used, or radiation might even be expressed in units no longer accepted as standard S.I. units, such as calories cm^{-2} day^{-1}.

In the 24-hour FAO-PM ET_o equation (Equation 2.4), radiation expressed in MJ m^{-2} day^{-1} is converted (Box 3.2) to equivalent evaporation in mm day^{-1} by using a conversion factor equal to the inverse of the latent heat of vaporization ($1/\lambda = 0.408$):

$$\text{equivalent evaporation [mm day}^{-1}] = 0.408 \times \text{Radiation [MJ m}^{-2}\text{ day}^{-1}] \quad (3.17a)$$

or, for hourly or shorter time periods,

$$\text{equivalent evaporation [mm hr}^{-1}] = 0.408 \times \text{Radiation [MJ m}^{-2}\text{ hr}^{-1}]. \quad (3.17b)$$

BOX 3.2
Conversion from energy values to equivalent evaporation

The conversion from energy values to depths of water or vice versa is given by:

$$\text{Radiation [depth of water]} = \frac{\text{Radiation [energy/surface]}}{\lambda\, \rho_w}$$

where λ latent heat of vaporization [MJ kg^{-1}],
ρ_w density of water, i.e. 1 000 kg m^{-3},
[depth of water] is expressed in m,
[energy/surface] is expressed in MJ m^{-2}.

By using a single value of 2.45 MJ kg^{-1} for λ (see section on atmospheric parameters) and multiplying the above equation by 1 000 to obtain mm:

$$\text{Radiation [mm day}^{-1}] \approx \frac{\text{Radiation [MJ m}^{-2}\text{ day}^{-1}]}{2.45} = 0.408\ \text{Radiation [MJ m}^{-2}\text{ day}^{-1}]$$

Common units used to express energy received on a unit surface per unit time, and conversion factors are summarized in Table 3.1.

TABLE 3.1
Conversion factors for radiation

	Multiplier to obtain energy received on a unit surface per unit time				Equivalent evaporation
	MJ m^{-2} day^{-1}	J cm^{-2} day^{-1}	cal cm^{-2} day^{-1}	W m^{-2}	mm day^{-1}
1 MJ m^{-2} day^{-1}	1	100	23.9	11.57	0.408
1 cal cm^{-2} day^{-1}	4.1868 10^{-2}	4.1868	1	0.485	0.0171
1 W m^{-2}	0.0864	8.64	2.06	1	0.0353
1 mm day^{-1}	2.45	245	58.5	28.4	1

3.3.3 Measurement

Solar radiation can be measured with pyranometers, radiometers or solarimeters. The instruments contain a sensor installed on a horizontal surface that measures the intensity of the total solar radiation, i.e. both direct and diffuse radiation from clear sky and/or cloudy conditions. Solar radiation is most commonly measured with pyranometers. Pyranometers measure the shortwave incoming radiation over a solid angle in the shape of a hemisphere oriented upwards.

In the most common "glassed dome" pyranometers, a thermopile is used within the instrument as the sensor, where thermal gradients are measured across hot and cold areas (black and white) or as conducted into the instrument base (solid black). The radiation intensity is proportional to the temperature differences between the sensing areas. Accuracy depends upon the sensitivity and age of the material used in the sensors, the response time and the distortion characteristics of the material constituting the dome covering the sensors.

A second type of pyranometer that is less expensive and that is gaining acceptance is the silicon diode instrument where electric current is generated by a photo sensitive diode in proportion to solar intensity. Ordinarily, silicon diode pyranometers are not fully sensitive to the full spectrum of visible light, and the calibration of the instrument is only valid for upward-facing solar measurements.

When a pyranometer is oriented downwards it measures the reflected shortwave radiation, and is thus called an albedometer. When two pyranometers are associated they must have full sensitivity to all spectra of visible light. This is important since together, with one oriented upwards and the other downwards, the net short-wave radiation can be measured by subtracting one reading from the other. The instrument is then called a net pyranometer. A point of caution is that any instrument used as an albedometer or net pyranometer composition of reflected light from vegetation is biased toward green. Therefore, most albedometers must be of the glass domed thermopile type and not the photo diode type.

Where pyranometers are not available, solar radiation can be estimated from the duration of bright sunshine. Sunshine duration, n, is most commonly recorded with the Campbell-Stokes heliograph. This instrument records periods of bright sunshine by using a glass globe that acts as a lens. The sun rays are concentrated at a focal point that burns a hole in a specially treated card mounted concentrically with the sphere. The movement of the sun changes the focal point throughout the day and a trace is burned into the card. If the sun is obscured, the trace is interrupted. The hours of bright sunshine are indicated by the

lengths of the line segments. Care is required to avoid the accumulation of rainwater on the paper. The heliograph must be oriented South in the northern hemisphere and North in the southern hemisphere. In China, another type of heliograph is used, where the solar beam penetrates through an orifice and traces a recording onto paper treated with a sensible chemical substance. Electronic records of sunshine duration can be obtained using photo-electric or rotating optical fibre sunshine recorders. Sunshine recorders are much less used today as compared to less expensive and more readily automated pyranometers. For purposes of establishing R_s, pyranometers are much preferred.

Net longwave and net shortwave radiation can be measured by recording the difference in output between sensors facing upward and downward, with both covering a solid angle in the shape of a hemisphere. In a net radiometer where a single output of R_n is produced, the glass domes are replaced by polyethylene domes that have a transmission range for both shortwave and longwave radiation. The sensors for net radiation are made from several thermocouples sensing heat generated by radiation from all wavelengths. All radiometers transform the radiation energy into thermal energy, a portion of which is transformed into an electric voltage gradient that provides appropriate conditions for continuous recording using dataloggers.

Four component radiometers measure incoming shortwave, reflected shortwave, incoming longwave and emitted longwave radiation all separately. Four component radiometers are generally considered to be more accurate than integrated net radiometers. They have the advantage that their four component measurements can each be quality assured separately. In addition, upward and downward facing components can be swapped periodically, for a few moments, by inverting the instrument. This provides a verification that both shortwave sensors have similar readings and that both longwave sensors have similar readings.

The black bodies of radiometers can lose their sensing capabilities with time, so that these instruments require regular and frequent calibration. Other net radiometers have been comprised of ventilated differential thermopiles, but they are seldom utilized.

The quantity of heat conducted into the soil, G, can be measured with systems of soil heat flux plates and thermocouples or thermistors.

3.3.4 Calculation procedures

Extraterrestrial radiation for daily periods (R_a)

The extraterrestrial radiation, R_a, for each day of the year and for different latitudes can be estimated from the solar constant, the solar declination, and the time of the year by:

$$R_a = \frac{24\,(60)}{\pi} G_{sc}\, d_r\, [\omega_s \sin(\varphi) \sin(\delta) + \cos(\varphi) \cos(\delta) \sin(\omega_s)] \qquad (3.18)$$

where
- R_a extraterrestrial radiation [MJ m^{-2} day^{-1}],
- G_{sc} solar constant = 0.0820 MJ m^{-2} min^{-1},
- d_r inverse of the square of the relative distance Earth-Sun (Equation 3.20),
- ω_s sunset hour angle (Equation 3.22 or 3.23) [rad],
- φ latitude (Equation 3.19) [rad],
- δ solar declination (Equation 3.21) [rad].

R_a is expressed in the above equation in MJ m^{-2} day^{-1}. The corresponding equivalent evaporation in mm day^{-1} is obtained by multiplying R_a by 0.408 (Equation 3.17a). The latitude, φ, expressed in radians is positive for the northern hemisphere and negative for the southern hemisphere (Example 3.6). The conversion from decimal degrees to radians (Example 3.6) is given by:

$$[\text{Radians}] = \frac{\pi}{180} [\text{decimal degrees}] \quad (3.19)$$

EXAMPLE 3.6
Conversion of latitude in degrees and minutes to radians

Express the latitudes of Bangkok (Thailand) at 13°44'N and Rio de Janeiro (Brazil) at 22°54'S in radians.

Latitude	Bangkok (northern hemisphere)	Rio de Janeiro (southern hemisphere)
degrees & minutes	13°44'N	22°54'S
decimal degrees	13 + 44/60 = 13.73	(–22) + (–54/60) = – 22.90
radians	(π/180) 13.73 = + 0.240	(π/180) (–22.90) = – 0.400

The latitudes of Bangkok and Rio de Janeiro are respectively + 0.240 and – 0.400 radians.

Example 3.7 refers to the computation of R_a for 24-hour periods (see also Box 3.3).

The inverse relative distance Earth-Sun squared, d_r, and the solar declination, δ, are given by:

$$d_r = 1 + 0.033 \cos\left(\frac{2\pi}{365} J\right) \quad (3.20)$$

$$\delta = 0.409 \sin\left(\frac{2\pi}{365} J - 1.39\right) \quad (3.21)$$

where J is the number of the day in the year between 1 (1 January) and 365 or 366 (31 December). Values for J can be determined for each day (D) of month (M) in a spreadsheet by using the expression J = "DD/MM/YYYY - DATE (YEAR (DD/MM/YYYY), 1, 0)" where M is month and D is the day. Note that the format of the date should be in accordance with the computer settings. For ten-day calculations, compute J for day D = 5, 15 and 25. For monthly calculations, J at the middle of the month is approximately given by J = INT (30.4 M – 15). J for all days of the year and an equation for estimating J are given in Annex 2 (Table A2.5).

The sunset hour angle, ω_s, is given by:

$$\omega_s = \arccos [-\tan (\varphi) \tan (\delta)] \quad (3.22)$$

As the arccos function is not available in all computer languages, the sunset hour angle can also be computed using the arctan function:

$$\omega_s = \frac{\pi}{2} - \arctan\left[\frac{-\tan (\phi) \tan (\delta)}{X^{0.5}}\right] \quad (3.23)$$

where $X = 1 - [\tan (\varphi)]^2 [\tan (\delta)]^2$ with $X = 0.00001$ if $X \leq 0$ (3.24)

Values for R_a for different latitudes are given in Annex 2 (Table A2.6). These values represent R_a on the 15th day of each month. These values deviate from values that are averaged over each day of the month by less than 1 percent for all latitudes during non-frozen periods and are included for simplicity of calculation. These values deviate slightly from the values in the Smithsonian Tables. For the winter months in latitudes greater than 55° (N or S), the equations for R_a have limited validity. Reference should be made to the Smithsonian Tables to assess possible deviations. An example of the computation of R_a for 24-hour periods is given in Example 3.7, and a calculation sheet is presented in Box 3.3.

EXAMPLE 3.7
Determination of extraterrestrial radiation

Determine the extraterrestrial radiation (R_a) for 3 September at 20°S.

From Eq. 3.19	20°S or $\phi = (\pi/180)(-20) =$ (the value is negative for the southern hemisphere)	− 0.35	rad
From Table A2.5:	The number of day in the year, J =	246	days
From Eq. 3.20	$d_r = 1 + 0.033 \cos(2\pi(246)/365) =$	0.985	
From Eq. 3.21	$\delta = 0.409 \sin(2\pi(246)/365 - 1.39) =$	0.120	rad
From Eq. 3.22:	$\omega_s = \arccos[-\tan(-0.35)\tan(0.120)] =$	1.527	rad
Then:	$\sin(\varphi)\sin(\delta) = \sin(-0.35)\sin(0.120) =$	− 0.041	-
and:	$\cos(\varphi)\cos(\delta) = \cos(-0.35)\cos(0.120) =$	0.933	-
From Eq. 3.18	$R_a = 24(60)/\pi\ (0.0820)(0.985)$ $[1.527(-0.041)+0.933\sin(1.527)] =$	32.2	MJ m^{-2} d^{-1}
From Eq. 3.17	expressed as equivalent evaporation = 0.408 (32.2) =	13.1	mm day^{-1}

The extraterrestrial radiation is 32.2 MJ m^{-2} day^{-1}.

Extraterrestrial radiation for hourly or shorter periods (R_a)

For hourly or shorter periods, the solar time angle at the beginning and end of the period should be considered when calculating R_a:

$$R_a = \frac{12\,(60)}{\pi} G_{sc} d_r [(\omega_2 - \omega_1) \sin(\varphi) \sin(\delta) + \cos(\varphi) \cos(\delta) (\sin(\omega_2) - \sin(\omega_1))] \quad (3.25)$$

where
- R_a extraterrestrial radiation in the hour (or shorter) period [MJ m^{-2} $hour^{-1}$],
- G_{sc} solar constant = 0.0820 MJ m^{-2} min^{-1},
- d_r inverse relative distance Earth-Sun squared (Equation 3.20),
- δ solar declination [rad] (Equation 3.21),
- φ latitude [rad] (Equation 3.19),
- ω_1 solar time angle at the beginning of period [rad] (Equation 3.26),
- ω_2 solar time angle at the end of the period [rad] (Equation 3.27).

The solar time angles at the beginning and end of the period are given by:

$$\omega_1 = \omega - \frac{\pi t_1}{24} \quad (3.26)$$

$$\omega_2 = \omega + \frac{\pi t_1}{24} \quad (3.27)$$

where ω solar time angle at the midpoint of hourly or shorter period [rad],

t_1 length of the calculation period [hour]: i.e. 1 for an hourly period or 0.5 for a 30-minute period.

The solar time angle at the midpoint of the period is:

$$\omega = \frac{\pi}{12}[(t + 0.06667(L_z - L_m) + S_c) - 12] \quad (3.28)$$

where t standard clock time at the midpoint of the period [hour]. For example, for a period between 14.00 and 15.00 hours, t = 14.5,

L_z longitude of the centre of the local time zone [degrees west of Greenwich]. For example, L_z = 75, 90, 105 and 120° for the Eastern, Central, Rocky Mountain and Pacific time zones (United States) and L_z = 0° for Greenwich (United Kingdom), 330° for Cairo (Egypt), and 255° for Bangkok (Thailand),

L_m longitude of the measurement site [degrees west of Greenwich],

S_c seasonal correction for solar time [hour].

When $\omega < -\omega_s$ or $\omega > \omega_s$ (Equation 3.28) indicate that the sun is below the horizon so that, by definition, R_a is zero.

The seasonal correction for solar time is:

$$S_c = 0.1645 \sin(2b) - 0.1255 \cos(b) - 0.025 \sin(b) \quad (3.29)$$

$$b = \frac{2\pi (J - 81)}{364} \quad (3.30)$$

where J is the number of the day in the year.

Daylight hours (N)

The daylight hours, N, are given by:

$$N = \frac{24}{\pi}\omega_s \quad (3.31)$$

where ω_s is the sunset hour angle in radians given by Equation 3.22 or 3.23. Mean values for N (15th day of each month) for different latitudes are given in Annex 2, Table A2.7.

An example of the computation of the daylight hours, N, is given in Example 3.8, and the calculation is presented in Box 3.3.

EXAMPLE 3.8
Determination of daylight hours

Determine the daylight hours (N) for 3 September at 20°S.

From Example 3.7:	ω_s = arccos[−tan (−0.35) tan (0.120)] =	1.527	rad
From Eq. 3.31:	N = 24/π (1.527) =	11.7	hour

The number of daylight hours is 11.7 hours.

BOX 3.3
Calculation sheet for extraterrestrial radiation (R_a) and daylight hours (N)

Latitude Degrees and minutes are + positive for northern hemisphere and - negative for southern hemisphere

Degrees		°	---------->		°
Minutes		‘	---------- / 60 ---------->		°
Decimal degrees = Sum(degrees+minutes/60)					°
$\varphi = \pi/180 \times$ [decimal degrees] Eq. 3.19					rad

Day of the year

Day					
Month			J Table A2.5 (Annex 2)		
$d_r = 1+0.033 \cos(2\pi J/365)$ Eq. 3.20					
$\delta = 0.409 \sin(2\pi J/365 - 1.39)$ Eq. 3.21					rad
$\sin(\varphi)\sin(\delta)$					
$\cos(\varphi)\cos(\delta)$					
$\omega_s = \arccos[-\tan(\varphi)\tan(\delta)]$ Eq. 3.22					rad
$(24\ (60)/\pi)\ G_{sc}$				**37.59**	MJ m^{-2} day^{-1}

Extraterrestrial radiation: R_a

$R_a = \frac{24\ (60)}{\pi} G_{sc}\ d_r\ [\omega_s \sin(\varphi) \sin(\delta) + \cos(\varphi) \cos(\delta) \sin(\omega_s)]$ Eq. 3.18		MJ m^{-2} day^{-1}

Daylight hours: N

$N = \frac{24}{\pi}\omega_s$ Eq. 3.31		hour per day

Solar radiation (R_s)

If the solar radiation, R_s, is not measured, it can be calculated with the Angstrom equation (Example 3.9), which relates solar radiation to extraterrestrial radiation and relative sunshine duration:

$$R_s = \left(a_s + b_s \frac{n}{N}\right) R_a \qquad (3.32)$$

where
- R_s solar or shortwave radiation [MJ m^{-2} day^{-1}],
- n actual duration of sunshine [hour],
- N maximum possible duration of sunshine or daylight hours [hour],
- n/N relative sunshine duration [-],
- R_a extraterrestrial radiation [MJ m^{-2} day^{-1}],
- a_s regression constant, expressing the fraction of extraterrestrial radiation reaching the earth on overcast days (when n = 0),
- a_s+b_s fraction of extraterrestrial radiation reaching the earth on clear sky days (when n = N).

R_s is expressed in the above equation in MJ m^{-2} day^{-1}. The corresponding equivalent evaporation in mm day^{-1} is obtained by multiplying R_s by 0.408 (Equation 3.17). Depending on atmospheric conditions (humidity, dust) and solar declination (latitude and month), the Angstrom values a_s and b_s will vary. Where no actual solar radiation data are available and no calibration has been carried out for improved a_s and b_s parameters, the values a_s = 0.25 and b_s = 0.50 are recommended.

The extraterrestrial radiation, R_a, and the maximum possible duration of sunshine, N, are given by Equations 3.18 and 3.31. Values for R_a and N for different latitudes are also listed in Annex 2 (Tables A2.6 and A2.7).

EXAMPLE 3.9
Determination of solar radiation from measured duration of sunshine

In Rio de Janeiro (Brazil) at a latitude of 22°54'S, 220 hours of sunshine were recorded in May. Determine the solar radiation.

From Eq. 3.19:	latitude = 22°54'S = 22.90°S		
	or π/180 (– 22.90) =	– 0.40	rad
From Table A2.5:	for 15 May, the day in the year (J) =	135	--
From Eq. 3.18 or Table A2.6:	R_a =	25.1	MJ m^{-2} day^{-1}
From Eq. 3.31 or Table A2.7	N =	10.9	hours day^{-1}
	n = 220 hours / 31 days =	7.1	hours day^{-1}
From Eq. 3.32:	R_s = [0.25 + 0.50 (7.1/10.9)] R_a =		
	= 0.58 R_a = 0.58 (25.1) =	14.5	MJ m^{-2} day^{-1}
From Eq. 3.17:	expressed as equivalent evaporation		
	= 0.408 (14.5) =	5.9	mm day^{-1}

The estimated solar radiation is 14.5 MJ m^{-2} day^{-1}.

Clear-sky solar radiation (R_{so})

The calculation of the clear-sky radiation, R_{so}, when n = N, is required for computing net longwave radiation.

- For near sea level or when calibrated values for a_s and b_s are available:

$$R_{so} = (a_s + b_s) R_a \tag{3.33}$$

where R_{so} clear-sky solar radiation [MJ m^{-2} day^{-1}],
a_s+b_s fraction of extraterrestrial radiation reaching the earth on clear-sky days (n = N).

- When calibrated values for a_s and b_s are not available an approximate estimate that is based only upon elevation representing average air mass is:

$$R_{so} = (0.75 + 2 \times 10^{-5} z) R_a \tag{3.34}$$

where z is station elevation above sea level [m].

A more accurate estimate for R_{so} is obtained by considering the sun's angle through the atmosphere and the amount of water vapor, which is variable from day to day and is an effective absorber of solar radiation. The angle of the sun, which changes during the day, defines the relative length of the atmosphere that radiation must pass through and undergo absorption and scattering. In this procedure, the estimate is separated into the components of beam and diffuse radiation:

$$R_{so} = (K_B + K_D) R_a \tag{3.35}$$

where K_B the clearness index for direct beam radiation [-],
K_D the corresponding index for diffuse radiation [-].

Equation 3.36 for K_B uses improved coefficients and considers the effects of air pressure, sun angle and precipitable water:

$$K_B = 0.98 \exp\left[\frac{-0.00146\ P}{K_t \sin\beta} - 0.075\left(\frac{W}{\sin\beta}\right)^{0.4}\right] \tag{3.36}$$

where K_t a clearness coefficient that is affected by turbidity and haze ($0 < K_t \leq 1.0$),
P atmospheric pressure [kPa],
β angle of the sun above the horizon [rad],
W precipitable water in the atmosphere [mm].

The ratio $P / K_t \sin\beta$ represents the effective amount of air mass that the solar beam passes through. $K_t = 1.0$ for clean air and $K_t \approx 0.5$ for turbid, dusty, or polluted air. The value for K_t may vary with the time of year and with the cleansing of the atmosphere by precipitation. General values for K_t for a region can be determined using a pristine pyranometer that has a calibration traceable to the national or international solar standard. In general, for routine estimation of R_n and R_{so} envelopes using $K_t = 1.0$ is recommended, especially for agricultural areas where the air is likely to be free of dust. An exception might be during the pre-monsoon season on the Indian subcontinent when substantial haze can exist. Aerosols and dust can substantially increase the haziness of the atmosphere, especially when coupled with high humidity levels.

Haze reduces the beam component of solar radiation but may increase the diffuse component through increased scattering of radiation. Under hazy conditions, a value $K_t = 0.5$ essentially doubles the effective thickness of the atmosphere, and a value $K_t = 0.25$ quadruples the effective thickness.

Precipitable water is estimated as:

$$W = 0.14\ e_a\ P + 2.1 \tag{3.37}$$

where W precipitable water in the atmosphere [mm],
e_a actual vapor pressure of the air (at a height of approximately 2 m [kPa],
P mean atmospheric pressure [kPa].

The diffuse radiation index K_D is estimated from K_B:

$$K_D = 0.35 - 0.36\ K_B \qquad \text{for } K_B \geq 0.15 \tag{3.38a}$$

$$K_D = 0.18 + 0.82\ K_B \qquad \text{for } K_B < 0.15 \tag{3.38b}$$

The diffusive component increases as the direct beam component decreases when $K_B > 0.15$ due to the increased amount of scatter of the direct beam component. For clear sky conditions, K_B is always > 0.15 for daily data and, nearly always, is > 0.15 for hourly periods.

For daily (24-hour) time periods, the mean daily sun angle, weighted according to R_a, can be estimated as:

$$\sin \beta_{24} = \sin\left[0.85 + 0.3\ \varphi \sin\left(\frac{2\pi}{365} J - 1.39\right) - 0.42\ \varphi^2\right] \tag{3.39}$$

where β_{24} the average β during the daylight period, weighted according to R_a [rad],
φ latitude [rad],
J day of the year [1-366].

The sin (β_{24}) variable is used in place of sin(β) in Equation 3.36 for 24-hour time steps and represents the average sun angle during daylight hours. The value for β_{24} should be limited to positive values.

For hourly or shorter periods, the sun angle β is calculated as:

$$\sin \beta = \sin \varphi \sin \delta + \cos \varphi \cos \delta \cos \omega \tag{3.40}$$

where φ latitude [rad],
δ solar declination [rad],
ω solar time angle at the midpoint of the hourly or shorter period [rad].

Equations from 3.35 to 3.40 produce quite accurate estimates for R_{so}. However, reasons may exist for measured R_s on cloud-free days to deviate from the theoretical R_{so} curve. These reasons include air turbidity and haziness caused by dust and aerosols that are not reflected by the equations, effects of nearly invisible clouds high overhead, and late afternoon clouding. The inclusion of precipitable water in Equation 3.36 instructs

the R_{so} estimate to increase on dry days due to less water vapor to absorb shortwave radiation. Reflection of solar radiation by adjacent clouds can occasionally increase measured hourly R_s to levels that are above R_{so}. The relationship $R_{so} = (K_B + K_D) R_a$ is useful for verifying correct operation of pyranometers as described in Chapter 2.

Net solar or net shortwave radiation (R_{ns})

The net shortwave radiation resulting from the balance between incoming and reflected solar radiation is given by:

$$R_{ns} = (1 - \alpha) R_s \tag{3.41}$$

where R_{ns} net solar or shortwave radiation [MJ m^{-2} day^{-1}],

α albedo or canopy reflection coefficient, which is 0.23 for the hypothetical grass reference crop [dimensionless],

R_s the incoming solar radiation [MJ m^{-2} day^{-1}].

R_{ns} is expressed in the above equation in MJ m^{-2} day^{-1} for 24-h time steps.

Net longwave radiation (R_{nl})

The net longwave radiation is defined as the difference between longwave radiation emitted and reflected from the surface and the longwave radiation emitted to the surface from the atmosphere. The rate of longwave energy emission is proportional to the absolute temperature of the surface raised to the fourth power. This relation is expressed quantitatively by the Stefan-Boltzmann law. The net energy flux is less than that emitted from the earth's surface due to the absorption and downward radiation from the sky. Water vapour, clouds, carbon dioxide and dust are absorbers and emitters of longwave radiation. Their concentrations should be known when assessing the net outgoing flux. In general, the longwave radiation emitted from the surface is greater than the longwave radiation emitted to the surface from the sky. This is because, under sunlit conditions, the surface tends to be warmer than the effective atmospheric temperature. As humidity and cloudiness play an important role, the Stefan-Boltzmann law is corrected by these two factors when estimating the net outgoing flux of longwave radiation. It is assumed that the concentrations of the other absorbers are constant resulting:

$$R_{nl} = \sigma \left[\frac{T_{max,K}^4 + T_{min,K}^4}{2} \right] (0.34 - 0.14 \sqrt{e_a}) \left(1.35 \frac{R_s}{R_{so}} - 0.35\right) \tag{3.42}$$

where R_{nl} net outgoing longwave radiation [MJ m^{-2} day^{-1} (or $hour^{-1}$)],

σ Stefan-Boltzmann constant [4.903×10^{-9} MJ K^{-4} m^{-2} day^{-1} for 24-hour timesteps, or 2.041×10^{-10} MJ K^{-4} m^{-2} $hour^{-1}$ for hourly timesteps],

$T_{max,K}$ maximum absolute temperature [K = °C + 273.15],

$T_{min,K}$ minimum absolute temperature [K = °C + 273.15],

e_a actual vapour pressure [kPa],

R_s/R_{so} relative shortwave radiation (limited to ≤ 1.0),

R_s measured or calculated solar radiation [MJ m^{-2} day^{-1} (or $hour^{-1}$)],

R_{so} calculated (Equation 3.35) clear-sky radiation [MJ m^{-2} day^{-1} (or $hour^{-1}$)].

An average of the maximum air temperature to the fourth power and the minimum air temperature to the fourth power is commonly used in the Stefan-Boltzmann equation for 24-hour time steps (Example 3.10).

It should be noted that:

- The term $(0.34–0.14\sqrt{e_a})$ expresses the adjustment for air humidity impacts on atmospheric emission of longwave radiation, and R_{nl} will be smaller if the humidity increases due to more downwelling longwave radiation.
- The effect of cloudiness is expressed by $(1.35\ R_s/R_{so}–0.35)$. The term becomes smaller if the cloudiness increases and hence R_s decreases. Because cloud temperatures tend to be warmer than the clear atmosphere, as cloudiness increases, the magnitude of downwelling longwave radiation increases and R_{nl} becomes smaller.
- The smaller the adjustment terms, the larger the flux of longwave radiation emitted to the surface from the atmosphere and accordingly, the smaller the net outgoing flux of longwave radiation. It should be noted that the R_s/R_{so} term in Equation 3.42 must be limited so that $R_s/R_{so} \leq 1.0$.

Where measurements of incoming and outgoing short and longwave radiation during bright sunny and overcast hours are available, calibration of the coefficients in Equation 3.42 can be carried out. Annex 2 (Table A2.8) lists values for σT_K^4 for different air temperatures.

EXAMPLE 3.10
Determination of net longwave radiation

In Rio de Janeiro (Brazil) at a latitude of 22°54'S (= –22.90°), elevation of 60 m, 220 hours of bright sunshine, a mean monthly daily maximum and minimum air temperature of 25.1 and 19.1 °C and a vapour pressure of 2.1 kPa were recorded in May. Determine the net longwave radiation.

From Example 3.9: From Eq. 3.34 and Ex 3.9	R_s = $R_{so} = 0.751\ R_a = 0.751\ (25.1)$ =	14.5 18.9	MJ m^{-2} day^{-1} MJ m^{-2} day^{-1}
From Table A2.8 or for: Then: and:	σ = T_{max} = 25.1 °C = $\sigma T_{max, K}^4$ =	4.9×10^{-9} 298.3 38.8	MJ K^{-4} m^{-2} day^{-1} K MJ m^{-2} day^{-1}
and: and:	T_{min} = 19.1 °C = $\sigma\ T_{max, K}^4$ = 35.7 MJ m^{-2} day^{-1}	292.3 35.7	K MJ m^{-2} day^{-1}
and: and: and: -	e_a = $0.34 – 0.14\ \sqrt{e_a}$ = R_s/R_{so} = (14.5)/(18.8) 1.35(0.77)–0.35 =	2.1 0.14 0.77 0.69	kPa - - -
From Eq. 3.42:	R_{nl} = [(38.8 + 35.7)/2] (0.14) (0.69) =	3.60	MJ m^{-2} day^{-1}
From Eq. 3.17:	expressed as equivalent evaporation = 0.408 (3.6) =	 1.47	 mm day^{-1}

The net longwave radiation is 3.6 MJ m^{-2} day^{-1}.

Net radiation (R_n)

The net radiation (R_n) is the difference between the incoming net shortwave radiation (R_{ns}) and the outgoing net longwave radiation (R_{nl}) (Box 3.4 and Example 3.11):

$$R_n = R_{ns} - R_{nl} \quad (3.43)$$

BOX 3.4
Calculation sheet for net radiation (R_n)

Latitude		°			
Day			R_a (Box 3.3 or Table A2.6)		MJ m^{-2} day^{-1}
Month			N (Box 3.3 or Table A2.7)		hours
n		hours	(in absence of R_s) n/N		

Net solar radiation: R_{ns}

If n is measured instead of R_s: $R_s = (0.25+0.50\ n/N)\ R_a$ Eq. 3.32		MJ m^{-2} day^{-1}
$R_{so} = [0.75 + 2\ (\text{Altitude})/\ 100\ 000]\ R_a$ Eq. 3.34 or 3.35		MJ m^{-2} day^{-1}
R_s/R_{so} (≤ 1.0)		
$R_{ns} = 0.77\ R_s$ Eq. 3.41		MJ m^{-2} day^{-1}

Net longwave radiation: R_{nl}

with $\sigma = 4.903\ 10^{-9}$ MJ K^{-4} m^{-2} day^{-1}
and $T_K = T\ [°C] + 273.15$

T_{max}		°C	$T_{max,K} = T_{max} + 273.15$		K
T_{min}		°C	$T_{min,K} = T_{min} + 273.15$		K
			$\sigma T_{max,K}^4$ (Table A2.8)		MJ m^{-2} day^{-1}
			$\sigma T_{min,K}^4$ (Table A2.8)		MJ m^{-2} day^{-1}
			$(\sigma T_{max,K}^4 + \sigma T_{min,K}^4)/2$		MJ m^{-2} day^{-1}
e_a		kPa	$(0.34-0.14\sqrt{e_a})$		
R_s/R_{so}			$(1.35\ R_s/R_{so} - 0.35)$		
$R_{nl} = (\sigma T_{max,K}^4 + \sigma T_{min,K}^4)/2\ (0.34-0.14\sqrt{e_a})(1.35R_s/R_{so}-0.35)$ Eq. 3.42					MJ m^{-2} day^{-1}

Net radiation: R_n

$R_n = R_{ns} - R_{nl}$ Eq. 3.43		MJ m^{-2} day^{-1}

EXAMPLE 3.11
Determination of net radiation

Determine the net radiation in Rio de Janeiro in May with the data from previous examples.

From Example 3.9:	R_s =	14.5	MJ m^{-2} day^{-1}
From Eq. 3.41:	$R_{ns} = (1 - 0.23)\ R_s$ =	11.2	MJ m^{-2} day^{-1}
From Example 3.10:	R_{nl} =	3.6	MJ m^{-2} day^{-1}
From Eq. 3.43:	$R_n = 11.2 - 3.6$ =	7.6	MJ m^{-2} day^{-1}
From Eq. 3.17:	expressed as equivalent evaporation = 0.408 (7.6) =	3.1	mm day^{-1}

Soil heat flux (G)

Complex models are available to describe soil heat flux. Because soil heat flux over 24-hour time periods is small compared to R_n, particularly when the surface is covered by vegetation such as with the grass reference, a simple calculation procedure is presented here for calculation time steps 24 hours or longer, based on the idea that the soil temperature follows air temperature:

$$G = c_s \frac{T_i - T_{i-1}}{\Delta t} \Delta z \qquad (3.44)$$

where
- G soil heat flux [MJ m^{-2} day^{-1}],
- c_s soil heat capacity [MJ m^{-3} °C^{-1}],
- T_i air temperature at time i [°C],
- T_{i-1} air temperature at time i-1 [°C],
- Δt length of time interval [day],
- Δz effective soil depth that is warmed or cooled [m].

As the soil temperature lags air temperature, the average temperature for a period should be considered when assessing the daily soil heat flux, i.e. Δt should exceed one day. The depth of penetration of the temperature wave is determined by the length of the time interval. The effective soil depth, Δz, is only 0.10-0.20 m for a time interval of one or a few days but might be 2 m or more for monthly periods. The soil heat capacity is related to its mineral composition and water content.

- For day and ten-day periods:

As the magnitude of the day or ten-day soil heat flux beneath the grass reference surface is relatively small, it may be ignored and thus:

$$G_{day} \approx 0 \qquad (3.45)$$

- For monthly periods:

When assuming a constant soil heat capacity of 2.1 MJ m^{-3} °C^{-1} and an appropriate soil depth, Equation 3.44 can be used to derive G for monthly periods (Example 3.12):

$$G_{month,i} = 0.7\ (T_{month,\ i+1} - T_{month,\ i-1}) \qquad (3.46)$$

or, if $T_{month, i+1}$ is unknown:

$$G_{month, i} = 0.14 (T_{month, i} - T_{month, i-1}) \quad (3.47)$$

where $T_{month, i}$ mean air temperature of month i [°C],
$T_{month, i-1}$ mean air temperature of previous month [°C],
$T_{month, i+1}$ mean air temperature of next month [°C].

- For hourly or shorter periods:

For hourly (or shorter) calculations, G beneath a dense cover of grass does not correlate well with air temperature. Hourly G can be approximated during daylight periods as:

$$G_{hr} = 0.1 R_n \quad (3.48)$$

and during nighttime periods as:

$$G_{hr} = 0.5 R_n \quad (3.49)$$

Where the soil is warming, the soil heat flux G is positive. The amount of energy required for this process is subtracted from R_n when estimating evapotranspiration.

EXAMPLE 3.12
Determination of soil heat flux for monthly periods

Determine the soil heat flux in April in Algiers (Algeria) when the soil is warming. The mean monthly temperatures of March, April and May are 14.1, 16.1, and 18.8°C respectively.

From Eq. 3.46	for the month of April: $G_{month} = 0.07 (18.8 - 14.1) =$	0.33	MJ m^{-2} day^{-1}
From Eq. 3.17	expressed as equivalent evaporation = 0.408 (0.33) =	0.13	mm day^{-1}

The soil heat flux is 0.33 MJ m^{-2} day^{-1}

3.4 WIND SPEED

3.4.1 Measurement

Wind is characterized by its direction and velocity. Wind direction refers to the direction from which the wind is blowing. For the computation of evapotranspiration, wind speed is the relevant variable. As wind speed at a given location varies with time, it is necessary to express it as an average over a given time interval. Wind speed is given in meters per second (m s^{-1}) or kilometres per day (km day^{-1}). For the standardized calculation of evapotranspiration with the PM equation, wind speed measured at 2 m or transformed to an equivalent wind speed at 2 m above the surface is required.

Wind speed is measured with anemometers. The anemometers commonly used in weather stations are composed of cups or propellers that are turned by the force of the wind. By counting the number of revolutions over a given time period, the average

wind speed over the measuring period is computed. In some modern weather stations, sonic anemometers are used. These have the advantage of having no moving parts or bearings that require maintenance.

Measurements by cup and propeller anemometers are reliable provided that maintenance ensures appropriate functioning of the mechanical parts. Older designs of anemometers utilize mechanical counters as the output device. Modern anemometers may be equipped with generators giving a voltage signal that is proportional to the wind speed. Other anemometers may be equipped with small magnetic reed switches or with opto-electronic couplers that generate electric impulses in proportion to the wind speed. The electronic devices are utilized in automatic weather stations.

Anemometers that collect wind speed data to be used to calculate ET_o are always placed at a height not less than 2 m above the ground, following recommendations by WMO. The accuracy of wind speed measurements depends on the upwind fetch as much as on instrumentation. A large upwind fetch that is free of buildings and trees is definitely required for representative measurements. In many agrometeorological weather stations, it may be preferred to place the anemometer at 3 to 4 m above the ground surface so that the anemometer has less opportunity to be obstructed or influenced by tall crops, such as maize, growing in the vicinity. The wind speed measurement is then adjusted, numerically, to an equivalent wind speed at 2 m (Equation 3.50).

3.4.2 Wind profile relationship

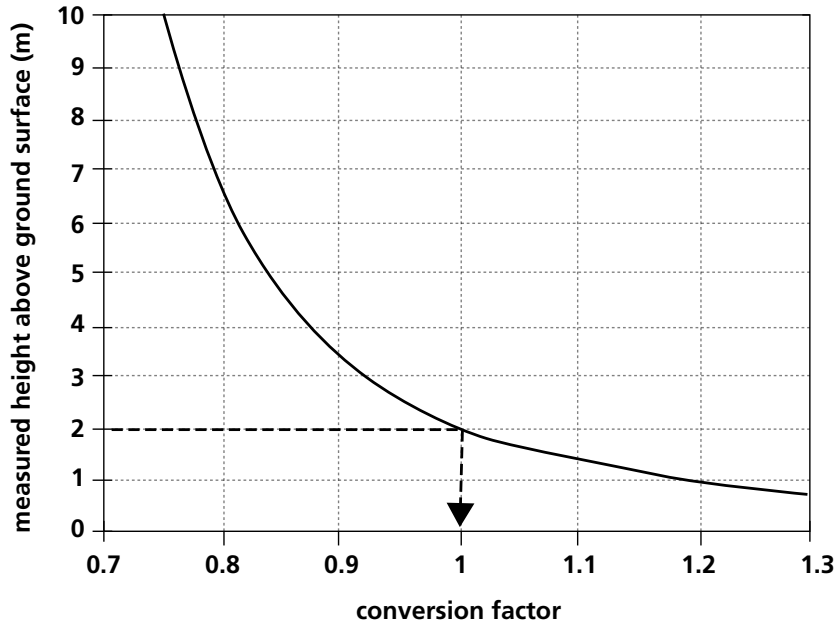
Surface friction tends to slow down wind passing over it. Therefore, wind speed is slowest at the surface and increases with height. For this reason, anemometers are placed at a chosen standard height, i.e. 10 m in meteorology and 2, 3 or 4 m in agrometeorology. To adjust wind speed data obtained from instruments placed at elevations other than the standard 2 m height to the 2 m height, a logarithmic wind speed profile may be used for measurements above a short grassed surface (Example 3.13):

$$u_2 = u_Z \frac{4.87}{\ln (67.8\ z - 5.42)} \qquad (3.50)$$

where u_2 wind speed at 2 m above ground surface [m s^{-1}],
u_z measured wind speed at z m above ground surface [m s^{-1}],
z height of measurement above ground surface [m].

The corresponding multipliers or conversion factors are given in Annex 2 (Table A2.9) and are plotted in Figure 3.7. Equation 3.50 assumes a neutral or nearly neutral boundary layer of air for some distance above the surface. This condition is common for well-watered grassed surfaces where ET consumes most of the energy available so that sensible heat flux, H, and thus the temperature gradient between surface and air, and thus buoyancy effects are small.

FIGURE 3.7
Conversion factor to convert wind speed measurement at a certain height above ground level to wind speed at the standard height (2 m)



Source: FAO, 1998

EXAMPLE 3.13
Adjusting wind speed data to the standard height

Determine the wind speed at the standard height of 2 m, from a measured wind speed of 3.2 m s^{-1} at 10 m above the soil surface.

For:	u_z =	3.2	m s^{-1}
And:	z =	10	m
Then:	Conversion factor = 4.87 / ln (67.8 10 - 5.42) =	0.75	-
From Eq. 3.50:	u_2 = 3.2 (0.75) =	2.4	m s^{-1}

3.5 CLIMATIC DATA ACQUISITION

Meteorological data are recorded at various types of weather stations. Agrometeorological stations are intended to be sited in cropped areas where instruments are exposed to atmospheric conditions similar to those for the surrounding crops. In these stations, air temperature and humidity, wind speed and solar radiation or sunshine duration are typically measured at 2 m above an extensive surface of grass or short crop. Where needed and feasible, the cover of the station is irrigated. Guidelines for the establishment and maintenance of agrometeorological stations are given in the FAO Irrigation and Drainage Paper No. 27. The World Meteorological Organization (WMO) has published, since 1950, the Guide to Instruments and Methods of Observation (WMO-No. 8). The 2021 edition includes several volumes: I: Measurement of Meteorological Variables, 2021 edition; II: Measurement of Cryospheric Variables, 2018 edition; III: Observing Systems, 2021 edition; IV: Space-based Observations, 2018 edition; V: Quality Assurance and Management of Observing Systems, 2018 edition.

Data collected at stations other than agrometeorological stations require a careful analysis of their validity before their use. For example, in aeronautic stations, data relevant to aviation are measured. As airports are often situated near urban conditions and away from irrigated areas, temperatures are often higher than those found in rural agricultural areas as explained in Chapter 2. Wind speed is commonly measured at 10 m height above the ground surface and needs to be adjusted to the standard 2 m height for the ET_o estimate (Figure 3.7).

The country's national meteorological services should be contacted for information on the climatic data collected at various types of weather stations in the country. National services commonly publish meteorological bulletins listing processed climatic data from the various stations. As described in Chapter 2, gridded weather data sets are becoming commonly available. It is essential that those data be assessed for quality and representativeness for use in estimating ET_o.

3.6 ESTIMATING MISSING CLIMATIC DATA

The assessment of the reference evapotranspiration with the FAO-PM ET_o method is developed in Chapter 4. The calculation of ET_o with the FAO-PM ET_o equation requires hourly, daily, ten-day, or monthly maximum and minimum air temperature (T_{max} and T_{min}), actual vapour pressure (e_a), net radiation (R_n calculated from R_s) and wind speed standardized to 2 m (u_2).

If some of the required weather data are missing or cannot be calculated, it is strongly recommended that the user estimate the missing climatic data with one of the following procedures and continue to use the FAO-PM ET_o equation. *The use of an alternative ET_o calculation procedure, requiring only limited meteorological parameters, is less recommended.* Procedures to estimate missing humidity, solar radiation and wind speed data are given in this section.

3.6.1 Estimating missing humidity data

Where humidity data are lacking or are of questionable quality, an estimate of actual vapour pressure, ea, can be obtained by assuming that the dewpoint temperature (T_{dew}) is near the daily minimum temperature (T_{min}). This statement implicitly assumes that at sunrise, when the air temperature is close to T_{min}, that the near-surface air over a well-watered, transpiring surface is nearly saturated with water vapour and the relative humidity is nearly 100 percent. Reasons for this are described in Chapter 2 and procedures to estimate T_{dew} are presented in Chapter 2.

3.6.2 Estimating missing radiation data

Net radiation measuring devices, requiring professional control, have rarely been installed in agrometeorological stations. In the absence of direct measurement, longwave and net radiation can be derived from more commonly observed weather parameters, i.e., solar radiation or sunshine hours, air temperature and vapour pressure. Where solar radiation is not measured, it can be estimated from measured hours of bright sunshine. However, where daily sunshine hours (n) are not available, solar radiation data cannot be computed with the calculation procedures previously presented. This section presents additional alternative methods to estimate solar radiation data.

Solar radiation data from a nearby weather station

This method relies on the fact that for the same month and often for the same day, the variables affecting incoming solar radiation, R_s, and sunshine duration, n, are similar throughout a given region. This implies or requires that: (i) the size of the region is small; (ii) the air masses governing rainfall and cloudiness are nearly identical within

parts of the region; and (iii) the physiography of the region is almost homogenous. Differences in relief should be negligible as they strongly influence the movement of air masses. Under such conditions, radiation data observed at nearby stations can be used.

Caution should be used when applying this method to mountainous and coastal areas where differences in exposure and altitude could be important or where cloudiness and rainfall are variable due to convective conditions. Moreover, data from a station located nearby but situated on the other side of a mountain may not be transferable as conditions governing cloud formation and radiation may be different. The user should observe climatic conditions in both locations and obtain information from local persons concerning general differences in cloud cover and type.

Where the north-south distance to a weather station within the same homogeneous region exceeds 50 km so that the value for R_a changes, the R_s measurement should be adjusted using the ratio of the solar to extraterrestrial radiation, R_s/R_a:

$$R_S = \frac{R_{s,reg}}{R_{a,reg}} R_a \qquad (3.51)$$

where $R_{s,reg}$ solar radiation at the regional location [MJ m^{-2} day^{-1}],
$R_{a,reg}$ extraterrestrial radiation at the regional location [MJ m^{-2} day^{-1}].

Once the solar radiation has been derived from the radiation data of a nearby station, the net longwave radiation (Equation 3.42) and the net radiation (Equation 3.43) can be calculated. The estimation method of Equation 3.51 is recommended for monthly calculations of ET_o. If using the method for daily estimates of ET_o, a more careful analysis of weather data in the importing and exporting meteorological stations has to be performed to verify whether both stations are in the same homogeneous climatic region and are close enough to experience similar conditions within the same day. The analysis should include the comparison of daily weather data from both stations, particularly the maximum and minimum air temperature and humidity. In fact, similar cloudiness and sunshine durations are related to similarities in temperature and humidity trends. Related statistical procedures are described in Annex 3.

Generally, daily calculations of ET_o with estimated radiation data are justified when utilized as a sum or an average over a several-day period. This is the case for the computation of the mean evapotranspiration demand between successive irrigations or when planning irrigation schedules. Under these conditions, the relative error for one day often counterbalances the error for another day of the averaging period. Daily estimates should not be utilized as true daily estimates but only in averages over the period under consideration.

Solar radiation data derived from air temperature difference

The difference between the maximum and minimum air temperature is related to the degree of cloud cover in a location. Clear-sky conditions result in high temperatures during the day (T_{max}), because the atmosphere is transparent to the incoming solar radiation, and in low temperatures during the night (T_{min}), because less downward longwave radiation is emitted by the atmosphere. On the other hand, in overcast conditions, T_{max} is relatively smaller because a significant part of the incoming solar radiation never reaches the earth's surface and is absorbed and reflected by the clouds. Similarly, T_{min} will be relatively higher because the cloud cover acts as a blanket and decreases the net outgoing longwave radiation. Therefore, the difference between the maximum and minimum air temperature ($T_{max} - T_{min}$) can be used as an indicator of the fraction of extraterrestrial radiation that reaches the earth's surface. This principle has been utilized by Hargreaves and Samani to develop estimates of ET_o using only air temperature data.

The Hargreaves-Samani radiation formula, adjusted and validated at several weather stations in a variety of climate conditions, becomes:

$$R_s = k_{Rs}\sqrt{(T_{max} - T_{min})}\, R_a \quad (3.52)$$

where R_a extraterrestrial radiation [MJ m^{-2} d^{-1}],
T_{max} maximum air temperature [°C],
T_{min} minimum air temperature [°C],
k_{Rs} empirical adjustment coefficient [$°C^{-0.5}$].

The fraction of extraterrestrial radiation that reaches the earth's surface, R_s/R_a, ranges from about 0.25 on a day with dense cloud cover to about 0.75 to 0.80 on a cloudless day with a clear sky, depending on elevation. *R_s estimated by Equation 3.52 should be limited to ≤ R_{so} from Equation 3.34 or 3.35.*

The square root of the temperature difference is closely related to the existing daily solar radiation in a given location. The adjustment coefficient k_{Rs} was originally recommended by Hargreaves and Samani as 0.16 for land locations, not influenced by large water bodies, and 0.19 for "coastal" locations. However, k_{Rs} is empirical and relates to the atmospheric conditions that influence the availability of incoming solar radiation at crop and natural surfaces, and so with the availability of energy for evaporation. Thus, k_{Rs} has been shown to vary spatially with site elevation and the distance to large water bodies. In addition, its value may increase with the aridity of the site and with wind speed. A useful approach is to estimate k_{Rs} values using various long-term (ideally > 30-years) average weather variables that allow for characterizing the site location: the difference between the maximum and minimum air temperature ($TD = T_{max} - T_{min}$), RH and u_2. In practice, the site averages are computed from available local observations or from a nearby weather station, where climates should be similar, when such data are not available locally (Example 3.14).

For all types of climates, k_{Rs} may be estimated from a global multiple linear regression as follows:

$$k_{Rs} = 0.3648 - 0.0099\, TD_{avg} + 0.0194\, u_{2\,avg} - 0.0017\, RH_{avg} \quad (3.53)$$

where k_{Rs} empirical radiation adjustment coefficient [$°C^{-0.5}$],
TD_{avg} long-term average of the seasonal difference between T_{max} and T_{min} ($TD = T_{max} - T_{min}$) [°C],
$u_{2\,avg}$ long-term seasonal average (or seasonal estimate) of daily wind speed at 2 m height [m s^{-2}],
RH_{avg} long-term seasonal average (or seasonal estimate) of daily relative humidity [%].

Alternatively, the k_{Rs} estimation can be refined according to the local climate group as defined by the UNEP aridity index (AI) characteristic of the region, defined in Box 2.4 in Chapter 2. This is done using a range of multiple linear regression equations. In this approach, k_{Rs} becomes:

- For Hyper-arid and Arid locations (AI < 0.20)

$$k_{Rs} = 0.2169 - 0.0042\, TD_{avg} + 0.0352\, u_{2\,avg} - 0.0011\, RH_{avg} \quad (3.54a)$$

- For Semiarid locations (0.20 ≤ AI < 0.50)

$$k_{Rs} = 0.3880 - 0.0095\, TD_{avg} + 0.0224\, u_{2\,avg} - 0.0022\, RH_{avg} \quad (3.54b)$$

- For Moist and dry sub-humid locations (0.50 ≤ AI < 1.00)

$$k_{Rs} = 0.3958 - 0.0105\ TD_{avg} + 0.0186\ u_{2\,avg} - 0.0021\ RH_{avg} \quad (3.54c)$$

- For Humid climates (AI ≥ 1.00)

$$k_{Rs} = 0.5191 - 0.0104\ TD_{avg} + 0.0188\ u_{2\,avg} - 0.0035\ RH_{avg} \quad (3.54d)$$

EXAMPLE 3.14
Determination of solar radiation from temperature data

Determine the solar radiation from the temperature data of July in Lyon (France) at a latitude of 45°43'N and at 200 m above sea level. In July, the mean monthly maximum and minimum air temperatures are 26.6 and 14.8 °C respectively. Using the long-term FAO-CLIMWAT weather database it was possible to estimate for that location the average (T_{max}–T_{min}) = 9.34 °C, u_2 = 2.14 m s^{-1} and RH = 79%. The Aridity index = 1.17 (Box 2.4) thus, because AI≥1.0, that location presents a humid climate type.

	latitude = 45°43' = +45.72° decimal degrees =	0.80	radian
From Table A2.5: From Eq. 3.18 or Annex 2 Table A2.6:	The day of the year for 15 July is R_a =	196 40.6	- MJ m^{-2} day^{-1}
From Eq. 3.52 (same latitude):	$R_s = k_{Rs} [\sqrt{(26.6 - 14.8)}] R_a$	-	MJ m^{-2} day^{-1}
From Eq. 3.54d:	k_{Rs} = 0.5191 – 0.0104 (9.34) + 0.0188 (2.14) – 0.0035 (79)	0.19	$°C^{-0.5}$
	$R_s = 0.19 [\sqrt{(26.6-14.8)}] R_a = 0.65$ (40.6) =	26.4	MJ m^{-2} day^{-1}
From Eq. 3.17:	equivalent evaporation = 0.408 (26.4) =	10.8	mm day^{-1}

In July, the estimated solar radiation, R_s, is 26.4 MJ m^{-2} day^{-1}

Figure 3.8 presents an example of the use of Equation 3.53 for estimating R_s. Comparisons with measurements show a regression coefficient very close to 1.0 for this location, thus a small bias in estimation.

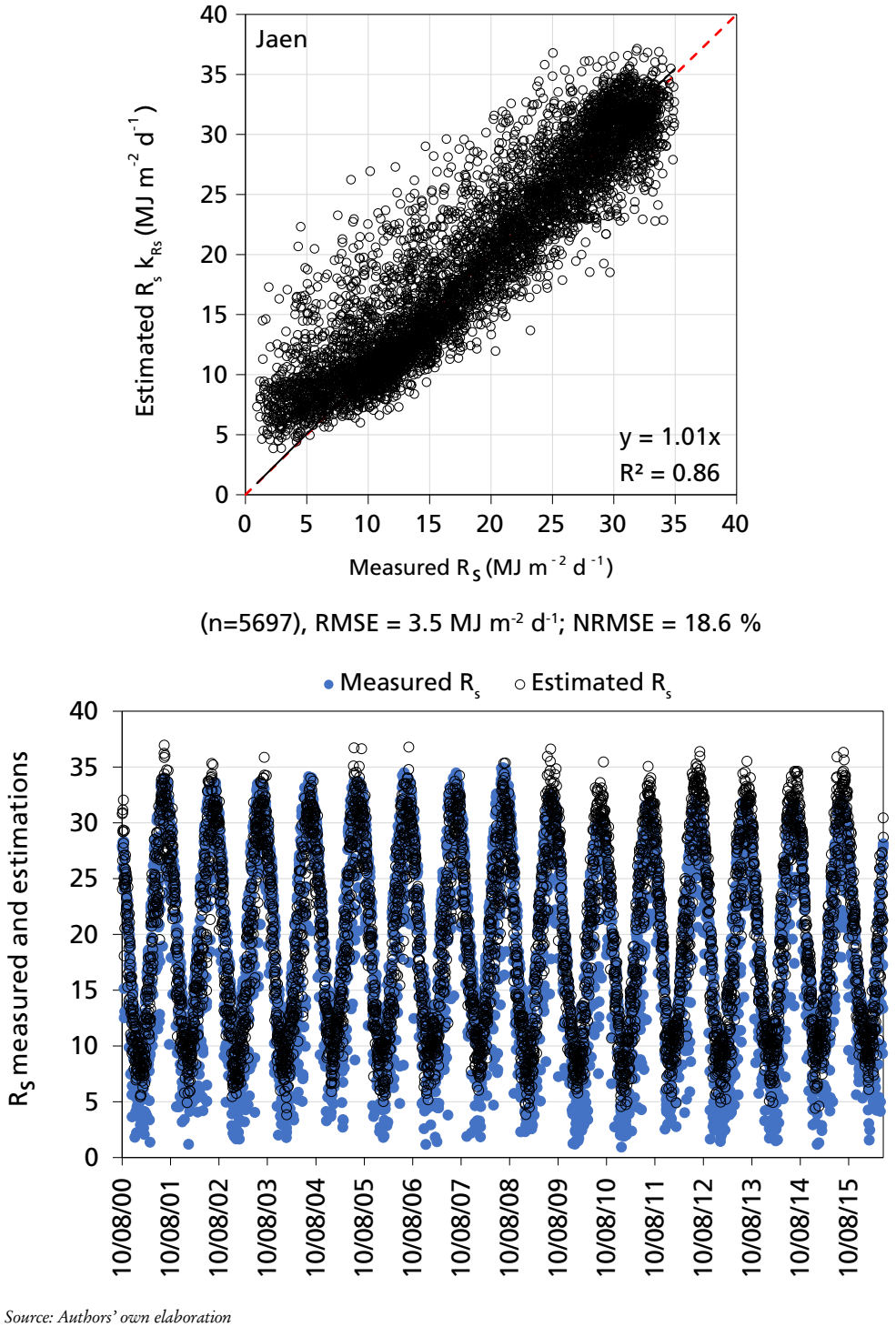
The temperature difference method is recommended for locations where it is not appropriate to import radiation data from a regional station, either because homogeneous climate conditions do not occur, or because data for the region are lacking (Example 3.15). Annex 3 includes the statistical methods to assess and correct the homogeneity of time series.

Caution is required when daily computations of ET_o are needed. The advice given for Equation 3.52 fully applies.

Solar radiation data derived from geostationary satellite gridded data

As discussed in Chapter 2, nowadays there is the possibility of using diverse gridded sources of weather data. Figure 3.9 indicates good adherence of the satellite-based solar radiation data relative to the weather station observations. The accuracy of the R_s estimates by satellite is smaller for the island locations due to high winds and frequent cloudy conditions in the humid islands of the Azores archipelago which were used for the example. Nevertheless, over-estimation errors are relatively small when compared with average daily values, thus indicating the feasibility of using the gridded estimates for island locations that are not very often covered by clouds, as for Santa Cruz, but poor for Granja where cloudiness and wind are higher.

FIGURE 3.8
Daily R_s estimated with the temperature differences (Eq. 3.52) and k_{Rs} computed from Equation 3.53 compared with measured values at a location near Jaen, Spain



Source: Authors' own elaboration

EXAMPLE 3.15
Determination of net radiation in the absence of radiation data

Calculate the net radiation for Bangkok (13°44'N) by using T_{max} and T_{min}. The station is located at the coast at 2 m above sea level. In April, the monthly average of the daily maximum temperature, daily minimum temperature and daily vapour pressure are 34.8 °C, 25.6 °C and 2.85 kPa respectively. Using the FAO-CLIMWAT weather database it was possible to estimate for that location the average (T_{max}–T_{min}) = 8.7 °C, u_2 = 3 m s^{-1} and RH = 80%. The AI = 0.68, thus 0.65≤AI<1.00, the location has a moist sub-humid climate.

For Latitude 13°44'N = +13.73° decimal degrees = 0.24 radian and for 15 April, J =105:			
From Eq. 3.18 or Table A2.6:	R_a =	38.1	MJ m^{-2} d^{-1}
	(in coastal location) k_{Rs} =	0.19	
	$(T_{max} - T_{min})$ = (34.8 – 25.6) =	9.2°C	°C
From Eq. 3.54c	k_{Rs} = 0.3958 – 0.0105 (8.7) + 0.0186 (3) – 0.0021 (80)	0.19	$°C^{-0.5}$
From Eq. 3.52:	$R_s = 0.19 \sqrt{(9.2)}\ R_a$	21.9	MJ m^{-2} day^{-1}
From Eq. 3.34:	$R_{so} = 0.75\ R_a$	28.5	MJ m^{-2} day^{-1}
From Eq. 3.41:	$R_{ns} = 0.77\ R_s$	16.9	MJ m^{-2} day^{-1}
	σ =	4.903 10^{-9}	MJ K^{-4} m^{-2} day^{-1}
	T_{max} =	34.8	°C
	$\sigma T_{max,K}^4$ =	44.1	MJ m^{-2} day^{-1}
	T_{min} =	25.6	°C
	$\sigma T_{min,K}^4$ =	39.1	MJ m^{-2} day^{-1}
	$(\sigma T_{max,K}^4 + \sigma T_{min,K}^4)/2$ = (44.1+39.1)/2	41.6	MJ m^{-2} day^{-1}
For:	e_a = 2.85 kPa	2.85	kPa
	$(0.34 - 0.14\sqrt{e_a})$ =	0.10	-
For:	R_s/R_{so} =	0.77	-
Then:	$(1.35(R_s/R_{so}) - 0.35)$ =	0.69	-
From Eq. 3.42:	R_{nl} = 41.6 (0.10) 0.69 =	2.87	MJ m^{-2} day^{-1}
From Eq. 3.43:	R_n = (16.9 – 2.87) =	14.0	MJ m^{-2} day^{-1}
From Eq. 3.17:	equivalent evaporation= 0.408 (14.0) =	5.7	mm day^{-1}

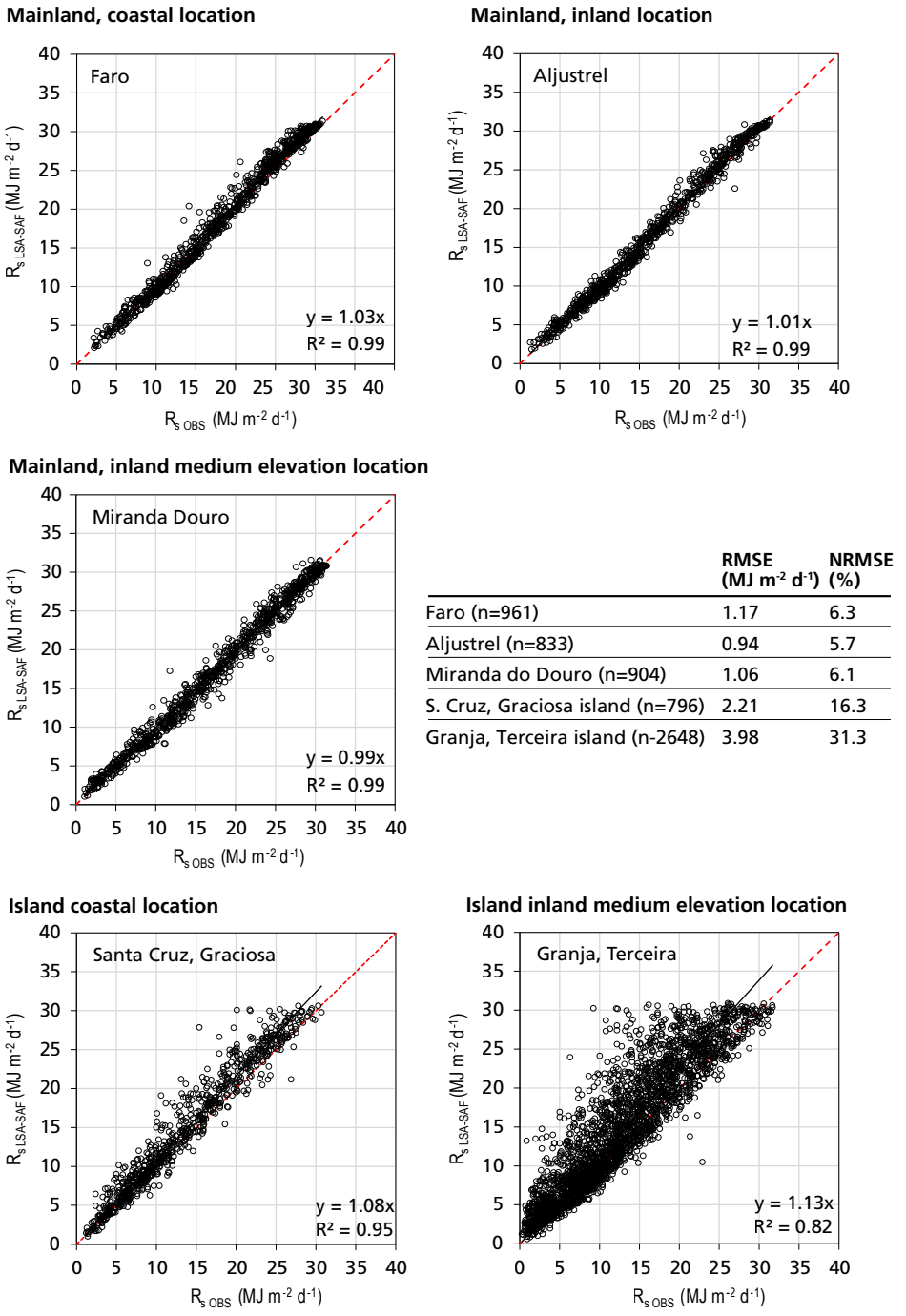
The estimated net radiation is 14.0 MJ m^{-2} day^{-1}

Solar radiation data derived from reanalysis weather data

Other alternative sources of weather data are provided by reanalysis products as detailed in Chapter 2. Figure 3.10 indicates good adherence of the reanalysis solar radiation data relative to the weather station observations. However, estimates have lower accuracy when compared with the satellite-based R_s analyzed in Figure 3.9. As per the examples in Figure 3.10, the accuracy of the R_s estimates by reanalysis ERA5 is good for the estimates relative to mainland locations, with regression coefficients through the origin close to 1.00 and normalized errors of estimation representing less than 20 percent of the mean values observed. For the coastal island location, Santa Cruz, the goodness of fit results are similar but the coefficient of determination is lower. Differently, for the inland location, Granja, results are poor, likely due to cloudiness and varied winds.

FIGURE 3.9

Comparing observed shorth radiation ($R_{s\ OBS}$) with gridded shortware radiation derived from the Meteosat Second Generation geostationary satellite ($R_{s\ LSA\text{-}SAF}$) for various locations in mainland Portugal and the Azores islands



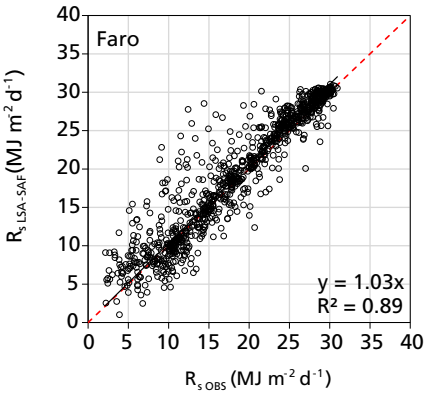
	RMSE (MJ m^{-2} d^{-1})	NRMSE (%)
Faro (n=961)	1.17	6.3
Aljustrel (n=833)	0.94	5.7
Miranda do Douro (n=904)	1.06	6.1
S. Cruz, Graciosa island (n=796)	2.21	16.3
Granja, Terceira island (n-2648)	3.98	31.3

Source: Authors' own elaboration

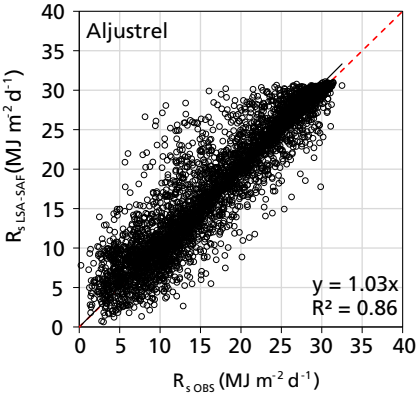
FIGURE 3.10

Point observed shorthwave radiation ($R_{s\ OBS}$) and gridded shortwave radiation derived from the reanalysis ERA5 ($R_{s\ ERA5}$) for a set of location in mainland Portugal and the islands of Azores

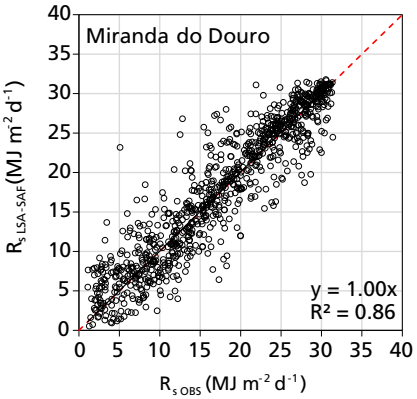
Mainland, coastal location



Mainland, inland location

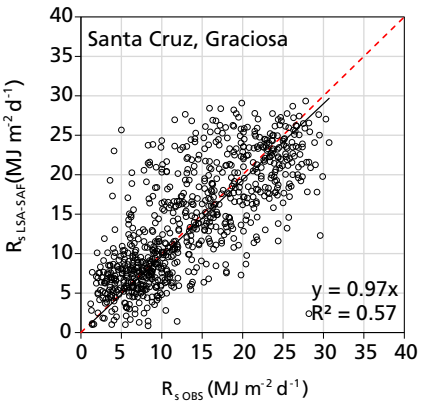


Mainland, medium altitude inland location

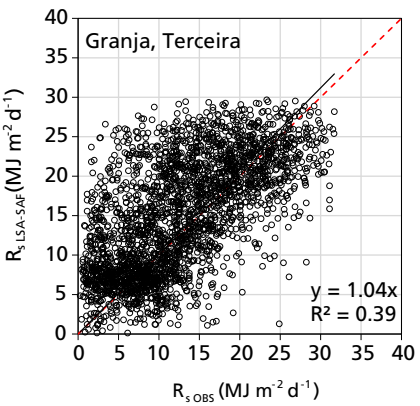


	RMSE (MJ m^{-2} d^{-1})	NRMSE (%)
Faro (n=961)	2.84	15.3
Aljustrel (n=833)	3.24	18.6
Miranda do Douro (n=904)	3.17	18.5
S. Cruz, Graciosa island (n=796)	2.21	16.3
Granja, Terceira island (n-2648)	3.98	31.3

Island location, humid climate



Altitude island location, humid climate



Source: Authors' own elaboration

Results shown in Figure 3.10 comparing R_s estimated from ERA5 reanalysis data have more outliers in estimates than those obtained from the gridded satellite-based data for mainland locations. Nevertheless, the regression coefficients are close to 1.0, and errors are much smaller than the daily means of the observed values. The quality of predictions depends upon the data ingested, which may lead to future improvements when more data is available for the reanalysis. When averaged over longer periods, error in estimates and occurrence of outlying values should decrease.

3.6.3 Missing wind speed data

Wind speed data from a nearby weather station

Importing wind speed data from a nearby station, as for radiation data, relies on the fact that the air flow above a "homogeneous" region may have relatively large variations through the course of a day but small variations when referring to longer periods or the total for the day. Data from a nearby station may be imported where air masses are of the same origin or where the same fronts govern air flows in the region and where the relief is similar.

When importing wind speed data from another station, the regional climate, trends in the variation of other meteorological parameters and relief should be comparable. Strong winds are often associated with low relative humidity and light winds are common with high relative humidity. Thus, trends in variation of daily maximum and minimum relative humidities should be similar in both locations. In mountainous areas, data should not necessarily be imported from the nearest station but from nearby stations with similar elevation and exposure to the dominant winds. The paired stations may even vary from one season to another, depending on the dominant winds.

Imported wind speed data can be used when making monthly estimates of evapotranspiration. Daily calculations are justified when utilized as a sum or average over a several-day period, such as a week or decade.

Empirical estimates of monthly wind speed

As the variation in wind speed averaged over monthly periods is relatively small and fluctuates around average values, monthly values of wind speed may be estimated. The "average" wind speed estimates may be selected from information available for the regional climate but should take seasonal changes into account. General values are suggested in Table 3.2.

TABLE 3.2
General classes of monthly wind speed data

Description	Mean monthly wind speed at 2 m
light wind	... ≤ 1.0 m s^{-1}
light to moderate wind	1 - 3 m s^{-1}
moderate to strong wind	3 - 5 m s^{-1}
strong wind	... ≥ 5.0 m s^{-1}

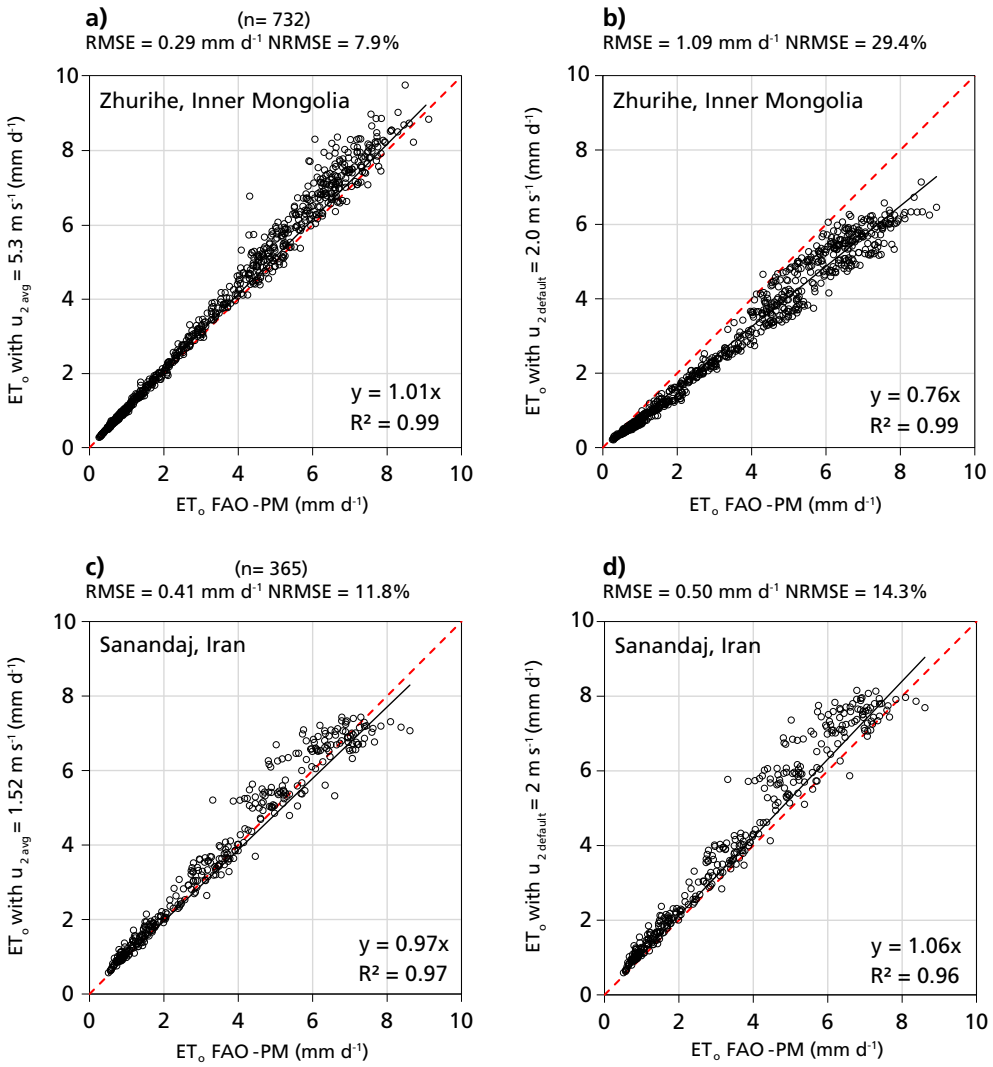
Where no wind data are available within the region, a value of 2 m s^{-1} can be used as a temporary estimate. This value is the average over 2 000 weather stations around the globe. Figure 3.11 shows examples of using the regional average u_2 and the world standard average (i.e. default value of u_2 = 2 m s^{-1}) to estimate ET_o, one for a strong wind location and the other for a light-moderate wind location. The large underestimation

of ET_o when using the default average value is due to overall higher u_2 values observed at the location ($u_{2\,avg}$ = 5.3 m s^{-1}). The results are different at locations characterized by light-moderate wind speeds with an overestimation tendency when the default u_2 value is used in ET_o estimation.

In general, wind speed at 2 m, u_2, should be limited to about $u_2 \geq 0.5$ m s^{-1} when used in the FAO-PM-ET_o equation. This is necessary to account for the effects of boundary layer instability and buoyancy of air in promoting exchange of vapour at the surface when air is calm. This effect occurs when the wind speed is small, and buoyancy of warm air induces air exchange at the surface; $u_2 \geq 0.5$ m s^{-1} in the ET_o equation improves the estimation accuracy.

FIGURE 3.11

Comparing daily FAO-PM ET_o computed with complete weather datasets observed at a strong wind location in Inner Mongolia, China, and a light-moderate wind location in Iran with ET_o computed with incomplete data sets where wind velocity was replaced by (a, c) the regional wind speed average and (b, d) the world default value of 2 m s^{-1}



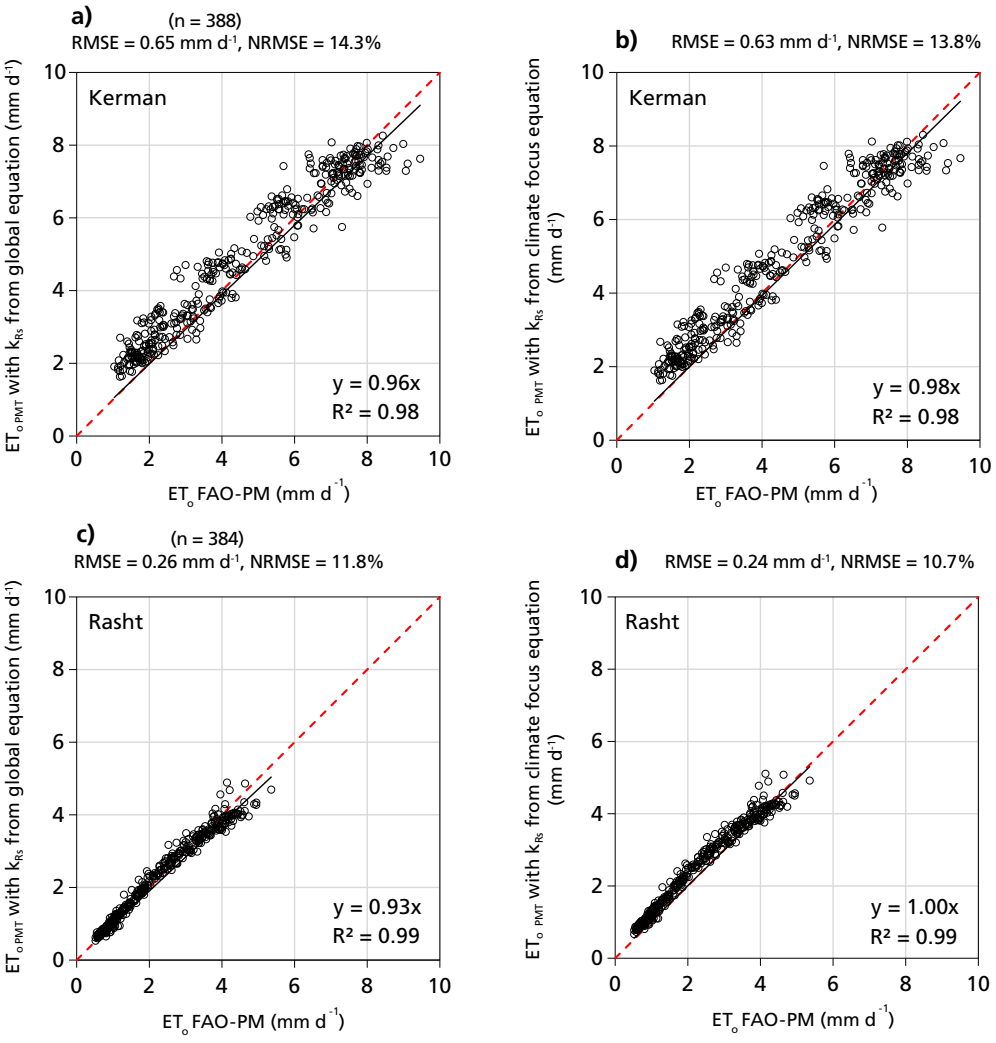
Source: Authors' own elaboration

3.6.4 Minimum data requirements: the Penman-Monteith temperature approach (PMT)

The PMT approach applies the FAO standardized Penman-Monteith equation where solar radiation, vapour pressure and wind data may be estimated when missing or when found to be faulty or nonrepresentative. Many of the suggested procedures rely upon maximum and minimum air temperature measurements. Unfortunately, there is no dependable way to estimate air temperature when it is missing. Therefore, it is suggested that maximum and minimum daily air temperature data are the minimum data requirements necessary to apply the FAO-PM equation. Figure 3.12 presents an example of the successful application of PMT, when using Equation 3.52 for estimating R_s and Equations 3.53 and 3.54a for estimating k_{Rs} and using the default u_2 value of 2 m/s. As expected, using the climate focused multiple linear regression equation for k_{Rs}, the regression coefficient for ET_o is closer to 1.0 relative to the global regression Equation 3.53, and RMSE is also smaller. This condition has been observed in most of the analyzed applications.

FIGURE 3.12

Daily FAO-PM ET_o calculated with full data sets compared with ET_o calculated with the PMT approach, using the default u_2 value of 2 m s^{-1}, the actual vapor pressure e_a estimated with the Equation 3.9, and R_s estimated with Equation 3.52 using k_{Rs} computed with a) and c) global equation (Equation 3.53) and b) and d) climate focused equations (Equation 3.54a) for the arid location of Kerman, Iran and Eq. 3.54d for the humid location of Rasht, Iran



Source: Authors' own elaboration

As recommended in the section above, the FAO Penman-Monteith Temperature method, PMT, is recommended over other empirical ET_o methods, even when daily maximum and minimum air temperature data are the only data available. This is because the PM equation contains appropriate and physically based relationships among weather variables and because the missing weather variables can be relatively accurately estimated using informed procedures described in the previous section and in Chapter 2. Therefore, the PM equation is likely to produce ET_o estimates that are sensitive to the expected behaviours of the missing weather variables. This is in contrast to a stand-alone temperature-based method such as the Hargreaves-Samani method that has no means to change its behaviour under different general weather conditions such as high wind locations vs. low wind locations, or humid vs. arid locations. With this in mind, however, the Hargreaves-Samani ET_o equation is provided in the following section as a backup procedure.

3.6.5 An alternative equation for computing ET_o when key weather data are missing: the Hargreaves-Samani equation

When solar radiation data, relative humidity data and/or wind speed data are missing, they should be estimated using the procedures presented in the previous section. As an alternative, ET_o can be estimated using the Hargreaves-Samani (HS) ET_o equation where:

$$ET_{o\,HS} = 0.0135\, k_{Rs\,HS} \frac{R_a}{\lambda} (T_{max} - T_{min})^{0.5} (T_{mean} + 17.8) \quad (3.55)$$

where
- T_{max} maximum of daily air temperature relative to the time step calculation [°C],
- T_{min} minimum of daily air temperature relative to the time step calculation [°C],
- T_{mean} mean air of daily temperature relative to the time step calculation [°C],
- λ latent heat of vaporization [2.45 MJ kg^{-1}],
- $k_{Rs\,HS}$ the radiation adjustment coefficient for the Hargreaves-Samani equation [$°C^{-0.5}$],
- R_a the extraterrestrial radiation [MJ m^{-2} day^{-1}],
- 0.0135 factor for conversion of units from the American to the International system,
- 17.8 an empirical parameter.

The default $k_{Rs\,HS}$ value is 0.17 in the original formulation. As per Equations 3.53 and 3.54, the empirical $k_{Rs\,HS}$ adjustment coefficient may be estimated using a multiple linear regression equation using the long-term average weather variables ($TD=T_{max}-T_{min}$, RH and u_2) that allow characterization of the site location. Thus, the general equation is:

$$k_{Rs\,HS} = 0.3023 - 0.0049\, TD_{avg} + 0.0151\, u_{2\,avg} - 0.0017\, RH_{avg} \quad (3.56)$$

where parameters are defined as for Equation 3.53. Similarly, for improving the $k_{Rs\,HS}$ estimation according to the local climate as defined by the UNEP aridity index (AI) (see Box 2.4 and Example 2.1), a range of multiple linear regression equations may be applied. Thus, $k_{Rs\,HS}$ becomes:

- For Hyper-arid and Arid locations (AI < 0.20)

$$k_{Rs\,HS} = 0.2073 - 0.0023\, TD_{avg} + 0.0224\, u_{2\,avg} - 0.0009\, RH_{avg} \quad (3.57a)$$

- For Semiarid locations (0.20 ≤ AI< 0.50)

$$k_{Rs\,HS} = 0.2962 - 0.0049\, TD_{avg} + 0.0117\, u_{2\,avg} - 0.0014\, RH_{avg} \quad (3.57b)$$

- For Moist and dry sub-humid locations ($0.50 \leq AI < 1.00$)

$$k_{Rs\ HS} = 0.3396 - 0.0059\ TD_{avg} + 0.0125\ u_{2\ avg} - 0.0020\ RH_{avg} \quad (3.57c)$$

- For Humid climates ($AI \geq 1.00$)

$$k_{Rs\ HS} = 0.3695 - 0.0066\ TD_{avg} + 0.0127\ u_{2\ avg} - 0.0024\ RH_{avg} \quad (3.57d)$$

Equation 3.55, using the estimated $k_{Rs\ HS}$, should be verified in each new region by comparing with estimates by the FAO-PM ET_o equation (Equation 2.4) at weather stations where solar radiation, air temperature, humidity, and wind speed are measured and QAQC'ed and where the station environments are well-watered. If necessary, Equation 3.56 can be calibrated on a monthly or annual basis by determining the empirical coefficients of the regression equation:

$$ET_o = a + b\ ET_{o\ HS\ Eq.3.55} \quad (3.58)$$

where the "Eq. 3.55" subscript refers to ET_o estimated using Equation 3.55

Figure 3.13 presents an example of the application of Equation 3.55 for estimating ET_o and using Equation 3.56 for the $k_{Rs\ HS}$ estimation (Example 3.16). In the same example the ET_o estimation with the PMT approach is also presented for the same locations. Results for the applications to a humid and a semi-arid location are good since the regression coefficients are quite close to 1.0 indicating that predictions have a small bias of estimation, and estimation errors RMSE are much smaller than the average values of ET_o. These goodness-of-fit indicators, also observed for many other applications, suggest that the approach is feasible. However, results for the PMT method are slightly better than those for the HS equation.

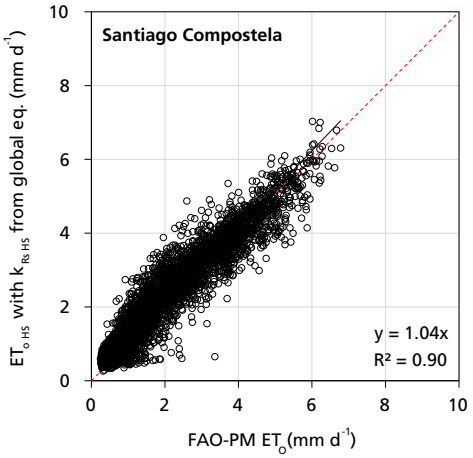
In general, estimating solar radiation, vapor pressure and wind speed as described in Equations 3.52 and 2.6 or 2.7, and Table 3.2 and then utilizing these estimates in Equation 2.4 (the FAO-PM ET_o equation) should provide somewhat more accurate estimates as compared to estimating ET_o directly using Equation 3.55 This is due to the ability of the estimation equations to incorporate general climatic characteristics such as high or low wind speed or high or low relative humidity into the ET_o estimate made using Equation 2.4.

FIGURE 3.13

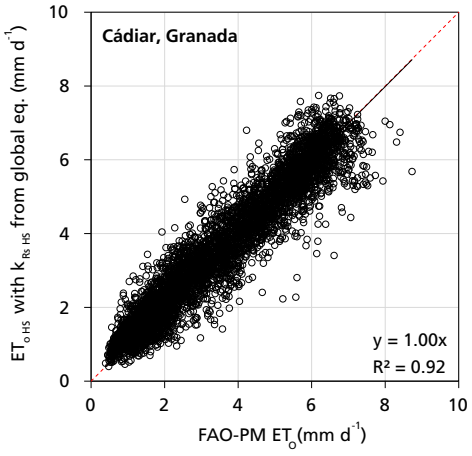
Daily ET_o calculated with FAO-PM ET_o equation when compared with i) ET_o calculated with Hargreaves-Samani equation, computing R_s with $k_{Rs\ HS}$ derived with Equation 3.56 (global) and Equation 3.57 (climate focused), and ii) ET_o calculated with the PMT approach, computing R_s with k_{Rs} derived with Equation 3.53 (global) and climate focused Equation 3.54 for Santiago de Compostela, Spain (humid climate) and for Cádiar, Granada, Spain (semi-arid climate)

i) ET_o computed with the Hargreaves Samani equation vs. FAO-PM ET_o

k_{Rs} from global equation

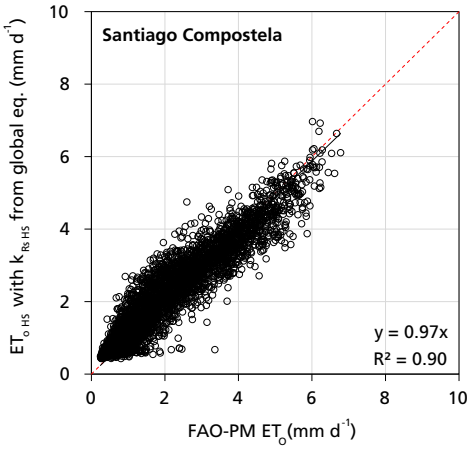


(n = 3877)
RMSE = 0.50 mm d^{-1}, NRMSE = 24.5%

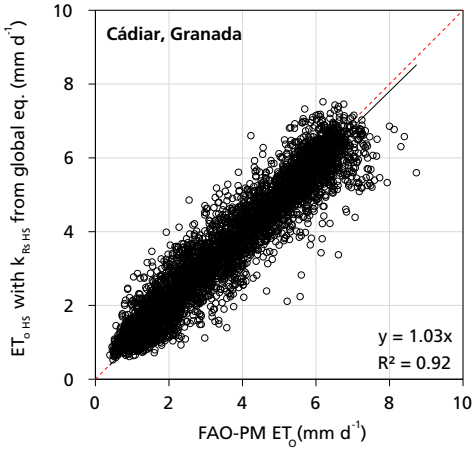


(n = 5637)
RMSE = 0.55 mm d^{-1}, NRMSE = 17.1%

k_{Rs} from climate focus equation



RMSE = 0.46 mm d^{-1}, NRMSE = 22.7%



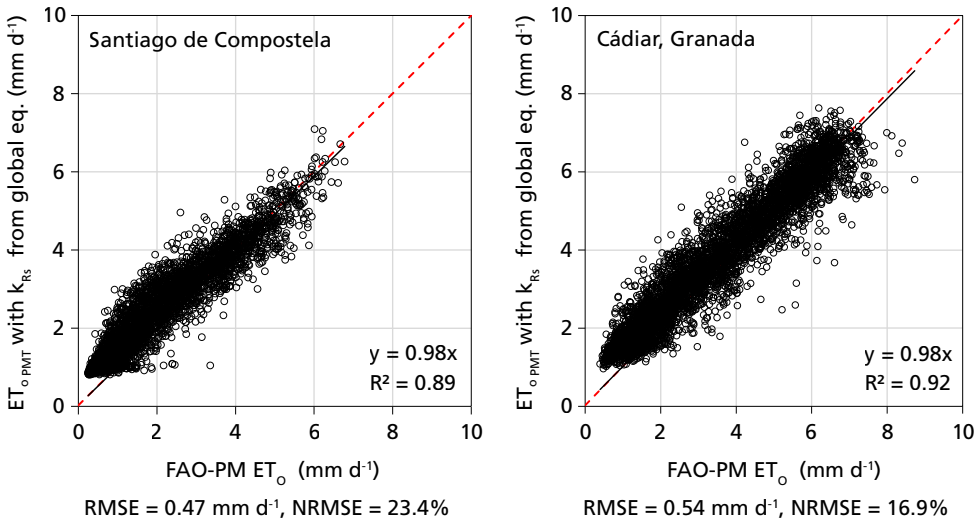
RMSE = 0.57 mm d^{-1}, NRMSE = 17.9%

Source: Authors' own elaboration

FIGURE 3.13 (CONTINUED)

ii) ET_o computed with the PMT approach vs. FAO-PM ET_o

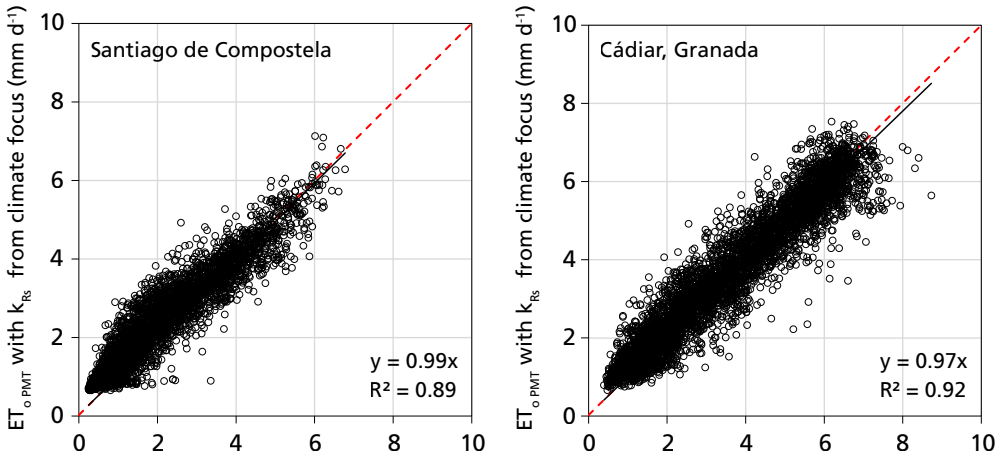
k_{Rs} from global equation



RMSE = 0.47 mm d^{-1}, NRMSE = 23.4%

RMSE = 0.54 mm d^{-1}, NRMSE = 16.9%

k_{Rs} from climate focus equation



RMSE = 0.48 mm d^{-1}, NRMSE = 23.5%

RMSE = 0.54 mm d^{-1}, NRMSE = 16.9%

Source: Authors' own elaboration

EXAMPLE 3.16
Determination of Hargreaves-Samani ET_o from air temperature data

Determine the ET_o from the air temperature data of July in Lyon (France) at a latitude of 45°43'N and at 200 m above sea level. In July, the mean monthly maximum and minimum air temperatures are 26.6 and 14.8 °C respectively. Using the long term FAO-CLIMWAT weather database it was possible to estimate for that location the $(T_{max}\text{-}T_{min})_{avg}$ = 9.34 °C, $u_{2\ avg}$ = 2.14 m s^{-1} and RH_{avg} = 79%. The Aridity index = 1.17 thus, because AI≥1.0, that location behaves as a humid climate type.

From Table A2.5: From Eq. 3.18 or Annex 2, Table A2.6:	latitude = 45°43' = +45.72° decimal degrees = The day of the year for 15 July is R_a =	0.80 196 40.6	radian - MJ m^{-2} day^{-1}
From Eq. 3.54d	k_{Rs} = 0.3695 – 0.0066 (9.34) + 0.0127 (2.14) – 0.0024 (79) =	0.15	°$C^{-0.5}$
From Eq. 3.55:	$(T_{max} - T_{min})$ T_{mean} ET_o	11.8 20.7 4.44	°C °C mm day^{-1}

In summary, the application of the FAO-PM ET_o equation with estimated weather parameters is recommended over the use of Hargreaves-Samani Equation 3.55. Equation 3.55 using the default $k_{Rs\ HS}$ value tends to underpredict ET_o under high wind conditions (u_2 > 3 m s^{-1}) and to overpredict under conditions of high relative humidity.

Chapter 4
Determination of ET_o

This chapter demonstrates how crop reference evapotranspiration (ET_o) is determined either from meteorological data or from pan evaporation.

The FAO Penman-Monteith method is maintained as the sole standard method for the computation of ET_o from meteorological data. The method itself is introduced in Chapter 2, and the computation of all data required for the calculation of ET_o is described and discussed in Chapter 3. This chapter presents guidelines to calculate ET_o with different time steps, ranging from hours to months, and with missing climatic data. The ET_o calculation can be done by hand with the help of a calculation sheet, or by means of a computer. The use of computers is highly recommended.

ET_o can also be approximated from the evaporation loss from a water surface. The procedure to obtain ET_o from pan evaporation and the coefficients for different types of pans are presented in this chapter.

4.1 FAO PENMAN-MONTEITH ET_o EQUATION

From the original Penman-Monteith equation and the equations of the aerodynamic and canopy resistance, the FAO Penman-Monteith equation has been derived in Chapter 2. For 24-hour calculation time steps:

$$ET_o = \frac{0.408\,\Delta\,(R_n - G) + \gamma\,\frac{900}{T+273}\,u_2\,(e_s - e_a)}{\Delta + \gamma\,(1 + 0.34\,u_2)} \qquad \text{(2.4 bis)}$$

where
- ET_o reference evapotranspiration [mm day^{-1}],
- R_n net radiation at the crop surface [MJ m^{-2} day^{-1}],
- G soil heat flux density [MJ m^{-2} day^{-1}],
- T air temperature at 2 m height [°C],
- u_2 wind speed at 2 m height [m s^{-1}],
- e_s saturation vapour pressure [kPa],
- e_a actual vapour pressure [kPa],
- $e_s - e_a$ saturation vapour pressure deficit [kPa],
- Δ slope vapour pressure curve [kPa $°C^{-1}$],
- γ psychrometric constant [kPa $°C^{-1}$].

The FAO Penman-Monteith ET_o (FAO-PM ET_o) equation determines the evapotranspiration from the hypothetical grass reference surface and provides a standard to which evapotranspiration in different periods of the year or in other regions can be compared and to which the evapotranspiration from other crops can be related.

4.1.1 Calculation procedure

Calculation sheet

ET_o can be estimated for daily (24-hour) calculation time steps by means of the calculation sheet presented in Box 4.1. The calculation sheet refers to tables in Annex 2 for the determination of some of the climatic parameters. The calculation procedure consists of the following steps:

1. Derivation of some climatic parameters from the daily maximum (T_{max}) and minimum (T_{min}) air temperature, altitude (z) and mean wind speed (u_2).
2. Calculation of the vapour pressure deficit ($e_s - e_a$). The saturation vapour pressure (e_s) is derived from T_{max} and T_{min}, while the actual vapour pressure (e_a) can be derived from the dewpoint temperature (T_{dew}), from maximum (RH_{max}) and minimum (RH_{min}) relative humidity, or from the maximum (RH_{max}), or from mean relative humidity (RH_{mean}).
3. Determination of the net radiation (R_n) as the difference between the net shortwave radiation (R_{ns}) and the net longwave radiation (R_{nl}). In the calculation sheet, the effect of soil heat flux (G) is ignored for daily calculations as the magnitude of the flux in this case is relatively small. The net radiation, expressed in MJ m^{-2} day^{-1}, is converted to mm day^{-1} (equivalent evaporation) in the FAO-PM ET_o equation by using 0.408 as the conversion factor within the equation.
4. ET_o is obtained by combining the results of the previous steps.

Examples 4.1, 4.2, and 4.3 present typical cases using the calculation sheet for monthly, daily, and hourly calculation timesteps. It is highly recommended that computerized software be verified by comparing, step-by-step, with component calculations in these examples.

Computerized calculations

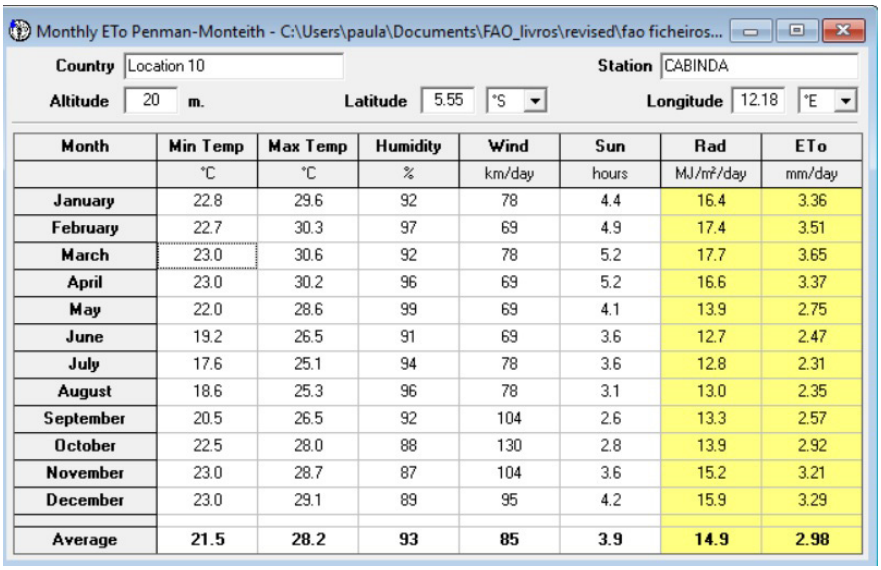
Calculations of the reference crop evapotranspiration (ET_o) are nearly always computerized. The calculation procedures of all data required for the calculation of ET_o by means of the FAO-PM ET_o equation are presented in Chapter 3. Typical sequences in which the calculations can be executed are given in the calculation sheets. The procedures presented in Boxes 3.1 (vapour pressure deficit), 3.3 (extraterrestrial radiation and daylight hours), 3.4 (net radiation) and 4.1 (ET_o) can be used when developing a spreadsheet or computer program to calculate ET_o. Examples 4.1, 4.2 and 4.3 can be used to confirm computerized calculations for monthly, daily, and hourly timesteps, respectively.

Many software packages already use the FAO-PM ET_o equation to assess the reference evapotranspiration. As an example, the output of CROPWAT, the FAO software for irrigation scheduling, is presented in Figure 4.1

4.1.2 ET_o calculated with different time steps

The selection of the time step with which ET_o is calculated depends on the purpose of the calculation, the accuracy required and the time step of the climatic data available.

FIGURE 4.1
Daily ET_o computed by CROPWAT 8.0


Monthly ETo Penman-Monteith - C:\Users\paula\Documents\FAO_livros\revised\fao ficheiros...

Country: Location 10 Station: CABINDA

Altitude: 20 m. Latitude: 5.55 °S Longitude: 12.18 °E

Month	Min Temp	Max Temp	Humidity	Wind	Sun	Rad	ETo
	°C	°C	%	km/day	hours	MJ/m²/day	mm/day
January	22.8	29.6	92	78	4.4	16.4	3.36
February	22.7	30.3	97	69	4.9	17.4	3.51
March	23.0	30.6	92	78	5.2	17.7	3.65
April	23.0	30.2	96	69	5.2	16.6	3.37
May	22.0	28.6	99	69	4.1	13.9	2.75
June	19.2	26.5	91	69	3.6	12.7	2.47
July	17.6	25.1	94	78	3.6	12.8	2.31
August	18.6	25.3	96	78	3.1	13.0	2.35
September	20.5	26.5	92	104	2.6	13.3	2.57
October	22.5	28.0	88	130	2.8	13.9	2.92
November	23.0	28.7	87	104	3.6	15.2	3.21
December	23.0	29.1	89	95	4.2	15.9	3.29
Average	21.5	28.2	93	85	3.9	14.9	2.98

Source: FAO, 1998

BOX 4.1
Calculation sheet for ET_o (FAO PM-ET_o) for daily calculation time steps using meteorological tables of Annex 2

Parameters

T_{max}		°C			
T_{min}		°C	$T_{mean} = (T_{max}+T_{min})/2$		°C
T_{mean}		°C	Δ (Table A2.4 of Annex 2)		kPa °C^{-1}
Altitude		m	γ (Table A2.2 of Annex 2)		kPa °C^{-1}
u_2		m s^{-1}	$(1 + 0.34\ u_2)$		
$\Delta/[\Delta + \gamma\ (1 + 0.34\ u_2)]$					
$\gamma/[\Delta + \gamma\ (1 + 0.34\ u_2)]$					
$[900/(T_{mean} + 273)]\ u_2$					

Vapour pressure deficit

T_{max}		°C	$e°(T_{max})$ (Table A2.3)		kPa
T_{min}		°C	$e°(T_{min})$ (Table A2.3)		kPa
Saturation vapour pressure $e_s = [(e°(T_{max}) + e°(T_{min})]/2$					kPa
e_a derived from dewpoint temperature:					
T_{dew}		°C	$e_a = e°(T_{dew})$ (Table A2.3)		kPa
OR e_a derived from maximum and minimum relative humidity:					
RH_{max}		%	$e°(T_{min})\ RH_{max}/100$		kPa
RH_{min}		%	$e°(T_{max})\ RH_{min}/100$		kPa
			e_a = Average of the results above		kPa

BOX 4.1 (CONT.)

OR e_a derived from maximum relative humidity: (recommended if there are errors in RH_{min})

RH_{max}			%	$e_a = e°(T_{min})\ RH_{max}/100$		kPa

OR e_a derived from mean relative humidity: (less recommended due to non-linearities)

RH_{mean}			%	$e_a = e_s\ RH_{mean}/100$		kPa
Vapour pressure deficit				$(e_s - e_a)$		kPa

Radiation

Latitude		degree			
Day			R_a (Table A2.6)		MJ m^{-2} day^{-1}
Month			N (Table A2.7)		hours
n		hours	n/N		
If no R_s data available: $R_s = (0.25 + 0.50\ n/N)\ R_a$					MJ m^{-2} day^{-1}
$R_{so} = [0.75 + 2\ (Altitude)/100\ 000]\ R_a$					MJ m^{-2} day^{-1}
R_s / R_{so}					
$R_{ns} = 0.77\ R_s$					MJ m^{-2} day^{-1}
T_{max}			$\sigma\ T_{max,\ K}^4$ (Table A2.8)		MJ m^{-2} day^{-1}
T_{min}			$\sigma\ T_{min,\ K}^4$ (Table A2.8)		MJ m^{-2} day^{-1}
$(\sigma T_{max,\ K}^4 + \sigma T_{min,\ K}^4)/2$					MJ m^{-2} day^{-1}
e_a		kPa	$(0.34 - 0.14\ \sqrt{e_a})$		
R_s/R_{so}			$(1.35\ R_s/R_{so} - 0.35)$		
$R_{nl} = (\sigma T_{max,\ K}^4 + \sigma T_{min,K}^4)/2\ (0.34 - 0.14\ \sqrt{e_a})\ (1.35\ R_s/R_{so} - 0.35)$					MJ m^{-2} day^{-1}
$R_n = R_{ns} - R_{nl}$					MJ m^{-2} day^{-1}
T_{month}		°C	G_{day} (assume) = 0		MJ m^{-2} day^{-1}
$T_{month-1}$		°C	$G_{month} = 0.14\ (T_{month} - T_{month-1})$		MJ m^{-2} day^{-1}
$R_n - G$					MJ m^{-2} day^{-1}
$0.408\ (R_n - G)$					mm day^{-1}

Grass reference evapotranspiration

$$\left[\frac{\Delta}{\Delta + \gamma\,(1 + 0.34\,u_2)}\right]\,[\,0.408\,(R_n - G)] \qquad \text{mm day}^{-1}$$

$$\left[\frac{\gamma}{\Delta + \gamma\,(1 + 0.34\,u_2)}\right]\left[\frac{900}{T + 273}\right]\,u_2[(e_s - e_a)] \qquad \text{mm day}^{-1}$$

$$ET_o = \frac{0.408\,\Delta\,(R_n - G) + \gamma\,\dfrac{900}{T + 273}\,u_2\,(e_s - e_a)}{\Delta + \gamma\,(1 + 0.34\,u_2)} \qquad \text{mm day}^{-1}$$

Ten-day or monthly time step

Notwithstanding the non-linearity in the FAO-PM ET_o equation and some weather parameter methods, mean ten-day or monthly weather data can be used to compute the mean ten-day or monthly values for the reference evapotranspiration. The value of the reference evapotranspiration calculated with mean monthly weather data is indeed very similar to the average of the daily ET_o values calculated with daily average weather data for that month.

The meteorological data consist of:

- Air temperature: ten-day or monthly average daily maximum (T_{max}) and average daily minimum temperature (T_{min}).
- Air humidity: ten-day or monthly average of the daily actual vapour pressure (e_a) derived from psychrometric, dewpoint or relative humidity data.
- Wind speed: ten-day or monthly average of daily wind speed data transformed to a 2 m height (u_2).
- Radiation: ten-day or monthly average of daily net radiation (R_n) computed from the mean ten-day or monthly measured shortwave radiation or from actual duration of daily sunshine hours (n). The extraterrestrial radiation (R_a) and daylight hours (N) for a specific day of the month can be computed using Equations 3.18 and 3.31 in Chapter 3, or can be selected from Tables A2.5 and A2.6 in Annex 2.

When the soil is warming (spring) or cooling (autumn), the soil heat flux (G) for monthly periods may become significant relative to the mean monthly R_n. In these cases G cannot be ignored and its value should be determined from the mean monthly air temperatures of the previous and next month. Chapter 3 outlines the calculation procedure (Equations 3.46 and 3.47).

EXAMPLE 4.1
Determination of ET_o with mean monthly data

Given the monthly average climatic data of April of Bangkok (Thailand) located at 13 °44'N and at an elevation of 2 m:

-	Monthly average daily maximum temperature (T_{max}) =	34.8	°C
-	Monthly average daily minimum temperature (T_{min}) =	25.6	°C
-	Monthly average daily vapour pressure (e_a) =	2.85	kPa
Measured at 2 m	Monthly average daily wind speed (u_2) =	2	$m\ s^{-1}$
-	Monthly average sunshine duration (n) =	8.5	$hours\ day^{-1}$
For April	Mean monthly average temperature ($T_{month, i}$) =	30.2	°C
For March	Mean monthly average temperature ($T_{month, i-1}$) =	29.2	°C
Determination according to the outline of Box 4.1 (calculation sheet ET_o)			
Parameters			
-	$T_{mean} = [(T_{max} = 34.8) + (T_{min} = 25.6)]/2$ =	30.2	°C
From Table A2.4 or Eq. 3.8:	Δ =	0.246	$kPa\ °C^{-1}$
From Table A2.1 and Table A2.2 or Eq. 3.1 and Eq. 3.3:	Altitude =	2	m
	P =	101.3	kPa
	γ =	0.0673	$kPa\ °C^{-1}$
-	$(1 + 0.34\ u_2)$ =	1.68	-
-	$\Delta/[\Delta+\gamma(1+0.34u_2)] = 0.246/[(0.246 + 0.0673\ (1.68)]$ =	0.685	-

EXAMPLE 4.1 (CONT.)

-	$\gamma/[\Delta+\gamma(1+0.34u_2)] = 0.0673/[0.246 + 0.0673\ (1.68)] =$	0.188	-
-	$900/(T_{mean}+273)\ u_2 =$	5.94	-
Vapour pressure deficit			
From Table A2.3 or Eq. 3.6:	$T_{max} =$ $e°(T_{max}) =$	34.8 5.56	°C kPa
From Table A2.3 or Eq. 3.6:	$T_{min} =$ $e°(T_{min}) =$	25.6 3.28	°C kPa
Eq. 3.7: Given	$e_s = (5.56 + 3.28)/2 =$ $e_a =$	4.42 2.85	kPa kPa
-	Vapour pressure deficit $(e_s - e_a) = (4.42 - 2.85) =$	1.57	kPa
Radiation (for month = April)			
From Table A2.6 or A2.5 or Eq. 3.18:	J = (for 15 April) Latitude = 13°44'N = (13 + 44/60) = $R_a =$	105 13.73 38.06	- °N MJ m^{-2} day^{-1}
N (Table A2.7 or Eq. 3.31):	Daylength N =	12.31	hours
- From Eq. 3.32: From Eq. 3.34 or 3.35: - From Eq. 3.41:	$n/N = (8.5/12.31) =$ $R_s = [0.25 + 0.50\ (0.69)]\ 38.06 =$ $R_{so} = (0.75 + 2\ (2)/100\ 000)\ 38.06 =$ $R_s/R_{so} = (22.65/28.54) =$ $R_{ns} = 0.77\ (22.65) =$	0.69 22.65 28.54 0.79 17.44	- MJ m^{-2} day^{-1} MJ m^{-2} day^{-1} - MJ m^{-2} day^{-1}
From Table A2.8:	$T_{max} =$ $T_{max,\ K} = 34.8 + 273.15 =$ $\sigma T_{max,\ K}^{\ 4} =$	34.8 307.95 44.07	°C K MJ m^{-2} day^{-1}
From Table A2.8:	$T_{min} =$ $T_{min,K} = 25.6 + 273.15 =$ $\sigma T_{min,K}^{\ 4} =$	25.6 298.75 39.03	°C K MJ m^{-2} day^{-1}
-	$(\sigma T_{max,\ K}^{\ 4} + \sigma T_{min,K}^{\ 4})/2 = (44.07 + 39.03)/2 =$	41.55	MJ m^{-2} day^{-1}
For: Then	$e_a =$ $(0.34 - 0.14\sqrt{e_a}) =$	2.85 0.104	kPa -
For: Then:	$R_s/R_{so} =$ $(1.35\ R_s/R_{so} - 0.35) =$	0.79 0.72	- -
From Eq. 3.42: From Eq. 3.43: From Eq. 3.47: - -	$R_{nl} = 41.55\ (0.104)\ 0.72 =$ $R_n = (17.44 - 3.11) =$ $G = 0.14\ (30.2 - 29.2) =$ $(R_n - G) = (14.33 - 0.14) =$ $0.408\ (R_n - G) =$	3.11 14.33 0.14 14.19 5.79	MJ m^{-2} day^{-1} MJ m^{-2} day^{-1} MJ m^{-2} day^{-1} MJ m^{-2} day^{-1} mm day^{-1}
Grass reference evapotranspiration			
- - - - -	$0.408\ (R_n - G)\ \Delta/[\Delta+\gamma\ (1+0.34u_2)] =$ $(5.79)\ 0.685 =$ $900u_2/(T+273)\ (e_s - e_a)\ \gamma/[\Delta+\gamma\ (1+0.34u_2)] =$ $5.94\ (1.57)\ 0.188 =$ $ET_o = (3.97 + 1.75) =$	3.97 1.75 5.72	mm day^{-1} mm day^{-1} mm day^{-1}

The grass reference evapotranspiration is 5.7 mm day^{-1} for the month of April.

Daily time step

Calculation of ET_o with the FAO-PM ET_o equation on 24-hour time scales will generally provide accurate results. The required meteorological data consists of:

- Air temperature: maximum (T_{max}) and minimum (T_{min}) daily air temperatures.
- Air humidity: mean daily actual vapour pressure (e_a) derived from psychrometric, dewpoint temperature or relative humidity data.
- Wind speed: daily average for 24 hours of wind speed transformed to the 2 m height (u_2).
- Radiation: net radiation (R_n) measured or computed from solar and longwave radiation or from the actual duration of sunshine (n). The extraterrestrial radiation (R_a) and daylight hours (N) for a specific day of the month should be computed using Equations 3.18 and 3.31. As the magnitude of daily soil heat flux (G) beneath the reference grass surface is relatively small, it may be ignored for 24-hour time steps.

EXAMPLE 4.2
Determination of ET_o with daily data

Given the meteorological data as measured on 6 July in Uccle (Brussels, Belgium) located at 50°48'N and at 100 m above sea level:

-	Maximum air temperature (T_{max}) =	21.5	°C
-	Minimum air temperature (T_{min}) =	12.3	°C
-	Maximum relative humidity (RH_{max}) =	84	%
-	Minimum relative humidity (RH_{min}) =	63	%
-	Wind speed measured at 10 m height =	10	km h^{-1}
-	Actual hours of sunshine (n) =	9.25	hours
Conversion of wind speed			
At 10 m height	Wind speed = 10 km h^{-1} or u_z =	2.78	m s^{-1}
From Eq. 3.50, with z = 10 m:	At standard 2 m height, u_2 = 0.748 (2.78) =	2.078	m s^{-1}
Parameters			
From Eq. 3.1, for:	altitude =	100	m
	P =	100.1	kPa
From Eq. 3.4:	T_{mean} = (21.5 + 12.3)/2 =	16.9	°C
From Eq. 3.8, for:	T_{mean} =	16.9	°C
	Δ =	0.122	kPa $°C^{-1}$
From Eq. 3.3, for:	P =	100.1	kPa
	γ =	0.0666	kPa $°C^{-1}$
-	(1 + 0.34 u_2) =	1.71	-
-	$\Delta/[\Delta+\gamma(1+0.34u_2)]$ =		
-	0.122/[(0.122 + 0.0666 (1.71)] =	0.518	-
-	$\gamma/[\Delta+\gamma(1+0.34u_2)]$ =		
-	0.0666/[0.122 + 0.0666 (1.71)] =	0.282	-
-	900/(T_{mean}+273) u_2 =	6.450	-

EXAMPLE 4.2 (CONT.)

Vapour pressure deficit			
From Eq. 3.6, for: Then:	T_{max} = $e°(T_{max})$ =	21.5 2.564	°C kPa
From Eq. 3.6, for: Then:	T_{min} = $e°(T_{min})$ =	12.3 1.431	°C kPa
-	e_s = (2.564 + 1.431) =	1.997	kPa
Given relative humidity data	RH_{max} = RH_{min} =	84 63	% %
From Eq. 3.13:	e_a = [1.431 (0.84) + 2.564 (0.63)]/2 =	1.409	kPa
-	Vapour pressure deficit ($e_s - e_a$) = (1.997 –1.409) =	0.589	kPa
Radiation			
From Table A2.5:	Month 7, Day = 6 J =	 187	 -
From Eq. 3.18:	Latitude = 50°48'N = J = R_a =	50.80 187 41.09	°N - MJ m^{-2} day^{-1}
From Eq. 3.31:	Latitude = 50°48'N = J = N = n/N = 9.25/16.3 =	50.80 187 16.1 0.57	°N - hours -
From Eq. 3.32 From Eq. 3.34 - From Eq. 3.41	R_s = [0.25 + 0.50 (0.57)] 41.09 R_{so} = (0.75 + 2 (100)/100 000) 41.09 = R_s/R_{so} = R_{ns} = 0.77 (22.07) =	22.07 30.90 0.71 17.00	MJ m^{-2} day^{-1} MJ m^{-2} day^{-1} - MJ m^{-2} day^{-1}
For: Then:	T_{max} = $T_{max, K}$ = 21.5 + 273.15 = $\sigma T_{max, K}^4$ =	21.5 294.65 36.93	°C K MJ m^{-2} day^{-1}
For: Then: -	T_{min} = $T_{min, K}$ = 12.3 + 273.15 = $\sigma T_{min, K}^4$ = $(\sigma T_{max, K}^4 + \sigma T_{min, K}^4)/2$ = (36.93 + 32.53)/2 =	12.3 285.45 32.53 34.73	°C K MJ m^{-2} day^{-1} MJ m^{-2} day^{-1}
- -	$(0.34 - 0.14\sqrt{e_a})$ = $(1.35(R_s/R_{so}) - 0.35)$ =	0.174 0.614	- -
From Eq. 3.42 From Eq. 3.43 From Eq. 3.45 - -	R_{nl} = 34.73 (0.174) 0.614 = R_n = (17.00 – 3.71) = G = $(R_n - G)$ = (13.29 – 0) = 0.408 $(R_n - G)$ =	3.71 13.29 0 13.29 5.42	MJ m^{-2} day^{-1} MJ m^{-2} day^{-1} MJ m^{-2} day^{-1} MJ m^{-2} day^{-1} mm day^{-1}
Grass reference evapotranspiration			
- - -	0.408 $(R_n - G)$ $\Delta/[\Delta+\gamma(1+0.34u_2)]$ = $900/(T+273)$ u_2 $(e_s - e_a)$ $\gamma/[\Delta+\gamma(1+0.34u_2)]$= ET_o (Eq. 2.4) = 2.81 + 1.07 = 3.88 ≈	2.81 1.07 3.88	mm day^{-1} mm day^{-1} mm day^{-1}

The grass reference evapotranspiration is 3.9 mm day^{-1} for 6 July.

Hourly time step

In areas where substantial changes in wind speed, dewpoint or cloudiness occur during the day, calculation of the ET_o equation using hourly time steps is generally better than using 24-hour calculation time steps. Such weather changes can cause 24-hour means to misrepresent the evaporative power of the environment during parts of the day and may introduce errors in the calculations. However, under most conditions, the application of the FAO-PM ET_o equation with 24-hour data produces accurate results.

With the advent of electronic, automated weather stations, weather data are increasingly reported for hourly or shorter periods. Therefore, in situations where calculations are computerized, the FAO-PM ET_o equation can be applied on an hourly basis with good results. When applying the FAO-PM ET_o equation on an hourly or shorter time scale, the equation and some of the procedures for calculating meteorological data should be adjusted for the smaller time step.

The FAO-PM ET_o equation for:

- ***hourly time steps during daytime*** (when $R_n \geq 0$):

$$ET_o = \frac{0.408\,\Delta\,(R_n - G) + \gamma\,\frac{37}{T + 273}\,u_2\,(e_s - e_a)}{\Delta + \gamma\,(1 + 0.24\,u_2)} \qquad \text{(2.5a bis)}$$

where
- ET_o reference evapotranspiration [mm $hour^{-1}$],
- R_n net radiation at the grass surface [MJ m^{-2} $hour^{-1}$],
- G soil heat flux density [MJ m^{-2} $hour^{-1}$],
- T mean hourly air temperature at 2 m height [°C],
- u_2 average hourly wind speed at 2 m height [m s^{-1}],
- e_s average hourly saturation vapour pressure [kPa],
- e_a average hourly actual vapour pressure [kPa],
- $e_s - e_a$ saturation vapour pressure deficit [kPa],
- Δ slope vapour pressure curve [kPa $°C^{-1}$],
- γ psychrometric constant [kPa $°C^{-1}$].

- ***hourly time steps during night time*** (when $R_n < 0$):

$$ET_o = \frac{0.408\,\Delta\,(R_n - G) + \gamma\,\frac{37}{T + 273}\,u_2\,(e_s - e_a)}{\Delta + \gamma\,(1+0.96\,u_2)} \qquad \text{(2.5b bis)}$$

In Equation 2.5a for hourly timesteps, the coefficient in the numerator is 37 rather than 900 that is used for 24-hour timesteps and the coefficient in the denominator is 0.24 rather than 0.34 that is used for 24-hour. During nighttime (when $R_n < 0$) the 0.24 coefficient in the denominator is changed to 0.96 in Equation 2.5b (hourly).

Given relative humidity measurements, the hourly actual vapour pressure is determined as:

$$e_a = e°\,(T_{hr})\left(\frac{RH_{hr}}{100}\right) \qquad (4.1)$$

where e_a average hourly actual vapour pressure [kPa],
$e°(T_{hr})$ saturation vapour pressure at air temperature T_{hr} [kPa] (Equation 3.6),
RH_{hr} average hourly relative humidity [%].

The net radiation is the difference between the net shortwave radiation (R_{ns}) and the net longwave radiation (R_{nl}) at the hourly time steps. Consequently:

- If R_{ns} and R_{nl} need to be calculated, the extraterrestrial radiation value (R_a) for the hourly period (Equation 3.25) should be used.
- In the computation of R_{nl} by means of Equation 3.42, $(\sigma T_{max, K}^4 + \sigma T_{min, K}^4)/2$ is replaced by $\sigma T_{hr, K}^4$ and the Stefan-Boltzman constant becomes

$$\sigma = (4.903/24)\ 10^{-9} = 2.043\ 10^{-10}\ \text{MJ m}^{-2}\ \text{h}^{-1}$$

Since the ratio R_s/R_{so} is used to represent cloud cover, when calculating R_{nl} for hourly periods during the nighttime, R_{so} becomes zero, so that the ratio R_s/R_{so} does not apply. That ratio in Equation 3.42 can be set equal to the R_s/R_{so} calculated for a time period occurring 2-3 hours before sunset, before the sun angle becomes small. This will generally serve as a good approximation of cloudiness occurring during the subsequent nighttime.

The hourly period that is 2 to 3 hours before sunset can be identified during the computation of R_a as the period where ω, calculated from Equation 3.28, is within the range $(\omega_s - 0.79) \le \omega \le (\omega_s - 0.52)$, where ω_s is calculated using Equation 3.22. As a more approximate alternative, one can assume $R_s/R_{so} = 0.4$ to 0.6 during nighttime periods in humid and subhumid climates and $R_s/R_{so} = 0.7$ to 0.8 in arid and semiarid climates. A value of $R_s/R_{so} = 0.3$ presumes total cloud cover.

Soil heat flux is important for hourly calculations. Equations 3.48 and 3.49 can be used to derive G for the hourly periods.

The required meteorological data consists of:

- Air temperature: mean hourly temperature (T_{hr}).
- Air humidity: average hourly relative humidity (RH_{hr}) or hourly dewpoint temperature.
- Wind speed: average hourly wind speed data measured at or transformed to 2 m height (u_2).
- Radiation: total hourly solar (R_s) or net radiation (R_n).

Because of the need for standardization, the constants in Equations 2.5 are based on a constant surface resistance (r_s) of 50 s m^{-1} during daytime periods, i.e., when $R_n > 0$. The $r_s = 50$ s m^{-1} is represented by the 0.24 coefficient in the denominator of Equation 2.5a. This constant resistance best represents the behaviour of the clipped grass reference during daytime hours. The constant resistance is increased to 200 s m^{-1} for night-time periods, when $R_n < 0$, and is represented by using a constant 0.96 in the denominator of Equation 2.5b instead of the constant 0.24.

When the calculations of hourly ET_o from Equation 2.5a and 2.5b are summed over 24-hour periods to produce 24-hour ET_o, the hourly sums are generally equivalent to calculations of ET_o made on a 24-hour time step.

Precise estimates of ET_o for specific hourly periods may require the use of aerodynamic stability functions and functions for modifying the value for r_s based on levels of radiation, humidity and temperature. Application of these complex functions is not

normally required when hourly calculations are to be summed to 24-hour totals. Therefore, those functions are not described here.

Equations 2.5a and 2.5b can be applied to time steps shorter or longer than hourly timesteps, for example, to 0.5-hour timesteps or 3-hour timesteps as long as all units per time are expressed as units per hour.

Due to the added complexity of hourly calculation timesteps, it is highly recommended that a computerized software be verified by comparing, step-by-step, with component calculations in Example 4.3. The example shows calculations for both nighttime and daytime hourly periods.

EXAMPLE 4.3

Determination of ET_o with hourly data

Given mean average hourly data between 02.00 and 03.00 hours (during nighttime) and 14.00 and 15.00 hours on 1 October in N'Diaye (Senegal) at 16°13'N and 16°15'W and 8 m above sea level.

Measured climatic data		**02.00-03.00 h**	**14.00-15.00 h**	Units
T_{hr}: mean hourly temperature =		28	38	°C
RH_{hr}: mean hourly relative humidity =		90	52	%
u_2: mean hourly wind speed =		1.9	3.3	m s^{-1}
R_s: total solar radiation =		-	2.450	MJ m^{-2} $hour^{-1}$
Parameters				
From Eq. 3.1, for:	altitude =	8	8	m
	P =	101.2	101.2	kPa
From Eq. 3.8	Δ =	0.220	0.358	kPa $°C^{-1}$
From Eq. 3.3	γ =	0.0673	0.0673	kPa $°C^{-1}$
Vapour pressure deficit				
From Eq. 3.6	$e°(T)$ =	3.780	6.625	kPa
From Eq. 3.13	e_a =	3.402	3.445	kPa
-	$e_s - e_a$ =	0.378	3.180	kPa
Extraterrestrial radiation		**02.00-03.00 h**	**14.00-15.00 h**	Units
From Table A2.5 for 1 October:	J = 274			-
From Eq. 3.19:	$\phi = \pi/180$ (16.22) = 0.2830			rad
From Eq. 3.20:	d_r = 1.0001			-
From Eq. 3.21:	δ = – 0.0753			rad
From Eq. 3.30:	b = 3.3315			-
From Eq. 3.29:	S_c = 0.1889			hour
-	L_z = 15			degrees
-	L_m = 16.25			degrees
-	t =	2.5	14.5	hour
From Eq. 3.28:	ω =	–2.46	0.682	rad
-	t_1 =	1	1	hour
From Eq. 3.26:	ω_1 =	-	0.5512	rad
From Eq. 3.27:	ω_2 =	-	0.8130	rad
From Eq. 3.25:	R_a =	0	3.543	MJ m^{-2} $hour^{-1}$

EXAMPLE 4.3 (CONT.)

Radiation				
Given	R_s =	0	2.450	MJ m^{-2} $hour^{-1}$
From Eq. 3.34 (or 3.35):	R_{so} =	0	2.658	MJ m^{-2} $hour^{-1}$
From Eq. 3.41:	R_{ns} =	0	1.887	MJ m^{-2} $hour^{-1}$
-	σT_K^4 =	1.679	1.913	MJ m^{-2} $hour^{-1}$
-	$(0.34-0.14\sqrt{e_a})$ =	0.082	0.080	-
-	R_s/R_{so} =	0.8 (assumed)	0.922	-
-	$(1.35\ R_s/R_{so} - 0.35)$ =	0.730	0.894	-
From Eq. 3.42:	R_{nl} =	0.100	0.137	MJ m^{-2} $hour^{-1}$
From Eq. 3.43:	R_n =	– 0.100	1.749	MJ m^{-2} $hour^{-1}$
From Eq. 3.49, 3.48:	G =	– 0.050	0.175	MJ m^{-2} $hour^{-1}$
-	$(R_n - G)$ =	– 0.050	1.574	MJ m^{-2} $hour^{-1}$
-	$0.408(R_n - G)$ =	– 0.020	0.642	mm $hour^{-1}$
Grass reference evapotranspiration		**Nighttime**	**Daytime**	
From Eq. 2.5a:	Radiative term =		0.481	mm $hour^{-1}$
From Eq. 2.5b:	Radiative term =	– 0.011		mm $hour^{-1}$
From Eq. 2.5a:	Aerodynamic term =		0.175	mm $hour^{-1}$
From Eq. 2.5b:	Aerodynamic term =	0.014		mm $hour^{-1}$
Eq. 2.5a:	ET_o =		0.656	mm $hour^{-1}$
Eq. 2.5b:	ET_o =	0.004		mm $hour^{-1}$

The grass reference evapotranspiration is 0.00 mm $hour^{-1}$ between 02.00 and 03.00 hours and 0.66 mm $hour^{-1}$ between 14.00 and 15.00 hours.

4.2 CALCULATION PROCEDURES WITH MISSING DATA

4.2.1 Calculation alternatives

The meteorological data, required to estimate ET_o by means of the FAO-PM ET_o equation, consists of air temperature, air humidity, wind speed and radiation. *Where some of these data are missing or cannot be calculated, it is strongly recommended that the user estimate the missing climatic data with one of the procedures presented in Chapters 2 and 3 and that the FAO-PM method be used for the calculation of ET_o. The use of an alternative ET_o calculation procedure, requiring only limited meteorological parameters, is less recommended.*

Example 4.4 illustrates the estimation of monthly ET_o with the FAO-PM for a data set containing only maximum and minimum air temperature. The procedures given in Chapters 2 and 3 to estimate missing humidity, radiation and wind speed data should be validated by comparing ET_o calculated with full and with limited data sets for weather stations in the region having complete data sets.

EXAMPLE 4.4

Determination of monthly ET_o with missing data

Given the monthly average daily maximum and average daily minimum air temperature of July from a station near Foggia, Italy (41°28' N, altitude 74 m). No other climatic data were recorded. Using the long term CLIMWAT weather database, the Aridity Index (AI) was computed from Chapter 2, Box 2.4, as the ratio between the mean annual precipitation (mm) and the mean annual climatic evaporation index (mm), resulting in AI = 0.51, thus, indicating that the location has a dry sub-humid climate type. In addition, using the same data set it was possible to estimate the average (T_{max}–T_{min}) = 13.0 °C, u_2 = 2.42 m s^{-1} and RH = 71%.

-	Monthly average daily maximum temperature (T_{max}) =	32.2	°C
-	Monthly average daily minimum temperature (T_{min}) =	19.2	°C

Determination following Box 4.1 (calculation sheet for ET_o)

Estimation of wind speed:

2.42 m s^{-1} is used as the estimator of u_2 because it constitutes the average local value. Due to the relatively short crop height of 0.12 m of the reference crop and the appearance of u_2 in both the nominator and denominator of the FAO Penman-Monteith equation, ET_o is not highly sensitive to normal ranges of wind speed.

Parameters:			
-	T_{mean} = (32.2 + 19.2)/2 =	25.7	°C
From Table A2.4 or Eq. 3.8:	T_{mean} = Δ =	25.7 0.196	°C kPa °C^{-1}
From Eq. 3.1: From Table A2.2 or Eq. 3.3: - - - -	Altitude = P = γ = (1 + 0.34 u_2) = (1 + 0.34 (2.42)) = Δ/[Δ+γ (1+0.34u_2)] = 0.196/[0.196 + 0.067 (1.82)] = γ/[Δ+γ (1+0.34u_2)] = 0.067/[0.196 + 0.067 (1.82)] = (900/(T_{mean}+273)) u_2 =	74 100.4 0.067 1.82 0.617 0.210 7.29	m kPa kPa °C^{-1} - - - -
Estimation of humidity data:			
Assume (Eq. 2.6 for AI = 0.51): Thus (Eq. 3.6 or Table A2.3) for: Then	T_{dew} ≈ T_{min} – 1 = T_{dew} = e_a =	18.2 18.2 2.09	°C °C kPa
From Table A2.3 or Eq. 3.6, for: Then:	T_{max} = $e°(T_{max})$ =	32.2 4.81	°C kPa
From Table A2.3 or Eq. 3.6, for: Then:	T_{min} = $e°(T_{min})$ =	19.2 2.22	°C kPa
Eq. 3.7: -	e_s = (4.81 + 2.22)/2 = (e_s-e_a) = (3.52 – 2.09) =	3.52 1.43	kPa kPa

EXAMPLE 4.4 (CONT.)

Estimation of radiation data:			
R_s can be derived from the difference between T_{max} and T_{min}:			
From Eq. 3.52 From Eq. 3.54 (for AI = 0.51)	$R_s = k_{Rs}\sqrt{(32.2-19.2)}\ R_a$ k_{Rs}=0.396-0.011(10.1)+0.019(2.42)-0.002(71) $R_s = 0.69\ R_a$	- 0.19	MJ m^{-2} hour^{-1} MJ m^{-2} hour^{-1}
Table A2.6 for: Then:	For Day 15, Month = July, J = Latitude = 41°28'N = R_a =	196 41.47 40.76	- °N MJ m^{-2} hour^{-1}
- - - -	$R_s = 0.69\ R_a = 0.69\ (40.76)$ = $R_{so} = (0.75 + 2\ (200)/100\ 000)\ 40.76$ = R_s/R_{so} = $R_{ns} = 0.77\ (28.12)$ =	28.12 30.63 0.92 21.65	MJ m^{-2} hour^{-1} MJ m^{-2} hour^{-1} - MJ m^{-2} hour^{-1}
Table 2.8, for: Then:	T_{max} = $T_{max, K} = 32.2 + 273.15$ = $\sigma T_{max, K}^4$	32.2 305.35 42.60	°C K MJ m^{-2} hour^{-1}
Table 2.8, for: Then: -	T_{min} = $T_{min, K} = 19.2 + 273.15$ $\sigma T_{min, K}^4$ $(\sigma T_{max, K}^4 + \sigma T_{min, K}^4)/2 = (42.60 + 35.79)/2$ =	19.2 292.35 35.79 39.23	°C K MJ m^{-2} hour^{-1} MJ m^{-2} hour^{-1}
For: Then:	e_a = $(0.34-0.14\sqrt{e_a})$ =	2.09 0.14	kPa -
For: Then: - -	R_s/R_{so} = $(1.35\ R_s/R_{so} - 0.35)$ = $R_{nl} = 39.23\ (0.14)\ 0.89$ = $R_n = (21.65 - 4.89)$ =	0.92 0.89 4.89 16.76	- - MJ m^{-2} hour^{-1} MJ m^{-2} hour^{-1}
Assume: - -	G = $(R_n - G) = (16.76 - 0)$ = $0.408\ (R_n - G)$ =	0 16.76 6.84	MJ m^{-2} hour^{-1} MJ m^{-2} hour^{-1} mm day^{-1}
Grass reference evapotranspiration:			
- - -	$0.408\ (R_n\text{-}G)\ \Delta/[\Delta+\gamma(1+0.34u_2)]$ = $900/(T+273)u_2(e_s\text{-}e_a)\gamma/[\Delta+\gamma(1+0.34\ u_2)]$ = $ET_o = (4.21 + 2.20)$ =	4.21 2.20 6.41	mm day^{-1} mm day^{-1} mm day^{-1}

The estimated grass reference evapotranspiration is 6.41 mm day^{-1}. If instead of 2.42 m s^{-1}, the wind speed is estimated as 1 or 3 m s^{-1}, ET_o would have been 1 percent lower (5.70 mm day^{-1}) or 1 percent higher (6.65 mm day^{-1}) respectively, due to the relatively high RH at the location.

4.2.2 Use of Artificial Intelligence (AI) for missing climatic data and for calculating ET_o

Artificial Intelligence (AI) and machine learning processes can be useful for recognizing and reproducing patterns found in large amounts of data. AI is highly effective in analyzing time series data. These attributes make these processes potentially valuable for efficiently forecasting and understanding complex time-dependent phenomena. However, precautions must be made prior to and during their application, especially in ET data assessment and extension.

The performance of AI models relies heavily on the quality and availability of data. AI can not readily correct for faulty data that may be ingested into the AI process, but are not recognized as faulty. Nor does AI provide a replacement for physics-based relationships. These guidelines wish to be quite clear that the FAO-PM equation does NOT need to be replaced by an AI-generated simpler equation, even when some weather data are missing. The FAO-PM equation is a physics-based equation that contains the internal, physical responses and sensitivities to impacts of the four primary weather parameters that drive the ET process. These parameters are solar radiation (and therefore net radiation), air temperature, air humidity and wind speed. These parameters have strong interactions within the ET process that utilizes knowledge of available environmental energy and aerodynamic transfer processes to estimate the total transfer and transformation of energy into the vapourization of liquid water.

Many practitioners are learning to use AI, but they may not be learning or adhering to the physics that govern ET processes. In addition, users of AI may not first gain or exercise an ability to recognize or to question faulty data prior to ingestion into an AI procedure. As a result, they may create abstract and incorrect relationships between ET_o and weather data or between ET_o and ground-measurements of reference ET that may not have the appropriate structures, accuracies and sensitivities. Good quality assessment and quality control (QAQC) is fundamental to the process. The QAQC step requires overview, correction and concurrence by a knowledgeable human.

These guidelines make it clear that it is best to remain with the use of a physics-based reference equation, the FAO-Penman-Monteith equation, and make best estimates for missing weather data, rather than to employ an AI engine that may not adhere to all physical laws that govern ET. The result of the latter process can be the production of tangential errors that violate physical principles. Precaution, coupled with human oversight and review, is critical.

A well-designed and trained AI process may be useful for recognizing deviation of weather data from that occurring under reference (well-watered) conditions due to aridity in or surrounding a weather station. The process can then make necessary adjustments to the data using procedures similar to those addressed previously in Chapter 2.

4.3 PAN EVAPORATION METHOD

4.3.1 Pan evaporation

The evaporation rate from pans filled with water is easily obtained. In the absence of rain, the amount of water evaporated during a period (mm day^{-1}) corresponds with the decrease in water depth in that period. Pans provide a measurement of the integrated effect of radiation, wind, temperature and humidity on the evaporation from an open water surface. Although the pan responds in a similar fashion to the same climatic factors affecting crop transpiration, several factors produce significant differences in loss of water from a water surface and from a cropped surface. The reflection of solar radiation from water in the shallow pan might be different from the assumed 23 percent for the grass reference surface. Storage of heat within the pan can be appreciable and may cause significant evaporation during the night while most crops transpire only during the daytime. There are also differences in turbulence, temperature and humidity of the air immediately above the respective surfaces. Heat transfer through the sides of the pan occurs and affects the energy balance.

Notwithstanding the difference between pan-evaporation and the evapotranspiration of cropped surfaces, the use of pans to predict ET_o for periods of 10 days or longer may be warranted. The pan evaporation is related to the reference evapotranspiration by an empirically derived pan coefficient:

$$ET_o = K_p \, E_{pan} \tag{4.2}$$

where ET_o reference evapotranspiration [mm day^{-1}],
K_p pan coefficient [-],
E_{pan} pan evaporation [mm day^{-1}].

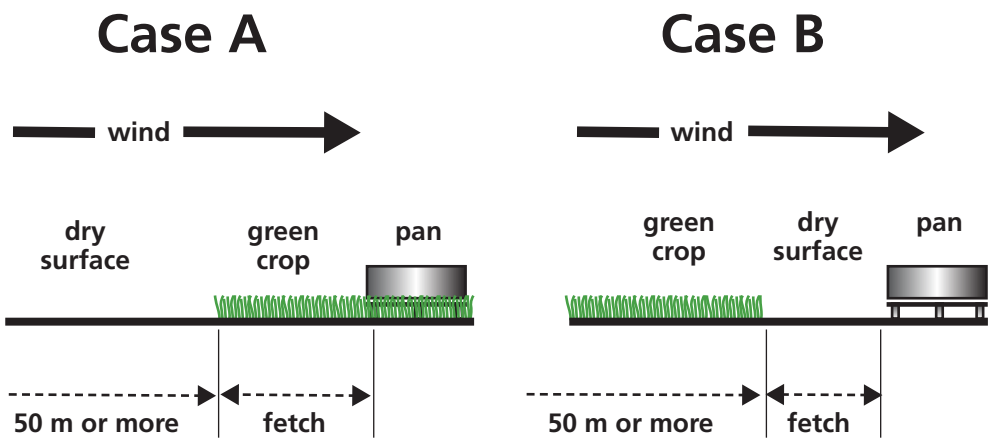
4.3.2 Pan coefficient (K_p)

Pan types and environment

Different types of pans exist. Descriptions of Class A and Colorado sunken pans are given in Boxes 4.2 and 4.3. As the colour, size, and field position of the pan have a significant influence on the measured results, the pan coefficients are pan specific.

In selecting the appropriate pan coefficient, not only the pan type, but also the ground cover in the station, its surroundings as well as the general wind and humidity conditions, should be checked. The siting of the pan and the pan environment also influence the results. This is particularly so where the pan is placed in fallow rather than cropped fields. Two cases are commonly considered: Case A where the pan is sited on a short green (grass) cover and surrounded by fallow soil; and Case B where the pan is sited on fallow soil and surrounded by a green crop (Figure 4.2).

FIGURE 4.2
Two cases of evaporation pan siting and their environment



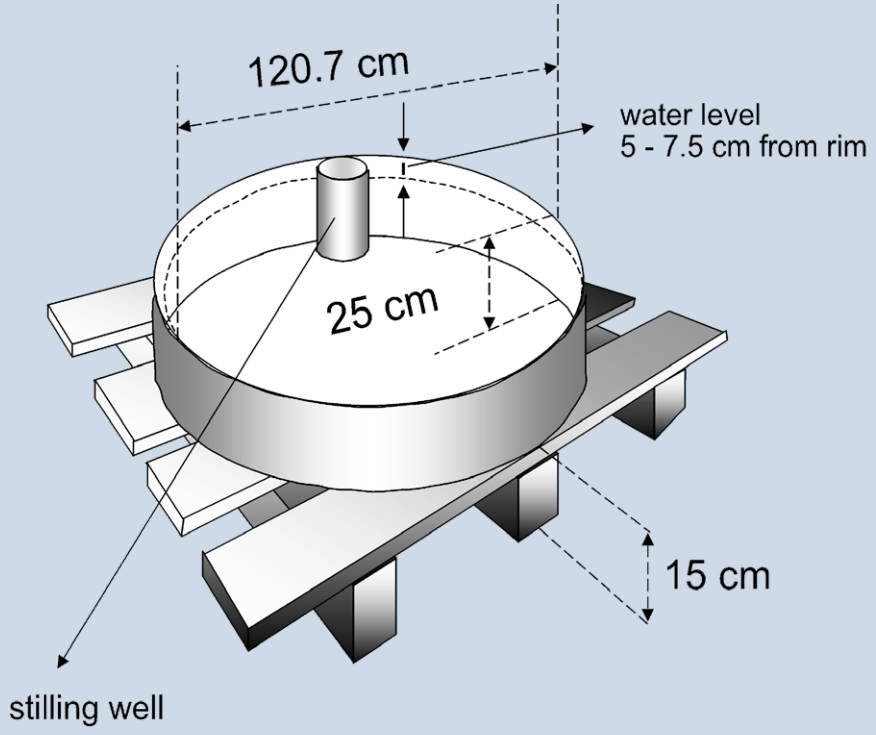
Source: FAO56, 1998

BOX 4.2
Description of Class A pan

The Class A Evaporation pan is circular, 120.7 cm in diameter and 25 cm deep. It is made of galvanized iron (22 gauge) or Monel metal (0.8 mm). The pan is mounted on a wooden open frame platform which is 15 cm above ground level. The soil is built up to within 5 cm of the bottom of the pan. The pan must be level. It is filled with water to 5 cm below the rim, and the water level should not be allowed to drop to more than 7.5 cm below the rim. The water should be regularly renewed, at least weekly, to eliminate extreme turbidity. The pan, if galvanized, is painted annually with aluminum paint. Screens over the pan are not a standard requirement and should preferably not be used. Pans should be protected by fences to keep animals from drinking.

The site should preferably be under grass, 20 by 20 m (at least 15 by 15), open on all sides to permit free circulation of the air. It is preferable that stations be located in the centre or on the leeward side of large cropped fields.

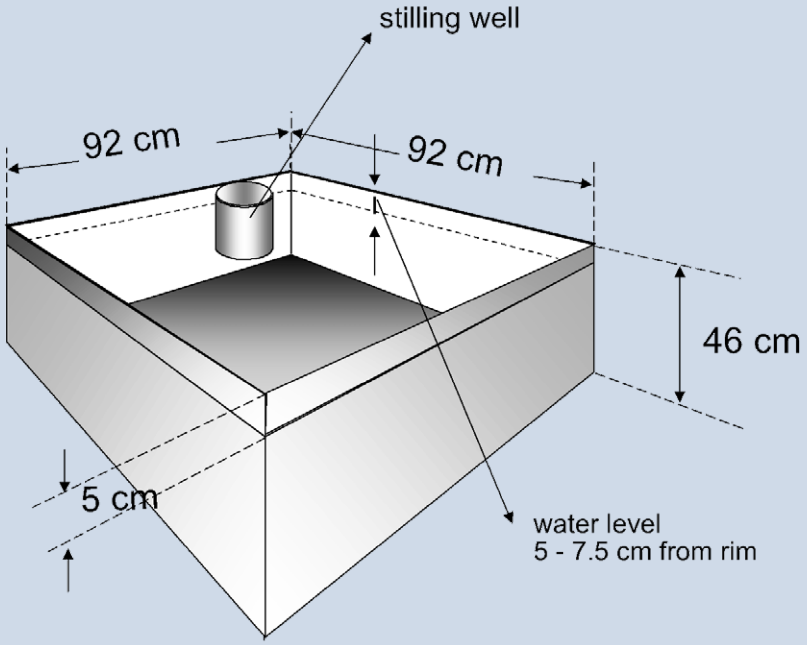
Pan readings are taken daily in the early morning at the same time that precipitation is measured. Measurements are made in a stilling well that is situated in the pan near one edge. The stilling well is a metal cylinder of about 10 cm in diameter and some 20 cm deep with a small hole at the bottom.



BOX 4.3
Description of Colorado sunken pan

The Colorado sunken pan is 92 cm (3 ft) square and 46 cm (18 in) deep, made of 3 mm thick iron, placed in the ground with the rim 5 cm (2 in) above the soil level. Also, the dimensions 1 m square and 0.5 m deep are frequently used. The pan is painted with black tar paint. The water level is maintained at or slightly below ground level, i.e. 5-7.5 cm below the rim.

Measurements are taken similarly to those for the Class A pan. Siting and environment requirements are also similar to those for the Class A pan. Sunken Colorado pans are sometimes preferred in crop water requirements studies, as these pans give a better direct estimation of the reference evapotranspiration than does the Class A pan. The disadvantage is that maintenance is more difficult and leaks are not visible.



Pan coefficients

Depending on the type of pan and the size and state of the upwind buffer zone (fetch), pan coefficients will differ. The larger the upwind buffer zone, the more the air moving over the pan will be in equilibrium with the buffer zone. At equilibrium with a large fetch, the air contains more water vapour and less heat in Case A than in Case B (Figure 4.2). Pan coefficients for the Class A pan and for the Colorado sunken pan for different ground cover, fetch and climatic conditions are presented in Tables 4.1 and 4.2. For extensive areas of bare-fallow soils and no agricultural development, reduce K_p by 20 percent under hot, windy conditions; by 5-10 percent for moderate wind, temperature and humidity conditions.

Regression equations derived from the tables are presented in Table 4.3. Where measured data from other types of sunken pans are available, such data should first be related to Colorado sunken pan data or to the FAO Penman-Monteith equation to develop K_p. Ratios between evaporation from sunken pans and from the Colorado sunken pan for different climatic conditions and pan environment are given in Table 4.4.

TABLE 4.1

Pan coefficients (K_p) for Class A pan for different pan siting and environment and different levels of mean relative humidity and wind speed (FAO Irrigation and Drainage Paper No. 24)

Class A pan	Case A: Pan placed in short green cropped area				Case B: Pan placed in dry fallow area			
RH mean (%)		low < 40	medium 40 -70	high > 70		low < 40	medium 40 -70	high > 70
Wind speed ($m\ s^{-1}$)	**Windward side distance of green crop (m)**				**Windward side distance of dry fallow (m)**			
Light < 2	1 10 100 1 000	.55 .65 .7 .75	.65 .75 .8 .85	.75 .85 .85 .85	1 10 100 1 000	.7 .6 .55 .5	.8 .7 .65 .6	.85 .8 .75 .7
Moderate 2-5	1 10 100 1 000	.5 .6 .65 .7	.6 .7 .75 .8	.65 .75 .8 .8	1 10 100 1 000	.65 .55 .5 .45	.75 .65 .6 .55	.8 .7 .65 .6
Strong 5-8	1 10 100 1 000	.45 .55 .6 .65	.5 .6 .65 .7	.6 .65 .7 .75	1 10 100 1 000	.6 .5 .45 .4	.65 .55 .5 .45	.7 .65 .6 .55
Very strong > 8	1 10 100 1 000	.4 .45 .5 .55	.45 .55 .6 .6	.5 .6 .65 .65	1 10 100 1 000	.5 .45 .4 .35	.6 .5 .45 .4	.65 .55 .5 .45

TABLE 4.2

Pan coefficients (K_p) for Colorado sunken pan for different pan siting and environment and different levels of mean relative humidity and wind speed (FAO Irrig. and Drain. Paper No. 24)

Colorado sunken pan	Case A: Pan placed in short green cropped area				Case B: Pan placed in dry fallow area			
RH mean (%)		low < 40	medium 40 -70	high > 70		low < 40	medium 40 -70	high > 70
Wind speed ($m\ s^{-1}$)	**Windward side distance of green crop (m)**				**Windward side distance of dry fallow (m)**			
Light < 2	1 10 ≥ 100	.75 1.0 1.1	.75 1.0 1.1	.8 1.0 1.1	1 10 100 1 000	1.1 .85 .75 .7	1.1 .85 .75 .7	1.1 .85 .8 .75
Moderate 2-5	1 10 ≥ 100	.65 .85 .95	.7 .85 .95	.7 .9 .95	1 10 100 1 000	.95 .75 .65 .6	.95 .75 .65 .6	.95 .75 .7 .65
Strong 5-8	1 10 ≥ 100	.55 .75 .8	.6 .75 .8	.65 .75 .8	1 10 100 1 000	.8 .65 .55 .5	.8 .65 .6 .55	.8 .65 .65 .6
Very strong > 8	1 10 ≥ 100	.5 .65 .7	.55 .7 .75	.6 .7 .75	1 10 100 1 000	.7 .55 .5 .45	.75 .6 .55 .5	.75 .65 .6 .55

Where data are missing, wind speed can be estimated by taking a global value of u_2 = 2 m s^{-1}, or as indicated in Table 3.2. RH_{mean} can be approximated from observations at a nearby station as referred to in Chapter 3 relative to missing data.

TABLE 4.3
Pan coefficients (K_p): regression equations derived from Tables 4.1 and 4.2

Class A pan with green fetch	$K_p = 0.108 - 0.0286\,u_2 + 0.0422\,\ln(FET) + 0.1434\,\ln(RH_{mean}) - 0.000631\,[\ln(FET)]^2\,\ln(RH_{mean})$
Class A pan with dry fetch	$K_p = 0.61 + 0.00341\,RH_{mean} - 0.000162\,u_2\,RH_{mean} - 0.00000959\,u_2\,FET + 0.00327\,u_2\,\ln(FET) - 0.00289\,u_2\,\ln(86.4\,u_2) - 0.0106\,\ln(86.4\,u_2)\,\ln(FET) + 0.00063\,[\ln(FET)]^2\,\ln(86.4\,u_2)$
Colorado sunken pan with green fetch	$K_p = 0.87 + 0.119\,\ln(FET) - 0.0157\,[\ln(86.4\,u_2)]^2 - 0.0019\,[\ln(FET)]^2\,\ln(86.4\,u_2) + 0.013\,\ln(86.4\,u_2)\,\ln(RH_{mean}) - 0.000053\,\ln(86.4\,u_2)\,\ln(FET)\,RH_{mean}$
Colorado sunken pan with dry fetch	$K_p = 1.145\text{-}0.080\,u_2 + 0.000903\,(u_2)^2\,\ln(RH_{mean}) - 0.0964\,\ln(FET) + 0.0031\,u_2\,\ln(FET) + 0.0015\,[\ln(FET)]^2\,\ln(RH_{mean})$
Coefficients and parameters	K_p pan coefficient [ ] u_2 average daily wind speed at 2 m height (m s^{-1}) RH_{mean} average daily relative humidity [%] = $(RH_{max} + RH_{min})/2$ FET fetch, or distance of the identified surface type (grass or short green agricultural crop for case A, dry crop or bare soil for case B upwind of the evaporation pan)
Range for variables	1 m ≤ FET ≤ 1 000 m (these limits must be observed) 30% ≤ RH_{mean} ≤ 84% 1 m s^{-1} ≤ u_2 ≤ 8 m s^{-1}

TABLE 4.4
Ratios between the evaporation from sunken pans and a Colorado sunken pan for different climatic conditions and environments (FAO Irrigation and Drainage Paper No. 24)

Climate		**Ratio E_{pan} mentioned and E_{pan} Colorado**			
		Humid-temperate climate		**Arid to semi-arid (dry season)**	
Ground cover surrounding pan (50 m or more)	**Pan area (m^2)**	**Short green cover**	**Dry fallow**	**Short green cover**	**Dry fallow**
GGI 20 diameter 5 m, depth 2 m (former Soviet Union)	19.6	1.0	1.1	1.05	1.25
Sunken pan diameter 12 ft (3.66 m) depth 3.3 ft (Israel)	10.5	↓	↓	↓	↓
BPI diameter 6 ft (1.83 m), depth 2 ft (0.61 m) (USA)	2.6				
Kenya pan diameter 4 ft (1.22 m) depth 14 in (0.356 m)	1.2				
Australian pan diameter 3 ft (0.91 m) depth 3 ft (0.91 m)	0.7		1.0		1.0
Symmons pan 6 ft2 (0.56 m^2) depth 2 ft (0.61 m)	0.6		↓		↓
Aslyng pan 0.33 m^2, depth 1 m (Denmark)	0.3			1.0	
GGI 3 000 diameter 0.618 cm, depth 60-80 cm (former Soviet Union)	0.3	↓		↓	
Sunken pan diameter 50 cm, depth 25 cm (Netherlands)	0.2	1.0	0.95	1.0	0.95

The arrows indicate possible variation of values between the first values and the next values; the user selects the interpolation mode.

Adjustments

Under some conditions not accounted for in the tables, the presented K_p coefficients may need some adjustment. This is the case in areas with no agricultural development, or where the pans are enclosed by tall crops. Not maintaining the standard colour of the pan or installing screens can affect the pan readings and will require some adjustment of the pan coefficient.

In areas with no agricultural development and extensive areas of bare soils (large fetch, Case B), as found under desert or semi-desert conditions, the listed values for K_p given for arid, windy areas may need to be reduced by up to 20 percent; for areas with moderate levels of wind, temperature and relative humidity, the listed values may need to be reduced by 5-10 percent; no or little reduction in K_p is needed in humid, cool conditions.

Where pans are placed in a small enclosure but surrounded by tall crops, for example 2.5 m high maize, the listed pan coefficients will need to be increased by up to 30 percent for dry windy climates, whereas only a 5-10 percent increase is required for calm, humid conditions.

Painting the pans may affect the pan evaporation. The pan coefficients presented apply to galvanized pans annually painted with aluminium and to stainless steel pans. Little difference in E_{pan} will occur where the inside and outside surfaces of the pan are painted white. An increase in E_{pan} of up to 10 percent may occur when they are painted black. The material from which the pan is made may account for variations of only a few percent.

The level at which the water is maintained in the pan is important; resulting errors may be up to 15 percent when water levels in the Class A pan fall 10 cm below the accepted standard of between 5 and 7.5 cm below the rim. Screens mounted over pans will reduce E_{pan} by up to 10 percent. In an attempt to avoid pans being used by birds for drinking, pans filled to the rim with water can be placed near the Class A pan; birds may prefer to use the fully filled pan. The evaporation pan should be placed in a large, secure, wire enclosure to prevent animals from entering and drinking. The turbidity of the water in the pan usually does not affect E_{pan} by more than 5 percent. The overall variation in E_{pan} is not constant with time because of aging, deterioration and repainting.

Recommendations

The above considerations and adjustments indicate that the use of tables or the corresponding equations may not be sufficient to consider all local environmental factors influencing K_p and that local adjustment may be required. To do so, an appropriate calibration of E_{pan} against ET_o computed with the Penman-Monteith method is recommended.

It is recommended that the pan should be installed inside a short green cropped area with a size of a square of at least 15 by 15 m. The pan should not be installed in the centre but at a distance of at least 10 m from the green crop edge in the general upwind direction.

Where observations of wind speed and relative humidity, required for the computation of K_p, are not available at the site, estimates of the weather variables from a nearby station have to be utilized. It is then recommended that these variables be averaged for the computation period and that E_{pan} be averaged for the same period.

Notice from the Example 4.5 that the for pan sittings produce the same pan evaporation 7.9 mm per day but necessarily selecting distinct pan coefficients leads to very different results for the reference evapotranspiration, thus much care is required in selecting the pan coefficient.

EXAMPLE 4.5
Determination of ET_o from pan evaporation using tables

Given the daily evaporation data for the first week of July for a Class A pan installed in a green area surrounded by short irrigated field crops: 8.2, 7.5, 7.6, 6.8, 7.6, 8.9 and 8.5 mm day^{-1}. In that period the mean wind speed is 1.9 m s^{-1} and the daily mean relative humidity is 73 percent. Determine the 7-day average reference evapotranspiration.

Pan is installed on a green surface: Case A

Pan is surrounded by irrigated crops:	$fetch_{max}$ =	1 000	m
Wind speed is light:	u <	2	ms^{-1}
Relative humidity is high:	RH_{mean} >	70	%
From Table 4.1 (for above conditions):	K_p =	0.85	-
From Eq. 4.2:	E_{pan} = (8.2 + 7.5 + 7.6 + 6.8 + 7.6 + 8.9 + 8.5)/7 =	7.9	mm day^{-1}
	ET_o = 0.85 (7.9) =	6.7	mm day^{-1}

The 7-day average of the crop reference evapotranspiration is 6.7 mm day^{-1}

EXAMPLE 4.6

Determination of ET_o from pan evaporation using equations

Given the 7-day average evaporation measurement from Example 4.5, estimate the ET_o for the two types of pans and two types of fetch (FET) conditions represented by equations in Table 4.3. Assume that fetch (FET) is 1000 m in both fetch cases (green and dry) u_2= 1.9 m s^{-1}, and RH_{mean} = 73%.

Class A pan with green fetch	$K_p = 0.108\text{-}0.0286\ u_2 + 0.0422\ \ln(FET) + 0.1434\ \ln(RH_{mean}) - 0.000631\ [\ln(FET)]^2\ \ln(RH_{mean})$ E_{pan} = ET_o =	$K_p = 0.108 - 0.0286\ (1.9) + 0.0422\ \ln(1000) + 0.1434\ \ln(73) - 0.000631\ [\ln(1000)]^2\ \ln(73)$ $ET_o = 0.83\ (7.9)$	0.83 7.9 6.6	- mm/day mm/day
Class A pan with dry fetch	$K_p = 0.61 + 0.00341\ RH_{mean} - 0.000162\ u_2\ RH_{mean} - 0.00000959\ u_2\ FET + 0.00327\ u_2 \ln(FET) - 0.00289\ u_2 \ln(86.4u_2) - 0.0106 \ln(86.4u_2) \ln(FET) + 0.00063\ [\ln(FET)]^2 \ln(86.4u_2)$ E_{pan} = ET_o =	$K_p = 0.61 + 0.00341\ (73) - 0.000162\ (1.9)\ (73) - 0.00000959\ (1.9)\ (1000) + 0.00327\ (1.9) \ln(1000) - 0.00289\ (1.9) \ln(86.4\ (1.9)) - 0.0106 \ln(86.4(1.9)) \ln(1000) + 0.00063\ [\ln(1000)]^2 \ln(86.4\ (1.9)) =$ $ET_o = 0.61\ (7.9)$	0.61 7.9 4.8	- mm/day mm/day
Colorado sunken pan with green fetch	$K_p = 0.87 + 0.119\ \ln(FET) - 0.0157\ [\ln(86.4\ u_2)]^2 - 0.0019\ [\ln(FET)]^2\ \ln(86.4\ u_2) + 0.013\ \ln(86.4\ u_2)\ \ln(RH_{mean}) - 0.000053\ \ln(86.4\ u_2)\ \ln(FET)\ RH_{mean}$ E_{pan} = ET_o =	$K_p = 0.87 + 0.119\ \ln(100) - 0.0157\ [\ln(86.4\ (1.9))]^2 - 0.0019\ [\ln(1000)]^2\ \ln(86.4\ (1.9)) + 0.013\ \ln(86.4\ (1.9)) \ln(73) - 0.000053\ \ln(86.4\ (1.9))\ \ln(1000)\ (73) =$ $ET_o = 0.97\ (7.9)$	0.97 7.9 7.7	- mm/day mm/day
Colorado sunken pan with dry fetch	$K_p = 1.145\text{-}0.080\ u_2 + 0.000903\ (u_2)^2\ \ln(RH_{mean}) - 0.0964\ \ln(FET) + 0.0031\ u_2\ \ln(FET) + 0.0015\ [\ln(FET)]^2\ \ln(RH_{mean})$ E_{pan} = ET_o =	$K_p = 1.145 - 0.080\ (1.9) + 0.000903\ (1.9)^2\ \ln(73) - 0.0964\ \ln(1000) + 0.0031\ (1.9)\ \ln(1000) + 0.0015\ [\ln(1000)]^2\ \ln(73) =$ $ET_o = 0.69\ (7.9)$	0.69 7.9 5.4	- mm/day mm/day
The 7-day average of the crop reference evapotranspiration for the four pan/fetch conditions is 6.6, 4.8, 7.7, and 5.4 mm/day				

Chapter 5

Introduction to crop evapotranspiration (ET_c) and its measurement

This chapter outlines the crop coefficient (K_c) approach for calculating the crop evapotranspiration under standard conditions (ET_c). The standard conditions refer to crops grown in large fields under excellent agronomic and soil water conditions. The crop evapotranspiration differs distinctly from the reference evapotranspiration (ET_o) as the ground cover, canopy properties and aerodynamic resistance of the crop are different from grass. The effects of characteristics that distinguish field crops from grass are integrated into the crop coefficient. In the crop coefficient approach, crop evapotranspiration is calculated by multiplying ET_o by K_c.

Differences in evaporation and transpiration between field crops and the reference grass surface can be integrated into a single crop coefficient (K_c) or separated into two coefficients: a basal crop (K_{cb}) and a soil evaporation coefficient (K_e), i.e. $K_c = K_{cb} + K_e$. The approach to follow should be selected as a function of the purpose of the calculation, the accuracy required and the data available. A brief review of ET_c measurement using ground and remote observations and sensors is added.

5.1 CALCULATION PROCEDURES

5.1.1 Direct calculation

The evapotranspiration rate from a cropped surface can be directly measured by the mass transfer or the energy balance method. It can also be derived from studies of the soil water balance determined from cropped fields or from lysimeters.

Crop evapotranspiration can also be derived from meteorological and crop data by means of the FAO-PM ET_o equation (Equation 2.1). By adjusting the albedo and the aerodynamic and canopy surface resistances to the growing characteristics of the specific crop, the evapotranspiration rate can be directly estimated using the Penman-Monteith combination equation (Equation 2.3). The albedo and resistances are, however, difficult to estimate accurately as they may vary continually during the growing season as climatic conditions change, as the crop develops, and with the wetness of the soil surface. The canopy resistance will further be influenced by the soil water availability, and it increases strongly if the crop is subjected to water stress.

As there is still a considerable lack of consolidated information on the aerodynamic and canopy resistances for the various cropped surfaces, the FAO-PM ET_o equation is used in these guidelines only for estimating ET_o, the evapotranspiration from a well-watered (hypothetical) grass surface having fixed crop height, albedo, and surface resistance.

5.1.2 Crop coefficient approach

In the crop coefficient approach the crop evapotranspiration, ET_c, is calculated by multiplying the reference crop evapotranspiration, ET_o, by a crop coefficient, K_c:

$$ET_c = K_c\ ET_o \tag{5.1}$$

where ET_c crop evapotranspiration [mm d^{-1}],
K_c crop coefficient [dimensionless],
ET_o reference crop evapotranspiration [mm d^{-1}].

Most of the effects of the various weather conditions are incorporated into the ET_o estimate. Therefore, as ET_o represents an index of climatic demand, K_c varies predominately with the specific crop characteristics and only to a limited extent with climate. This enables the transfer of standard values for K_c between locations and between climates. This has been a primary reason for the global acceptance and usefulness of the crop coefficient approach and the K_c standard values developed in past studies.

The reference ET_o is defined and calculated using the FAO-PM ET_o equation (Chapters 2 and 4). The crop coefficient, K_c, is basically the ratio of the crop ET_c to the reference ET_o, and it represents an integration of the effects of four primary characteristics that distinguish the crop from the reference grass. These characteristics are:

- Crop height: the crop height influences the aerodynamic resistance term, r_a, of the FAO-PM ET_o equation and the turbulent transfer of vapour from the crop into the atmosphere. The r_a term appears twice in the full form of the FAO-PM ET_o equation.
- Albedo (reflectance) of the crop-soil surface: the albedo is affected by the fraction of ground covered by vegetation and by the soil surface wetness. The albedo of the crop-soil surface influences the net radiation of the surface, R_n, which is the primary source of the energy exchange for the evaporation process.
- Canopy resistance: the resistance of the crop to vapour transfer is affected by leaf area (number of stomata), leaf age and condition, and the degree of stomatal control. The canopy resistance influences the surface resistance, r_s.
- Evaporation from soil, especially exposed soil.

The soil surface wetness and the fraction of ground covered by vegetation influence the surface resistance, r_s. Following soil wetting, the vapour transfer rate from the soil is high, especially for crops having incomplete ground cover. The combined surface resistance of the canopy and the soil determines the (bulk) surface resistance, r_s. The surface resistance term in the Penman-Monteith combination Equation 2.3 represents the resistance to vapour flow from within plant leaves and from beneath the soil surface.

The K_c in Equation 5.1 predicts ET_c under standard conditions. The standard condition represents an upper envelope of crop evapotranspiration and represents conditions where no limitations are placed on crop growth or evapotranspiration due to water shortage, low crop density, or disease, weed, insect, or salinity pressures. The ET_c estimated by K_c is adjusted if necessary to non-standard conditions. The non-standard actual condition, which is assumed to represent the actual ET condition, is termed $ET_{c\ act}$. $ET_{c\ act}$ is estimated and applied when any environmental condition or characteristic is known to have a negative impact on, or sets a limit on ET_c. Factors for correcting ET_c to $ET_{c\ act}$ are described in Chapter 8 (Part C).

5.2 FACTORS DETERMINING THE CROP COEFFICIENT

The crop coefficient integrates the effect of characteristics that distinguish a typical field crop from the grass reference, where the standardized grass reference has a constant appearance and a complete ground cover. Consequently, different crops will have different K_c coefficients. The changing characteristics of the crop over the growing season affect the K_c coefficient. Finally, as evaporation is an integrated part of crop evapotranspiration, conditions affecting soil evaporation will also have an effect on K_c.

5.2.1 Crop type

Due to differences in albedo, crop height, aerodynamic properties, and leaf and stomata properties, the evapotranspiration from full-grown, well-watered crops differs from ET_o.

The close spacings of plants and taller canopy height and roughness of many full-grown agricultural crops cause these crops to have K_c factors that are larger than 1. The K_c factor is often 5-10 percent higher than the reference (where $K_c = 1.0$), and even 15-20 percent greater for some tall crops such as maize, sorghum or sugar cane (Figure 5.1). Typical values for the crop coefficient for full-grown crops ($K_{c\ mid}$) are listed in Tables 6.1 – 6.5 of Chapter 6.

In most species, the stomata open as irradiance increases. In addition to the stomatal response to the environment, the position and number of the stomata and the resistance of the cuticula to vapour transfer determine the water loss from the crop. Species with stomata on only the lower side of the leaf and/or large leaf resistances will have relatively smaller K_c values. This is the case for citrus and most deciduous fruit trees. Transpiration control and spacing of the trees, providing only 70 percent ground cover for mature trees, may cause the K_c of those trees, if cultivated without a ground cover crop, to be smaller than 1.0 (Figure 5.1).

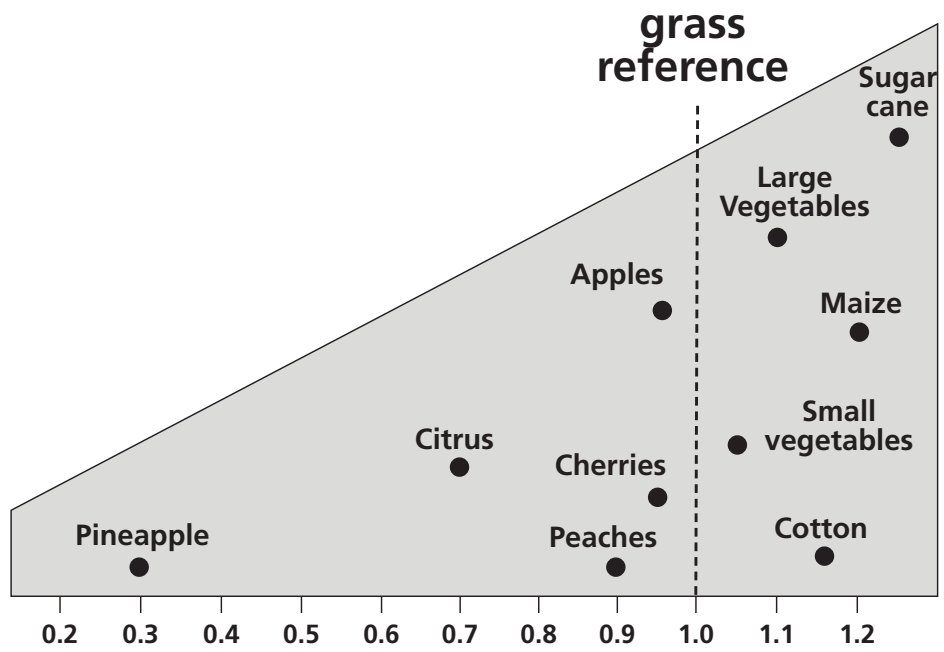
5.2.2 Climate

The K_c values of Tables 6.1 – 6.5 of Chapter 6 are typical values expected for average K_c under a standard climatic condition, which is defined as a sub-humid climate with average daytime minimum relative humidity (RH_{min}) ≈ 45% and having calm to moderate wind speeds averaging $u_2 = 2$ m s^{-1}.

Variations in wind speed alter the aerodynamic resistance of the crops and may alter their crop coefficients, especially for those crops that are substantially taller than the hypothetical grass reference. The effect of the difference in aerodynamic properties between the grass reference surface and agricultural crops is not only crop-specific but it also varies with the climatic conditions and crop height. Because aerodynamic properties are greater for many agricultural crops as compared to the grass reference, the ratio of ET_c to ET_o (i.e. K_c) for many crops increases as wind speed increases and as relative humidity decreases. More arid climates and conditions of greater wind speed will tend to have higher values for K_c. More humid climates and conditions of lower wind speed will have lower values for K_c.

The relative impact of the climate on K_c for full-grown crops is illustrated in Figure 5.2. The upper bounds represent extremely arid and windy conditions, while the lower bounds are valid under very humid and calm weather conditions. The ranges expected in K_c as climate and weather conditions change are quite small for short crops but are large for tall crops. Guidelines for the adjustment of K_c to the climatic conditions as a function of crop height are given in Chapter 6.

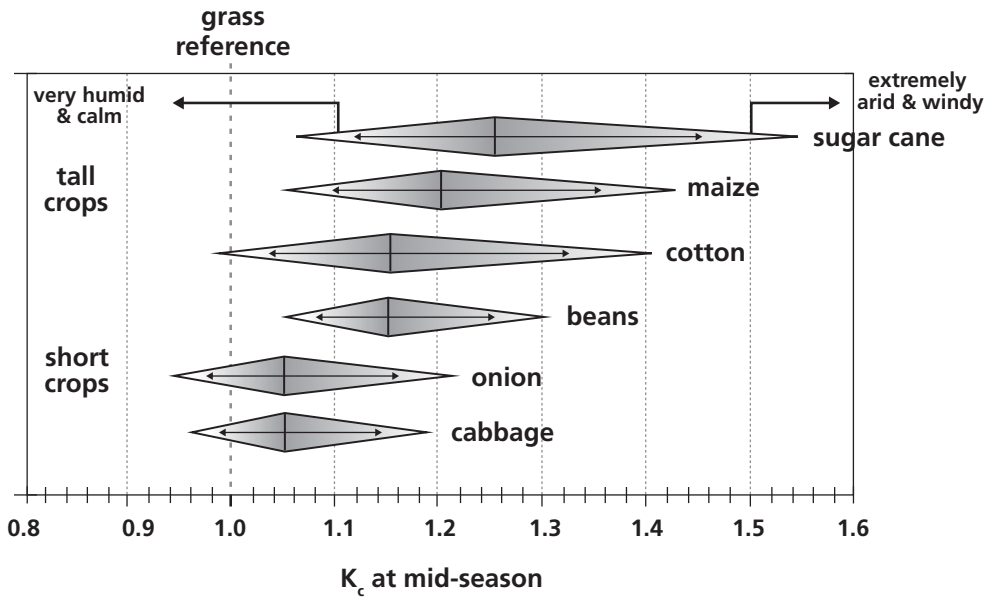
FIGURE 5.1
Indicative typical K_c for different types of full-grown crops
(K_c values are just indicative as from the first version of FAO 56, 1998)



Source: FAO, 1998

FIGURE 5.2
Indicative extreme ranges expected in K_c for full-grown crops as climate and weather change

(K_c values are not updated but are from the first version of FAO 56, 1998)



Source: FAO, 1998

Under humid and calm wind conditions, K_c becomes less dependent on the differences between the aerodynamic components of ET_c and ET_o and the K_c values for "full-cover" agricultural crops do not exceed 1.0 by more than about 0.05. This is because full-cover agricultural crops and the reference crop of clipped grass both provide for nearly maximum absorption of shortwave radiation, which is the primary energy source for evaporation under humid and calm conditions. Generally, the albedos, α, are similar over a wide range of full-cover agricultural crops, including the reference crop. Because the vapour pressure deficit $(e_s - e_a)$ is small under humid conditions, differences in ET caused by differences in aerodynamic resistance, r_a, between the agricultural crop and the reference crop are also small, especially with low to moderate wind speed.

Under arid conditions, the effect of differences in r_a between the agricultural crop and the grass reference crop on ET_c becomes more pronounced because the $(e_s - e_a)$ term may be relatively large. The larger magnitudes of $(e_s - e_a)$ amplify differences in the aerodynamic term in the numerator of the Penman-Monteith combination equation (Equation 2.1) for both the crop and the reference crop. Hence, K_c will be larger under arid conditions when the agricultural crop has a leaf area and roughness height that are greater than that of the grass reference.

Because the $1/r_a$ term in the numerator of the FAO-PM ET_o equation (Equation 2.4 and 2.5a, b) is multiplied by the vapour pressure deficit $(e_s - e_a)$, the ET from tall crops increases proportionately more relative to ET_o than does ET from short crops when relative humidity is low. The K_c for tall crops, such as those 2-3 m in height, can be as much as 30 percent higher in a windy, arid climate as compared with a calm, humid climate. The increase in K_c is due to the influence of the larger aerodynamic roughness of the tall crop relative to grass on the transport of water vapour from the surface.

5.2.3 Soil evaporation

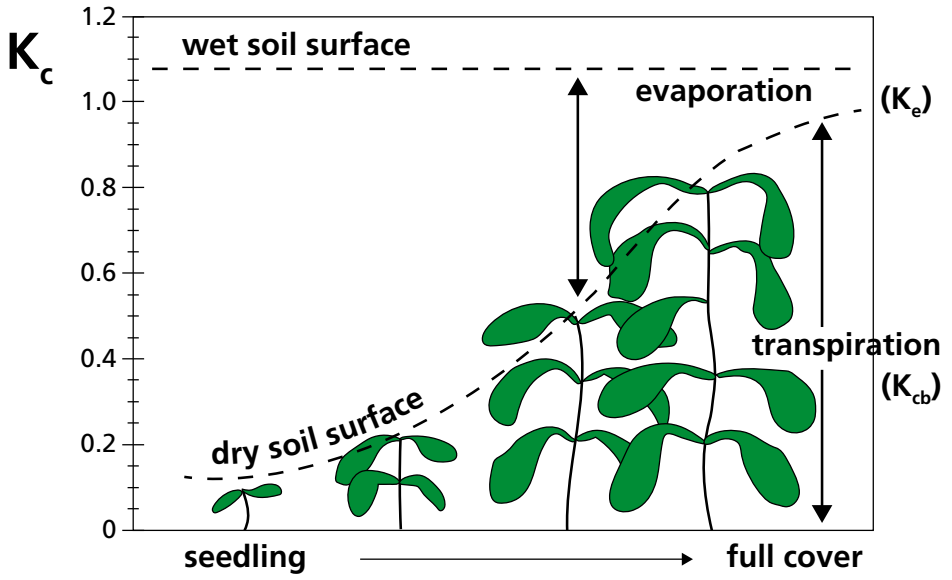
Differences in soil evaporation and crop transpiration between field crops and the reference surface are integrated within the crop coefficient. The K_c coefficient for full-cover crops primarily reflects differences in transpiration as the contribution of soil evaporation is relatively small. After rainfall or irrigation, the effect of evaporation is predominant when the crop is small and scarcely shades the ground. For such low-cover conditions, the K_c coefficient is determined largely by the frequency with which the soil surface is wetted. Where the soil is wet for most of the time from irrigation or rain, the evaporation from the soil surface will be considerable and K_c may exceed 1. On the other hand, where the soil surface is dry, evaporation is restricted, and K_c will be small and might even drop to less than 0.20 (Figure 5.3).

Differences in soil evaporation between the field crop and the reference surface can be forecast more precisely by using a dual crop coefficient.

5.2.4 Crop growth stages

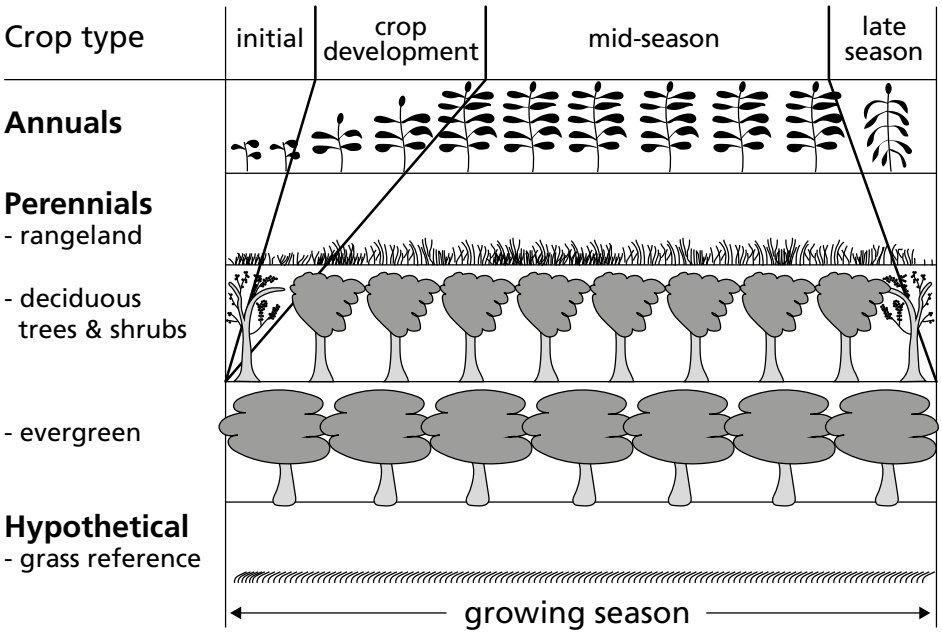
As the crop develops, the ground cover, crop height and leaf area change. Due to differences in evapotranspiration during the various growth stages, the K_c for a given crop will vary over the growing period. The growing period can be divided into four distinct growth stages: initial, crop development, mid-season, and late season. Figure 5.4 illustrates the general sequence and proportions of these stages for different types of crops.

FIGURE 5.3
The effect of soil evaporation on K_c. The horizontal dashed line represent K_c when the soil surface is kept continously wet. The curved dashed line corresponds to K_c when the soil surface is kept dry but the crop receives sufficient water to sustain full transpiration



Source: FAO, 1998

FIGURE 5.4
Crop growth stages for different types of crops



Source: FAO, 1998

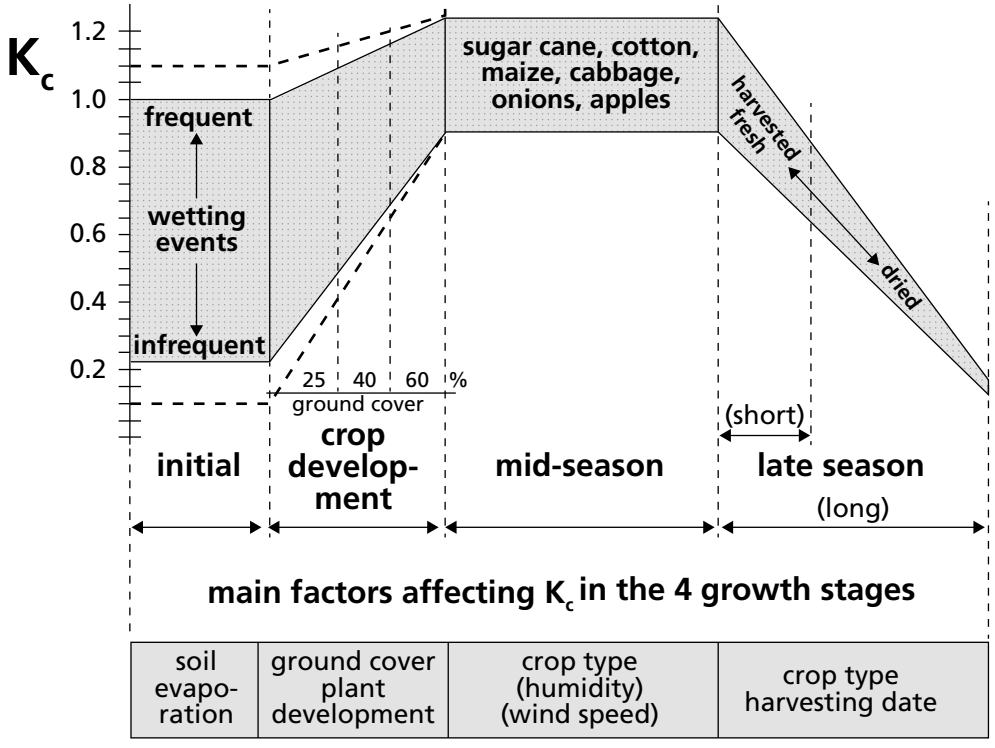
Initial stage

The initial stage runs from the planting date to approximately 10 percent of ground cover. The length of the initial period is highly dependent on the crop, the crop variety, the planting date, and the climate. The end of the initial period is determined as the time

when approximately 10 percent of the ground surface is covered by green vegetation. For perennial crops, the planting date is replaced by the "greenup" date, i.e. the time when the initiation of new leaves occurs, or in other cases by the "sprouting" date, i.e. the time when the dormancy phase of the sprout buds ends.

During the initial period, the leaf area is small, and evapotranspiration is predominantly in the form of soil evaporation. Therefore, the K_c during the initial period ($K_{c\ ini}$) is large when the soil is wet from irrigation and rainfall and is low when the soil surface is dry (Figure 5.5). The time for the soil surface to dry is determined by the time interval between wetting events, the evaporation power of the atmosphere (ET_o) and the depth of the wetting event. General estimates for $K_{c\ ini}$ as a function of the frequency of wetting and ET_o are given in Table 5.1. The data assume a medium textured soil. The procedure for estimating $K_{c\ ini}$ is presented in Chapter 6.

FIGURE 5.5
Typical ranges expected in the K_c for the four growth stages



Source: FAO, 1998

TABLE 5.1
Approximate values for $K_{c\ ini}$ for medium wetting events (10-40 mm) and medium textured soil

wetting interval	evaporating power of the atmosphere (ET_o)			
	low 1 - 3 mm day^{-1}	moderate 3 - 5 mm day^{-1}	high 5 - 7 mm day^{-1}	very high > 7 mm day^{-1}
less than weekly	1.2 - 0.8	1.1 - 0.6	1.0 - 0.4	0.9 - 0.3
weekly	0.8	0.6	0.4	0.3
longer than once per week	0.7 - 0.4	0.4 - 0.2*	0.3 - 0.2*	0.2* - 0.1*

Values derived from Figures 6.3 and 6.4.
(*) Note that irrigation intervals may be too large to sustain full transpiration for some young annual crops.

Crop development stage

The crop development stage runs from 10 percent ground cover to effective full cover. Effective full cover for many crops occurs at the initiation of flowering. For row crops where rows commonly interlock leaves such as beans, sugar beets, potatoes and maize, effective cover can be defined as the time when some leaves of plants in adjacent rows begin to intermingle so that soil shading becomes nearly complete, or when plants reach nearly full size if no intermingling occurs. For some crops, especially those taller than 0.5 m, the average fraction of the ground surface covered by vegetation (f_c) at the start of effective full cover is about 0.7-0.8. Fractions of sunlit and shaded soil and leaves do not change significantly with further growth of the crop beyond $f_c \approx 0.7$ to 0.8. It is understood that the crop or plant can continue to grow in both height and leaf area after the time of effective full cover. Because it is difficult to visually determine when densely sown vegetation such as winter and spring cereals and some grasses reach effective full cover, the more easily detectable stage of heading (flowering) is generally used for these types of crops.

For dense grasses, effective full cover may occur at about 0.10-0.15 m in height. For thin stands of grass (dry rangeland), grass height may approach 0.3-0.5 m before effective full cover is reached. Densely planted forages such as alfalfa and clover reach effective full cover at about 0.3-0.4 m. Another way to estimate the occurrence of effective full cover is when the leaf area index (LAI) reaches three in case of very dense low crops. LAI is defined as the average total area of leaves (one side) per unit area of ground surface.

In the case of evergreen woody perennials, it is difficult to distinguish when crop development started and when it ends. Leaves are there and shade the ground through the full crop season. Differences in K_c mostly result from the frequency of wettings by rain and irrigation covering the full surface. Partial cover of irrigation produces small impacts. In the case of deciduous woody perennials, full development of leaves is attained quickly and rain and irrigation impacts are minor.

As the crop develops and shades more and more of the ground, evaporation becomes more restricted, and transpiration gradually becomes the major process. During the crop development stage, the K_c value corresponds to the amounts of ground cover and plant development. Typically, if the soil surface is dry, $K_c = 0.5$ corresponds to about 25-40 percent of the ground surface covered by vegetation due to the effects of shading and due to microscale transport of sensible heat from the soil into the vegetation. A $K_c = 0.7$ often corresponds to about 40-60 percent ground cover. These values will vary, depending on the crop, frequency of wetting and whether the crop uses more water than the reference crop at full ground cover (e.g. depending on its canopy architecture and crop height relative to clipped grass).

Mid-season stage

The mid-season stage runs from effective full cover to the start of maturity. The start of maturity is often indicated by the beginning of the aging, yellowing or senescence of leaves, initial leaf drop, or the browning of fruit to the degree that the crop evapotranspiration is reduced relative to the reference ET_o. The mid-season stage is the longest stage for perennials and many annuals, but it may be relatively short for vegetable crops that are harvested fresh for their green vegetation.

At the mid-season stage the K_c reaches its maximum value. The value for K_c ($K_{c\ mid}$) is relatively constant for most growing and cultural conditions, namely for both evergreen and deciduous woody perennials. Deviation of the $K_{c\ mid}$ of annuals from the reference value "1.0" is primarily due to differences in crop height and resistance between the grass reference surface and the agricultural crop, and weather conditions.

Late season stage

The late season stage runs from the start of maturity to harvest or full senescence. The calculation for K_c and ET_c is presumed to end when the crop is harvested, dries out naturally, reaches full senescence, or experiences leaf drop.

For some perennial vegetation in frost-free climates, crops may grow year-round so that the date of termination may be taken as the same as the date of planting or green-up. This is the case of evergreen woody perennials, which K_c value may not change; however, for deciduous woody perennials K_c decreases until leaf dropping or is kept relatively high if autumn irrigation is required to ensure physiological processes that determine next year yield.

The K_c value at the end of the late season stage ($K_{c\ end}$) reflects crop and water management practices (Figure 5.5). The $K_{c\ end}$ value is high if the crop is frequently irrigated until harvested fresh. If the crop is allowed to senesce and to dry out in the field before harvest, the $K_{c\ end}$ value will be small. Senescence is usually associated with less efficient stomatal conductance of leaf surfaces due to the effects of aging, thereby causing a reduction in K_c.

5.3 CROP EVAPOTRANSPIRATION (ET_c)

Crop evapotranspiration is calculated by multiplying ET_o by K_c, a coefficient expressing the difference in evapotranspiration between the cropped and the reference grass surface. The difference can be combined into one single coefficient, or it can be split into two factors describing separately the differences in evaporation and transpiration between both surfaces. The selection of the approach depends on the purpose of the calculation, the accuracy required, the climatic data available and the time step with which the calculations are executed. Table 5.2 presents the general selection criteria.

TABLE 5.2
General selection criteria for the single and dual crop coefficient approaches

	Single crop coefficient K_c	Dual crop coefficient $K_{cb} + K_e$
Purpose of calculation	• irrigation planning and design • irrigation management • basic irrigation schedules • real-time irrigation scheduling for non-frequent water applications (surface and sprinkler irrigation)	• research • real time irrigation scheduling • irrigation scheduling for high-frequency water application (micro-irrigation and automated sprinkler irrigation) • supplemental irrigation • detailed soil and hydrologic water balance studies
Time step	daily, 10-day, monthly (data and calculation)	daily (data and calculation)
Solution method	graphical pocket calculator computer	computer

5.3.1 Single and dual crop coefficient approaches

Single crop coefficient approach (K_c)

In the single crop coefficient approach, the effect of crop transpiration and soil evaporation are combined into a single coefficient. The coefficient K_c integrates differences in the soil evaporation and crop transpiration rate between the crop and the grass reference surface. *As soil evaporation may fluctuate daily as a result of rainfall or irrigation, the single crop coefficient expresses only the time-averaged (multi-day) effects of crop evapotranspiration.*

As the single K_c coefficient averages soil evaporation and transpiration, the approach is used to compute ET_c for weekly or longer time periods, although calculations may proceed on a daily time step. The single, time-averaged K_c is used for planning studies and irrigation system design where the time-averaging effects of soil wetting are acceptable and relevant. This is the case for surface irrigation and set sprinkler systems where the time interval between successive irrigation is of several days, often ten days or more. For typical irrigation management, the time-averaged single K_c is valid.

Dual crop coefficient approach (K_{cb} + K_e)

In the dual crop coefficient approach, the effects of crop transpiration and soil evaporation are determined separately. Two coefficients are used: the basal crop coefficient (K_{cb}) to describe plant transpiration, and the soil water evaporation coefficient (K_e) to describe evaporation from the soil surface. The single K_c coefficient is replaced by:

$$K_c = K_{cb} + K_e \tag{5.2}$$

where K_{cb} basal crop coefficient,
K_e soil water evaporation coefficient.

The basal crop coefficient, K_{cb}, is defined as the ratio of ET_c to ET_o when the soil surface layer is dry but where the average soil water content of the root zone is adequate to sustain full plant transpiration. The K_{cb} represents the baseline standard K_c in the absence of the additional effects of soil wetting by irrigation or precipitation.

The basal crop coefficient, K_{cb}, can also be defined as the ratio of transpiration, T_c to ET_o. Therefore, it is also called the transpiration coefficient. The soil evaporation coefficient, K_e, describes the evaporation component from the soil surface. If the soil is wet, following rain or irrigation, K_e may be large. However, the sum of K_{cb} and K_e can never exceed a maximum value, $K_{c\ max}$, determined by the energy available for evapotranspiration at the soil surface. As the soil surface becomes drier, K_e becomes smaller and falls to zero when no water is left for evaporation. The estimation of K_e requires a daily water balance computation for the calculation of the soil water content remaining in the upper topsoil.

The dual coefficient approach requires more numerical calculations than the procedure using the single time-averaged K_c coefficient. The dual procedure is best for real-time irrigation scheduling, for soil water balance computations, and for research studies where the effects of day-to-day variations in soil surface wetness and the resulting impacts on daily ET_c, the soil water profile, and deep percolation fluxes are important. This is the case for high-frequency irrigation with micro-irrigation systems, namely drippers, or lateral move systems such as centre pivots and linear move systems.

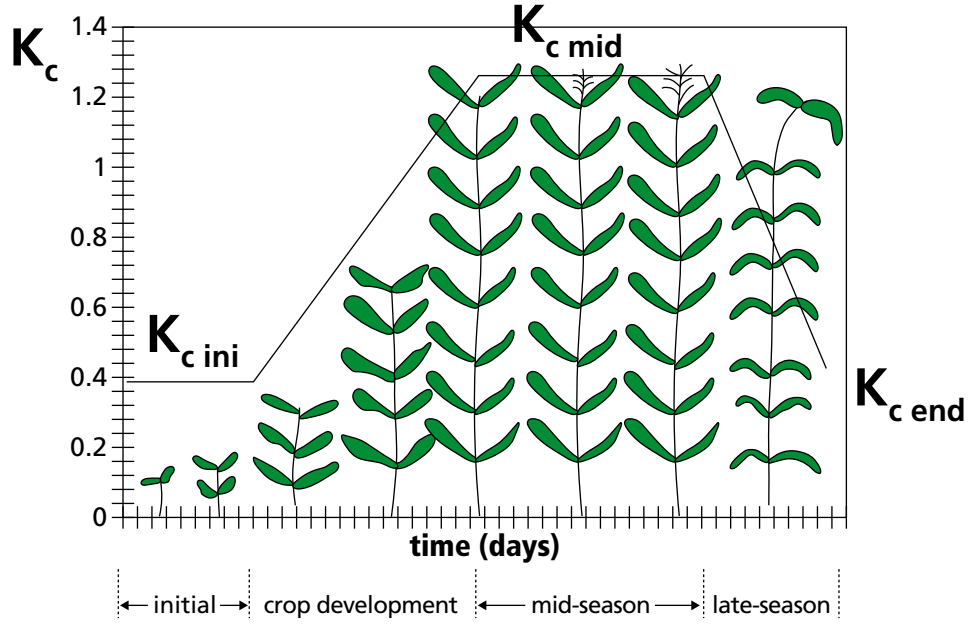
5.3.2 Crop coefficient curve

After the selection of the calculation approach, the determination of the lengths for the crop growth stages and the corresponding crop coefficients, a crop coefficient curve can be constructed. *The curve represents the changes in the crop coefficient over the length of the growing season. The shape of the curve represents the changes in the vegetation and ground cover during plant development and maturation that affect the ratio of ET_c to ET_o.* From the curve, the K_c coefficient and, hence, ET_c can be derived for any period within the growing season.

Single crop coefficient

The generalized crop coefficient curve is shown in Figure 5.6. Shortly after the planting of annuals or shortly after the initiation of new leaves for perennials, the value for K_c is small, often less than 0.4. The K_c begins to increase from the initial K_c value, $K_{c\ ini}$, at the beginning of rapid plant development and reaches a maximum value, $K_{c\ mid}$, at the time of maximum or near-maximum plant development. During the late season period, as leaves begin to age and senesce due to natural or cultural practices, the K_c begins to decrease until it reaches a lower value at the end of the growing period equal to $K_{c\ end}$.

FIGURE 5.6
Generalized crop coefficient curve for the single crop coefficient approach



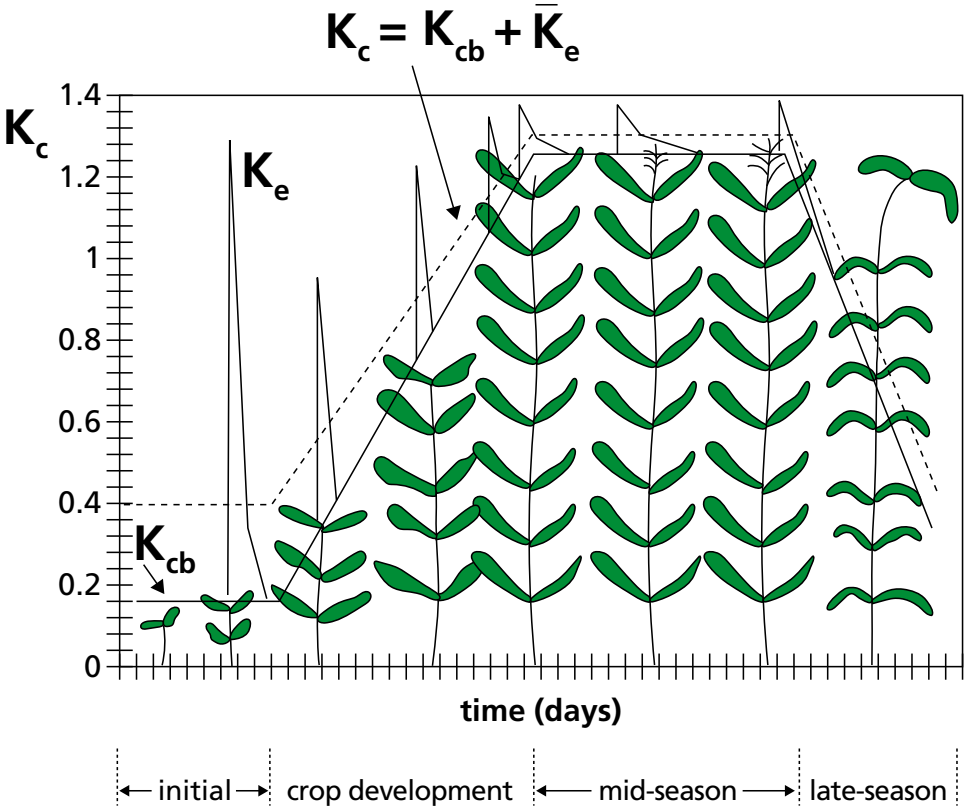
Source: FAO, 1998

Dual crop coefficient

The single "time-averaged" K_c curve in Figure 5.6 incorporates averaged wetting effects into the K_c factor. The value for $K_{c\ mid}$ is relatively constant for most growing and cultural conditions. However, the values for $K_{c\ ini}$ and $K_{c\ end}$ can vary considerably on a daily basis, depending on the frequency of wetting by irrigation and rainfall. The dual crop coefficient approach calculates the actual increases in K_c for each day as a function of plant development and the wetness of the soil surface. *As the single K_c coefficient includes averaged effects of evaporation from the soil, the basal crop coefficient, K_{cb}, describing only plant transpiration, lies below the K_c value* (Figure 5.7).

The largest difference between K_c and K_{cb} is found in the initial growth stage where evapotranspiration is predominantly in the form of soil evaporation and crop transpiration is still small. Because crop canopies are near or at full ground cover during the mid-season stage, soil evaporation beneath the canopy has less effect on crop evapotranspiration and the value for K_{cb} in the mid-season stage will be nearly the same as K_c. *Depending on the ground cover, the basal crop coefficient during the mid-season may be only 0.05-0.10 lower than the K_c value. Depending on the frequency with which the crop is irrigated during the late season stage, K_{cb} will be similar to (if infrequently irrigated) or less than the K_c value.*

FIGURE 5.7
Crop coefficient curves showing the basal K_{cb} (continuous line), soil evaporation K_e (line with several peaks) and the corresponding single $K_c = K_{cb} + K_e$ curve (dashed line)



Source: FAO, 1998

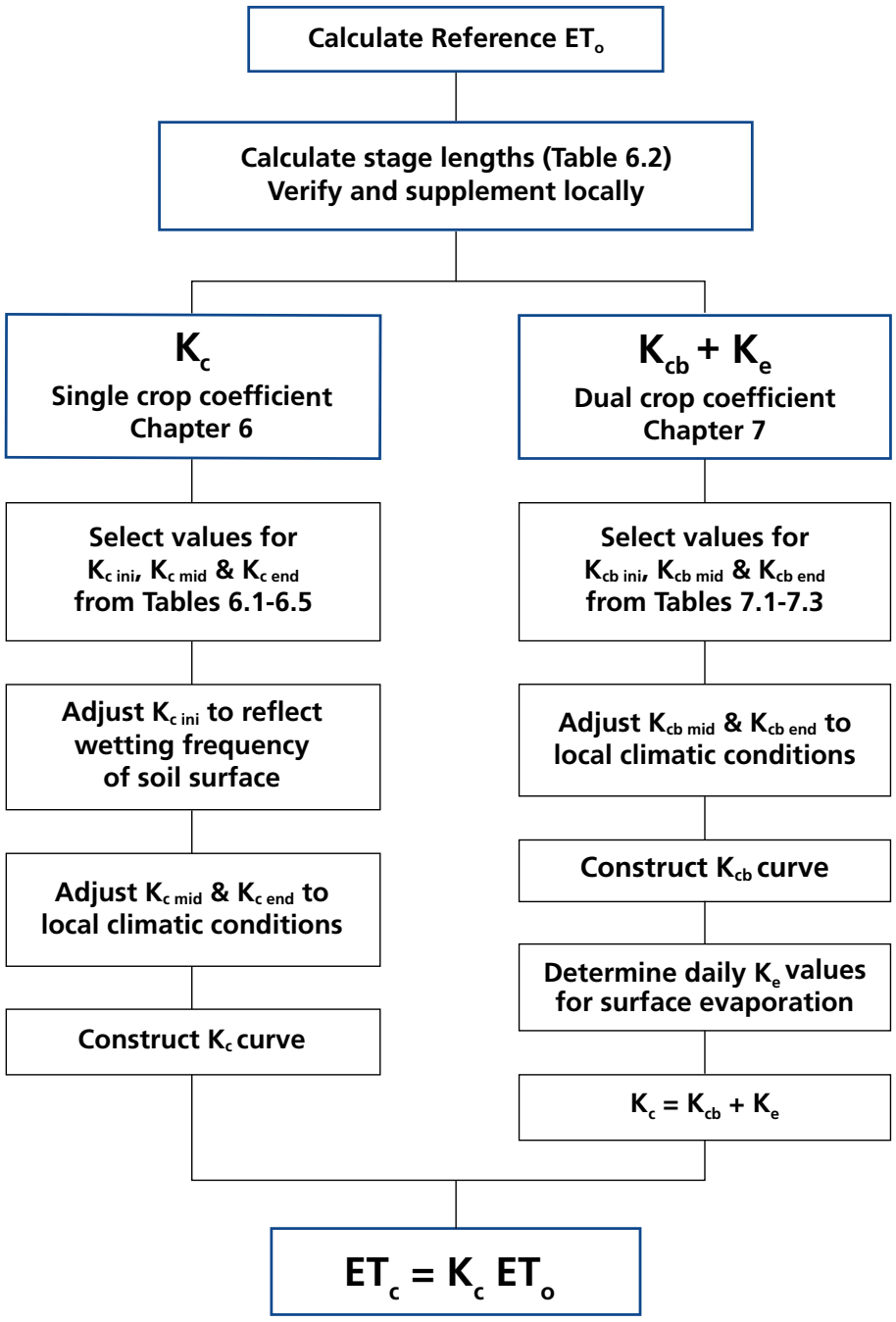
Figure 5.7 presents typical shapes for the K_{cb}, K_e and single K_c curves. The K_{cb} curve in the figure represents the minimum K_c for conditions of adequate soil water and dry soil surface. The K_e "spikes" in the figure represent increased soil evaporation when precipitation or irrigation has wetted the soil surface and has temporarily increased total ET_c. These wet soil evaporation spikes decrease as the soil surface layer dries. The spikes generally reach a maximum value of 1.0-1.2, depending on the climate, the magnitude of the wetting event and the portion of the soil surface wetted.

Summed together, the values for K_{cb} and for K_e represent the single crop coefficient, K_c. The total K_c curve, shown as the dashed line in Figure 5.7, illustrates the effect of averaging $K_{cb} + K_e$ over time and is displayed as a "smoothed" curve. It is this smoothed curve that is represented by the single K_c calculation procedure. The K_c curve lies above the K_{cb} curve, with potentially large differences during the initial and development stages, depending on the frequency of soil wetting.

5.4 FLOW CHART OF THE CALCULATIONS

The calculation procedures required for the crop coefficient approaches are developed in the following chapters. In Chapter 6, a single time-averaged crop coefficient is used to calculate ET_c. Guidelines for estimating lengths for the four growth stages are given in Chapter 6 using cumulative growing degree days (GDD). The dual approach using two coefficients to describe the effects of crop and soil separately is outlined in Chapter 7. Figure 5.8 presents the general calculation procedures.

FIGURE 5.8
General procedure for calculating ET_c



Source: FAO, 1998

5.5 DETERMINATION OF CROP EVAPOTRANSPIRATION AND CROP COEFFICIENTS

The main methods for measuring actual and standard ET_c are described in the last section of Chapter 1, namely the soil water balance in lysimeters or the field, and the surface energy balance performed with ground and remote sensing approaches. That section is intended to draw the readers' attention to the need for accuracy in the determination of crop ET, thus discussing some critical aspects that affect measurement accuracy and the main pitfalls to be avoided. Over the last 25 years, hundreds of scientific papers have been published reporting crop ET measurements or estimates, and new (updated) K_c and K_{cb} values for nearly all vegetable and field crops, as well as for fruit trees and vines. As a follow-up, the current section presents several examples of procedures for determining crop coefficients using ground and remote sensing techniques.

5.5.1 K_c and K_{cb} curves from ground measurements

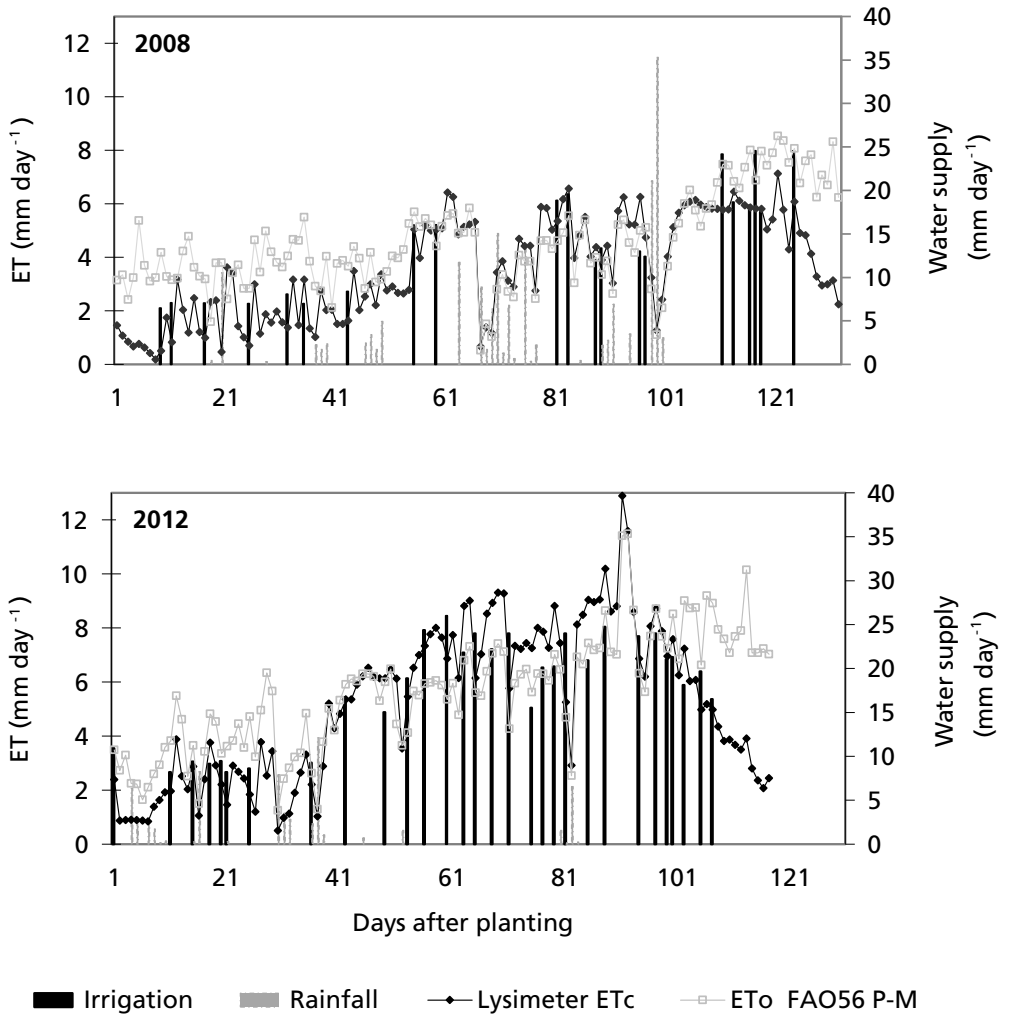
Figure 5.9 shows daily canola (*Brassica napus* L. cv. "Kabel") ET_c data measured by a large weighing lysimeter, 2.7 × 2.3 × 1.7 m in size, near Albacete (SE Spain) under pristine, well-watered conditions during two growing seasons. All agronomic practices - fertilization, weeds, pest and disease control - were performed aiming at optimizing crop growth conditions. ET_c values increased quickly as the crop development period progressed due to the fast increase of the canopy cover facilitated by favourable spring temperatures. Peak ET_c was reached during the mid-season period coinciding with a maximum fraction of ground covered by vegetation.

The peak ET periods had measured ET_c approaching or even exceeding the values for ET_o. ET_c declined during the late season stage as green canopy cover decreased. Seasonal lysimeter-based canola ET_c was 472 and 602 mm in 2008 and 2012, respectively. The 28 percent higher ET_c value in 2012 was due to higher evaporative demand during the growing season, mainly during the period of maximum canola water requirements. The midseason period was also 5 days longer in 2012. The total ET_o was 74 percent higher in 2012 than in 2008, because the midseason period occurred about 15 days later in 2012 than in 2008, when ET_o was higher, coinciding with the high temperatures of late spring and early summer. The higher irrigation requirements by 2012 were due to lower precipitation in that year relative to 2008. The larger number of irrigation events in 2012 increased the total ET_c through increased evaporation from the soil.

K_c values, shown in Figure 5.10, were calculated as the ratio of the lysimeter-measured ET_c to ET_o, that was computed with the FAO-PM ET_o equation. The K_c values were adjusted to the FAO56 segmented crop coefficient curve relative to the four canola growth stages. K_c values during crop development and late season stages were linearly interpolated between $K_{c\ ini}$ and $K_{c\ mid}$. Lengths of the growth stages were determined so that the five-day K_c averages followed the linear segments as closely as possible, with similar numbers of values above and below the curve segments. The maximum average K_c values of 1.15 in 2008 and 1.24 in 2012 defined the $K_{c\ mid}$ for the mid-season stage. After adjusting to the standardized subhumid climate by solving Equation 6.16 for $K_{c\ (tab)}$, the estimated $K_{c\ mid}$ values were 1.13 and 1.17. The $K_{c\ mid}$ value was higher in 2012 than in 2008, primarily due to the drier weather conditions in 2012 (lower minimum relative humidity).

FIGURE 5.9

Daily values for ET_o calculated using the FAO-PM ET_o equation and daily canola ET_c values measured by a large weighing lysimeter near Albacete (SE Spain) during two canola seasons. Daily irrigation (black vertical bars) and rainfall (grey vertical bars) are also represented



Source: Authors' own elaboration

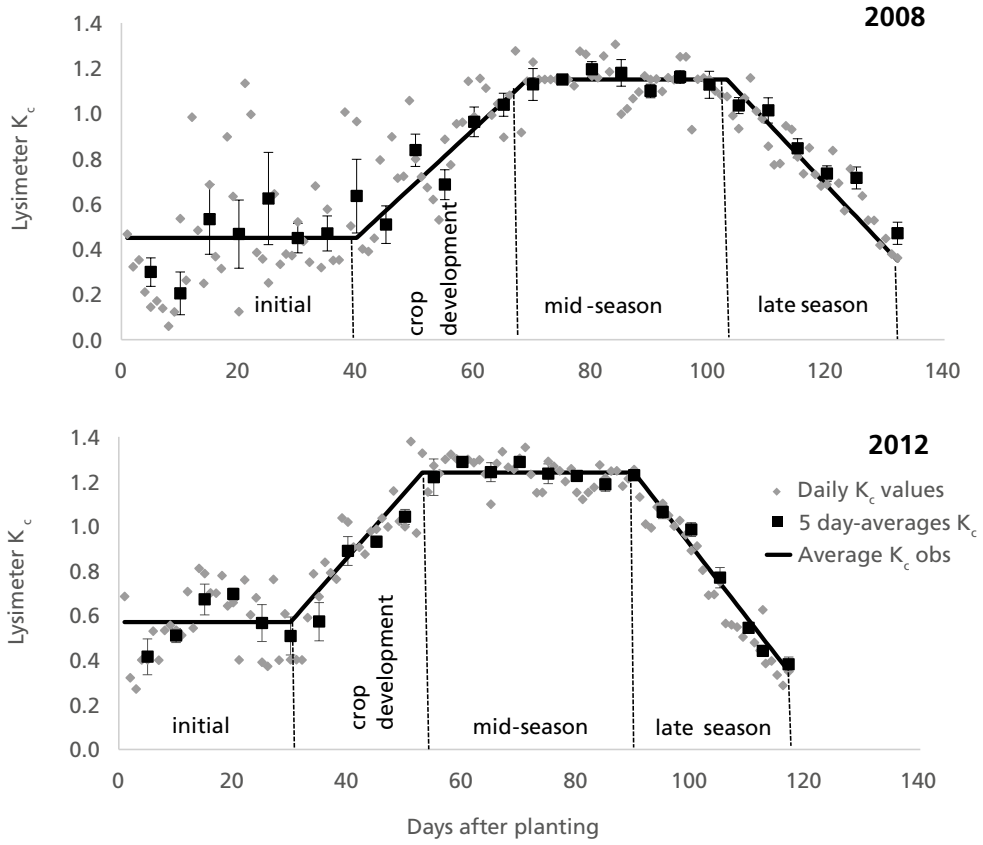
The importance of the soil evaporation component was evaluated by applying the dual crop coefficient procedure to separate crop transpiration, estimated with the basal crop coefficient, K_{cb}, and soil evaporation, estimated with the soil evaporation coefficient, K_e. Figure 5.11 shows K_e values, which were greater during the initial and beginning of the crop development stages when the soil surface was wet and a high energy flux reached the surface, and water was thereby available for evaporation. As the crop canopy expanded during the development period, K_{cb} values increased while the K_e values decreased. The K_e values increased during the late season period, due to the decreasing values of the fraction of surface covered by vegetation (f_c) during that period caused by crop senescence.

As with the single K_c, K_{cb} values during development and late-season crop growth periods were linearly interpolated with time, and period lengths were adjusted so that the line segments fell between higher and lower K_{cb} values. Figure 5.11 also shows measured K_{cb} values relative to the four-segment crop coefficient curve (average $K_{cb\ (obs)}$). Values for $K_{cb\ mid}$ were estimated to be 1.11 and 1.20 for 2008 and 2012. After adjusting to the

standardized subhumid climate by solving Equation 6.16 for $K_{cb\ (tab)}$, the estimated $K_{cb\ mid}$ values were 1.09 and 1.13. The latter values differed by 3.6 percent for the two years studied. That difference is well within the expected error of lysimeter measurements and expected uncertainty in the ET_o calculations used to determine the $K_{cb\ mid}$ values.

FIGURE 5.10

Single crop coefficient for irrigated canola during 2008 and 2012 growing seasons. Diamonds represent daily K_c values, and squares indicate 5-day K_c averages calculated. Bars indicate the standard deviation for the 5-day averages. The standard FAO K_c segments curve that comprises four general crop growth stages is superposed (average $K_{c\ obs}$)



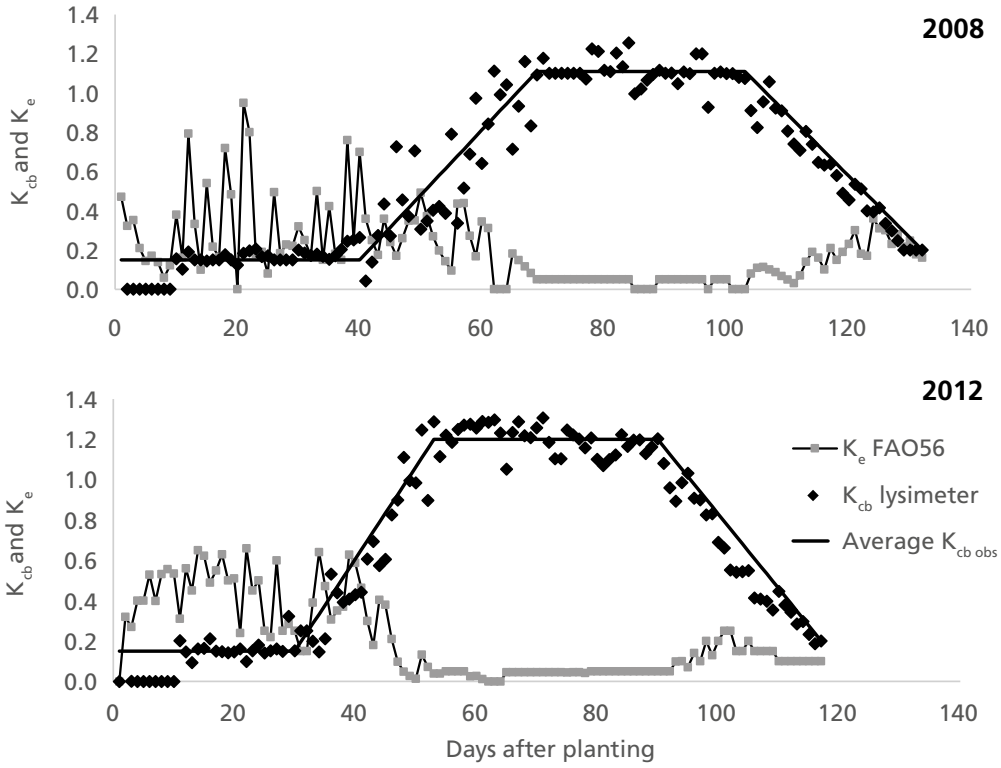
Source: Authors' own elaboration

The consistency of K_{cb} values during the midseason period, especially during 2008, indicates dependable and consistent accuracy by the weighing lysimeter combined with consistency in the performance of the FAO-PM equation. The FAO-PM equation produced daily ET_o that was used as the divisor to convert measured ET_c into the K_c curves. Some day-to-day variations in K_{cb} and K_c are expected due to combined random errors in lysimeter measurements, K_e estimations, weather measurements and ET_o estimates.

Table 5.3 presents a summary of the crop coefficient values K_c and K_{cb} as developed from the lysimeter measurements for mid and end-season periods (´obs´ subscripts), and respective values adjusted to the standard subhumid climate (RH_{min} = 45% and u_2 = 2 m s^{-1}, ´std´subscripts). Values for $K_{c\ end\ (obs)}$ and $K_{cb\ end\ (obs)}$ did not require adjustment to standard climate conditions as they remained below 0.45 for both years. Table 5.3 includes observed ancillary crop parameters for irrigated canola, namely, maximum root depth ($Z_{r\ max}$), maximum crop height (h_{max}) and maximum fraction of ground cover ($f_{c\ max}$).

FIGURE 5.11

Dual crop coefficient data and curves for irrigated canola during 2008 and 2012 growing seasons. The evaporation coefficient, K_e, was calculated using the FAO 56 daily K_e procedure and the transpiration component, K_{cb}, was estimated iteratively from lysimeter K_c minus calculated K_e. The general FAO 56 four growth stage K_{cb} segments are superposed



Source: Authors' own elaboration

TABLE 5.3

Dual crop coefficients for irrigated canola during 2008 and 2012 growing seasons. The evaporation coefficient, K_e, was calculated using the FAO56 procedure and the transpiration component, K_{cb}, was estimated from lysimeter K_c minus calculated K_e

Season	$K_{c\ mid\ (obs)}$ ($K_{c\ mid\ (std)}$)	$K_{c\ end\ (obs)}$	$K_{cb\ mid\ (obs)}$ ($K_{cb\ mid\ (std)}$)	$K_{cb\ end\ (obs)}$	*$Z_{r\ max}$ (m)	h_{max} (m)	$f_{c\ max}$
2008	1.15 (1.13)	0.35	1.11 (1.09)	0.20	0.40	1.0	0.95
2012	1.24 (1.17)	0.35	1.20 (1.13)	0.20	0.40	1.0	0.97

*limited by the soil depth

Note: Both different standard values for both years were obtained from adjustments to climate

5.5.2 Determination of K_c using spectral reflectance from remote sensing

Actual K_{cb} and K_c values can be estimated using remote sensing based on field, aerial or satellite platforms, using empirical approaches based on spectral reflectance. Remote observations of K_{cb} and K_c are valuable when field information, such as planting dates or rates of vegetation development, are not well known. The estimation of K_{cb} from reflectance-based vegetation indices (VI) is possible because of the generally close correspondence between vegetation amount and transpiration, where, as vegetation cover increases, leaf area increases and transpiration and K_{cb} increase. The crop coefficients obtained from reflectance represent the actual ET conditions of the field (ideal or nonideal) and are termed $K_{cb\ act\ RS}$ and $K_{c\ act\ RS}$.

Challenges with VI-based estimation of the single K_c, that includes the evaporation component, is that spectral-reflectance-based observations generally cannot detect the presence of wet soil. Therefore, independent methods that use knowledge of specific wetting events are needed for estimating the evaporation from bare soil following precipitation and irrigation events. Wetting information is often difficult to obtain for specific fields, so that an empirically-based adjustment or coefficient is required to convert remotely-based K_{cb} into an equivalent K_c.

Remotely-sensed $K_{cb\ act\ RS}$ and $K_{c\ act\ RS}$ may also have difficulty in capturing the reduced ET associated with water stress caused by short-term soil-water shortage because the short-term water stress is generally not indicated by a significant change in VI. Under water shortage, a thermally-based method described in the next section may be required.

Vegetation indices are mathematical combinations of surface reflectances obtained for different spectral bands. VIs are used to determine canopy biophysical variables such as the fraction of surface covered by vegetation (f_c), crop height, LAI, and crop coefficients. A common VI is the Normalized Difference Vegetation Index (NDVI) which uses surface reflectance values obtained from the Near Infra-Red (NIR) band (~0.7–1.3 µm) and the Red band (~0.6–0.7 µm), thus where:

$$NDVI = \frac{r_{NIR} - r_{red}}{r_{NIR} + r_{red}} \tag{5.3}$$

where r_{NIR} surface reflectance in the near-infrared band (NIR) [ ],

r_{red} surface reflectance in the red band [ ].

It is important that NDVI is based on reflectance data rather than based on "radiance" data or "digital number" forms of data to maintain consistency in NDVI estimates over time.

Another index widely applied is the Soil Adjusted Vegetation Index (SAVI) based on a similar definition as for NDVI but including a parameter L = 0.5 to reduce the influence of the variation in soil colour and brightness.

A common usage is to estimate actual K_{cb} directly from NDVI:

$$K_{cb} = a_{cb} NDVI + b_{cb} \tag{5.4}$$

where a_{cb} and b_{cb} are generalized regression coefficients for estimating K_{cb}, which may be specific to a crop type or a general crop group.

The relationship in Equation 5.4 estimates the basal K_{cb} only, because K_{cb} represents primarily the transpiration component, which is most closely related to vegetation amount and therefore to the VI. However, the single actual K_c can be approximated using a similar relationship:

$$K_c = a_c NDVI + b_c \tag{5.5}$$

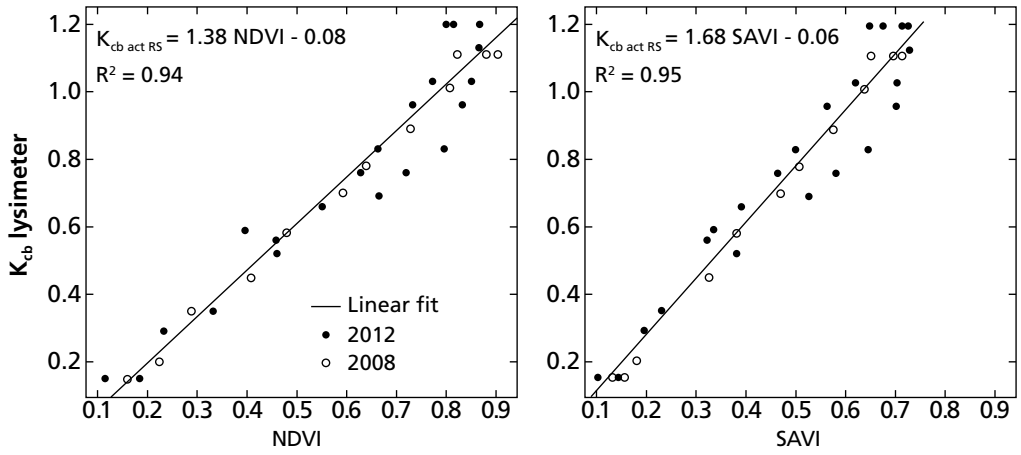
where a_c and b_c are generalized coefficients for estimating $K_{c\ act}$, that may be specific to a crop or to a crop group, and also to expected amounts of evaporation from wet soil. The accuracy of the $ET_{c\ act}$ estimate is reduced when using Equation 5.5 for agricultural fields that have higher or lower wetting frequency than that of the original conditions when a_c and b_c coefficients were derived.

Sometimes, a more refined method is used where first estimates from the VI refer to the fraction of ground cover (f_c), and f_c is then used to determine K_{cb} or K_c using additional information such as crop height. Detailed guidelines and discussion for this approach are provided in Chapter 8.

Figure 5.12 shows the linear relationships between NDVI and SAVI for the same canola K_{cb} values shown earlier, which were obtained from the lysimeter during a 2-year study carried out near Albacete (SE Spain).

In the case of the estimation of actual K_{cb} based on NDVI and SAVI obtained using a ground-based radiometer, the SAVI-based K_{cb} function resulted in slightly better accuracy according to the coefficient of regression and the interception coefficient of the linear regression as compared to the NDVI. The K_{cb} vs. NDVI and K_{cb} vs. SAVI relationships show good linearity and consistency between the two years, indicating a likelihood for good accuracy when using the developed relationships for other years and locations.

FIGURE 5.12
Linear fit of lysimeter-based basal crop coefficient (K_{cb}) vs. vegetation indices NDVI (left) and SAVI (right) for a canola crop observed during two years near Albacete, Spain



Source: Authors' own elaboration

5.5.3 Determination of K_c using energy balance from remote sensing

The remote sensing surface energy balance (RS-SEB) method was introduced and described in Chapter 1. Over the last 25 years, various remote sensing approaches based on surface energy balance have been developed and widely used. RS-SEB methods take advantage of the physical fact that conversion of liquid water to vapour requires very substantial amounts of energy. This conversion of liquid to vapour during the evaporation process involves the same physical requirements whether the process is the transpiration of water from inside a plant leaf or the evaporation of water from a wet soil surface. The energy consumption by the conversion of liquid to vapour cools the evaporating surface, thus variations in surface temperature can be used to estimate the net radiation and sensible heat exchange at the surface. These estimates can then be used with the energy balance equation (Equation 1.1) to estimate the crop evapotranspiration.

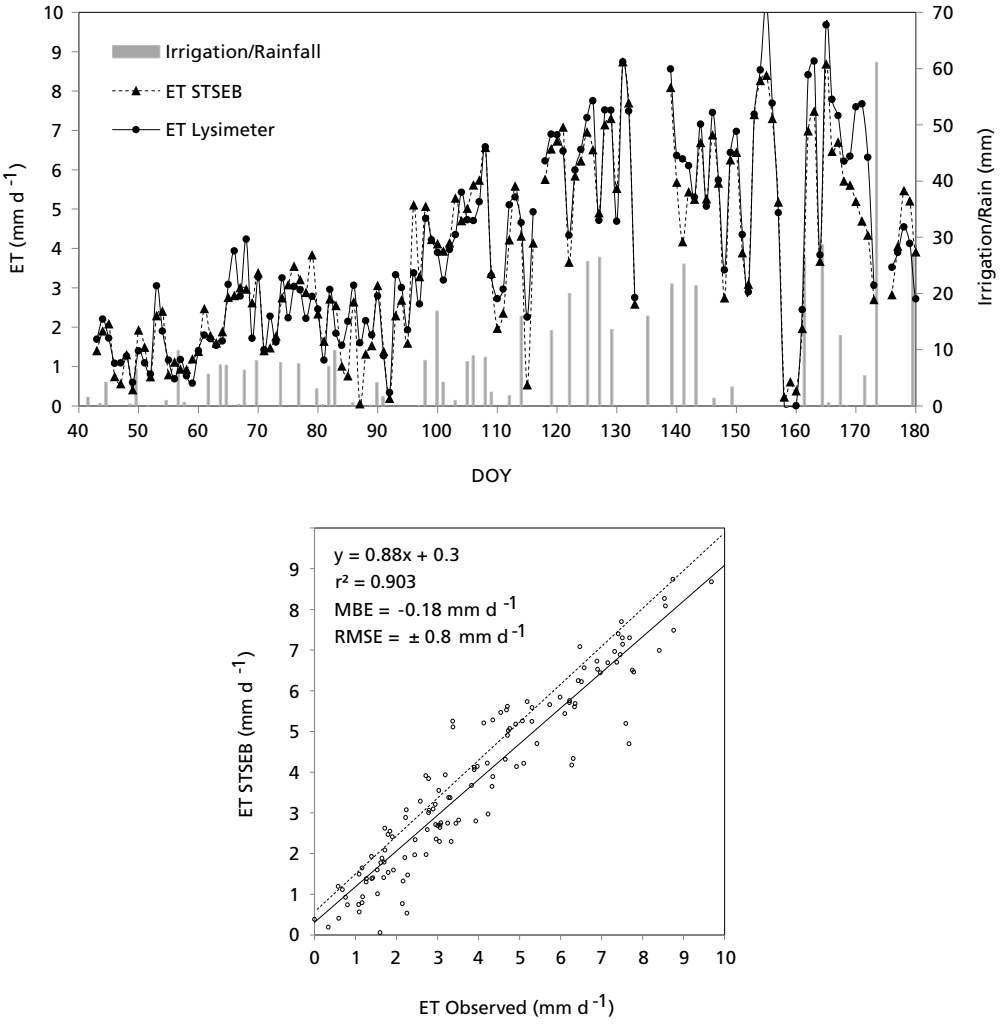
As an example of a local application of RS-SEB, Figure 5.13 shows data from a field experiment conducted near Albacete (SE Spain), where lysimeter observations and ET_c estimates using a Simplified Two-Source Energy Balance (STSEB) model are compared for a sprinkler-irrigated wheat field cropped under near-pristine conditions.

A quantitative analysis comparing modelled (STSEB) and observed (lysimeter) daily ET_c shows an average underestimation of ET_c by the RS-SEB model of 0.18 mm day^{-1}, with a determination coefficient (r^2) of 0.903. The combination of the two-source energy balance with local radiometric temperatures has shown to effectively estimate actual

ET_c values for a variety of crops and environmental conditions, which can be useful in developing actual $K_{c\ act}$ values, especially for conditions of water and salt stresses. The main limitation to its operational application is the retrieval of land surface temperature from satellite data with sufficient spatial and temporal resolution. The spatial resolution of thermal data can be especially limiting in agricultural areas composed of cropped fields smaller than 1 ha.

FIGURE 5.13

Evolution of modelled and observed daily ET_c values over a wheat growth season near Albacete (SE Spain). Irrigation and rainfall amounts are plotted in the upper figure. The daily ET_c estimated by the SEB model STSEB are compared with lysimeter-based ET_c in both figures



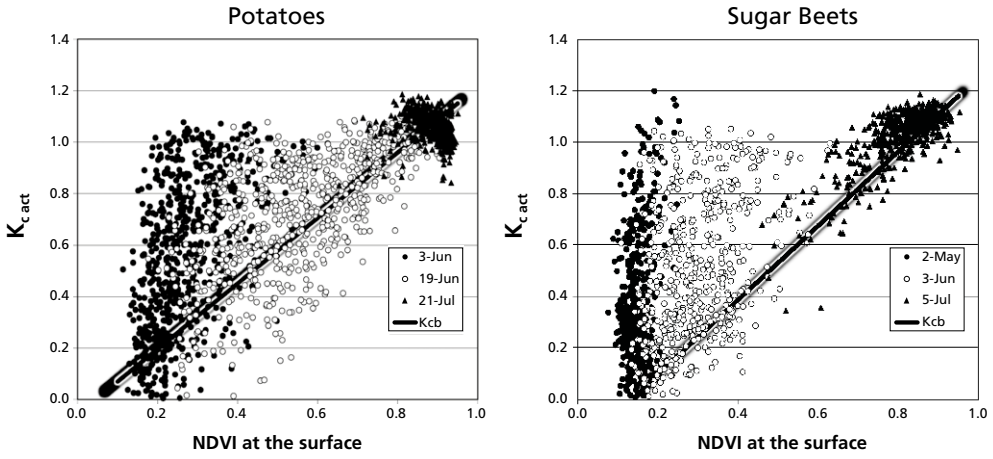
Source: Authors' own elaboration

Figure 5.14 shows an example of $K_{c\ act}$ determined with the METRIC model using Landsat imagery for hundreds of potato and sugar beet fields in Idaho, USA. The $K_{c\ act}$ is plotted against NDVI determined from atmospherically corrected surface reflectance. The figure demonstrates two important points. A large number of $K_{c\ act}$ values lie well above the drawn K_{cb} line when NDVI is low, indicating that those fields were wet, with evaporation from exposed soil due to irrigation at the time of the satellite overpass. This was especially the case early in the growing season. Differently, later in

the growing season, when nearly full ground cover is attained as indicated by NDVI greater than about 0.8, nearly all $ET_{c\ act}$ and $K_{c\ act}$ values represent transpiration from the full leaf cover, with little evaporation from the shaded soil. During that time (July 21 for potatoes and July 5 for sugar beets) nearly all fields had $K_{c\ act}$ in a relatively tight cluster of points, indicating that nearly all fields were well-watered and transpiring at near-maximum rates. Under those conditions, the observed $K_{c\ act}$ was essentially the same as the standard K_c. Values for $K_{c\ act}$ that lie below the approximated standard K_{cb} line were experiencing some water stress.

FIGURE 5.14

Actual $K_{c\ act}$ determined by the METRIC SEB model for hundreds of potato and sugar beet fields in Idaho, USA on three Landsat overpass dates for each crop, showing the impacts of wet soil on the $K_{c\ act}$. Also shown an approximate K_{cb} vs. NDVI line. The NDVI was calculated using atmospherically corrected surface reflectance



Source: Authors' own elaboration

A second important point that is illustrated in Figure 5.14 is that there can be a wide variability of observed $K_{c\ act}$ among fields, especially during periods of low vegetation density (during the initial and development periods). This variability was caused by the large variation in the dates of planting and of wetness of fields by irrigation due to variability in irrigation scheduling applied. Such large variability can make it difficult to develop and use a single K_c vs. NDVI relationship, unless values are averaged to represent multiple-day periods that include both wetness and dryness of exposed soil. During those longer periods, the $K_{c\ act}$ for different fields will be expected to lie closer together.

It is of interest to note that maximum values for $K_{c\ act}$ approached 1.0 for some fields over the whole range of NDVI for both crop types, just increasing when NDVI values are high, indicating high $ET_{c\ act}$ by then. As described in Chapter 7, high $K_{c\ act}$ values are predicted early in the growing season due to high soil evaporation, thus a high evaporation coefficient K_e when the soil is wet. It is to be noted that the $K_{c\ act}$ values in Figure 5.14 represent values of only three single days during the growing season. Thus, fields having high values on one of the satellite overpass days would be expected to reduce its $K_{c\ act}$ as the soil dries during the subsequent days, before the next irrigation.

Figure 5.14 demonstrates the value of remote sensing in producing information on the variability in $ET_{c\ act}$ among fields having the same crop. The variability is both on a daily base and for the total growing season $ET_{c\ act}$. The RS-based $ET_{c\ act}$ data can be used to create histograms showing the space and time variability in ET among populations of the same crop, which may be useful to support farm advising.

5.5.4 Strengths and limitations of remote sensing for evapotranspiration

The VI-based and SEB-based approaches can be considered a complement to the FAO 56 K_c approach because they can capture within-season sudden climatic, environmental and management stresses, thus assisting in identifying potential problems and developing corrective actions.

Advantages of RS-based crop coefficients estimation include: Quick analyses can be performed by mid-level technicians; large areas can be covered; relationships VI-K_{cb} can be calibrated; the spatial resolution can be high, and costs can be relatively low. Disadvantages refer mainly to: latter relationships may vary with the stomatal control that impacts the ET vs. greenness indices and K_{cb} vs. VI relationships, thus requiring that crop type be known; when short of water, trees can exhibit more stomatal control than annual crops and, therefore, cause a shift in the K_{cb} vs VI relationship; VI-based relationships tend to overestimate ET during conditions of acute water shortage because there is a VI time lag in any reduction in vegetation amount during periods of water shortage, whereas vegetation can quickly reduce stomatal conductance and therefore transpiration. The estimation of the soil evaporation component is less accurate because there is not a direct relationship with the vegetation amount, and VI's may not identify or quantify multi-storeyed canopies, especially for dense vegetation. Advantages and disadvantages of RS are given in Tables 5.4 and 5.5. A robust procedure is required to interpolate $K_{c\ act}$ information from satellite or aerial overpass dates over intervening periods.

TABLE 5.4
Advantages and disadvantages of RS-based crop coefficients (using spectral reflectance)

Advantages of RS-based crop coefficients estimation	Disadvantages of RS-based crop coefficients estimation
• Quick analyses can be performed by mid-level technicians; • Large areas can be covered; • Relationships VI-K_{cb} can be calibrated; and • The spatial resolution can be high, and costs can be relatively low.	• Relationships may vary with the stomatal control that impacts the ET vs. greenness indices and K_{cb} vs. VI relationships, thus requiring that crop type be known. • When short of water, trees can exhibit more stomatal control than annual crops and, therefore, cause a shift in the K_{cb} vs VI relationship. • VI-based relationships tend to overestimate ET during conditions of acute water shortage because there is a VI time lag in any reduction in vegetation amount during periods of water shortage, whereas vegetation can quickly reduce stomatal conductance and therefore transpiration. • The estimation of the soil evaporation component is less accurate because there is not a direct relationship with the vegetation amount. • VI's may not identify or quantify multi-storeyed canopies, especially for dense vegetation.

TABLE 5.5
Advantages and disadvantages of RS Energy Balance based crop coefficient estimation

Advantages of RS Energy Balance crop coefficients estimation	Disadvantages of RS Energy Balance crop coefficients estimation
• $ET_{c\ act}$ can be determined through the observed variation in surface temperature; • Both transpiration and evaporation from the soil are incorporated into the $ET_{c\ act}$ and $K_{c\ act}$ observations, providing estimates of total water consumption; • Large areas can be covered and spatial resolution can be high; • The variability of $K_{c\ act}$ among fields having the same crop type can be observed. • Costs for producing estimates can be low for RSSEB systems that are becoming automated and with data made freely available or at low cost to the public.	• Thermal sensors are required onboard satellites or aerial platforms but satellites equipped with thermal sensors having field-scale spatial resolution are not as common as satellites that observe only the spectral reflectance. • Time intervals between thermal-satellite overpasses can be eight days (e.g. Landsat). • Cloud cover impacts the thermal signal and prevents the use of image information from clouded areas, thereby extending the period between $K_{c\ act}$ information. • The calibration of the RS-SEB processes used by models may include some uncertainties or bias. • Satellite and aerial imagery must be supplemented with ground-based meteorological or gridded meteorological data which can introduce some bias into estimates.

5.5.5 Complementary use of Remote Sensing Vegetation Index and Surface Energy Balance for determination of $K_{c\ act}$

As previously described, VI-based methods perform best for estimating $K_{cb\ act}$ due to the close ties between transpiration, as represented by the K_{cb}, and the amount of vegetation cover as represented by the VI. In contrast, RS-SEB models produce estimates for total $K_{c\ act}$ by viewing the combined processes of transpiration and evaporation. As a result, RS-SEB models can be used to develop relationships between $K_{c\ act}$ and VI that can then be employed using the more frequently available VI-based imagery. Another use of paired RS-SEB and VI-based estimates is to estimate whether water shortage or drought is impacting crop production. This is done by noting periods when RS-SEB-based $ET_{c\ act}$ is lower than the VI-based ET_c estimate. Determinations of reduced ET from water stress can be made difficult, however, by uncertainties, inaccuracies and biases in both modeling approaches.

5.5.6 Use of Artificial Intelligence (AI) for developing K_c data and precautions

Artificial Intelligence (AI) algorithms can ingest data from remote sensing, weather stations, and ground measurements to learn complex relationships observed among different factors and ET rates. AI models can help reduce uncertainty in ET estimations by learning from various data sources and accounting for land cover dynamics. This can lead to better water management practices and a more comprehensive understanding of the water cycle. However, AI can not directly recognize nor correct for faulty data that is ingested into the process. This is most true with K_c data, where substantial errors and biases in both ground-based and remote-sensing-based measurements can frequently occur. *The application of powerful cognition and ingestion of large amounts of data does not offset nor cure the impacts of faulty, biased data. In fact, AI may magnify the influences of biased or faulty ET data on derived relationships. The result can be a worse estimation of ET as compared to using direct K_c curves or remotely-sensed ET procedures alone.* Therefore, strong precautions should be taken, along with a required precursor involving a study of the physics of the evapotranspiration processes, impacts of agronomic practices and plant physiology.

Some advantages of employing AI can include the classification of crop type in near real time. Near-real-time classification, for example using satellite imagery, can facilitate the forecasting of near- and mid-term, field-specific ET for irrigation management. Given

accurate K_c estimation from remote sensing models, AI can help to analyze and develop ranges of K_c within populations of the same crop type. AI can also be employed to develop or expand the use of growing degree days for estimating K_c curve development.

However, and again, products from AI are only as good as the data ingested. *ET data ingested into AI need substantial quality assessment and quality control (QAQC).* The QAQC step requires overview, correction and concurrence by a knowledgeable human. Practitioners who operate AI and machine learning need to have knowledge of the physics of ET processes and general knowledge of physiology and phenologies involved in plant growth. It is helpful to know what crops "look like" on the ground regarding their density, row spacing, architecture and height and consequently their behaviour regarding ET. Application of AI is facilitated by a knowledge of agronomic culture and irrigation practices.

5.5.7 Requirements for transferability of standard K_c and K_{cb}

Thousands of papers have been published during the last 25 years reporting on the use and obtention of K_c values and K_c curves for essentially all cultivated crops. However, the objectives and conditions of the various studies have been quite diverse resulting in sometimes contradictory and competing values for K_c. This presents a clear disadvantage when assessing the transferability of K_c values to different environments as detailed in Box 5.1.

BOX 5.1
Limitations to the transferability of K_c data

Limitations to the transferability of K_c data can be due to a variety of factors, including:

1. Using an ET_o estimation method different than the standard FAO-PM ET_o equation;
2. Using non-standard cultivation conditions, such as mulch, plastic tunnels, intercropping and greenhouses/screen houses;
3. Adopting crop management practices that may cause various types of stresses, e.g., relative to the use of fertilizers, chemicals, or wastewater;
4. Insufficient description of the experiments, namely of the plant density, height, and vigor, and the instrumentation used and its accuracy;
5. Insufficient calibration and validation of models when used for ET estimation.
6. Cropping under soil and water salinity/sodicity stresses;
7. Using non-optimal plant density and/or planting dates;
8. Adopting deficit irrigation;
9. Insufficient size of experimental plots;
10. Inadequate lysimeter management;
11. Using energy balance instrumentation without adopting an appropriate energy balance closure;
12. Lack of appropriate estimation of soil water fluxes that affect the accuracy of soil water balance (SWB), particularly the amount of water passing through the bottom boundary of the control volume, deep percolation and capillary rise;
13. Lack of consideration of possible occurrence of local advection;
14. Lack of correction of weather data collected over non-transpiring or low-transpiring surfaces; and
15. Using non-calibrated vegetation indices when remote sensing is adopted.

In addition, reported K_c curves often do not follow the defined procedure for the FAO segmented curves, or K_c results are presented as time-dependent polynomial equations, or are only referenced to weekly, 10-day or, more often, monthly periods. Crop growth stages are sometimes defined contradictorily relative to definitions adopted in FAO 56 and previously in FAO 24.

Information provided in those papers may be sufficient relative to the objectives of the studies reported but is insufficient to provide for the transferability of reported K_c into different environments. *These limitations oblige the user to conduct a careful review of published material to determine when derived K_c are only of local interest and application and/or represent non-standard experimental conditions.* This is in contrast to when the reported K_c represents a standard K_c that adheres to recognizable potential evaporative crop conditions that are expected to result from optimal, pristine cropping practices, and are consequently transferable to other environments when adequately adjusted to the standardized climate.

The above concerns and observations were taken into account when compiling the K_c information contained in the Tables of Chapters 6 and 7. Intending to provide accurate standard and updated K_c values, the compilation procedure required that the ET data reported in the literature were exempt from difficult-to-determine biases and mistakes that could compromise the accuracy and representativeness of the data. Chapter 1 describes the primary methods for measuring actual ET_c or indirectly deriving ET_c estimates, emphasizing the accuracy requirements in measurements and the main pitfalls to avoid. These requirements were considered when assessing K_c and ET_c data from the literature.

Chapter 6
ET_c and single crop coefficients (K_c)

This chapter deals with the calculation of crop evapotranspiration (ET_c) under standard, pristine, non-stressed cropping conditions. No limitations are placed on crop growth and evapotranspiration from soil water and salinity stress, reduced crop density, pests and diseases, weed infestation, or low fertility and nutrients (see Figure 1.4 bis). ET_c is determined by the crop coefficient approach whereby the effect of the various weather conditions are incorporated into ET_o and the crop characteristics into the standard K_c coefficient:

$$ET_c = K_c\ ET_o \tag{6.1}$$

The single K_c coefficient incorporates crop characteristics and averaged effects of evaporation from the soil. For normal irrigation planning and management purposes, for the development of basic irrigation schedules, and for most hydrologic water balance studies, average crop coefficients are relevant and may be more convenient than the K_c computed on a daily time step using the dual K_c approach where a separate crop transpiration and soil evaporation coefficient are determined (Chapter 7). Only when values for K_c are needed on a daily basis for specific fields of crops and/or for specific years, or where wetting frequencies from precipitation or irrigation are frequent or varied, must a separate transpiration and evaporation coefficient ($K_{cb} + K_e$) be considered.

The calculation procedure for crop evapotranspiration, ET_c, consists of:

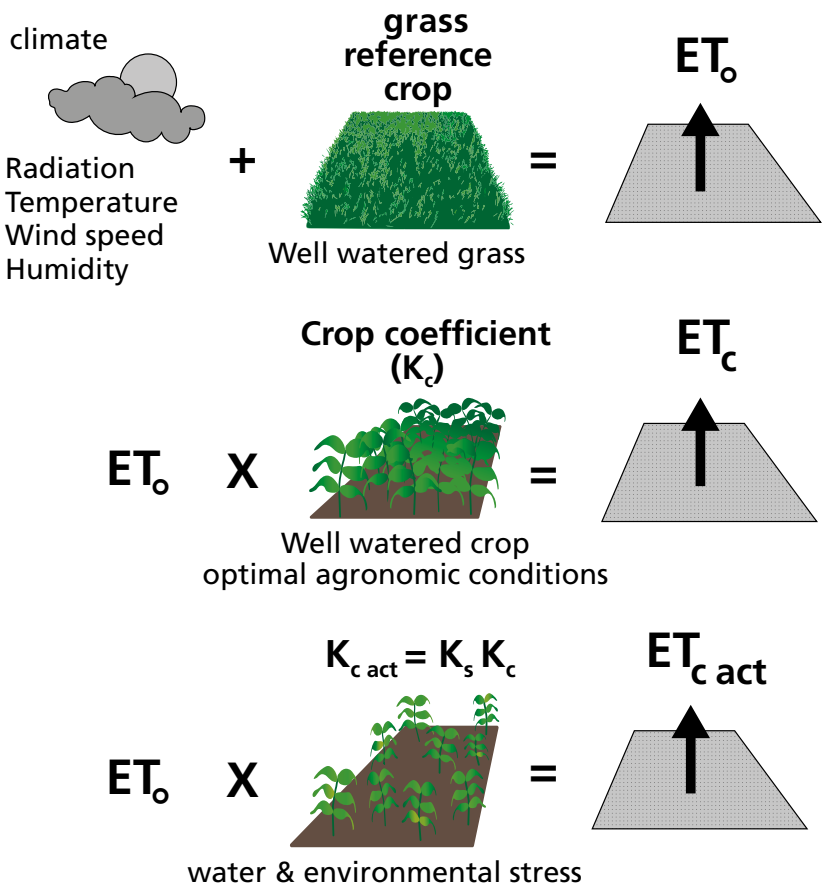
1. identifying the crop growth stages, selecting the corresponding K_c coefficients and determining their stage lengths;
2. adjusting the selected K_c coefficients for frequency of wetting or climatic conditions during each stage;
3. constructing the crop coefficient curve, i.e. providing to determine K_c values for any period during the growing season; and
4. calculating ET_c as the product of ET_o and K_c.

6.1 CROP COEFFICIENTS. TABULATED VALUES

Changes in vegetation and ground cover imply that the crop coefficient K_c varies during the growing season. The trends in K_c during the growing season are represented in the crop coefficient segmented curve. Only three values for K_c are required to describe and construct the crop coefficient curve: those during the initial stage ($K_{c\ ini}$), the mid-season stage ($K_{c\ mid}$) and at the end of the late season stage ($K_{c\ end}$).

The standard values for $K_{c\ ini}$, $K_{c\ mid}$, and $K_{c\ end}$, presented as single-valued K_c for agricultural crops and natural vegetation, are listed in Table 6.1 for vegetable crops, Table 6.2 for field crops, Table 6.3 for fruit trees and vines, Table 6.4 for grasslands and grasses, and in Table 6.5 for riparian and non-riparian wetland ecosystems. The crop coefficients are organized by group type and alphabetic order inside the groups to assist readers in locating the crop in the tables. In addition to K_c values, maximum crop heights (h, m) and ranges of maximum root depths (Z_r, m) are listed in the Tables.

FIGURE 1.4 (BIS)
Reference evapotranspiration (ET_o), standard crop evapotranspiration (ET_c) and actual (non-standard) crop evapotranspiration ($ET_{c\ act}$)



Source: FAO56, 1998

The tabulated standard coefficients are single values that integrate the effects of both transpiration and evaporation over time. The effects of the integration over time represent an average wetting frequency for a "standard" crop under typical growing conditions in an irrigated setting. The vegetable crops and the orchard and vineyard crops generally represent evaporation associated with drip irrigation, thus having frequent applications of small irrigation depths, while field crops and grasses, when irrigated, are mostly considered to be under surface or sprinkler irrigation. Naturally, wetlands and riparian vegetation are not irrigated.

The values for K_c (Tables 6.1 through 6.5) during the initial and crop development stages can be subject to the effects of large variations in wetting frequencies by rainfall (and irrigation) and, therefore, refinements to the value used for $K_{c\ ini}$ should always be made with consideration of the impacts of rainfall and/or irrigation on soil evaporation, especially when the presence of bare soil can dominate. For frequent wettings such as with high frequency sprinkler and drip irrigation, or rainfall, the values for $K_{c\ ini}$ may increase substantially. The tabulated K_c, particularly the $K_{c\ ini}$, may be substantially reduced when the soil is mulched using plastic or organic mulches, which control and decrease evaporation from the soil. These effects vary with the wetting frequency by rainfall and irrigation, soil characteristics, soil-water conservation practices, and the fraction of soil covered by the vegetation. Thus, the tabulated standard K_c does not include the effects of soil mulching, which shall be considered specifically according to the type and amount of mulches used as dealt with in Chapter 10.

TABLE 6.1

Single (time-averaged) crop coefficients, K_c, and ranges of maximum plant heights and root depths for non-stressed, well-managed and highly productive vegetable crops in the standardized subhumid climate ($RH_{min} \approx 45\%$, $u_2 \approx 2$ m s^{-1}) for use with the FAO-PM ET_o equation

Crop	$K_{c\ ini}$	$K_{c\ mid}$	$K_{c\ end}$	Maximum crop height (h, m)	Maximum root depths (Z_r, m)
a. Roots, tubers, bulbs, and stem vegetable crops					
Asparagus (*Asparagus officinalis*)	**0.50**	**1.00**	**0.30**	0.80	1.20 – 1.80
Carrots (*Daucus carota*)	**0.70**	**1.00**	**0.85**	0.30	0.30 – 0.50
Cassava (*Manihot esculenta*)					
Year 1	**0.30**	**1.00**	**0.60**	1.00	0.50 – 0.80
Year 2	**0.60**	**1.10**	**0.60**	1.50	0.70 – 1.00
Celeriac, celery root (*Apium graveolens* group *rapaceum*)	**0.60**	**1.05**	**1.00**	0.60 – 0.90	0.20 – 0.50
Chufa, tigernut (*Cyperus esculentus*)	**0.50**	**1.10**	**0.60**	0.60	0.30 – 0.50
Garlic (*Allium sativum*)	**0.70**	**1.05**	**0.70**	0.50	0.30 – 0.50
Leek (*Allium porrum*)	**0.70**	**1.00**	**0.90**	0.50	0.30 – 0.90
Mashua (*Tropaeolum tuberosum*)	**0.50**	**1.05**	**0.75**	0.60	0.30 – 0.60
Onions (*Allium cepa*)					
Dry	**0.70**	**1.05**	**0.70**	0.45	0.30 – 0.60
Green	**0.70**	**1.00**	**0.90**	0.45	0.30 – 0.60
Seed	**0.70**	**1.00**	**0.50**	0.45	0.30 – 0.60
Parsnip (*Pastinaca sativa*)	**0.50**	**1.05**	**1.00**	0.40	0.50 – 1.00
Potato (*Solanum tuberosum*)					
Long season	**0.50**	**1.10**	**0.40**	0.60	0.40 – 0.60
Short season	**0.50**	**1.10**	**0.60**	0.60	0.40 – 0.60
Radish (*Raphanus sativus*)	**0.70**	**0.95**	**0.85**	0.30	0.30 – 0.50
Rutabaga (*Brassica napobrassica*)	**0.50**	**1.10**	**1.00**	0.60	0.50 – 1.00
Sweet potato (*Ipomoea batatas*)	**0.50**	**1.10**	**0.65**	0.50	1.00 – 1.20
Table beets (*Beta vulgaris*)	**0.50**	**1.05**	**1.00**	0.40	0.45 – 1.00
Turnip (*Brassica rapa cv. rapa*)	**0.50**	**1.05**	**1.00**	0.60	0.50 – 0.90
Yam, Taro (*Colocasia esculenta*)	**0.70**	**1.05**	**1.00**	1.20	0.30 – 0.40
b. Leaves and flowers vegetable crops					
African cabbage, spiderwisp (*Cleome gynandra)*	**0.60**	**1.00**	**0.90**	0.30 – 0.45	0.50 – 0.80
African spinach (*Amaranthus cruentus*)	**0.60**	**1.10**	**1.00**	0.50 – 1.20	0.30 – 0.60
Artichoke (*Cynara scolymus*)	**0.50**	**1.00**	**0.95**	0.80	0.60 – 1.20
Arugula, rucula (*Eruca sativa* or *E. vesicaria*)	**0.70**	**1.00**	**1.00**	0.45	0.30 – 0.60
Broccoli (*Brassica oleracea cv. italica*)	**0.70**	**1.10**	**1.10**	0.60	0.40 – 0.90
Brussel sprouts (*B. oleracea* cv. *gemmifera*)	**0.70**	**1.05**	**1.05**	0.75	0.40 – 0.90
Cabbage					
Common cabbage (*B. oleracea* cv. *capitata*)	**0.70**	**1.05**	**1.00**	0.40	0.30 – 0.50
Kale, leaf cabbage (*B. oleracea* var *sabellica*), and Collard green (*B. oleracea* var. *acephala*)					
1st year	**0.60**	**0.95**	**0.90**	0.80 – 1.20	0.50 – 0.80
2nd year	**0.70**	**0.95**	**0.85**	0.80 – 1.20	0.50 – 0.80
Chinese cabbage, bok choy (*B. rapa cv. chinensis*)					
1st year	**0.60**	**1.05**	**1.00**	0.40	0.50
2nd year	**0.70**	**1.05**	**0.95**	0.40	0.50
Chinese cabbage, pak choi (*B. rapa* ssp. *pekinensis*)					
1st year	**0.60**	**1.00**	**0.95**	0.30 – 0.40	0.50 – 0.80
2nd year	**0.70**	**1.00**	**0.90**	0.30 – 0.40	0.50 – 0.80
Cauliflower (*B. oleracea* cv. *botrytis*)	**0.70**	**1.05**	**1.00**	0.40	0.40 – 0.90
Celery (*Apium graveolens* var. *dulce*)	**0.70**	**1.05**	**1.00**	0.60	0.30 – 0.90
Endive, escarola *(Cichorium endivia)*	**0.40**	**0.90**	**0.85**	0.30 – 0.50	0.15 – 0.30
Jute mallow for leaves (*Corchorus olitorius*)	**0.50**	**1.05**	**0.95**	1.10 – 1.50	0.50 – 0.70
Lettuce (*Lactuca sativa*)	**0.70**	**1.05**	**1.05**	0.35	0.30 – 0.50
Purslane (*Portulaca oleracea*)	**0.60**	**1.00**	**1.00**	0.40	0.30 – 0.50
Spinach (*Spinacia oleracea*)	**0.70**	**1.05**	**1.00**	0.35	0.30 – 0.50

TABLE 6.1 (CONT.)

Crop	$K_{c\ ini}$	$K_{c\ mid}$	$K_{c\ end}$	Maximum crop height (h, m)	Maximum root depths (Z_r, m)
c. Fruit vegetable crops					
Banana (*Musa acuminata*)					
1st cycle	**0.50**	**1.05**	**1.00**	2.00 – 3.30	0.50 – 1.20
2nd cycle	**1.00**	**1.15**	**1.10**	2.50 – 4.00	0.50 – 1.20
Bell pepper (*Capsicum annuum*)	**0.60**	**1.10**	**1.00**	0.70	0.50 – 1.00
Bitter gourd, bitter melon (*Momordica charantia*)	**0.60**	**1.00**	**0.70**	1.00 – 2.50	0.40 – 0.80
Cape gooseberry (*Physalis peruviana*)					
1st year	**0.45**	**0.50**	**0.40**	1.0	0.50 – 1.20
Following years*	**0.55**	**0.95**	**0.65**	1.0 – 2.0	0.50 – 1.20
Chili habanero (*Capsicum chinense*)	**0.60**	**1.15**	**0.75**	0.80	0.50 – 1.20
Chili pepper (*Capsicum annuum*)	**0.60**	**1.10**	**0.80**	0.75	0.50 – 1.20
Chili tabasco pepper (*C. frutescens*)					
First harvest	**0.60**	**1.15**	**0.75**	0.80	0.50 – 1.20
Second harvest	**0.70**	**1.05**	**0.65**	0.80	0.50 – 1.20
Cucumber (*Cucumis sativus*)					
For processing	**0.60**	**1.00**	**0.90**	0.40	0.60 – 1.20
Fresh market	**0.60**	**1.00**	**0.75**	0.40	0.60 – 1.20
Fresh market, on trellis	**0.60**	**1.00**	**0.75**	0.80 – 1.20	0.60 – 1.20
Eggplant (*Solanum melongena*)	**0.60**	**1.05**	**0.95**	0.80	0.40 – 1.00
Fennel (*Foeniculum vulgare*)	**0.35**	**1.00**	**0.40**	0.45 – 0.70	0.40 – 0.80
Melon (*Cucumis melo*)	**0.50**	**1.00**	**0.80**	0.30	0.60 – 1.20
Melon cantaloupe (*C. melo* cv. *cantalupensis*)	**0.50**	**1.00**	**0.80**	0.30	0.60 – 1.20
Nightshade, sunberry (*Solanum retroflexum*)	**0.70**	**1.00**	**0.95**	0.75	0.60 – 0.80
Okra (*Abelmoschus esculentus*)	**0.50**	**0.95**	**0.80**	0.90	0.60 – 0.80
Pineapple (*Ananas comosus*)	**0.40**	**0.90**	**0.90**	0.40	0.30 – 0.60
Pumpkin, winter squash (*Cucurbita pepo*)	**0.50**	**0.95**	**0.70**	0.40	0.90 – 1.50
Strawberries (*Fragaria* × *ananassa*)	**0.50**	**0.80**	**0.75**	0.20	0.30 – 0.50
Tomato (*Solanum lycopersicum*)					
For processing	**0.60**	**1.10**	**0.85**	0.60	0.60 – 1.20
Fresh market	**0.60**	**1.10**	**1.00**	0.60	0.60 – 1.20
Fresh market, with trellis	**0.60**	**1.15**	**0.90**	1.50	0.60 – 1.20
Watermelon (*Citrullus lanatus*)	**0.40**	**1.05**	**0.70**	0.40	0.80 – 1.50
Zucchini, Squash (*Cucurbita pepo*)	**0.50**	**1.00**	**0.70**	0.50	0.60 – 1.10
d. Spicy and medicinal herbs					
d1. Root, tuber and rhizome herbs					
Chicory and bitter chicory (*Cichorium intybus* and *C. intybus* var *intybus*)	**0.40**	**0.90**	**0.85**	0.45 – 0.90	0.50 – 0.70
Ginger (*Zingiber officinale*)	**0.60**	**1.00**	**0.65**	0.50 – 0.90	0.30 – 0.50
Licorice (*Glycyrrhiza glabra*)	**0.50**	**1.00**	**0.90**	1.00	0.90
Nutgrass, purple nutsedge (*Cyperus rotundus*)	**0.50**	**0.90**	**0.60**	1.00 – 1.20	0.30 – 0.50
Turmeric, golden curry (*Curcuma longa*)	**0.40**	**1.05**	**0.90**	0.90 – 1.10	0.20 – 0.40
d2. Leafy, flower and stem herbs					
Absinthe, wormwood, (*Artemisia absinthium*)	**0.45**	**0.80**	**0.40**	0.80 – 1.20	0.60 – 1.00
Basil (*Ocimum basilicum*)	**0.60**	**1.05**	**1.00**	0.50 – 1.00	0.30 – 0.50
Cardamom, large & black (*Amomum subulatum*)	**0.90**	**0.95**	**0.95**	1.50 – 3.00	1.00 – 1.50
Chamomile (*Chamomilla recutita*)	**0.35**	**0.85**	**0.50**	0.50 – 0.90	0.30 – 0.50
Coriander (*Coriandrum sativum*)	**0.60**	**1.10**	**1.05**	0.20 – 0.40	0.30 – 0.50
Fenugreek (*Trigonella foenum-graecum*)	**0.60**	**1.00**	**0.75**	0.60 – 1.40	0.30 – 0.80
Lavender (*Lavandula angustifolia*)	**0.35**	**0.95**	**0.80**	0.20 – 0.60	0.50 – 1.00
Lemon balm, balm (*Melissa officinalis*)	**0.60**	**1.05**	**1.00**	0.50	0.30 – 0.50

* Cultivated as a perennial for two or more consecutive years to maximize productivity.

TABLE 6.1 (CONT.)

Crop	$K_{c\ ini}$	$K_{c\ mid}$	$K_{c\ end}$	Maximum crop height (h, m)	Maximum root depths (Z_r, m)
Mint Japanese mint, wild mint (*Mentha arvensis*)	**0.50**	**1.10**	**1.05**	0.30 – 0.80	0.40 – 0.70
Peppermint (*Mentha* x *piperita*) Single harvest Multiple harvests	 **0.55** **0.55**	 **1.10** **1.00**	 **1.05** **0.95**	 0.30 – 0.50 0.30 – 0.50	 0.30 – 0.50 0.30 – 0.50
Spearmint, common mint (*Mentha spicata*)	**0.50**	**1.00**	**0.95**	0.30 – 0.50	0.30 – 0.50
Oregano (*Origanum vulgare*)	**0.40**	**0.90**	**0.80**	0.40 – 0.50	0.30 – 0.70
Parsley (*Petroselinum crispum*)	**0.60**	**1.00**	**0.90**	0.30 – 0.45	0.30 – 0.50
Plumed cockscomb (*Celosia argentea*)	**0.45**	**1.00**	**0.40**	0.50 – 0.60	0.30 – 0.50
Roselle (*Hibiscus sabdariffa*)	**0.70**	**1.05**	**0.80**	0.50 – 0.70	0.30 – 0.65
Rosemary, rosmarin (*Rosmarinus officinalis*)	**0.30**	**1.00**	**0.85**	0.45 – 0.80	0.40 – 1.00
Saffron (*Crocus sativus*)	**0.60**	**0.95**	**0.40**	0.10 – 0.15	0.30 – 0.60
Stevia, candyleaf, sweetleaf (*Stevia rebaudiana*)	**0.45**	**0.95**	**0.70**	0.30 – 0.80	0.30 – 0.60
Summer savory (*Satureja hortensis)*	**0.60**	**1.10**	**1.10**	0.50	0.30 – 0.60
Thyme (T*hymus vulgaris*)	**0.30**	**0.95**	**0.80**	0.30 – 0.50	0.20 – 0.30
Yarrow (*Achillea millefolium)*	**0.50**	**1.00**	**0.80**	0.90 – 1.00	1.00
d3. Fruit, pod and seed herbs					
Ajowan (*Trachyspermum ammi)*	**0.70**	**0.90**	**0.40**	0.65 – 1.00	0.50 – 0.80
Black cumin (*Nigella sativa*)	**0.50**	**1.05**	**0.50**	0.40	0.50 – 0.80
Chia (*Salvia hispanica*)	**0.50**	**1.00**	**0.40**	0.40 – 1.20	0.30 – 0.50
Cumin (*Cuminum cyminum*)	**0.40**	**0.95**	**0.40**	0.15 – 0.40	0.10 – 0.40
Milk thistle (*Silybum marianum* or *Carduus m.*)	**0.35**	**0.85**	**0.70**	1.00 – 1.85	0.40 – 0.80
Lesquerella, popweed (*Physaria fendleri*)	**0.30**	**0.85**	**0.30**	0.30 – 0.50	0.15 – 0.35
Vanilla (*Vanilla planifolia*) including tutor and shadow plants	**1.00**	**1.00**	**1.00**	1.50 – 2.00	0.40 – 0.60

TABLE 6.2

Single (time-averaged) crop coefficients, K_c, and ranges of maximum plant heights and root depths for non-stressed, well-managed and highly productive field crops in the standardized subhumid climate ($RH_{min} \approx 45\%$, $u_2 \approx 2$ m s^{-1}) for use with the FAO-PM ET_o equation

Crop	$K_{c\ ini}$	$K_{c\ mid}$	$K_{c\ end}$	Maximum crop height (h, m)	Maximum root depths (Z_r, m)
a. Grain legumes					
Bean					
Common bean (*Phaseolus vulgaris*)					
Green	**0.50**	**1.05**	**0.95**	0.50 – 0.70	0.50 – 0.90
Dry	**0.40**	**1.05**	**0.40**	0.50 – 0.70	0.50 – 0.90
Butter bean, lima bean (*Phaseolus lunatus*)	**0.40**	**1.00**	**0.40**	0.60 – 1.80	0.65
Mat bean, moth bean (*Vigna aconitifolia*)	**0.55**	**1.05**	**0.35**	0.15 – 0.25	0.80 – 1.40
Black and green gram (*Vigna mungo*)					
Black gram (dry)	**0.40**	**1.10**	**0.35**	0.50 – 0.70	0.60 – 1.00
Green gram	**0.40**	**1.10**	**0.40**	0.60 – 0.90	0.40 – 1.00
Chickpea, garbanzo (*Cicer arietinum*)	**0.40**	**1.05**	**0.35**	0.50 – 0.70	0.70 – 1.00
Clusterbean, guar (*Cyamopsis tetragonoloba*)					
For green beans	**0.40**	**1.00**	**0.90**	0.80 – 1.50	0.50 – 1.00
For guar gum	**0.40**	**1.00**	**0.40**	0.80 – 1.50	0.50 – 1.00
Cowpea (*Vigna unguiculata*)					
Green	**0.40**	**1.05**	**0.60**	0.60 – 0.80	0.60 – 1.30
Dry	**0.40**	**1.05**	**0.35**	0.60 – 0.80	0.60 – 1.30
Fava bean (*Vicia faba*)					
Fresh	**0.50**	**1.10**	**1.00**	0.80	0.50 – 0.70
Dry	**0.40**	**1.10**	**0.40**	0.80	0.50 – 0.70
Groundnut (peanut) (*Arachis hypogaea*)	**0.40**	**1.05**	**0.60**	0.50	0.50 – 1.00
Lentil (*Lens culinaris*)	**0.40**	**1.05**	**0.30**	0.50	0.60 – 0.80
Pea (*Pisum sativum*)					
Fresh	**0.70**	**1.15**	**1.10**	0.60	0.60 – 1.00
Dry	**0.60**	**1.10**	**0.30**	0.80	0.60 – 1.00
Soybean (*Glycine max*)	**0.40**	**1.10**	**0.50**	0.80	0.60 – 1.40
b. Fiber Crops					
Cotton (*Gossypium hirsutum*)	**0.40**	**1.10**	**0.50**	1.20	1.00 – 1.70
Flax (Linseed) (*Linum usitatissimum*)	**0.45**	**1.05**	**0.25**	1.00	1.00 – 1.50
Jute mallow for fibre (*Corchorus olitorius*)	**0.50**	**1.00**	**1.00**	1.10 – 2.70	0.50 – 0.70
Kenaf (*Hibiscus cannabinus*)	**0.40**	**1.10**	**0.70**	3.00 – 4.00	0.50 – 0.80
Sisal (*Agave sisalana*)	**0.25**	**0.55**	**0.55**	1.50	0.50 – 1.00
Sunn hemp (*Crotalaria juncea*)	**0.50**	**1.10**	**1.00**	1.10	0.50 – 0.70
c. Oil Crops					
Camelina (*Camelina sativa*)	**0.20**	**1.10**	**0.45**	0.80	1.00 – 1.50
Canola (*Brassica napus*)	**0.50**	**1.10**	**0.35**	1.00 – 1.50	0.80 – 1.30
Castorbean (*Ricinus communis*)	**0.35**	**1.05**	**0.40**	1.00 – 1.50	0.80 – 1.30
Linseed (Flax) (*Linum usitatissimum*)	**0.35**	**0.95**	**0.25**	0.90	1.00 – 1.50
Mustard (*Brassica juncea*)	**0.35**	**1.10**	**0.40**	1.50 – 2.00	0.50 – 1.10
Safflower (*Carthamus tinctorius*)	**0.35**	**1.05**	**0.25**	1.10	1.00 – 2.00
Sesame (*Sesamum indicum*)	**0.35**	**1.05**	**0.25**	1.30	1.00 – 1.50
Sunflower (*Helianthus annuus*)	**0.35**	**1.15**	**0.30**	2.00	0.80 – 2.00

TABLE 6.2 (CONT.)

Crop	$K_{c\ in}$	$K_{c\ mid}$	$K_{c\ end}$	Maximum crop height (h, m)	Maximum root depths (Z_r, m)
d. Sugar crops					
Sugar beet (*Beta vulgaris*)	**0.35**	**1.10**	**0.75**	0.50	0.70 – 1.20
Sugar cane (*Saccharum officinarum*)	**0.40**	**1.20**	**0.80**	3.00 – 4.00	1.00 – 1.50
e. Bioenergy non–edible crops					
Cattail (*Typha latifolia*)	**0.70**	**1.15**	**0.85**	1.50 – 2.00	1.00 – 1.50
Cordgrass (*Spartina foliosa, S. alterniflora*)	**1.00**	**1.15**	**1.15**	2.00 – 2.50	1.00 – 1.50
Giant miscanthus (*Miscanthus x giganteus*)	**1.00**	**1.25**	**1.25**	2.00 – 3.00	1.00 – 2.00
Giant reed (*Arundo donax*)	**1.00**	**1.20**	**0.90**	2.00 – 3.00	1.00 – 1.50
Reed canary grass (*Phalaris arundinacea*)	**1.00**	**1.15**	**0.90**	1.50 – 2.00	1.00 – 2.00
Sedge (*Carex riparia, C. acutiformis*)	**0.85**	**1.15**	**0.85**	1.00 – 2.00	1.00 – 1.50
Switch grass (*Panicum virgatum*)	**0.90**	**0.95**	**0.95**	2.00 – 2.50	1.50 – 3.00
f. Cereals and pseudocereals					
Amaranth grain (*Amaranthus sp.*)	**0.30**	**1.10**	**0.25**	2.00	0.50 – 1.50
Barley (*Hordeum vulgare*)	**0.30**	**1.10**	**0.25**	0.70 – 0.90	0.60 – 1.20
Buckwheat (*Fagopyrum esculentum*)	**0.30**	**1.05**	**0.25**	1.00 – 1.20	0.60 – 1.20
Maize (*Zea mays*)					
Grain, low grain moisture at harvest	**0.30**	**1.20**	**0.30**	2.50 – 3.50	0.60 – 1.50
Grain, high grain moisture at harvest	**0.30**	**1.20**	**0.65**	2.50 – 3.50	0.60 – 1.50
Silage	**0.30**	**1.15**	**0.95**	2.50 – 3.20	0.60 – 1.50
Sweet	**0.30**	**1.15**	**1.05**	1.50 – 2.50	0.60 – 1.50
Oats (*Avena sativa*)	**0.30**	**1.05**	**0.25**	0.80 – 1.10	1.00 – 1.50
Pearl Millet (*Pennisetum glaucum*)	**0.30**	**1.10**	**0.35**	2.00	1.00 – 2.00
Quinoa (*Chenopodium quinoa*)	**0.40**	**1.10**	**0.50**	1.00 – 1.20	0.60 – 1.20
Rye (*Secale cereale*)	**0.30**	**1.00**	**0.25**	0.90	0.60 – 1.20
Sorghum (*Sorghum bicolor*)					
Grain	**0.30**	**1.05**	**0.45**	1.50 – 2.00	1.00 – 1.50
Silage	**0.30**	**1.05**	**0.90**	2.00 – 3.00	1.00 – 1.50
Sweet	**0.30**	**1.10**	**0.95**	3.00 – 4.00	0.60 – 1.50
Teff (*Eragrostis tef*)	**0.30**	**1.05**	**0.30**	1.10	0.60 – 1.20
Wheat, common (*Triticum aestivum*)					
Winter, low grain moisture at harvest	**0.70**	**1.15**	**0.25**	0.70 – 1.10	1.00 – 1.50
Winter, high grain moisture at harvest	**0.70**	**1.15**	**0.55**	0.70 – 1.10	1.00 – 1.50
Spring, low grain moisture at harvest	**0.30**	**1.15**	**0.25**	0.70 – 1.10	1.00 – 1.50
Spring, high grain moisture at harvest	**0.30**	**1.15**	**0.45**	0.70 – 1.10	1.00 – 1.50
Wheat, durum (*Triticum durum*)					
Winter, low grain moisture at harvest	**0.50**	**1.05**	**0.25**	0.70 – 1.10	1.00 – 1.50
Winter, high grain moisture at harvest	**0.50**	**1.05**	**0.55**	0.70 – 1.10	1.00 – 1.50
g. Rice (*Oryza sativa*)					
Flooded	**1.05**	**1.20**	**1.05**	1.00	0.50
Flooded, dry seeding	**0.85**	**1.20**	**1.05**	1.00	0.50
Flooded, anticipated cut–off	**1.05**	**1.20**	**0.80**	1.00	0.50
Intermittent irrigation	**0.95**	**1.20**	**1.00**	1.00	0.70
Aerobic, sprinkler irrigation	**0.90**	**1.10**	**0.80**	0.80	1.00
Aerobic, surface irrigation	**0.90**	**1.10**	**0.85**	0.80	1.00
Rainfed	**0.80**	**1.00**	**0.80**	0.70	0.80 – 1.20

TABLE 6.3

Single (time-averaged) crop coefficients, K_c, and ranges of the fraction of the ground cover and of maximum plant heights for non-stressed, well-managed and highly productive fruit trees, shrubs and vines for use with the FAO-PM ET_o equation

Degree of ground cover or field system, training, and plant density	Fraction of ground cover (f_c)	Maximum crop height (h, m)	$K_{c\ ini}$	$K_{c\ mid}$	$K_{c\ end}$
a. Mediterranean and warm temperate climates fruit trees and vines					
a.1. Olive *(Olea europea)*					
Young (< 7 years traditional, < 4 years intensive)	0.15 – 0.30	1.5 – 2.0	**0.40**	**0.35**	**0.40**
Traditional, non–irrigated; low density; 100 – 200 pl ha^{-1}	0.15 – 0.30	2.5 – 3.5	**0.45**	**0.45**	**0.45**
Traditional, non–irrigated; medium density; 200 – 300 pl ha^{-1}	0.20 – 0.40	3.0 – 4.5	**0.50**	**0.50**	**0.50**
Intensive, hedge–prune; low to medium density; 300 – 800 pl ha^{-1}	0.30 – 0.40	3.0 – 3.5	**0.50**	**0.60**	**0.60**
Super–Intensive; medium density, hedgerow; 800 – 1500 pl ha^{-1}	0.35 – 0.45	3.0 – 3.5	**0.55**	**0.65**	**0.65**
Super–intensive; high density, hedgerow; 1 500 – 2 000 pl ha^{-1}	0.45 – 0.55	3.0 – 4.0	**0.60**	**0.70**	**0.70**
a.2. Table grapes *(Vitis vinifera)*					
Low (Young); diverse trellis and training	< 0.40	< 1.5	**0.35**	**0.65**	**0.55**
Medium; T–trellis, Y–trellis and vertical shoot positioned (VSP); 1 200 – 1 700 pl ha^{-1}	0.40 – 0.60	1.5 – 2.2	**0.35**	**0.95**	**0.70**
High; Y–trellis and overhead system; 1 200 – 1 700 pl ha^{-1}	0.60 – 0.95	2.0 – 2.5	**0.45**	**1.10**	**0.80**
a.3. Wine grapes *(Vitis vinifera)*					
Very low (Young < 5 years); diverse trellis and trainings; 2 000 – 3 300 pl ha^{-1}	< 0.15	< 1.5	**0.30**	**0.35**	**0.30**
Low; diverse trellis and training; 2 000 – 3 300 pl ha^{-1}	0.15 – 0.35	1.5 – 2.0	**0.35**	**0.60**	**0.40**
Medium; VSP, single & double Guyot, GDC, single & bilateral cordon, Lyre, Y–trellis; 2 000 – 3 300 pl ha^{-1}	0.35 – 0.50	1.5 – 2.0	**0.40**	**0.80**	**0.55**
High ; VSP, GDC, Lyre, Y–trellis, T–trellis, QCT, Pergola; 2 000 – 4 300 pl ha^{-1}	0.50 – 0.60	1.5 – 2.5	**0.40**	**0.95**	**0.65**
Very high; Y–trellis, GDC and overhead system; 2 000 – 4 300 pl ha^{-1}	> 0.60	1.8 – 2.5	**0.45**	**1.05**	**0.70**
a.4. Citrus trees					
a.4.1. Clementine *(Citrus clementina),* **Mandarin** *(C. reticulata),* **Lime** *(C. aurantifolia)*					
Young (< 5 years); vase	0.10 – 0.20	< 1.5	**0.50**	**0.50**	**0.55**
Low; vase; < 400 pl ha^{-1}	0.20 – 0.35	1.5 – 2.5	**0.55**	**0.60**	**0.60**
Medium; vase; 400 – 550 pl ha^{-1}	0.35 – 0.60	2.5 – 3.0	**0.70**	**0.75**	**0.75**
High; vase; > 550 pl ha^{-1}	> 0.60	3.0 – 4.0	**0.80**	**0.85**	**0.85**
a.4.2. Lemon *(C. limon)*					
Young (< 5 years); vase	0.10 – 0.20	< 1.5	**0.50**	**0.40**	**0.55**
Low density; vase; < 200 pl ha^{-1}	0.20 – 0.50	1.5 – 2.5	**0.55**	**0.60**	**0.65**
Medium density; vase; 200 – 400 pl ha^{-1}	0.50 – 0.70	2.5 – 3.0	**0.75**	**0.75**	**0.80**
High density; vase; > 400 pl ha^{-1}	> 0.70	3.0 – 4.0	**0.75**	**0.80**	**0.80**
Hedgerow (Industry); > 1 000 pl ha^{-1}	> 0.60	> 2.5	**0.80**	**0.85**	**0.85**
a.4.3. Orange *(C. sinensis),* **Grapefruit** *(C. paradisi),* **Tangelo** *(C. tangelo)*					
Young (< 5 years); vase	0.10 – 0.20	< 2.0	**0.50**	**0.45**	**0.50**
Low; vase; < 400 pl ha^{-1}	0.20 – 0.40	2.0 – 3.0	**0.55**	**0.60**	**0.60**
Medium; vase; 400 – 600 pl ha^{-1}	0.40 – 0.70	3.0 – 4.0	**0.70**	**0.70**	**0.70**
High; vase; 600 – 950 pl ha^{-1}	> 0.70	> 4.0	**0.70**	**0.80**	**0.75**
Hedgerow (Industry); > 1 250 pl ha^{-1}	> 0.60	> 2.5	**0.70**	**0.75**	**0.75**
a.5. Avocado *(Persea americana)*					
Young (< 3 years); vase	< 0.20	< 2.0	**0.50**	**0.50**	**0.50**
Low ground cover; vase; < 300 pl ha^{-1}	0.20 – 0.35	2.0 – 3.0	**0.60**	**0.65**	**0.65**
Medium to high; vase; 300 – 400 pl ha^{-1}	0.35 – 0.60	3.0 – 4.0	**0.80**	**0.85**	**0.85**
Very high; vase; > 400 pl ha^{-1}	> 0.60	> 4.0	**0.90**	**1.00**	**0.90**
Hedgerow; medium density; 300 – 400 pl ha^{-1}	0.35 – 0.60	> 2.5	**0.80**	**0.85**	**0.80**
Hedgerow; high density; > 400 pl ha^{-1}	0.60 – 0.85	> 3.0	**0.90**	**0.95**	**0.85**

* Vase: when after prunning the branches take a form like a vase, thus opened in the center.

TABLE 6.3 (CONT.)

Degree of ground cover or field system, training, and plant density	Fraction of ground cover (f_c)	Maximum crop height (h, m)	$K_{c\ ini}$	$K_{c\ mid}$	$K_{c\ end}$
a.6. Fig tree *(Ficus carica)*					
Young (< 5 years)	< 0.20	< 1.5	**0.35**	0.45	0.35
Low, < 400 pl ha^{-1}	0.20 – 0.30	1.5 – 2.0	**0.35**	0.60	0.35
Medium to high, 800 – 1 000 pl ha^{-1}	0.30 – 0.50	2.0 – 2.5	**0.40**	0.80	0.40
a.7. Loquat *(Eriobotrya japonica)*					
Young (< 4 years); vase	< 0.25	< 2.5	**0.50**	**0.55**	**0.55**
Low to medium; vase; 450 – 600 pl ha^{-1}	0.25 – 0.50	2.5 – 3.5	**0.55**	**0.75**	**0.70**
High; vase; > 600 pl ha^{-1}	> 0.50	> 3.5	**0.60**	**0.85**	**0.75**
a.8. Persimmon *(Diospyros kaki)*					
Young (< 5 years)	< 0.15	< 2.0	**0.35**	**0.40**	**0.25**
Low to medium, 300 – 500 pl ha^{-1}	0.15 – 0.40	2.0 – 2.5	**0.30**	**0.65**	**0.35**
High, 500 – 800 pl ha^{-1}	0.40 – 0.60	2.5 – 3.0	**0.35**	**0.90**	**0.50**
a.9. Pomegranate *(Punica granatum)*					
Young (< 5 years)	< 0.25	< 2.0	**0.30**	**0.40**	**0.30**
Low; vase; 500 – 800 pl ha^{-1}	0.25 – 0.40	2.0 – 2.5	**0.35**	**0.55**	**0.35**
Medium; vase; 500 – 800 pl ha^{-1}	0.40 – 0.60	2.5 – 3.0	**0.45**	**0.70**	**0.45**
High; vase; > 800 pl ha^{-1}	> 0.60	> 3.0	**0.45**	**0.90**	**0.60**
a.10. Tea *(Camellia sinensis)*					
Young (< 2 years); low hedgerow; 7 000 – 13 500 pl ha^{-1}	< 0.50	< 0.70	**0.80**	**0.90**	**0.90**
Mature; hedgerow; 7 000 – 13 500 pl ha^{-1}	0.50 – 0.90	> 0.70	**1.05**	**1.05**	**1.05**
b. Temperate climate fruit trees, vines and shrubs					
b.1. Pome trees					
b.1.1. Apple *(Malus domestica)*					
Young (< 4 years); central leader	< 0.25	< 3.0	**0.30**	**0.55**	**0.35**
Low; central leader; < 1 500 pl ha^{-1}	0.25 – 0.40	3.0 – 4.0	**0.35**	**0.65**	**0.40**
Medium; central leader; 1 500 – 3 000 pl ha^{-1}	0.40 – 0.50	3.0 – 4.0	**0.45**	**0.85**	**0.45**
High, central leader; 1 500 – 3 000 pl ha^{-1}	0.50 – 0.70	3.0 – 4.5	**0.50**	**0.95**	**0.60**
Very high, central leader; > 3 000 pl ha^{-1}	> 0.70	> 3.0	**0.55**	**1.00**	**0.75**
b.1.2. Pear *(Pyrus communis)*					
Young (< 5 years); all training systems	< 0.35	< 2.5	**0.35**	**0.60**	**0.45**
Low; vase;< 1 000 pl ha^{-1}	0.35 – 0.40	2.0 – 2.5	**0.40**	**0.75**	**0.50**
Medium; central leader; 1 000 – 1 600 pl ha^{-1}	0.35 – 0.60	2.5 – 4.0	**0.50**	**0.95**	**0.60**
High, central leader, other trellis systems; > 1500 pl ha^{-1}	> 0.60	> 3.5	**0.65**	**1.05**	**0.70**
b.2. Stone fruit trees					
b.2.1. Apricot *(Prunus armeniaca),* **plums** *(P. domestica and P. salicina)*					
Young, vase; (< 5 years)	< 0.40	< 2.5	**0.35**	**0.60**	**0.40**
Low; vase; < 200 pl ha^{-1}	0.40 – 0.65	2.5 – 3.0	**0.45**	**0.90**	**0.45**
Medium; vase; 200 – 400 pl ha^{-1}	0.65 – 0.70	3.0 – 4.5	**0.50**	**1.00**	**0.55**
High; vase; > 400 pl ha^{-1}	0.70 – 0.80	3.0 – 5.5	**0.55**	**1.05**	**0.60**
Very high; vase; > 400 pl ha^{-1}	> 0.80	3.0 – 5.5	**0.60**	**1.05**	**0.65**
b.2.2. Cherry *(Prunus avium)*					
Young (< 5 years)	< 0.30	< 3.0	**0.40**	**0.60**	**0.50**
Low; vase; < 800 pl ha^{-1}	0.25 – 0.40	3.0 – 4.0	**0.45**	**0.65**	**0.50**
Medium to High (vigorous rootstock); vase; < 800 pl ha^{-1}	0.40 – 0.70	> 4.0	**0.50**	**0.85**	**0.70**
Low to Medium (semi vigorous rootstock); central leader; 800 – 1 250 pl ha^{-1}	0.40 – 0.50	4.0 – 5.0	**0.50**	**0.85**	**0.60**
Medium to High (semi–dwarf rootstock); central leader; > 1 200 pl ha^{-1}	0.40 – 0.70	> 2.5	**0.55**	**0.90**	**0.65**
High (dwarf rootstock), central leader > 1 200 pl ha^{-1}	0.70 – 0.90	2.0 – 3.5	**0.60**	**1.00**	**0.85**

TABLE 6.3 (CONT.)

Degree of ground cover or field system, training, and plant density	Fraction of ground cover (f_c)	Maximum crop height (h, m)	$K_{c\ ini}$	$K_{c\ mid}$	$K_{c\ end}$
b.2.3. Peach *(Prunus persica)*					
Young (< 4 years)	< 0.30	< 2.0	**0.35**	**0.60**	**0.40**
Low (high vigour), vase; 400 – 600 pl ha^{-1}	0.30 – 0.50	2.0 – 3.5	**0.35**	**0.80**	**0.50**
Medium (medium vigour); vase; 400 – 700 pl ha$_{-1}$	0.50 – 0.65	3.5 – 4.5	**0.45**	**0.95**	**0.60**
High (low vigour), vase; > 700 pl ha^{-1}	> 0.65	> 4.0	**0.45**	**1.05**	**0.70**
Medium (medium vigour), trellis; > 1 100 pl ha^{-1}	0.30 – 0.60	3.5 – 4.5	**0.40**	**0.85**	**0.55**
High (low vigour), trellis; > 1 500 pl ha^{-1}	0.60 – 0.75	> 3.5	**0.45**	**1.10**	**0.65**
Very high (low vigour), trellis; > 2 000 pl ha^{-1}	> 0.75	> 3.5	**0.50**	**1.15**	**0.75**
b.3. Nut fruit trees					
b.3.1. Almond *(Prunus dulcis)*					
Young (< 4 years)	< 0.30	< 3.0	**0.30**	**0.50**	**0.45**
Low (high vigour), vase; < 400 pl ha^{-1}	0.20 – 0.40	2.0 – 4.0	**0.35**	**0.55**	**0.45**
Medium (medium vigour), vase; <400 pl ha^{-1}	0.40 – 0.55	4.0 – 4.5	**0.45**	**0.85**	**0.55**
High (low vigour), vase; > 200 pl ha^{-1}	0.55 – 0.65	4.5 – 5.0	**0.45**	**1.00**	**0.75**
Very high (low vigour), vase; > 200 pl ha^{-1}	> 0.65	4.0 – 5.5	**0.55**	**1.10**	**0.85**
Hedgerow/Super–intensive (dwarf rootstock); > 2500 pl ha^{-1}	0.30 – 0.45	2.0 – 2.5	**0.35**	**0.90**	**0.50**
b.3.2. Hazelnut *(Corylus avellana)*					
Young (< 6 years)	< 0.35	< 2.0	**0.35**	**0.65**	**0.45**
Low to medium; multiple stem–shrub system; < 500 pl ha^{-1}	0.35 – 0.55	2.0 – 3.5	**0.40**	**0.80**	**0.50**
Medium to high; multiple stem–shrub system; 500 – 700 pl ha^{-1}	0.55 – 0.70	3.5 – 4.5	**0.45**	**0.90**	**0.55**
Very high; multiple stem–shrub system > 700 pl ha^{-1}	> 0.70	> 4.5	**0.50**	**1.00**	**0.60**
Hedgerow/Super–intensive, > 1 000 pl ha^{-1}	>0.55	>2.5	**0.50**	**1.05**	**0.55**
b.3.3. Pecans *(Carya illinoinensis)*					
Young (< 8 years)	< 0.40	< 8.0	**0.40**	**0.70**	**0.50**
Low; vase; > 100 pl ha^{-1}	0.40 – 0.60	8.0 – 10.0	**0.45**	**0.75**	**0.55**
Medium to high; vase; 100 – 250 pl ha^{-1}	0.60 – 0.80	10.0 – 13.0	**0.50**	**0.90**	**0.70**
Very high; vase; 250 – 300 pl ha^{-1}	> 0.80	> 13.0	**0.55**	**0.95**	**0.70**
b.3.4. Pistachio *(Pistacia vera)*					
Young (< 10 years)	< 0.35	< 3.0	**0.30**	**0.50**	**0.35**
Low; vase; > 100 pl ha^{-1}	0.35 – 0.50	3.0 – 4.0	**0.35**	**0.80**	**0.45**
Medium to high; vase; 150 – 350 pl ha^{-1}	0.50 – 0.70	4.0 – 4.5	**0.40**	**0.90**	**0.55**
Very high; vase; > 300 pl ha^{-1}	> 0.70	> 4.5	**0.50**	**1.00**	**0.65**
Hedgerow/Super-intensive, > 1 000 pl ha^{-1}	> 0.30	> 2.5	**0.35**	**0.95**	**0.50**
b.3.5. Walnut *(Juglans regia)*					
Young, vase (< 7 years)	< 0.55	< 6.0	**0.45**	**0.80**	**0.55**
Low; vase; < 200 pl ha^{-1}	0.55 – 0.75	6.0 – 10.0	**0.45**	**1.05**	**0.60**
Medium to very high; vase; 200 – 550 pl ha^{-1}	0.75 – 0.95	10.0 – 12.0	**0.50**	**1.10**	**0.70**
b.4. Other temperate vines and shrubs					
b.4.1. Kiwifruit *(Actinidia deliciosa)*					
Young (< 6 years); trellis	< 0.60	< 2.0	**0.35**	**0.75**	**0.65**
Medium density; T–bar trellis; < 800 pl ha^{-1}	0.60 – 0.80	2.0 – 2.2	**0.40**	**0.95**	**0.75**
High density; pergola trellis; > 800 pl ha^{-1}	0.60 – 0.95	2.0 – 2.5	**0.45**	**1.00**	**0.85**
b.4.2. Hop *(Humulus lupulus)*					
Common density; trellis; < 3 500 pl ha^{-1}	0.08 – 0.12	6.0 – 7.0	**0.30**	**1.00**	**0.90**
b.4.3. Highbush blueberries *(Vaccinium corymbosum)*					
Young (< 3 years)	< 0.50	< 1.50	**0.30**	**0.65**	**0.45**
Common density, 2 500 – 5 000 pl ha^{-1}	0.50 – 0.80	1.50 – 2.00	**0.55**	**0.90**	**0.75**
b.4.4. Lowbush blueberries *(Vaccinium angustifolium)*					
Young (< 2 years)	< 0.50	< 0.60	**0.30**	**0.65**	**0.55**
Common density, > 5 000 pl ha^{-1}	0.50 – 0.80	0.60 – 1.00	**0.45**	**0.80**	**0.65**

TABLE 6.3 (CONT.)

Degree of ground cover or field system, training, and plant density	Fraction of ground cover (f_c)	Maximum crop height (h, m)	$K_{c\ ini}$	$K_{c\ mid}$	$K_{c\ end}$
b.4.5. Rabbiteye blueberry *(Vaccinium ashei)*					
Young (< 3 years)	< 0.50	< 0.60	**0.30**	**0.65**	**0.55**
Common density, 2 500 – 5 000 pl ha^{-1}	0.50 – 0.80	0.60 – 1.00	**0.45**	**0.75**	**0.65**
b.4.6. Blackberry *(Rubus subgenus rubus)*					
Young (< 3 years)	< 0.50	< 1.00	**0.30**	**0.45**	**0.35**
Common density, 2 000 – 5 000 pl ha^{-1}	0.50 – 0.85	1.00 – 2.00	**0.35**	**0.90**	**0.50**
c. Sub–tropical and tropical orchards and plantations					
c.1. Sub–tropical and tropical small fruit trees and shrub					
c.1.1. Acerola *(Malpighia emarginata)*					
Young (< 5 years)	< 0.50	< 2.0	**0.50**	**0.60**	**0.55**
Medium–High, 250 – 700 pl ha^{-1}	0.50 – 0.70	2.0 – 4.0	**0.70**	**0.85**	**0.70**
c.1.2. Carambola *(Averrhoa carambola)*					
Young (< 4 years)	< 0.40	< 2.00	**0.75**	**1.00**	**0.90**
Common density, 150 - 300 pl ha^{-1}	0.40 – 0.60	2.0 – 4.00	**1.00**	**1.15**	**1.10**
c.1.3. Cherimoya *(Annona cherimola)*					
Young (< 3 years)	< 0.15	< 1.5	**0.35**	**0.45**	**0.40**
Low to medium, 125 – 400 pl ha^{-1}	0.15 – 0.35	1.5 – 3.5	**0.45**	**0.60**	**0.50**
Medium to high, 500 – 1 250 pl ha^{-1}	> 0.35	> 3.0	**0.65**	**0.85**	**0.70**
c.1.4. Cocoa *(Theobroma cacao)*					
Young (< 4 years), shading	< 0.50	< 2.0	**0.60**	**0.80**	**0.75**
Medium shading, < 1 100 pl ha^{-1}	0.50 – 0.70	2.0 – 2.5	**0.90**	**0.95**	**0.95**
High shading, < 1 500 pl ha^{-1}	> 0.70	> 2.0	**1.00**	**1.05**	**1.05**
c.1.5. Coffee *(Coffea arabica)*					
Young (< 4 years)	< 0.20	< 2.0	**0.40**	**0.55**	**0.50**
Low to Medium, < 5 500 pl ha^{-1}	0.20 – 0.50	1.5 – 3.0	**0.65**	**0.80**	**0.80**
High (including hedgerow), > 5 500 pl ha^{-1}	> 0.50	> 2.0	**0.90**	**1.00**	**1.00**
c.1.6. Dragon fruit *(Hylocereus undatus)*					
Young (< 3 years)	< 0.20	< 2.0	**0.30**	**0.40**	**0.40**
Medium, < 1 800 pl ha^{-1}	0.20 – 0.30	2.0 – 2.5	**0.40**	**0.65**	**0.55**
High, > 1 800 pl ha^{-1}	> 0.30	2.0 – 2.5	**0.65**	**0.90**	**0.75**
High (hedgerow), > 2000 pl ha^{-1}	> 0.25	2.0 – 2.5	**0.60**	**0.90**	**0.70**
c.1.7. Guava *(Psidium guajava)*					
Young (< 8 years)	< 0.20	< 1.5	**0.40**	**0.45**	**0.50**
Low to medium, < 300 pl ha^{-1}	0.20 – 0.40	1.5 – 2.0	**0.60**	**0.75**	**0.70**
High, < 350 pl ha^{-1}	0.40 – 0.60	2.0 – 2.5	**0.75**	**0.90**	**0.85**
Very high, > 300 pl ha^{-1}	> 0.60	> 2.5	**0.90**	**1.05**	**1.00**
c.1.8. Jujube *(Ziziphus jujuba)*					
Young (< 3 years)	< 0.25	< 2.0	**0.30**	**0.45**	**0.35**
Low to Medium, < 1 100 pl ha^{-1}	0.25 – 0.40	2.0 – 3.0	**0.35**	**0.55**	**0.40**
High, > 1 100 pl ha^{-1}	> 0.40	> 3.0	**0.40**	**0.80**	**0.55**
c.1.9. Papaya *(Carica papaya)*					
Young (< 1 year)	< 0.40	< 1.5	**0.50**	**0.75**	**0.65**
Medium (single row), 3 000 – 4 500 pl ha^{-1}	0.40 – 0.70	1.5 – 3.0	**0.65**	**0.95**	**0.70**
High (single or double row), < 7 000 pl ha^{-1}	0.70 – 0.85	> 2.5	**0.80**	**1.05**	**0.85**
c.1.10. Passionfruit *(Passiflora edulis)*					
Young (< 0.5 years); Vertical shoot position	< 0.15	1.5 – 2.0	**0.45**	**0.50**	**0.45**
Medium; VSP; 500 pl ha^{-1}	0.15 – 0.35	1.5 – 2.0	**0.60**	**0.75**	**0.65**
High; VSP; 1 000 pl ha^{-1}	0.35 – 0.45	1.5 – 2.0	**0.85**	**1.00**	**0.85**
Medium (< 0.5 years); hedgerow; 1000 pl ha^{-1}	< 0.35	1.5 – 2.0	**0.60**	**0.65**	**0.65**
High; hedgerow; 1000 pl ha^{-1}	> 0.35	1.5 – 2.0	**0.90**	**1.00**	**0.90**
Young (< 0.5 years); overhead trellis	< 0.60	1.8 – 2.2	**0.70**	**0.80**	**0.75**
Very high; overhead trellis; 2 000 pl ha^{-1}	> 0.60	1.8 – 2.2	**1.00**	**1.05**	**1.00**

TABLE 6.3 (CONT.)

Degree of ground cover or field system, training, and plant density	Fraction of ground cover (f_c)	Maximum crop height (h, m)	$K_{c\ ini}$	$K_{c\ mid}$	$K_{c\ end}$
c.2. Tropical fruit trees					
c.2.1. Cashew *(Anacardium occidentale),* **dwarf**					
Young (< 5 years)	< 0.50	< 3.0	**0.50**	**0.60**	**0.55**
Common density, 150 - 330 pl ha^{-1}	0.50 – 0.65	3.0 – 5.0	**0.65**	**0.65**	**0.65**
c.2.2. Jaboticaba *(Plinia peruviana)*					
Young (< 5 years)	< 0.50	< 2.0	**0.50**	**0.65**	**0.55**
Common density, 250 – 500 pl ha^{-1}	> 0.50	> 2.5	**0.70**	**0.95**	**0.80**
c.2.3. Longan *(Dimocarpus longan)*					
Young (<5 years)	<0.30	1.0 – 2.0	**0.50**	**0.60**	**0.60**
Common density, 250 - 300 pl ha^{-1}	>0.30	2.5 – 4.0	**0.80**	**0.95**	**0.90**
c.2.4. Lychee *(Litchi chinensis)*					
Young (< 5 years); vase	< 0.20	< 2.5	**0.55**	**0.60**	**0.60**
Low to medium; vase; 100 – 200 pl ha^{-1}	0.20 – 0.30	2.5 – 4.0	**0.65**	**0.75**	**0.70**
High; vase; 200 – 400 pl ha^{-1}	> 0.30	> 4.0	**0.80**	**1.00**	**0.95**
c.2.5. Macadamia *(Macadamia integrifolia)*					
Young (< 10 years)	< 0.50	< 2.0	**0.55**	**0.55**	**0.50**
Medium–High, 300 – 500 pl ha^{-1}	> 0.50	2.0 – 4.5	**0.75**	**0.75**	**0.70**
c.2.6. Mango *(Mangifera indica)*					
Young (< 4 years)	< 0.25	< 2.0	**0.45**	**0.60**	**0.55**
Low to medium,100 – 250 pl ha^{-1}	0.25 – 0.40	2.0 – 3.0	**0.60**	**0.85**	**0.75**
High, 400 – 600 pl ha^{-1}	0.40 – 0.50	2.5 – 4.0	**0.75**	**0.95**	**0.85**
c.3. Palm, fiber, fodder, oil and rubber plantations					
c.3.1. Cactus pear *(Opuntia ficus–indica)*					
Young (< 3 years)	< 0.30	< 1.5	**0.20**	**0.25**	**0.25**
Medium, 350 – 800 pl ha^{-1}	0.30 – 0.50	1.5 – 2.0	**0.30**	**0.40**	**0.30**
High, > 800 pl ha^{-1}	> 0.50	> 2.0	**0.35**	**0.50**	**0.40**
c.3.2. Coconut *(Cocos nucifera)* **and Açaí palm** *(Euterpe oleracea)*					
Young (< 15 years/coconut, < 3 years/açaí)	< 0.30	< 2.5	**0.45**	**0.55**	**0.55**
Medium, 100 – 200 pl ha^{-1}	0.30 – 0.50	2.5 – 5.0	**0.75**	**0.85**	**0.85**
High, > 200 pl ha^{-1}	> 0.50	> 5.0	**0.95**	**1.00**	**1.00**
c.3.3. Date palm *(Phoenix dactylifera)*					
Young (< 10 years)	< 0.20	< 4.0	**0.45**	**0.45**	**0.45**
Medium, 100 – 130 pl ha^{-1}	0.20 – 0.50	4.0 – 10.0	**0.70**	**0.70**	**0.70**
High, 130 – 200 pl ha^{-1}	0.50 – 0.70	5.0 – 12.0	**0.85**	**0.85**	**0.85**
Very high, > 200 pl ha^{-1}	> 0.70	> 10.0	**0.95**	**0.95**	**0.95**
c.3.4. Jatropha *(Jatropha curcas)*					
Young (< 2 years); vase	< 0.15	< 1.0	**0.45**	**0.45**	**0.45**
Low to medium; vase; 1 100 – 1 500 pl ha^{-1}	0.15 – 0.35	1.0 – 2.0	**0.50**	**0.70**	**0.55**
High; vase; > 1 500 pl ha^{-1}	> 0.35	> 2.0	**0.65**	**0.90**	**0.75**
c.3.5. Oil palm *(Elaeis guineensis)*					
Young (< 3 years)	< 0.30	< 2.5	**0.45**	**0.55**	**0.55**
Medium, 100 – 150 pl ha^{-1}	0.30 – 0.50	2.5 – 5.0	**0.75**	**0.85**	**0.85**
High, > 150 pl ha^{-1}	> 0.50	> 5.0	**0.95**	**1.00**	**1.00**
c.3.6. Peach palm *(Bactris gasipaes)* **for producing heart of palm, "palmito"**					
Young (≤ 2.5 years)	< 0.30	< 2.0	**0.60**	**0.80**	**0.70**
Common density, 3 000 – 5 000 pl ha^{-1}	> 0.30	> 2.0	**0.80**	**0.95**	**0.90**

TABLE 6.3 (CONT.)

Degree of ground cover or field system, training, and plant density	Fraction of ground cover (f_c)	Maximum crop height (h, m)	$K_{c\ ini}$	$K_{c\ mid}$	$K_{c\ end}$
c.3.7. Ramie *(Boehmeria nivea)*					
Young (< 2 years)	< 0.80	0.70 – 2.0	**0.25**	**0.65**	**0.60**
Common density, 40 000 – 60 000 pl ha^{-1}	0.80 – 0.95	2.5	**0.30**	**1.05**	**1.00**
c.3.8. Rubber tree *(Hevea brasiliensis)*					
Young (< 6 years)	< 0.75	< 8.0	**0.55**	**0.75**	**0.65**
Common density, 400 – 700 pl ha^{-1}	0.75 – 0.90	8.0 – 15.0	**0.85**	**0.95**	**0.85**

The tabulated standard K_c values result from an in-depth review of crop coefficients published in scientific and technical papers. This review led to the selection of only those publications that have reported consistent field research approaches to derive the K_c values, as well as adequate scientific computational and writing skills. The first sources of information and data were the previous FAO Irrigation and Drainage Papers Number 24 and 33 and the original version of the current Irrigation and Drainage Paper Number 56. The updated reviews were published in various books or reports and, mainly, in peer reviewed journal articles. Information from reviews was assembled and tabulated in purposeful journal papers listed in the bibliography of the current guidelines. These papers provide detailed information not included herein, and that could be accessed in the original articles, which are of great interest to readers.

The standard values for $K_{c\ mid}$ and $K_{c\ end}$ in Tables 6.1 and 6.2 represent those for a standardized sub-humid climate having an average daytime minimum relative humidity (RH_{min}) of about 45 percent and with calm to moderate wind speeds averaging u_2 = 2 m s^{-1}. For more humid or arid conditions, or for more or less windy conditions, the K_c values for the mid-season and end-season stages should be corrected for climate as described hereafter in this chapter. *The values for standard K_c in Tables 6.1, 6.2 and 6.3 represent values for non-stressed crops cultivated under excellent agronomic and water management conditions and achieving maximum crop yield (standard conditions).* Mild and controlled water stress that may favour yield quality, called eustress, also leads to small water savings. It may be particularly interesting for tree and vine fruit crops, which are often subjected to controlled and regulated mild water stress.

The tabulated values of K_c for grasslands and grasses (Table 6.4.) include those for both semi-natural vegetation and planted grasses, mostly pasturelands for grazing, often with cuts for hay as is the case of mountain meadows, irrigated or not. They should be used as standard crop coefficients. Planted grasslands may include forages for grazing and, less frequently, for hay and silage. Typical but wide groups of such grasslands are therefore identified, and K_c values are also assigned to them. Other numerous semi-natural grasses used for grazing are those characterizing given biomes or ecosystems, including in regions marked by aridity; K_c values are assigned to those typical groups. Finally, there is a large number of cultivated grasses having specialized uses such as hay, grazing or landscape, the latter referred to as turf grasses. K_c values are assigned to groups of somewhat similar grass crops, but alfalfa is individually considered. Grasslands are kept low if used for grazing and landscape, or with a variable height during every cutting cycle. Their K_c values were not adjusted to climate.

Users should compare the local climate with the standardized climate (RH_{min} of about 45 percent and with calm to moderate wind speeds averaging u_2 = 2 m s^{-1}) and adjust K_c values when required. Moreover, the K_c values should be reduced by 10 to 30 percent when vegetation conditions or cultivation practices are not optimal and depart from the standard.

Considering the worldwide need for water saving, the K_c values for irrigated crops in Tables 6.1 through 6.4 are the maximal values to be adopted when computing ET_c for irrigation scheduling. For water savings in irrigation, these tabulated $K_{c\ mid}$ values can be decreased by 10 to 30 percent to correspond with soil water depletion thresholds associated with moderating yield losses resulting from water stress. Moreover, where pristine cropping is not possible, i.e. when stand density, height, leaf area or cultivation practices are inferior to those applied under pristine conditions, the value for $K_{c\ mid}$ and, for most crops, for $K_{c\ end}$, will need to be reduced (Chapters 8, 9 and 10).

TABLE 6.4

Single (time-averaged) crop coefficients, K_c for non-stressed well-managed semi-natural and planted grasslands, biome semi-natural grasslands, and grasses for grazing, for hay and for landscapes to be used with the FAO-PM ET_o equation

Grasses and grasslands	$K_{c\ ini}$	$K_{c\ mid}$	$K_{c\ end}$	$K_{c\ avg}$
a. Seminatural and planted grasslands				
High mountain, semi–natural, no killing frost	**0.40**	**1.00**	**0.35**	0.70–0.90
High mountain, semi–natural, killing frost	**0.40**	**1.10**	**0.95**	0.80–1.00
Non–irrigated, for grazing (or seed)	**0.55**	**0.95**	**0.50**	0.75–0.90
Non–irrigated, for hay (typical cut cycles)	**0.45**	**1.15**	**1.15**	0.95–1.10
Irrigated, for grazing (or seed)	**0.55**	**1.05**	**0.55**	0.85–1.00
Irrigated, for hay (typical cut cycles)	**0.55**	**1.05**	**1.00**	0.95–1.05
b. Biome and ecosystem grasslands				
Savanna grasslands	**0.35**	**0.90**	**0.35**	0.50–0.70
Steppe grasslands	**0.30**	**0.75**	**0.30**	0.40–0.60
Meadows and pastures in high mountain	**0.40**	**1.00**	**0.45**	0.70–0.90
Cold and temperate grassland ecosystems	**0.40**	**0.65**	**0.45**	0.45–0.60
Mixed grasslands and forests	**0.35**	**0.70**	**0.40**	0.45–0.60
Shrublands	**0.25**	**0.70**	**0.40**	0.40–0.60
c. Grasses for hay, grazing and landscape				
Alfalfa for hay; typical multiple cutting cycles	**0.50**	**1.20**	**1.15**	1.00–1.10
Alfalfa for seed	**0.40**	**1.10**	**0.65**	0.75–0.95
Grasses for grazing, high height of grazing cuts	**0.90**	**1.05**	**0.90**	0.90–1.00
Grasses for grazing, short height of grazing cuts	**0.85**	**0.95**	**0.85**	0.85–0.90
Grasses for grazing with large cut cycles	**0.90**	**0.95**	**0.95**	0.90–0.95
Grasses for grazing with short cut cycles	**0.85**	**0.95**	**1.00**	0.90–1.00
Grasses for seed production	**0.30**	**0.90**	**0.65**	0.60–0.75
Grasses for hay; typical cut cycles	**0.55**	**1.15**	**1.05**	0.90–1.05
Turf grasses, gulf courses (cut h<0.01 m)	**0.80**	**0.80**	**0.80**	0.75–0.85
Turf grasses, lawns (cut h<0.10 m)	**0.50**	**0.70**	**0.50**	0.55–0.65
Turf grasses, urban	**0.65**	**0.90**	**0.50**	0.65–0.80

Wetlands are among the main protected vegetation and ecosystems, and their management requires estimation of evapotranspiration and water availability for their vegetation and for external use. Wetlands consist of areas where water covers the soil or is present either at near the soil surface all year or for varying periods of time during the year, namely during part of the growing season. Water saturation of the soils, together with climate, largely determines the soil evolution and hydric soil characteristics, and the types of plant and animal communities living in and on the soil. Wetlands may support both aquatic and terrestrial species. The continued presence of water favours conditions adequate for the growth of hydrophyte plants and characteristic ecosystems. There are varied types or categories of wetlands due to differences in

soils, topography, climate, hydrology, water chemistry, vegetation, as well as human disturbance. Wetlands occur worldwide except in Antarctica. The non-permanent wetlands consist essentially of riparian ecosystems created by river networks and groundwater, which may have very important ecosystem services, often essential for neighbour populations. *The knowledge of wetlands K_c may be used to estimate their evapotranspiration, thus supporting the computation of the water balance and therefore supporting their conservation and rehabilitation.*

K_c for typical wetland and riparian ecosystems, which are given in Table 6.5 (a and b), may be used as standard values for the various ecosystems. Users should compare the ecosystems under study with those tabulated, select the one that is most similar and adapt/use the proposed K_c. The K_c values of the respective main vegetation are listed in Table 6.5 (c and d) and should be used in the same way searching for and evaluating the similarity between vegetation tabulated and observed. The K_c values in Table 6.5 were obtained from the available studies on evapotranspiration of worldwide wetland and riparian ecosystems, using appropriate methods as described in Chapters 1 and 5. Those studies required quite demanding field work. Tabulation herein pretends to ease research on those ecosystems for their characterization, protection and conservation, as well as for maintaining and improving their ecosystem services; highly precise K_c is not essential for these purposes but using approximately correct values.

TABLE 6.5

Single (time-averaged) crop coefficients, K_c, for typical wetland and riparian ecosystems and associated primary vegetation for use with the FAO-PM ET_o equation

a. Wetland ecosystems	$K_{c\ ini}$	$K_{c\ mid}$	$K_{c\ end}$	$K_{c\ avg}$
High mountain (approximately > 3 000 m), wet meadow, with soil freezing in winter	**0.40**	**0.90**	**0.40**	0.75
High mountain (approximately > 2 000 m), wet grassland, cold winter without soil freezing	**0.40**	**0.85**	**0.40**	0.60
Low mountain (approximately 2 000 to 1 000 m), reeds and shrubs, cold winter without soil freezing	**0.65**	**1.05**	**0.40**	0.80
Transition from mountain to lowland (1 000 to 100 m), reed and shrubs, cold winter	**0.50**	**0.80**	**0.70**	0.70
Lowlands and flood plains, grasses, temperate climates	**0.55**	**0.85**	**0.55**	0.75
Coastal lowlands, reed, temperate climates	**0.70**	**1.00**	**0.70**	0.90
Salt marshes, grasses and shrubs, warm climates	**0.20**	**0.60**	**0.35**	0.45
Wet savanna, large grasses, warm climates	**0.60**	**1.00**	**0.65**	0.80
Peat bogs	**0.50**	**1.05**	**0.80**	0.85
b. Typical riparian ecosystems				
Lowlands, trees and shrubs, cold winter	**0.40**	**0.80**	**0.40**	0.70
Lowland, grasses, cold winter	**0.90**	**1.10**	**0.90**	1.00
Lowland, trees, cold winter, less water	**0.20**	**0.40**	**0.10**	0.35
Lowland, trees, cold winter, more water	**0.60**	**0.90**	**0.70**	0.75
Trees, temperate and hot climates, water available	**0.45**	**0.90**	**0.50**	0.70
Trees and shrubs, temperate and hot climates, dry	**0.20**	**0.45**	**0.20**	0.35
Trees and shrubs, temperate and hot climates, humid	**0.85**	**0.95**	**0.80**	0.90
c. Common wetland vegetation				
Sedges (*Carex spp.*)	**0.70**	**1.10**	**0.70**	0.80
Foxtail (*Ceratophylum demersum*)	**0.75**	**1.05**	**0.80**	0.85
Sawgrass (*Cladium jamaicense*)	**0.70**	**1.05**	**0.70**	0.90
Saltgrass (*Distichlis spicata*)	**0.25**	**0.55**	**0.40**	0.45
Common reed (*Phragmites australis*)	**0.70**	**1.10**	**0.75**	1.00
Peach willow (*Salix amigdaloides*)	**0.60**	**1.05**	**0.60**	0.90
Bulrush (*Scirpus lacustris*)	**0.65**	**1.05**	**0.50**	0.85
Cattail (*Typha latifolia*)	**0.60**	**1.05**	**0.65**	0.85

TABLE 6.5 (CONT.)

d. Main plants of riparian ecosystems				
Big sagebrush (*Artemisia tridentata*)	**0.35**	**0.80**	**0.35**	0.50
Sedge (*Carex spp.*)	**0.40**	**0.85**	**0.75**	0.65
Russian olive (*Elaeagnus angustifolia*)	**0.65**	**0.80**	**0.65**	0.70
Arrowweed (*Pluchea serice*)	**0.30**	**0.60**	**0.20**	0.40
Blue grass (*Poa pratensis*)	**0.30**	**0.65**	**0.20**	0.50
Euphrates poplar (*Populus eufratica*)	**0.60**	**0.70**	**0.60**	0.65
Cottonwood, Alamo (*P. deltoides*)	**0.55**	**0.80**	**0.60**	0.70
Frémont's cottonwood (*P. fremontii*)	**0.45**	**0.75**	**0.45**	0.55
Gansu's cottonwood (*P. gansuensis*)	**0.55**	**0.85**	**0.60**	0.65
Mesquite (*Prosopis velutina*)	**0.15**	**0.40**	**0.40**	0.35
Willow (*Salix spp.*)	**0.55**	**0.95**	**0.65**	0.75
Saltcedar (*Tamarix ramosissima*) wet	**0.45**	**0.90**	**0.60**	0.70
Saltcedar (*T. ramosissima*) dry	**0.30**	**0.45**	**0.30**	0.40

6.2 CROP COEFFICIENT FOR THE INITIAL STAGE ($K_{c\ ini}$)

6.2.1 Calculation procedure

The values for $K_{c\ ini}$ in Tables 6.1 – 6.5 are only approximations and, for irrigation of crops, should only be used for estimating ET_c during preliminary or planning studies. For all crops and vegetation, the value for $K_{c\ ini}$ that is listed is considered representative of the crop under typical irrigation water management, or under typical water supply conditions in cases of semi-natural non-planted grasslands (Table 6.4) and wetlands and riparian vegetation (Table 6.5). More accurate estimates of $K_{c\ ini}$ can be obtained by considering:

a. *The time interval between wetting events*

Evapotranspiration during the initial stage for annual crops is predominately in the form of evaporation from exposed soil due to low levels of vegetation. Therefore, accurate estimates for $K_{c\ ini}$ should consider the frequency with which the soil surface is wetted during the initial period. Where the soil is frequently or continuously wet from irrigation, rain, or a natural surface or groundwater source, the evaporation from the soil surface can be considerable and $K_{c\ ini}$ will be large, e.g. flooded rice and wetlands vegetation. On the other hand, where the soil surface is dry, evaporation is restricted and the $K_{c\ ini}$ will be small (Table 5.1).

b. *Evaporation power of the atmosphere*

The value for $K_{c\ ini}$ is affected by the evaporating power of the atmosphere, i.e. which is indicated by ET_o. The higher the evaporation power of the atmosphere, the quicker the soil will dry between water applications and the smaller the time-averaged $K_{c\ ini}$ will be for any particular period. In contrast, $K_{c\ ini}$ will generally be high for inundated vegetation.

c. *Magnitude of the wetting events*

The amount of water available in the topsoil for evaporation and hence the time for the soil surface to dry is a function of the magnitude of the wetting event. Therefore, $K_{c\ ini}$ will be smaller for light wetting events than for large wettings. Depending on the time interval between wetting events, the magnitude of the wetting event and the evaporation power of the atmosphere, $K_{c\ ini}$ can vary between 0.10 and 1.15. In general, $K_{c\ ini}$ for flooded vegetation is not affected by precipitation events.

6.2.2 Time interval between wetting events under irrigation

For estimating $K_{c\ ini}$, a mean time interval between wetting events is estimated by counting all rainfall and irrigation events occurring during the initial period that are greater than a few millimetres. Wetting events occurring on adjacent days can be counted as one event. The mean wetting interval is estimated by dividing the time length (days) of the initial period by the number of events (Example 6.1).

EXAMPLE 6.1
Estimation of the time interval between wetting events from mean monthly rainfall data

Estimate, from mean monthly rainfall data, the interval between rains during the rainy season for a station in a temperate climate (Paris, France: 50 mm/month), dry climate (Gafsa, Tunisia: 20 mm/month) and tropical climate (Calcutta, India: 300 mm/month).

Station	monthly rain (mm $month^{-1}$)	typical rainfall (mm)	number of rainy days	interval between rains
Paris	50	3	17	~ 2 days
Gafsa	20	5	4	weekly
Calcutta	300	20	15	~ 2 days

Where only monthly rainfall values are available without any information on the number of rainy days, the number of events within the month can be estimated by dividing the monthly rainfall depth by the depth of a typical rain event. The typical depth, if it exists, can vary widely from climate to climate, region to region and from season to season. Table 6.6 presents some information on the typical rainfall depths. After deciding what rainfall is typical for the region and time of the year, the number of rainy days and the mean wetting interval can be estimated.

TABLE 6.6
Classification of rainfall depths

Rainfall event	depth
Very light (drizzle)	≤ 3 mm
Light (light showers)	5 mm
Medium (showers)	≥ 10 mm
Heavy (rainstorms)	≥ 40 mm

When rainfall is insufficient, irrigation is needed to keep a crop well-watered. Even where irrigation has not yet been developed, the mean interval between future irrigations should be estimated to obtain the required frequency of wetting necessary to keep the crop stress free. The interval might be as small as a few days for small vegetables, or just one or two days in the case of very frequent irrigation by drippers or center pivots, but up to a week or longer for surface irrigated cereals, however, depending on the climatic conditions. Where no estimate of the interval can be made, the user may refer to the values for $K_{c\ ini}$ of Tables 6.1 – 6.5.

6.2.3 Determination of $K_{c\ ini}$

The crop coefficient for the initial growth stage can be derived from Figures 6.1 and 6.2 which provide estimates for $K_{c\ ini}$ as a function of the average interval between wetting events, the evaporation power ET_o, and the magnitude of the wetting event.

Light wetting events (infiltration depths of 10 mm or less): rainfall and high frequency irrigation systems

Figure 6.1 is used for all soil types when wetting events are light. When wetting during the initial period is only by precipitation, one will usually use Figure 6.1 to determine $K_{c\ ini}$. The graph can also be used when irrigation is by high frequency systems, such as micro-irrigation and center-pivot, and light applications of about 10 mm or less per wetting event are applied (Example 6.2).

EXAMPLE 6.2
Graphical determination of $K_{c\ ini}$

A silt loam soil receives irrigation every two days during the initial growth stage via a center pivot irrigation system. The average depth applied by the centre pivot system is about 12 mm per event and the average ET_o during the initial stage is 4 mm day^{-1}. Estimate the crop evapotranspiration during that stage

From Figure 6.1 using the 2-day interval curve:	$K_{c\ ini}$ =	0.85	-
-	$ET_c = K_c\ ET_o$ = 0.85 (4.0) =	3.4	mm day^{-1}

The average crop evapotranspiration during the initial growth stage is 3.4 mm day^{-1}

Heavy wetting events (infiltration depths of 40 mm or more): surface and sprinkler irrigation

Figure 6.2 is used for heavy wetting events when infiltration depths are greater than 40 mm, such as for when wetting is primarily by periodic irrigation by sprinkler or surface irrigation. Following a wetting event, the amount of water available in the topsoil for evaporation is considerable, and the time for the soil surface to dry might be significantly increased. Consequently, the average K_c factor is larger than for light wetting events. As the time for the soil surface to dry is, apart from the evaporation power and the frequency of wetting, determined by the water storage capacity of the topsoil, a distinction is made between soil types.

Figure 6.2a is used for coarse textured soils and Figure 6.2b is used for fine and medium textured soils. Coarse textured soils include sands and loamy sand textured soils. Medium textured soils include sandy loam, loam, silt loam and silt textured soils. Fine textured soils include silty clay loam, silty clay and clay textured soils.

Average wetting events (infiltration depths between 10 and 40 mm):

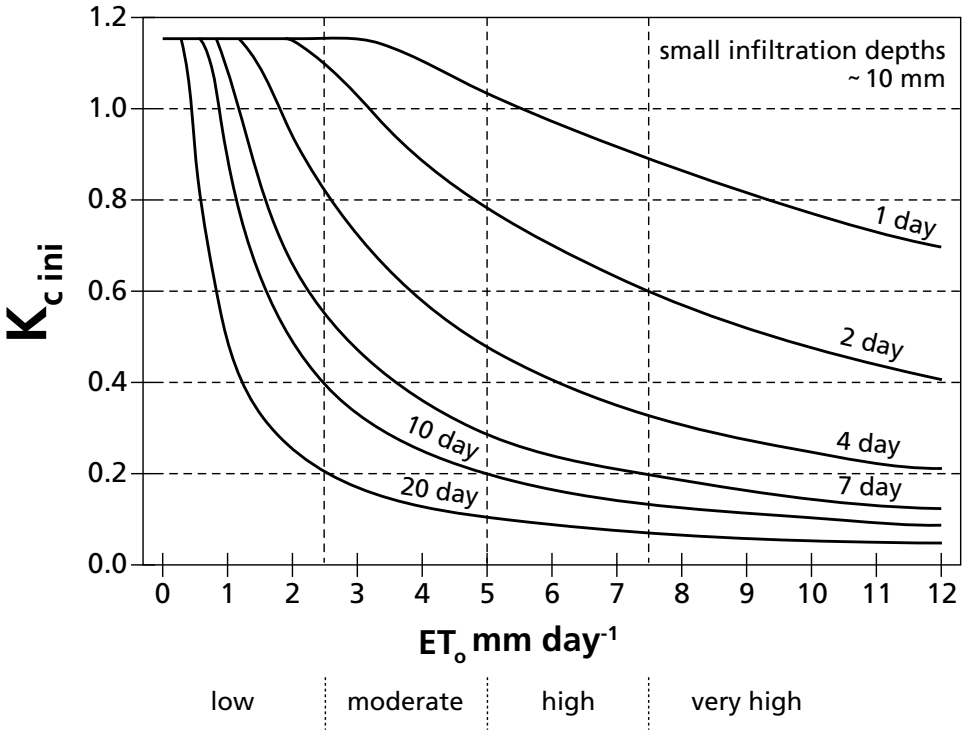
Where average infiltration depths are between 10 and 40 mm, the value for $K_{c\ ini}$ can be estimated using interpolation from Figures 6.1 and 6.2 (Example 6.3):

$$K_{c\ ini} = K_{c\ ini\ (Fig.\ 6.1)} \frac{(Inf - 10)}{(40 - 10)} \left[K_{c\ ini\ (Fig.\ 6.2)} - K_{c\ ini\ (Fig.\ 6.1)} \right] \qquad (6.2)$$

where $K_{c\ ini\ (Fig.\ 6.1)}$ value for $K_{c\ ini}$ from Figure 6.1,
$K_{c\ ini\ (Fig.\ 6.2)}$ value for $K_{c\ ini}$ from Figure 6.2,
Inf average infiltration depth [mm].

The values 10 and 40 in Equation 6.2 are the average depths of infiltration (millimetres) upon which Figures 6.1 and 6.2 are based.

FIGURE 6.1
Average $K_{c\ ini}$ as related to the level of ET_o and the interval between irrigation and/or significant rain during the initial growth stage for all soil types when wetting events are light to medium (3-10 mm per event)



Source: FAO, 1998

EXAMPLE 6.3
Interpolation between light and heavy wetting events

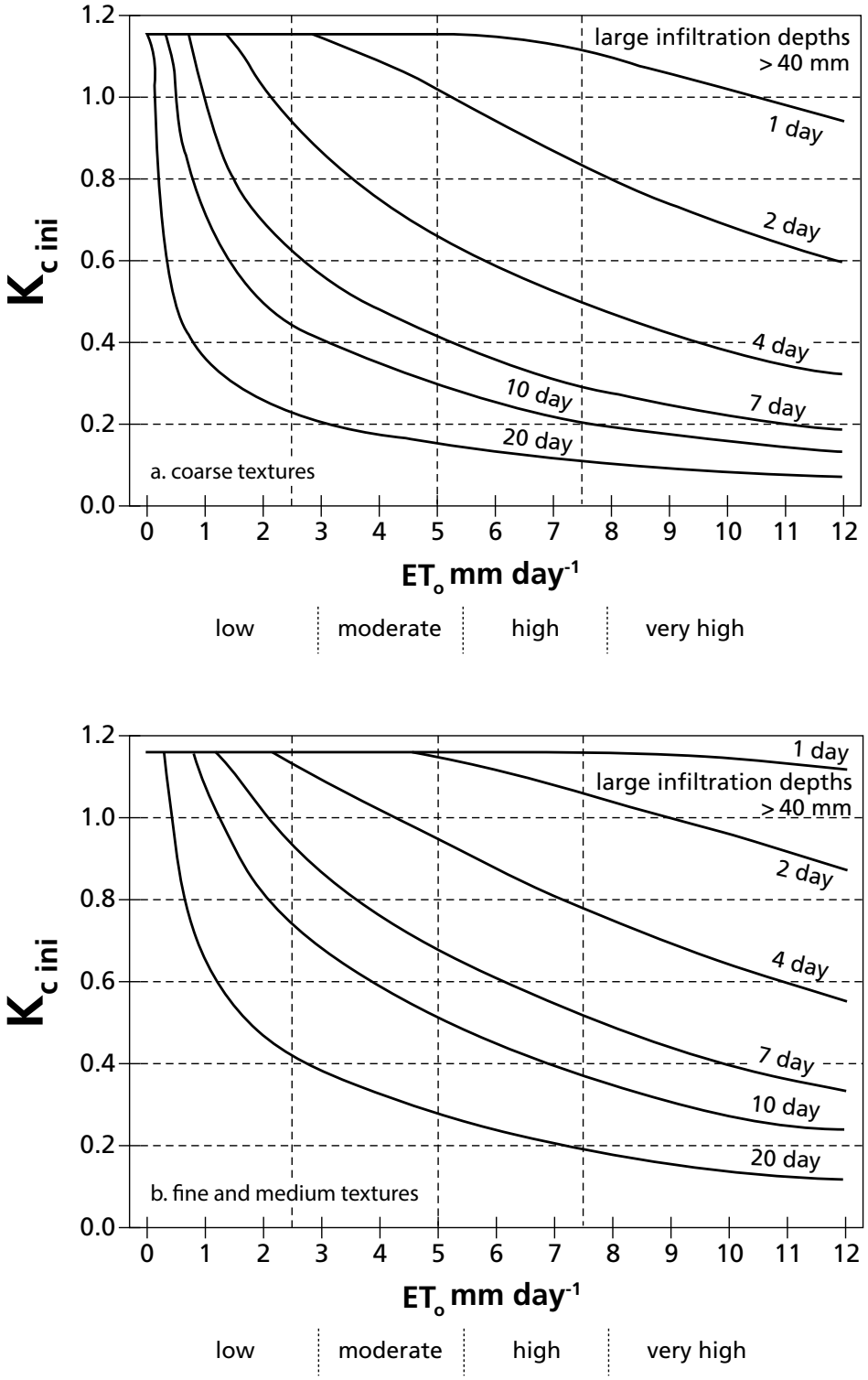
Small vegetables cultivated in a dry area on coarse textured soil receive 20 mm of water twice a week by means of a sprinkler irrigation system. The average ET_o during the initial stage is 5 mm day^{-1}. Estimate the crop evapotranspiration during that stage.

For:	7 days/2 times per week = ET_o = and a coarse textured soil	3.5 5	day interval mm day^{-1}
From Fig. 6.1:	$K_{c\ ini}$ (Fig. 6.1) ≈	0.55	-
From Fig. 6.2.a:	$K_{c\ ini}$ (Fig. 6.2a) ≈	0.70	-
For: From Eq. 6.3: From Eq. 6.1:	I = $K_{c\ ini}$ = 0.55 + [(20 – 10)/(40 – 10)] (0.7 – 0.55) = 0.55 + 0.33 (0.15) = ET_c = 0.60 (5) =	20 0.60 3.0	mm - mm day^{-1}

The average crop evapotranspiration during the initial growth stage for small vegetables is 3.0 mm day^{-1}.

FIGURE 6.2

Average $K_{c\ ini}$ as related to the level of ET_o and the interval between irrigation greater than or equal to 40 mm per wetting event, during the initial growth stage for a) coarse textured soils; b) fine and medium textured soils



Source: FAO, 1998

6.2.4 Adjustment for partial wetting by irrigation

Many types of irrigation systems wet only a fraction of the soil surface. For example, for a trickle irrigation system, the fraction of the surface wetted, f_w, may only be 0.4. For furrow irrigation systems, the fraction of the surface wetted may range from 0.3 to 0.8. Common values for the fraction of the soil surface wetted by irrigation or precipitation are given in Table 7.3 (next chapter). When only a fraction of the soil surface is wetted, the value for $K_{c\ ini}$ obtained from Tables 6.1 – 6.5 or from Figures 6.1 or 6.2 should be multiplied by the fraction of the surface wetted to adjust for the partial wetting (Example 6.4):

$$K_{c\ ini} = f_w\ K_{c\ ini\ (Tab,\ Fig)} \tag{6.3}$$

where f_w the fraction of surfaced wetted by irrigation or rain [0 - 1],
$K_{c\ ini\ (Tab,\ Fig)}$ the value for $K_{c\ ini}$ from Tables 6.1 – 6.5 or Figure 6.1 or 6.2.

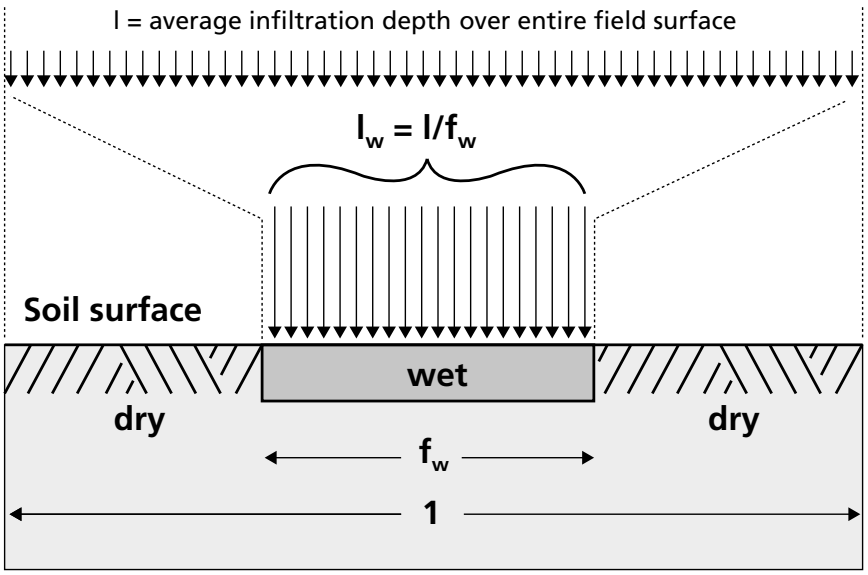
In addition, in selecting which figure to use (i.e. Figure 6.1 or 6.2), the average infiltrated depth, expressed in millimetres over the entire field surface, should be divided by f_w to represent the true infiltrated depth of water that is concentrated over the part of the surface that is wetted (Figure 6.3):

$$I_w = \frac{I}{f_w} \tag{6.4}$$

where I_w irrigation depth for the part of the surface that is wetted [mm],
f_w fraction of surface wetted by irrigation,
I the irrigation depth averaged over the field [mm].

When irrigation of part of the soil surface and precipitation over the entire soil surface both occur during the initial period, f_w should represent the average of f_w for each type of wetting (irrigation and precipitation), weighted according to the total infiltration depth received by each wetting type.

FIGURE 6.3
Partial wetting by irrigation



Source: FAO, 1998

EXAMPLE 6.4
Determination of $K_{c\ ini}$ for partial wetting of the soil surface

Determine the evapotranspiration of the crop in Example 6.2 if it had been irrigated using a trickle system every two days (with 12 mm each application expressed as an equivalent depth over the field area), and where the average fraction of surface wet was 0.4, and where little or no precipitation occurred during the initial period.

The average depth of infiltration per event in the wetted fraction of the surface:

From Eq. 6.4:	$I_w = I / f_w$ = 12 mm / 0.4 =	30	mm
Therefore, one can interpolate between Fig. 6.1 representing light wetting events (~10 mm per event) and Fig. 6.2.b representing medium textured soil and large wetting events (~40 mm per event).			
For:	ET_o = 4 mm day^{-1}	4	mm day^{-1}
and:	a 2-day wetting interval:	-	-
Fig. 6.1 produces:	$K_{c\ ini}$ = 0.85	0.85	-
Fig. 6.2.b produces:	$K_{c\ ini}$ = 1.15	1.15	-
From Eq. 6.2:	$K_{c\ ini}$ = 0.85 + [(30 – 10)/(40 – 10)] (1.15 – 0.85) =	1.05	-

Because the fraction of soil surface wetted by the trickle system is 0.4, the actual $K_{c\ ini}$ for the trickle irrigation is calculated as:

From Eq. 6.3:	$K_{c\ ini} = f_w\ K_{c\ ini\ Fig}$ = 0.4 (1.05) =	0.42	-
	This value (0.42) represents the $K_{c\ ini}$ as applied over the entire field area.		
-	$ET_c = K_{c\ ini}\ ET_o$ = 0.42 (4) =	1.7	mm day^{-1}

The average crop evapotranspiration during the initial growth stage for this trickle-irrigated crop is 1.7 mm day^{-1}.

6.2.5 $K_{c\ ini}$ for trees, shrubs and semi-natural grasses

$K_{c\ ini}$ for trees, shrubs and dryland semi-natural grasses should reflect the ground condition prior to leaf emergence or initiation, and the ground condition during the dormancy or low active period for evergreen trees, shrubs and grasses. The $K_{c\ ini}$ depends upon the amount of grass or weed cover, frequency of soil wetting, tree density and mulch density. For a deciduous orchard in frost-free climates, the $K_{c\ ini}$ can be as high as 0.8 or 0.9 where grass ground cover exists, and as low as 0.3 or 0.4 when the soil surface is kept bare and wetting is infrequent. The $K_{c\ ini}$ for an evergreen orchard with a dormant period has less variation from $K_{c\ mid}$, as exemplified for citrus in Table 6.3. For 50 percent canopy or less, the $K_{c\ ini}$ also reflects ground cover conditions (bare soil, mulch or active grass or weed cover). Relative to grasses, values of $K_{c\ ini}$ in Table 6.4 are good examples.

6.2.6 $K_{c\ ini}$ for paddy rice and flooded wetlands and riparian ecosystems

For rice growing in flooded paddy fields, the ET_c during the initial stage mainly consists of evaporation from the standing water. The $K_{c\ ini}$ in Table 6.2 may be adjusted for the local climate as indicated in Table 6.7, which may also be used for wetlands.

TABLE 6.7
$K_{c\ ini}$ for flooded rice and wetlands for various climatic conditions

Climate, humidity	Wind speed		
	light	moderate	strong
arid/semi-arid	1.10	1.15	1.20
sub-humid/humid	1.05	1.10	1.15
very humid	1.00	1.05	1.10

6.2.7 Background and computations for K_c for the initial stage of annual crops

Evaporation from bare soil (E_s) can be characterized as occurring in two distinct stages. Stage 1 is termed the "energy-limited" stage. During this stage, moisture is transported to the soil surface at a rate sufficient to supply the potential rate of evaporation (E_{so}), which, in turn, is governed by energy availability at the soil surface. In this procedure, E_{so} is estimated from

$$E_{so} = 1.15\ ET_o \tag{6.5}$$

where E_{so} is the potential rate of evaporation [mm d^{-1}] and ET_o is the mean ET_o during the initial period [mm d^{-1}]. The value 1.15 represents increased evaporation potential due to the low albedo of wet soil and the possibility of heat stored in the surface layer during previous dry periods.

Stage 2 is termed the "soil water limited" stage, where hydraulic transport of subsurface water to the soil surface is unable to supply water at the potential evaporation rate. During stage 2, the soil surface appears partially dry and a portion of the evaporation occurs from below the soil surface. The energy required for subsurface evaporation is supplied by the transport of heat from the soil surface into the soil profile. The evaporation rate during stage 2 drying decreases as soil water content decreases as shown in Figure 6.4. The evaporation rate can therefore be expressed as being proportional to the water remaining in the evaporation layer relative to the maximum depth of water that can be evaporated from the same soil layer during stage 2 drying. The background of this procedure is the same as for the computation of the evaporation coefficient K_e in the dual crop coefficient approach dealt with in Chapter 7, where further details are provided.

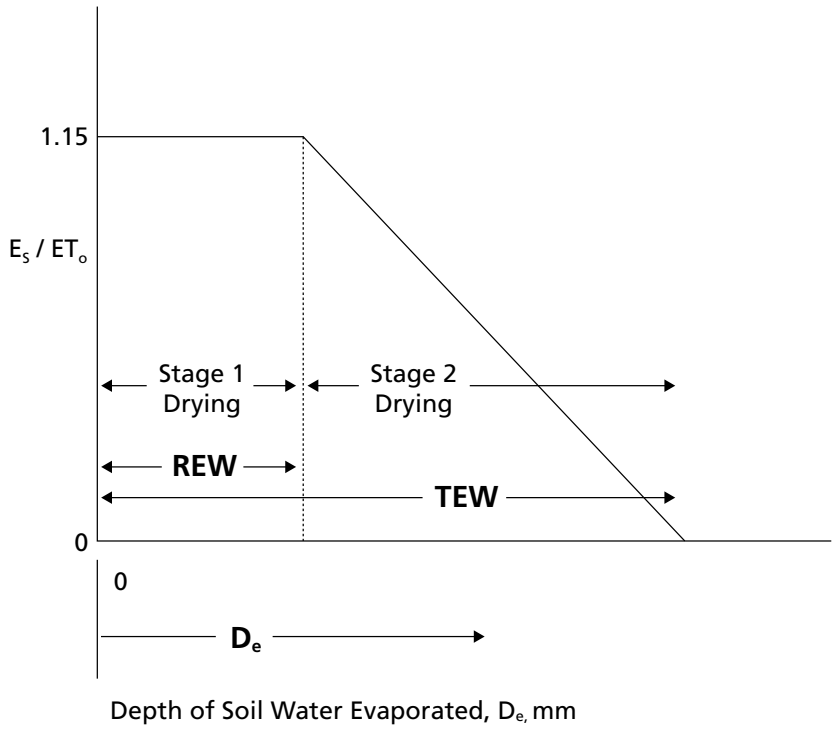
The maximum total depth of water that can be evaporated from the surface soil layer is termed “total evaporable water” or TEW. In turn, the maximum total depth of water that can be evaporated during stage 1 is termed “readily evaporable water” or REW, which is discussed in Chapter 7.

If the evaporation rate during stage 2 drying is assumed to be linearly proportional to the equivalent depth of water remaining in the evaporation layer, as shown in Figure 6.4, then the average soil water evaporation rate during stage 2 can be estimated as

$$E_s = E_{so}\left[\frac{TEW - D_e}{TEW - REW}\right] \quad \text{when } D_e > REW \tag{6.6}$$

where E_s is the actual evaporation rate [mm d^{-1}] at any particular time when the depletion from the soil surface layer equals D_e. E_{so} is the potential rate of evaporation [mm d^{-1}], D_e is the water depletion from the surface layer [mm], TEW and REW are, respectively, the total and the readily evaporable water in the surface layer [mm]. The length of time required to complete stage 1 drying (t_1) is $t_1 = REW/E_{so}$.

FIGURE 6.4
Two stage model for soil evaporation during the initial period



Source: FAO, 1998

6.2.8 General equation for $K_{c\ ini}$

Equation 6.6 can be integrated over the range REW to TEW, resulting in the basic equation for $K_{c\ ini}$ *during stage 2, for* $t_w > t_1$*:*

$$K_{c\ in\ i} = \frac{TEW - (TEW - REW)\ \exp\left(\frac{-(t_w - t_1)\,E_{so}\left(1+\frac{REW}{TEW-REW}\right)}{TEW}\right)}{t_w\ ET_o} \qquad (6.7)$$

where $K_{c\ ini} = E_s/ET_o$, t_w is the mean interval between wetting events [days], and t_1 is the time when stage 1 drying is completed ($t_1 = REW/E_{so}$) [days]. The "exp" represents the exponential of the value contained within the parenthesis following it. The $K_{c\ ini}$ calculated from Equation 6.7 is limited to $K_{c\ ini} \leq 1.15$.

When $t_w < t_1$, i.e. the entire process resides within stage 1, so that:

$$K_{c\ ini} = \frac{E_{so}}{ET_o} = 1.15 \text{ (for } t_w < t_1) \qquad (6.8)$$

Where furrow or trickle irrigation is practiced, and only a portion of the soil surface is wetted, the value calculated for $K_{c\ ini}$ in Equations 6.7 and 6.8 should be reduced in proportion to the average fraction of surface wetted, f_w [0 – 1]. Indicative values for f_w are tabulated in Chapter 7. Equation 6.3 is used to make the adjustment:

$$K_{c\ ini} = f_w\ K_{c\ ini\ (fw\,=\,1)} \qquad (6.9)$$

where f_w is the fraction of surfaced wetted by irrigation or rain [0 – 1], and $K_{c\ ini\ (fw = 1)}$ is the value for $K_{c\ ini}$ for $f_w = 1$ from Equation 6.7 or 6.8. Accordingly, the value for the infiltration depth from irrigation (I_w) should be adjusted using Equation 6.4:

$$I_w = \frac{I}{f_w} \qquad (6.4\ bis)$$

where I_w is the depth of irrigation water that is infiltrated over the part of the surface that is wetted [mm] and I is the depth of water infiltrated from irrigation, expressed as one-dimensional depth over the entire surface area [mm].

BOX 6.1
Equations for Figures 6.1 and 6.2

Figures 6.1 and 6.2 can be reproduced numerically by applying Equation 6.7 under the following conditions. For all applications:

$t_1 = REW_{cor} / E_{so}$ and $E_{so} = 1.15\ ET_o$

If $t_1 < t_w$ then $K_{c\ ini} = 1.15$

Otherwise, apply Equation 6.7 using the following parameters (with TEW_{cor} and REW_{cor} used in place of TEW and REW in Equation 6.7):

For Figure 6.1 (all soil textures having light infiltration depths. < 10 mm):

$TEW_{cor} = 10$ mm

$REW_{cor} = \min (\max (2.5, 6/(ET_o)^{0.5}), 7)$

For Figure 6.2a (coarse soil textures having large infiltration depths, ≥40 mm):

$TEW_{cor} = \min (15, 7\ (ET)^{0.5})$

$REW_{cor} = \min (6, TEW_{cor} - 0.01)$

For Figure 6.2b (medium and fine soil textures having large infiltration depths, ≥ 40 mm):

$TEW_{cor} = \min (28, 13\ (ET)^{0.5})$

$REW_{cor} = \min (9, TEW_{cor} - 0.01)$

6.2.9 Total evaporable water

The value for TEW is the maximum depth of water that can be evaporated from the soil following wetting. The value for TEW is governed by the depth of the soil profile contributing to soil water evaporation and by the soil water holding properties within the evaporating layer. In addition, the value for TEW is affected by the unsaturated hydraulic conductivity, by the presence of a hydraulically limiting layer beneath the evaporating layer, and by the conduction of sensible heat into the soil to supply energy for subsurface evaporation. An approximation for the maximum value of TEW for initial periods having $ET_o \geq 5$ mm d^{-1} is:

$$TEW = 1\ 000\ (\theta_{FC} - 0.5\ \theta_{WP})\ Z_e \qquad (6.10)$$

where TEW has units of mm, θ_{FC} is soil water content at field capacity [$m^3\ m^{-3}$], θ_{WP} is soil water content at wilting point [$m^3\ m^{-3}$], and Z_e is the depth of the soil surface layer that is subject to drying by evaporation [0.10 to 0.15 m]. If unknown, a value of $Z_e = 0.15$ m is recommended. Typical values for θ_{FC} and θ_{WP} are given in Chapter 7.

During winter and other cool season months, less radiation energy is available to penetrate the soil surface and to evaporate water from within a drying soil, and TEW may be less. Therefore, when $ET_o < 5$ mm d^{-1}, TEW for use in Equation 6.7 is estimated as:

$$TEW = 1\,000\,(\theta_{FC} - 0.5\,\theta_{WP})\,Z_e\sqrt{\frac{ET_o}{5}} \qquad (6.11)$$

where ET_o is reference ET in mm d^{-1}. Equation 6.11 is intended to correct TEW for use during the initial stage with mostly bare soil. It is not intended for use with the dual K_c procedure. REW is limited so that REW ≤ TEW.

6.2.10 Number of wetting events and average depth

Estimating the number of wetting events and the corresponding time between wetting events during the initial period was previously described. The number of wetting events (both from precipitation and irrigation) occurring during the initial period is determined by considering that two wetting events occurring on adjoining days can be counted as one event, and individual wetting events of less than 0.2 ET_o can be ignored.

The average time between wetting events during the initial period (t_w) is approximated as:

$$t_w = \frac{L_{ini}}{n_w + 0.5} \qquad (6.12)$$

where t_w is in days, L_{ini} is the length of the initial period [days], and n_w is the number of wetting events during the initial period.

The average depth of water added to the evaporating layer at each wetting event is determined by dividing the sum of the precipitation and irrigation infiltration occurring during all wetting events by the number of events, thus:

$$P_{mean} = \frac{\Sigma\, p_n + \Sigma\, I_w}{n_w} \qquad (6.13)$$

where P_{mean} is the average depth of infiltrated water per wetting event [mm], P_n is the depth of infiltrated precipitation occurring during the initial period, and I_w is the infiltrated irrigation depth for the part of the surface that is wetted [mm] (Equation 6.4). Each individual value of P_n and I_w must be limited in Equation 6.13, so that $P_n \le TEW$ and $I_w \le TEW$ where TEW is from Equation 6.10 or 6.11.

6.2.11 Limitations on TEW and REW

In the case of wetting depths (P_{mean}) that are smaller than the TEW, the evaporation process, including stage 1 drying, may terminate sooner than expected. The actual values for TEW and REW must be corrected according to P_{mean}. Therefore, TEW and REW are calculated according to the average total water available during each drying cycle:

$$TEW_{cor} = \min\left(TEW,\, P_{mean} + \frac{W_{ini}}{n_w}\right) \qquad (6.14)$$

and

$$REW_{cor} = REW \left[\min \left(\frac{P_{mean} + \frac{W_{ini}}{n_w}}{T\,EW}, 1 \right) \right] \quad (6.15)$$

where "min ()" is a function to select the minimum value of those in braces that are separated by the comma, and where TEW is from Equation 6.10 or 6.11. W_{ini} is the equivalent depth of water [mm] in the evaporation layer (of thickness Z_e) at the time of planting (beginning of the initial period). W_{ini} has a maximum value of TEW when the initial soil water content of the evaporation layer is at field capacity. Values for TEW_{cor} and REW_{cor} from Equations 6.14 and 6.15 are used in place of TEW and REW in Equation 6.7. Example 6.5 shows the numerical application of this equation to small vegetables.

EXAMPLE 6.5
Application of Equation 6.7 to an example for small vegetables

As in Example 6.3, small vegetables are cultivated in a dry area on coarse textured soil and receive 20 mm of water twice a week by means of a sprinkler irrigation system. The average ET_o during the initial stage is 5 mm day^{-1}. Estimate the crop evapotranspiration during that stage.

For:	t_w = 7 days per week/2 times per week =	3.5	day interval
	ET_o =	5	mm day^{-1}
	E_{so} = 1.15 ET_o = 1.15 (5) =	5.75	mm day^{-1}
For Fig. 6.1:	TEW_{cor} =	10	mm
	REW_{cor} = min(max(2.5, 6/($5^{0.5}$), 7) =	2.7	mm
	t_1 = REW/E_{so} = 2.7/5.75 =	0.47	days
	since $t_w > t_1$, use Eq. 6.7: $K_{c\ ini\ (Fig.\ 6.1)}$ = (10 – (10–2.7) exp[–(3.5–0.47)(5.75)(1+2.7/(10–2.7))/10])/(3.5(5)) =	0.57	-
For Fig. 6.2a:	TEW_{cor} = min (15, 7($5^{0.5}$)) =	15	mm
	REW_{cor} = min (6, 15–0.001) =	6	mm
	t_1 = REW/E_{so} = 6/5.75 =	1.04	days
	since $t_w > t_1$, use Eq. 6.7: $K_{c\ ini\ (Fig.\ 6.2a)}$ = (15–(15–6) exp[–(3.5–1.04)(5.75)(1+6/(15–6))/15])/(3.5(5)) =	0.75	-
For:	I =	20	mm
From Eq. 6.2:	$K_{c\ ini}$ = 0.57 + [(20–10)/(40–10)] (0.75–0.57) = 0.57 + 0.33(0.12) =	0.63	-
From Eq. 6.1:	ET_c = 0.63 (5) =	3.2	mm day^{-1}

The average crop evapotranspiration during the initial growth stage for the small vegetables is 3.2 mm day^{-1}. The values in this example agree relatively closely with those obtained from Example 6.3.

The numerical application of Equation 6.7 using the parameters and constraints listed here will fully reproduce Figures 6.1 and 6.2a and b. However, calculations made in the vicinity of ET_o = 5 mm d^{-1} may deviate from the curves in Figures 6.2a and b, because the curves were smoothed before plotting. The smoothing caused small, insignificant differences between the figures and the numerical procedure. The parameters listed above are reduced from Equations 6.10 through 6.15 and use typical values for θ_{FC} and θ_{WP}. In situations where wetting events are not equally spaced during the initial period, the dual K_c approach presented in Chapter 7, along with a daily soil water balance, can provide more accurate results.

6.3 CROP COEFFICIENT FOR THE MID-SEASON STAGE ($K_{c\ mid}$)

6.3.1 Illustration of the climatic effect

As discussed in Chapter 5, the effect of the difference in aerodynamic properties between the grass reference surface and agricultural crops is not only crop specific but also varies with the climatic conditions and crop height (Figures 5.2 and 5.4). More arid climates and conditions of greater wind speed will have higher values for $K_{c\ mid}$. More humid climates and conditions of lower wind speed will have lower values for $K_{c\ mid}$. These variations are caused by the effects of greater aerodynamic roughness for many vegetation types as compared to the standardized grass reference ET_o.

The relative impact of climate on $K_{c\ mid}$ is illustrated in Figure 6.5 where the adjustments to the values from Tables 6.1 – 6.5 are shown for various types of climates, mean daily wind speed and various crop heights. It must be noted that $K_{c\ mid}$ increases are progressively reduced when crop height increases, and that $K_{c\ mid}$ increases are larger when wind speed is higher. As an example, expected variations for $K_{c\ mid}$ for wheat in response to regional climatic conditions are presented in Box 6.2.

6.3.2 Adjustment of $K_{c\ mid}$ for climate

For specific adjustment in climates where RH_{min} differs from 45 percent, or where u_2 is larger or smaller than 2.0 m s^{-1}, the $K_{c\ mid}$ values from Table 6.1 – 6.5 are adjusted as:

$$K_{c\ mid} = K_{c\ mid\ (Tab)} + [0.04\ (u_2 - 2) - 0.004\ (RH_{min} - 45)] \left(\frac{h}{3}\right)^{0.3} \qquad (6.16)$$

where $K_{c\ mid\ (Tab)}$ value for $K_{c\ mid}$ taken from Tables 6.1 – 6.5,

u_2 mean value for daily wind speed at 2 m height over grass during the mid-season growth stage [m s^{-1}], for 1 m $s^{-1} \leq u_2 \leq 6$ m s^{-1},

RH_{min} mean value for daily minimum relative humidity during the mid-season growth stage [%], for 20% $\leq RH_{min} \leq$ 80%,

h mean plant height during the mid-season stage [m] for 0.1 m < h < 10 m.

The $K_{c\ mid}$ and $K_{c\ end}$ values determined with equations 6.16 and 6.19 are average adjustments for the mid-season and late season periods. The values for parameters u_2 and RH_{min} should be accordingly taken as averages for these periods. When observed meteorological data are not available, approximate values may be used, such as from Tables 6.8 and 6.9. The limits expressed for parameters u_2, RH_{min} and h should be observed.

BOX 6.2
Demonstration of the effect of climate on $K_{c\ mid}$ for wheat crop grown under field conditions

From Table 6.2 for wheat: $K_{c\ mid}$ = 1.15 and h = 1.0 m and using Figure 6.5 or Eq. 6.16:

For semi-arid to arid conditions

- for strong wind (4 m s^{-1}) $K_{c\ mid}$ = 1.15 + 0.10 = 1.25
- for moderate wind (2 m s^{-1}) $K_{c\ mid}$ = 1.15 + 0.05 = 1.20
- for calm wind (1 m s^{-1}) $K_{c\ mid}$ = 1.15 + 0.00 = 1.15

For sub-humid conditions

- for strong wind (4 m s^{-1}) $K_{c\ mid}$ = 1.15 + 0.05 = 1.20
- for moderate wind (2 m s^{-1}) $K_{c\ mid}$ = 1.15 + 0.00 = 1.15
- for calm wind (1 m s^{-1}) $K_{c\ mid}$ = 1.15 – 0.05 = 1.10

For humid and very humid conditions

- for strong wind (4 m s^{-1}) $K_{c\ mid}$ = 1.15 – 0.05 = 1.10
- for moderate wind (2 m s^{-1}) $K_{c\ mid}$ = 1.15 – 0.10 = 1.05
- for calm wind (1 m s^{-1}) $K_{c\ mid}$ = 1.15 – 0.15 = 1.00

Depending on the aridity of the climate and the wind conditions, the crop coefficient for wheat during the mid-season stage ranges from 1.00 (humid and calm wind) to 1.25 (arid and strong wind). This represents a range of 20 percent in $K_{c\ mid}$ across climate conditions.

EXAMPLE 6.6
Determination of $K_{c\ mid}$

Calculate $K_{c\ mid}$ for maize crops near Taipei, Taiwan and near Mocha, Yemen. The average mean daily wind speed (u_2) during the mid-season stage at Taipei is about 1.3 m s^{-1} and the minimum relative humidity (RH_{min}) during this stage averages 75 percent. The average u_2 during the mid-season near Mocha is 4.6 m s^{-1} and the RH_{min} during this stage averages 44 percent.

From Table 6.2, the value for $K_{c\ mid}$ is 1.20 for maize. The value for h from Table 6.2 is 2.5 – 3.5 m. However, for the variety grown here, the midseason height is 2 m. Therefore h = 2 m is used here. Using Equation 6.16

For Taipei (humid climate):

$$K_{c\ mid} = 1.20 + [0.04\ (1.3 - 2) - 0.004\ (75 - 45)]\left(\frac{2}{3}\right)^{0.3} = 1.07$$

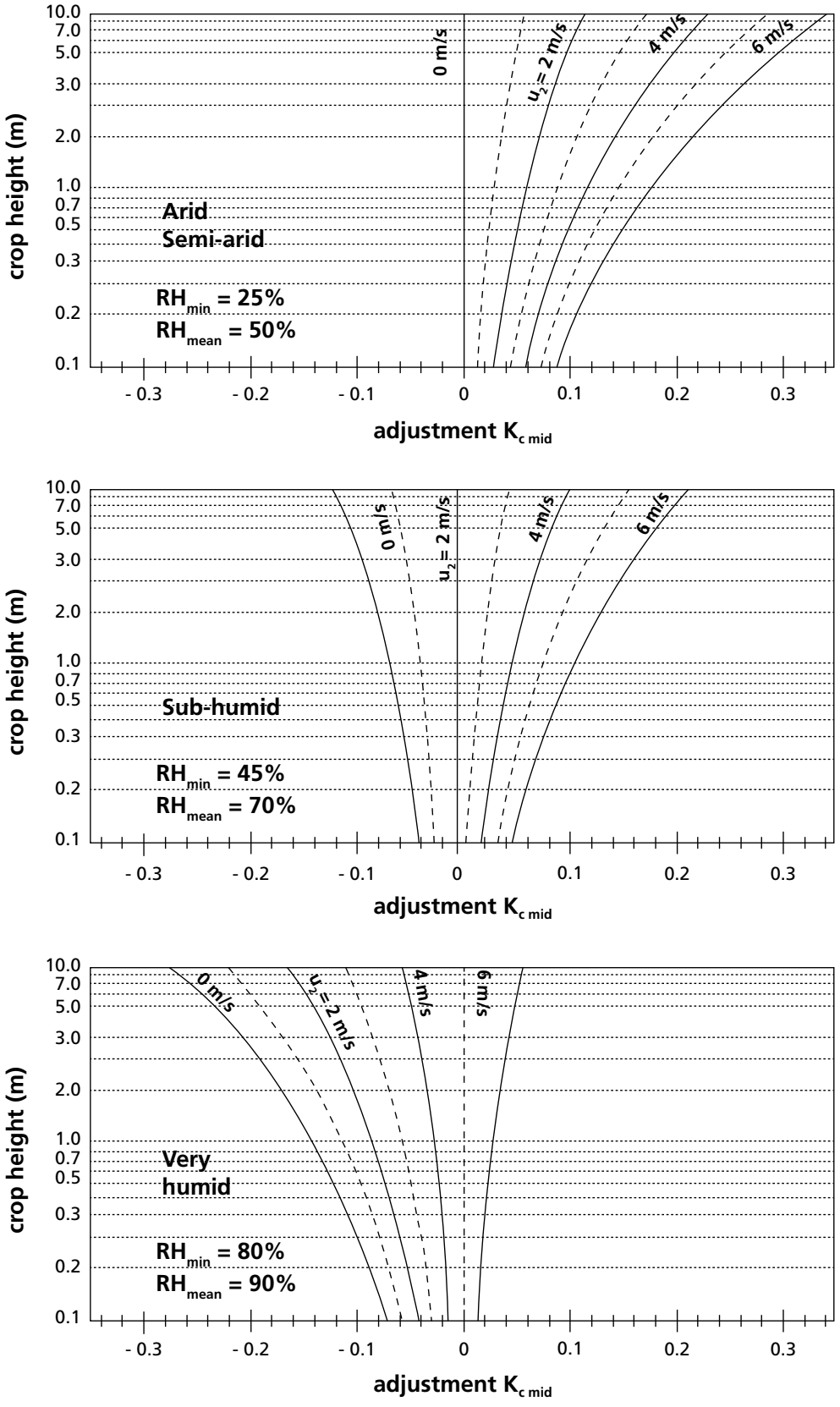
For Mocha (arid climate):

$$K_{c\ mid} = 1.20 + [0.04\ (4.6 - 2) - 0.004\ (44 - 45)]\left(\frac{2}{3}\right)^{0.3} = 1.30$$

The average crop coefficient predicted for the mid-season stage is 1.07 for Taipei and 1.30 for Mocha.

FIGURE 6.5

Adjustment (additive) to the $K_{c\ mind}$ values from Tables 6.1 - 6.5 for different crop heights and mean daily wind speeds (u_2) for different humidity conditions



Source: FAO, 1998

RH_{min} is used in Equation 6.16 rather than RH_{mean} because it is easier to approximate RH_{min} from T_{max} where relative humidity data are unavailable. Moreover, under the common condition where T_{min} approaches T_{dew} (i.e. $RH_{max} \approx 100$ percent), the vapour pressure deficit ($e_s - e_a$), with e_s from Equation 3.7 and e_a from Equation 3.13, becomes $[(100–RH_{min})/200]\ e°(T_{max})$, where $e°(T_{max})$ is saturation vapour pressure at maximum daily air temperature. This indicates that RH_{min} better reflects the impact of vapour pressure deficit on K_c than does RH_{mean}. RH_{min} is calculated on a daily or even an average monthly basis as:

$$RH_{min} = \frac{e°\,(T_{dew})}{e°\,(T_{max})}\ 100 \qquad (6.17)$$

where T_{dew} is mean dewpoint temperature and T_{max} is mean daily maximum air temperature during the mid-season growth stage. Where dewpoint temperature or other hygrometric data are not available or are of questionable quality, RH_{min} can be estimated by substituting mean daily minimum air temperature, T_{min}, for T_{dew}. However, in the case of arid and semi-arid climates, T_{min} in equation (6.18) should be assessed as to whether they need to be adjusted as indicated in Chapter 2 (Equations 2.13 and 2.14) by reducing values for T_{min} to better approximate T_{dew} Then:

$$RH_{min} = \frac{e°\,(T_{min})}{e°\,(T_{max})}\ 100 \qquad (6.18)$$

The values for u_2 and RH_{min} need only be approximate for the mid-season growth stage. This is because Equation 6.16 is not strongly sensitive to these values, changing 0.04 per 1 m s^{-1} change in u_2 and per 10 percent change in RH_{min} for a 3 m tall crop. Measurements, calculation and estimation of missing wind and humidity data are provided in Chapter 2. Wind speed measured at other than 2 m height should be adjusted to reflect values for wind speed at 2 m over grass using Equation 3.50. Where no data on u_2 or RH_{min} are available, the general classification for wind speed and humidity data given in Tables 6.8 and 6.9 can be used.

TABLE 6.8
Empirical estimates of monthly wind speed data

Description	mean monthly wind speed at 2 m
Light wind	...≤ 1.0 m s^{-1}
Light to moderate wind	2.0 m s^{-1}
Moderate to strong wind	4.0 m s^{-1}
Strong wind	... ≥ 5.0 m s^{-1}
General global conditions	**2 m s^{-1}**

TABLE 6.9
Typical values for RH_{min} compared with RH_{mean} for general climatic classifications

Climatic classification	RH_{min} (%)	RH_{mean} (%)
Arid	20	45
Semi-arid	30	55
Sub-humid	45	70
Humid	70	85
Very humid	80	90

Equation 6.16 is valid for vegetable and field crops with mean plant heights up to around 3 m. For plant heights shorter than 0.1 m, vegetation will behave aerodynamically similar to grass reference and Equation 6.16 should not be applied. In case of fruit trees and vines, the correction for climate is less necessary unless transfers of $K_{c\ mid}$ values are done for very different climates because orchard architecture and crop training play a large role in plant's aerodynamic behaviour. Example values for h are listed in Tables 6.1 – 6.3 for various crops but the mean plant height can vary greatly with crop cultivar and cultural practices. Therefore, wherever possible, h should be obtained from general field observations. However, the presence of the 0.3 exponent in Equation 6.16 makes the equation relatively insensitive to small errors in the value used for h. Generally, only a single value for h is used in Equation 6.16 to represent the mid-season period. Example 6.6 compares applications of Equation 6.16 for the same crop in different climates.

6.3.3 Adjustment for frequency of wetting

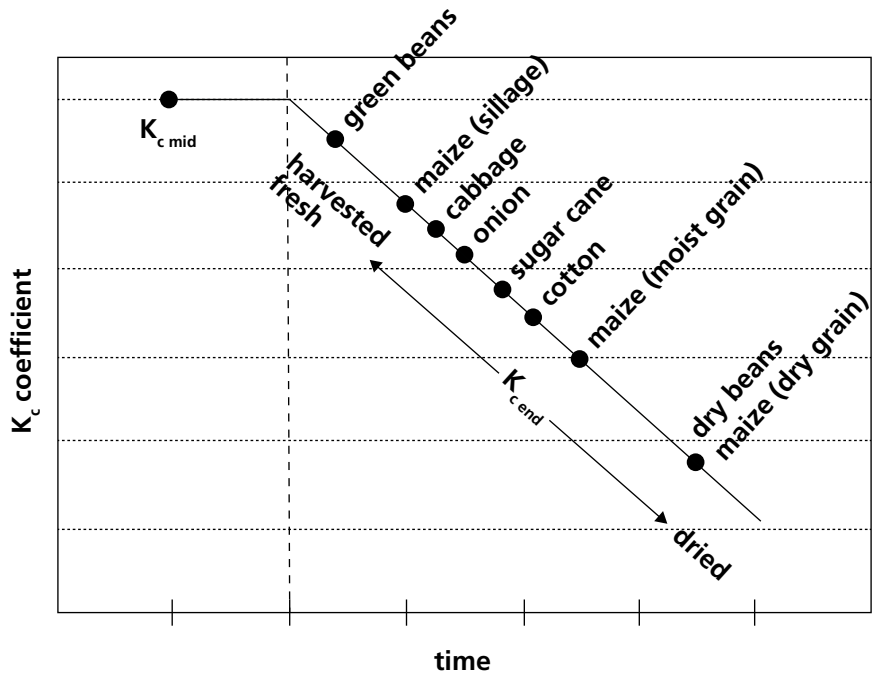
$K_{c\ mid}$ is less affected by wetting frequency than is $K_{c\ ini}$, as vegetation during this stage is generally near full ground cover so that the effects of surface evaporation on K_c are smaller. For frequent irrigation of crops (more frequently than every 3 days) and where the $K_{c\ mid}$ of Tables 6.1 – 6.2 is less than 1.0, the value can be replaced by approximately 1.1-1.3 to account for the additional combined effects of continuously wet soil, evaporation due to interception (sprinkler irrigation) and roughness of the vegetation, especially where the irrigation system moistens an important fraction of the soil surface ($f_w > 0.3$).

6.4 CROP COEFFICIENT FOR THE END OF THE LATE SEASON STAGE ($K_{c\ end}$)

Typical values for the crop coefficient at the end of the late season stage, $K_{c\ end}$, are listed in Tables 6.1 – 6.5 for various agricultural crops and land covers. The values given for $K_{c\ end}$ reflect crop and water management practices particular to those crops and vegetation. If the crop is irrigated frequently until harvested fresh, the topsoil remains wet and the $K_{c\ end}$ value will be relatively high. On the other hand, crops that are allowed to senesce and dry out in the field before harvest receive less frequent irrigation or no irrigation at all during the late season stage. Consequently, both the soil surface and vegetation are dry and the value for $K_{c\ end}$ will be relatively small (Figure 6.6).

Where the local water management and harvest timing practices are known to deviate from the typical ones, the user should make adequate adjustments to the values for $K_{c\ end}$. Some guidance on the adjustment of K_c values for wetting frequency is provided in Chapter 7. For premature harvest, the user can construct a K_c curve using the $K_{c\ end}$ value provided in Tables 6.1 – 6.5 and a late season length typical of a normal harvest date; and then the user should terminate the constructed curve early, corresponding to the time of the early harvest. The $K_{c\ end}$ values in Tables 6.1 – 6.5 are typical values expected for the average $K_{c\ end}$ under the standard climatic conditions.

FIGURE 6.6
Ranges expected for $K_{c\ end}$



Source: FAO, 1998

For specific adjustment in climates where RH_{min} differs from 45 percent and/or where u_2 is larger or smaller than 2.0 m s^{-1}, and when $K_{c\ end\ (Tab)} > 0.45$, the following equation similar to Equation 6.16 can be used:

$$K_{c\ end} = K_{c\ end\ (Tab)} + [0.04\ (u_2 - 2) - 0.004\ (RH_{min} - 45)] \left(\frac{h}{3}\right)^{0.3} \quad (6.19)$$

where $K_{c\ end\ (Tab)}$ value for $K_{c\ end}$ taken from Tables 6.1 – 6.5, and when $K_{c\ end\ (Tab)} > 0.45$,

u_2 mean value for daily wind speed at 2 m height over grass during the late season growth stage [m s^{-1}], for 1 m $s^{-1} \leq u_2 \leq 6$ m s^{-1},

RH_{min} mean value for daily minimum relative humidity during the late season stage [%], for $20\% \leq RH_{min} \leq 80\%$,

h mean plant height [m] during the late season, for 0.1 m ≤ h ≤ 15 m.

Equation 6.19 reduces the $K_{c\ end}$ with increasing RH_{min}. This reduction in $K_{c\ end}$ occurs for crops that are harvested "green" or before becoming dry, i.e. when $K_{c\ end} \geq 0.45$. When crops are allowed to senesce and dry in the field ($K_{c\ end} < 0.45$), u_2 and RH_{min} have less effect on $K_{c\ end}$ and no adjustment is necessary. In fact, $K_{c\ end}$ may decrease with decreasing RH_{min} for crops that are ripe and dry at the time of harvest, as lower relative humidity accelerates the drying process. Due to the characteristics of wetlands and riparian vegetation, and of semi-natural rainfed grasslands, the adjustment for climate may only seldom apply to $K_{c\ end}$ of these types of vegetation independently of the respective $K_{c\ end}$ in Tables 6.4 and 6.5.

The tabulated $K_{c\ end}$ values for fruit trees and shrubs listed are typical values for standard climatic conditions and may need to be adjusted by using Equations 6.16 and 6.19 where RH_{min} or u_2 largely differ from the standard climate. Seasonal adjustments for climate may not be required when variations in ET_c depend mostly on variations in ET_o.

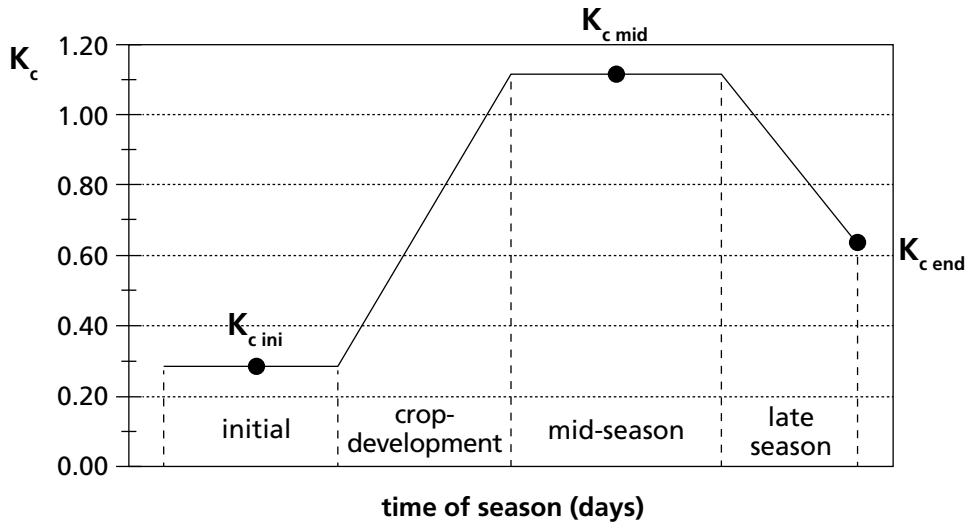
6.5 CONSTRUCTION OF THE K_c CURVE

6.5.1 Annual crops

Only three-point values for K_c are required to describe and to construct the K_c curve. The curve such as that shown in Figure 6.7 is constructed using the following steps:

1. Divide the growing period into four general growth stages that describe crop phenology or development (initial, crop development, mid-season, and late season stage), determine the lengths of the growth stages, and identify the three K_c values that correspond to $K_{c\ ini}$, $K_{c\ mid}$ and $K_{c\ end}$ from Tables 6.1 – 6.5.
2. Adjust the K_c values to the frequency of wetting and/or climatic conditions of the growth stages as outlined in the previous section.
3. Construct a curve (following Chapter 5) by connecting straight line segments through each of the four growth stages. Horizontal lines are drawn through $K_{c\ ini}$ in the initial stage and through $K_{c\ mid}$ in the mid-season stage. Diagonal lines are drawn from $K_{c\ ini}$ to $K_{c\ mid}$ within the course of the crop development stage and from $K_{c\ mid}$ to $K_{c\ end}$ within the course of the late season stage (Figure 6.7).
4. Consider a downward adjustment to the K_c values in cases of inadequate soil water and consequent water stress, as well as in case of stress due to salinity, lack of fertility and inadequacy of other crop practices, as indicated early in this chapter or as described in Chapters 8 and 9.

FIGURE 6.7
Crop coefficient curve

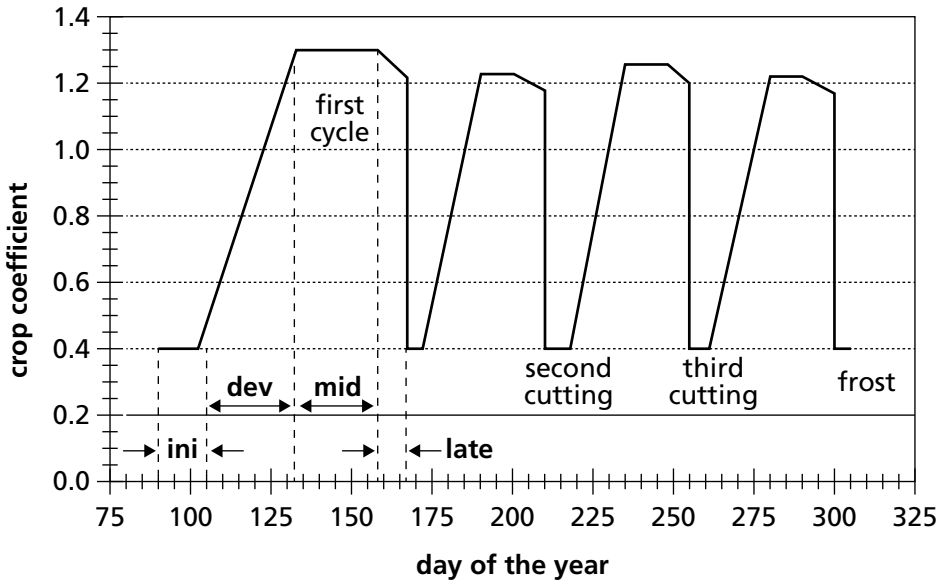


Source: FAO, 1998

6.5.2 K_c curves for forage crops

Many crops grown for forage or hay are harvested/cut several times during the growing season. Each cut essentially terminates a "sub" growing cycle and an associated K_c curve, and initiates a new "sub" growing cycle and associated K_c curve. The resulting K_c curve for the entire growing season is the aggregation of a series of K_c curves associated with each sub-cycle. Figure 6.8 presents a K_c curve for the entire growing season constructed for alfalfa grown for hay in southern Idaho, USA. A larger number of cuts may be considered for other locations depending on forage species, management of the forage or grassland and on climate (see Table 6.4).

FIGURE 6.8
Constructed curve for K_c alfalfa hay in southern Idaho, the United State using values from Table 6.4 and adjusted using Equation 6.18 and 6.21 (data from J.L. Wright, USDA-ARS (ret.) 1990)



Source: FAO, 1998

In the southern Idaho climate, greenup (leaf initiation) begins in the spring on about day 90 of the year. The crop is usually cut for hay three or four times during the growing season. Therefore, Figure 6.8 shows four K_c sub-cycles or cutting cycles: sub-cycle 1 follows greenup in the spring and the three other K_c sub-cycles follow cuttings. These lengths were based on observations. In the southern Idaho climate, frosts terminate the growing season sometime in the fall, usually around day 280-290 of the year (early to mid-October). Cuttings create a ground surface with less than 10 percent vegetation cover. In climates with a very cold winter, cutting cycle 1 is often longer in duration than the following cycles due to lower air and soil temperatures during this period that reduce crop growth rates.

The magnitudes of the K_c values during the mid-season periods of each cutting cycle shown in Figure 6.8 vary from cycle to cycle due to adjusting the values for $K_{c\,mid}$ and $K_{c\,end}$ for each cutting cycle period using Equations 6.16 and 6.19. In applying these two adjustment equations, the u_2 and RH_{min} values were averaged for the mid-season and late season stages within each cutting cycle. However, in practice, these adjustments may not be required for the tabulated values of hay grasses in Table 6.4 due to the wide range of varieties. Basal K_{cb} curves similar to Figure 6.8 can be constructed for forage or hay crops, following procedures presented in Chapter 7 and as tabulated there.

6.5.3 $K_{c\ mid}$ when effects of individual cutting periods are averaged

The user may wish to average the effects of cuttings for a forage crop over the course of the growing season. When cutting effects are averaged, then only a single value for $K_{c\ mid}$ and only a single K_c curve need to be employed for the whole growing season. When this is the case, a "single" K_c curve is constructed as in Figure 6.7, where only one midseason period is considered for the forage crop. The $K_{c\ mid}$ for this total midseason period must average the effects of grazing or hay cuttings. The value that is used for $K_{c\ mid}$ is therefore an average of the K_c curve for the time period starting at the first attainment of full cover and ending at the beginning of the final late season period, often near dormancy or frost. The value used for $K_{c\ mid}$ under these averaged conditions may be only about 80 percent of the K_c value that represents full ground cover as in Figure 6.8. For example, for alfalfa hay, the average, full-season $K_{c\ mid}$ often is 1.05, whereas the $K_{c\ mid}$ for an individual cutting period is 1.20.

6.6 CALCULATING ET_c

The K_c value for any period during the growing period can be graphically or numerically determined from the crop coefficient curve. Once the K_c values have been derived, the crop evapotranspiration, ET_c, can be calculated by multiplying the K_c values by the corresponding ET_o values.

6.6.1 Graphical determination of K_c

Weekly, ten-day or monthly values for K_c are necessary when ET_c calculations are made on weekly, ten-day or monthly time steps. A general procedure is to construct the K_c curve, overlay the curve with the lengths of the weeks, decades or months, and to derive graphically from the curve the K_c value for the period under consideration (Figure 6.7). Since all decades have a duration of 10 days, adopting decades as time periods facilitates the derivation of K_c and introduces little error in the calculation of ET_c. The constructed K_c curve in Figure 6.9 relative to a dry bean crop (detailed in Box 6.3), was used to construct the curve in Figure 6.10. This curve has been overlaid with the lengths of the decades. K_c values of 0.15, 1.19 and 0.35 and the actual lengths for growth stages equal to 25, 25, 30 and 20 days were used. The crop was planted at the beginning of the last decade of May and was harvested 100 days later.

BOX 6.3

Case study of a dry bean crop at Kimberly, Idaho, the United States (single crop coefficient)

An example application for using the K_c procedure under average soil wetness conditions is presented for a dry bean crop planted on 23 May 1974 at Kimberly, Idaho, the United States (latitude = 42.4°N). The initial, development, mid-season and late season stage lengths are for a continental climate as 20, 30, 40 and 20 days. Initial values for $K_{c\ ini}$, $K_{c\ mid}$ and $K_{c\ end}$ are selected from Table 6.2 as 0.4, 1.05, and 0.40. The $K_{c\ mid}$ = 1.05 was adjusted to $K_{c\ mid}$ = 1.15 to account for the very dense spacing of the dry bean crop.

The mean RH_{min} and u_2 during both the mid-season and late season growth stages were 30 percent and 2.2 m s^{-1}. The maximum height suggested in Table 6.2 for dry beans is 0.50-0.70 m. However, local observations indicate a height of only 0.4 m that is used herein. Therefore, $K_{c\ mid}$ is adjusted for climate using Equation 6.16 as:

$$K_{c\ mid} = 1.15 + [0.04\ (2.2 - 2) - 0.004\ (30 - 45)]\left(\frac{0.4}{3}\right)^{0.3} = 1.19$$

BOX 6.3 (CONT.)

As $K_{c\ end}$ = 0.40 is less than 0.45, no adjustment is made to $K_{c\ end}$. The value for $K_{c\ mid}$ is not significantly different from the initial $K_{c\ mid}$ = 1.15 as $u_2 \approx 2$ m/s, RH_{min} is just 15 percent lower than the 45 percent represented in Table 6.2, and the height of the beans is relatively short. The initial K_c curve for dry beans in Idaho can be drawn, for initial, planning purposes, as shown by the dotted line in the graph of Figure 6.9, where $K_{c\ ini}$, $K_{c\ mid}$ and $K_{c\ end}$ are 0.4, 1.19, and 0.35 and the four lengths of growth stages are initially estimated to be 20, 30, 40 and 20 days. Note that the $K_{c\ ini}$ = 0.4 taken from Table 6.2 serves only as an initial, approximate estimate for $K_{c\ ini}$.

Adjustment of $K_{c\ ini}$

$K_{c\ ini}$ can be more accurately estimated using the approach described in this chapter. ET_o during the initial period at Kimberly (late May - early June, 1974) averaged 5.3 mm day^{-1}, and the wetting interval during this period was approximately 14 days (2 rainfall events occurred averaging 5 mm per event). Therefore, as the wetting events were light (< 10 mm each), Figure 6.1 is used. The soil texture at Kimberly, Idaho is silt loam. From Figure 6.1, $K_{c\ ini}$ for the 14-day wetting interval and ET_o = 5.3 mm day^{-1} is about 0.15. This value is substantially less than the general 0.4 value suggested by Table 6.2, and emphasizes the need to utilize local, actual precipitation and irrigation data when determining $K_{c\ ini}$.

Comparison of constructed curves with measurements (see Figure 6.9)

Because the ET_c data for the dry bean crop at Kimberly, Idaho were measured using a precision lysimeter system during 1974 by Wright (1990), the actual K_c measurements can be compared with the constructed K_c curves, where actual K_c was calculated by dividing lysimeter measurements of ET_c by daily ET_o estimated using the FAO Penman-Monteith equation.

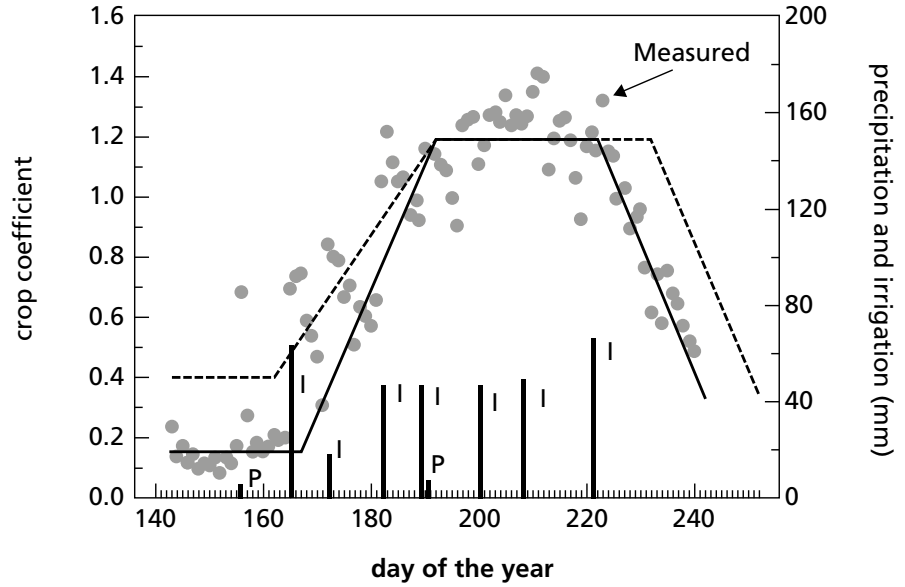
As illustrated in the graph of Figure 6.9, the mid-season length as taken for the general, continental climate overestimated the true mid-season length for dry beans in southern Idaho, which averaged only about 30 days rather than 40 days as formerly suggested. This illustrates the importance of using the local observation of 30 days for mid-season period length rather than a general value.

The final, best estimate for the K_c curve for the dry bean crop is plotted (lower curve in graph of Figure 6.9) using K_c values of 0.15, 1.19, and 0.35 and the actual observed lengths of growth stages equal to 25, 25, 30 and 20 days. Note the impact that the error in estimating mid-season length has on the area under the K_c curve. This supports the need to obtain local observations of growth stage dates and lengths.

The value calculated for $K_{c\ mid}$ (1.19) appears to have underestimated the measured value for K_c during portions of the mid-season period at Kimberly. Some of this effect was due to the impacts of increased soil water evaporation following four irrigations during the 1974 mid-season, which increased the effective K_c. This is illustrated in Box 7.1 of Chapter 7, where the basal $K_{cb} + K_e$ approach is introduced and demonstrated for this same example.

The 0.15 value calculated for $K_{c\ ini}$ using Figure 6.1 agrees closely with measured K_c during the initial period. Measured K_c during the development period exceeded the final K_c curve during days on or following wetting events. The day-to-day variation in the lysimeter measured K_c (Figure 6.9) is normal and is caused by day-to-day variations in weather, in wind direction, by errors in the prediction of R_n and ET_o, and by some random errors in the lysimeter measurements and weather measurements.

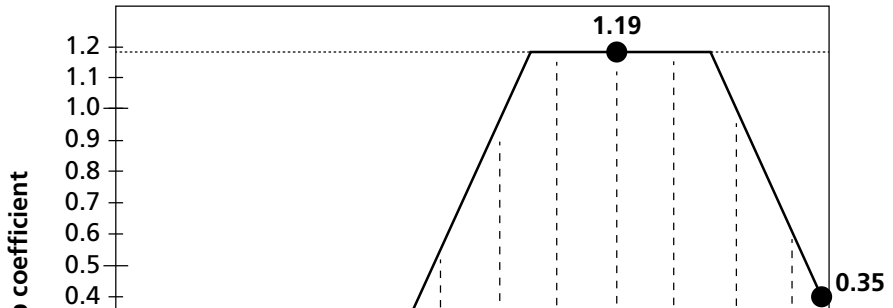
FIGURE 6.9
Comparing the constructed K_c curve using the tabulated values (dotted line) with those modified using $K_{c\ ini}$ from Figure 6.1 and the observed L_{ini} = 25, L_{dev} = 25, L_{mid} = 30, and L_{late} = 20 days (heavy line) for dry beans at Kimberly, Idaho (see Box 6.3), which correspond to the daily measured K_c (lysimeter data from J.L. Wright, USDA-ARS (ret.) 1990)



Source: FAO, 1998

For all decades the K_c values can be derived directly from the curve (Figure 6.10). The value at the middle of the decade is considered to be the average K_c of that 10-day period. Only the second decade of June, where the K_c value changes abruptly (Figure 6.10), requires some calculation. During the first five days of that decade, K_c = 0.15, while during the second part of the decade K_c varies from 0.15 to 0.36 at the end of day 10. The K_c for that decade is consequently: 5/10 (0.15) + 5/10 (0.15+0.36)/2 = 0.20.

FIGURE 6.10
K_c curve and ten-day values for K_c derived from the graph for the dry bean crop example (Box 6.3)



K_c	0.15	0.15	0.20	0.57	0.98	1.19	1.19	1.19	0.98	0.56	
ET_o	3.0	3.4	4.0	4.2	4.5	5.1	5.6	6.0	5.5	5.2	mm/day
ET_c	0.5	0.5	0.8	2.4	4.4	6.1	6.7	7.1	5.4	2.9	mm/day

Source: FAO, 1998

6.6.2 Numerical determination of K_c

The K_c coefficient for any period of the growing season can be derived by considering that during the initial and mid-season stages K_c is constant and equal to the K_c value of the growth stage under consideration (see Example 6.7). During the crop development and late season stage, K_c varies linearly between the K_c at the end of the previous stage ($K_{c\ prev}$) and the K_c at the beginning of the next stage ($K_{c\ next}$), which is $K_{c\ end}$ in the case of the late season stage:

$$K_{c\,i} = K_{c\ prev} + \left[\frac{i - \Sigma(L_{prev})}{L_{stage}}\right](K_{c\ next} - K_{c\ prev}) \qquad (6.20)$$

where i day number within the growing season,
$K_{c\,i}$ crop coefficient on day i,
L_{stage} length of the stage under consideration [days],
$\Sigma(L_{prev})$ sum of the lengths of all previous stages [days].

Equation 6.20 applies to all four stages.

EXAMPLE 6.7
Numerical determination of K_c

Determine K_c at day 20, 40, 70 and 95 for the dry bean crop (Figure 6.10).

Crop growth stage	Length (days)	K_c
initial	25	$K_{c\ ini} = 0.15$
crop development	25	0.15 ... 1.19
mid-season	30	$K_{c\ mid} = 1.19$
late season	20	1.19 .. $K_{c\ end} = 0.35$

At i = 20:	initial stage, $K_c = K_{c\ ini}$ =	0.15	-
At i = 40	Crop development stage,		
For:	$\Sigma(L_{prev}) = L_{ini}$ =	25	days
and:	$L_{stage} = L_{dev}$ =	25	days
From Eq. 6.20:	K_c = 0.15+[(40–25)/25](1.19–0.15)=	0.77	-
At i = 70:	mid-season stage, $K_c = K_{c\ mid}$ =	1.19	-
At i = 95	late season stage,		
For:	$\Sigma(L_{prev}) = L_{ini} + L_{dev} + L_{mid}$ = (25+25+30)=	80	days
and:	$L_{stage} = L_{late}$ =	20	days
From Eq. 6.20:	K_c = 1.19+[(95–80)/20](0.35 –1.19)=	0.56	-

The crop coefficients at day 20, 40, 70 and 95 for the dry bean crop are 0.15, 0.77, 1.19 and 0.56 respectively.

6.7 ALFALFA-BASED CROP COEFFICIENTS

As two reference crop definitions (grass and alfalfa) are in use in various parts of the world, two families of K_c curves for agricultural crops have been developed in the literature. These are the alfalfa-based K_c curves and the grass-based curves in FAO 24 and in the original version of the current FAO 56, and those reported herein. The user must exercise caution to avoid mixing grass-based K_c values with alfalfa reference ET and vice versa. Usually, a K_c based on the alfalfa reference can be "converted" for use with a grass reference by multiplying by a factor ranging from about 1.05 to 1.35, depending on the climate (1.05 for humid, calm conditions, and 1.2 for semi-arid, moderately windy conditions, and 1.35 for arid, windy conditions):

$$K_{c\ (grass)} = K_{ratio}\ K_{c\ (alfalfa)} \tag{6.21}$$

where $K_{c\ (grass)}$ grass-based K_c (these Guidelines),
$K_{c\ (alfalfa)}$ alfalfa-based K_c,
K_{ratio} conversion factor (1.05 ... 1.35).

A reference conversion ratio can be established for any climate by using the $K_{c\ mid} = 1.20$ listed for alfalfa in Table 6.4 and then adjusting this $K_{c\ mid}$ for the climate using Equation 6.16. For example, at Kimberly, Idaho, the United States, where $RH_{min} = 30$ percent and $u_2 = 2.2$ m s^{-1} are average values during the summer months, a reference conversion ratio between alfalfa and grass references using Equation 6.16 is approximately:

$$K_{ratio} = 1.2 + [0.04\ (2.2 - 2) - 0.004\ (30 - 45)]\left(\frac{0.5}{3}\right)^{0.3} = 1.24 \tag{6.22}$$

where h = 0.5 m is the standard height for the alfalfa reference.

6.8 LENGTHS OF GROWTH STAGES

6.8.1 Thermal lengths of the crop growth stages

The construction of the crop coefficient curve requires the estimation of the lengths of the four stages that connect or represent the three K_c values. These stages are the initial (ini), development (dev), mid-season (mid) and late-season (end) as in Figure 6.7 and are defined in Chapter 5.

In the absence of limiting factors such as water or nutrients, a crop's development essentially depends on the temperature accumulation over time. Thus, the use of thermal time or growing degree days (GDD) to estimate the crop growth stage lengths has the merits of allowing one to adequately extrapolate from one year to the next, to diverse locations, and to take into consideration the impacts of climate change, particularly in temperature, on crop growth cycles. The use of GDD is important for studies relating to the assessment of climate change impacts on water and irrigation requirements. Therefore, in this updated publication, the use of GDD to determine the lengths of crop growth stages and the shape of the seasonal K_c curve is provided rather than static stage lengths as provided in the 1998 FAO56 publication. Stage lengths, expressed in days, are available in the previous version of these guidelines.

The time of emergence of vegetation and the time of effective full cover can be estimated using sophisticated plant growth models or by cumulative GDD regression equations. While those models should be verified or validated for the local area or for a specific crop variety using local observations, the use of generalized GDD is a more simplified but reliable method that can be applied using simple computer software or spreadsheets.

Crop growth and development of the K_c curve is assumed to occur on any day when the mean air temperature exceeds a lower temperature threshold termed the base temperature (T_{base}). The value of T_{base} depends on the crop and variety. In conjunction with T_{base}, when the mean daily air temperature exceeds an upper temperature threshold (T_{upper}), crop development is assumed to be limited to the GDD below that limit. Thus, the estimation of the GDD for a certain crop is based upon the daily average between the daily maximum and minimum temperatures and values for T_{base} and T_{upper} as follows:

$$GDD = \frac{T_{max} + T_{min}}{2} - T_{base} \quad \text{if} \quad T_{base} < \frac{T_{max} + T_{min}}{2} < T_{upper} \tag{6.23a}$$

$$GDD = T_{upper} - T_{base} \quad \text{if} \quad \frac{T_{max} + T_{min}}{2} \geq T_{upper} \tag{6.23b}$$

$$GDD = 0 \quad \text{if} \quad \frac{T_{max} + T_{min}}{2} \leq T_{base} \tag{6.23c}$$

Table 6.10 presents the base and the upper limit temperatures for vegetable crops, field crops and grasses, fruit tree and vine crops, and wetland vegetation. The number of tabulated crops are fewer than those included in the crop coefficient tables despite the large number of literature references, because many papers did not fully and clearly explain how GDD values were obtained, thus were not usable.

Cumulative GDD values for various crops and respective growth stages are given in Table 6.11. These values were obtained from focused computations using field observed data and may be considered as reliable indicative values. However, users must use this information with caution since the diverse cultivars of every crop have different cumulative GDD requirements, particularly when they are long or short season, or they are adapted to more warm or fresh environments. For this reason, in some cases, values are given for short and long season cultivars, or for early and late maturity cultivars, that relate to low and high seasonal GDD requirements. Therefore, better use is expected when users know the characteristics of considered cultivars, i.e. when they approach the lower or upper limit, not just averaging tabulated values.

TABLE 6.10

Indicative base (T_{base}) and upper (T_{upper}) temperature thresholds used to calculate the growing degree days (GDD) for the diverse vegetable, field and woody fruit crops in Tables 6.11 and 6.12

Crop	T_{base} (°C)	T_{upper} (°C)
Vegetable crops		
Roots, tubers, bulbs and steam vegetable crops		
Asparagus (*Asparagus officinalis*)	4.5	35
Carrots (*Daucus carota*)	6	30
Cassava (*Manihot esculenta*)	10	40
Celery (*Apium graveolens*)	4	24
Garlic (*Allium sativum*)	4	30
Onions (*Allium cepa*)	4.5	35
Potato (*Solanum tuberosum*)	2	30
Radish (*Raphanus sativus*)	4.5	30
Sweet potato (*Ipomoea batatas*)	12	30
Table beets (*Beta vulgaris*)	4.5	30
Turnip (*Brassica rapa*)	4	35
Yam, Taro (*Colocasia esculenta*)	10	35
Leaves and flowers vegetable crops		
Artichoke (*Cynara scolymus*)	5	32
Broccoli (*Brassica oleracea* var. italica)	4.5	30
Brussel sprouts (*B. oleracea* cv. gemmifera)	4.5	30
Cabbage (*B. oleracea*)	4.5	30
Cauliflower (*B. oleracea* cv. botrytis)	4.5	30
Lettuce (*Lactuca sativa*)	4	28
Spinach (*Spinacia oleracea*)	4	25
Fruit vegetable crops		
Banana (*Musa accuminata*)	14	37
Bell & Chili pepper (*Capsicum annuum*)	10	35
Cucumber (*Cucumis sativus*)	10	32
Eggplant (*Solanum melongena*)	10	35
Melon (*Cucumis melo*)	10	38
Melon, cantaloupe (*Cucumis melo*)	10	38
Okra (*Abelmoschus esculentus*)	10	35
Pineapple (*Ananas comosus*)	13	25
Pumpkin (*Cucurbita argyrosperma*)	10	32
Squash, Zucchini (*Cucurbita pepo*)	10	32
Strawberries (*Fragaria* × *ananassa*)	3	30
Tomato (*Solanum lycopersicum*)	7	28
Watermelon (*Citrullus lanatus*)	10	35
Spicy and medicinal herbs		
Basil (*Ocimum basilicum*)	10	30
Black cumin (*Nigella sativa*)	5	30
Black pepper (*Piper nigrum*)	10	40
Coriander (*Coriandrum sativum*)	4	30
Fenugreek (*Trigonella foenum-graecum*)	4	35
Lemon balm (*Melissa officinalis*)	5	35
Mint (*Mentha spicata*)	4	35
Oregano (*Origanum vulgare*)	5	30
Parsley (*Petroselinum crispum)*	4	27
Saffron (*Crocus sativus*)	10	30

TABLE 6.10 (CONT.)

Crop	T_{base} (°C)	T_{upper} (°C)
Field crops		
Grain legumes		
Bean (*Phaseolus vulgaris*)	10	32
Mungbean (*Vigna radiata*)	8	40
Black and green gram (*Vigna mungo*)	10	40
Chickpea, garbanzo (*Cicer arietinum*)	5	32
Cowpea (*Vigna unguiculata*)	10	35
Fava bean (*Vicia faba*)	5	30
Groundnut, peanut (*Arachis hypogaea*)	10	35
Lentil (*Lens culinaris*)	2	35
Pea (*Pisum sativum*)	5	27
Soybean (*Glycine max*)	10	40
Fiber crops		
Cotton (*Gossypium hirsutum*)	12.5	35
Flax, Linseed (*Linum usitatissimum*)	0	36
Sisal (*Agave sisalana*)	10	35
Oil crops		
Camelina (*Camelina sativa*)	4	30
Canola (*Brassica napus*)	2	30
Castorbean (*Ricinus communis*)	10	30
Guar (*Cyamopsis tetragonoloba*)	15	40
Mustard (*Brassica juncea*)	5	40
Safflower (*Carthamus tinctorius*)	5	35
Sesame (*Sesamum indicum*)	10	40
Sunflower (*Helianthus annuus*)	8	30
Sugar crops		
Sugar beet (*Beta vulgaris*)	5	30
Sugar cane (*Saccharum officinarum*)	10	45
Cereals and pseudocereals		
Amaranth grain (*Amaranthus sp.*)	7	40
Barley (*Hordeum vulgare*)	0	30
Maize (*Zea mays*)	10	32
Oats (*Avena sativa*)	0	30
Pearl Millet (*Pennisetum glaucum*)	10	40
Quinoa (*Chenopodium quinoa*)	2	25
Rye (*Secale cereale*)	4	35
Sorghum (*Sorghum bicolor*)	10	30
Teff (*Eragrostis tef*)	10	30
Wheat, common (*Triticum aestivum*)	0	35
Wheat, durum (*Triticum durum*)	0	35
Rice (*Oryza sativa*)	12	35
Grasses		
Alfalfa (*Medicago sativa*)	0	30
Bermuda (*Cynodon dactylon*)	10	30
Berseem clover (*Trifolium alexandrinum*)	4	30
Common vetch (*Vicia sativa*)	0	30
Elephant grass (*Pennisetum sp.*)	15	40
Grass pastures	4	30

TABLE 6.10 (CONT.)

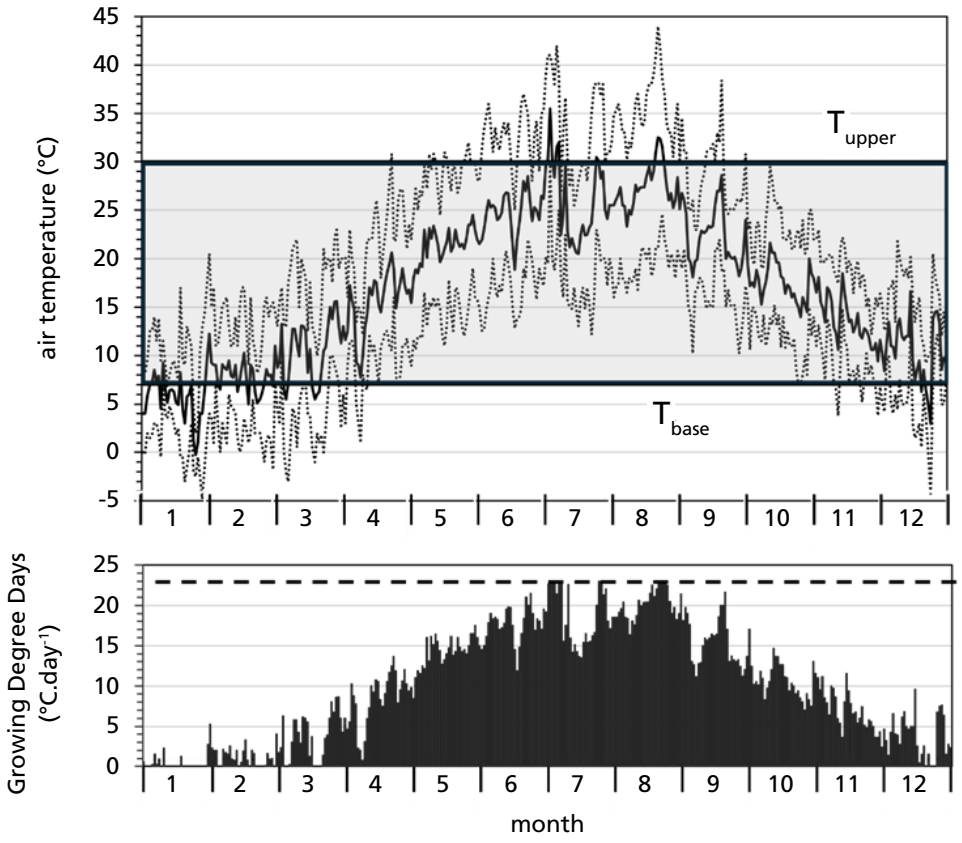
Crop	T_{base} (°C)	T_{upper} (°C)
Guinea grass (*Panicum maximum*)	14	40
Hairy vetch (*Vicia villosa*)	2	25
Italian ryegrass (*Lolium multiflorum*)	2	25
Persian clover (*Trifolium resupinatum*)	6	35
Sudan grass (Sorghum × drummondii)	10	35
Trees, shrubs, and vines		
Mediterranean and warm temperate climates fruit trees and vines		
Olive (*Olea europaea*)	7	35
Wine and table grapes (*Vitis vinifera*)	10	35
Citrus (*Citrus* sp.)	13	30
Avocado (*Persea americana*)	6	33
Loquat (*Eriobotrya japonica*)	4	42
Pomegranate (*Punica granatum*)	10	35
Temperate climate fruit trees, vines and shrubs		
Pomme fruit trees		
Apple (*Malus domestica*)	4	35
Pear (*Pyrus communis*)	4	35
Stone fruit trees		
Apricot (*Prunus armeniaca*)	7	25
Cherry (*Prunus avium*)	4.5	30
Peach and nectarine (*Prunus persica*)	7	35
Plum (*P. domestica* and *P. salicina*)	4.5	25
Nut fruit trees		
Almond (*Prunus dulcis*)	4.5	35
Pistachio (*Pistacia vera*)	9	35
Walnut (*Juglans regia*)	4.5	30
Other temperate vines and shrubs		
Kiwifruit (*Actinidia deliciosa*)	6	30
Hop (*Humulus lupulus*)	5	30
Berries (*Vaccinium* sp.)	6	30
Sub–tropical and tropical small fruit trees and shrub		
Coffee (*Coffea arabica*)	10	32
Guava (*Psidium guajava*)	12	35
Papaya (*Carica papaya*)	15	35
Passionfruit (*Passiflora edulis*)	10	35
Tropical fruit trees		
Lychee (*Litchi chinensis*)	10	30
Mango (*Mangifera indica*)	12	32
Palm plantations		
Date palm (*Phoenix dactylifera*)	10	35
Oil palm (*Elaeis guineensis*)	15	40

Values in Table 6.11 are approximate values that can be used as a starting point to define starting and ending dates for the four growth stage periods used for the definition of the crop coefficient curve. The values should be checked and adjusted using local observations or data. Actual GDD values and dates for growth stage periods can highly vary due to the occurrence of limiting factors such as water, nutrients, pests

and diseases, weeds, as well as agricultural practices, plant density and crop variety. In addition, variability may be due to soil wetness/dryness, and more/less favourable weather. Local information should be obtained by interviewing farmers, ranchers, agricultural extension agents and local researchers, by conducting local surveys, or by remote sensing. When determining stage dates from local observations, related guidelines and visual descriptions may be helpful.

FIGURE 6.11

Example of daily minimum and maximum air temperature (dotted lines) and mean air temperature (full line) for Foggia, Italy (year 2000), with an indication of T_{base} (7°C) and T_{upper} (30°C), and the corresponding calculated Growing Degree Days (Equation 6.23) for an olive orchard



Source: Authors' own elaboration

The cumulative GDD that defines the length of the crop stages and the crop season are shown in Table 6.11 for various annual and perennial crops:

- GDD_{ini} cumulative growing degree days from the planting of the annual crop or greenup of the perennial crop until the start of the development period, thus the length of the initial period;
- GDD_{dev} cumulative growing degree days from the start of the development period until the beginning of the midseason period, thus the length of the development period;
- GDD_{mid} cumulative growing degree days from the start of the midseason period to its end, i.e., length of the midseason period;

- GDD_{late} cumulative growing degree days from the end of the midseason period until harvest, full maturity or plant death, i.e., length of the late season period;
- Total GDD total cumulative growing degree days, from planting or greenup until harvest or end of the growing season, thus crop season length.

The length for each of the four growth stage periods is determined by summing GDD on a daily basis, using Equations 6.23. The number of days occurring can represent the length of the stage in days.

The rate at which vegetation cover develops and the time at which it attains effective full cover are affected by weather conditions in general and by mean daily air temperature in particular. Therefore, the length of time between planting and effective full cover will vary with climate, latitude, elevation, and planting date. It will also vary with cultivar (crop variety). Generally, once the effective full cover for a plant canopy has been reached, the rate of further phenological development (flowering, fruit development, ripening, and senescence) is more dependent on plant genotype and less dependent on weather.

The end of the mid-season and beginning of the late season is usually marked by senescence of leaves, often beginning with the lower leaves of plants. The length of the late season period may be relatively short (less than 10 days) for vegetation killed by frost (for example, maize at high elevations in latitudes > 40°N) or for agricultural crops that are harvested fresh (for example, table beets and small vegetables). Moisture stress or other environmental stresses will usually accelerate the rate of crop maturation and can shorten the mid and late season growing periods, however affecting yields.

Information in Table 6.11 refers to trusted observations only. It is possible to find more GDD in the literature for a large number of crops, but it is often not clear which values of T_{base} and T_{upper} temperatures were used as compared to those in Table 6.10, what specific GDD calculations were applied, or if the crop growth stages were defined according to those in FAO56 as adopted in the current study. To avoid such problems, GDD values in Table 6.11 were recalculated from literature sources containing trusted observations and data analyses. Users of Table 6.11 are encouraged to, first, perform their own observations using the T_{base} and T_{upper} temperatures in Table 6.10 or, secondly, to directly use data from Table 6.11 for the crops listed, or, thirdly and indirectly, by comparing the considered crop with another, selected by the user, listed in Table 6.11.

Table 6.11 summarises GDD data that are based on published and refereed literature sources. That Table is complemented by Table 6.12 that lists approximate minimum and maximum values for GDD that do not appear in Table 6.11. The GDD values in Table 6.12 are associated with the range of original growth stage lengths reported in the original Table 11 of the 1998 FAO56 publication. Table 6.12 can be used as a general indication of expected ranges in values for GDD. The values for GDD in Table 6.12 were developed by associating general climatic and cultural regions listed for the crops in the original Table 11 of the 1998 publication with specific representative locations on the globe. The representative locations were then used to calculate GDD values that are associated with the lengths of stages in the original table. The calculation of associated GDD used the ERA5 (fifth generation ECMWF atmospheric reanalysis of the global climate) gridded weather data set using normal air temperature over the 1991-2020 period. The minimum and maximum GDD values or the "common" GDD values should be considered to be only approximate values and can serve as a starting point and perhaps as bounds on locally determined values or for new values to be found in the literature.

TABLE 6.11

Field observed indicative cumulative growing degree days required to complete each crop growth stage and the total crop season for non-stressed and well-managed vegetable, field, and woody fruit crops estimated using the base and upper temperature thresholds given in Table 6.10

Crop	Type of season	Initial GDD_{ini} (°C)	Develop. GDD_{dev} (°C)	Mid-season GDD_{mid} (°C)	Harvest GDD_{late} (°C)	Total GDD (°C)
Roots, tubers, bulbs vegetable crops						
Carrots (*Daucus carota*)	Common	320	465	500	290	1 575
Garlic (*Allium sativum*)	Short season	150	340	335	315	1 140
	Long season	580	615	315	240	1 750
Onions (*Allium cepa*)	Common	460	470	880	480	2 290
Potato (*Solanum tuberosum*)	Short season	260	490	525	325	1 600
	Long season	405	530	490	835	2 260
Leaves and flowers vegetable crops						
Broccoli (*Brassica oleracea* cv. *italica*)	Short season	195	250	210	100	755
	Long season	295	350	525	110	1 280
Lettuce (*Lactuca sativa*)	Short season	365	385	215	20	985
	Long season	360	455	455	20	1 290
Fruit vegetable crops						
Bell pepper (*Capsicum annuum*)	Common	445	1180	745	45	2 415
Melon (*Cucumis melo*)	Short season	185	520	315	155	1 175
	Long season	140	545	460	455	1 600
Tomato (*Solanum lycopersicum*)	Industry	280	520	880	220	1 900
	Market	325	660	880	200	2 065
Grain legumes						
Bean, seed (*Phaseolus vulgaris*)	Short season	160	260	360	180	960
	Long season	320	410	400	220	1 350
Pea, fresh (*Pisum sativum*)	Industry	105	255	510	35	905
	Market	155	350	575	320	1 400
Soybean (*Glycine max*)	Short season	300	350	650	350	1 660
	Long season	310	410	680	560	1 960
Fiber crops & oil crops						
Cotton (*Gossypium hirsutum*)	Short season	280	520	800	350	1 950
	Long season	345	905	615	285	2 150
Canola (*Brassica napus*)	Short season	330	310	450	595	1 685
	Long season	460	555	905	595	2 515
Sunflower (*Helianthus annuus*)	Short season	345	555	715	175	1 790
	Long season	385	570	685	265	1 905
Cereals						
Barley (*Hordeum vulgare*)	Short season	290	455	345	360	1 450
	Long season	300	675	690	660	2 325
Maize, grain (*Zea mays*)	Short season	200	380	500	340	1 420
	Long season	295	420	775	465	1 955
Maize, silage (*Zea mays*)	Short season	200	380	430	120	1 130
	Long season	250	500	600	200	1 550

TABLE 6.11 (CONT.)

Crop	Type of season	Initial GDD_{ini} (°C)	Develop. GDD_{dev} (°C)	Mid-season GDD_{mid} (°C)	Harvest GDD_{late} (°C)	Total GDD (°C)
Maize, sweet (*Zea mays*)	Short season	200	310	360	115	985
	Long season	355	500	430	200	1 485
Oats (*Avena sativa*)	Common	450	595	550	255	1 850
Sorghum (*Sorghum bicolor*)	Short season	280	300	570	230	1380
	Long season	280	310	650	280	1520
Wheat, Winter (*Triticum aestivum*)	Short season	395	445	440	365	1 645
	Long season	560	405	730	410	2 105
Wheat, Spring (*Triticum aestivum*)	Short season	360	480	590	190	1 620
	Long season	360	680	720	540	2 300
Rice (*Oryza sativa*)	Short season	235	465	665	295	1 660
	Long season	255	490	740	320	1805
Fruit trees and vines						
Almond (*Prunus dulcis*)	Early maturation	250	300	2260	815	3 625
	Late maturation	280	325	2730	475	3 810
Citrus (Oranges, clementines, mandarin)	Common	35	220	1 580	60	1 895
Olive (*Olea europaea*)	Early maturation	50	430	1385	800	2 665
	Late maturation	380	350	2 055	720	3 505
Pistachio (*Pistacia vera*)	Common	225	555	1 130	210	2 120
Pomegranate (*Punica granatum*)	Common	95	185	1 900	265	2 445
Wine grapes (*Vitis vinifera*)	Early maturation	435	1 225	505	160	2 325
	Late maturation	465	1 145	800	590	3 000

TABLE 6.12

Approximate ranges for cumulative growing degree days required to complete each crop growth stage and the total crop season for non-stressed and well-managed vegetable, field, and woody fruit crops for crops not found in Table 6.11. These values were derived from the original Table 11 of the 1998 FAO56 publication

Crop	Total Length of season	Initial GDD_{ini} (°C)	Develop. GDD_{dev} (°C)	Mid-season GDD_{mid} (°C)	Harvest GDD_{late} (°C)	Total GDD (°C)
Roots, tubers, bulbs vegetable crops						
Asparagus (*Asparagus officinalis*)	Min length	380	350	2 510	480	3 720
	Max length	670	540	2 670	1 480	5 360
Cassava (*Manihot esculenta*), 1st y	Common	260	510	1 260	850	2 880
Celery roots (*Apium graveolens*)	Min length	400	370	510	190	1 470
	Max length	130	420	1 730	400	2 680
Sweet potato (*Ipomoea batatas*)	Min length	150	300	480	430	1 360
	Max length	170	320	560	370	1 420
Table beets (*Beta vulgaris*)	Min length	30	130	180	100	440
	Max length	40	150	180	100	470
Leaves and flowers vegetable crops						
Artichoke (*Cynara scolymus*), 1st y	Common	290	510	2 100	80	2 980
Artichoke (*Cynara scolymus*), 2nd y	Common	120	230	2 560	50	2 960
Asparagus (*Asparagus officinalis*)	Min length	380	350	2 510	480	3 720
	Max length	670	540	2 670	1 480	5 360
Cabbage (*B. oleracea*)	Common	790	620	220	80	1 710
Cauliflower (*B. oleracea var. botrytis*)	Common	700	610	200	60	1 570
Mint (*Mentha*)	Common	120	290	2 510	820	3 740
Spinach (*Spinacia oleracea*)	Min length	130	190	180	70	570
	Max length	250	200	140	20	610
Fruit and Pod vegetable crops						
Bean, green (*Phaseolus vulgaris*)	Min length	20	110	220	100	450
	Max length	230	390	400	150	1 170
Cucumber (*Cucumis sativus*)	Min length	80	170	450	270	970
	Max length	370	510	520	120	1 520
Eggplant (*Solanum melongena*)	Min length	280	440	380	80	1 180
	Max length	230	540	480	180	1 430
Pumpkin & Winter squash (*Cucurbita pepo*)	Min length	170	380	370	150	1 070
	Max length	230	440	430	200	1 300
Squash, Zucchini (*Cucurbita pepo*)	Min length	90	200	210	130	630
	Max length	160	370	360	260	1 150
Grain legumes						
Black & green gram (*Vigna mungo*)	Common	110	240	420	200	970
Chickpea (*Cicer arietinum*)	Min length	80	200	400	220	900
	Max length	120	250	430	210	1 010
Cowpea (*Vigna unguiculata*)	Min length	220	350	440	230	1 240
	Max length	310	450	550	300	1 610

TABLE 6.12 (CONT.)

Crop	Total Length of season	Initial GDD_{ini} (°C)	Develop. GDD_{dev} (°C)	Mid-season GDD_{mid} (°C)	Harvest GDD_{late} (°C)	Total GDD (°C)
Fava bean (*Vicia faba*)	Min length	80	200	400	220	900
	Max length	120	250	430	210	1 010
Groundnut, peanut (*Arachis hypogaea*)	Min length	240	470	760	460	1 930
	Max length	430	800	600	310	2 140
Lentil (*Lens culinaris*)	Min length	130	270	830	620	1 850
	Max length	120	300	860	650	1 930
Fiber & oil crops						
Flax (Linseed) (*Linum usitatissimum*)	Min length	210	450	840	730	2 230
	Max length	760	710	1 530	1 150	4 150
Castorbean (*Ricinus communis*)	Min length	280	530	640	330	1 780
	Max length	190	490	880	550	2 110
Safflower (*Carthamus tinctorius*)	Min length	110	410	800	470	1 790
	Max length	220	520	910	580	2 230
Sesame (*Sesamum indicum*)	Common	170	400	570	190	1 330
Sugar crops						
Sugar beet (*Beta vulgaris*)	Min length	300	290	790	520	1 900
	Max length	490	690	1770	120	3 070
Sugar cane (*Saccharum officinarum*) - virgin	Min length	350	810	3 480	1 510	6 150
	Max length	830	2 110	4 990	3 450	11 380
Sugar cane (*Saccharum officinarum*) - ratoon	Min length	250	910	2 630	770	4 560
	Max length	350	1 730	3 500	730	6 310
Pseudo-cereals						
Teff (*Eragrostis tef*)	Min length	140	310	750	280	1 480
	Max length	500	700	1 070	620	2 890
Vines						
Hops (*Humulus lupulus*)	Common	90	360	1 430	170	2 050
Forages and hay						
Alfalfa (*Medicago sativa*)- first cutting cycle	Min length	130	270	290	160	850
	Max length	70	310	380	180	940
Alfalfa (*Medicago sativa*)- middle cycles	Min length	80	170	180	100	530
	Max length	70	340	190	220	820
Bermuda (*Cynodon dactylon*) - for hay, several cuttings	Min length	30	260	510	450	1 250
	Max length	80	260	490	660	1 490
Grass Pasture	Common	30	260	510	400	1 200
Sudan grass	Min length	50	290	250	160	750
	Max length	320	410	290	210	1 230

6.8.2 Converting the thermal time of the growth stages into calendar time

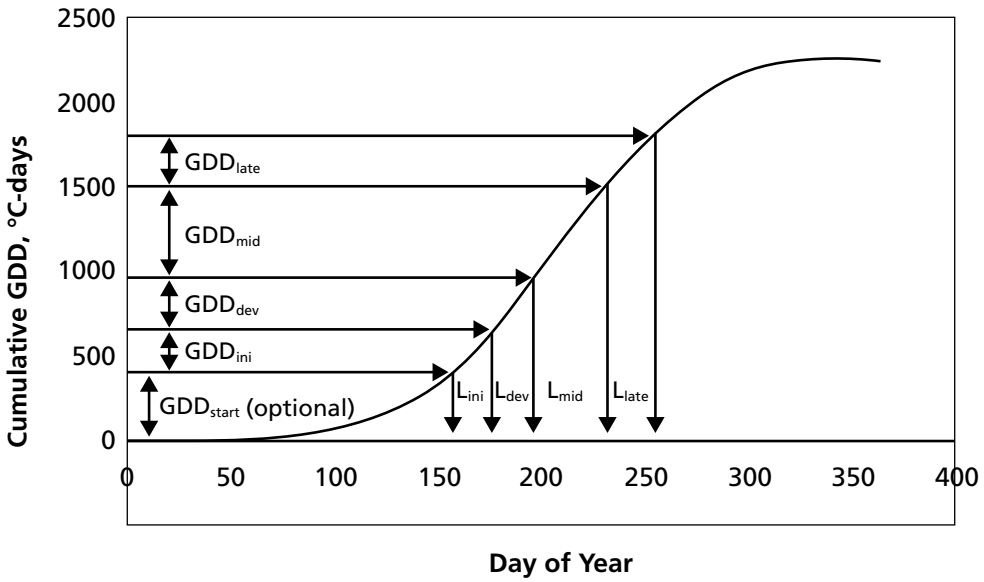
Once the planting date has been determined for annual crops, the length of the growth stages in thermal time can be converted to an equivalent length in days.

A graphic showing the transformation of GDD into lengths of growth stages, in days, is shown in Figure 6.12. When using a spreadsheet, the GDD can be accumulated from the date of planting and a lookup type of function can be employed to select the date for each stage. Figure 6.12 applies the procedure for transforming GDD values into lengths for growth periods for a crop of maize near Holdrege, NE. USA., using air temperature data from 2017. The estimated planting date occurred relatively soon after the beginning of the rapid accumulation of GDD in the spring. Harvest (end) occurred right before a transition to cold periods where there was little additional accumulation of GDD.

Figure 6.13 shows an example of the variation in lengths of the growth period of a maize cultivar for several locations and for different sowing dates. The figure represents the average length of the crop cycle using a weather data set of at least 10 years. This example clearly shows the advantage of using GDD instead of time for establishing the duration of the crop stages, particularly in locations that experience greater temperature variability.

FIGURE 6.12

Transformation of cumulative growing degree days for the four growth periods into equivalent lengths in days. The GDD_{start} is optional and can be used, if desired, to estimate planting date for crops growing in cold temperate climates using GDD since 1 January

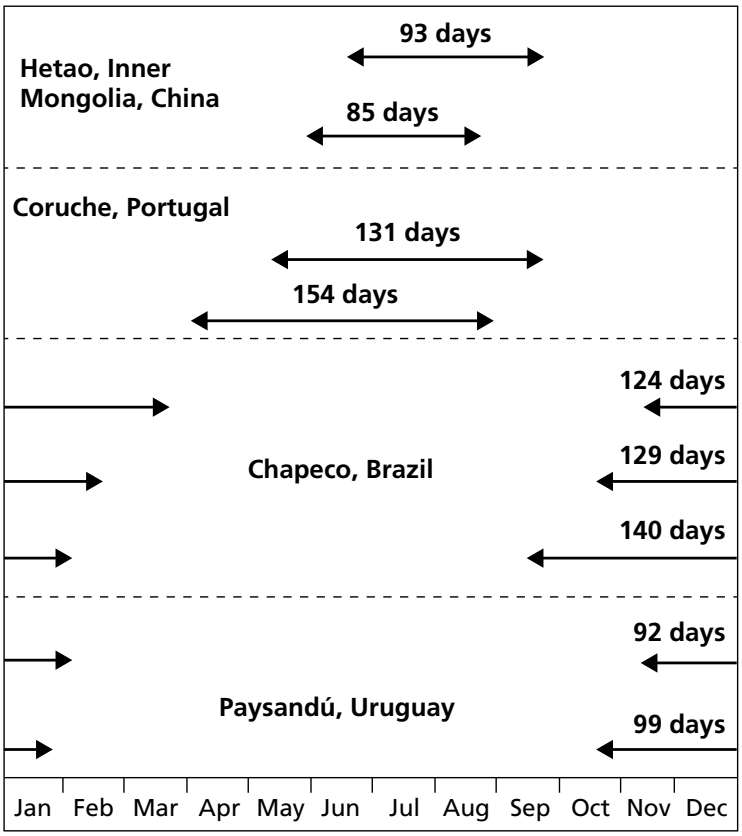


Source: Authors' own elaboration

FIGURE 6.13

Variation in the length of the growing period of maize (FAO600 cycle with a seasonal GDD of 1687 °C) sown at various months of the year at different locations in the world

Note that for the southern hemisphere the initiation is in the Autumn



Source: Authors' own elaboration

Chapter 7

ET_c and dual crop coefficient ($K_c = K_{cb} + K_e$)

This chapter, complementing Chapter 6, deals with the calculation of crop evapotranspiration (ET_c) under standard, non-stressed cropping conditions where no limitations are placed on crop growth or evapotranspiration. This chapter presents the procedure for predicting the effects of specific wetting events on the value of the crop coefficient K_c. The solution consists of splitting K_c into two separate coefficients, one for crop transpiration, i.e. the basal crop coefficient (K_{cb}), and one for soil evaporation (K_e):

$$ET_c = (K_{cb} + K_e)\, ET_o \qquad (7.1)$$

The dual crop coefficient approach is more complicated and more computationally intensive than the single crop coefficient approach (K_c) dealt with in Chapter 6. The procedure is conducted on a daily basis and is intended for applications using computers. It is recommended that the approach be followed when improved estimates for K_c are needed, for example, to schedule irrigations for individual fields on a daily basis, mainly in case of incomplete cover crops, or to increase accuracy in hydrologic studies.

The calculation procedure for crop evapotranspiration, ET_c, consists of:

1. identifying the lengths of crop growth stages, and selecting the corresponding basal K_{cb} coefficients;
2. adjusting the selected K_{cb} coefficients for climatic conditions during the respective crop growth stage;
3. constructing the basal crop coefficient curve (allowing one to determine K_{cb} values for any day or timestep during the crop season);
4. determining daily the soil evaporation coefficient K_e values for surface evaporation; and
5. calculating ET_c as the product of ET_o and ($K_{cb} + K_e$).

7.1 TRANSPIRATION COMPONENT (K_{cb} ET_O)

7.1.1 Basal crop coefficient (K_{cb})

The basal crop coefficient (K_{cb}) is defined as the ratio of the crop evapotranspiration over the grass reference evapotranspiration (ET_c/ET_o) when the soil surface is dry but transpiration is occurring at a potential rate, i.e. water is not limiting transpiration (Figure 5.3). Therefore, "K_{cb} ET_o" represents primarily the transpiration component of ET_c and K_{cb} is often called transpiration coefficient. The K_{cb} ET_o does include a small, residual diffusive evaporation component supplied by soil water below the dry surface and by soil water from beneath dense vegetation.

As the K_c values of Chapter 6 include averaged effects of evaporation from the soil surface, the K_{cb} values lie below the K_c values as illustrated in Figure 5.7 and a separate table for K_{cb} is required. Recommended values for K_{cb} are listed in Tables 7.1 to 7.3 for the same crops listed in Tables 6.1 to 6.3. Values tabulated in Tables 7.1 to 7.3 do not include the influence of mulches, which are dealt with in Chapter 10.

As with Tables 6.1 to 6.3, the values for K_{cb} in Tables 7.1 to 7.3 represent K_{cb} for a standardized sub-humid climate and with moderate wind speed. For specific adjustment in climates where RH_{min} differs from 45 percent or where the wind speed is larger or smaller than 2 m s^{-1}, the $K_{cb\ mid}$ must be adjusted. In addition, $K_{cb\ end}$ values larger than 0.45 are also adjusted. Adjustment is made using the following equation:

$$K_{cb} = K_{cb\ (Tab)} + [0.04\ (u_2 - 2) - 0.004\ (RH_{min} - 45)]\left(\frac{h}{3}\right)^{0.3} \qquad (7.2)$$

where $K_{cb\ (Tab)}$ the value for $K_{cb\ mid}$ or $K_{cb\ end}$ (if ≥ 0.45) taken from Tables 7.1 to 7.3,

u_2 the mean value for daily wind speed at 2 m height over grass during the mid- or late season growth stage [m s^{-1}] for 1 m $s^{-1} \leq u_2 \leq$ 6 m s^{-1},

RH_{min} the mean value for daily minimum relative humidity during the mid- or late season growth stage [%] for 20% ≤ RH_{min} ≤ 80%,

h the mean plant height during the mid- or late season stage [m] (from Tables 6.1 to 6.3) for 20% ≤ RH_{min} ≤ 80%.

For a full discussion on the impact of the climatic correction, and the numerical determination of $K_{cb\ mid}$ and $K_{cb\ end}$, the user is referred to the discussions for $K_{c\ mid}$ and $K_{c\ end}$ in Chapter 6.

Table 7.4 summarizes the general guidelines that were used in deriving K_{cb} values from the K_c values listed in Tables 6.1 to 6.3. Where local research results are available, values for K_{cb} from Tables 7.1 to 7.3 can be modified to reflect the effects of local conditions, cultural practices or crop varieties on K_{cb}. Where local K_{cb} values are used, no adjustment for climate using Equation 7.2 is necessary.

Local values for K_{cb} should not be expected to deviate by more than 0.2 from the values in Tables 7.1 to 7.3. A greater deviation should signal the need to investigate or evaluate the local research technique, equipment and cultural practices. Thus, tabulated K_{cb} values consist of reference values for the locally derived K_{cb} values, i.e. serving to control the accuracy of locally derived values.

TABLE 7.1

Basal crop coefficients, K_{cb}, and ranges of maximum plant heights and of maximum fraction of ground cover for non-stressed, well-managed vegetable crops in subhumid climates ($RH_{min} \approx 45\%$, $u_2 \approx 2$ m s^{-1}) for use with the FAO Penman-Monteith ET_o equation

Crop	$K_{cb\ ini}$	$K_{cb\ mid}$	$K_{cb\ end}$	Maximum crop height (h, m)	Maximum ground cover fraction (f_c)
a. Roots, tubers, bulbs, and stem vegetable crops					
Asparagus (*Asparagus officinalis*)	**0.15**	**0.80**	**0.20**	0.80	0.90
Carrots (*Daucus carota*)	**0.15**	**0.90**	**0.75**	0.30	0.90
Cassava (*Manihot esculenta*)					
Year 1	**0.15**	**0.95**	**0.50**	1.00	0.80
Year 2	**0.50**	**1.00**	**0.50**	1.50	0.80
Celeriac, celery root (*Apium graveolens* group *rapaceum*)	**0.15**	**0.95**	**0.85**	0.60 – 0.90	0.95
Chufa, tigernut (*Cyperus esculentus*)	**0.15**	**1.00**	**0.50**	0.60	0.98
Garlic (*Allium sativum*)	**0.15**	**1.00**	**0.50**	0.50	0.85
Leek (*Allium porrum*)	**0.15**	**0.95**	**0.85**	0.50	0.80
Mashua (*Tropaeolum tuberosum*)	**0.15**	**1.00**	**0.65**	0.60	0.85
Onions (*Allium cepa*)					
Dry	**0.15**	**1.00**	**0.60**	0.45	0.85
Green	**0.15**	**0.95**	**0.80**	0.45	0.75
Seed	**0.15**	**0.95**	**0.45**	0.45	0.75
Parsnip (*Pastinaca sativa*)	**0.15**	**1.00**	**0.95**	0.40	0.85
Potato (*Solanum tuberosum*)					
Long season	**0.15**	**1.05**	**0.30**	0.60	0.90
Short season	**0.15**	**1.05**	**0.50**	0.60	0.90
Radish (*Raphanus sativus*)	**0.15**	**0.90**	**0.75**	0.30	0.80
Rutabaga (*Brassica napobrassica*)	**0.15**	**1.05**	**0.90**	0.60	0.85
Sweet potato (*Ipomoea batatas*)	**0.15**	**1.05**	**0.50**	0.50	0.95
Table beets (*Beta vulgaris*)	**0.15**	**1.00**	**0.90**	0.40	0.85
Turnip (*Brassica rapa* cv. *rapa*)	**0.15**	**1.00**	**0.95**	0.60	0.85
Yam, Taro (*Colocasia esculenta*)	**0.15**	**1.00**	**0.95**	1.20	0.80
b. Leaves and flowers vegetable crops					
African cabbage, spiderwisp (*Cleome gynandra*)	**0.15**	**0.90**	**0.75**	0.30 – 0.45	0.85
African spinach, (*Amaranthus cruentus*)	**0.15**	**1.00**	**0.90**	0.50 – 1.20	0.95
Artichoke (*Cynara scolymus*)	**0.15**	**0.85**	**0.80**	0.80	0.75
Arugula, rocket (*Eruca vesicaria, E. sativa*)	**0.15**	**0.95**	**0.95**	0.45	0.85
Broccoli (*Brassica oleracea* cv. *italica*)	**0.15**	**1.00**	**1.00**	0.60	0.90
Brussel sprouts (*B. oleracea* cv. *gemmifera*)	**0.15**	**1.00**	**1.00**	0.75	0.90
Cabbage					
Cabbage (*B. oleracea cv. capitata*)	**0.15**	**1.00**	**0.90**	0.40	0.85
Kale, leaf cabbage (*B. oleracea* var. *sabellica*) and Collard green (*B. oleracea* var. *acephala*)					
1st year	**0.15**	**0.90**	**0.80**	0.80 – 1.20	0.85
2nd year	**0.55**	**0.90**	**0.75**	0.80 – 1.20	0.85
Chinese cabbage, bok choy (*B. rapa cv. chinensis*)					
1st year	**0.15**	**1.00**	**0.90**	0.40	0.85
2nd year	**0.55**	**1.00**	**0.80**	0.40	0.85
Chinese cabbage, pak choi (*B. rapa* ssp. *pekinensis*)					
1st year	**0.15**	**0.95**	**0.80**	0.30 – 0.40	0.90
2nd year	**0.50**	**0.95**	**0.75**	0.30 – 0.40	0.90

TABLE 7.1 (CONT.)

Crop	$K_{cb\ ini}$	$K_{cb\ mid}$	$K_{cb\ end}$	Maximum crop height (h, m)	Maximum ground cover fraction (f_c)
Cauliflower (*B. oleracea cv. botrytis*)	**0.15**	**1.00**	**0.95**	0.40	0.85
Celery (*Apium graveolens*)	**0.15**	**1.00**	**0.95**	0.60	0.80
Endive, escarola (*Cichorium endivia*)	**0.15**	**0.80**	**0.75**	0.30 – 0.50	0.95
Jute mallow for leaves (*Corchorus olitorius*)	**0.15**	**0.95**	**0.85**	1.10 – 1.50	0.95
Lettuce (*Lactuca sativa*)	**0.15**	**1.00**	**1.00**	0.35	0.80
Purslane (*Portulaca oleracea*)	**0.15**	**0.90**	**0.85**	0.40	0.90
Spinach (*Spinacia oleracea*)	**0.15**	**0.95**	**0.95**	0.35	0.80
c. Fruit vegetable crops					
Banana (*Musa acuminata*)					
1st cycle	**0.15**	**1.00**	**0.90**	2.00 – 3.30	0.95
2nd cycle	**0.60**	**1.05**	**1.00**	2.50 – 4.00	0.85
Bell pepper (*Capsicum annuum*)	**0.15**	**1.00**	**0.70**	0.70	0.90
Bitter gourd, bitter melon (*Momordica charantia*)	**0.15**	**0.85**	**0.60**	1.00 – 2.50	0.95
Cape gooseberry (*Physalis peruviana*)					
1st year	**0.25**	0.40	0.30	1.0	0.30
Following years*	**0.35**	0.85	0.50	1.0 – 2.0	0.30-0.50
Chili habanero (*Capsicum chinense*)	**0.15**	**1.05**	**0.65**	0.80	0.85
Chili pepper (*Capsicum annuum*)	**0.15**	**1.05**	**0.70**	0.75	0.85
Chili tabasco pepper (*Capsicum frutescens*)					
First harvest	**0.15**	**1.05**	**0.65**	0.80	0.85
Second harvest	**0.55**	**1.00**	**0.60**	0.80	0.85
Cucumber (*Cucumis sativus*)					
For processing	**0.15**	**0.95**	**0.80**	0.40	0.90
Fresh market	**0.15**	**0.95**	**0.70**	0.40	0.90
Fresh market, on trellis	**0.15**	**0.95**	**0.70**	0.80 – 1.20	0.90
Eggplant (*Solanum melongena*)	**0.15**	**1.00**	**0.90**	0.80	0.75
Fennel (*Foeniculum vulgare*)	**0.15**	**0.90**	**0.30**	0.45 – 0.70	0.85
Melon (*Cucumis melo*)	**0.15**	**0.95**	**0.75**	0.30	0.95
Melon cantaloupe (*C. melo* cv. *cantalupensis*)	**0.15**	**0.95**	**0.70**	0.30	0.90
Nightshade, sunberry (*Solanum retroflexum*)	**0.15**	**0.85**	**0.80**	0.75	0.80
Okra (*Abelmoschus esculentus*)	**0.15**	**0.90**	**0.75**	0.90	0.90
Pineapple (*Ananas comosus*)	**0.15**	**0.85**	**0.80**	0.40	0.90
Pumpkin, winter squash (*Cucurbita pepo*)	**0.15**	**0.90**	**0.55**	0.40	0.90
Strawberries (*Fragaria* × *ananassa*)	**0.15**	**0.75**	**0.65**	0.20	0.75
Tomato (*Solanum lycopersicum*)					
For processing	**0.15**	**1.05**	**0.75**	0.60	0.90
Fresh market	**0.15**	**1.05**	**0.90**	0.60	0.90
Fresh market, with trellis	**0.15**	**1.10**	**0.80**	1.50	0.90
Watermelon (*Citrullus lanatus*)	**0.15**	**1.00**	**0.60**	0.40	0.85
Zucchini, Squash (*Cucurbita pepo*)	**0.15**	**0.90**	**0.60**	0.50	0.90

* Cultivated as a perennial for two or more consecutive years to maximize productivity.

TABLE 7.1 (CONT.)

Crop	$K_{cb\ ini}$	$K_{cb\ mid}$	$K_{cb\ end}$	Maximum crop height (h, m)	Maximum ground cover fraction (f_c)
d. Spicy and medicinal herbs					
d1. Root, tuber and rhizome herbs					
Chicory and bitter chicory (*Cichorium intybus* and *C. intybus* var. *intybus*)	**0.15**	**0.85**	**0.75**	0.45 – 0.90	0.85
Ginger (*Zingiber officinale*)	**0.15**	**0.75**	**0.50**	0.50 – 0.90	0.95
Licorice (*Glycyrrhiza glabra*)	**0.15**	**0.90**	**0.75**	1.00	0.90
Nutgrass, purple nutsedge, (*Cyperus rotundus*)	**0.15**	**0.80**	**0.40**	1.00 – 1.20	0.90
Turmeric, golden curry (*Curcuma longa*)	**0.15**	**0.95**	**0.75**	0.90 – 1.10	0.90
d2. Leafy, flower and stem herbs					
Absinthe, wormwood, (*Artemisia absinthium*)	**0.15**	**0.70**	**0.30**	0.80 – 1.20	0.85
Basil (*Ocimum basilicum*)	**0.15**	**0.95**	**0.90**	0.50 – 1.00	0.95
Cardamom, large & black (*Amomum subulatum*)	**0.15**	**0.85**	**0.85**	1.50 – 3.00	0.80
Chamomile (*Chamomilla recutita*)	**0.15**	**0.80**	**0.40**	0.50 – 0.90	0.95
Coriander (*Coriandrum sativum*)	**0.15**	**1.00**	**0.95**	0.20 – 0.40	0.95
Fenugreek (*Trigonella foenum-graecum*)	**0.15**	**0.90**	**0.60**	0.60 – 1.40	0.95
Lavender (*Lavandula angustifolia*)	**0.15**	**0.85**	**0.65**	0.20 – 0.60	0.85
Lemon balm, balm (*Melissa officinalis*)	**0.15**	**1.00**	**0.90**	0.50	0.80
Mint					
Japanese mint, wild mint (*Mentha arvensis*)	**0.15**	**1.05**	**1.00**	0.30 – 0.80	0.95
Peppermint (*Mentha x piperita*)					
Single harvest	**0.15**	**1.05**	**0.95**	0.30 – 0.50	0.95
Multiple harvest	**0.15**	**0.90**	**0.85**	0.30 – 0.50	0.90
Spearmint, common mint (*Mentha spicata*)	**0.15**	**0.95**	**0.85**	0.30 – 0.50	0.95
Oregano (*Origanum vulgare*)	**0.15**	**0.80**	**0.65**	0.40 – 0.50	0.95
Parsley (*Petroselinum crispum*)	**0.15**	**0.95**	**0.85**	0.30 – 0.45	0.95
Plumed cockscomb (*Celosia argentea*)	**0.15**	**0.85**	**0.30**	0.50 – 0.60	0.95
Roselle (*Hibiscus sabdariffa*)	**0.15**	**0.95**	**0.70**	0.50 – 0.70	0.80
Rosemary, rosmarin (*Rosmarinus officinalis*)	**0.15**	**0.80**	**0.60**	0.45 – 0.80	0.85
Saffron (*Crocus sativus*)	**0.15**	**0.85**	**0.30**	0.10 – 0.15	0.30
Stevia, candyleaf, sweetleaf (*Stevia rebaudiana*)	**0.15**	**0.85**	**0.60**	0.30 – 0.80	0.90
Summer savory (*Satureja hortensis*)	**0.15**	**1.00**	**0.95**	0.50	0.90
Thyme (*Thymus vulgaris*)	**0.15**	**0.85**	**0.70**	0.30 – 0.50	0.90
Yarrow (*Achillea millefolium*)	**0.15**	**0.90**	**0.70**	0.90 – 1.00	0.75
d3. Fruit, pod and seed herbs					
Ajowan (*Trachyspermum ammi*)	**0.15**	**0.70**	**0.20**	0.65 – 1.00	0.50 – 0.80
Black cumin (*Nigella sativa*)	**0.15**	**0.95**	**0.35**	0.40	0.95
Chia (*Salvia hispanica*)	**0.15**	**0.90**	**0.30**	0.40 – 1.20	0.95
Cumin (*Cuminum cyminum*)	**0.15**	**0.80**	**0.30**	0.15 – 0.40	0.95
Milk thistle (*Silybum marianum* or *Carduus m.*)	**0.15**	**0.75**	**0.55**	1.00 – 1.85	0.75
Lesquerella, popweed (*Physaria fendleri*)	**0.15**	**0.75**	**0.20**	0.30 – 0.50	0.90
Vanilla (*Vanilla planifolia*), including tutor and shadow plants	**0.95**	**0.95**	**0.95**	1.50 – 2.00	0.95

TABLE 7.2

Basal crop coefficients, K_{cb}, and ranges of maximum plant heights and of maximum fraction of ground cover for non-stressed, well-managed field crops in subhumid climates (RH_{min}≈45%, u_2≈2 m s^{-1}) for use with the FAO Penman-Monteith ET_o equation

Crop	$K_{cb\ ini}$	$K_{cb\ mid}$	$K_{cb\ end}$	Maximum crop height (h, m)	Maximum ground cover fraction (f_c)
a. Grain legumes					
Bean					
Common bean (*Phaseolus vulgaris*)					
Green	**0.15**	**1.00**	**0.85**	0.50 – 0.70	0.80
Dry	**0.15**	**1.00**	**0.30**	0.50 – 0.70	0.80
Butter bean, lima bean (*Phaseolus lunatus*)	**0.15**	**0.90**	**0.25**	0.60 – 1.80	0.60
Mat bean, moth bean (*Vigna aconitifolia*)	**0.15**	**0.95**	**0.20**	0.15 – 0.25	0.80
Black and green gram (*Vigna mungo*)					
Black gram (dry)	**0.15**	**1.05**	**0.25**	0.50 – 0.70	0.95
Green gram	**0.15**	**1.05**	**0.30**	0.60 – 0.90	0.95
Chickpea, garbanzo (*Cicer arietinum*)	**0.15**	**1.00**	**0.25**	0.50 – 0.70	0.90
Clusterbean, guar (*Cyamopsis tetragonoloba*)					
For green beans	**0.15**	**0.90**	**0.80**	0.80 – 1.50	0.85
For guar gum	**0.15**	**0.90**	**0.30**	0.80 – 1.50	0.85
Cowpea (*Vigna unguiculata*)					
Green	**0.15**	**1.00**	**0.50**	0.60 – 0.80	0.80
Dry	**0.15**	**1.00**	**0.25**	0.60 – 0.80	0.80
Fava bean (*Vicia faba*)					
Fresh	**0.15**	**1.05**	**0.95**	0.80	0.85
Dry	**0.15**	**1.05**	**0.30**	0.80	0.85
Groundnut, peanut (*Arachis hypogaea*)	**0.15**	**1.00**	**0.50**	0.50	0.90
Lentil (*Lens culinaris*)	**0.15**	**1.00**	**0.20**	0.50	0.85
Pea (*Pisum sativum*)					
Fresh	**0.15**	**1.10**	**1.05**	0.60	0.98
Dry	**0.15**	**1.05**	**0.20**	0.80	0.98
Soybean (*Glycine max*)	**0.15**	**1.05**	**0.35**	0.80	0.98
b. Fiber Crops					
Cotton (*Gossypium hirsutum*)	**0.15**	**1.05**	**0.40**	1.20	0.90
Flax (Linseed) (*Linum usitatissimum*)	**0.15**	**1.00**	**0.20**	1.00	0.85
Jute mallow for fibre (*Corchorus olitorius*)	**0.15**	**0.95**	**0.55**	1.10 – 2.70	0.90
Kenaf (*Hibiscus cannabinus*)	**0.15**	**1.05**	**0.60**	3.00 – 4.00	0.95
Sisal (*Agave sisalana*)	**0.15**	**0.50**	**0.50**	1.50	0.85
Sunn hemp (*Crotalaria juncea*)	**0.15**	**1.00**	**0.90**	1.10	0.85
c. Oil Crops					
Camelina (*Camelina sativa*)	**0.15**	**1.05**	**0.35**	0.80	0.95
Canola (*Brassica napus*)	**0.15**	**1.05**	**0.25**	1.00 – 1.50	0.95
Castorbean (*Ricinus communis*)	**0.15**	**1.00**	**0.30**	1.00 – 1.50	0.95
Linseed, Flax (*Linum usitatissimum*)	**0.15**	**0.90**	**0.15**	0.90	0.90
Mustard (*Brassica juncea*)	**0.15**	**1.05**	**0.30**	1.50 – 2.00	0.95
Safflower (*Carthamus tinctorius*)	**0.15**	**1.00**	**0.15**	1.10	0.90
Sesame (*Sesamum indicum*)	**0.15**	**1.00**	**0.20**	1.30	0.85
Sunflower (*Helianthus annuus*)	**0.15**	**1.10**	**0.20**	2.00	0.90
d. Sugar crops					
Sugar beet (*Beta vulgaris*)	**0.15**	**1.05**	**0.65**	0.50	0.90
Sugar cane (*Saccharum officinarum*)	**0.15**	**1.15**	**0.70**	3.00 – 4.00	0.95

TABLE 7.2 (CONT.)

Crop	$K_{cb\ ini}$	$K_{cb\ mid}$	$K_{cb\ end}$	Maximum crop height (h, m)	Maximum ground cover fraction (f_c)
e. Cereals and pseudocereals					
Amaranth grain (*Amaranthus* sp.)	**0.15**	**1.05**	**0.20**	2.00	0.80
Barley (*Hordeum vulgare*)	**0.15**	**1.05**	**0.20**	0.70 – 0.90	0.90
Buckwheat (*Fagopyrum esculentum*)	**0.15**	**1.00**	**0.20**	1.00 – 1.20	0.90
Maize (*Zea mays*)					
Grain, low grain moisture at harvest	**0.15**	**1.15**	**0.25**	2.50 – 3.50	0.95
Grain, high grain moisture at harvest	**0.15**	**1.15**	**0.55**	2.50 – 3.50	0.95
Silage	**0.15**	**1.10**	**0.80**	2.50 – 3.20	0.90
Sweet	**0.15**	**1.10**	**1.00**	1.50 – 2.50	0.85
Oats (*Avena sativa*)	**0.15**	**1.00**	**0.20**	0.80 – 1.10	0.90
Pearl Millet (*Pennisetum glaucum*)	**0.15**	**1.05**	**0.20**	2.00	0.85
Quinoa (*Chenopodium quinoa*)	**0.15**	**1.05**	**0.40**	1.00 – 1.20	0.80
Rye (*Secale cereale*)	**0.15**	**0.95**	**0.20**	0.90	0.90
Sorghum (*Sorghum bicolor*)					
Grain	**0.15**	**1.00**	**0.35**	1.50 – 2.00	0.85
Silage	**0.15**	**0.95**	**0.85**	2.00 – 3.00	0.80
Sweet	**0.15**	**1.05**	**0.90**	3.00 – 4.00	0.95
Teff (*Eragrostis tef*)	**0.15**	**1.00**	**0.20**	1.10	0.80
Wheat, common (*Triticum aestivum*)					
Winter, low grain moisture at harvest	**0.15**	**1.10**	**0.20**	0.70 – 1.10	0.95
Winter, high grain moisture at harvest	**0.15**	**1.10**	**0.45**	0.70 – 1.10	0.95
Spring, low grain moisture at harvest	**0.15**	**1.10**	**0.20**	0.70 – 1.10	0.95
Spring, high grain moisture at harvest	**0.15**	**1.10**	**0.40**	0.70 – 1.10	0.95
Wheat, durum (*Triticum durum*)					
Winter, low grain moisture at harvest	**0.15**	**1.00**	**0.20**	0.70 – 1.10	0.95
Winter, high grain moisture at harvest	**0.15**	**1.00**	**0.45**	0.70 – 1.10	0.95
f. Rice (*Oryza sativa*)					
Flooded	**0.15**	**1.15**	**0.90**	1.00	0.95
Flooded, dry seeding	**0.15**	**1.15**	**0.90**	1.00	0.95
Flooded, anticipated cut–off	**0.15**	**1.15**	**0.70**	1.00	0.95
Intermittent irrigation	**0.15**	**1.15**	**0.85**	1.00	0.95
Aerobic, sprinkler irrigation	**0.15**	**1.05**	**0.70**	0.80	0.95
Aerobic, surface irrigation	**0.15**	**1.05**	**0.70**	0.80	0.95
Rainfed	**0.15**	**0.95**	**0.70**	0.70	0.90

TABLE 7.3

Basal crop coefficients, K_{cb}, and ranges of maximum plant heights and of maximum fraction of ground cover for non-stressed, well-managed fruit trees and vines for use with the FAO Penman-Monteith ET_o equation

Degree of ground cover or field system, training, and plant density	Fraction of ground cover (f_c)	Maximum crop height (h, m)	$K_{cb\ ini}$	$K_{cb\ mid}$	$K_{cb\ end}$
a. Mediterranean and warm temperate climates fruit trees and vines					
a.1. Olive *(Olea europea)*					
Young (< 7 years traditional, < 4 years intensive)	0.15 – 0.30	1.5 – 2.0	**0.20**	**0.30**	**0.20**
Traditional, non–irrigated; low density; 100 – 200 pl ha^{-1}	0.15 – 0.30	2.5 – 3.5	**0.25**	**0.30**	**0.30**
Traditional, non–irrigated; medium density; 200 – 300 pl ha^{-1}	0.20 – 0.40	3.0 – 4.5	**0.30**	**0.35**	**0.35**
Intensive, hedge–prune; low density; 300 – 800 pl ha^{-1}	0.30 – 0.40	3.0 – 3.5	**0.40**	**0.50**	**0.45**
Super–Intensive; medium density; hedgerow; 800 – 1 500 pl ha^{-1}	0.35 – 0.45	3.0 – 3.5	**0.45**	**0.60**	**0.55**
Super–intensive; high density, hedgerow; 1 500 – 2 000 pl ha^{-1}	0.45 – 0.55	3.0 – 4.0	**0.50**	**0.65**	**0.60**
a.2. Table grapes *(Vitis vinifera)*					
Low (Young); diverse trellis and trainings	< 0.40	< 1.5	**0.20**	**0.55**	**0.45**
Medium; T–trellis, Y–trellis and VSP; 1 200 – 1 700 pl ha^{-1}	0.40 – 0.60	1.5 – 2.2	**0.25**	**0.85**	**0.60**
High; Y–trellis and overhead system; 1 200 – 1 700 pl ha^{-1}	0.60 – 0.95	2.0 – 2.5	**0.35**	**1.05**	**0.75**
a.3. Wine grapes (*Vitis vinifera*)					
Very low (Young < 5 years); diverse trellis and trainings; 2 000 – 3 300 pl ha^{-1}	< 0.15	< 1.5	**0.10**	**0.20**	**0.15**
Low; diverse trellis and trainings; 2 000 – 3 300 pl ha^{-1}	0.15 – 0.35	1.5 – 2.0	**0.20**	**0.45**	**0.25**
Medium; VSP, single & double Guyot, GDC, single & bilateral cordon, Lyre, Y–trellis; 2000 – 3300 pl ha^{-1}	0.35 – 0.50	1.5 – 2.0	**0.25**	**0.70**	**0.45**
High ; VSP, GDC, Lyre, Y–trellis, T–trellis, QCT, Pergola; 2 000 – 4 300 pl ha^{-1}	0.50 – 0.60	1.5 – 2.5	**0.30**	**0.85**	**0.55**
Very high; Y–trellis, GDC and overhead system; 2 000 – 4 300 pl ha^{-1}	> 0.60	1.8 – 2.5	**0.35**	**0.95**	**0.60**
a.4. Citrus trees					
a.4.1. Clementine *(Citrus clementina)*, **Mandarin** *(C. reticulata)*, **Lime** *(C. aurantifolia)*					
Young (< 5 years); vase	0.10 – 0.20	< 1.5	**0.35**	**0.35**	**0.35**
Low; vase; < 400 pl ha^{-1}	0.20 – 0.35	1.5 – 2.5	**0.40**	**0.50**	**0.50**
Medium; vase; 400 – 550 pl ha^{-1}	0.35 – 0.60	2.5 – 3.0	**0.60**	**0.65**	**0.65**
High; vase; > 550 pl ha^{-1}	> 0.60	3.0 – 4.0	**0.70**	**0.75**	**0.75**
a.4.2. Lemon *(C. limon)*					
Young (< 5 years); vase	0.10 – 0.20	< 1.5	**0.35**	**0.35**	**0.35**
Low density; vase; < 200 pl ha^{-1}	0.20 – 0.50	1.5 – 2.5	**0.40**	**0.55**	**0.55**
Medium density; vase; 200 – 400 pl ha^{-1}	0.50 – 0.70	2.5 – 3.0	**0.65**	**0.70**	**0.70**
High density; vase; > 400 pl ha^{-1}	> 0.70	3.0 – 4.0	**0.70**	**0.75**	**0.75**
Hedgerow (Industry); > 1 000 pl ha^{-1}	> 0.60	> 2.5	**0.70**	**0.75**	**0.75**
a.4.3. Orange *(C. sinensis)*, **Grapefruit** *(C. paradisi)*, **Tangelo** *(C. tangelo)*					
Young (< 5 years); vase	0.10 –0.20	< 2.0	**0.35**	**0.35**	**0.35**
Low; vase; < 400 pl ha^{-1}	0.20 – 0.40	2.0 – 3.0	**0.40**	**0.50**	**0.45**
Medium; vase; 400 – 600 pl ha^{-1}	0.40 – 0.70	3.0 – 4.0	**0.60**	**0.65**	**0.60**
High; vase; 600 – 950 pl ha^{-1}	> 0.70	> 4.0	**0.65**	**0.75**	**0.70**
Hedgerow (Industry); > 1 250 pl ha^{-1}	> 0.60	> 2.5	**0.65**	**0.70**	**0.70**
a.5. Avocado *(Persea americana)*					
Young (< 3 years); vase	< 0.20	< 2.0	**0.30**	**0.35**	**0.35**
Low ground cover; vase; < 300 pl ha^{-1}	0.20 – 0.35	2.0 – 3.0	**0.45**	**0.55**	**0.50**
Medium to high; vase; 300 – 400 pl ha^{-1}	0.35 – 0.60	3.0 – 4.0	**0.65**	**0.75**	**0.70**
Very high; vase; > 400 pl ha^{-1}	> 0.60	> 4.0	**0.80**	**0.90**	**0.80**
Hedgerow; medium density; 300 – 400 pl ha^{-1}	0.35 – 0.60	> 2.5	**0.70**	**0.75**	**0.70**
Hedgerow; high density; > 400 pl ha^{-1}	0.60 – 0.85	> 3.0	**0.85**	**0.90**	**0.80**

TABLE 7.3 (CONT.)

Degree of ground cover or field system, training, and plant density	Fraction of ground cover (f_c)	Maximum crop height (h, m)	$K_{cb\ ini}$	$K_{cb\ mid}$	$K_{cb\ end}$
a.6. Fig tree *(Ficus carica)*					
Young (< 5 years)	< 0.20	< 1.5	**0.15**	**0.25**	**0.15**
Low, < 400 pl ha^{-1}	0.20 – 0.30	1.5 – 2.0	**0.20**	**0.50**	**0.20**
Medium to high, 800 – 1 000 pl ha^{-1}	0.30 – 0.50	2.0 – 2.5	**0.25**	**0.75**	**0.30**
a.7. Loquat *(Eriobotrya japonica)*					
Young (< 4 years); vase	< 0.25	< 2.5	**0.30**	**0.35**	**0.35**
Low to medium; vase; 450 – 600 pl ha^{-1}	0.25 – 0.50	2.5 – 3.5	**0.40**	**0.65**	**0.60**
High; vase; > 600 pl ha^{-1}	> 0.50	> 3.5	**0.50**	**0.75**	**0.65**
a.8. Persimmon *(Diospyros kaki)*					
Young (< 5 years)	< 0.15	< 2.0	**0.15**	**0.25**	**0.10**
Low to medium, 300 – 500 pl ha^{-1}	0.15 – 0.40	2.0 – 2.5	**0.15**	**0.55**	**0.20**
High, 500 – 800 pl ha^{-1}	0.40 – 0.60	2.5 – 3.0	**0.20**	**0.80**	**0.45**
a.9. Pomegranate *(Punica granatum)*					
Young (< 5 years)	< 0.25	< 2.0	**0.15**	**0.30**	**0.20**
Low; vase; 500 – 800 pl ha^{-1}	0.25 – 0.40	2.0 – 2.5	**0.20**	**0.45**	**0.25**
Medium; vase; 500 – 800 pl ha^{-1}	0.40 – 0.60	2.5 – 3.0	**0.30**	**0.65**	**0.35**
High; vase; > 800 pl ha^{-1}	> 0.60	> 3.0	**0.35**	**0.85**	**0.50**
a.10. Tea *(Camellia sinensis)*					
Young (< 2 years); low hedgerow; 7 000 – 13 500 pl ha^{-1}	< 0.50	< 0.70	**0.60**	**0.70**	**0.70**
Mature; hedgerow; 7 000 – 13 500 pl ha^{-1}	0.50 – 0.90	> 0.70	**0.95**	**0.95**	**0.95**
b. Temperate climate fruit trees, vines and shrubs					
b.1. Pome trees					
b.1.1. Apple *(Malus domestica)*					
Young (< 4 years); central leader	< 0.25	< 3.0	**0.20**	**0.45**	**0.25**
Low; central leader; < 1 500 pl ha^{-1}	0.25 – 0.40	3.0 – 4.0	**0.25**	**0.55**	**0.30**
Medium; central leader; 1 500 – 3 000 pl ha^{-1}	0.40 – 0.50	3.0 – 4.0	**0.35**	**0.75**	**0.35**
High, central leader; 1 500 – 3 000 pl ha^{-1}	0.50 – 0.70	3.0 – 4.5	**0.40**	**0.90**	**0.50**
Very high, central leader; > 3 000 pl ha^{-1}	> 0.70	> 3.0	**0.45**	**0.95**	**0.50**
b.1.2. Pear *(Pyrus communis)*					
Young (< 5 years); all training systems	< 0.35	< 2.5	**0.25**	**0.50**	**0.35**
Low; vase; < 1 000 pl ha^{-1}	0.35 – 0.40	2.0 – 2.5	**0.30**	**0.70**	**0.40**
Medium; central leader; 1 000 – 1 600 pl ha^{-1}	0.35 – 0.60	2.5 – 4.0	**0.40**	**0.90**	**0.50**
High, central leader, other trellis systems; > 1 500 pl ha^{-1}	> 0.60	> 3.5	**0.55**	**1.00**	**0.60**
b.2. Stone fruit trees					
b.2.1. Apricot *(Prunus armeniaca)*, **plum** (*P. domestica* and *P. salicina*)					
Young, vase; (< 5 years)	< 0.40	< 2.5	**0.20**	**0.50**	**0.25**
Low; vase; < 200 pl ha^{-1}	0.40 – 0.65	2.5 – 3.0	**0.30**	**0.70**	**0.35**
Medium; vase; 200 – 400 pl ha^{-1}	0.65 – 0.70	3.0 – 4.5	**0.40**	**0.85**	**0.45**
High; vase; > 400 pl ha^{-1}	0.70 – 0.80	3.0 – 5.5	**0.45**	**0.95**	**0.55**
Very high; vase; > 400 pl ha^{-1}	> 0.80	3.0 – 5.5	**0.50**	**1.00**	**0.60**
b.2.2. Cherry *(Prunus avium)*					
Young (< 5 years)	< 0.30	< 3.0	**0.25**	**0.50**	**0.35**
Low; vase; < 800 pl ha^{-1}	0.25 – 0.40	3.0 – 4.0	**0.30**	**0.55**	**0.40**
Medium to High (vigorous rootstock); vase; < 800 pl ha^{-1}	0.40 – 0.70	> 4.0	**0.35**	**0.80**	**0.45**
Low to Medium (semi vigorous rootstock); central leader; 800 – 1 250 pl ha-1	0.40 – 0.50	4.0 – 5.0	**0.35**	**0.75**	**0.50**
Medium to High (semi–dwarf rootstock); central leader; > 1 200 pl ha^{-1}	0.40 – 0.70	> 2.5	**0.45**	**0.85**	**0.55**
High (dwarf rootstock), central leader; > 1 200 pl ha-^{1}	0.70–0.90	2.0 – 3.5	**0.50**	**0.95**	**0.60**

TABLE 7.3 (CONT.)

Degree of ground cover or field system, training, and plant density	Fraction of ground cover (f_c)	Maximum crop height (h, m)	$K_{cb\ ini}$	$K_{cb\ mid}$	$K_{cb\ end}$
b.2.3. Peach *(Prunus persica)*					
Young (< 4 years)	< 0.30	< 2.0	**0.20**	**0.50**	**0.25**
Low (high vigour), vase; 400 – 600 pl ha^{-1}	0.30 – 0.50	2.0 – 3.5	**0.25**	**0.70**	**0.40**
Medium (medium vigour); vase; 400 – 700 pl ha^{-1}	0.50 – 0.65	3.5 – 4.5	**0.30**	**0.90**	**0.50**
High (low vigour), vase; > 700 pl ha^{-1}	> 0.65	> 4.0	**0.35**	**1.00**	**0.60**
Medium (medium vigour), trellis; > 1 100 pl ha^{-1}	0.30 – 0.60	3.5 – 4.5	**0.30**	**0.80**	**0.45**
High (low vigour), trellis; > 1 500 pl ha^{-1}	0.60 – 0.75	> 3.5	**0.35**	**1.05**	**0.55**
Very high (low vigour), trellis; > 2 000 pl ha^{-1}	> 0.75	> 3.5	**0.40**	**1.10**	**0.65**
b.3. Nut fruit trees					
b.3.1. Almond *(Prunus dulcis)*					
Young (< 4 years)	< 0.30	< 3.0	**0.15**	**0.40**	**0.30**
Low (high vigour), vase; < 400 pl ha^{-1}	0.20 – 0.40	2.0 – 4.0	**0.25**	**0.50**	**0.35**
Medium (medium vigour), vase; < 400 pl ha^{-1}	0.40 – 0.55	4.0 – 4.5	**0.30**	**0.80**	**0.45**
High (low vigour), vase; > 200 pl ha^{-1}	0.55 – 0.65	4.5 – 5.0	**0.35**	**0.95**	**0.55**
Very high (low vigour), vase; > 200 pl ha^{-1}	> 0.65	4.0 – 5.5	**0.35**	**1.05**	**0.60**
Hedgerow/Super–intensive (dwarf rootstock), > 2 500 pl ha^{-1}	0.30 – 0.45	2.0 – 2.5	**0.25**	**0.80**	**0.40**
b.3.2. Hazelnut *(Corylus avellana)*					
Young (< 6 years)	< 0.35	< 2.0	**0.20**	**0.55**	**0.35**
Low to medium; multiple stem–shrub system; < 500 pl ha^{-1}	0.35 – 0.55	2.0 – 3.5	**0.25**	**0.70**	**0.40**
Medium to high; multiple stem–shrub system, 500 – 700 pl ha^{-1}	0.55 – 0.70	3.5 – 4.5	**0.35**	**0.80**	**0.45**
Very high; multiple stem–shrub system, > 700 pl ha^{-1}	> 0.70	> 4.5	**0.40**	**0.90**	**0.50**
Hedgerow/Super–intensive, > 1 000 pl ha^{-1}	>0.55	>2.5	**0.40**	**0.90**	**0.50**
b.3.3. Pecans *(Carya illinoinensis)*					
Young (< 8 years)	< 0.40	< 8.0	**0.25**	**0.60**	**0.40**
Low; vase; > 100 pl ha^{-1}	0.40 – 0.60	8.0 – 10.0	**0.30**	**0.65**	**0.45**
Medium to high; vase; 100 – 250 pl ha^{-1}	0.60 – 0.80	10.0 – 13.0	**0.40**	**0.80**	**0.55**
Very high; vase; 250 – 300 pl ha^{-1}	> 0.80	> 13.0	**0.45**	**0.90**	**0.60**
b.3.4. Pistachio *(Pistacia vera)*					
Young (< 10 years)	< 0.35	< 3.0	**0.15**	**0.40**	**0.25**
Low; vase; > 100 pl ha^{-1}	0.35 – 0.50	3.0 – 4.0	**0.25**	**0.70**	**0.35**
Medium to high; vase; 150 – 350 pl ha^{-1}	0.50 – 0.70	4.0 – 4.5	**0.30**	**0.85**	**0.45**
Very high; vase; > 300 pl ha^{-1}	> 0.70	> 4.5	**0.35**	**0.95**	**0.55**
Hedgerow/Super-intensive, > 1 000 pl ha^{-1}	> 0.30	> 2.5	**0.30**	**0.90**	**0.45**
b.3.5. Walnut *(Juglans regia)*					
Young, vase (< 7 years)	< 0.55	< 6.0	**0.25**	**0.70**	**0.45**
Low; vase; < 200 pl ha^{-1}	0.55 – 0.75	6.0 – 10.0	**0.30**	**1.00**	**0.50**
Medium to very high; vase; 200 – 550 pl ha^{-1}	0.75 – 0.95	10.0 – 12.0	**0.40**	**1.05**	**0.60**
b.4. Other temperate trees and shrubs					
b.4.1. Kiwifruit *(Actinidia deliciosa)*					
Young (< 6 years); trellis	< 0.60	< 2.0	**0.20**	**0.65**	**0.40**
Medium density; T–bar trellis; < 800 pl ha^{-1}	0.60 – 0.80	2.0 – 2.2	**0.25**	**0.85**	**0.70**
High density; pergola trellis; > 800 pl ha^{-1}	0.60 – 0.95	2.0 – 2.5	**0.30**	**0.90**	**0.75**
b.4.2. Hop *(Humulus lupulus)*					
Common density; trellis; < 3 500 pl ha^{-1}	0.08 – 0.12	6.0 – 7.0	**0.15**	**0.95**	**0.80**
b.4.3. Highbush blueberries *(Vaccinium corymbosum)*					
Young (< 3 years)	< 0.50	< 1.50	**0.20**	**0.55**	**0.35**
Common density, 2 500 – 5 000 pl ha^{-1}	0.50 – 0.80	1.50 – 2.00	**0.45**	**0.85**	**0.65**

TABLE 7.3 (CONT.)

Degree of ground cover or field system, training, and plant density	Fraction of ground cover (f_c)	Maximum crop height (h, m)	$K_{cb\ ini}$	$K_{cb\ mid}$	$K_{cb\ end}$
b.4.4. Lowbush blueberries *(Vaccinium angustifolium)*					
Young (< 2 years)	< 0.50	< 0.60	**0.20**	**0.55**	**0.40**
Common density, > 5 000 pl ha^{-1}	0.50 – 0.80	0.60 – 1.00	**0.35**	**0.75**	**0.60**
b.4.5. Rabbiteye blueberry *(Vaccinium ashei)*					
Young (< 3 years)	< 0.50	< 0.60	**0.20**	**0.55**	**0.40**
Common density, industry, 2 500 – 5 000 pl ha^{-1}	0.50 – 0.80	0.60 – 1.00	**0.35**	**0.70**	**0.60**
b.4.6. Blackberry *(Rubus L. subgenus Rubus)*					
Young (< 3 years)	< 0.50	< 1.00	**0.20**	**0.35**	**0.25**
Common density, 2 000 – 5 000 pl ha^{-1}	0.50 – 0.85	1.00 – 2.00	**0.25**	**0.85**	**0.40**
c. Sub–tropical and tropical orchards and plantations					
c.1. Sub–tropical and tropical small fruit trees and shrub					
c.1.1. Acerola *(Malpighia emarginata)*					
Young (< 5 years)	< 0.50	< 2.0	**0.35**	**0.45**	**0.40**
Medium–High, 250 – 700 pl ha^{-1}	0.50 – 0.70	2.0 – 4.0	**0.60**	**0.65**	**0.60**
c.1.2. Carambola *(Averrhoa carambola)*					
Young (< 4 years)	< 0.40	<2.0	**0.55**	**0.90**	**0.70**
Common density, 156 – 278 pl ha^{-1}	0.40 – 0.60	2.0 – 4.0	**0.85**	**1.05**	**1.00**
c.1.3. Cherimoya *(Annona cherimola)*					
Young (< 3 years)	< 0.15	< 1.5	**0.15**	**0.25**	**0.20**
Low to medium, 125 – 400 pl ha^{-1}	0.15 – 0.35	1.5 – 3.5	**0.30**	**0.45**	**0.40**
Medium to high, 500 – 1250 pl ha^{-1}	> 0.35	> 3.0	**0.55**	**0.80**	**0.65**
c.1.4. Cocoa *(Theobroma cacao)*					
Young (< 4 years), shading	< 0.50	< 2.0	**0.45**	**0.65**	**0.60**
Medium shading, < 1 100 pl ha^{-1}	0.50 – 0.70	2.0 – 2.5	**0.80**	**0.85**	**0.85**
High shading, < 1 500 pl ha^{-1}	> 0.70	> 2.0	**0.90**	**0.95**	**0.95**
c.1.5. Coffee *(Coffea arabica)*					
Young (< 4 years)	< 0.20	< 2.0	**0.30**	**0.35**	**0.35**
Low to Medium, < 6 000 pl ha^{-1}	0.20 – 0.50	1.5 – 3.0	**0.60**	**0.70**	**0.70**
High (including hedgerow), > 6 000 pl ha^{-1}	> 0.50	>2.0	**0.85**	**0.90**	**0.90**
c.1.6. Dragon fruit *(Hylocereus undatus)*					
Young (< 3 years)	< 0.20	< 2.0	**0.15**	**0.25**	**0.25**
Medium, < 1 800 pl ha^{-1}	0.20 – 0.30	2.0 – 2.5	**0.25**	**0.50**	**0.40**
High, > 1 800 pl ha^{-1}	> 0.30	2.0 – 2.5	**0.50**	**0.80**	**0.60**
High (hedgerow), > 2 000 pl ha^{-1}	> 0.25	2.0 – 2.5	**0.45**	**0.80**	**0.55**
c.1.7. Guava *(Psidium guajava)*					
Young (< 8 years)	< 0.20	< 1.5	**0.15**	**0.30**	**0.25**
Low to medium, < 300 pl ha^{-1}	0.20 – 0.40	1.5 – 2.0	**0.40**	**0.60**	**0.55**
High, < 350 pl ha^{-1}	0.40 – 0.60	2.0 – 2.5	**0.65**	**0.85**	**0.80**
Very high, > 300 pl ha^{-1}	> 0.60	> 2.5	**0.80**	**0.90**	**0.85**
c.1.8. Jujube *(Ziziphus jujuba)*					
Young (< 3 years)	< 0.25	< 2.0	**0.15**	**0.30**	**0.20**
Low to Medium; < 1 100 pl ha^{-1}	0.25 – 0.40	2.0 – 3.0	**0.20**	**0.45**	**0.30**
High; > 1 100 pl ha^{-1}	> 0.40	> 3.0	**0.25**	**0.70**	**0.45**
c.1.9. Papaya *(Carica papaya)*					
Young (< 1 year)	< 0.40	< 1.5	**0.35**	**0.60**	**0.55**
Medium (single row), 3 000 – 4 500 pl ha^{-1}	0.40 – 0.70	1.5 – 3.0	**0.55**	**0.85**	**0.60**
High (single or double row), < 7 000 pl ha^{-1}	0.70 – 0.85	> 2.5	**0.75**	**1.00**	**0.80**

TABLE 7.3 (CONT.)

Degree of ground cover or field system, training, and plant density	Fraction of ground cover (f_c)	Maximum crop height (h, m)	$K_{cb\ ini}$	$K_{cb\ mid}$	$K_{cb\ end}$
c.1.10. Passionfruit *(Passiflora edulis)*					
Young (< ½ year), vertical shoot position (VSP)	< 0.15	1.5 – 2.0	**0.25**	**0.30**	**0.25**
Medium; VSP; 500 pl ha^{-1}	0.15 – 0.35	1.5 – 2.0	**0.45**	**0.60**	**0.50**
High; VSP; 1 000 pl ha^{-1}	0.35 – 0.45	1.5 – 2.0	**0.70**	**0.85**	**0.75**
Young (< ½ year), hedgerow	< 0.35	1.5 – 2.0	**0.50**	**0.60**	**0.55**
High; hedgerow; 1 000 pl ha^{-1}	> 0.35	1.5 – 2.0	**0.80**	**0.95**	**0.80**
Young (< ½ year), overhead trellis	< 0.60	1.8 – 2.2	**0.60**	**0.75**	**0.70**
Very high; overhead trellis; 2 000 pl ha^{-1}	> 0.60	1.8 – 2.2	**0.95**	**1.00**	**0.95**
c.2. Tropical fruit trees					
c.2.1. Cashew, dwarf *(Anacardium occidentale)*					
Young (< 5 years)	< 0.50	< 3.0	**0.30**	**0.50**	**0.40**
Common density, 150 – 330 pl ha^{-1}	0.50 – 0.65	3.0 – 5.0	**0.55**	**0.60**	**0.55**
c.2.2. Jaboticaba *(Plinia peruviana)*					
Young (< 5 years)	< 0.50	< 2.0	**0.35**	**0.45**	**0.40**
Common density, 250 – 500 pl ha^{-1}	> 0.50	> 2.5	**0.55**	**0.80**	**0.65**
c.2.3. Longan *(Dimocarpus longan)*					
Young (< 5 years)	< 0.30	1.0 – 2.0	**0.30**	**0.45**	**0.45**
Common density, 250 - 300 pl ha^{-1}	> 0.30	2.5 – 4.0	**0.70**	**0.90**	**0.85**
c.2.4. Lychee *(Litchi chinensis)*					
Young (< 5 years); vase	< 0.20	< 2.5	**0.35**	**0.40**	**0.40**
Low to medium; vase; 100 – 200 pl ha^{-1}	0.20 – 0.30	2.5 – 4.0	**0.45**	**0.60**	**0.55**
High; vase; 200 – 400 pl ha^{-1}	> 0.30	> 4.0	**0.70**	**0.90**	**0.80**
c.2.5. Macadamia *(Macadamia integrifolia)*					
Young (< 10 years)	< 0.50	< 2.0	**0.35**	**0.50**	**0.40**
Medium–High; 300 – 500 pl ha^{-1}	> 0.50	2.0 – 4.5	**0.55**	**0.60**	**0.55**
c.2.6. Mango *(Mangifera indica)*					
Young (< 4 years)	< 0.25	< 2.0	**0.25**	**0.40**	**0.35**
Low to medium,100 – 250 pl ha^{-1}	0.25 – 0.40	2.0 – 3.0	**0.45**	**0.70**	**0.60**
High, 400 – 600 pl ha^{-1}	0.40 – 0.50	2.5 – 4.0	**0.65**	**0.85**	**0.75**
c.3. Palm, fiber, fodder, oil and rubber plantations					
c.3.1. Cactus pear *(Opuntia ficus–indica)*					
Young (< 3 years)	< 0.30	< 1.5	**0.15**	**0.20**	**0.20**
Medium, 350 – 800 pl ha^{-1}	0.30 – 0.50	1.5 – 2.0	**0.20**	**0.30**	**0.25**
High, > 800 pl ha^{-1}	> 0.50	> 2.0	**0.30**	**0.45**	**0.35**
c.3.2. Coconut *(Cocos nucifera)* and **Açaí palm** *(Euterpe oleracea)*					
Young (< 15 years/coconut, < 3 years/açaí)	< 0.30	< 2.5	**0.30**	**0.35**	**0.35**
Medium, 100 – 200 pl ha^{-1}	0.30 – 0.50	2.5 – 5.0	**0.60**	**0.65**	**0.65**
High, > 200 pl ha^{-1}	> 0.50	> 5.0	**0.90**	**0.90**	**0.90**
c.3.3. Date palm *(Phoenix dactylifera)*					
Young (< 10 years)	< 0.20	< 4.0	**0.25**	**0.35**	**0.35**
Medium, 100 – 130 pl ha^{-1}	0.20 – 0.50	4.0 – 10.0	**0.50**	**0.60**	**0.60**
High, 130 – 200 pl ha^{-1}	0.50 – 0.70	5.0 – 12.0	**0.80**	**0.80**	**0.80**
Very high, > 200 pl ha^{-1}	> 0.70	> 10.0	**0.85**	**0.90**	**0.90**
c.3.4. Jatropha *(Jatropha curcas)*					
Young (< 2 years); vase	< 0.15	< 1.0	**0.25**	**0.25**	**0.25**
Low to medium; vase; 1 100 – 1 500 pl ha^{-1}	0.15 – 0.35	1.0 – 2.0	**0.35**	**0.55**	**0.40**
High; vase; > 1 500 pl ha^{-1}	> 0.35	> 2.0	**0.55**	**0.80**	**0.70**

TABLE 7.3 (CONT.)

Degree of ground cover or field system, training, and plant density	Fraction of ground cover (f_c)	Maximum crop height (h, m)	$K_{cb\ ini}$	$K_{cb\ mid}$	$K_{cb\ end}$
c.3.5. Oil palm *(Elaeis guineensis)*					
Young (< 3 years)	< 0.30	< 2.5	**0.30**	**0.35**	**0.35**
Medium, 100 – 150 pl ha^{-1}	0.30 – 0.50	2.5 – 5.0	**0.60**	**0.70**	**0.70**
High, > 150 pl ha^{-1}	> 0.50	> 5.0	**0.85**	**0.95**	**0.95**
c.3.6. Peach palm *(Bactris gasipaes)*					
Young (≤ 2.5 years)	< 0.30	< 2.0	**0.25**	**0.50**	**0.40**
Common density, 3 000 – 5 000 pl ha^{-1} (for production of heart of palm, "palmito")	> 0.30	> 2.0	**0.55**	**0.75**	**0.70**
c.3.7. Ramie *(Boehmeria nivea)*					
Young (< 2 years)	< 0.80	0.70 – 1.0	**0.15**	**0.55**	**0.50**
Common density, 40 000 – 60 000 pl ha^{-1}	0.80 – 0.95	1.0 – 2.0	**0.20**	**1.00**	**0.95**
c.3.8. Rubber tree *(Hevea brasiliensis)*					
Young (< 6 years)	< 0.75	< 8.0	**0.40**	**0.60**	**0.50**
Common density; 400 – 700 pl ha^{-1}	0.75 – 0.90	8.0 – 15.0	**0.75**	**0.90**	**0.80**

TABLE 7.4

General guidelines to derive K_{cb} from the K_c values listed in Tables 6.1 to 6.3 when not indicated in the selected publications

Growth stage	Ground condition, irrigation and cultural practices	K_{cb}	further adjustment
Initial	Annual–crop - (nearly) bare soil surface	0.15	-
	Perennial–crop - (nearly) bare soil surface	0.15 - 0.20	-
	Grasses, brush and trees - killing frost	0.30 - 0.40	-
	Perennial–crop - some ground cover or leaf cover		
	- infrequently irrigated (sprinkler or surface irrigated)	$K_{c\ ini}$ (Tab.6.3) - 0.1	-
	- frequently irrigated (micro- or drip-irrigated)	$K_{c\ ini}$ (Tab. 6.1) - 0.2	-
Mid-season	Ground cover more than ≈ 80%	$K_{c\ mid}$ (Tab.6.1-6.3) - 0.05	Climate (Eq. 7.2)
	Ground cover less than ≈ 80%	$K_{c\ mid}$ (Tab.6.1-6.3) - 0.10	Climate (Eq. 7.2)
At the end of the season	Infrequently irrigated or wetted during late season	$K_{c\ end}$ - 0.05	-
	Frequently irrigated or wetted during late season	$K_{c\ end}$ - 0.10	Climate (Eq. 7.2)

Climate: adjustment for climate using Eq. 7.2 where $K_{cb} > 0.45$

7.1.2 Determination of daily K_{cb} values

As outlined in Chapter 6, only three point values are required to describe and to construct the crop coefficient curve. After dividing the growing period into the four general growth stages and selecting and adjusting the K_{cb} values corresponding to the initial ($K_{cb\ ini}$), mid-season ($K_{cb\ mid}$) and end of the late season stage ($K_{cb\ end}$) as per Example 7.1, the crop coefficient curve can be drawn (Figure 7.1) and the K_{cb} coefficients can be derived (Example 7.2).

EXAMPLE 7.1
Selection and climate adjustment of basal crop coefficients, K_{cb}

Select $K_{cb\ ini}$, $K_{cb\ mid}$ and $K_{cb\ end}$ for the dry bean crop of Box 6.3.

$K_{cb\ ini}$, $K_{cb\ mid}$ and $K_{cb\ end}$ can be selected directly from Table 7.2 for dry beans as 0.15, 1.00 and 0.30 with height h equal to 0.5 m. When adjusted for a climate where u_2= 2.2 m s^{-1} and RH_{min} = 30%, using Eq. 7.2:

$K_{cb\ ini} = 0.15$

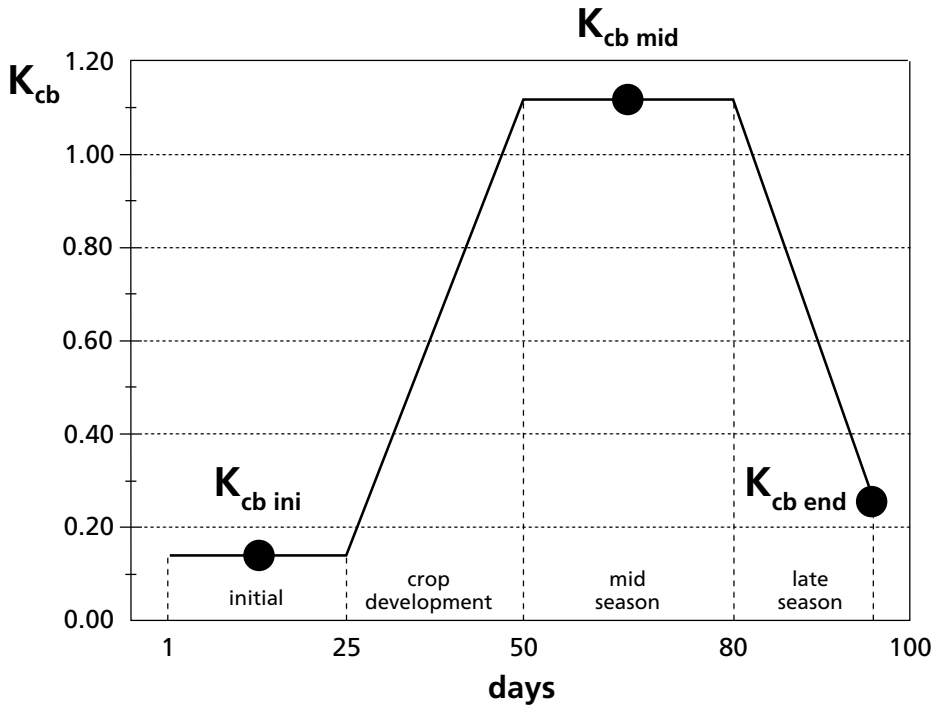
$K_{cb\ mid} = 1.00 + (0.04\ (2.2 - 2) - 0.004\ (30 - 45))(0.50/3)^{0.3} = 1.04$

$K_{cb\ end} = 0.30$ (as $K_{cb} < 0.45$)

Height for beans was taken from Table 7.2 as 0.50 m.

The corresponding K_{cb} curve is shown in Figure 7.1. However, the $K_{cb\ mid}$ was changed to 1.14 for Figure 7.1 and later in Example 7.8 to agree with observed data for the example.

FIGURE 7.1
Constructed basal crop coefficient (K_{cb}) curve for a dry bean crop (Example 7.1) using growth stage lengths of 25, 25, 30 and 20 days
(Note that the $K_{cb\ mid}$ value was changed from 1.04, as obtained using Table 7.2, to 1.14 to agree with field-based observations compared later in Example 7.8.)



Source: FAO, 1998

EXAMPLE 7.2
Determination of daily values for K_{cb} of a dry bean crop

Calculate the basal crop coefficient for the dry bean crop (Example 7.1, Figure 7.1) at the middle of each of the four growth stages.

Initial stage	(L_{ini} = 25 days), at day 12 of the growing period: $K_{cb} = K_{cb\ ini} = 0.15$
Crop development stage	(L_{dev} = 25 days), at day (25 + 25/2) = 37 of the growing period, using Eq. 6.20: $K_{cb} = 0.15 + [(37 - 25)/25]\ (1.04 - 0.15) = 0.58$
Mid-season stage	(L_{mid} = 30 days), at day (25 + 25 + 30/2) = 65 of the growing period: $K_{cb} = K_{cb\ mid} = 1.04$
Late season stage	(L_{late} = 20 days), at day (25 + 25 + 30 + 20/2) = 90 of the growing period, Eq. 6.20: $K_{cb} = 1.04 + [(90 - (25 + 25 + 30))/20]\ (0.30 - 1.04) = 0.60$

The basal crop coefficients, K_{cb}, at days 12, 37, 65 and 90 of the growing period are 0.15, 0.58, 1.04 and 0.67 respectively.

7.2 EVAPORATION COMPONENT (K_e ET_o)

The soil evaporation coefficient, K_e, describes the evaporation component of ET_c. Where the topsoil is wet, following rain or irrigation, K_e is maximal. Where the soil surface is dry, K_e is small and even zero if no water remains near the soil surface for evaporation. Two calculation procedures are presented, the FAO56 original and the skin layer evaporation approach, which consists of a refinement of the former to increase accuracy, mainly for small wetting depths.

7.2.1 FAO 56 original calculation procedure

Where the soil is wet, evaporation from the soil occurs at the maximum rate. However, the crop coefficient ($K_c = K_{cb} + K_e$) can never exceed a maximum value, $K_{c\ max}$. This value is determined by the energy available for evapotranspiration at the soil and plant surfaces, thus ($K_{cb} + K_e \leq K_{c\ max}$), or $K_e \leq (K_{c\ max} - K_{cb})$.

When the topsoil dries out less water is available for evaporation and a reduction in evaporation begins to occur in proportion to the amount of water remaining in the surface soil layer, or:

$$K_e = K_r\ (K_{c\ max} - K_{cb}) \leq f_{ew}\ K_{c\ max} \tag{7.3a}$$

or

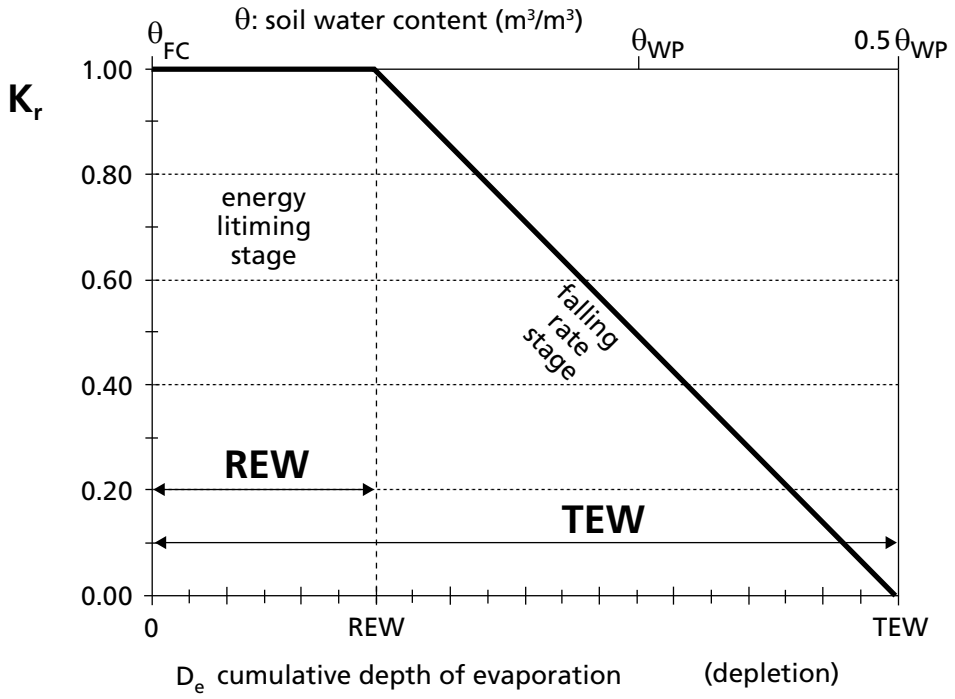
$$K_e = \min\ [K_r\ (K_{c\ max} - K_{cb}),\ f_{ew}\ K_{c\ max}] \tag{7.3b}$$

where
- K_e soil evaporation coefficient,
- K_{cb} basal crop coefficient,
- $K_{c\ max}$ maximum value of K_c following rain or irrigation,
- K_r dimensionless evaporation reduction coefficient dependent on the cumulative depth of water depleted (evaporated) from the topsoil,
- f_{ew} fraction of the soil that is both exposed and wetted, i.e. the fraction of soil surface from which most evaporation occurs.

When calculations refer to a crop under stress, the K_{cb} in Equation 7.3 is replaced by the actual basal crop coefficient, $K_{cb\ act} = K_s\ K_{cb}$, where K_s is a dimensionless water stress coefficient [0.01–1.0] relative to the root zone, which is developed and presented in Chapter 8.

Following rain or irrigation, K_r is 1.0, and evaporation is only determined by the energy available for evaporation. As the soil surface dries, K_r becomes less than 1.0 and evaporation is reduced. K_r becomes zero when no water is left for evaporation in the upper soil layer (Figure 7.2).

FIGURE 7.2
Soil evaporation reduction coefficient, K_r



Source: FAO, 1998

Evaporation occurs predominantly from the exposed soil fraction. Hence, evaporation is restricted at any moment by the energy available at the exposed soil fraction, i.e. K_e cannot exceed $f_{ew}\ K_{c\ max}$, where f_{ew} is the fraction of exposed soil from which most evaporation occurs, i.e. the fraction of the soil not covered by vegetation and that is wetted by irrigation or precipitation.

The calculation procedure consists of determining:

- the upper limit $K_{c\ max}$;
- the soil evaporation reduction coefficient K_r; and
- the exposed and wetted soil fraction f_{ew}.

The estimation of K_r requires a daily water balance computation for the surface soil layer.

Upper limit $K_{c\ max}$

$K_{c\ max}$ represents an upper limit on the evaporation and transpiration from any cropped surface and is imposed to reflect the natural constraints placed on available energy represented by the energy balance difference $R_n - G - H$ (Equation 1.1). *$K_{c\ max}$ ranges from about 1.05 to 1.30* when using the grass reference ET_o:

$$K_{c\ max} = \max\left(\left\{1.2 + [0.04\ (u_2 - 2) - 0.004\ (RH_{min} - 45)]\left(\frac{h}{3}\right)^{0.3}\right\}, \{K_{cb} + 0.05\}\right) \quad (7.4)$$

where h mean maximum plant height during the period of calculation (initial, development, mid-season, or late-season) [m],

K_{cb} basal crop coefficient,

max( ) maximum value of the parameters in braces { } that are separated by the comma.

Equation 7.4 ensures that $K_{c\ max}$ is always greater than or equal to the sum K_{cb} + 0.05. This requirement suggests that wet soil will always increase the value for K_{cb} by 0.05 following complete wetting of the soil surface, even during periods of full ground cover. A value of 1.2 is used for $K_{c\ max}$ in Equation 7.4 because of the effect of increased aerodynamic roughness of surrounding crops during development, mid-season and late season growth stages that can increase the turbulent transfer of vapour from the exposed soil surface. The "1.2" coefficient also reflects the impact of the reduced albedo of wet soil and the contribution of heat stored in dry soil prior to the wetting event. All of these factors can contribute to an increased evaporation relative to the reference.

The "1.2" coefficient in Equation 7.4 represents the effects of wetting intervals that are greater than 3 or 4 days. If irrigation or precipitation events are more frequent, for example daily or every two days, then the soil has less opportunity to absorb heat between wettings, and the "1.2" coefficient in Equation 7.4 can be reduced, e.g. to about 1.1. The time step to compute $K_{c\ max}$ may vary from daily to monthly.

Soil evaporation reduction coefficient (K_r)

Soil evaporation from the exposed soil can be assumed to take place in two stages: an energy limiting stage, and a falling rate stage (Figure 7.2). When the soil surface is wet, K_r is 1.0. When the water content in the upper soil becomes limiting, K_r decreases and becomes zero when the total amount of water that can be evaporated from the topsoil is depleted.

Maximum amount of water that can be evaporated

In the simple evaporation procedure, it is assumed that the water content of the evaporating layer of the soil is at field capacity, θ_{FC}, shortly following a major wetting event and that the soil's evaporating layer can dry to a soil water content level that is halfway between oven dry (no water left) and wilting point, θ_{WP}. *The amount of water that can be depleted by evaporation during a complete drying cycle can hence be estimated as:*

$$TEW = 1\ 000\ (\theta_{FC} - 0.5\ \theta_{WP})\ Z_e \quad (7.5)$$

where TEW total evaporable water = maximum depth of water that can be evaporated from the soil when the topsoil has been initially completely wetted [mm],

θ_{FC} soil water content at field capacity [$m^3\ m^{-3}$],

θ_{WP} soil water content at wilting point [$m^3\ m^{-3}$],

Z_e depth of the surface soil layer that is subject to drying by way of evaporation [0.10-0.15 m].

Where unknown, a value for Z_e, the effective depth of the soil evaporation layer, of 0.10-0.15 m is recommended. Typical values for θ_{FC}, θ_{WP}, REW and TEW are given in Table 7.5. When water has to be maintained in the soil root zone above field capacity as for rice and wetland vegetation, the TEW Equation 7.5 may be modified by replacing θ_{FC} with the wet threshold θ_{WT} adopted by the user and REW may also be adjusted for a value above the one in Table 7.5.

TABLE 7.5
Typical soil water characteristics for different soil types

Soil type (USA Soil Texture Classification)	Soil water characteristics			Evaporation parameters	
	θ_{FC}	θ_{WP}	$(\theta_{FC}-\theta_{WP})$	Amount of water that can be depleted by evaporation	
				stage 1 REW	stages 1 and 2 TEW (Z_e = 0.10 m)
	$m^3\ m^{-3}$	$m^3\ m^{-3}$	$m^3\ m^{-3}$	mm	mm
Sand	0.07 - 0.17	0.02 - 0.07	0.05 - 0.11	2 - 7	6 - 12
Loamy sand	0.11 - 0.19	0.03 - 0.10	0.06 - 0.12	4 - 8	9 - 14
Sandy loam	0.18 - 0.28	0.06 - 0.16	0.11 - 0.15	6 - 10	15 - 20
Loam	0.20 - 0.30	0.07 - 0.17	0.13 - 0.18	8 - 10	16 - 22
Silt loam	0.22 - 0.36	0.09 - 0.21	0.13 - 0.19	8 - 11	18 - 25
Silt	0.28 - 0.36	0.12 - 0.22	0.16 - 0.20	8 - 11	22 - 26
Silt clay loam	0.30 - 0.37	0.17 - 0.24	0.13 - 0.18	8 - 11	22 - 27
Silty clay	0.30 - 0.42	0.17 - 0.29	0.13 - 0.19	8 - 12	22 - 28
Clay	0.32 - 0.40	0.20 - 0.24	0.12 - 0.20	8 - 12	22 - 29

Stage 1: energy limiting stage

At the start of a drying cycle, following heavy rain or irrigation, the soil water content in the topsoil is at field capacity and the amount of water depleted by evaporation, D_e, is zero. *During stage 1 of the drying process, the soil surface remains wet and it is assumed that evaporation from soil exposed to the atmosphere will occur at the maximum rate limited only by energy availability at the soil surface*. This stage holds until the cumulative depth of evaporation, D_e, is such that the hydraulic properties of the upper soil become limiting, and water cannot be transported to the soil surface at a rate that can supply the potential demand. During stage 1 drying, $K_r = 1.0$.

The cumulative depth of evaporation, D_e, at the end of stage 1 drying is REW (readily evaporable water, which is the maximum depth of water that can be evaporated from the topsoil layer without restriction during stage 1). The depth normally ranges from 5 to 12 mm and is generally highest for medium and fine textured soils. Typical values for REW are given in Table 7.5.

Stage 2: falling rate stage

The second stage (where the evaporation rate is reducing) is termed the "falling rate stage" evaporation and starts when depletion D_e exceeds REW. At this point, the soil surface is visibly dry, and the evaporation from the exposed soil decreases in proportion to the amount of water remaining in the surface soil layer:

$$K_r = \frac{TEW - D_{e,\,i-1}}{TEW - REW} \qquad \text{for } D_{e,\,i-1} > REW \tag{7.6}$$

where K_r dimensionless evaporation reduction coefficient dependent on the soil water depletion (cumulative depth of evaporation) from the topsoil layer ($K_r = 1$ when $D_{e,\,i-1} \leq REW$),

$D_{e, i-1}$ cumulative depth of evaporation (depletion) from the soil surface layer at the end of day i-1 (the previous day) [mm],

TEW maximum cumulative depth of evaporation (depletion) from, the soil surface layer when $K_r = 0$ (TEW = total evaporable water) [mm],

REW cumulative depth of evaporation (depletion) at the end of stage 1 (REW = readily evaporable water) [mm].

Exposed and wetted soil fraction

f_{ew}: calculation procedure

In crops with incomplete ground cover, evaporation from the soil often does not occur uniformly over the entire surface but is greater between plants where exposure to sunlight occurs and where more air ventilation is able to transport vapour from the soil surface to above the canopy. This is especially true where only part of the soil surface is wetted by irrigation.

It is recognized that both the location and the fraction of the soil surface exposed to sunlight change to some degree with the time of the day and depending on row orientation. The procedure presented here predicts a general averaged fraction of the soil surface from which the majority of evaporation occurs. Diffusive evaporation from the soil beneath the crop canopy is very small and assumed to be largely included in the basal K_{cb} coefficient.

Where the complete soil surface is wetted, as by precipitation or sprinkler, then the fraction of soil surface from which most evaporation occurs, f_{ew}, is essentially defined as $(1–f_c)$, where f_c is the average fraction of soil surface covered by vegetation and $(1–f_c)$ is the approximate fraction of soil surface that is exposed (Example 7.3). However, for irrigation systems where only a fraction of the ground surface is wetted, f_{ew} must be limited to f_w, the fraction of the soil surface wetted by irrigation (Figure 7.3). Therefore, f_{ew} is calculated as:

$$f_{ew} = \min (1 - f_c, f_w) \quad (7.7)$$

where $1 – f_c$ average exposed soil fraction not covered (or shaded) by vegetation [0.01-1],

f_w average fraction of soil surface wetted by irrigation or precipitation [0.01-1].

The "min()" function selects the lowest value of the "$1 – f_c$" and "f_w" values. Figure 7.3 illustrates the relation of f_{ew} to $(1 – f_c)$ and f_w. The limitation imposed by Equation 7.7 assumes that the fraction of soil wetted by irrigation occurs within the fraction of soil exposed to sunlight and wind. This is generally the case, except perhaps with drip irrigation (Figure 7.3). In the case of drip irrigation, the majority of soil wetted by irrigation is beneath the canopy and shaded. More complex models may be required to more accurately estimate total evaporation from the soil. The value for f_w may need to be reduced to about one-half to one-third of that given in Table 7.6 to account for the effects of shading of emitters by the plant canopy (Example 7.6). A general simple approach could be to multiply tabled f_w by $[1 – (2/3)f_c]$ for drip irrigation.

f_w: fraction of soil surface wetted by irrigation or precipitation

Table 7.6 presents typical values for f_w. Nevertheless, users should preferably observe that fraction instead of just adopting the tabulated values. Where a mixture of irrigation and precipitation occurs within the same drying period or on the same day, the value for f_w should be based on a weighted average of the f_w for precipitation ($f_w = 1$) and the f_w for the irrigation system. The weighting should be approximately proportional to the infiltration depths from each water source.

EXAMPLE 7.3
Determination of the evapotranspiration from bare soil

Determine the evapotranspiration from a bare loamy soil surface ($K_{cb} \approx 0.15$) for ten successive days following a heavy rain. The reference evapotranspiration during the drying period is $ET_o = 4.5$ mm day^{-1}, and the climate is subhumid with light wind.

From Table 7.5 — For Loam: TEW ≈ 20 mm and REW ≈ 9 mm

For rain on bare soil — $f_{ew} = 1$

From Eq. 7.4 — $K_{c\,max} = 1.20$

(1) Day	(2) $D_{e\,start}$ mm	(3) Stage	(4) K_r	(5) K_e	(6) $K_e\,ET_o$ mm day^{-1}	(7) $D_{e\,end}$ mm	(8) ET_c mm day^{-1}
1	0.00	1	1	1.05	4.73	4.73	5.4
2	4.73	1	1	1.05	4.73	9.45	5.4
3	9.45	2	(20–9.45)/(20–9)=0.96	1.01	4.53	13.98	5.2
4	13.98	2	(20–13.98)/(20–9)=0.55	0.57	2.59	16.57	3.3
5	16.57	2	(20–16.57)/(20–9)=0.31	0.33	1.47	18.04	2.1
6	18.04	2	(20–18.04)/(20–9)=0.18	0.19	0.84	18.88	1.5
7	18.88	2	(20–18.88/(20–9)=0.10	0.11	0.48	19.36	1.2
8	19.36	2	(20–19.36)/(20–9)=0.06	0.06	0.27	19.64	0.9
9	19.64	2	(20–19.64)/(20–9)=0.03	0.03	0.16	19.79	0.8
10	19.79	2	(20–19.79)/(20–9)=0.02	0.02	0.09	19.88	0.8

The example evidences that the estimation of K_r requires a daily water balance calculation. This is further developed in the section on the daily calculation of K_e.

Legend:

(1) Day number.

(2) Depletion at the beginning of the day (= depletion at the end of the previous day).

(3) Soil evaporation stage (stage 2 starts if $D_e >$ REW = 9 mm).

(4) K_r ($K_r = 1$ for stage 1. Use Eq. 7.6 for stage 2).

(5) From Eq. 7.3: $K_e = K_r\,(K_{c\,max} - K_{cb}) = K_r\,(1.20–0.15) = 1.05\,K_r \leq 1.20$.

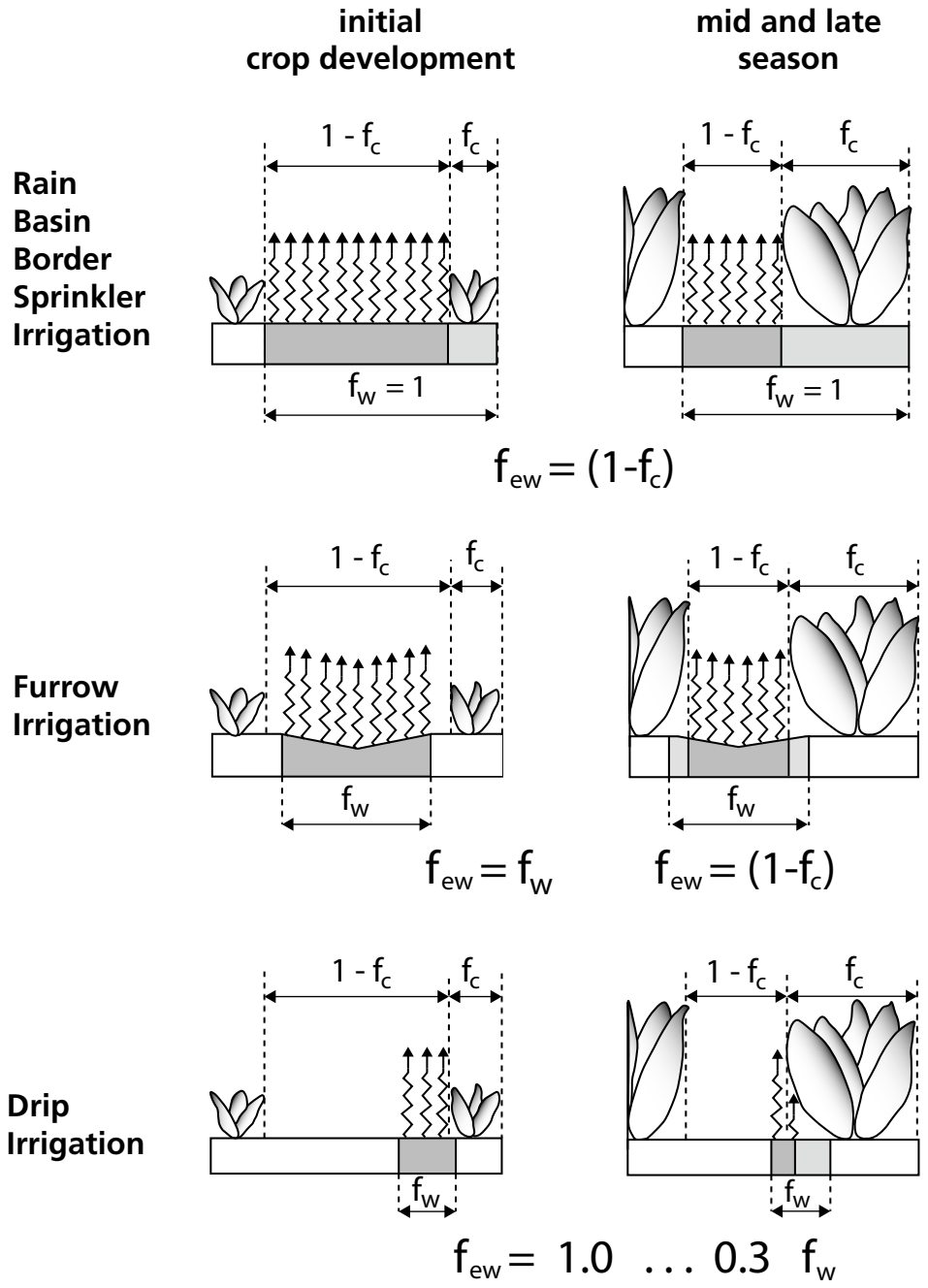
(6) Evaporation component: $K_e\,ET_o = K_e$ (4.5 mm day^{-1}).

(7) Depletion at the end of the day = (2) - (6).

(8) $ET_c = (K_{cb} + K_e)\,ET_o = (0.15 + K_e)\,ET_o = (0.15 + K_e)$ 4.5 mm day^{-1}.

where $K_{cb}\,ET_o = (0.15\,ET_o) \approx 0.7$ mm day^{-1} is basal, "diffusive" evaporation from the soil, possibly from beneath the Z_e depth (~ 0.10 to 0.15 m). Since the soil in this situation is bare, one could set the K_{cb} equal to zero so that maximum K_e becomes $K_e = K_{c\,max} = 1.20$. Then all of the evaporation would be deducted from the surface soil layer.

FIGURE 7.3
Determination of variable f_{ew} (dark areas identified with $1–f_c$) as a function of the fraction of ground surface coverage (f_c) and the fraction of the surface wetted (f_w)



Source: FAO, 1998

TABLE 7.6
Common values of fraction f_w of soil surface wetted by irrigation or precipitation

Wetting event	f_w
Precipitation	1.0
Sprinkler irrigation, field crops	1.0
Sprinkler irrigation, orchards	0.7 … 1.0
Basin irrigation	1.0
Border irrigation	1.0
Furrow irrigation (every furrow), narrow bed	0.6 ... 1.0
Furrow irrigation (every furrow), wide bed	0.4 ... 0.6
Furrow irrigation (alternated furrows)	0.3 ... 0.5
Micro-spray irrigation, orchards	0.5 … 0.8
Drip irrigation	0.3 ... 0.4

Alternatively, on each day of the application, the following rules can be applied to determine f_w for that and subsequent days in a more simplified manner:

- Surface is wetted by irrigation and rain: f_w is the f_w for the irrigation system.
- Surface is wetted by irrigation only: f_w is the f_w for the irrigation system.
- Surface is wetted by significant rain (i.e. > 3 to 4 mm) with no irrigation: $f_w = 1$.
- Where there is neither irrigation nor significant precipitation: f_w is the f_w of the previous day or timestep calculation.

1–f_c: exposed soil fraction

The fraction of the soil surface that is covered by vegetation is termed f_c. Therefore, $(1 - f_c)$ represents the fraction of the soil that is exposed to sunlight and air ventilation and from which the majority of soil evaporation originates. The value for f_c is numerically limited to < 0.99 and should be observed in the field. The user should assume appropriate values for the various growth stages. Typical values for f_c and $(1 - f_c)$ for field and vegetable crops are given in Table 7.7.

TABLE 7.7
Common values of fractions covered by annual crops vegetation (f_c) and exposed to sunlight ($1 - f_c$)

Crop growth stage	f_c	$1 - f_c$
Initial stage	0.0 - 0.1	1.0 - 0.9
Crop development stage	0.1 - 0.8	0.9 - 0.2
Mid-season stage	0.8 - 1.0	0.2 - 0.0
Late season stage	0.8 - 0.2	0.2 - 0.8

Where f_c is not measured, f_c can be estimated using the relationship:

$$f_c = \left(\frac{K_{cb} - K_{c\,min}}{K_{c\,max} - K_{c\,min}} \right)^{(1 + 0.5h)} \tag{7.8}$$

where f_c the effective fraction of soil surface covered by vegetation [0-0.99],
K_{cb} the value for the basal crop coefficient for the particular day or period,
$K_{c\,min}$ the minimum K_c for dry bare soil with no ground cover [≈ 0.15-0.20],
$K_{c\,max}$ the maximum K_c immediately following wetting (Equation 7.4),
h mean plant height [m].

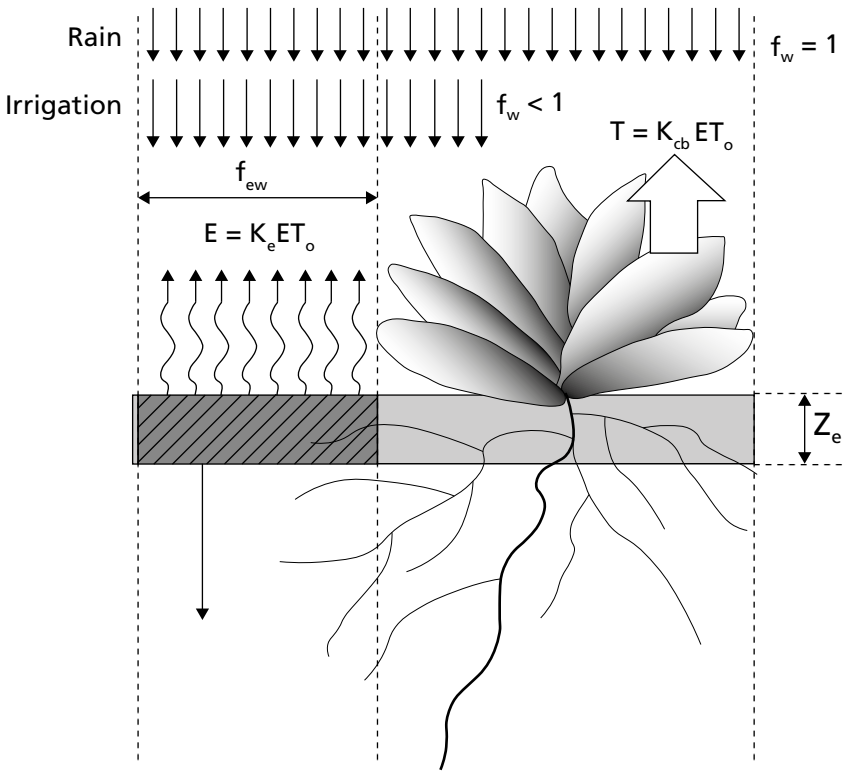
This equation should be used with some caution and validated from local field observations. $K_{c\ min}$ is the minimum crop coefficient for dry bare agricultural soil when transpiration and evaporation from the soil are near baseline (diffusive) levels. $K_{c\ min} \approx 0.15 - 0.20$ is recommended. The value of $K_{c\ min}$ is an integral part of all K_{cb} coefficients. $K_{c\ min}$ ordinarily has the same value as the $K_{cb\ ini}$ used for annual crops under nearly bare soil conditions (0.15 - 0.20). The reduction of the value for $K_{c\ min}$ for very dry soils is described in Chapter 8.

Equation 7.8 assumes that the value for K_{cb} is largely affected by the fraction of soil surface covered by vegetation. This is a good assumption for most vegetation and conditions. The "1+0.5h" exponent in the equation represents the effect of plant height on shading the soil surface and in increasing the value for K_{cb} given a specific value for f_c. The user should limit the difference $K_{cb} - K_{c\ min}$ to ≥ 0.01 for numerical stability. The value for f_c will change daily as K_{cb} changes. Therefore, the above equation 7.8 is applied daily (see Examples 7.4, 7.5 and 7.6 where K_{cb} is computed for, respectively, sprinkler, surface and drip irrigation).

Daily water balance for the evaporation layer

The estimation of K_e in the calculation procedure requires a daily water balance computation for the surface soil layer for the calculation of the cumulative evaporation or depletion from the wet condition.

FIGURE 7.4
Water balance of the topsoil layer



Source: FAO, 1998

The daily soil water balance equation for the exposed and wetted soil fraction f_{ew} is (Figure 7.4):

$$D_{e,i} = D_{e,\,i-1} - (P_i - RO_i) - \frac{I_i}{f_w} + \frac{E_i}{f_{ew}} + T_{ew,\,i} + DP_{e,\,i} \qquad (7.9)$$

where
- $D_{e,\,i-1}$ cumulative depth of evaporation following complete wetting from the exposed and wetted fraction of the topsoil at the end of day i-1 [mm],
- $D_{e,\,i}$ cumulative depth of evaporation (depletion) following complete wetting at the end of the day i [mm],
- P_i precipitation on day i [mm],
- RO_i precipitation runoff from the soil surface on the day i [mm],
- I_i irrigation depth on the day i that infiltrates the soil [mm],
- E_i evaporation on the day i (i.e. $E_i = K_e\ ET_o$) [mm],
- $T_{ew,\,i}$ depth of transpiration from the exposed and wetted fraction of the soil surface layer on the day i [mm],
- $DP_{e,\,i}$ deep percolation loss from the topsoil evaporation layer on the day i if soil water content exceeds field capacity [mm],
- f_w fraction of soil surface wetted by irrigation [0.01-1],
- f_{ew} exposed and wetted soil fraction [0.01-1].

The deep percolation from the topsoil layer is expected to occur after a heavy rain or irrigation event, because the soil water content in the surface layer may exceed field capacity for a short period of time until excess water moves into deeper layers such as the root zone. Thus, deep percolation ($DP_{e,\,i}$) of water from the surface layer is calculated as:

$$DP_{e,\,i} = (P_i - RO_i) + \frac{I_i}{f_w} - D_{e,\,i-1} \qquad (7.10)$$

where
- $DP_{e,\,i}$ deep percolation loss from the topsoil evaporation layer on the day i if soil water content exceeds field capacity [mm],
- P_i precipitation on day i [mm],
- RO_i precipitation runoff on day i [mm],
- I_i irrigation depth that infiltrates the soil on day i [mm],
- f_w fraction of soil surface wetted by irrigation [0.01-1],
- $D_{e,\,i-1}$ cumulative depth of evaporation from the exposed and wetted fraction of the topsoil at the end of day i-1 [mm].

As long as the soil water content in the evaporation layer is below the soil water content at field capacity (i.e., $D_e = 0$), the surface layer is considered not to be draining and DP_e is null.

EXAMPLE 7.4

Calculation of the dual crop coefficient ($K_{cb} + K_e$) under sprinkler irrigation

A field of cotton has just been sprinkler irrigated. The K_{cb} for the specific day (during the development period) has been computed using Table 7.2 and Eq. 7.2 and then interpolated from the K_{cb} curve as 0.9. The ET_o = 7 mm day^{-1}, u_2 = 3 m s^{-1} and RH_{min} = 20%. Estimate the crop coefficient ($K_{cb} + K_e$)

Assuming h = 1 m, from Eq. 7.4, $K_{c\,max}$ for this arid climate is:

$$K_{c\,max} = \max\left(\left\{1.2 + [0.04\,(3-2) - 0.004\,(20-45]\left(\frac{1}{3}\right)^{0.3}\right\}, \{0.9 + 0.05\}\right) = 1.30$$

From Eq. 7.8, where $K_{c\,min}$ = 0.15:

$$f_c = [(K_{cb} - K_{c\,min})/(K_{c\,max} - K_{c\,min})]^{(1 + 0.5h)}$$

$$= [(0.9 - 0.15)/(1.3 - 0.15)]^{(1 + 0.5\,(1))} = 0.53$$

As the field was sprinkler irrigated, f_w = 1.0 and from Eq. 7.7:

$$f_{ew} = \min\,(1 - f_c, f_w)$$

$$= \min\,(1 - 0.53, 1.0) = 0.47$$

Assuming that the irrigation was sufficient to fill the evaporating layer to field capacity, so that:

K_r = 1, thus evaporation would be in stage 1.

From Eq. 7.3: $K_e = 1.00\,(1.30–0.90) = 0.40$

The value is compared against the upper limit ($f_{ew}\,K_{c\,max}$) to ensure that it is less than the upper limit:

$f_{ew}\,K_{c\,max}$ = 0.47 (1.30) = 0.61, which is greater than the value for K_e. Therefore, the value for K_e can be used with no limitation.

The total K_c for the field, assuming no moisture stress that would be caused by a dry soil profile, is

$$K_c = K_{cb} + K_e = 0.9 + 0.40 = 1.30$$

This value is large because of the very wet soil surface, the relatively tall rough crop as compared to the grass reference, and the arid climate (u_2 = 3 m s^{-1} and RH_{min} = 20%). In this situation, K_c happens to equal $K_{c\,max}$, as the field has just been wetted by sprinkler irrigation.

EXAMPLE 7.5

Calculation of the dual crop coefficient (K_{cb} + K_e) under furrow irrigation

The cotton field in the previous example (Ex. 7.4) has been irrigated by furrow irrigation of alternate rows rather than by sprinkler, and the fraction of the field surface wetted by the irrigation is 0.3.

The f_{ew} in this case is calculated from Eq. 7.7 as:

$f_{ew} = \min (1 - f_c, f_w) = \min (1 - 0.53, 0.3) = 0.3$

Assuming that the irrigation was sufficient to fill the f_{ew} portion of the evaporating layer to field capacity, so that $K_r = 1$, evaporation would be in stage 1.

From Eq. 7.3: $K_e = 1.00 (1.30 - 0.9) = 0.40$

The value is compared to the upper limit f_{ew} $K_{c\ max}$ which is 0.30 (1.30) = 0.39. Because 0.40 > 0.39, K_e from the f_{ew} surface area is constrained to 0.39.

The total K_c for the furrow irrigated field, assuming no moisture stress that would be caused by dry soil, is

$K_c = K_{cb} + K_e = 0.9 + 0.39 = 1.29.$

This value is essentially the same as for the previous example (Example 7.4) because the procedure assumes that the soil between alternate rows is the portion that is wetted by the irrigation, so that the majority of the field surface has either vegetation cover or wet soil.

EXAMPLE 7.6

Calculation of the dual crop coefficient (K_{cb} + K_e) under drip irrigation

The cotton field in the previous example (Ex. 7.4) has been irrigated by drip irrigation, where the emitters are placed beneath the cotton canopy. The fraction of the field surface wetted by the irrigation is 0.3.

The f_{ew} in this case is calculated from Eq. 7.7 as $f_{ew} = \min (1-f_c, f_w)$. Because the emitters are beneath the canopy so that less energy is available for evaporation, the value for f_w is reduced by multiplying by 1 - (2/3) f_c, so that:

$f_{ew} = \min [(1-f_c), (1 - 0.67\ f_c)\ f_w)] = \min [(1 - 0.53), (1 - 0.67\ (0.53))\ (0.3)] = 0.19$

Assuming that the irrigation was sufficient to fill the f_w portion of the evaporating layer to field capacity, so that $K_r = 1$, evaporation would be in stage 1.

From Eq. 7.3: $K_e = 1.00 (1.30 - 0.90) = 0.40.$

The value is compared to the upper limit f_{ew} $K_{c\ max} = 0.19 (1.30) = 0.25$. Because 0.25 < 0.40, K_e from the f_w fraction of the surface area is constrained by the available energy. Therefore $K_e = 0.25$.

The total K_c for the drip irrigated field, assuming no moisture stress due to dry soil, is:

$K_c = K_{cb} + K_e = 0.9 + 0.25 = 1.15.$

This K_c value is less than that for sprinkler and furrow irrigation (Examples 7.4 and 7.5).

7.2.2 Calculation procedure using the skin layer approach

The use of the fraction of the time step calculation (day or hour) that resides in stage 1 evaporation, $A_{stage\ 1}$, is an extension to the original FAO 56 approach to provide more precise definition of the transition from stage 1 to stage 2 drying during a time step. The improved definition can be important when using daily calculation time steps, especially for coarse soils having small REW. The skin layer approach improves daily estimates for K_e when the depths of wetting events are small. Small depths of wetting occur when precipitation or sprinkler irrigation events are less than about 10 mm. If small wetting events occur when the total 100 to 150 mm deep soil evaporation layer is dry ($D_{e,i}$ ~ TEW), then much of the added water remains close to the top of the soil evaporation layer and tends to evaporate quickly. The original procedure implicitly assumes that small water additions are mixed throughout the soil evaporation layer. As a result, the original procedure tends to extend the evaporation of the added water over a number of days at a relatively low rate due to small values calculated for K_r. In contrast, the skin evaporation extension retains water close to the surface in the REW zone, so the E_s rate is generally higher in value and shorter in time. The skin evaporation extension tends to reproduce the relatively short-lived and higher magnitude Stage 1 spikes of E_s and K_e that are observed in field monitoring.

It is important to note that the skin evaporation extension and the original K_e computation procedure will generally produce similar total depths of evaporation over an extended period. This is because both methods adhere to a water balance of the topsoil and to the conservation of mass. The difference is in the duration and magnitude of K_e. These differences in daily estimates may not matter when estimating total water consumption (ET) over extended periods. The skin evaporation extension may be most important when comparing ET estimates to ground or remote sensing observations where the timing and magnitude of evaporation on a specific day is important. The skin evaporation extension can be applied to daily or hourly calculation timesteps.

As for the original calculation procedure, $K_{c\ max}$ is determined by the energy available for evapotranspiration at the soil and plant surfaces, thus resulting $K_{cb} + K_e \leq K_{c\ max}$ or $K_e \leq K_{c\ max} - K_{cb}$. When the topsoil dries out, less water is available for evaporation and a reduction in evaporation begins to occur in proportion to the amount of water remaining in the surface soil layer, or:

$$K_e = K_r\ (K_{c\ max} - K_s\ K_{cb}) \leq f_{ew}\ K_{c\ max} \qquad (7.3a\ bis)$$

where
- K_e soil evaporation coefficient,
- $K_s\ K_{cb}$ actual basal crop coefficient, where K_s is a dimensionless water stress coefficient relative to the root zone [0.01-1.0], which is developed in Chapter 8,
- $K_{c\ max}$ maximum value of K_c following rain or irrigation,
- K_r dimensionless evaporation reduction coefficient dependent on the cumulative depth of water depleted (evaporated) from the topsoil,
- f_{ew} fraction of the soil that is both exposed and wetted, i.e. the fraction of soil surface from which most evaporation occurs.

Concepts relative to the reduction coefficient are the same as for the original procedure, i.e. the amounts of water TEW and REW that can be evaporated from the soil are defined the same way (Equation 7.5) and depletion during the energy limiting stage is defined the same way. When calculations refer to a crop under stress, then K_{cb}

in Equation 7.6 is replaced by the actual basal crop coefficient, $K_{cb\,act} = K_s\, K_{cb}$. Changes refer to the placement of evaporation into Stage 1 or Stage 2, the falling rate stage, when D_e exceeds REW, particularly to K_r that is now defined by Equation 7.11:

$$K_r = \max\left[\min\left(\frac{TEW - D_{e,\,i-1}}{TEW - REW}, 1.0\right), 0.0\right](1 - A_{stage\ 1}) + A_{stage\ 1} \qquad (7.11)$$

where K_r dimensionless evaporation reduction coefficient ($K_r = 1.0$ when $D_{e,\,i-1} \le REW$),

$D_{e,\,i-1}$ cumulative depth of evaporation (depletion) from the soil surface layer at the end of day i-1 (the previous day) [mm],

TEW maximum cumulative depth of evaporation (depletion) from the soil surface layer at the time that $K_r = 0$ (TEW = total evaporable water [mm],

REW cumulative depth of evaporation (depletion) at the end of Stage 1 (REW = readily evaporable water) [mm],

$A_{stage\ 1}$ the fraction of the time step (day or hour) that resides in stage 1 evaporation $0 \le A_{stage\ 1} \le 1.0$,

$1 - A_{stage\ 1}$ the fraction of the time step that resides in stage 2.

The max () function determines the greater of the two values in the brackets that are separated by the comma and the min () function determines the lesser of the two values in the parentheses that are separated by the comma. The max and min functions effectively limit the value for K_r to $0 \le K_r \le 1.0$. Setting $A_{stage\ 1}$ to 0 causes Equation 7.11 to revert to the original FAO 56-1998 form.

$A_{stage\ 1}$ is estimated for a time step length of one day or one hour as:

$$A_{stage\ 1} = \frac{REW - D_{REW_{i-1}}}{K_{c\,max}\, ET_o} \qquad \text{where } 0 \le A_{stage\ 1} \le 1.0 \qquad (7.12)$$

where $D_{REW\ i-1}$ the depletion of the REW soil surface layer during stage 1 drying, mm, at the end of time step i-1 (Equation 7.14),

$K_{c\,max}$ the value for K_e expected during stage 1 drying.

Water balance for the evaporation skin layer

Estimation of K_r in Equation 7.11 requires a water balance conducted for the f_{ew} fraction of the surface soil layer from which TEW depth can be evaporated. The water balance can be calculated using a daily or hourly or shorter time step:

$$D_{e,\,i} = D_{e,\,i-1} - (P_i - RO_i) - \frac{I_i}{f_w} + \frac{E_i}{f_{ew}} \qquad \text{with } 0 \le D_{e,\,i} \le TEW \qquad (7.13)$$

where i time step (daily or hourly or shorter time),

$D_{e,i}$ cumulative depletion depth of the Z_e depth of soil that contributes to evaporation [mm] at the end of timestep i and $0 \le D_{e,\,i} \le TEW$,

$D_{e,\,i-1}$ cumulative depletion depth [mm] at the end of timestep i-1 and $0 \le D_{e,\,i-1} \le TEW$,

P_i precipitation during timestep i [mm],

RO_i precipitation runoff from the soil surface during timestep i [mm],

I_i irrigation depth during timestep i that infiltrates the soil [mm],
E_i evaporation during timestep i ($E_i = K_e\ ET_o$) [mm],
f_w fraction of soil surface wetted by irrigation [0.01-1] (Figure 7.3),
f_{ew} exposed and wetted soil fraction [0.01-1] (Figure 7.3).

Assuming that the surface layer is at field capacity following heavy rain or irrigation, the minimum value for $D_{e,i}$ is zero. $D_{e,i}$ and $D_{e,i-1}$ are constrained to limits of $0 \le D_{e,i} \le$ TEW. Any P or I additions in excess of $D_{e,i}$ [or $D_{REW,i}$ in Equation 7.14] are assumed to infiltrate to depths in the soil below Z_e (or below the REW soil layer in the case of $D_{REW,i}$). The irrigation depth I_i is divided by f_w to approximate the infiltration depth concentrated to the f_w portion of the soil surface. Similarly, E_i is divided by f_{ew} because all E_i (other than residual evaporation implicit to the K_{cb} coefficient) is assumed to be taken from the f_{ew} fraction of the surface layer.

The parameter $D_{REW\ i-1}$ in Equation 7.12 is the depletion of water from the REW layer, referred to as the soil "skin" layer, at the end of the prior i-1 time step. For light wetting events, stage 1 evaporation may last for only part of a single time step. The water balance equation for determining D_{REW} is:

$$D_{REW,i} = D_{REW,i-1} - (P_i - RO_i) - \frac{I_i}{f_w} + \frac{E_i}{f_{ew}} \quad \text{with } 0 \le D_{REW,i} \le REW \qquad (7.14)$$

where $D_{REW,i}$ cumulative depletion depth of the REW soil layer that contributes to stage 1 evaporation at the end of timestep i [mm] and $0 \le D_{REW,i} \le$ REW,
$D_{REW,i-1}$ cumulative depletion depth at the end of timestep i-1 [mm] and $0 \le D_{REW,i-1} \le$ REW.

All other terms are the same as for Equation 7.13. $D_{REW,i}$ and $D_{REW,i-1}$ are constrained to limits of $0 \le D_{REW,i} \le$ REW. The advantage of the calculation of $A_{stage\ 1}$ and D_{REW} is that small irrigation or precipitation events may only wet the REW ("skin") of the soil, but not the entire Z_e depth of the evaporation layer. The inclusion of $A_{stage\ 1}$ and D_{REW} will cause the estimate for K_e to "spike" towards 1.0 following a light wetting event for a short period of time. This is quite common where light wetting events wet only the upper skin of soil and evaporate off relatively quickly. Those events may be less effective in replenishing the root zone.

The calculation of $D_{e,i}$ and $D_{REW,i}$ requires that first K_r is calculated from Equation 7.11. Therefore, the calculation of K_r is based on the ending depletion of the previous day or time period, i-1. As a result, the precipitation and irrigation (P_i and I_i) occurring on day or time period i will not impact the K_r calculation for day or time period i until the following day or time period. This also has the delayed impact on the K_e value for day or time period i (Example 7.7). This outcome works well when a precipitation or irrigation event occurs late in a day or time period so that its impact on evaporation is delayed until the following day or time period. In cases where wetting events occur early in the day or during midday, then some evaporation effects should be expected to occur on that day (i). In order to initiate this action, the subscript "i" in Equations 7.13 and 7.14 for P, RO and I can be set to "i+1" indicating that the following day's or time period's wetting event is to be considered in estimating the D_e and D_{REW} depletions at the end of day or time period "i". The subscript for E_i in Equations 7.13 and 7.14 should remain as "i". The use of an i+1 subscript was utilized in Example 7.8. Example 7.9 provides for comparing the original and the skin layer methods.

EXAMPLE 7.7

Determination of the daily evapotranspiration from bare soil previously wetted with heavy rain using the skin layer approach

Determine the evapotranspiration from a bare loamy soil surface ($K_{cb} \approx 0.15$) for ten successive days following a heavy rain. The reference evapotranspiration during the drying period is $ET_o = 4.5$ mm day^{-1}, and the climate is subhumid with light wind.

From Table 7.5 For loam: TEW ≈ 20 mm and REW ≈ 9 mm

For rain on bare soil $f_{ew} = 1.0$

From Eq. 7.4 $K_{c\ max} = 1.20$

(1) Day	(2) $D_{REW\ start}$ ($D_{REW\ i-1}$) mm	(3) $D_{e\ start}$ ($D_{e\ i-1}$) mm	(4) $A_{Stage\ 1}$	(5) Stage	(6) K_r	(7) K_e	(8) $K_e\ ET_o$ mm day^{-1}	(9) $D_{REW\ end}$ ($D_{REW\ i}$) mm	(10) $D_{e\ end}$ ($D_{e\ i}$) mm	(11) ET_c mm day^{-1}
1	0	0.00	1.00	1	1.00	1.05	4.73	4.73	4.73	5.4
2	4.73	4.73	0.79	1	1.00	1.05	4.73	9.00	9.45	5.4
3	9.00	9.45	0.00	2	0.96	1.01	4.53	9.00	13.98	5.2
4	9.00	13.98	0.00	2	0.55	0.57	2.59	9.00	16.57	3.3
5	9.00	16.57	0.00	2	0.31	0.33	1.47	9.00	18.04	2.1
6	9.00	18.04	0.00	2	0.18	0.19	0.84	9.00	18.88	1.5
7	9.00	18.88	0.00	2	0.10	0.11	0.48	9.00	19.36	1.2
8	9.00	19.36	0.00	2	0.06	0.06	0.27	9.00	19.64	0.9
9	9.00	19.64	0.00	2	0.03	0.03	0.16	9.00	19.79	0.8
10	9.00	19.79	0.00	2	0.02	0.02	0.09	9.00	19.88	0.8

Legend:

(1) Day number.

(2) Depletion of REW layer at the beginning of the day or period (= depletion at the end of the previous day or period).

(3) Depletion of Z_e layer at the beginning of the day or period (= depletion at the end of the previous day or period).

(4) Fraction of the time step (day or hour) that resides in stage 1 evaporation ($0 \le A_{stage\ 1} \le 1.0$) (Eq. 7.12).

(5) Soil evaporation stage 2 starts when $D_{e\ i-1} > REW = 9$ mm.

(6) From Eq. 7.6: K_r ($K_r = 1$ for stage 1).

(7) From Eq. 7.3: $K_e = K_r\ (K_{c\ max} - K_{cb}) = K_r\ (1.20 - 0.15) = 1.05\ K_r \le 1.20$.

(8) Evaporation component: $K_e\ ET_o = K_e$ (4.5 mm day^{-1}).

(9) From Eq. 7.14: Depletion of REW layer at end of the day or period ($D_{REW\ i}$).

(10) From Eq. 7.13: Depletion of Z_e layer at end of the day or period ($D_{e\ i}$).

(11) From Eq. 7.1: $ET_c = (K_{cb} + K_e)\ ET_o = (0.15 + K_e)\ ET_o = (0.15 + K_e)$ 4.5 mm day^{-1}.

where $K_{cb}\ ET_o = (0.15\ ET_o) \approx 0.7$ mm day^{-1} is basal, "diffusive" evaporation from the soil, possibly from beneath the Z_e depth (~ 0.10 to 0.15 m). Since the soil in this situation is bare, one could set the K_{cb} equal to zero so that maximum K_e becomes $K_e = K_{c\ max} = 1.20$. Then all of the evaporation would be deducted from the surface soil layer.

The example demonstrates that the estimation of K_r requires a daily (or hourly) water balance calculation. This is further developed in the section on the daily or hourly calculation of K_e.

EXAMPLE 7.8

Determination of the daily evapotranspiration from bare soil with a previous wetting event and an evening light irrigation of 10 mm on day 6 using the skin layer approach

The following example is the same as Example 7.3 except with a light irrigation event on Day 6.

Determine the evapotranspiration from a bare loamy soil surface (K_{cb} » 0.15) for ten successive days following a heavy rain. The reference evapotranspiration during the drying period is ET_o = 4.5 mm day^{-1}, and the climate is subhumid with light wind.

From Table 7.5 For Loam: TEW » 20 mm and REW » 9 mm

For rain on bare soil $f_{ew} = 1.0$

From Eq. 7.4 $K_{c\,max} = 1.20$

(1) Day	(2) $D_{REW\ start}$ ($D_{REW\ i-1}$) mm	(3) $D_{e\ start}$ ($D_{e\ i-1}$) mm	(4) $A_{Stage\ 1}$	(5) Stage	(6) K_r	(7) K_e	(8) $K_e\ ET_o$ mm day^{-1}	(9) $D_{REW\ end}$ ($D_{REW\ i}$) mm	(10) $D_{e\ end}$ ($D_{e\ i}$) mm	(11) ET_c mm day^{-1}	(12) I mm
1	0	0	1	1	1.00	1.05	4.73	4.73	4.73	5.4	0
2	4.73	4.73	0.79	1	1.00	1.05	4.73	9.00	9.45	5.4	0
3	9.00	9.45	0.00	2	0.96	1.01	4.53	9.00	13.98	5.2	0
4	9.00	13.98	0.00	2	0.55	0.57	2.59	9.00	16.57	3.3	0
5	9.00	16.57	0.00	2	0.31	0.33	1.47	9.00	18.04	2.1	0
6	9.00	18.04	0.00	2	0.18	0.19	0.84	0.00	8.88	1.5	10
7	0.00	8.88	1.00	1	1.00	1.05	4.73	4.73	13.61	5.4	0
8	4.73	13.61	0.79	1	0.91	0.96	4.31	9.00	17.92	5.0	0
9	9.00	17.92	0.00	2	0.19	0.20	0.89	9.00	18.81	1.6	0
10	9.00	18.81	0.00	2	0.11	0.11	0.51	9.00	19.32	1.2	0

Legend:

(1) Day number.

(2) Depletion of REW layer at the beginning of the day or period (= depletion at the end of the previous day or period).

(3) Depletion of Z_e layer at the beginning of the day or period (= depletion at the end of the previous day or period).

(4) Fraction of the time step (day or hour) that resides in stage 1 evaporation ($0 \le A_{stage\ 1} \le 1.0$) (Eq. 7.12).

(5) Soil evaporation stage (stage 2 starts when $D_{e\ i-1}$ > REW = 9 mm).

(6) From Eq. 7.11: K_r (K_r = 1 for stage 1).

(7) From Eq. 7.3: $K_e = K_r\ (K_{c\,max} - K_{cb}) = K_r\ (1.20 - 0.15) = 1.05\ K_r \le 1.20$.

(8) Evaporation component: $K_e\ ET_o = K_e$ (4.5 mm day^{-1}).

(9) From Eq. 7.14: Depletion of REW layer at end of the day or period ($D_{REW\ i}$).

(10) From Eq. 7.13: Depletion of Z_e layer at end of the day or period ($D_{e\ i}$).

(11) From Eq. 7.1: $ET_c = (K_{cb} + K_e)\ ET_o = (0.15 + K_e)\ ET_o = (0.15 + K_e)$ 4.5 mm day^{-1}.

where $K_{cb}\ ET_o = (0.15\ ET_o) \approx 0.7$ mm day^{-1} is basal, "diffusive" evaporation from the soil, possibly from beneath the Z_e depth (~ 0.10 to 0.15 m). Since the soil in this situation is bare, one could set the K_{cb} equal to zero so that maximum K_e becomes $K_e = K_{c\,max} = 1.20$. Then all of the evaporation would be deducted from the surface soil layer.

(12) Irrigation depth (mm) over the f_{ew} fraction of soil

The example demonstrates that the estimation of K_r requires a daily (or hourly) water balance calculation. This is further developed in the section on the daily or hourly calculation of K_e.

7.2.3 Daily calculation of K_e

Limits on $D_{e,i}$

By assuming that the topsoil is at field capacity following heavy rain or irrigation, the minimum value for the depletion $D_{e,i}$ is zero. As the soil surface dries, $D_{e,i}$ increases and in the absence of any wetting event will steadily reach its maximum value TEW (Equation 7.5). At that moment no water is left for evaporation in the upper soil layer, K_r becomes zero, and the value for $D_{e,i}$ remains at TEW until the topsoil is wetted once again. The limits imposed on $D_{e,i}$ are consequently:

$$0 \leq D_{e,i} \leq TEW \tag{7.15}$$

Initial depletion

To initiate the water balance for the evaporating layer, the user can assume that the topsoil is near field capacity following a heavy rain or irrigation, i.e. $D_{e,i-1} = 0$. Where a long period of time has elapsed since the last wetting, the user can assume that all evaporable water has been depleted from the evaporation layer at the beginning of calculations, i.e. $D_{e,i-1} = TEW = 1\,000\,(\theta_{FC} - 0.5\,\theta_{WP})\,Z_e$.

Precipitation and runoff

P_i is equivalent to daily precipitation. Daily precipitation in amounts less than about 0.2 ET_o is normally entirely evaporated and can usually be ignored in the K_e and water balance calculations. However, small amounts of precipitation are also correctly dealt with by Equation 7.11 – 7.15. Light precipitation events will generally have little or no runoff. The amount of rainfall lost by runoff depends on: the intensity of rainfall; the slope of land; the soil type, its hydraulic conditions and antecedent moisture content; and the land use and cover. For general situations where many irrigated fields have small slopes, RO_i can be assumed to be zero or can be accounted for by considering only a certain percentage of P_i. This is especially true for the water balance of the topsoil layer, because almost all precipitation events that would have intensities or depths large enough to cause runoff would probably still replenish the water content of the topsoil layer to field capacity. Therefore, the impact of the runoff component can be ignored.

Irrigation

I_i is generally expressed as a depth of water that is equivalent to the mean infiltrated irrigation depth distributed over the entire field. Therefore, the value I_i/f_w is used to describe the actual concentration of the irrigation volume over the fraction of the soil that is wetted (Figure 6.3).

Evaporation

Evaporation beneath the vegetation canopy is assumed to be included in K_{cb} and is therefore not explicitly quantified. The computed evaporation is fully concentrated in the exposed, wetted topsoil. The evaporation E_i is given by $K_e\,ET_o$. The E_i/f_{ew} provides for the actual concentration of the evaporation over the fraction of the soil that is both exposed and wetted.

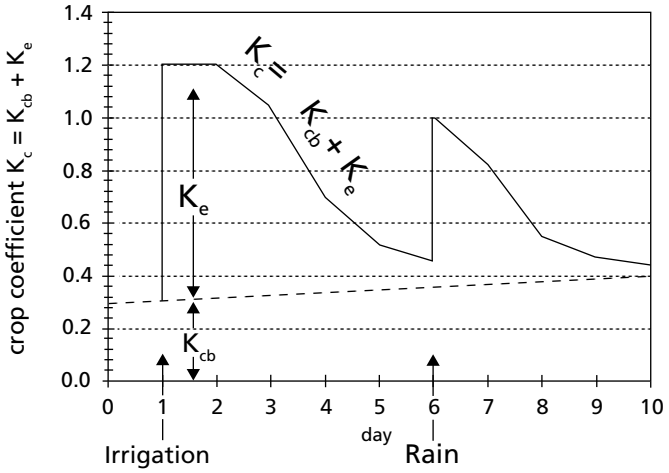
Transpiration

Except for shallow rooted crops (i.e. where the depth of the maximum rooting zone is < 0.5 to 0.6 m), the amount of transpiration from the evaporating soil layer (T_{ew}) is small and can be ignored (i.e. $T_{ew} = 0$). In addition, for row crops, most of the water extracted by the roots may be extracted from beneath the vegetation canopy. Therefore, T_{ew} from the f_{ew} fraction of soil surface can be assumed to be zero in these cases.

7.2.4 Daily water balance calculation

The daily water balance calculation for the surface layer, even for shallow rooted crops, is not usually sensitive to extractions for transpiration from the surface layer (T_{ew}) for the first 3-5 days following a wetting event. T_{ew} can, therefore, generally be ignored. The effects of the reduction of the water content of the evaporating soil layer due to T_{ew} can be accounted for if needed by decreasing the value for Z_e, for example from 0.15 to 0.12 m or from 0.10 to 0.08 m.

FIGURE 7.5
Impact of irrigation and precipitation events on the evaporation coefficient and crop coefficient



Source: FAO, 1998

EXAMPLE 7.9
Estimation of crop evapotranspiration with the dual crop coefficient approach when using both the original and evaporation skin layer for K_e estimation

Estimate the crop evapotranspiration, ET_c for ten successive days. It is assumed that:

- the soil is a sandy loam soil, characterized by $\theta_{FC} = 0.23$ m^3 m^{-3} and $\theta_{WP} = 0.10$ m^3 m^{-3}
- the depth of the surface soil layer that is subject to drying by way of evaporation, Z_e, is 0.1m
- during the period, the height of the vegetation h=0.30 m, the average wind speed $u_2 = 1.6$ m s^{-1}, and $RH_{min} = 35\%$
- the K_{cb} on day 1 is 0.30 and increases to 0.40 by day 10
- all evaporable water has been depleted from evaporation layer at the beginning of calculation ($D_{e, i-1}$ = TEW)
- irrigation occurs at the beginning of day 1 (I = 40 mm), and the fraction of soil surface wetted by irrigation, $f_w = 0.8$
- a rain of 6 mm occurred at the beginning of day 6

From Tab. 7.5 REW ≈ 8 mm

From Eq. 7.7 TEW = 100 (0.23 - 0.5 (0.10)) 0.1 = 18 mm

From Eq. 7.4 $K_{c\,max} = 1.2 + [0.04\,(1.6 - 2) - 0.004\,(35 - 45)]^{0.3} = 1.21$

All evaporable water has been depleted at the beginning of calculations, $D_{e, i-1}$ = TEW = 18 mm

EXAMPLE 7.9 (CONT.)

a. ET_c computation with the FAO56 original procedure

(1a) Day	(2a) ET_o	(3a) P-RO	(4a) I/f_w	(5a) $1-f_c$	(6a) f_w	(7a) f_{ew}	(8a) K_{cb}	(9a) $D_{e,\,i\,start}$
	mm day^{-1}	mm	mm					mm
start	-	-	-	-	-	-	-	-
1	4.5	0	50	0.92	0.8	0.80	0.30	0
2	5.0	0	0	0.91	0.8	0.80	0.31	5
3	3.9	0	0	0.91	0.8	0.80	0.32	11
4	4.2	0	0	0.90	0.8	0.80	0.33	14
5	4.8	0	0	0.89	0.8	0.80	0.34	16
6	2.7	6	0	0.89	1	0.89	0.36	11
7	5.8	0	0	0.88	1	0.88	0.37	13
8	5.1	0	0	0.87	1	0.87	0.38	16
9	4.7	0	0	0.87	1	0.87	0.39	17
10	5.2	0	0	0.86	1	0.86	0.40	18

(10a) K_r	(11a) K_e	(12a) E/f_{ew}	(13a) DP_e	(14a) $D_{e,i\,end}$	(15a) E	(16a) K_c	(17a) ET_c
		mm	mm	mm	mm day^{-1}		mm day^{-1}
-	-	-	-	18	-	-	-
1.00	0.91	5.1	32	5	4.1	1.21	5.5
1.00	0.90	5.6	0	11	4.5	1.21	6.1
0.70	0.62	3.0	0	14	2.8	1.04	4.0
0.40	0.35	1.8	0	16	1.5	0.70	2.9
0.20	0.18	1.1	0	17	0.8	0.52	2.5
0.75	0.64	2.0	0	13	1.7	1.00	2.7
0.53	0.45	3.0	0	16	2.6	0.82	4.7
0.20	0.17	1.0	0	17	0.9	0.55	2.8
0.09	0.08	0.4	0	18	0.4	0.47	2.2
0.05	0.04	0.2	0	18	0.2	0.44	2.3

Total ET_c for days 1-10 is 35.7 mm.

(1a) Day number.

(2a) ET_o is given. Note that ET_o would be forecast values in real time irrigation scheduling but are known values after the occurrence of the day, during an update of the calculations.

(3a) (P - RO) are known values after the occurrence of the day, during an update of the calculations.

(4a) Net irrigation depth for the part of the soil surface wetted by irrigation.

(5a) $(1 - f_c)$ is given (interpolated between 0.92 on day 1 and 0.86 on day 10).

(6a) If significant rain: $f_{w,i} = 1.0$ (Tab. 7.6)

If irrigation: $f_{w,i} = 0.8$ (given),

otherwise: $f_{w,i} = f_{w,i-1}$.

(7a) Eq. 7.7. Fraction of soil surface from which most evaporation occurs.

(8a) K_{cb} is given (interpolated between 0.30 on day 1 and 0.40 on day 10).

(9a) $D_{e,i\,start}$ (depletion at start of day)

If precipitation and irrigation occur early in the day then the status of depletion from the soil surface layer (at the start of the day) should be updated:

$= \text{Max}\,(D_{e,i-1} - I_{n,i}/f_{wi} - (P - RO)_i, \text{ or } 0)$.

EXAMPLE 7.9 (CONT.)

where $D_{e,i-1}$ is taken from column 14 of previous day.

If precipitation and irrigation occur late in the day, then column 6 should be set equal to $D_{e,i-1}$ (column 14 of previous day).

(10a) If $D_{e,i} \leq REW$ $K_r = 1$; If $D_{e,i} > REW$ K_r = Eq. 7.6.

(11a) Eq. 7.3 where $K_e = K_r (K_{c\,max} - K_{cb}) \leq f_{ew} K_{c\,max}$ (e.g. $K_e = \min (K_r (K_{c\,max} - K_{cb}), f_{ew} K_{c\,max})$.

(12a) Evaporation from the wetted and exposed fraction of the soil surface = $(K_e ET_o)/f_{ew}$.

(13a) Eq. 7.9 where $DP_e \geq 0$. (This is deep percolation from the evaporating layer).

(14a) $D_{e,i}$ (depletion at end of day) is from Eq. 7.9 where $D_{e,i-1}$ is value in column 14 of previous day.

(15a) Mean evaporation expressed as distributed over the entire field surface = $K_e ET_o$.

(16a) $K_c = K_{cb} + K_e$.

(17a) Eq. 7.1.

b. ET_c computation procedure using the skin layer approach for K_e estimation

(1b) Day	(2b) ET_o	(3b) P-RO	(4b) I/f_w	(5b) f_c	(6b) $1-f_c$	(7b) f_w	(8b) f_{ew}	(9b) K_{cb}	(10b) $D_{REW,\,i-1\,start}$
	mm d^{-1}	mm	mm						mm
start	-	-	-	-	-	-	0.80	0.30	8.0
1	4.5	0	50	0.11	0.89	0.8	0.80	0.30	0.0
2	5.0	0	0	0.11	0.89	0.8	0.80	0.31	5.1
3	3.9	0	0	0.12	0.88	0.8	0.80	0.32	8.0
4	4.2	0	0	0.13	0.87	0.8	0.80	0.33	8.0
5	4.8	0	0	0.14	0.86	0.8	0.80	0.34	8.0
6	2.7	6	0	0.16	0.84	1	0.84	0.36	3.2
7	5.8	0	0	0.16	0.84	1	0.84	0.37	5.9
8	5.1	0	0	0.17	0.83	1	0.83	0.38	8.0
9	4.7	0	0	0.18	0.82	1	0.82	0.39	8.0
10	5.2	0	0	0.19	0.81	1	0.81	0.40	8.0
(11b) $D_{e,\,i-1\,start}$	(12b) $A_{stage\,1}$	(13b) K_r	(14b) K_e	(15b) E/f_{ew}	(16b) $D_{REW,i\,end}$	(17b) $D_{e,i\,end}$	(18b) E	(19b) K_c	(20b) ET_c
mm				mm d^{-1}	mm	mm	mm d^{-1}		mm d^{-1}
18.0	0	0.00	0	0.00	0.0	0.0	-	-	-
0.0	1.0	1.00	0.91	5.12	5.1	5.1	4.10	1.21	5.4
5.1	0.5	1.00	0.90	5.63	8.0	10.7	4.50	1.21	6.1
10.7	0	0.73	0.65	3.15	8.0	13.9	2.52	0.97	3.8
13.9	0	0.41	0.36	1.90	8.0	15.8	1.52	0.69	2.9
15.8	0	0.22	0.19	1.15	3.2	10.9	0.92	0.53	2.6
10.9	1.0	1.00	0.85	2.72	5.9	13.7	2.30	1.21	3.3
13.7	0.3	0.61	0.51	3.53	8.0	17.2	2.95	0.88	5.1
17.2	0	0.08	0.07	0.41	8.0	17.6	0.34	0.45	2.3
17.6	0	0.04	0.03	0.19	8.0	17.8	0.15	0.42	2.0
17.8	0	0.02	0.02	0.11	8.0	17.9	0.09	0.42	2.2

Total ET_c for days 1-10 is 35.7 mm. This is the same total as with the original procedure, although day to day values vary.

EXAMPLE 7.9 (CONT.)

(1b) Day number.

(2b) ET_o is given. Note that ET_o would be forecast values in real time irrigation scheduling but are known values after the occurrence of the day, during an update of the calculations.

(3b) (P - RO) are known values after the occurrence of the day, during an update of the calculations.

(4b) Net irrigation depth for the part of the soil surface wetted by irrigation.

(5b) f_c is from Eq. 7.8 using $K_{c\,min} = 0.15$ and the calculated $K_{c\,max} = 1.21$.

(6b) $(1 - f_c)$ is used in Eq. 7.7 to calculate f_{ew}.

(7b) If significant rain: $f_{w,i} = 1.0$ (Tab. 7.6)
If irrigation: $f_{w,i} = 0.8$ (given),
otherwise: $f_{w,i} = f_{w,i-1}$.

(8b) Eq. 7.7 Fraction of soil surface from which most evaporation occurs (fraction of exposed and wetted soil).

(9b) K_{cb} is given (interpolated between 0.30 on day 1 and 0.40 on day 10).

(10b) $D_{REW,i\,start}$ (depletion of the REW ("skin") layer at the start of day i (and at the end of day i-1)

(11b) $D_{e,i\,start}$ (depletion at start of the day (and at the end of day i-1))

(12b) $A_{stage\,1}$ Fraction of the time step (day or hour) that resides in stage 1 evaporation ($0 \le A_{stage\,1} \le 1.0$) (Eq. 7.12)

(13b) Evaporation reduction coefficient. If $D_{e,\,i-1} \le REW$ then $K_r = 1$
If $D_{e,\,i-1} > REW$ then K_r = Eq. 7.12.

(14b) Eq. 7.3 where $K_e = K_r (K_{c\,max} - K_s K_{cb}) \le f_{ew} K_{c\,max}$
(e.g. $K_e = \min (K_r (K_{c\,max} - K_s K_{cb}), f_{ew} K_{c\,max})$. Here, K_s (Chapter 8), is assumed to be 1.0.

(15b) Evaporation from the wetted and exposed fraction of the soil surface = $(K_e\, ET_o)/_{few}$.

(16b) $D_{REW,i}$ (depletion of the REW ("skin") layer at the end of the day i) is from Eq. 7.15 where $D_{REW,i-1}$ is D_{REW} at the end of the previous day. If precipitation and irrigation occur early in the day, which is the assumption here, then the subscripts i in Eq. 7.14 for P, RO and I are changed to "i+1". If precipitation and irrigation occur late in the day, then Eq.7.14 is used with no changes to the subscripts.

(17b) $D_{e,i}$ (depletion at the end of the day i) is from Eq. 7.13 where $D_{e,i-1}$ is D_e at the end of the previous day. If precipitation and irrigation occur early in the day, which is the assumption here, then the subscripts i in Eq. 7.13 for P, RO and I are changed to "i+1". If precipitation and irrigation occur late in the day, then Eq. 7.13 is used with no changes to the subscripts.

(18b) Mean evaporation expressed as distributed over the entire field surface = $K_e\, ET_o$.

(19b) $K_c = K_{cb} + K_e$.

(20b) $ET_c = K_c\, ET_o$

7.3 ORDER OF CALCULATION

In making calculations for the $K_{cb} + K_e$ procedure, for example when using a spreadsheet, the calculations should generally proceed in the order shown in Example 7.8: K_{cb}, h, $K_{c\,max}$, f_c, f_w, f_{ew}, $A_{stage\,1}$, K_r, K_e, E, D_{REW}, D_e, I (if being predicted during irrigation scheduling), E, K_c, and ET_c. If the original approach is used $A_{stage\,1}$ and D_{REW} are not required.

The sequential procedure for calculating crop evapotranspiration is presented in Box 7.1. Dual crop coefficient case studies for a dry bean crop at Kimberly, USA, for a pea crop for industry at Alpiarça, Portugal, and a super-high density (hedgerow) olive orchard at Viana do Alentejo, Portugal, are presented with summaries of calculation procedures in Boxes 7.2, 7.3 and 7.4.

BOX 7.1
Calculating ET_c

The calculation procedure lends itself to application by computer, either in the form of electronic spreadsheets (as calculated for Example 7.9) or in the form of structured programming languages. The calculation procedure consists of determining:

a. Reference evapotranspiration, ET_o: Estimate ET_o using the procedure given in Chapter 4.

b. Growth stages: Determine the locally adjusted lengths of the four growth stages by performing field observations (for general information on cumulative growing degree days used to estimate growth stage lengths, consult Table 6.11):

- initial growth stage: L_{ini} [days];
- crop development stage: L_{dev} [days];
- mid-season stage: L_{mid} [days]; and
- late season stage: L_{late} [days].

c. Basal crop coefficient, K_{cb}: Calculate basal crop coefficients for each day of the growing period:

- select $K_{cb\ ini}$, $K_{cb\ mid}$ and $K_{cb\ end}$ from Tables 7.1 through 7.3;
- adjust $K_{cb\ mid}$ and $K_{cb\ end}$ to the local climatic conditions (Equation 7.2); and
- determine the daily K_{cb} values
  - ◊ initial growth stage: $K_{cb} = K_{cb\ ini}$,
  - ◊ crop development stage: from $K_{cb\ ini}$ to $K_{cb\ mid}$ (Equation 6.20),
  - ◊ mid-season stage: $K_{cb} = K_{cb\ mid}$,
  - ◊ late season stage: from $K_{cb\ mid}$ to $K_{cb\ end}$ (Equation 6.20).

d. Evaporation coefficient, K_e: Calculate the maximum value for K_c, i.e. the upper limit $K_{c\ max}$ (Equation 7.4), and determine for each day of the growing period:

- the fraction of soil covered by vegetation, f_c (Tables 7.1-7.3, Table 7.7 or Equation 7.8);
- the fraction of soil surface wetted by irrigation or precipitation, f_w (Table 7.6);
- the fraction of soil covered by vegetation, f_c (Tables 7.1-7.3, Table 7.7 or Equation 7.8);
- the fraction of soil surface wetted by irrigation or precipitation, f_w (Table 7.6);
- the fraction of soil surface from which most evaporation occurs, f_{ew} (Equation 7.7);
- the cumulative depletion from the REW layer, D_{REW}, and from the evaporating soil layer, D_e, determined by means of a daily (or shorter) soil water balance of the topsoil (Equation 7.14 and 7.15);
- the corresponding evaporation reduction coefficient, K_r (Equation 7.12); and
- the soil evaporation coefficient, K_e (Equation 7.3).

e. Crop evapotranspiration, ET_c: Calculate $ET_c = (K_{cb} + K_e)\ ET_o$ (Equation 7.1).

BOX 7.2

Case study for the dual crop coefficient applied to a dry bean crop at Kimberly, Idaho, the United States

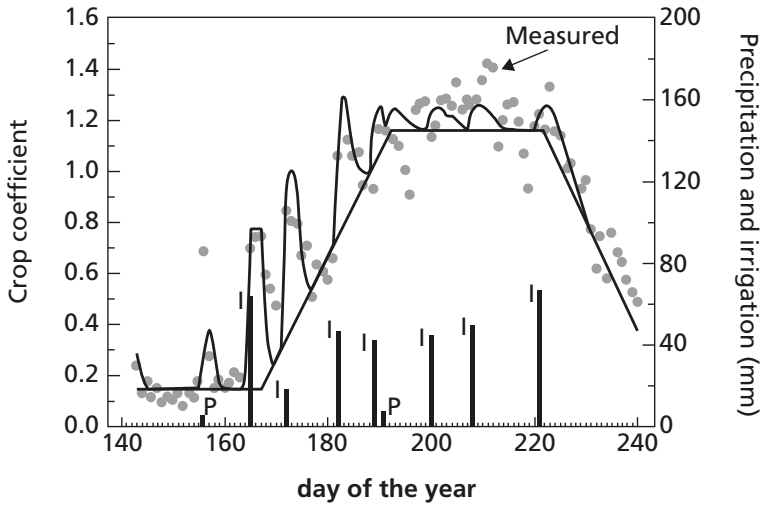
Results from applying the $K_{cb} + K_e$ procedure for a snap bean crop harvested as dry seed are shown in Figure 7.6. This example uses the same data set that were used in the case study of Box 6.2. The observed ET_c data were measured using a precision lysimeter system at Kimberly, Idaho. Values for $K_{cb\ ini}$, $K_{cb\ mid}$, and $K_{cb\ end}$ were calculated in Example 7.1 and Table 7.2 as 0.15, 1.04, and 0.30. However, the $K_{cb\ mid}$ is set here to 1.14 to agree best with the observed ET_c data for the particular bean crop, as demonstrated in the figure below. The time lengths of growth stages were 25, 25, 30, and 20 days. The K_{cb} values are plotted in Figure 7.1. The value for $K_{c\ max}$ from Eq. 7.4 for the mid-season period averaged 1.24, based on $u_2 = 2.2$ m s^{-1} and $RH_{min} = 30\%$ for Kimberly. The soil at Kimberly was a silt loam texture. Assuming that the depth of the evaporation soil layer, Z_e, was 0.1 m, values for TEW = 22 mm and REW = 9 mm, based on Eq. 7.5 and using soil data from Table 7.5.

The occurrence and magnitudes of individual wetting events are shown in the figure below. Nearly all wetting events were from irrigation. Because the irrigation was by furrow irrigation of alternate rows, the value for f_w was set equal to 0.5. Irrigation events occurred at about midday or during early afternoon.

The agreement between the estimated values for daily $K_{cb} + K_e$ (thin continuous line) and actual 24-hour measurements (symbols) is relatively good. Measured and predicted $K_{cb}+K_e$ became higher following wetting by rainfall or irrigation, as expected. The two wet soil evaporation "spikes" occurring during the late initial period and early development period (between days 160 and 180) were less than $K_{c\ max}$, because this evaporation was from a soil surface only partially wetted by furrow irrigation where $f_w = 0.5$. The value for f_{ew} was constrained to f_w by Eq. 7.7 during these two events, because during this period, $f_w < 1 - f_c$. Therefore, less than all of the "potential energy" was converted into evaporation due to the limitation on maximum evaporation per unit surface area that was imposed by Eq. 7.3.

FIGURE 7.6

Measured (dots) and predicted (thin solid line) daily crop coefficients ($K_{cb}+ K_e$) and the basal crop curve (thick solid line) for a dry bean crop at Kimberly, Idaho, USA. P in the figure denotes precipitation events and I denotes the irrigation events (data from Wright, 1990)



Source: FAO, 1998

BOX 7.3

Case study for dual crop coefficient applied to an industrial green pea crop, at Alpiarça, Portugal

The results of applying the ($K_{cb} + K_e$) method for a vining pea crop harvested early in the season as green seed for the industry are shown in Figure 7.7. The ET_c and its partitioning were estimated using the soil water balance model SIMDualKc, that employs the dual crop coefficient method, after its calibration with observed soil water content data. The values for $K_{cb\ ini}$, $K_{cb\ mid}$, and $K_{cb\ end}$ in Table 7.2 were used to initialize the model. After calibration and adjustment to climate conditions (Equation 7.2) the K_{cb} values were 0.15, 1.13, and 1.09 respectively for the initial, mid-season and end-season stages. The lengths of growth stages were observed in the field as 36 (105 GDD), 32 (255 GDD), 50 (505 GDD), and 2 days (35 GDD). GDD values were computed assuming a T_{base} = 5 °C and a T_{upper} = 27 °C. The late season lasted only a few days because the harvest took place immediately after the pea grain had reached maturity. The fraction of ground cover by the crop reached 0.98 at mid-season, and the maximum height was 0.45 m.

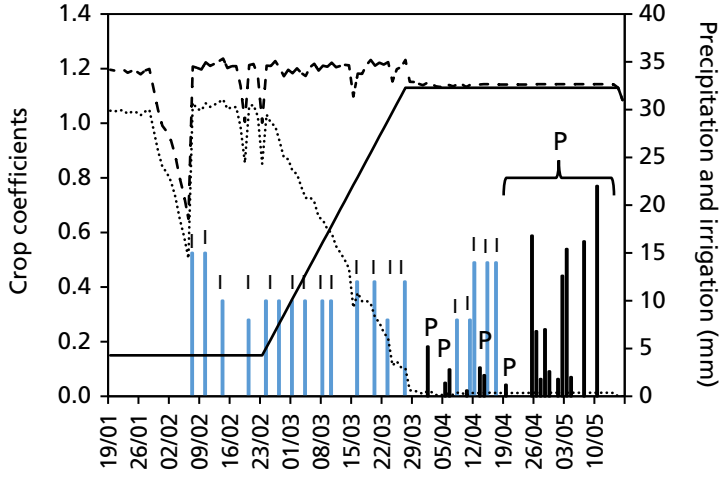
The value for $K_{c\ max}$ from Eq. 7.4 for the mid-season period averaged 1.20 considering u_2 = 1.8 m s^{-1} and RH_{min} = 53% for Alpiarça. The soil was a silty-loam texture. Assuming that the depth of the evaporation soil layer, Z_e, was 0.1 m, values for TEW = 24 mm and REW = 10 mm, based on Eq. 7.5 and using soil data from Table 7.5.

The occurrence and magnitudes of individual wetting events are shown in Figure 7.7. The number of wetting events from precipitation and irrigation was similar with a total number of 34 events. Gross irrigation depths ranged from 10 to 19 mm. Because the irrigation was by sprinkler, the value for f_w was set equal to 1.0. Irrigation events occurred at early morning or late afternoon to avoid wind drift evaporation losses.

K_e peaks occur in response to high soil water content at the beginning of the cropping season, as well as in response to rainfall and irrigation events. Several K_e peaks occur mainly in the initial and crop development stages when the fraction of ground cover by the crop was small and therefore solar energy was largely available at the soil surface for evaporation. From the middle of the development stage, the K_e peaks became smaller. From the mid-season to harvest, the peaks were very low because the crop almost completely covered the soil (f_c = 0.98) and evaporation became nearly null.

FIGURE 7.7

Predicted daily crop coefficient ($K_{cb} + K_e$) (dashed line), evaporation coefficient (K_e) (dotted line) and the basal crop coefficient curve (solid line) for a very dense green pea crop at Alpiarça, Portugal. P in the figure denotes a precipitation and I denotes an irrigation



Source: Authors' own elaboration

BOX 7.4

Case study for the application of the dual crop coefficient to a super-high density (hedgerow) olive orchard at Viana do Alentejo, Portugal

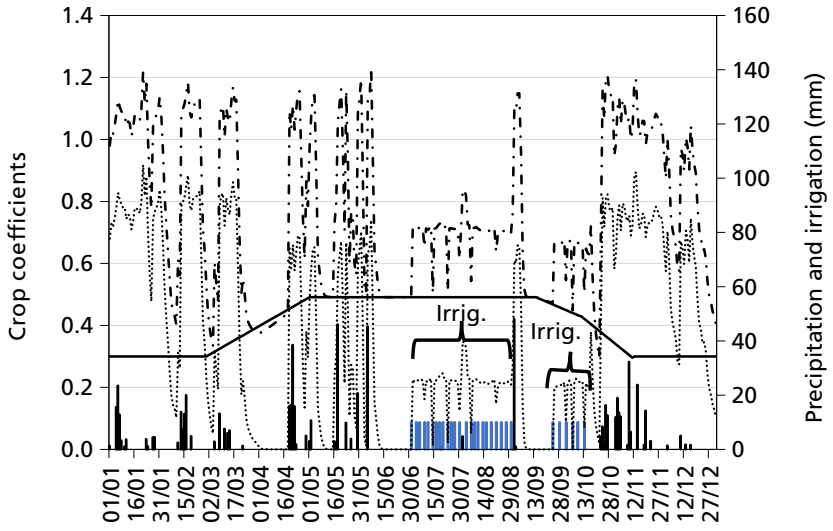
The results of applying the (K_{cb} + K_e) method for a super-high density (hedgerow) olive orchard are shown in Figure 7.8. The observed ET_c data were measured using an eddy covariance system and T_c was measured with the sap flow method. The figure below shows the daily dynamics of K_{cb} and K_e estimated using the soil water balance model SIMDualKc, that employs the dual crop coefficient method, after calibration with field observations. The values for $K_{cb\ ini}$, $K_{cb\ mid}$, and $K_{cb\ end}$ were estimated to be 0.30, 0.49, and 0.43, respectively. A K_{cb} for the non-growing period, when the crop was dormant, was set to 0.30. Growth stage lengths were 11, 51, 137, and 58 days as observed in the field. The non-growing period lasted 108 days. The value for $K_{c\ max}$ from Eq. 7.4 for mid-season averaged 1.21, based on u_2 = 2.2 m s^{-1} and RH_{min} = 45% for Viana do Alentejo, Portugal. The soil in the farmer's field has a sandy loam texture. The depth of the evaporation soil layer, Z_e, was 0.1 m, with TEW = 18 mm and REW = 9 mm, based on Eq. 7.5 and using the observed soil data. The fraction of ground cover by the trees reached a maximum of 0.38 by mid-season and the maximum tree height was 4 m.

The occurrence of individual precipitation and irrigation wetting events is shown in Figure 7.8. All irrigation-related wetting events occurred during the mid-season stage. Irrigation was via drip irrigation system. The observed value for f_w was 0.23. Irrigation events took place mainly in the early morning.

The predicted values of (K_{cb} + K_e) increased as expected after wetting by rain or irrigation. The K_c peaks resulting from irrigation in the mid-season were below $K_{c\ max}$ because evaporation was from a soil surface that was only partially wetted by drip irrigation (f_w = 0.23). The value for f_{ew} was constrained to f_w by Eq. 7.7 during these events, because during this period, $f_w < 1 - f_c$. Therefore, due to the limit on maximum evaporation per unit area imposed by Equation 7.3, less than all the potential energy was converted to soil evaporation.

FIGURE 7.8

Predicted daily crop coefficient (K_{cb}+K_e) (dashed/dotted line), evaporation coefficient (K_e) (dotted line) and the basal crop curve (solid line) for a hedgerow olive orchard at Viana do Alentejo, Portugal. Vertical lines in the figure denote precipitation or irrigation (Irrig.) events



Source: Authors' own elaboration

Chapter 8

ET_c under soil water and salinity stress conditions

Forces acting on the soil water decrease its potential energy and make it less available for plant root extraction. When the soil is wet, the water has a high potential energy, is relatively free to move and is easily taken up by the plant roots. In dry soils, the water has a low potential energy and is strongly bound by capillary and absorptive forces to the soil matrix and is less easily extracted by the crop.

When the potential energy of the soil water drops below a threshold value, the crop is said to be water stressed. The effects of soil water stress are described by multiplying the basal crop coefficient by the water stress coefficient, K_s:

$$ET_{c\ act} = (K_s\ K_{cb} + K_e)\ ET_o \tag{8.1}$$

For soil water limiting conditions, $K_s < 1$. Where there is no soil water deficit, $K_s = 1$. K_s describes the effect of water stress on crop transpiration. Where the single crop coefficient is used, the effect of water stress is incorporated into K_c as:

$$ET_{c\ act} = K_s\ K_c\ ET_o = K_{c\ act}\ ET_o \tag{8.2}$$

Salinity also decreases the potential energy of the soil water and makes water more difficult to be extracted by the plant roots. Thus Equations 8.1 and 8.2 also apply when salinity of the soil and the irrigation water affect extraction by the roots, with K_s becoming a salinity and water stress coefficient, i.e. salinity and water stress combine in $K_s < 1$ as developed later in this chapter. Because the water stress coefficient impacts only crop transpiration, rather than evaporation from soil, the application of K_s using Equation 8.1 is generally more accurate than its application using Equation 8.2. However, in situations where evaporation from soil is not a large component of ET_c, the use of Equation 8.2 will provide quite reasonable results.

8.1 SOIL WATER AVAILABILITY

8.1.1 Total available water (TAW)

Soil water availability refers to the capacity of a soil to retain water available to plants. After heavy rainfall or irrigation, the soil will drain until field capacity is reached. *Field capacity* is the amount of water that a well-drained soil should hold against gravitational forces, or the amount of water remaining when downward drainage has markedly decreased. In the absence of water supply, the water content in the root zone decreases as a result of water uptake by the crop. As water uptake progresses, the remaining water is held to the soil particles with greater force, lowering its potential energy and making it more difficult for the plant to extract it. Eventually, a point is reached where the crop can no longer extract the remaining water. The water uptake becomes zero when wilting point is reached. *Wilting point* is the water content at which plants will permanently wilt. As the water content above field capacity cannot be held

against the forces of gravity and will drain, and as the water content below wilting point cannot be extracted by plant roots, the *total available water* in the root zone is the difference between the water content at field capacity and wilting point:

$$TAW = 1000\ (\theta_{FC} - \theta_{WP})\ Z_r \tag{8.3}$$

where TAW the total available soil water in the root zone [mm],
θ_{FC} the water content at field capacity [$m^3\ m^{-3}$],
θ_{WP} the water content at wilting point [$m^3\ m^{-3}$],
Z_r the rooting depth [m].

TAW is the amount of water that a crop can extract from its root zone, and its magnitude depends on the type of soil and the rooting depth. Typical ranges for field capacity and wilting point are listed in Table 7.5 for various soil texture classes. Ranges of the maximum effective rooting depth for various crops are given in Tables 6.1 through 6.3.

8.1.2 Readily available water (RAW)

Although water is theoretically available in the root zone until the wilting point is attained, the crop water uptake is reduced well before the wilting point is reached. Where the soil is sufficiently wet, the soil supplies water fast enough to meet the atmospheric demand of the crop, and water uptake equals ET_c. As the soil water content decreases, water becomes more strongly bound to the soil matrix and is more difficult to extract. When the soil water content drops below a threshold value, soil water can no longer be transported quickly enough toward the roots to respond to the transpiration demand and the crop begins to experience stress. *The fraction of TAW that a crop can extract from the root zone without suffering water stress is the readily available soil water*:

$$RAW = p\ TAW \tag{8.4}$$

where RAW the readily available soil water in the root zone [mm],
p average fraction of total available soil water (TAW) that can be depleted from the root zone before moisture stress (reduction in ET) occurs [0-1].

Values for p are listed in Table 8.1 through 8.4 respectively for vegetable, field, woody fruit and grass crops. The factor p differs from one crop to another. The factor p normally varies from 0.20 for very shallow rooted plants at high rates of ET_c (> 8 mm d^{-1}) to 0.70 for deep rooted plants at low rates of ET_c (< 3 mm d^{-1}). A value of 0.50 for p is commonly used for many field crops, a lower value is often used for vegetable crops, and a higher value is used for woody crops with large roots and to grasses, generally less susceptible to water stress. For many woody crops there is little information and, thus, indicative values are given for each group. The p value is small for frequent and very frequent irrigation, and larger for surface irrigation where time intervals between irrigation events are large. Therefore, the values of p listed in Table 8.1 through 8.4 are for large time intervals between irrigation events, thus for surface irrigation. Users may reduce the tabulated p values for the irrigation method that they are using.

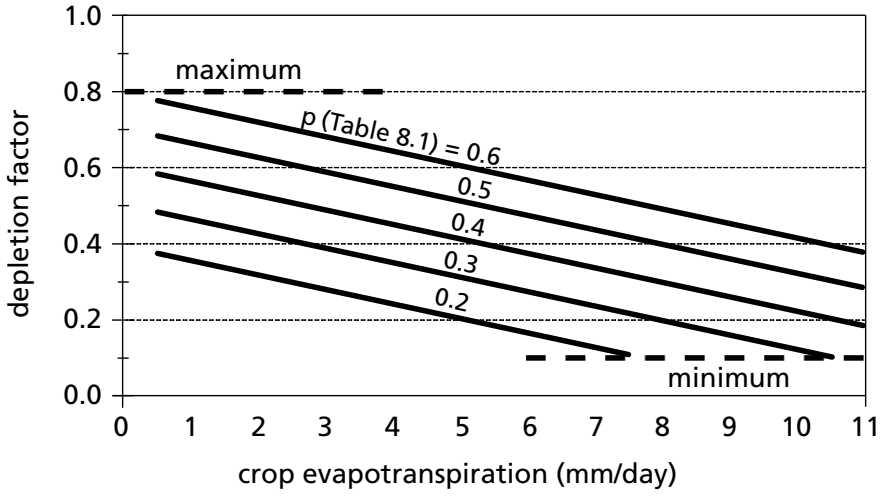
The fraction p is a function of the evaporation power of the atmosphere. At low rates of ET_c, the p values listed in Table 8.1 through 8.4 are higher than at high rates of ET_c. For hot dry weather conditions, where ET_c is high, p maybe 10-25 percent less than the tabulated values in Table 8.1 through 8.4, and the soil is still relatively wet when

the stress starts to occur. When the crop evapotranspiration is low, p may be up to 20 percent more than the listed values. Often, a constant value is used for p for a specific growing period, rather than varying the value each day. A numerical approximation for adjusting p for ET_c rate is:

$$p = p_{Table} + 0.04\ (5 - ET_c) \tag{8.5}$$

where the adjusted p is limited to $0.1 \leq p \leq 0.8$ and ET_c is in mm day^{-1}. The influence of the ET_c numerical adjustment on p is shown in Figure 8.1. Examples of estimation of RAW for various soils and crops are given in Example 8.1.

FIGURE 8.1
Depletion factor for different levels of crop evapotranspiration



Source: FAO, 1998

Expressing the tolerance of crops to water stress as a function of the fraction (p) of TAW is only an approximation. The rate of root water uptake is, in fact, influenced more directly by the potential energy level of the soil water (soil matric potential and the associated hydraulic conductivity) than by water content alone. As a certain soil matric potential corresponds in different soil types with different soil water contents, the value for p is also a function of the soil type.

Generally, it can be stated that for fine textured soils (clay) the p values listed in Table 8.1 can be reduced by 5-10 percent, while for more coarse textured soils (sand), they can be increased by 5-10 percent.

RAW is similar to the term Management Allowed Depletion (MAD) often used in irrigation management practice. However, values for MAD are influenced by management and economic factors in addition to the physical factors influencing p. Generally, MAD < RAW where there is risk aversion or uncertainty, and MAD > RAW where plant moisture deficit is an intentional part of soil water management.

TABLE 8.1
Soil water depletion fraction for no stress (p) for vegetable crops

Vegetable Crops	Depletion fraction p (for ET_c ≈ 5 mm day^{-1})
Root, tuber, bulb and stem vegetable crops	
Asparagus (*Asparagus officinalis*)	0.45
Carrots (*Daucus carota*)	0.30
Cassava (*Manihot esculenta*)	0.50
Celeriac, celery root (*Apium graveolens group rapaceum*)	0.30
Chufa, tigernut (*Cyperus esculentus*)	0.35
Garlic (*Allium sativum*)	0.30
Leek (Allium porrum)	0.30
Mashua (*Tropaeolum tuberosum*)	0.35
Onions (*Allium cepa*)	0.30
Parsnip (*Pastinaca sativa*)	0.40
Potato (*Solanum tuberosum*)	0.40
Radish (*Raphanus sativus*)	0.30
Rutabaga (*Brassica napus*)	0.40
Sweet potato (*Ipomoea batatas*)	0.45
Table beets (*Beta vulgaris*)	0.45
Turnip (*Brassica rapa*)	0.40
Yam, Taro (*Colocasia esculenta*)	0.20
Leafy and flower vegetable crops	
African cabbage, spiderwisp (*Cleome gynandra*)	0.40
African spinach, (*Amaranthus cruentus*)	0.30
Artichoke (*Cynara scolymus*)	0.45
Arugula, rucula (*Eruca sativa* or *E. vesicaria*)	0.35
Broccoli (*Brassica oleracea* cv. *italica*)	0.40
Brussel sprouts (*Brassica oleracea* cv. *gemmifera*)	0.40
Cabbage	
Common cabbage (*Brassica oleracea*)	0.35
Kale, leaf cabbage (*B. oleracea* var. *sabellica*), and Collard green (*B. oleracea* var. *Acephala*)	0.40
Chinese cabbage, bok choy (*B. rapa* cv. *chinensis*)	0.35
Chinese cabbage, pak choi (*B. rapa* ssp. *pekinensis*)	0.35
Cauliflower (*Brassica oleracea* cv. *botrytis*)	0.40
Celery (*Apium graveolens* var. *dulce*)	0.40
Endive, escarola Cichorium endivia	0.30
Jute mallow for leaves (*Corchorus olitorius*)	0.40
Lettuce (*Lactuca sativa*)	0.30
Purslane (*Portulaca oleracea*)	0.35
Spinach (*Spinacia oleracea*)	0.25
Fruit vegetable crops	
Banana (*Musa sativa*)	0.35
Bell and Chili pepper (*Capsicum annuum*)	0.40
Bitter gourd, bitter melon (*Momordica charantia*)	0.40
Cape gooseberry *(Physalis peruviana)*	0.50
Chilli tabasco pepper (*Capsicum frutescens*)	0.45
Cucumber (*Cucumis sativus*)	0.45
Eggplant (*Solanum melongena*)	0.40
Fennel (*Foeniculum vulgare*)	0.40
Melon (*Cucumis melo*)	0.40
Melon cantaloupe (*Cucumis melo*)	0.40

TABLE 8.1 (CONT.)

Vegetable Crops	Depletion fraction p (for $ET_c \approx 5$ mm day^{-1})
Nightshade, sunberry (*Solanum retroflexum*)	0.40
Okra (*Abelmoschus esculentus*)	0.40
Pineapple (*Ananas comosus*)	0.35
Pumpkin, winter squash (*Cucumis pepo*)	0.40
Strawberries (*Fragaria × ananassa*)	0.25
Tomato (*Solanum lycopersicum*)	0.40
Watermelon (*Citrullus lanatus*)	0.45
Zucchini, squash (*Cucumis pepo*)	0.45
Spicy and medicinal herbs	
Root, tuber and rhizome herbs	
Chicory and bitter chicory (*Cichorium intybus and C. intybus var intybus*)	0.40
Ginger (*Zingiber officinale*)	0.30
Licorice (*Glycyrrhiza glabra*)	0.45
Nutgrass, purple nutsedge (*Cyperus rotundus*)	0.40
Turmeric, golden curry (*Curcuma longa*)	0.30
Leafy, flower and stem herbs	
Absinthe, wormwood, (*Artemisia absinthium*)	0.45
Basil (*Ocimum basilicum*)	0.30
Cardamom, large & black (*Amomum subulatum*)	0.50
Chamomile (*Chamomilla recutita*)	0.30
Coriander (*Coriandrum sativum*)	0.30
Fenugreek (*Trigonella foenum-graecum*)	0.40
Lavender (*Lavandula angustifolia*)	0.45
Lemon balm, balm (*Melissa officinalis*)	0.30
Mint	
Japanese mint, wild mint (*Mentha arvensis*)	0.30
Peppermint (*Mentha x piperita*)	0.30
Spearmint, common mint (*Mentha spicata*)	0.30
Oregano (*Origanum vulgare*)	0.30
Parsley (*Petroselinum crispum*)	0.30
Plumed cockscomb (*Celosia argentea*)	0.40
Roselle (*Hibiscus sabdariffa*)	0.35
Rosemary, rosmarin (*Rosmarinus officinalis*)	0.50
Saffron (*Crocus sativus*)	0.20
Stevia, candyleaf, sweetleaf (*Stevia rebaudiana)*	0.35
Summer savory (*Satureja hortensis)*	0.35
Thyme (*Thymus vulgaris*)	0.25
Yarrow (*Achillea millefolium*)	0.50
Fruit, pod and seed herbs	
Ajowan (*Trachyspermum ammi*)	0.45
Black cumin (*Nigella sativa*)	0.40
Chia (*Salvia hispanica*)	0.35
Cumin (*Cuminum cyminum*)	0.25
Milk thistle (*Silybum marianum or Carduus m.*)	0.40
Lesquerella, popweed (*Physaria fendleri*)	0.35
Vanilla (*Vanilla planifolia*), including tutor and shadow plants	0.25

TABLE 8.2
Soil water depletion fraction for no stress (p) for field crops

Field Crops	Depletion fraction p (for ET ≈ 5 mm day^{-1})
Grain legumes	
Bean	
Common bean (*Phaseolus vulgaris*)	0.45
Butter bean, lima bean (*Phaseolus lunatus*)	0.45
Mat bean, moth bean (*Vigna aconitifolia*)	0.50
Black & green gram (*Vigna mungo*)	0.45
Chickpea, garbanzo (*Cicer arietinum*)	0.45
Clusterbean, guar (*Cyamopsis tetragonoloba*)	0.50
Cowpea (*Vigna unguiculata*)	0.50
Fava bean (*Vicia faba*)	0.45
Groundnut (peanut) (*Arachis hypogaea*)	0.50
Lentil (*Lens culinaris*)	0.50
Pea (*Pisum sativum*)	0.40
Soybean (*Glycine max*)	0.50
Fiber crops	
Cotton (*Gossypium hirsutum*)	0.60
Flax (Linseed) (*Linum usitatissimum*)	0.50
Jute mallow for fibre (*Corchorus olitorius*)	0.50
Kenaf (*Hibiscus cannabinus*)	0.55
*Sisal (*Agave sisalana)	0.80
Sunn hemp (*Crotalaria juncea*)	0.55
Oil crops	
Camelina (*Camelina sativa*)	0.60
Canola (*Brassica napus*)	0.50
Castorbean (*Ricinus communis*)	0.60
Linseed (Flax) (*Linum usitatissimum*)	0.60
Mustard (*Brassica juncea*)	0.55
Safflower (*Carthamus tinctorius*)	0.60
Sesame (*Sesamum indicum*)	0.60
Sunflower (*Helianthus annuus*)	0.50
Sugar crops	
Sugar beet (*Beta vulgaris*)	0.55
Sugar cane (*Saccharum officinarum*)	0.60
Cereal and pseudocereal crops	
Amaranth grain (*Amaranthus sp.*)	0.55
Barley (*Hordeum vulgare*)	0.55
Buckwheat (*Fagopyrum esculentum*)	0.55
Maize (*Zea mays*)	0.50
Oats (*Avena sativa*)	0.55
Pearl Millet (*Pennisetum glaucum*)	0.55
Quinoa (*Chenopodium quinoa*)	0.55
Rye (*Secale cereale*)	0.55
Sorghum (*Sorghum bicolor*)	0.55
Teff (*Eragrostis tef*)	0.50
Wheat, common (*Triticum aestivum*)	0.50
Wheat, durum (*Triticum durum*)	0.50
Rice (*Oryza sativa*) Flooded paddies, anaerobic rice Sprinkler and surface irrigated, aerobic rice	 Threshold = 0.80 of θ_{sat} Threshold = 0.90 of θ_{FC}

TABLE 8.3
Soil water depletion fraction for no stress (p) for fruit trees and shrubs

Fruit trees and shrubs	Depletion fraction[1] p (for ET ≈ 5 mm day^{-1})
Mediterranean and warm temperate climates fruit trees and vines	**(0.50)**
Olive (*Olea europaea*) – Traditional	0.65
– Intensive and super-intensive	0.45
Grapes (*Vitis vinifera*) – Table	0.40
– Wine	0.50
Citrus (*Citrus sp.*)	0.45
Avocado (*Persea americana*)	0.55
Fig tree (*Ficus carica*)	0.50
Loquat (*Eriobotrya japonica*)	0.65
Persimmon (*Diospyros kaki*)	0.50
Pomegranate (*Punica granatum*)	0.45
Tea (*Camellia sinensis*) – non-shaded	0.40
– shaded	0.45
Temperate climate fruit trees, vines and shrubs	
Pome trees	**(0.50)**
Apple (*Malus domestica*)	0.50
Pear (*Pyrus communis*)	0.50
Stone fruit trees	**(0.50)**
Apricot (*Prunus armeniaca*)	0.50
Cherry (*Prunus avium*)	0.50
Peach (*Prunus persica*)	0.50
Plum (*Prunus domestica* and *P. salicina*)	0.50
Nut fruit trees	**(0.45)**
Almond (*Prunus dulcis*)	0.45
Hazelnut (*Corylus avellana*)	0.50
Pecan (*Carya illinoinensis*)	0.55
Pistachio (*Pistacia vera*)	0.40
Walnut (*Juglans regia*)	0.50
Other temperate trees and shrubs	**(0.45)**
Kiwifruit (*Actinidia deliciosa*)	0.40
Hop (*Humulus lupulus*)	0.50
Berries (bushes) (*Vaccinium sp.*)	0.50
Sub–tropical and tropical orchards and plantations	
Sub–tropical and tropical small fruit trees and shrubs	**(0.50)**
Acerola (*Malpighia emarginata*)	0.50
Carambola (*Averrhoa carambola*)	0.50
Cashew (*Anacardium occidentale*), dwarf	0.60
Cherimoya (*Annona cherimola*)	0.45
Cocoa (*Theobroma cacao*)	0.30
Coffee (*Coffea arabica*)	0.40
Dragon fruit (*Hylocereus undatus*)	0.60
Guava (*Psidium guajava*)	0.50
Jujube (*Ziziphus jujuba*)	0.50
Papaya (*Carica papaya*)	0.40
Passionfruit (*Passiflora edulis*)	0.40
Tropical fruit trees	**(0.55)**
Jaboticaba (*Plinia peruviana*)	0.60
Longan (*Dimocarpus longan*)	0.55

TABLE 8.3 (CONT.)

Fruit trees and shrubs	Depletion fraction[1] p (for ET ≈ 5 mm day^{-1})
Lychee (*Litchi chinensis*)	0.55
Macadamia (*Macadamia integrifolia*)	0.65
Mango (*Mangifera indica*)	0.50
Palm, fiber, fodder, oil and rubber plantations	**(0.65)**
Cactus pear (*Opuntia ficus–indica*)	0.70
Coconut (*Cocos nucifera*) and Açaí palm (*Euterpe oleracea*)	0.65
Date Palm (*Phoenix dactylifera*)	0.50
Jatropha (*Jatropha curcas*)	0.50
Palm Trees	0.65
Oil palm (*Elaeis guineensis*)	0.65
Peach palm (*Bactris gasipaes*)	0.55
Ramie (*Boehmeria nivea*)	0.55
Rubber tree (*Hevea brasiliensis*)	0.70

EXAMPLE 8.1

Determination of readily available soil water for various crops and soil types

Estimate RAW for a full-grown onion, tomato, and maize crop. Assume that the crops are cultivated on loamy sand, silt, and silty clay soils.

From Tables 6.3 and 8.1

Onion $Z_r \approx 0.4$ m, p = 0.30

Tomato $Z_r \approx 0.8$ m, p = 0.40

Maize $Z_r \approx 1.0$ m, p = 0.50

From Table 7.5

Loamy sand $\theta_{FC} \approx 0.15$ m^3 m^{-3}, $\theta_{WP} \approx 0.06$ m^3 m^{-3}

1 000 ($\theta_{FC} - \theta_{WP}$) = 90 mm(water)/m(soil depth)

Silt $\theta_{FC} \approx 0.32$ m^3 m^{-3}, $\theta_{WP} \approx 0.15$ m^3 m^{-3}

1 000 ($\theta_{FC} - \theta_{WP}$) = 170 mm (water) / m (soil depth)

Silty clay $\theta_{FC} \approx 0.35$ m^3 m^{-3}, $\theta_{WP} \approx 0.23$ m^3 m^{-3}

1 000 ($\theta_{FC} - \theta_{WP}$) = 120 mm (water) / m (soil depth)

	Loamy sand		Silt		Silty clay	
	TAW (Eq. 8.3) mm	RAW (Eq. 8.4) mm	TAW (Eq. 8.3) mm	RAW (Eq. 8.4) mm	TAW (Eq. 8.3) mm	RAW (Eq. 8.4) mm
Onion	36	11	68	20	48	14
Tomato	72	29	136	54	96	38
Maize	90	45	170	85	120	60

TABLE 8.4
Soil water depletion fraction for no stress (p) for grass crops

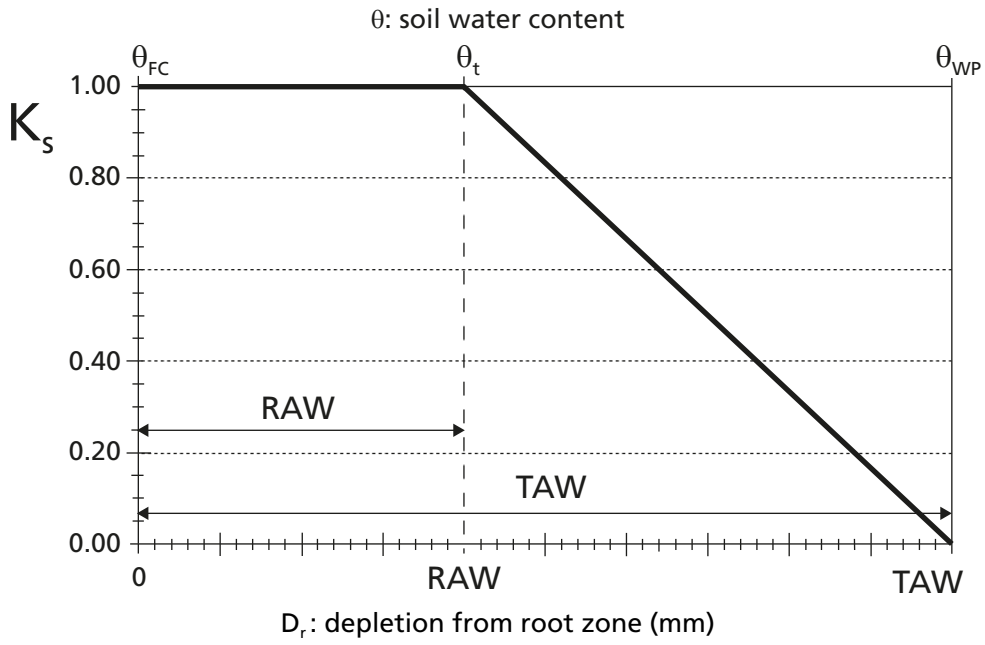
Forages and grasses	Depletion fraction p (for ET ≈ 5 mm day^{-1})
Alfalfa – for hay	0.55
– for seed	0.60
Bermuda for hay	0.55
Spring crop for seed	0.60
Clover and berseem, hay	0.50
Rye Grass for hay	0.60
Sudan Grass for hay	0.55
Grazing Pasture - rotated and extensive grazing	0.60
Grasses for seed	0.60
Turf grass – cool season grass	0.40
– warm season grass	0.50
Landscape grasses, green of golf courses (cut h<0.01 m)	0.20
Landscape grasses, golf courses and lawns (cut h<0.10 m)	0.40
Landscape grasses, urban	0.50

8.2 WATER STRESS COEFFICIENT (K_s)

The effects of soil water stress on crop ET are described by reducing the value for the crop coefficient. This is accomplished by multiplying the crop coefficient by the water stress coefficient, K_s (Equations 8.1 and 8.2).

Water content in the root zone can also be expressed by the root zone depletion, D_r, i.e. the water consumed relative to field capacity. At field capacity, the root zone depletion is zero ($D_r = 0$). When soil water is consumed by evapotranspiration the depletion is increasing, and stress will be induced when D_r reaches RAW. When the root zone depletion exceeds RAW (the water content drops below the threshold θ_t), the root zone depletion is high enough to limit evapotranspiration to less than potential values and the crop evapotranspiration begins to decrease in proportion to the amount of water remaining in the root zone (Figure 8.2 and Example 8.2).

FIGURE 8.2
Water stress coefficient, K_s



Source: FAO, 1998

For $D_r > RAW$, K_s is given by:

$$K_s = \frac{TAW - D_r}{TAW - RAW} = \frac{TAW - D_r}{(1 - p)\,TAW} \qquad (8.6)$$

where K_s transpiration reduction factor depending on the available soil water [0-1],
D_r root zone depletion [mm],
TAW total available soil water in the root zone [mm],
p fraction of TAW that a crop can extract without suffering water stress [-].

After the calculation of K_s, the actual evapotranspiration $ET_{c\,act}$ is computed by means of Equation 8.1 or 8.2. When the root zone depletion (D_r) is smaller than Readily Available Water (RAW), $K_s = 1$.

EXAMPLE 8.2
Effect of water stress on crop evapotranspiration

Estimate the effect of water stress on the evapotranspiration of a full-grown tomato crop (Z_r = 0.8 m and p = 0.40) cultivated on a silty soil (θ_{FC} = 0.32 m^3 m^{-3} and θ_{WP} = 0.12 m^3 m^{-3}) for the coming 10 days when the initial root zone depletion is 55 mm and neither rain nor irrigations are either forecasted or planned. The expected ET_o for the coming decade is 5 mm day^{-1} and K_c = 1.10.

From Eq. 8.3 TAW = 1 000 (0.32–0.12) 0.8 = 160 mm

From Eq. 8.4 RAW = 0.40 (160) = 64 mm

(1) Day	(2) ET_o	(3) K_c	(4) ET_c	(5) $D_{r,i\,start}$	(6) K_s	(7) $ET_{c\,act}$	(8) $D_{r,i\,end}$
	mm day^{-1}		mm day^{-1}	mm		mm day^{-1}	mm
start	-	-	-	-	-	-	55.0
1	5.0	1.1	5.5	55.0	1.00	5.5	60.5
2	5.0	1.1	5.5	60.5	1.00	5.5	66.0
3	5.0	1.1	5.5	66.0	0.98	5.4	71.4
4	5.0	1.1	5.5	71.4	0.92	5.1	76.5
5	5.0	1.1	5.5	76.5	0.87	4.8	81.2
6	5.0	1.1	5.5	81.2	0.82	4.5	85.8
7	5.0	1.1	5.5	85.8	0.77	4.3	90.0
8	5.0	1.1	5.5	90.0	0.73	4.0	94.0
9	5.0	1.1	5.5	94.0	0.69	3.8	97.8
10	5.0	1.1	5.5	97.8	0.65	3.6	100.8

Legend:

(1) Day number.

(2) Reference crop evapotranspiration.

(3) Crop coefficient.

(4) Eq. 6.1, crop ET if no water stress.

(5) Root zone depletion at the beginning of the day (column 8 of the previous day).

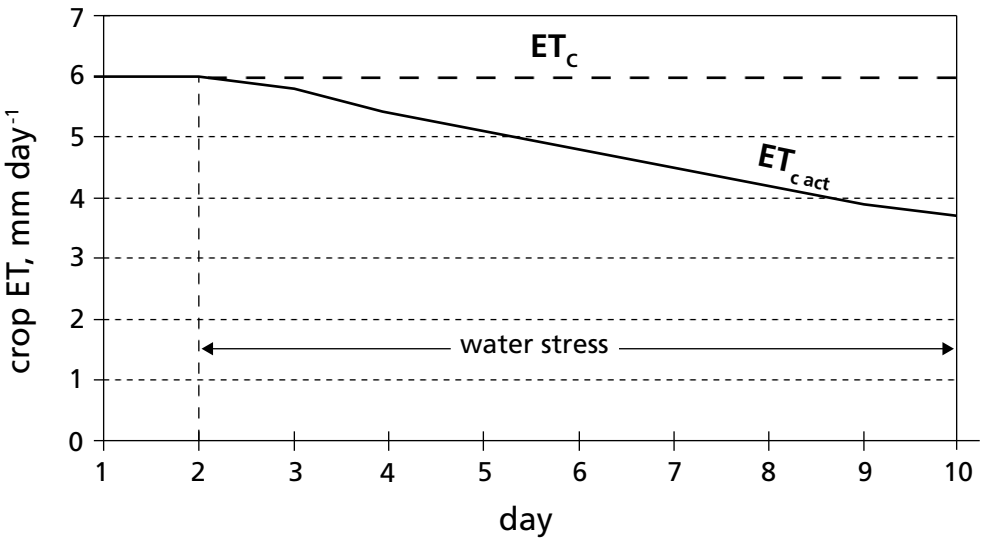
(6) Eq. 8.5 where K_s = 1 if $D_{r,i}$ < RAW.

(7) Eq. 8.1, crop ET under soil water stress conditions.

(8) Depletion at the end of the day.

The Figure 8.3 shows that when the crop entered into water stress after day 2 the K_s decreases the value of $ET_{c\,act}$ continuously from day 3.

FIGURE 8.3
Water stress coefficient effect on the evapotranspiration of a full-grown tomato crop



Source: FAO, 1998

8.3 SOIL WATER BALANCE

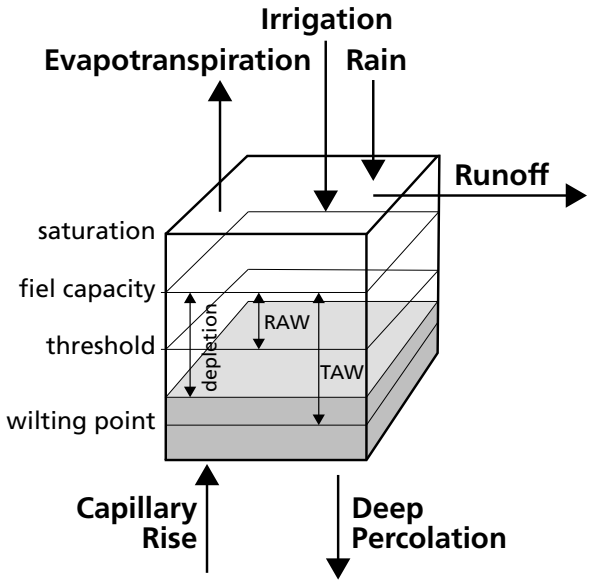
The estimation of K_s requires a daily water balance computation for the root zone. Schematically (Figure 8.4), the root zone can be presented by means of a container in which the water content may fluctuate. To express the water content as root zone depletion is useful. It makes the adding and subtracting of losses and gains straightforward as the various parameters of the soil water budget are usually expressed in terms of water depth. Rainfall, irrigation, and capillary rise of groundwater towards the root zone add water to the root zone and decrease the root zone depletion. Soil evaporation, crop transpiration and percolation losses remove water from the root zone and increase the depletion.

The daily water balance, expressed in terms of depletion at the end of the day is:

$$D_{r,i} = D_{r,i-1} - (P - RO)_i - I_i - CR_i + ET_{c\,act,i} + DP_i \qquad (8.7)$$

where
- $D_{r,i}$ root zone depletion at the end of the day i [mm],
- $D_{r,i-1}$ water content in the root zone at the end of the previous day, i-1 [mm],
- P precipitation [mm],
- RO runoff [mm],
- I applied irrigation depth [mm],
- CR capillary rise [mm],
- DP deep percolation or drainage depth [mm],
- $ET_{c\,act,i}$ (actual) crop evapotranspiration on day i [mm].

FIGURE 8.4
Water balance of the soil root zone



Source: FAO, 1998

8.3.1 Limits on $D_{r,i}$

In Figure 8.4 it is assumed that water can be stored in the root zone until field capacity is reached. Although following heavy rain or irrigation the water content might temporally exceed field capacity, the total amount of water above field capacity is assumed to be lost the same day by deep percolation, following any ET for that day. By assuming that the root zone is at field capacity following heavy rain or irrigation, the minimum value for the depletion $D_{r,i}$ is zero. As a result of percolation and evapotranspiration, the water content in the root zone will gradually decrease and the root zone depletion will increase. In the absence of any wetting event, the water content will steadily reach its minimum value θ_{WP}. At that moment no water is left for evapotranspiration in the root zone, K_s becomes zero, and the root zone depletion has reached its maximum value TAW. The limits imposed on $D_{r,i}$ are consequently $0 \leq D_{r,i} \leq TAW$.

8.3.2 Initial depletion

To initiate the water balance for the root zone, the initial depletion $D_{r,i-1}$ should be estimated. The initial depletion can be derived from measured soil water content by:

$$D_{r\,,\,i-1} = 1000\,(\theta_{FC} - \theta_{i-1})\,Z_r \qquad (8.8)$$

where θ_{i-1} is the average soil water content for the effective root zone. Following heavy rain or irrigation, the user can assume that the root zone is near field capacity, i.e. $D_{r,i-1} \approx 0$.

8.3.3 Precipitation (P), runoff (RO) and irrigation (I)

P_i is equivalent to daily precipitation. Daily precipitation in amounts less than about 0.2 ET_o is normally entirely evaporated and can usually be ignored in the water balance calculations especially when the single crop coefficient approach is being used. I_i is equivalent to the mean infiltrated irrigation depth expressed for the entire field surface. Runoff from the surface during precipitation can be predicted using standard procedures from hydrological texts.

8.3.4 Capillary rise (CR)

The amount of water transported upwards by capillary rise from the water table to the root zone depends on the soil type, the water table depth below the ground surface (WTD), the crop evapotranspiration and the wetness of the root zone. However, CR is primarily influenced by WTD as explained in Box 8.1.

The wetness of the root zone is represented by the actual soil water storage in the root zone (W_a). When WTD is close to the root zone, especially if $ET_{c\ act}$ is high, upward fluxes are expected to occur even when the soil water storage is close to field capacity. Under such WTD and $ET_{c\ act}$ conditions, CR fluxes are important, especially in silty soils. These fluxes depend mainly on WTD and may start at different water table depths and various values of soil water storage, depending on the soil type. This storage is termed the critical soil water storage (W_c) [mm] and varies between the soil water storage at field capacity and the storage when the soil water content corresponds to the depletion fraction p for no stress (Table 8.1). The critical soil water storage may be calculated as:

$$W_c = a_1\ WTD^{b1} \tag{8.9}$$

where WTD water table depth [m],
a_1 soil water storage for the maximum root depth at field capacity [mm],
b_1 adjustment parameter depending on soil physical characteristics, namely soil texture [-] (Table 8.5).

$ET_{c\ act}$ progressively depletes the soil water storage, and the upward fluxes attain a maximum value, called the potential capillary flux (CR_{max}), which is mainly dependent on WTD and less dependent on $ET_{c\ act}$. For shallow water table depths (WTD ≤ 3 m), a steady soil moisture profile is formed when CR_{max} is reached and the soil water storage assumes an almost constant value, termed the steady soil water storage (W_s), that is almost independent of ET_c. In contrast, when the water table is deeper (WTD > 3 m), the upward flow may no longer be sufficient to sustain $ET_{c\ act}$, which continuously decreases due to increased soil water deficit. Under such conditions steady soil water storage (W_s) varies with $ET_{c\ act}$ and CR_{max} is only reached when the soil water storage is low. The W_s, which is the total water stored in the root zone [mm], primarily depends upon the groundwater depth and therefore may be approximated similarly to W_c (Equation 8.9) as:

$$W_s = a_2 WTD^{b2} \qquad \text{for WTD} \le 3\ \text{m} \tag{8.10a}$$

$$W_s = 240\ \text{mm} \qquad \text{for WTD} > 3\ \text{m} \tag{8.10b}$$

$$a_2 = 1.1\ (W_{FC} + W_{WP})/2 \tag{8.10c}$$

where, a_2 empirical unsaturated soil water storage parameter [mm],
b_2 adjustment parameter depending on the soil texture [-],
W_{FC} and W_{WP} root zone water storage at field capacity and at the wilting point respectively [mm].

Indicative values for parameters a_2 and b_2 are given in Table 8.5.

TABLE 8.5
Parameters of the capillary rise parametric equations for three different soil types

Parameters	Silt loam soil	Sandy loam soil	Clay loam soil
a_1 [mm]	385	321	407
b_1	– 0.17	– 0.16	– 0.32
a_2 [mm]	320	303	289
b_2	– 0.27	– 0.54	– 0.16
a_3	– 1.3	– 0.15	– 1.4
b_3	6.6	2.1	6.8
a_4	4.6	7.55	1.11
b_4	– 0.65	– 2.03	– 0.98

The potential capillary flux, CR_{max}, depends upon the groundwater depth and the crop water demand, $ET_{c\ act}$, and varies daily with the fluctuations of WTD and $ET_{c\ act}$. A threshold value for WTD exists, so that when WTD is greater than the threshold, CR_{max} no longer increases. This critical WTD (WTD_c), corresponds to the water table depth when CR reaches the virtually constant value CR_{max}, i.e., the maximum potential groundwater contribution. It has been observed that the critical water table depth mainly depends on the $ET_{c\ act}$:

$$WTD_c = a_3\ ET_{c\ act} + b_3 \quad \text{When } ET_{c\ act} \leq 4 \text{ mm d}^{-1} \tag{8.11a}$$

$$WTD_c = 1.4 \text{ m} \quad \text{When } ET_{c\ act} > 4 \text{ mm d}^{-1} \tag{8.11b}$$

where indicative values for a_3 and b_3 are given in Table 8.5. When $WTD > WTD_c$, the flux CR_{max} decreases with WTD according to a power law. In contrast, at high water table depths ($WTD \leq WTD_c$), CR_{max} no longer depends on WTD_c but on ET rates. Thus, CR_{max} [mm d^{-1}] can be estimated daily, approximately, as:

$$CR_{max} = k\ ET_{c\ act} \quad \text{for } WTD \leq WTD_c \tag{8.12a}$$

$$CR_{max} = a_4\ WTD^{b4} \quad \text{for } WTD > WTD_c \tag{8.12b}$$

where k factor relating evapotranspiration with crop transpiration [-] (Eq. 8.12),
a_4 and b_4 empirical parameters related with the soil characteristics [-] (Table 8.5).

For $ET_{c\ act} \leq 4$ mm d^{-1}, the maximum value for CR_{max} approaches the potential crop transpiration (T_c) while for larger ET rates ($ET_c > 4$ mm d^{-1}) it is smaller than T_c because CR fluxes are insufficient to sustain the potential root extraction rates in the absence of precipitation or irrigation. Then the parameter k in Equation 8.12 is calculated as either a function of ET estimated from the leaf area index (LAI), or as a fixed dependency of $ET_{c\ act}$:

$$k = 1 - e^{-0.6\ LAI} \quad \text{for } ET_{c\ act} \leq 4 \text{ mm d}^{-1} \tag{8.13a}$$

$$k = 3.8/ET_{c\ act} \quad \text{for } ET_{c\ act} > 4 \text{ mm d}^{-1} \tag{8.13b}$$

CR is higher and closer to its potential maximum value when WTD is close to WTD_c and the soil water storage in the root zone is smaller, near the steady soil water storage, W_s (Equations 8.10). The fluxes CR may be approximated as a function of CR_{max} (Equations 8.14) as influenced by the soil water storage in the root zone:

$$CR = CR_{max} \quad \text{for } W_a < W_s \tag{8.14a}$$

$$CR = CR_{max} \left(\frac{W_c - W_a}{W_c - W_s} \right) \quad \text{for } W_s \leq W_a \leq W_c \tag{8.14b}$$

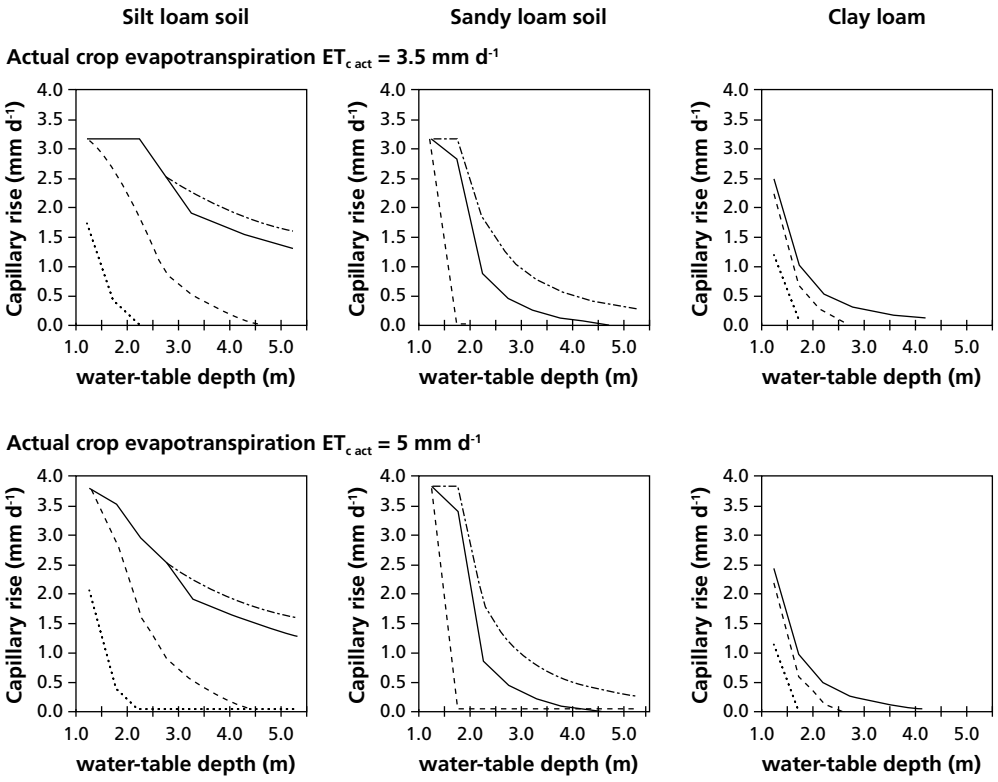
$$CR = 0 \quad \text{for } W_a > W_c \tag{8.14c}$$

WTD values in equations 8.9 to 8.12 should be limited to a depth within the root zone. The parameters a_1 through a_4 and b_1 through b_4, for accuracy of computations, may need to be optimized. If CR observations are available, parameters in Table 8.5 can be used to begin computations, with changes to those parameters made until differences between observed and estimated CR are small. If CR observations are not available for calibration of those parameters, the search for the best parameter values is done by comparing the results of the $ET_{c\ act}$ using the parameters of Table 8.5 with

those obtained in successive steps of the calibration process until differences between observed and estimated ET are small. An example application is presented for a case study in Inner Mongolia, China, using the soil water balance model SIMDualKc. The joint impact of soil type and ET is presented in Figure 8.5, which shows the CR fluxes, or groundwater contribution, for various soil types and $ET_{c\ act}$ of 3.5 and 5 mm d^{-1}. An application to various WTD is described in Box 8.1.

FIGURE 8.5

Groundwater contribution for three soil types as a function of crop evapotranspiration (mm d^{-1}), water table depth (m) and average soil water content in the root zone ($m^3\ m^{-3}$)



Legend: Soil water content at 0.20 $m^3\ m^{-3}$ (long dash double dot line), 0.25 $m^3\ m^{-3}$ (solid line), 0.30 $m^3\ m^{-3}$ (dash line), 0.35 $m^3\ m^{-3}$ (dot line)

Source: Authors' own elaboration

BOX 8.1

Example of the estimation of capillary rise from a shallow water table

The results of applying the CR parametric Equations 8.9 through 8.14 to a maize crop in Hetao, Inner Mongolia, China, are shown in the figure below. The observed $ET_{c\ act}$ and CR data were measured using five static water-table lysimeters when actual water table depths (WTD_a) were set at 1.25, 1.50, 1.75, 2.0 and 2.25 m below the soil surface. Precipitation events were prevented due to the use of a rain shelter. The occurrence of irrigation wetting events was the same in all the lysimeters. The soil in the lysimeters has a silty loam texture with an average (up to 2.60 m) soil water content of 0.38 and 0.15 $cm^3\ cm^{-3}$ at field capacity and at wilting point.

BOX 8.1 (CONT.)

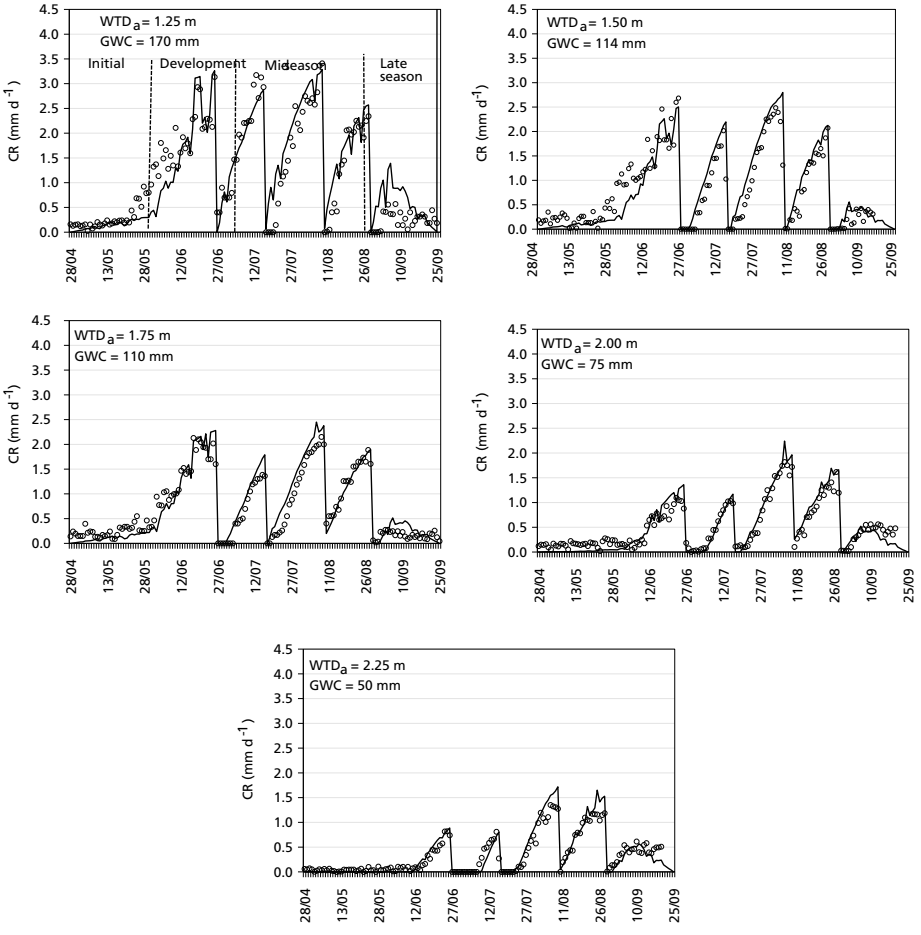
The Figure 8.6 shows the daily dynamics of CR observed in the diverse water-table lysimeters in the same season and estimated using the soil water balance model SIMDualKc after proper calibration against the previously mentioned observations. The growth stages identified in the first example as observed in the field were the same as those of the other lysimeters. The maximum root depth of maize was deeper when WTD was also deeper, $Z_{r\,max}$ was approximately 1.0 m for the WTD_a = 1.25 m, 1.25 m for the WTD_a = 1.50 m, 1.20 m for the WTD_a = 1.75 m, and 1.40 m for WTD_a of 2.00 and 2.25 m. The maximum leaf area index was 3.79, 3.89, 4.37, 4.64 and 4.41 $m^2\ m^{-2}$ for a WTD_a of 1.25, 1.50, 1.75, 2.0 and 2.25 m.

After calibration and validation of the model, the values for the CR parameters (Equations 8.9 to 8.14) were set at: a_1 ranging 385 to 425 mm, a_2 ranging 300 to 345 mm, b_1 = – 0.17, b_2 = – 0.27, a_3 = – 1.3, b_3 = 6.70, a_4 = 4.30, and b_4= – 0.70. The parameters a_1 and a_2 depend on the rooting depth and therefore changed according to the lysimeter.

The observed CR values increased as the crop developed, and reached a maximum in the mid-season when the crop LAI and ET were higher. CR decreased after irrigation and precipitation events and increased when soil water was depleted in the upper soil layers.

FIGURE 8.6

Example of predicted (continuous line) and measured (open dots) daily capillary rise (CR) for various actual water table depths (WTD_a) corresponding to different groundwater contributions (GWC)



8.3.5 Evapotranspiration (ET_c)

Where the soil water depletion is smaller than RAW, the crop evapotranspiration equals $ET_c = K_c\ ET_o$ or as $ET_c = (K_{cb} + K_e)\ ET_o$. As soon as $D_{r,i}$ exceeds RAW, the crop evapotranspiration is reduced and ET_c can be computed from Equation 8.1 or 8.2.

8.3.6 Deep percolation (DP)

Following heavy rain or irrigation, the soil water content in the root zone might exceed field capacity. In this simple procedure it is assumed that the soil water content is at θ_{FC} within the same day of the wetting event, so that the depletion $D_{r,i}$ in Equation 8.7 becomes zero. Therefore, following heavy rain or irrigation:

$$DP_i = (P_i - RO_i) + I_i - ET_{c\ act,i} - D_{r,\ i-1} \qquad \text{when } DP_i \geq 0 \qquad (8.15)$$

As long as the soil water content in the root zone is below field capacity (i.e. $D_{r,i} > 0$), it is assumed that the soil will not drain and $DP_i = 0$.

The decrease of the soil water storage in the root zone (W [mm]) due to internal drainage when actual soil water storage is initially above the water storage at field capacity ($W > W_{FC}$), may be represented by a decay function of time as:

$$W = a_D\ t^{bD} \qquad (8.16)$$

where t time after an irrigation or rain that increases storage to above field capacity [days],

a_D potential depth of water storage between field capacity and saturation over the effective depth of the root zone [mm],

b_D adjustment parameter referring to the potential rate of soil drainage.

DP (mm d^{-1}) is computed daily for successive days without rain or irrigation until the soil drains to field capacity. This is done using the daily storages W_i and W_{i+1} for days i and i+1 and the ET for the preceding day:

$$DP_{i+1} = (W_i - W_{i+1}) - ET_{c\ act,\ i} \qquad (8.17)$$

Table 8.6 presents indicative values for the parameters of Equation 8.16.

TABLE 8.6

Indicative values for the drainage parameters of Equation 8.16 for three primary soil types and associated water holding characteristics

Soil type	θ_{sat} [$m^3\ m^{-3}$]	θ_{FC} [$m^3\ m^{-3}$]	a_D [mm]	b_D [-]
Loam	0.50	0.35	340	–0.0220
Loamy sand	0.33	0.22	300	–0.0200
	0.40	0.24	246	–0.0200
Silt loam	0.45	0.33	370	–0.0173
	0.47	0.36	400	–0.0150
	0.50	0.39	380 to 410	–0.0173

BOX 8.2

Case study for the application of the deep percolation decay function to a malt barley crop

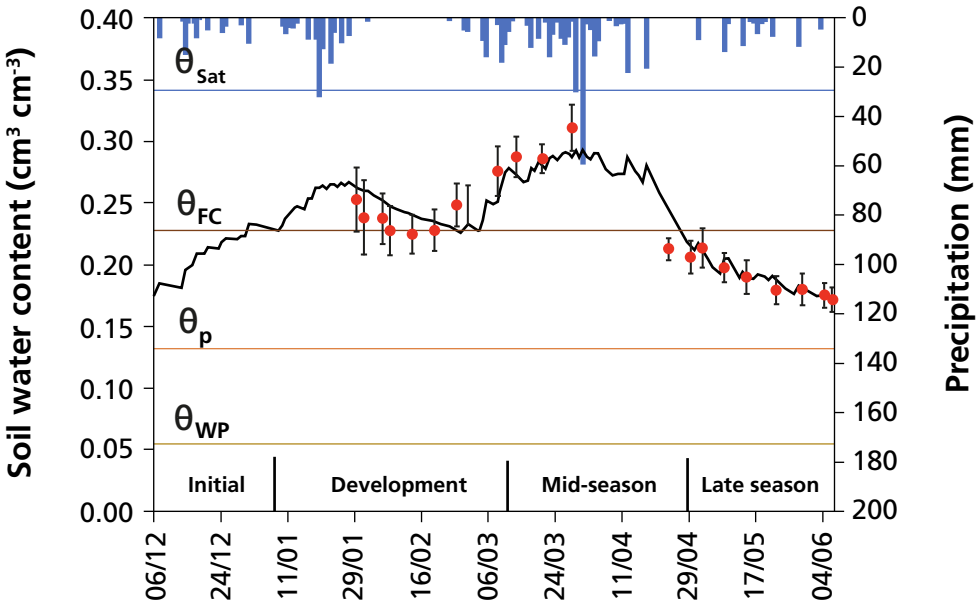
Results of the application of the deep percolation decay function (Eq. 8.16 and 8.17) for a loamy sand soil cropped with malt barley when soil water content is greater than field capacity are shown in Figure 8.7. The soil water content at saturation was 0.33 $m^3 m^{-3}$, at field capacity was 0.22 $m^3 m^{-3}$ and at the wilting point was 0.05 $m^3 m^{-3}$. The crop was rainfed, the seasonal precipitation was 569 mm and ET_c was 351 mm.

The soil water content was measured in a farmer's field with a calibrated frequency domain reflectometry (FDR) sensor; these observations were used for calibrating and validating the SIMDualKc soil water balance model. The calibrated parameter values for Equation 8.16 were a_D = 300 mm and b_D = -0.0020.

The dynamics of the soil water content in Figure 8.7 show that for most of the time up to the end of the mid-season, the soil water content was above field capacity, which was the result of very frequent precipitation events. Using the described decay function, the model was able to adequately simulate soil water and deep percolation. The estimated seasonal DP was 170 mm.

FIGURE 8.7

Example of daily observed (dots; vertical lines represent the standard deviation of the observations) and predicted (full line) soil water content in a loamy sand soil cropped with malt barley, Alpiarça, Portugal, using the model SIMDualKc. Vertical columns denote precipitation events



8.4 FORECASTING OR ALLOCATING IRRIGATIONS

Irrigation is required when rainfall is insufficient to compensate for the soil water used as evapotranspiration. The primary objective of irrigation is to apply water at the right time and in the right amount. The timing and depth of future irrigations can be planned by calculating the soil water balance of the root zone on a daily basis (Equation 8.7). To avoid crop water stress, irrigations should be applied before or at the moment when the readily available soil water is depleted so that $D_{r,i}$ is always ≤ RAW. To avoid deep percolation losses that may leach valuable nutrients out of the root zone, the net irrigation depth should be smaller than or equal to the root zone depletion ($I_i \leq D_{r,i}$) unless leaching of salts is needed.

Example 8.3 illustrates the application of a water balance of the root zone to predict irrigation dates to avoid water stress. The example utilizes various calculations for K_e from Example 7.8.

Boxes 8.3 and 8.4 show, respectively, an application of the dual K_c approach to a water-stressed soybean crop and to a hedgerow olive orchard under deficit irrigation management.

EXAMPLE 8.3
Irrigation scheduling to avoid crop water stress

Plan the irrigation applications for Example 7.9 and Figure 8.8. It is assumed that:

- Irrigations are to be applied when RAW is depleted.
- The depletion factor (p) is 0.6.
- All irrigations and precipitations occur early in the day.
- The depth of the root zone (Z_r) on day 1 is 0.3 m and increases to 0.35 m by day 10.
- The root zone depletion at the beginning of day 1 ($D_{r,i-1}$) is RAW.

From Eq. 8.3 $\quad TAW = 1\,000\,(0.23 - 0.10)\,Z_{r,i} = 130\,Z_{r,i}$ [mm]

From Eq. 8.4 $\quad RAW = 0.6\,TAW = 78\,Z_{r,i}$ [mm]

On day 1 $\quad$ when $Z_r = 0.3$ m: $D_{r,i-1} = RAW = 78\,(0.3) = 23$ mm

(1) Day	(2) ET_o	(3) Z_r	(4) RAW	(5) $D_{r,i\,start}$	(6) P-RO	(7) I	(8) K_s	(9) K_{cb}	(10) K_e	(11) K_c	(12) ET_c	(13) DP	(14) $D_{r,i\,end}$
	mm d^{-1}	m	mm	mm	mm	mm					mm	mm	mm
start	-	-	-	-	-	-	-	-	-	-	-	-	23
1	4.5	0.30	23	0	0	40	1	0.30	0.91	1.21	5.5	17	5
2	5.0	0.31	24	5	0	0	1	0.31	0.90	1.21	6.1	0	12
3	3.9	0.31	24	12	0	0	1	0.32	0.72	1.04	4.0	0	16
4	4.2	0.32	25	16	0	0	1	0.33	0.37	0.70	2.9	0	18
5	4.8	0.32	25	18	0	0	1	0.34	0.18	0.52	2.5	0	21
6	2.7	0.33	26	15	6	0	1	0.36	0.64	1.00	2.7	0	18
7	5.8	0.33	26	18	0	0	1	0.37	0.45	0.82	4.7	0	22
8	5.1	0.34	26	22	0	0	1	0.38	0.17	0.55	2.8	0	25
9	4.7	0.34	27	25	0	0	1	0.39	0.08	0.47	2.2	0	27
10	5.2	0.35	27	0	0	27	1	0.40	0.81	1.21	6.3	0	6

Description of the columns of the Table above:

(1) Day number.

(2) From Example 7.9.

(3) Z_r is given (interpolated between 0.3 m on day 1 and 0.35 m on day 10).

(4) Eq. 8.4.

EXAMPLE 8.3 (CONT.)

(5) $D_{r,i\ start}$ (root zone depletion at the beginning of the day)

If precipitation and irrigation occur early in the day then

$$D_{r,i\ start} = \text{Max}\ (D_{r,i-1\ end} - I - (P\text{-}RO), \text{or } 0)$$

If precipitation and irrigation occur late in the day, then

$$D_{r,i\ start} = D_{r,i-1end}$$

where $D_{r,i-1\ end}$ is taken from column 14 of the previous day

Since the depth of the root zone increases each day, the water content of the subsoil (θ_{sub}) has to be considered to update $D_{r,i}$

$$D_{r,i} = D_{r,i-1} + 1000\ (\theta_{FC} - \theta_{sub,i-1})\ \Delta Z_{r,i}$$

where $\Delta Z_{r,i}$ is the increase in rooting depth on day i.

In the example it is assumed that θ_{sub} is at field capacity (due to prior overirrigation and excessive rainfall on previous days). Therefore, a combination of the equations for $D_{r,i}$ can be utilized.

(6) From Example 7.9.

(7) Irrigation is required when $D_{r,i} \geq$ RAW.

On day 1, the irrigation depth (infiltrating the soil) is given (from Example 7.9: I = 40 mm)

On day 10, another irrigation is required. An irrigation with a net depth of 27 mm refills the root zone and avoids water loss by deep percolation (DP = 0 mm).

(8) Eq. 8.5, where $K_s = 1$ for $D_{r,i} \leq$ RAW.

(9) From Example 7.9.

(10) Day 1 to 9: From Example 7.9.

Day 10: Following the extra irrigation early in the day, the topsoil will be wet and K_r is 1; thus, from Eq. 7.3, results in $K_e = (1.21 - 0.40) = 0.81$.

(11) $K_c = K_s\ K_{cb} + K_e$.

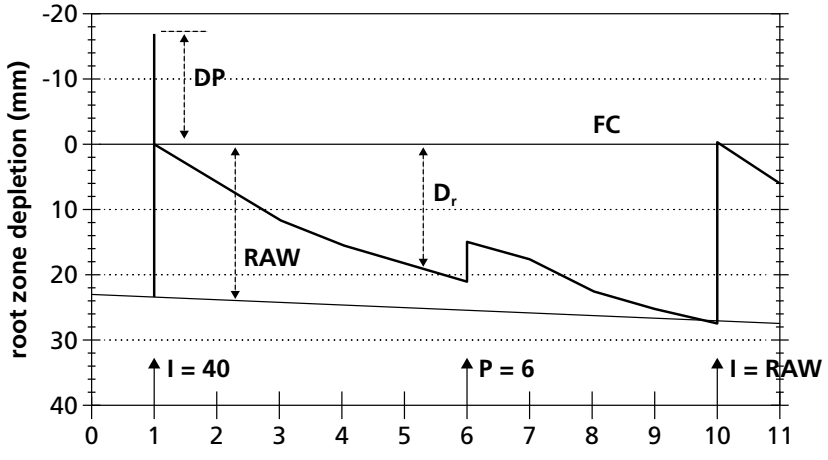
(12) Eq. 8.1.

(13) Eq. 8.9, where $D_{r,i-1}$ is taken from column 14 of the previous day.

(14) $D_{r,i}$ (root zone depletion at the end of one day) = the starting $D_{r,i}$ at the beginning of the next day (see column 5 of the Table above). From Eq. 8.7, where $D_{r,i-1}$ is taken from column 14 of the previous day (see Figure 8.8).

FIGURE 8.8

Example of the computed irrigation scheduling to avoid water stress (see daily values in the table of Example 8.3)



BOX 8.3

Case study on the application of the dual K_c approach to assess the irrigation scheduling used for a water-stressed soybean crop

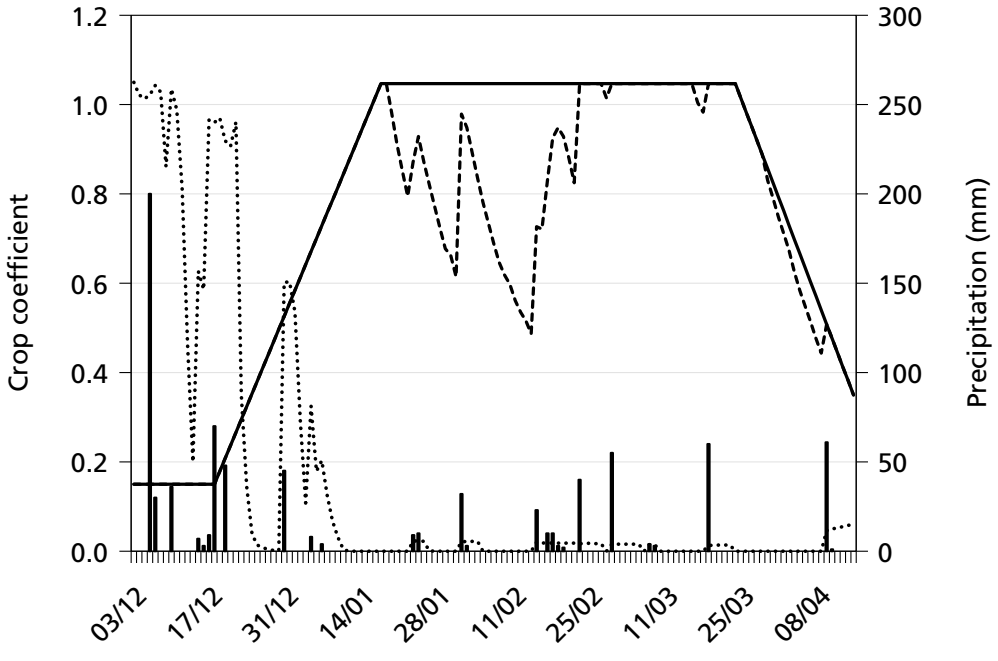
The results of applying the (K_{cb} + K_e) approach to a water-stressed soybean crop are shown in the Figure 8.9. The $ET_{c\ act}$ and its partitioning were estimated using the soil water balance model SIMDualKc after calibration with observed soil water content data. After calibration and adjustment to climate conditions (Equation 7.2) the potential K_{cb} values were 0.15, 1.10, and 0.35 respectively for the initial, mid-season and end-season stages while the p value was 0.50 for all the crop growth stages. The maximum fraction of ground covered by soybeans was 0.99.

The occurrence and magnitudes of individual wetting events from precipitation are shown in the Figure 8.9. The number of wetting events was 27 and the seasonal precipitation was 786 mm. However, almost 45% of this water was not available to the crop due to the occurrence of DP and runoff that commonly occur following large precipitation events.

The $K_{cb\ act}$ curve deviates from the potential K_{cb} curve due to water stress. A moderate water stress occurred during the first half of the mid-season, while a negligible water deficit occurred during the late season. High K_e peaks occurred in response to precipitation events when the soil water content of the evaporation layer was high at the beginning of the crop season. The K_e peaks during the mid-season stage are quite small because the fraction of ground covered by the crop was very high during that period.

FIGURE 8.9

Example of predicted daily basal crop coefficient curve (full line), actual basal crop coefficient (dashed line) and evaporation coefficient curve (dotted line) for a rainfed soybean crop at Paysandú, Uruguay. Vertical lines denote precipitation events (computations performed with the SIMDualKc water balance model are explained in Box 8.3)



BOX 8.4

Case study on the application of the dual K_c approach to assess deficit irrigation management used for a hedgerow drip irrigated olive orchard

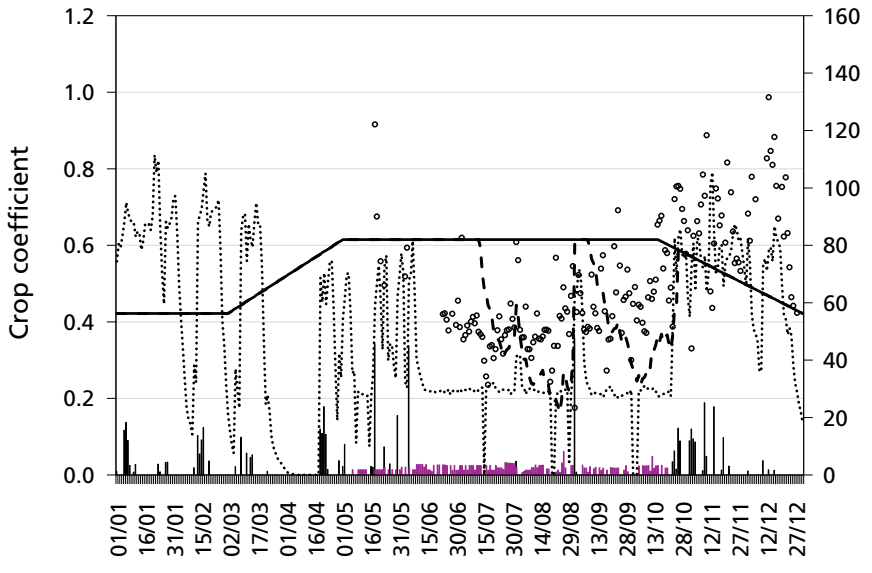
The results of applying the (K_{cb} + K_e) method for a super-high density (hedgerow) olive orchard are shown in the Figure 8.10. The observed $ET_{c\ act}$ data were measured using an eddy covariance system and actual T_c was measured using a sap flow method. This example uses the same data that were used in the case study of Box 7.4 but herein using the farmers' irrigation schedule. The Figure 8.10 shows the daily dynamics of potential and actual K_{cb} and K_e curves estimated using the soil water balance model SIMDualKc after calibration of K_{cb} against the previously mentioned observations. The potential values for $K_{cb\ ini}$, $K_{cb\ mid}$, and $K_{cb\ end}$ were estimated to be 0.30, 0.49, and 0.43, respectively. A K_{cb} for the non-growing period, when the crop is dormant, was set at 0.30. The soil of the farmer's field had a sandy loam texture. The depth of the evaporation soil layer, Z_e, was 0.1 m, with TEW = 18 mm and REW = 9 mm based on Eq. 7.5 and the observed soil data. The fraction of ground covered by the trees reached a maximum of 0.38 by mid-season and the maximum tree height was 4 m.

The occurrence of precipitation and irrigation events is shown in the Figure 8.10, which includes the out of season and dormant periods. All irrigation wetting events occurred during the mid-season stage when precipitation and soil water storage were not sufficient to maintain crop transpiration. This is observed by the open circles (observed total K_{cb} + K_e) and the dashed line (actual K_{cb} curve) below the solid line representing the potential K_{cb} curve. The soil evaporation K_e curve (dotted line) peaked after wetting events by rain or high-frequency irrigation.

The deviation of the actual K_{cb} curve (and observations) from the potential K_{cb} curve whenever a water deficit occurred in the mid-season stage indicates that the drip irrigation schedule adopted by the farmer was inadequate to ensure potential crop transpiration and represented a water deficit irrigation management.

FIGURE 8.10

Example of daily observed (open circles) and simulated (dashed line) actual basal crop coefficient curve, potential basal crop coefficient curve (full line), and evaporation coefficient curve (dotted line) for a drip irrigated hedgerow olive orchard at Viana do Alentejo, Portugal. Vertical lines denote precipitation or irrigation events (computations performed with the SIMDualKc water balance model are explained in Box 8.4)



8.5 YIELD-MOISTURE STRESS RELATIONSHIP

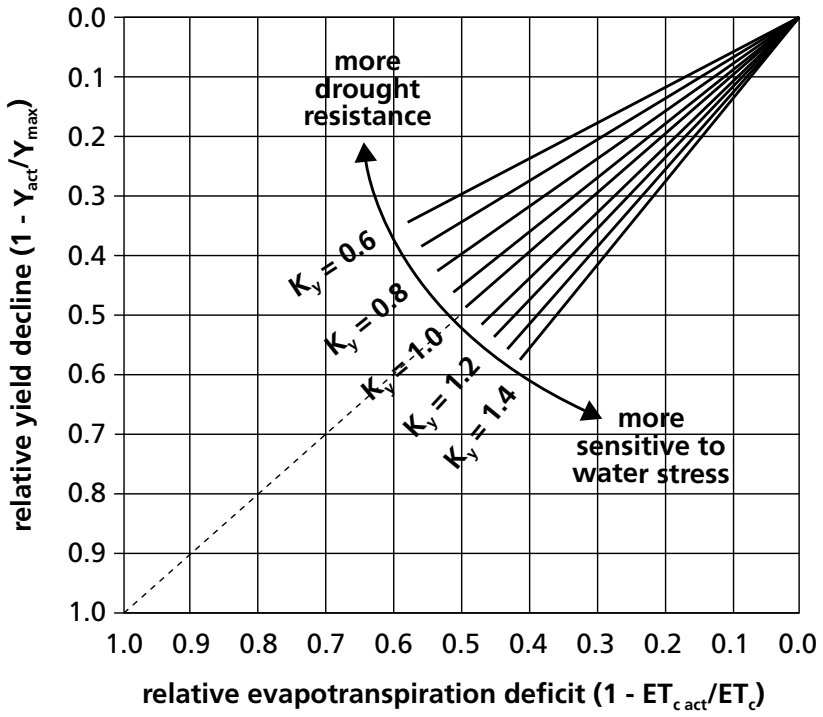
A simple, linear crop-water production function was proposed in the FAO Irrigation and Drainage Paper No33 to predict the reduction in crop yield when crop stress was caused by a shortage of soil water:

$$\left(1 - \frac{Y_{act}}{Y_{max}}\right) = K_y \left(1 - \frac{ET_{c\,act}}{ET_c}\right) \qquad (8.18)$$

where: Y_{act} actual crop yield [kg ha^{-1}],
Y_{max} potential (maximum) crop yield [kg ha^{-1}],
K_y a yield response factor [-],
$ET_{c\,act}$ actual crop evapotranspiration [mm d^{-1}],
ET_c crop evapotranspiration for standard, no-water-stress conditions [mm d^{-1}].

K_y is a factor that describes the reduction in relative yield according to the reduction in ET_c caused by soil water shortage. In FAO No33, K_y values are crop specific and may vary over the growing season, i.e. with the crop stage (Figure 8.11). In general, the decrease in yield due to water deficit during the vegetative and ripening period is relatively small, while during the flowering and yield formation periods it can be large. Values for K_y for individual growth periods and for the complete growing season have been included in FAO No33. Seasonal values for K_y are summarized in Table 8.7.

FIGURE 8.11
Relationship between yield decline and evapotranspiration deficit (water stress) for various yield response factors (K_y)



Source: Authors' own elaboration

TABLE 8.7
Seasonal yield response factor (K_y) for field and vegetable crops, fruit trees and vines

Crop	K_y	Crop	K_y
Vegetable crops		**Fiber crops**	
Banana	1.20-1.35	Cotton	0.85-1.25
Cabbage	0.95	**Oil Crops**	
Cucumber	0.95	Rapeseed	0.90
Eggplant	0.70-1.35	Safflower	0.80-0.93
Onion	1.10-1.25	Sunflower	0.95
Potato	1.10	**Sugar crops**	
Tomato	1.05-1.35	Sugarbeet	1.00-1.10
Watermelon	1.10-1.25	Sugarcane	1.20
Grain legumes		**Fruit trees and vines**	
Beans	1.15	Apple	1.20
Groundnut	0.70	Citrus	1.10-1.30
Pea	1.15	Grape	0.85
Pepper	1.10	Olive	0.70-0.80
Soybean	0.85-1.30	Peach	1.20
Cereals		**Grasses**	
Barley, malt	1.25	Alfalfa	1.10-1.30
Maize	1.25-1.32		
Sorghum	0.90		
Wheat, spring	1.00-1.15		
Wheat, winter	0.95-1.05		

Tabulated values for K_y cover relatively few crops. The selection of values to be used must be done carefully because the crop response to water stress depends on the crop cultivars, which may have a wide range of variation as tabulated for various crops. Recognizing the degree of tolerance to moisture stress for the cultivars being used is recommended. Also to be considered is the effect of cumulative stress and its timing during the season because stress impacts may be different when crops recover from stress during the first part of the cycle, or are subjected to stress at more critical stages such as flowering or fruiting. For the crops not included in Table 8.7, users are advised to consider K_y for crops of the same family or that have similar behaviour. Nevertheless, care must be used for this purpose.

8.6 STRESS EFFECTS OF SOIL SALINITY

Salts in the soil water solution can reduce evapotranspiration by making soil water less "available" for plant root extraction. Salts have an affinity for water and hence additional force is required for the crop to extract water from a saline soil. The presence of salts in the soil water solution reduces the total potential energy of the soil water solution. In addition, some salts may cause toxic effects in plants and can reduce plant metabolism, growth and yield. A function is presented hereafter that predicts the reduction in evapotranspiration caused by the salinity of soil water. The function is derived by combining yield-salinity equations from the FAO Irrigation and Drainage Paper No29 with yield-ET equations from FAO Irrigation and Drainage Paper No33. The resulting equation provides a first approximation to the reduction in evapotranspiration expected under various salinity conditions.

8.6.1 Salinity impacts and crops tolerance

There is evidence that crop yield and transpiration are not as sensitive to low osmotic potential as they are to low matric potential. Under saline conditions, many plants are able to partially compensate for the low osmotic potential of the soil water by

building up higher internal solute contents. This is done by absorbing ions from the soil solution and by synthesizing organic osmolytes. Both of these reactions reduce the impact of osmotic potential on water availability. However, the synthesis of organic osmolytes does require the expenditure of metabolic energy. Therefore, plant growth is often reduced under saline conditions. The reduced plant growth impacts transpiration by reducing ground cover, and a further reduction in transpiration is sometimes additionally caused by partial stomatal closure.

Other impacts of salts in the soil include direct sodium and chloride toxicities and induced nutrient deficiencies. These deficiencies reduce plant growth by reducing the rate of leaf elongation, the enlargement, and the division of cells in leaves. The modality depends on the method of irrigation. With sprinkler irrigation, adsorption of sodium and chloride through the leaf can result in toxic conditions for all crop species. With surface or trickle irrigation, direct toxic conditions generally occur only in vine and tree crops; however, high levels of sodium can induce calcium deficiencies for all crop species.

Since salt concentration changes as the soil water content changes, soil salinity is normally measured and expressed on the basis of the electrical conductivity of the saturation extract of the soil (EC_e). The EC_e is defined as the electrical conductivity of the soil water solution after the addition of a sufficient quantity of distilled water to bring the soil water content to saturation. EC_e is typically expressed in deciSiemens per meter (dS m^{-1}). Under optimum management conditions, crop yields remain at potential levels until a specific, threshold electrical conductivity of the saturation soil water extract ($EC_{e\ threshold}$) is reached. If the average EC_e of the root zone increases above this critical threshold value, the yield is presumed to begin to decrease linearly in proportion to the increase in salinity. The rate of decrease in yield with increase in salinity is usually expressed as a slope, b, having units of % reduction in yield per dS m^{-1} increase in EC_e.

All plants do not respond to salinity in a similar manner; some crops can produce acceptable yields at much higher soil salinity levels than others. This is because some crops are better able to make the needed osmotic adjustments that enable them to extract more water from a saline soil, or they may be more tolerant of some of the toxic effects of salinity.

Salt tolerance for many agricultural crops is provided in the FAO Irrigation and Drainage Papers No. 33 and 48. The $EC_{e\ threshold}$ and slope b from these and other sources are listed in Tables 8.8 through 8.11, respectively for vegetable, field, fruit tree and vine, and grass and forage crops. The data serve only as a guideline - tolerance vary depending on climate, soil conditions and cultural practices. Crops are often less tolerant during germination and seedling stage. As can be observed from the data in these Tables, there is an 8 to 11-fold range in salt tolerance of agricultural crops. The effect of soil salinity on yield and crop evapotranspiration is hence crop specific.

The $EC_{e\ threshold}$ and b parameters in Tables 8.8 through 8.11 were determined primarily in research experiments using nearly steady-state irrigation where soil water contents were maintained at levels close to field capacity. However, under most types of irrigation scheduling for sprinkler and surface irrigation, the soil water content is typically depleted to well below field capacity, so that the EC of the soil water solution, EC_{sw}, increases prior to irrigation, even though the EC of the saturation extract does not change.

The increased salt concentration in the soil water solution reduces the osmotic potential of the soil water solution (it becomes more negative), so that the plant must expend more metabolic energy and may exert more mechanical force to absorb water. In addition, the metabolic and toxic effects of salts on plants may become more pronounced as the soil dries and concentrations increase. However, the variation in soil water content during an irrigation interval has not been found to strongly influence crop evapotranspiration. This is because of the rise of soil water content to levels that are above that experienced under steady state irrigation early in a long irrigation interval.

TABLE 8.8

Salt tolerance of vegetable crops expressed as electrical conductivity of the soil saturation extract threshold when crop yield first reduces below the full yield potential ($EC_{e\ threshold}$), slope (b) of reduction in crop yield with increasing salinity beyond $EC_{e\ threshold}$, and tolerance rating

Crop	$EC_{e\ threshold}$ (dS m^{-1})	Yield reduction rate b (% / dS m^{-1})	Rating*
Root, tuber, bulb and stem vegetable crops			
Asparagus (*Asparagus officinalis*)	4.1	2.0	T
Carrot (*Daucus carota*)	1.0	14.0	S
Cassava (*Manihot esculenta*)	-	-	MS
Celeriac, celery root (*Apium graveolens var. rapaceum*)	1.8-2.5	6.2-13.0	MS
Garlic (*Allium sativum*)	3.9	14.3	MS
Jerusalem Artichoke (*Helianthus tuberosus*)	0.4	9.6	MS
Onion (*Allium cepa*)	1.2	16.0	S/MS
Parsnip (*Pastinaca sativa*)	-	-	S
Potato (*Solanum tuberosum*)	1.7	12.0	MS
Radish (*Raphanus sativus*)	1.2-2.0	7.6-13.0	MS
Sweet potato (*Ipomoea batatas*)	1.5-2.5	10.0-11.0	MS
Table beets (*Beta vulgaris*)	4.0	9.0	MT
Turnip (*Brassica rapa*)	0.9	9.0	MS
Leafy and flower vegetable crops			
Artichoke (*Cynara scolymus*)	6.1	11.5	MT
Broccoli (*Brassica oleracea cv. italica*)	1.3-2.8	9.2-15.8	MS
Brussel sprouts (*B. oleracea cv. gemmifera*)	1.8	9.7	MS
Cabbage (*B. oleracea*)	1.0-1.8	9.8-14.0	MS
Cabbage, Chinese (*B. campestris*)	3.2	10.0	MS
Cauliflower (*B. oleracea cv. botrytis*)	1.5-1.8	6.2-14.4	MS
Celery (*Apium graveolens*)	1.8-2.5	6.2-13.0	MS
Fennel (*Foeniculum vulgare*)	1.2	15.7	S
Lettuce (*Lactuca sativa*)	1.3-1.7	12.0-13.0	MS
Purslane (*Portulaca oleracea*)	6.3	9.6	MT
Spinach (*Spinacia oleracea*)	2.0-3.2	7.6-16.0	MS
Swiss chard (*Beta vulgaris subsp. vulgaris*)	7.0	5.7	T
Fruit vegetable crops			
Banana (*Musa* sp.)	-	-	MS
Bell and chili pepper (*Capsicum annuum*)	1.5-1.7	12.0-14.0	MS
Chili, tabasco pepper (*C. frutescens*)	-	-	MS
Cucumber (*Cucumis sativus*)	1.1-2.5	7.0-13.0	MS
Eggplant (*Solanum melongena*)	1.1	6.9	MS
Melon (*Cucumis melo*)	1.0	8.4	MS
Melon, cantaloupe (*C. melo cv. cantalupensis*)	1.3	5.7	MS
Okra (*Abelmoschus esculentus*)	-	-	MS
Pineapple (*Ananas comosus*)	-	-	MT
Pumpkin, winter squash (*Cucumis pepo*)	1.2	13.0	MS
Squash, Zucchini (*Cucurbita pepo*)	4.7-4.9	10.0-10.5	MT
Squash (scallop) (*C. pepo cv. melo pepo*)	3.2	16.0	MS
Strawberries (*Fragaria × ananassa*)	0.75-1.5	11.0-33.0	S
Tomato (*Solanum lycopersicum*)	0.9-2.5	9.0-9.9	MS
Watermelon (*Citrullus lanatus*)	-	-	MS

Root zone salinity is measured by the electrical conductivity of the saturation extract of the soil, reported in deciSiemens per meter (dS m^{-1}) at 25°C;
$EC_{e\ threshold}$ means the average root zone salinity at which yield starts to decline;
b is the percentage reduction in crop yield per 1 dS m^{-1} increase in EC_e beyond $EC_{e\ threshold}$
* *Ratings are: T = tolerant, MT = moderately tolerant, MS = moderately sensitive, and S = sensitive.*

TABLE 8.9

Salt tolerance of field crops expressed as electrical conductivity of the soil saturation extract threshold when crop yield first reduces below the full yield potential ($EC_{e\ threshold}$), slope (b) of reduction in crop yield with increasing salinity beyond $EC_{e\ threshold}$, and tolerance rating

Crop	$EC_{e\ threshold}$ (dS m^{-1})	Yield reduction rate b (% / dS m^{-1})	Rating*
Grain legumes			
Bean (*Phaseolus vulgaris*)	1.0	19.0	S
Bean, lima (*Phaseolus lunatus*)	6.0	14.5	MT
Bean, mung (*Vigna radiata*)	1.8	20.7	S
Black and green gram (*Vigna mungo*)	1.8	20.7	S
Chickpea (garbanzo) (*Cicer arietinum*)	-	-	MS
Cowpea (*Vigna unguiculata*)	4.9	12.0	MT
Fava bean (*Broadbean*) (*Vicia faba*)	1.5-1.6	9.6	MS
Groundnut (peanut) (*Arachis hypogaea*)	3.2	29.0	MS
Pea (*Pisum sativum*)	1.5-3.4	10.6-14.0	S/MS
Soybean (*Glycine max*)	5.0	20.0	MT
Fiber crops			
Cotton (*Gossypium hirsutum*)	7.7	5.2	T
Flax (Linseed) (*Linum usitatissimum*)	1.7	12.0	MS
Kenaf (*Hibiscus cannabinus*)	8.1	11.6	T
Oil crops			
Canola (*Brassica napus*)	11.0	13.0	T
Castorbean (*Ricinus communis*)	-	-	MS
Crambe (*Crambe abyssinica*)	2.0	6.5	MS
*Guar (*Cyamopsis tetragonoloba)	8.8	17.0	T
Lesquerella (*Lesquerella fenderli*)	6.1	19	MT
Linseed (Flax) (*Linum usitatissimum*)	1.7	12.0	MT
Rapeseed (*Brassica napus cv. napus*)	9.7-11.0	13-14	T
Safflower (*Carthamus tinctorius*)	-	-	MT
Sesame (*Sesamum indicum*)	-	-	S
Sunflower (*Helianthus annuus*)	4.8	5.0	MT
Sugar crops			
Sugar beet (*Beta vulgaris*)	7.0	5.9	T
Sugar cane (*Saccharum officinarum*)	1.7	5.9	MS
Cereals and pseudocereals			
Barley (*Hordeum vulgare*)	8.0	5.0	T
Maize (*Zea mays*)	1.7	12.0	MS
Oats (*Avena sativa*)	-	-	MT
Pearl Millet (*Pennisetum glaucum*)	-	-	MS
Rice (*Oryza sativa*)	3.0	12.0	S
Rye (*Secale cereale*)	11.4	10.8	T
Sorghum (*Sorghum bicolor*)	6.8	16.0	MT
Triticale (*X Triticosecale*)	6.1	2.5	T
Wheat, common (*Triticum aestivum*)	6.0	7.1	MT
Wheat, durum (*Triticum durum*)	5.7-5.9	3.8-5.5	T

Root zone salinity is measured by the electrical conductivity of the saturation extract of the soil, reported in deciSiemens per meter (dS m^{-1}) at 25°C;

$EC_{e\ threshold}$ means the average root zone salinity at which yield starts to decline;

b is the percentage reduction in crop yield per 1 dS m^{-1} increase in EC_e beyond $EC_{e\ threshold}$

* *Ratings are: T = tolerant, MT = moderately tolerant, MS = moderately sensitive, and S = sensitive.*

TABLE 8.10

Salt tolerance of fruit trees, shrubs and vines expressed as electrical conductivity of the soil saturation extract threshold when crop yield first reduces below the full yield potential ($EC_{e\ threshold}$), slope (b) of reduction in crop yield with increasing salinity beyond $EC_{e\ threshold}$, and tolerance rating

Crop	$EC_{e\ threshold}$ (dS m^{-1})	Yield reduction rate b (%/dS m^{-1})	Rating*
Mediterranean and warm temperate climates fruit trees and vines			
Olive (*Olea europaea*)	-	-	MT
Grape (*Vitis vinifera*)	1.5	9.6	MS
Grapefruit (*Citrus x paradisi*)	1.2-1.8	13.5-16.0	S
*Lemon (*Citrus limon)	1.5	12.8	S
Lime (*Citrus aurantiifolia*)	-	-	S
Orange (*Citrus sinensis*)	1.3-1.7	13.1-16.0	S
Pommelo (*Citrus maxima*)	-	-	S
Tangerine (Citrus reticulata)	-	-	S
Avocado (*Persea americana*)	-	-	S
Pomegranate *(Punica granatum)*	-	-	MT
Temperate climate fruit trees, vines and shrubs			
Pome trees			
Apple (*Malus communis*)	-	-	S
Pear (*Pyrus malus*)	-	-	S
Stone fruit trees			
Apricot (*Prunus armeniaca*)	1.6	24.0	S
Cherry (*Prunus spp.*)	-	-	S
Plum, prune (*Prunus domestica*)	1.5-2.6	18.0-31.0	S/MS
Peach (*Prunus persica*)	1.7	21.0	S
Nut fruit trees			
Almond (*Prunus dulcis*)	1.5	19.0	S
Pecan (*Carya illinoiensis*)	1.9	16.6	MS
Walnut *(Juglans sp.*)	1.7	16.1	MT
Temperate shrubs (Berries)	1.5	22.0	S
Sub–tropical and tropical small trees and shrubs			
Guava (*Psidium guajava*)	4.7	9.8	MT
Guayule (*Parthenium argentatum*)	8.7	11.5	MT
Palm plantations			
Coconut (*Cocos nucifera*)	-	-	MT
Date palm (*Phoenix dactylifera*)	4.0	3.6	T

Root zone salinity is measured by the electrical conductivity of the saturation extract of the soil, reported in deciSiemens per meter (dS m^{-1}) at 25°C; $EC_{e\ threshold}$ means the average root zone salinity at which yield starts to decline; b is the percentage reduction in crop yield per 1 dS m^{-1} increase in EC_e beyond $EC_{e\ threshold}$

* *Ratings are: T = tolerant, MT = moderately tolerant, MS = moderately sensitive, and S = sensitive.*

TABLE 8.11

Salt tolerance of grasses and forage crops expressed as electrical conductivity of the soil saturation extract threshold when crop yield first reduces below the full yield potential ($EC_{e\ threshold}$), slope (b) of reduction in crop yield with increasing salinity beyond $EC_{e\ threshold}$, and tolerance rating

Crop	$EC_{e\ threshold}$ (dS m^{-1})	Yield reduction rate b (% / dS m^{-1})	Rating*
Alfalfa (*Medicago sativa*)	2.0	7.3	MS
Barley (forage) (*Hordeum vulgare*)	6.0	7.1	MT
Bermuda grass (*Cynodon dactylon*)	6.9	6.4	T
Berseem clover (*Trifolium alexandrinum*)	1.5	5.7	MS
Canarygrass, (*Phalaris arundinacea*)	6.0	5.6	MT
Clover (*Trifolium sp*)	1.5	12.0	MS
Cowpea (forage) (*Vigna unguiculata*)	2.5	11.0	MS
Fescue, tall (*Festuca elatior*)	3.9	5.3-6.2	MT
Foxtail (*Alopecurus pratensis*)	1.5	9.6	MS
Harding grass (*Phalaris tuberosa* cv. *stenopteral*)	4.6	7.6	MT
Lovegrass (*Eragrosti ssp.*)	2.0	8.4	MS
Maize (forage) (*Zea mays*)	1.8	7.4	MS
Orchard grass (*Dactylis glomerata*)	1.5	6.2	MS
Rye (forage) (*Secale cereale*)	7.6	4.9	T
Ryegrass (*Lolium perenne*)	5.6	7.6	MT
Sesbania (*Sesbania exaltata*)	2.3	7.0	MS
Sphaerophysa (*Sphaerophysa salsula*)	2.2	7.0	MS
Sudangrass (*Sorghum sudanense*)	2.8	4.3	MT
Trefoil, narrowleaf birdsfoot (*Lotus corniculatus*)	5.0	10.0	MT
Trefoil, big (*Lotus pedunculatus*)	2.3	19.0	MS
Vetch (*Vicia angustifolia*)	3.0	11.0	MS
Wheat, common (forage) (*Triticum aestivum*)	4.5	2.6	MT
Wheat, durum (forage) (*Triticum turgidum*)	2.1	2.5	MT
Wheatgrass, tall (*Agropyron elongatum*)	7.5	4.2	T
Wheatgrass, fairway crested (*Agropyron cristatum*)	7.5	6.9	T
Wheatgrass, standard crested (*Elymus triticoides*)	3.5	4.0	MT
Wildrye, beardless (*Saccharum officinarum*)	2.7	6.0	MT

Root zone salinity is measured by the electrical conductivity of the saturation extract of the soil, reported in deciSiemens per meter (dS m^{-1}) at 25°C; $EC_{e\ threshold}$ means the average root zone salinity at which yield starts to decline; b is the percentage reduction in crop yield per 1 dS m^{-1} increase in EC_e beyond $EC_{e\ threshold}$

* *Ratings are: T = tolerant, MT = moderately tolerant, MS = moderately sensitive, and S = sensitive.*

There is a similar, counteractive decrease in soil water content later in a long irrigation interval. In addition, the distribution of salts in the root zone under low frequency irrigation can reduce salinity impacts during the first portion of the irrigation interval. Also, under high frequency irrigation of the soil surface, soil evaporation losses are higher. Consequently, given the same application depth, the leaching fraction is reduced. For these reasons, the length of irrigation interval and the change in EC of soil water during the interval have usually not been found to be factors in the reduction of ET, given that the same depths of water are infiltrated into the root zone over time.

Increased evaporation (K_e) under high frequency irrigation of the soil surface may counteract reductions in K_c caused by stress due to high EC_e of the root zone. Under these conditions, the total K_c and ET_c are not very different from the non-saline, standard conditions under less frequent irrigation, even though crop yields and crop transpiration are reduced. Because of this, under saline conditions, the K_s reducing factor should only be applied with the dual K_c approach.

The effects of soil salinity and water stress are generally additive in their impacts on crop evapotranspiration. Therefore, the same yield-ET functions may hold for both water shortage induced stress and for salinity induced stress.

8.6.2 Yield-Salinity relationship

A widely practiced approach for predicting the reduction in crop yield due to salinity has been described in the FAO Irrigation and Drainage Paper No29. The approach presumes that, under optimum management conditions, crop yields remain at potential levels until a specific, threshold electrical conductivity of the soil water solution is reached. When salinity increases beyond this threshold, crop yields are presumed to decrease linearly in proportion to the increase in salinity. The soil water salinity is expressed as the electrical conductivity of the saturation extract, EC_e. In equation form, the salinity response function is a linear equation. The procedure of FAO Irrigation and Drainage Paper No29 is followed:

$$\frac{Y_{act}}{Y_{max}} = 1 - (EC_e - EC_{e\ threshold}) \frac{b}{100} \qquad \text{when } EC_e > EC_{e\ threshold} \qquad (8.19)$$

where Y_{act} actual crop yield,

Y_{max} maximum expected crop yield when $EC_e < EC_{e\ threshold}$,

EC_e mean electrical conductivity of the root zone saturation extract [dS m^{-1}],

$EC_{e\ threshold}$ electrical conductivity of the saturation extract at the threshold of EC_e when crop yield first reduces below Y_m [dS m^{-1}],

b rate of yield reduction per unit of EC_e above $EC_{e\ threshold}$ [%/(dS m^{-1})].

Values for $EC_{e\ threshold}$ and b have been provided in the FAO Irrigation and Drainage Papers No29 and 48 as well as in the original version of FAO 56 and are updated and newly listed in Tables 8.7 through 8.10 for several agricultural crops. Relative to vegetable crops there is no published information on herbs, spices, and special vegetable crops, but it is commonly known that these crops are sensitive, or moderately sensitive, to soil and water salinity. There is also insufficient information about fruit tree and shrub crops, where tolerance varies much among crops. Generally, they should be considered as moderately sensitive (MS) at best.

Salinity-yield data from the FAO Irrigation and Drainage papers No 29 and 48 were mostly from studies where soil water content was held at about –30 kPa potential or higher at the 0.3 to 0.6 m soil depth, depending on the crop. However, these papers indicate that for most crops, the data are transferable to typical field situations where the readily available soil water (RAW) is depleted between irrigations. The accumulation of salts in terms of their osmotic effects is further modified depending on soil texture, agro-climatic conditions, ionic components of salinity, and soil-irrigation-crop management conditions.

8.6.3 Salinity-ET reduction relationship

Without water stress (D_r < RAW)

When salinity stress occurs without water stress, Equations 8.18 and 8.19 can be combined and solved for an equivalent K_s, where $K_s = ET_{c\ act}/ET_c$:

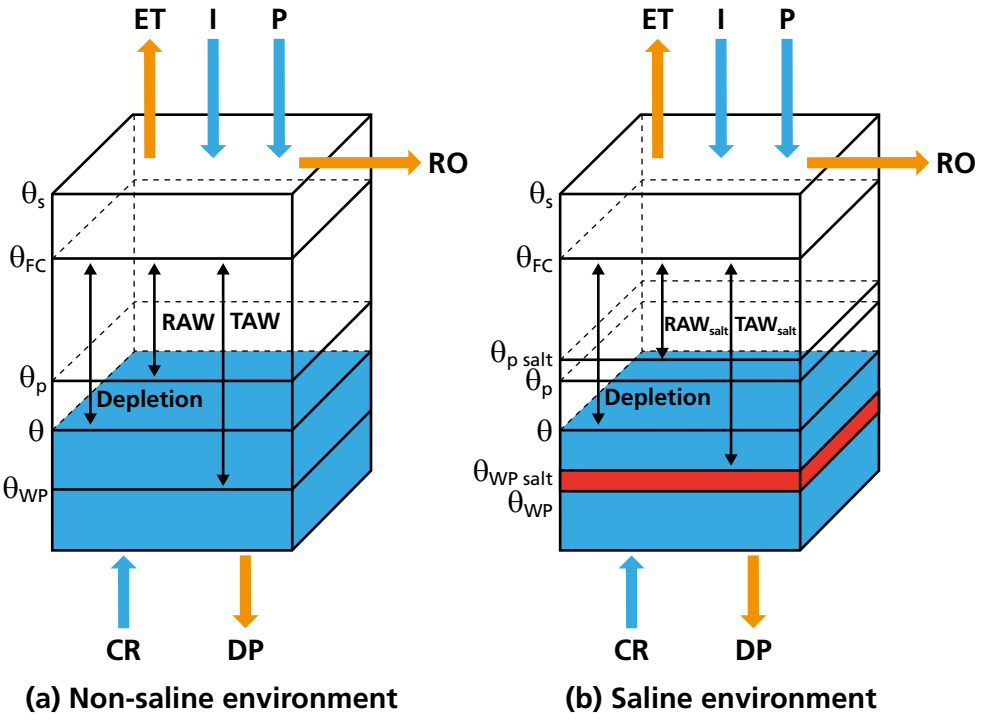
$$K_s = 1 - \frac{b}{K_y\ 100}(EC_e - EC_{e\ threshold}) \qquad \text{when } EC_e > EC_{e\ threshold} \qquad (8.20)$$

for conditions when $EC_e > EC_e$ threshold and soil water depletion is less than the readily available soil water depth (D_r < RAW). D_r and RAW are defined in a previous section.

Combined salinity-water stress (D_r > RAW)

When soil water stress occurs in addition to salinity stress, Equations 8.5, 8.19 and 8.20 need to be modified to appropriately perform the soil water balance as explained in Figure 8.12. Moreover, under salinity conditions, crop roots must overcome increased water retention energies due to both matric potential and decreased osmotic potential.

FIGURE 8.12
Water balance of the root zone under non- or low-saline environment and under saline environment



Source: Authors' own elaboration

This condition takes into account that the soil water content at the wilting point is higher in saline soils and therefore the available soil water is lower than under no salinity conditions. Therefore, an adjustment for the soil water content at the wilting point as a function of the actual salinity of the soil in relation to the specific salinity tolerance of the crops is required.

The soil water content at wilting point adjusted for saline conditions is increased relative to WP of non-saline soils as:

$$\theta_{WP\ salt} = \theta_{WP} + \frac{b}{100}\left(\frac{EC_e - EC_{e\ threshold}}{10}\right)(\theta_{FC} - \theta_{WP}) \quad (8.21)$$

where $\theta_{WP\ salt}$ soil water content at the wilting point adjusted for the salinity conditions [$m^3\ m^{-3}$],
θ_{WP} soil water content at the wilting point without salinity impacts [$m^3\ m^{-3}$],
θ_{FC} soil water content at field capacity [$m^3\ m^{-3}$],
b rate of reduction in yield per unit increase in EC_e [%/(dS m^{-1})],
EC_e electrical conductivity of the soil water saturation extract [dS m^{-1}],
$EC_{e\ threshold}$ threshold electrical conductivity when yield starts decreasing [dS m^{-1}].

Therefore, TAW (Equation 8.3) is reduced (Figure 8.12) to

$$TAW_{salt} = 1000\ (\theta_{FC} - \theta_{WP\ salt})\ Z_r \quad (8.22)$$

where TAW_{salt} adjusted value for TAW when θ_{WP} is increased due to soil salinity [mm],
θ_{FC} water content at field capacity [$m^3\ m^{-3}$],
$\theta_{WP\ salt}$ corrected value of soil water content at wilting point due to salinity [$m^3\ m^{-3}$],
Z_r crop root depth [m].

Under saline conditions, crops may be stressed for a depletion fraction smaller than p. Therefore, the depletion fraction p should also be adjusted under salinity conditions, thus decreased when $EC_e > EC_{e\ threshold}$ as a function of the percent crop yield reduction b:

$$p_{salt} = p - b\ (EC_e - EC_{e\ threshold})\ p \quad (8.23)$$

where p_{salt} adjusted value of p when the crop cultivated in a soil with a salinity EC_e [> 0],
p depletion fraction for no water stress [-],
EC_e electrical conductivity of the saturation soil water extract [dS m^{-1}],
$EC_{e\ threshold}$ threshold of EC_e when the crop starts yield reduction [dS m^{-1}].

Adjusting p implies decreasing RAW (Figure 8.12), thus Equation 8.4 is:

$$RAW_{salt} = p_{salt}\ TAW_{salt} \quad (8.24)$$

where RAW_{salt} corrected value for RAW for the considered crop and soil [mm],
p_{salt} adjusted value for p for the considered crop and soil [-],
TAW_{salt} adjusted total available water for the considered soil and crop [mm].

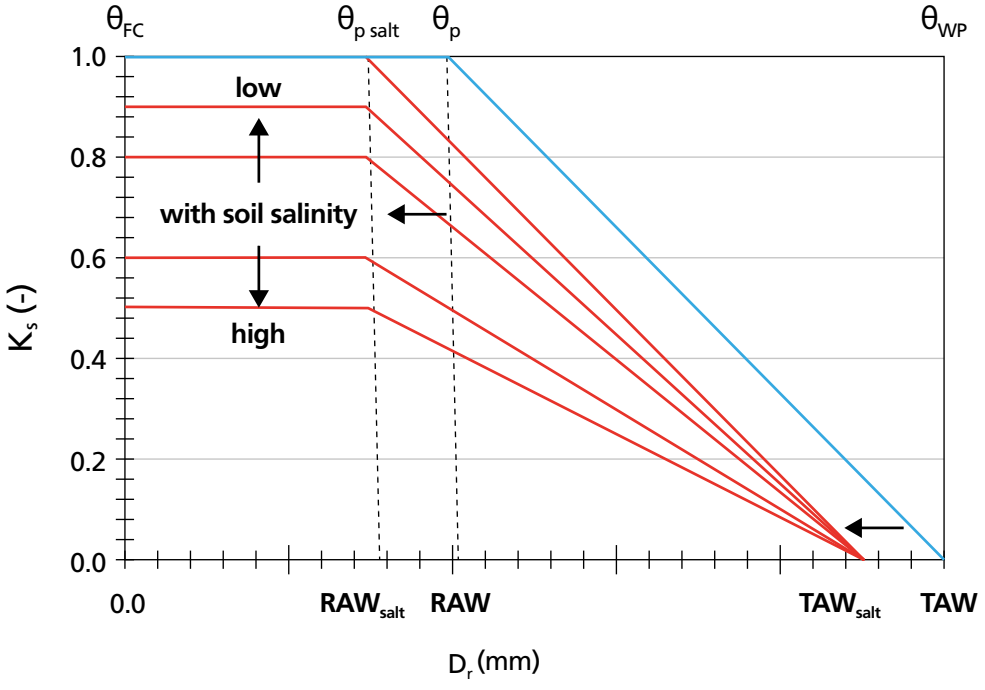
Adjusting the TAW and RAW to the saline cropping conditions (Equations 8.22 and 8.24) means that the stress coefficient must also be changed to better account for the impacts of salinity. Thus, when the cumulative depletion $D_{r,i} > RAW_{salt}$, the stress coefficient is:

$$K_s = \left(\frac{TAW_{salt} - D_{r,i}}{TAW_{salt} - RAW_{salt}} \right) \left(1 - \frac{b}{K_y \, 100} (EC_e - EC_{e\,threshold}) \right) \quad (8.25)$$

where
- TAW_{salt} adjusted total available water for the soil EC_e conditions [mm],
- RAW_{salt} readily available water for the soil EC_e conditions [mm],
- b rate of reduction in yield per unit increase in EC_e [%/(dS m^{-1})],
- K_y crop yield response factor [-],
- EC_e electrical conductivity of the saturation soil extract [dS m^{-1}],
- $EC_{e\,threshold}$ hreshold electrical conductivity when yield start decreasing [dS m^{-1}].

Figure 8.13 illustrates the changes in the water stress coefficient resulting from the implementation of Equations 8.22 to 8.25. This adjustment of the soil water balance to salinity is probably not important when soil salinity is low and the crop is tolerant but is relevant for less tolerant crops and more saline soils.

FIGURE 8.13
The effect of soil salinity on the water stress coefficient K_s (Equation 8.25)



Source: Authors' own elaboration

Limitations

Because the impact of salinity on plant growth and yield and on crop evapotranspiration is a time-integrated process, generally only the seasonal value for K_y is used in predicting the reduction in evapotranspiration. For many crops, the seasonal K_y is nearly 1.0 (Table 8.7). For crops where K_y is unknown, the user may assume $K_y = 1$ in Equations 8.19 and 8.25, or may select a K_y relative to a crop type that has similar behaviour.

Equations 8.19 and 8.25 are suggested as approximate estimates of salinity impacts on ET, and represent the general effects of salinity on evapotranspiration as occurring over an extended period of time (as measured in weeks or months). These equations are not expected to be accurate for predicting $ET_{c\ act}$ for specific days. Nor do they include other complicating effects such as specific ion toxicity. Application of Equations 8.19 and 8.25 presumes that the EC_e represents the average EC_e for the root zone.

The equations presented may not be valid at high salinity, where the linear relationships between EC_e, crop yield and K_s may not hold. The use of Equations 8.19 and 8.25 should usually be restricted to $EC_e < EC_{threshold} + 50/b$. In addition, care is required when the equations predict $Y_a = 0$ before $K_s = 0$ when $K_y > 1$ and vice versa.

As indicated earlier, a reduction in ET_c in the presence of soil salinity is often partially caused by reduced plant size and fraction of ground cover. These effects are largely included in the coefficient values. Therefore, where plant growth is affected by salinity, Equations 8.19 and 8.25 are applied and no other reductions in K_c are required.

8.6.4 Application

Under steady state conditions, the value for EC_e can be predicted as a function of EC of the irrigation water (EC_{iw}) and the leaching fraction, using a standard leaching formula. For example, the FAO-29 leaching formula $LR = EC_{iw} / (5\ EC_e - EC_{iw})$ predicts the leaching requirement when approximately a 40-30-20-10 percent water extraction pattern occurs from the upper to lower quarters of the root zone prior to irrigation. EC_{iw} is the electrical conductivity of the irrigation water. From this equation, EC_e is estimated as:

$$EC_e = \frac{1 + LF}{LF} \frac{EC_{iw}}{5} \tag{8.26}$$

where LF, the actual leaching fraction, is used in place of LR, the leaching requirement. Equation 8.26 predicts $EC_e = 1.5\ EC_{iw}$ under conditions where a 15-20 percent leaching fraction is used.

Other leaching fraction equations can be used in place of the FAO-29 equation to fit local characteristics. Equation 8.26 is only true if the irrigation water quality and the leaching fraction are constant over the growing season. Time is required to attain a salt equilibrium in the soil. If there are important winter rains of high-quality water and often excellent leaching, the salt balance in the soil will be quite different at the beginning of the season and with a lower average EC_e of the root zone than would be predicted from Equation 8.26. An appropriate local calibration of Equation 8.26 is desirable under these particular conditions.

EXAMPLE 8.4
Effect of soil salinity on crop evapotranspiration

A field of beans is cultivated on a silt loam soil and is irrigated during the midseason period using water having salinity EC_{iw} = 1.0 dS m^{-1}. A 15 percent leaching fraction is employed. The $EC_{e\ threshold}$ and slope from Table 8.5 are 1.0 dS m^{-1} and 19 %/(dS m^{-1}) respectively. The seasonal K_y from FAO Irrig. Drain. Pap. Nº 33 and Table 8.5 for beans is K_y = 1.15. Compare the effect on crop evapotranspiration for various levels of soil water depletion in the root zone under saline and nonsaline conditions. θ_{FC} = 0.22 $m^3\ m^{-3}$ and θ_{WP} = 0.11 $m^3\ m^{-3}$ and Z_r=1.0 m thus, TAW and RAW for the bean crop are 110 and 44 mm (for p = 0.4).

Since the leaching fraction is 0.15, EC_e is estimated from Equation 8.26 as EC_e = 1.5 EC_w = 1.5 (1) = 1.5 dS m^{-1}. The K_s in the presence of salinity stress and absence of moisture stress is:

$$K_s = \left(1 - \frac{b}{K_y\ 100}(EC_e - EC_{e\ threshold})\right) = \left(1 - \frac{19}{1.15\ (100)}(1.5 - 1.0)\right) = 0.92$$

The K_s in the presence of moisture stress, but in the absence of salinity stress is:

$$K_s = \left(\frac{TAW - D_r}{TAW - RAW}\right) = \left(\frac{110 - D_r}{110 - 44}\right) = \left(\frac{110 - D_r}{66}\right)$$

The K_s in the presence of both moisture stress and salinity stress (Equation 8.25) is:

$$K_{s,\,i} = \left(\frac{TAW_{salt} - D_{r,i}}{TAW_{salt} - RAW_{salt}}\right)\left(1 - \frac{b}{K_y\ 100}(EC_e - EC_{e\ threshold})\right) = \left(\frac{TAW_{salt} - D_{r,i}}{TAW_{salt} - RAW_{salt}}\right)0.92$$

From Equation 8.22 TAW_{salt}= 109 mm and from Equation 8.24 RAW_{salt} = 39.4 mm

The effect on crop evapotranspiration for various soil water depletions in the root zone (D_r) are:

D_r (mm)	K_s no soil salinity		K_s with soil salinity (EC_e = 1.5 dS m^{-1} (Eq. 8.26))		Additional reduction in potential ET_c due to salinity
0	1.00	no reduction in ET_c	0.92	8% reduction in ET_c	8%
35	1.00	no reduction in ET_c	0.92	8% reduction in ET_c	8%
40	1.00	no reduction in ET_c	0.91	9% reduction	9%
44	1.00	no reduction in ET_c	0.86	14% reduction	14%
50	0.91	9% reduction	0.78	22% reduction	13%
60	0.76	24% reduction	0.65	35% reduction	11%
70	0.61	39% reduction	0.51	49% reduction	10%
80	0.45	55% reduction	0.38	62% reduction	7%
90	0.30	70% reduction	0.25	75% reduction	5%
100	0.15	85% reduction	0.12	88% reduction	3%
110	0.00	ET_c = 0	0.00	ET_c = 0	--

BOX 8.5

Example of the combined effects of water and salinity stresses

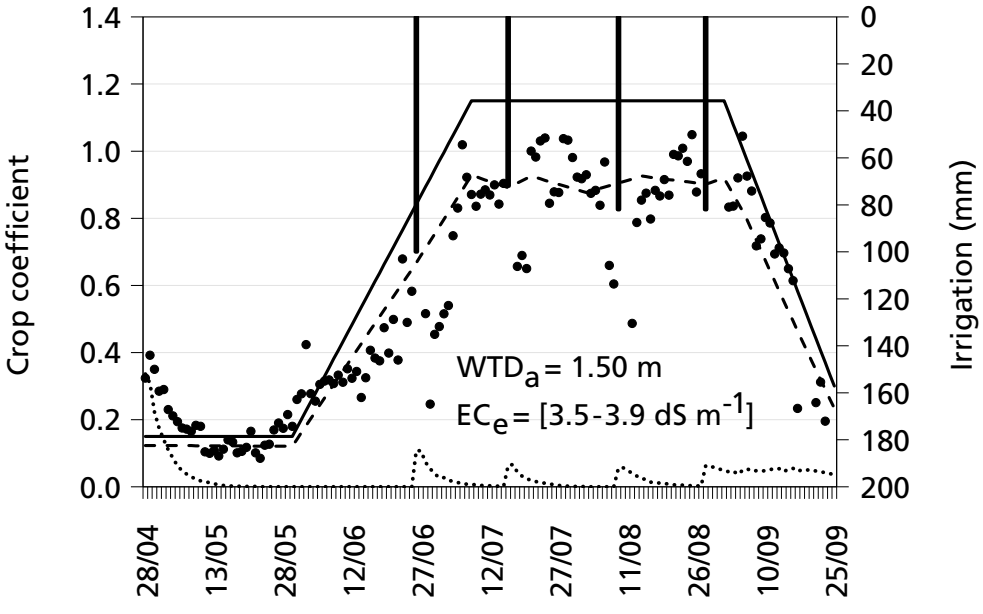
The results of applying the (K_{cb} + K_e) method for a maize crop under water and salinity stress are shown in the Figure 8.14. This example uses the same data set that were used in the case study of Box 8.1; however, only two examples were selected. The observed $ET_{c\ act}$, soil water content and CR data were measured using five static water-table lysimeters. The $ET_{c\ act}$ partitioning was estimated using the soil water balance model SIMDualKc after proper calibration with observed soil water content data. After calibration and adjustment to climate conditions (Equation 7.2) the K_{cb} values were 0.15, 1.15, and 0.30 respectively for the initial, mid-season and end-season stages while the depletion factor p was 0.55 for the initial, development and mid-season stages, and 0.80 for the end season stage.

All lysimeters were partially covered (75 percent of the soil surface) with plastic mulch, which led to a reduction of f_{ew} and constrained the $K_{c\ max}$ (Equation 7.4), thus limited the soil evaporation component (Equation 7.3). Precipitation was prevented due to the use of a rain shelter. The occurrence of irrigation wetting events was the same in all the lysimeters. Seasonal gross irrigation was 561–591 mm.

The $K_{cb\ act}$ curve deviates from the potential K_{cb} curve due to the combined effect of water and salinity stresses (Figure 8.14). Salinity stress occurred since the beginning of the season because EC_e was higher than the threshold limit of 1.7 dS m^{-1} (Table 8.2). Water stress occurred mainly during mid-season, resulting in a greater deviation of the $K_{cb\ act}$ curve from the standard K_{cb} curve. K_e peaks occurred in the K_e in response to high soil water content at the beginning of the crop season, and the large irrigation events practiced.

FIGURE 8.14

Example of observed decrease of actual crop coefficient (closed circles) due to salinity: predicted daily basal crop curve (thick line), actual basal crop coefficient (dashed line), evaporation coefficient (dotted line) for a surface irrigated maize cropped in a saline soil in Hetao, Inner Mongolia, China. The vertical lines in the figure denote irrigation events (computations performed with the SIMDualKc water balance model are detailed in Box 8.5)



Chapter 9

Actual crop ET ($ET_{c\ act}$) by agricultural and natural vegetation

Actual crop ET ($ET_{c\ act}$) refers to both agricultural and natural vegetation including non-pristine agricultural vegetation. The latter is defined, in the usage here, as vegetation having less than perfect growing conditions or stand characteristics (i.e. relatively poor conditions of density, height, leaf area, fertility, vitality and biomass yield) as compared to "pristine" maximal yield conditions. Non-typical cropping conditions are also included and refer to types or arrangements of agricultural crops that are not listed or described in Tables 6.1 to 6.4 and 7.1 to 7.3. However, non-typical cropping conditions are mainly dealt with in Chapter 10.

The approach whereby a crop is characterized by a crop coefficient, K_c, and the crop evapotranspiration is given by the product of K_c and the reference evapotranspiration ET_o, provides a simple and convenient way of also characterizing the evapotranspiration from natural vegetation and for diverse non-typical cultivation practices.

This chapter presents procedures for estimating actual K_c and K_{cb} values for natural vegetation and for agricultural vegetation independent of having K_c and K_{cb} values available. The section on estimating crop coefficients from fraction of ground cover and crop height is particularly important because it focuses on the estimation of $ET_{c\ act}$ aimed at its use in irrigation scheduling and water management in real time.

9.1 CALCULATION APPROACH, GENERAL ASPECTS

The first step in the K_c ET_o approach is the estimation of lengths of crop growth stages, which also applies to natural vegetation (Tables 6.4 and 6.5). The next step is the development of crop coefficient curves that represent the ratios of ET_c to ET_o during the various growth stages of the vegetation as described in Chapters 5 and 6.

9.1.1 Initial growth stage

The procedure to estimate crop coefficients for the initial growth stage for agricultural non-pristine and natural vegetation is identical to that described for agricultural crops in Chapter 6 (single crop coefficient $K_{c\ ini}$) or Chapter 7 (dual crop coefficient, $K_{cb\ ini}$ + K_e). The crop coefficient during this stage is primarily determined by the frequency with which the soil is wetted.

9.1.2 Mid and late season stages

The $K_{c\ act}$ during the mid-season period ($K_{c\ mid\ act}$ and $K_{cb\ mid\ act}$), and to a lesser extent the $K_{c\ act}$ during the late season period, differ from those described in previous chapters 6 and 7. As the ground cover for non-pristine vegetation is often reduced, the $K_{c\ act}$ is affected to a large extent by the frequency of precipitation and/or irrigation and by the amount of leaf area and ground cover. Natural vegetation, mainly when sparse, is affected by the frequency of precipitation and the $K_{c\ act}$ is dependent on the amount of leaf area and ground cover.

Dual crop coefficient approach

The determination of K_c for non-pristine and natural vegetation should ordinarily follow the approach described in Chapter 7 whereby separate transpiration (K_{cb}) and

evaporation (K_e) coefficients are used. The effects of evaporation from the soil surface can be directly estimated as such.

Two procedures that can be used to adjust the basal crop coefficient ($K_{cb\ mid\ act}$) for sparse vegetation are presented in this section. In these approaches, $K_{cb\ mid\ act}$ is estimated either from LAI or from the effective ground cover and height.

After the determination of $K_{cb\ mid\ act}$, the soil evaporation coefficient, K_e, should be determined as described in Chapter 7 to obtain the crop coefficient for the mid-season stage: $K_{c\ mid\ act} = K_{cb\ mid\ act} + K_e$. Even where the estimated $K_{cb\ mid\ act}$ is small, the total $K_{c\ mid\ act}$ ($= K_{cb\ mid\ act} + K_e$) following precipitation may sometimes be as high or higher than the K_c for pristine vegetation due to soil evaporation from among sparse vegetation.

Single crop coefficient approach

When the single crop coefficient K_c of Chapter 6 is used, the average effects of soil wetting are incorporated into a general mean K_c. Some guidelines for the estimation of $K_{c\ act}$ are given in the following sections. The single crop coefficient can also be derived from the adjusted K_{cb} by considering the frequency of soil wetting, i.e., $K_{c\ mid\ act} = K_{cb\ mid\ act} + 0.05$ for infrequent wetting and $K_{c\ mid\ act} = K_{cb\ mid\ act} + 0.10$ for wettings of up to once a week. For more frequent wettings, the dual crop coefficient approach should be used.

Alternatively, the procedure for adjustment for sparse vegetation described below in the next section can be used. Then, $K_{c\ min}$ can be set equal to $K_{c\ ini}$, where $K_{c\ ini}$ is estimated as described in Chapter 6 because $K_{c\ ini}$ incorporates the effects of soil evaporation and, therefore, may serve as a lower limit on the estimate for $K_{c\ mid}$.

9.1.3 Water stress conditions

Where rainfall or irrigation is low, water stress might be induced and the evapotranspiration will drop below the standard crop evapotranspiration, ET_c. The reduction in the value for K_c under conditions of low soil water availability is then determined using the stress coefficient K_s as described in Chapter 8. Similarly, when water stress is induced by salinity, then $K_{c\ act}$ is also determined using the salinity stress coefficient K_s described in Chapter 8 (Equations 8.20 and 8.25).

9.2 MID-SEASON STAGE–ADJUSTMENT FOR SPARSE VEGETATION

9.2.1 Adjustment from simple field observations

As a rough approximation for K_c during the mid-season stage for crops that usually nearly completely shade the soil under pristine conditions, but where cover is reduced due to disease, stress, pests, or planting density, the values for $K_{c\ mid}$ and $K_{cb\ mid}$ can be reduced by a factor depending on the actual vegetation development:

$$K_{c\ mid\ act} = K_{c\ mid} - A_{cm} \tag{9.1}$$

where $K_{c\ mid}$ the K_c from Table 6.1 to 6.5 ($K_{c\ mid}$) or 7.1 to 7.3 ($K_{cb\ mid}$) after adjusting it for climate (Equation 6.16 or 7.2),

$K_{c\ mid\ act}$ the actual K_c or K_{cb} ($K_{c\ mid\ act}$ or $K_{cb\ mid\ act}$),

A_{cm} empirical adjustment factor for mid-season unusual sparse crop cover.

The K_c adjustment using Equation 9.1 does not apply when crops are frequently wetted and soil evaporation increases due to the reduced ground cover. Under these conditions, increased evaporation can offset reduced transpiration from the reduced ground cover and Equation 9.1 should be applied only to K_{cb}.

The adjustment coefficient, A_{cm}, may be estimated from the LAI ratio:

$$A_{cm} = 1 - \left[\frac{LAI_{act}}{LAI_{dense}}\right]^{0.5} \quad (9.2)$$

where LAI_{act} is the field observed actual leaf area index (Box 9.1) and LAI_{dense} is the leaf area index expected for the same crop under normal dense conditions under standard crop management practices. The values for LAI in Equation 9.2 can be replaced by values for the ground cover fraction (f_c), actual and dense:

$$A_{cm} = 1 - \left[\frac{f_{c\ act}}{f_{c\ dense}}\right]^{0.5} \quad (9.3)$$

Example 9.1 shows an applied calculation.

EXAMPLE 9.1

First approximation of the crop coefficient for the mid-season stage for sparse vegetation

A tomato crop was grown at Davis, California, United States and only developed 50 percent ground cover during the midseason period. The height of the tomato crop was 0.55 m. The typical percentage of ground cover for tomatoes at the effective full cover at Davis is 85 to 90 percent and corresponds to the $K_{cb\ mid}$ listed in Table 7.1 for tomato. The mean values for wind speed and minimum relative humidity during the midseason period were u_2 = 1.1 m s^{-1} and RH_{min} = 30%. The latitude at Davis is 38.5 °N and the midpoint of the midseason occurs on July 20. What is an adjusted $K_{cb\ mid}$ for tomato that reflects the 50 percent ground cover condition?

From Tables 6.1 and 7.1 $K_{c\ mid}$ = 1.10 and $K_{cb\ mid}$ = 1.05

Following adjustments for climate (Eq. 6.16 and Eq. 7.2) where u_2 = 1.1 m s^{-1}, RH_{min} = 30% and mean crop height = 0.55 m,

$K_c = K_{c,Table} + [0.04(1.1 - 2) - 0.004(30 - 45)]\,(0.55/3)^{0.30} = K_{c,Table} + 0.01 =$
$K_{c\ mid}$ = 1.11 and $K_{cb\ mid}$ = 1.06.

The ground cover fraction implied in the tabulated values for tomatoes grown under pristine conditions is about 90 percent ($f_{c\ dense}$ = 0.90). For a sparse tomato crop where f_c = 0.50

From Eq. 9.3 $A_{cm} = 1 - (0.50/0.90)^{0.5} = 0.25$

The $K_{cb\ mid\ act}$ and $K_{c\ mid\ act}$ for 50 percent ground cover is

(from Eq. 9.1) $K_{cb\ mid\ act} = 1.06 - 0.25 = 0.81$

$K_{c\ mid\ act} = 1.11 - 0.25 = 0.86$

Compare the results with Example 9.4 where a more precise derivation of $K_{cb\ mid\ act}$ is made.

As a first estimate, the crop coefficient is expected to be 22 percent lower than the value under pristine conditions.

9.2.2 Estimation of $K_{cb\ mid}$ from Leaf Area Index (LAI)

Natural vegetation typically has less leaf area or fraction of ground cover than does agricultural vegetation that has been developed for full ground cover and for soil water conditions favouring vigorous growth. This is especially true in semi-arid and arid climates.

BOX 9.1
Measuring and estimating LAI

LAI is defined as the area of leaves per area of underlying ground surface averaged over a large area. Only one side of the leaves is counted. LAI can be measured directly by harvesting all green healthy leaves from vegetation over a measured or prescribed area, for example, 1 m^2 or 10 m^2, and then measuring and summing the areas of individual leaves using photometric methods or by measuring areas of several representative leaves, averaging, and then multiplying by the total number of leaves counted.

In the absence of measurements for LAI_{act}, it can be estimated for sparse, annual vegetation as:

$$LAI_{act} = LAI_{dense} \left[\frac{Population}{Population_{dense}} \right]^a$$

where LAI_{dense} LAI for the particular plant species under normal, "dense" or pristine growing conditions. LAI_{dense} can be obtained from various physiological sources and textbooks on crops and vegetation

Population number of plants per unit area of soil surface under the actual growing conditions [No. m^{-2}]

$Population_{dense}$ number of plants per unit area of soil surface under the "dense" or pristine growing conditions [No. m^{-2}]

a a = 0.5 when population is formed from vigorous growing plants; a = 1 when plants are less vigorous

The 0.5 exponent in the equation simulates the tendency for vegetation to compensate for reduced stand density by increasing the size and total leaf areas for individual plants. Therefore, LAI does not fall in direct proportion to plant population. Under conditions where the plant size does not increase with reduced stand density, the "a" exponent in the equation should be set to 1 (a = 1). These latter conditions may occur where soil fertility is poor or where soil salinity, soil water stress, or waterlogging inhibit both growth and stand density, so that the growth of individual plants is retarded.

The value for $K_{cb\ mid\ act}$ for non-pristine or natural vegetation should be reduced when plant density and/or leaf area are lower than for full cover conditions following the procedures described with Equations 9.1 to 9.3. Where LAI can be measured, or approximated (Box 9.1), a peak $K_{cb\ mid\ act}$ for natural or non-pristine agricultural vegetation can be approximated as:

$$K_{cb\ mid\ act} = K_{c\ min} + (K_{cb\ full\ mid} - K_{c\ min})\ (1 - \exp\ [-0.7\ LAI]) \qquad (9.4)$$

where $K_{cb\ mid\ act}$ estimated basal K_{cb} during the mid-season when plant density and/or leaf area are lower than for full cover conditions,

$K_{cb\,full\,mid}$ estimated basal K_{cb} during the mid-season (at peak plant size or height) for vegetation having full ground cover or LAI > 3 (Equation 9.5),
$K_{c\,min}$ the minimum K_c for bare soil ($K_{c\,min} \approx 0.15 - 0.20$),
LAI actual leaf area index [$m^2\ m^{-2}$].

Equation 9.4 is recommended for annual types of vegetation that are either natural or are in a non-pristine state due to sparse density or the effects on growth of some type of environmental stress. The LAI used in Equation 9.4 should be the "green" LAI representing only healthy leaves that are active in vapour transfer.

Mid-season $K_{cb\,full}$ ($K_{cb\,full\,mid}$) may be estimated as:

$$K_{cb\,full} = K_{cb,\,h} + [0.04(u_2 - 2) - 0.004(RH_{min} - 45)] \left(\frac{h}{3}\right)^{0.3} \quad (9.5)$$

where $K_{cb,\,h}$ $K_{cb\,mid}$ for full cover vegetation (LAI > 3) under sub-humid and calm wind conditions (RH_{min} = 45% and u_2 = 2 m s^{-1}), (Equation 9.6),
u_2 mean value for wind speed at 2 m height during the mid-season [m s^{-1}],
RH_{min} mean value for minimum daily relative humidity during the mid-season [%],
h mean maximum plant height [m].

The value for $K_{cb,\,h}$ is estimated, for h ≤ 2 m, as:

$$K_{cb,\,h} = 1.0 + 0.1\ h \quad (9.6)$$

where $K_{cb,\,h}$ is limited to ≤ 1.20 when h > 2 m.

The value of $K_{cb,\,h}$ = 1.20 represents a general upper limit on $K_{cb\,mid}$ for tall vegetation having full ground cover and LAI > 3 under the sub-humid and calm wind conditions. This limit of 1.20 is adjusted for other climatic conditions in Equation 9.5. Equations 9.5 and 9.6 produce a general approximation for the increase in $K_{cb\,full\,mid}$ with plant height and climate. The form of these equations adheres to trends represented in Equation 7.2.

9.3 ESTIMATING $ET_{c\,act}$ FROM ACTUAL TO MAXIMUM CROP YIELD RATIO

A simple, linear crop-water production function was adopted in the FAO Irrigation and Drainage Paper No. 33 to predict the reduction in crop yield when crop stress is caused by a shortage of soil water. This function, introduced in Chapter 8, is:

$$\left(1 - \frac{Y_{act}}{Y_{max}}\right) = K_y \left(1 - \frac{ET_{c\,act}}{ET_c}\right) \quad (8.18bis)$$

where Y_{act} actual yield of the crop [kg ha^{-1}] when ET is $ET_{c\,act}$,
Y_{max} maximum (expected) yield in the absence of environmental or water stresses [kg ha^{-1}] when ET is ET_c,
K_y yield response factor [ ],
ET_c potential (expected) seasonal crop evapotranspiration in the absence of water or environmental stresses ($K_c\ ET_o$) [mm],

$ET_{c\ act}$ actual seasonal crop evapotranspiration as impacted by environmental or water stresses [mm]

Values for K_y are reported in Table 8.7 for a wide range of crops for estimating the effect of water stress and associated reduction in $ET_{c\ act}$ on crop yield. Factors are presented there for predicting yield reductions when stress is evenly distributed throughout the growing period.

Many environmental stresses, such as water shortage, salinity, low fertility and diseases, impact yield by reducing the amount of $ET_{c\ act}$ relative to the potential amount ET_c. The same can be true when yields are reduced due to the use of low densities for plant populations as referred to before. Thus, for very general estimates of $ET_{c\ act}$, one can invert Equation 8.18bis and solve for the stress factor, K_s:

$$K_s = 1 - \frac{1}{K_y}\left(1 - \frac{Y_{act}}{Y_{max}}\right) \tag{9.7}$$

K_s is multiplied by K_{cb} or by K_c in Equation 8.1 or 8.2 to, respectively, predict $T_{c\ act}$ or $ET_{c\ act}$ in the presence of water stress, or other environmental stresses, or for low plant populations, or poor growth and yield conditions. The $ET_{c\ act}$ predicted using K_s from Equation 9.7 provides only a very general and approximate estimate of monthly (or even seasonal) evapotranspiration, $ET_{c\ act} = K_s\ K_c\ ET_o$. Equation 9.7 works best for forage or other indeterminate crops where the value for K_y is relatively constant during the season.

Equation 9.7 is generally only valid for estimating actual crop evapotranspiration for use in regional water balance studies, for studies of groundwater depletion and recharge, or for estimating historical water use. The procedure is not valid for estimating ET_c for daily or weekly time periods due to the very general nature of the K_y coefficient and the seasonal time scale of the crop yield. The procedures presented previously for adjusting ET_c using a daily soil water balance, salinity functions, or reductions in K_c based on leaf area or fraction of ground cover are recommended over the use of Equation 9.7.

EXAMPLE 9.2
Approximate estimation of K_s from crop yield data

An irrigation scheme (project) cultivates dry, edible beans. There is known to be a shortage of irrigation water and a corresponding reduction in crop yield. The reported yield for the scheme averages 1100 kg/ha. The potential yield for the region and variety of beans, in the absence of water or environmental stresses and with good soil fertility is 1800 kg/ha.

From FAO Irrigation and Drainage Paper No. 33 or Table 8.7, the K_y for dry beans, assuming that stresses are distributed uniformly through the growing season, is 1.15. Therefore, from Equation 9.7, the estimated K_s to apply with Equation 8.1 for the growing season is:

$$K_s = 1 - \frac{1}{1.15}\left(1 - \frac{1100}{1800}\right) = 0.66$$

Therefore, the $ET_{c\ act}$ for the season is estimated to be only 0.66 of the ET_c representing pristine growing and yielding conditions.

The seasonal $ET_{c\ act}$ is estimated to be $ET_{c\ act} = 0.66\ ET_c$ where ET_c is estimated as $ET_c = K_c\ ET_o$.

9.4 ESTIMATION OF CROP COEFFICIENTS FROM GROUND COVER AND HEIGHT

A popular and generally successful method for estimating K_c is to base the estimate on the fraction of ground covered by vegetation and the height of the vegetation. These two parameters are the primary characteristics of vegetation that control the rate of $ET_{c\,act}$ in terms of its ratio to the grass reference ET_o. This method, referred to as the Allen and Pereira (A&P) approach, is explained hereafter in Equations 9.8-9.14.

9.4.1 Concepts

For expanses of vegetation large enough that an equilibrium boundary layer is established, so that general one-dimensional equations such as the Penman-Monteith apply, a maximum upper limit on ET is established according to the law of conservation of energy. Thus, for expanses of vegetation larger than about 500 to 2 000 m², the K_c will have upper limits of 1.2-1.3 for the grass reference. However, under conditions of "clothesline effects" (where vegetation height exceeds that of the surroundings) or "oasis effects" (where vegetation has higher soil water availability than the surroundings), peak K_c may attain larger values. Caution is required when extrapolating ET measurements from small plots to large stands because the resulting ET may be overestimated.

An upper energy-constrained limit on K_c is the maximum value for K_c ($K_{c\,max}$) following rain or irrigation, which is governed by the amount of energy available for evaporation. The $K_{c\,max}$ varies with climate, ranging from about 1.05 to 1.30:

$$K_{c\,max} = \max\left(\left\{1.2 + [\,0.04\,(u_2 - 2) - 0.004\,(RH_{min} - 45)]\left(\frac{h}{3}\right)^{0.3}\right\}, \{K_{cb\,(Table)} + 0.05\}\right) \quad (9.8)$$

where u_2 mean daily wind speed at 2 m height during the growth stage [m s^{-1}],
RH_{min} mean daily minimum relative humidity during the growth stage [%],
$K_{cb\,(Table)}$ standard value for K_{cb} for the standardized climate (Tables 7.1 to 7.3),
h mean plant height (m) during the considered growth stage (initial, development, midseason, or late-season).

The basal K_{cb}, because it represents mostly transpiration, depends upon the amount of vegetation and can be expressed as a function of a density coefficient, K_d, as:

$$K_{cb} = K_{c\,min} + K_d\,(K_{cb\,full} - K_{c\,min}) \quad (9.9)$$

where K_{cb} approximate value of K_{cb} for the considered vegetation,
K_d density coefficient for the considered vegetation,
$K_{cb\,full}$ estimated basal K_{cb} at peak plant growth close to full ground cover,
$K_{c\,min}$ minimum basal K_{cb} for bare soil (0.15 under typical agricultural conditions where some subsurface soil moisture exists).

The approach of Equation 9.9 may be similarly applied to estimate a single K_c coefficient.

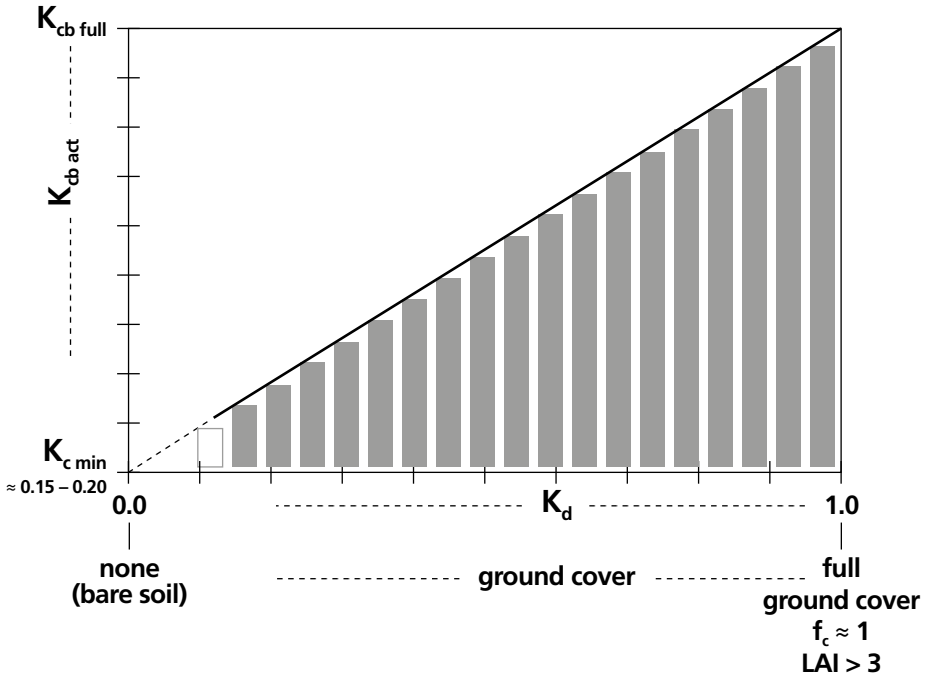
For tree crops, having a ground cover crop of grass or other, Equation 9.9 must consider the transpiration by the active ground cover ($K_{cb\,cover}$), thus becoming:

$$K_{cb} = K_{cb\,cover} + K_d\left(\max\left[K_{cb\,full} - K_{cb\,cover}, \frac{K_{cb\,full} - K_{cb\,cover}}{2}\right]\right) \quad (9.10)$$

where $K_{cb\,cover}$ is the K_{cb} of the active ground cover in the absence of tree foliage.

A simplified scheme is shown in Figure 9.1 that shows K_d growing with f_c.

FIGURE 9.1
The relationship between the mid-season basal crop coefficient ($K_{cb\ act}$) and the density coefficient for vegetation (K_d)



Source: Authors' own elaboration

The second term of the max function reduces the estimate for $K_{cb\ mid}$ by half the difference $K_{cb\ full} - K_{cb\ cover}$ when this difference is negative. This accounts for impacts of shading by vegetation when $K_{cb} < K_{cb\ cover}$ due to larger stomatal conductance of the ground cover. Equation 9.9 applies to estimating K_{cb} during any period, namely for any crop growth stage.

$K_{cb\ full}$ can be computed as a function of mean plant height and adjusted for climate:

$$K_{cb\ full} = F_r\left(\min\left(1.0 + k_h\, h,\ 1.20\right) + \left[0.04\,(u_2 - 2) - 0.004\,(RH_{min} - 45)\right]\left(\frac{h}{3}\right)^{0.3}\right) \qquad (9.11)$$

where h, u_2 and RH_{min} are defined in Equation 9.8 but herein refer to the mid-season,

F_r downward adjustment to $K_{cb\ full}$ ($F_r \leq 1.0$) for impacts of reduced stomatal conductance on transpiration,

k_h adjustment to the crop height,

$K_{cb\ full}$ general upper limit on $K_{cb\ mid}$.

Effects of crop height are considered through the sum $(1+k_h\, h)$, with $k_h = 0.1$ for tree and vine crops as well as tall field crops, and $k_h = 0.2$ for short crops and vegetables as described in the following sections. The value for $K_{cb\ full}$ represents a general upper limit on $K_{cb\ mid}$ for vegetation having full ground cover and LAI > 3 under full water supply.

Equation 9.11 suggests that an upper bound for $K_{cb\ full}$ is 1.20 plus adjustment for a nonstandard climate. Equation 9.11 produces increases in $K_{cb\ full}$ with plant height and with more arid climates. For small, isolated stand sizes, $K_{cb\ full}$ may need to be increased beyond the value given by Equation 9.11 due to the clothesline effect.

The parameter F_r applies a downward adjustment ($F_r \leq 1.0$) to $K_{cb\ full}$ and, consequently, to K_{cb}, if the vegetation exhibits more stomatal control on transpiration than is typical of agricultural crops. Examples of F_r are presented in the following sections. For the period when $K_{cb\ full}$ is computed, F_r may be estimated as:

$$F_r = \frac{\Delta + \gamma\,(1 + 0.34\,u_2)}{\Delta + \gamma\left(1 + 0.34\,u_2\,\frac{r_l}{r_{typ}}\right)} \tag{9.12}$$

where r_l mean leaf resistance for the vegetation under consideration [s m^{-1}],

r_{typ} typical leaf resistance representing pristine agricultural vegetation under well-watered conditions [s m^{-1}],

Δ slope of the saturation vapor pressure vs. air temperature curve [kPa $°C^{-1}$],

γ psychrometric constant [kPa $°C^{-1}$].

This factor, F_r, is multiplied against the estimate for $K_{cb\ full}$ in Equation 9.11 to adjust its value downward when the vegetation type adopts more stomatal control on transpiration than is typical. The adjustment using F_r is particularly important in the case of perennials and the late season. The standard value for F_r is 1.0 for most annual agricultural crops under pristine conditions, whose r_l value is often approximately 100 s m^{-1}. The value for r_l is limited to ≥ 100 s m^{-1}.

The density coefficient K_d used in Equations 9.9 and 9.10 describes the increase in K_{cb} with an increase in the amount of vegetation. Where estimates of the fraction of ground cover, f_c, are available, the K_d is estimated as:

$$K_d = \min\left(1,\ M_L\,f_{c\,eff},\ f_{c\,eff}^{\left(\frac{1}{1+h}\right)}\right) \tag{9.13}$$

where $f_{c\,eff}$ effective fraction of ground covered or shaded by vegetation [0.01–1],

M_L multiplier on $f_{c\,eff}$ relative to the canopy density or shading [1.0 – 2.0],

h mean height of the vegetation (m).

$f_{c\,eff}$ is often observed as the shaded area near noon, between 11.00 and 15.00. Estimation of $f_{c\,eff}$ for row crops is described in Box 9.2. $f_{c\,eff}$ is often determined from visual inspection, mainly using photographic equipment. Vegetation indices, such as the normalized difference vegetation index (NDVI), can also be used to estimate $f_{c\,eff}$ by remote sensing.

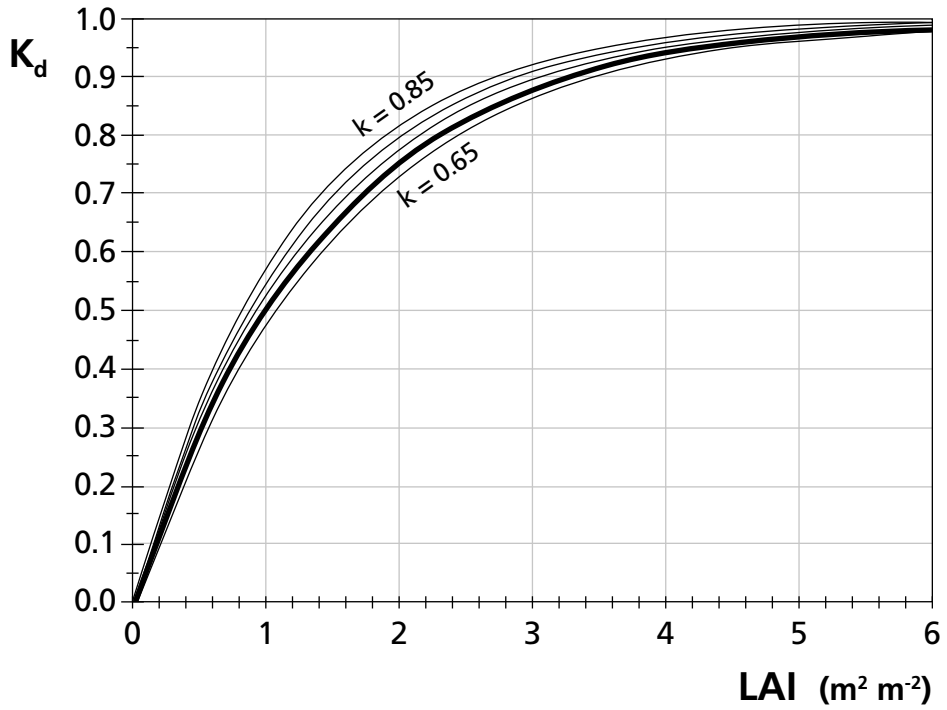
The M_L multiplier on $f_{c\,eff}$ in Equation 9.13 imposes an upper limit on the relative magnitude of transpiration per unit of ground area as represented by $f_{c\,eff}$ and is expected to mostly range from 1.0 to 2.0, depending on the canopy density and thickness. M_L may have a low value when the canopy porosity is high. The value for M_L can be modified to fit the specific vegetation and experimental data. M_L values for numerous crops are tabulated hereafter in the current chapter.

Where f_c is not available, but LAI is observed or estimated, K_d may be approximated as:

$$K_d = (1 - e^{[-0.7\ LAI]}) \quad (9.14)$$

where LAI is defined as the area of leaves per area of ground surface averaged over a large area ($m^2\ m^{-2}$) with consideration of only one side of "green", healthy leaves active in vapor transfer. The extinction coefficient k = 0.7 may change with the crop (Figure 9.2).

FIGURE 9.2
Relationship between the density coefficient (K_d) and the Leaf Area Index (LAI) for various extinction coefficients (k), thus for different crops



Source: Authors' own elaboration

BOX 9.2

Measuring and estimating $f_{c\ eff}$ for row crops

$f_{c\,eff}$ should normally represent the fraction of the soil surface that is shaded by vegetation. This value is generally larger than f_c, the actual fraction of the soil surface that is covered by vegetation as observed from directly overhead. The amount of shading represents the amount of solar radiation intercepted by plants for potential conversion into evapotranspiration. The total fraction of shading is a function of the sun angle and the horizontal size and shape of individual plants (or rows) relative to their height.

$f_{c\ eff}$ for "rectangular" shaped canopies such as most agricultural plant rows can be approximated as:

$$f_{c\,eff} = f_c \left[1 + \frac{HWR}{\tan(\eta)} \right] \leq 1$$

where f_c observed fraction of soil surface that is covered by vegetation as observed from nadir (overhead) [0.01–1],

HWR height to width ratio of individual plants or groups of plants when viewed from the east or from the west [ ],

tan(η) tangent of the mean angle of the sun, η, above the horizon during the period of maximum evapotranspiration (generally between 11.00 and 15.00 hours) [ ]. For most applications, η can be computed at solar noon (12.00 hours).

HWR is computed as:

$$HWR = \frac{h_{canopy}(\cos(\Gamma))}{Width}$$

where h_{canopy} mean vertical height of the canopy area of the plant [m],

Width mean width of the canopy of a plant or group of plants (e.g., row) [m],

Γ angle of plant row from east-west direction [rad] (for east-west rows, Γ = 0; for north-south rows, Γ = π/2)

For north-south rows, the HWR would be zero, as cos(π/2) = 0. This implies that rows of plants that run from north to south would have $f_{c\ eff} \approx f_c$ as all soil is exposed to the sun at various times of the day and as the shaded area is the same as the fraction of vegetation cover at midday.

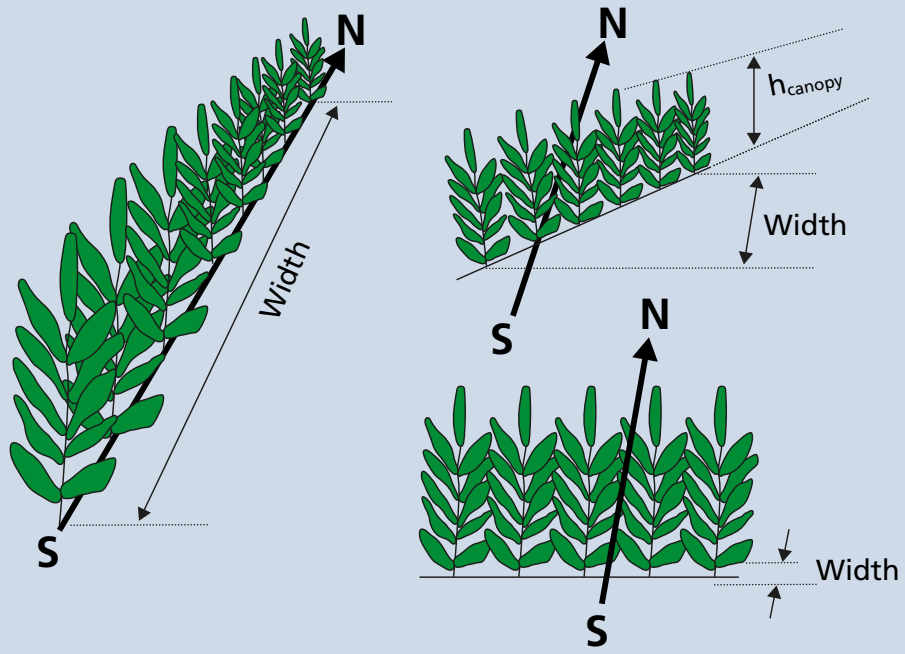
For trees or vegetation that do not have canopies that extend to the ground, h_{canopy} does not include the lower trunk length, but only the active canopy. Therefore, in these situations, $h_{canopy} < h$ where h is mean plant height.

For round or spherical shaped canopies such as trees, $f_{c\,eff}$ can be estimated as:

$$f_{c\,eff} = \frac{f_c}{\sin(\eta)} \leq 1$$

where sin(η) is the sine of the mean angle of the sun, η, above the horizon during the period of maximum evapotranspiration (generally between 11.00 and 15.00).

BOX 9.2 (CONT.)



Mean angle of the sun above the horizon during the period of maximum evapotranspiration (η)

The sine of η can be calculated for any specific time of day as:

$$\sin(\eta) = \sin(\varphi)\sin(\delta) + \cos(\varphi)\cos(\delta)\cos(\omega)$$

where φ latitude [rad] (negative for southern latitudes)

δ solar declination [rad] (Eq. 3.21)

ω solar time angle [rad] (Eq. 3.28)

Generally, $f_{c\ eff}$ can be calculated at solar noon (12.00), so that $\omega = 0$ and the above equation reduces to:

$$\sin(\eta) = \sin(\varphi)\sin(\delta) + \cos(\varphi)\cos(\delta)$$

The value for η can be obtained by taking the arcsine of the above equation.

9.4.2 Application

Examples of application of the computations presented above are given in the following: (a) the Example 9.3 refers to the estimation of the mid-season crop coefficient for a 2 m tall crop having low vegetation cover, $f_c = 0.30$; (b) the Example 9.4 relative to an application of the A&P approach for estimation of the mid-season actual crop coefficient of tomato in conditions of reduced ground cover; and (c) the Example 9.5 relative to an application of the A&P approach to compute the mid-season actual $K_{cb\ mid}$ for a traditional olive orchard from ground cover and height and considering effects of stomatal control.

EXAMPLE 9.3
Estimation of mid-season actual crop coefficient for a vegetation strip

Estimate $K_{cb\ mid}$ and $K_{c\ mid}$ for rectangular shaped 2 m tall vegetation that is as tall as it is wide, where 30 percent of the ground is covered by vegetation (f_c = 0.3) on 19 July (day 200 of the year) and at latitude 40°N. RH_{min} = 55% and u_2 = 1.5 m s^{-1} during the mid-season period.

On day J = 200 at latitude φ = 40 (π/180) = 0.70 radians (40°N), from Eq. 3.21, the solar declination δ = 0.36 radians.

At solar noon (ω = 0): $\sin(\eta) = \sin(\varphi)\sin(\delta) + \cos(\varphi)\cos(\delta) = 0.94$

The value for η by taking the arcsine of the above value is 1.24 radians and $\tan(\eta)$ = 2.8. If f_c = 0.3 and the HWR for the vegetation is 1, then $f_{c\ eff}$ from Box 9.2 is: 0.3(1+1/2.8) = 0.41.

From Eq. 9.6 $K_{cb,\ h} = 1.0 + 0.1(2) = 1.20$ (≤ 1.20, so OK)

From Eq. 9.5 $K_{cb\ full} = 1.20 + (0.04(1.5 - 2) - 0.004(55 - 45))\ (2/3)^{0.3} = 1.15$

Therefore, $K_{cb\ mid}$ estimated by Eq. 9.9 for $K_{c\ min}$ = 0.15 is

$K_{cb\ mid} = K_{c\ min} + (K_{cb\ full} - K_{c\ min})\ \min\ [1, M_L\ f_c, (f_{c\ eff})^{(1/(1\ +\ h))}]$, and for M_L = 2.0, then is

$K_{cb\ mid} = 0.15 + (1.15 - 0.15)\ \min[1, 2(0.3), (0.41)^{(1/(1+2))}] = 0.75$

This value does not need any further adjustment for climate.

$K_{c\ mid}$ (where $K_{c\ mid}$ includes average wetting effects) can be derived from $K_{cb\ mid}$ using the guidelines presented in the calculation procedures at the beginning of this chapter:

$K_{c\ mid} = K_{cb\ mid} + (0.05 \ldots 0.10) = 0.80 \ldots 0.85$

depending on the frequency of soil wetting.

The estimated crop coefficients for the mid-season stage are $K_{cb\ mid}$ = 0.75 and $K_{c\ mid}$ = 0.80 to 0.85.

EXAMPLE 9.4
Estimation of mid-season actual crop coefficient for tomato with reduced ground cover

A more precise estimate for $K_{cb\ mid}$ for the tomato crop of Example 9.1 that only developed 50% ground cover at Davis, California, the United States can be calculated if one knows that the tomato crop was planted in 1.52 m rows running east-west, that the crop reached a plant height of 0.75 m and that the height to width ratio of the tomato crop can be estimated as about 1.0 for the east-west rows. The latitude is 38.5°N and the midpoint of the mid-season occurs on July 20.

What is the adjusted $K_{cb\ mid}$ for tomatoes that reflects the 50% ground cover condition?

On day J = 201 (20 July) at latitude φ = 38.5 (π/180) = 0.67 radians (38.5°N), the solar declination (Eq. 3.21) is δ = 0.36 radians. At solar noon (ω = 0):

$\sin(\eta) = \sin(\varphi)\sin(\delta) + \cos(\varphi)\cos(\delta) = 0.95$

EXAMPLE 9.4 (CONT.)

The value for the angle η by taking the arcsine of the above value is 1.26 radians. Therefore, for the observed HWR = 1 and f_c = 0.5, the effective soil cover for the east-west rows can be calculated as (Box 9.2):

$$f_{c\ eff} = 0.5\ [1 + 1/\tan(1.26)] = 0.66$$

The $K_{cb\ mid}$ representing a full cover tomato crop may be estimated using Eq. 9.11 considering an average h for fully developed tomatoes (this variety) of about 0.55 m, a k_h of 0.2 and a F_r of 1.0. It results

$$K_{cb\ full} = 1.0 \left(\min (1.0 + 0.2\ (0.55), 1.20) + [0.04\ (1.1 - 2) - 0.004\ (30 - 45)] \left(\frac{0.55}{3} \right)^{0.3} \right) = 1.12$$

From Eq. 9.13 and using a M_L = 2.0, the density coefficient is:

$$K_d = \min \left(1, 2\ (0.66), 0.66^{1/(1+0.55)} \right) = 0.76$$

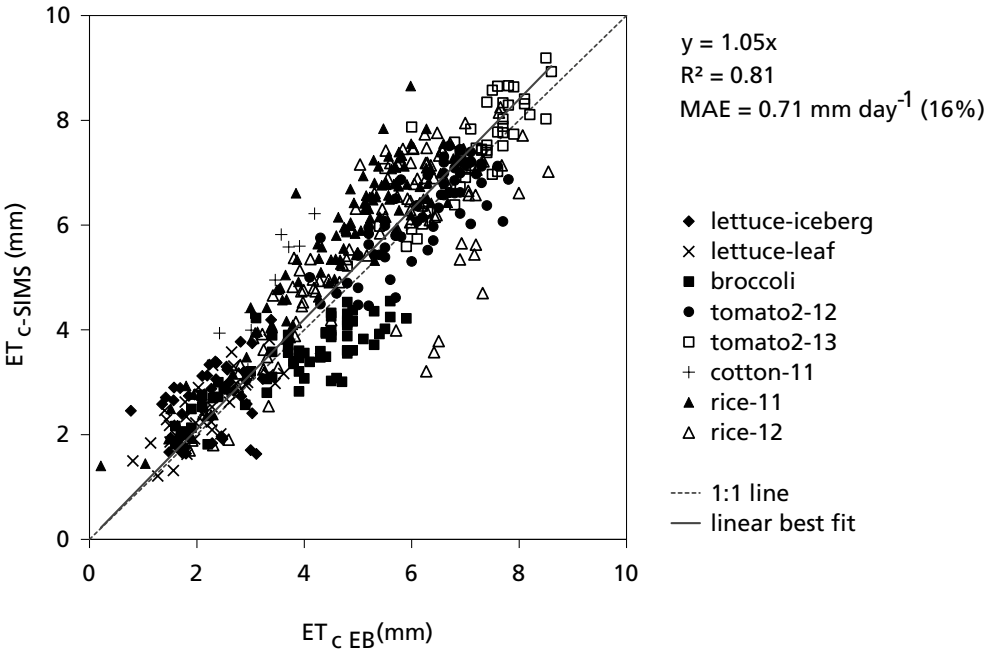
From Eq. 9.9 and using $K_{c\ min}$ = 0.15, the actual $K_{cb\ mid}$ for 50% ground cover would be:

$$K_{cb\ mid\ act} = 0.15 + 0.76\ (1.12 - 0.15) = 0.90$$

The results in $K_{cb\ mid\ act}$ = 0.90 for 50 percent ground cover, which is in line to the first estimate calculated in Example 9.1 and compares with the measured $K_{cb\ mid}$ near 0.90 as determined by precision lysimeter for the same location.

FIGURE 9.3

Comparing the actual crop evapotranspiration measured with the energy balance ($ET_{c\ EB}$) with the actual ET_c derived from remote sensing with the SIMS model ($ET_{c\text{-}SIMS}$) that uses ground cover estimates from remote sensing, with application to California, USA



Source: Authors' own elaboration

EXAMPLE 9.5
Estimation of actual $K_{cb\ mid}$ from ground cover and height considering effects of stomatal control

A traditional, non-irrigated orchard of olive (*Olea europaea*) trees has a tree spacing of 10 m. The horizontal diameter of the trees as viewed from overhead is 5 m. The tree height is 5 m. The lower 1.5 m of the trees have no foliage. The ground cover between the trees is bare. The mean u_2 during the mid-season growth stage is 2 m s^{-1} and the mean RH_{min} = 25%. The midpoint of the mid-season growth stage is on 29 June (i.e. day 180 of the year). The latitude of the location is 30°N.

On day J = 180 (29 June) at latitude ϕ = 30 (π/180) = 0.52 radians (30°N) and from Eq. 3.21 the solar declination δ = 0.405 radians. At solar noon (ω = 0):

$$\sin(\eta) = \sin(\phi)\sin(\delta) + \cos(\phi)\cos(\delta) = 0.99$$

As olive trees have somewhat round shapes, the effective fraction of ground cover (Box 9.2) can be estimated as $f_{c\ eff} = f_c/(\sin(\eta))$.

f_c = area of canopy/area tree spacing = (π $(5)^2$/4) / ((10) (10)) = 0.196

$f_{c\ eff}$ = 0.196/0.99 = 0.20

Assuming that olives have r_l of about 420 s m^{-1} and that r_{typ} equals 100 s m^{-1}, if average T_{mean} = 25 °C and that the elevation of the grove is 0 m (sea level), so that Δ = 0.189 kPa (Eq. 3.8) and γ = 0.0676 kPa (Eq. 3.3), F_r is estimated from Eq. 9.12 resulting:

$$F_r \approx \frac{0.189 + 0.0676\,(1 + 0.34\,(2))}{0.189 + 0.0676\,(1 + 0.34\,(2)(420/100))} = 0.67$$

From Eq. 9.13 and using M_L = 2.0:K_d=min(1, 2.0(0.2), $0.2^{(1/6)}$)=0.40

From Eq. 9.11 and using a k_h= 0.1:

$$K_{cb\ full}=0.67\left(\min(1.0 + 0.1(5), 1.20)+[0.04(2 - 2) - 0.004(25 - 45)]\left(\frac{5}{3}\right)^{0.3}\right) = 0.80$$

From Eq. 9.9 and using $K_{c\ min}$ = 0.15, results in $K_{cb\ mid\ act}$ = 0.15 + (0.74 – 0.15) 0.4) = 0.41

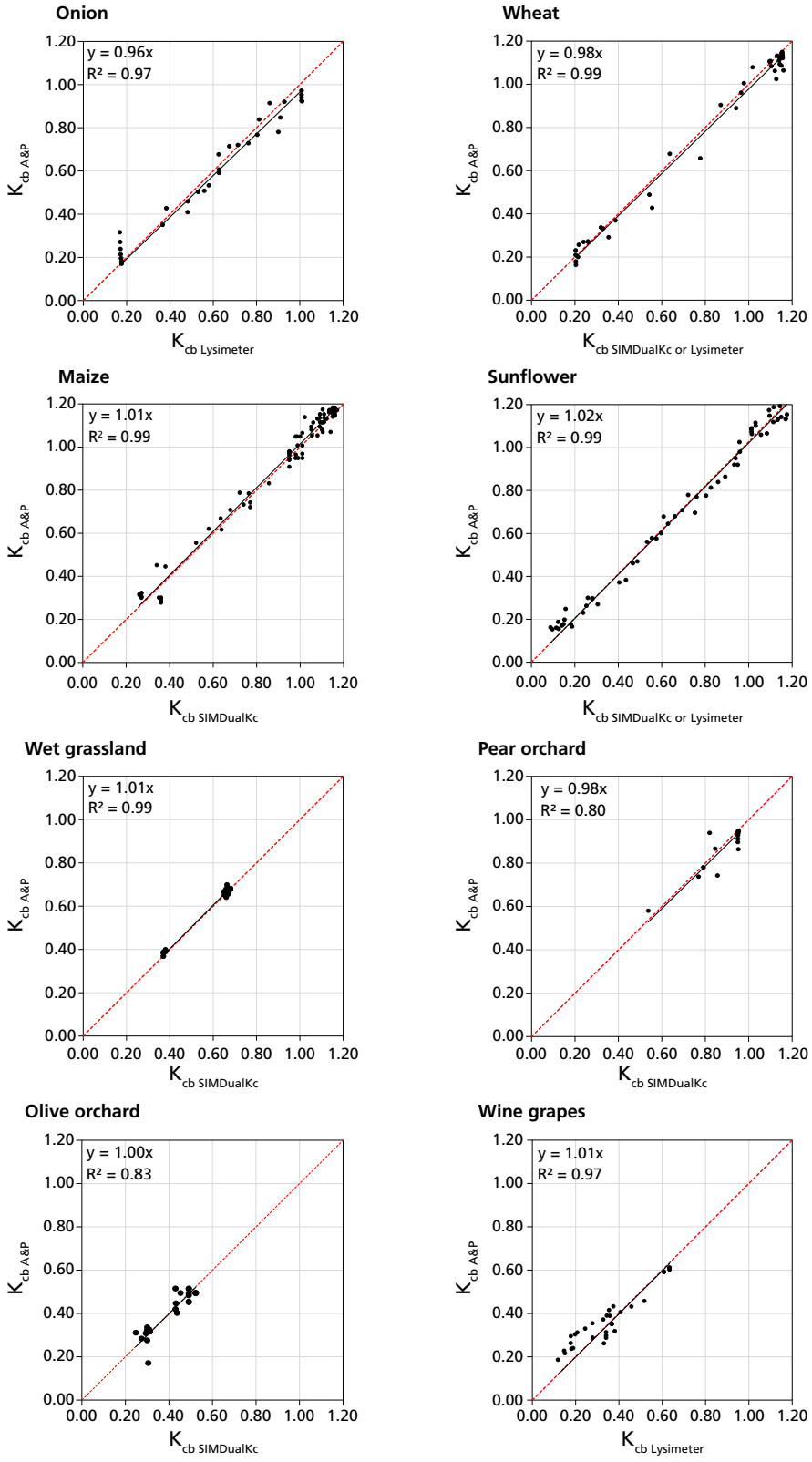
The value $K_{cb\ mid\ act}$= 0.41 estimated for f_c = 0.20 is higher than the value for $K_{cb\ mid}$ in Table 7.3 for olives with f_c = 0.20 to 0.40 and h = 3 to 4.5. The value from Table 7.3 is 0.35 which after adjustment for climate using Eq. 7.2 equals 0.44.

In addition to the examples above, Figure 9.3 presents a comparison of the actual ET_c measured with energy balance ($ET_{c\ EB}$) instrumentation with the actual ET_c derived from remote sensing with the SIMS model ($ET_{c\text{-}SIMS}$). The SIMS model uses ground cover and height estimates following the A&P approach, and is used to support irrigation scheduling in California, USA.

Figure 9.4 reports on tests of the A&P approach for various vegetable, field and grassland crops, as well as tree and vine crops. Tests consist of comparing field observed $K_{cb\ act}$ values obtained from measurements with a weighing lysimeter, with soil water content sensors, with sap-flow and/or with eddy covariance with $K_{cb\ act}$ estimated by performing the soil water balance with the model SIMDualKc, with $K_{cb\ A\&P}$ computed with the observed ground cover and vegetation height. All results show good agreement with measurements and with regression coefficients for regression forced through the origin close to 1.0, thus indicating the appropriateness of the approach.

FIGURE 9.4

Observed $K_{cb\ act}$ obtained from lysimeter measurements, or from soil water balance, sap-flow or eddy covariance measurements using the model SIMDualKc, compared with $K_{cb\ A\&P}$ computed with the A&P approach using the measured fraction of ground cover and crop height, for applications to various countries



Source: Authors' own elaboration

9.4.3 Parameterization of the ground cover fraction and height approach

The application of the A&P approach requires that parameters f_c, fraction of ground cover, and h, crop height, are observed or estimated from remote sensing, and that appropriate values for M_L, representing the intensity of canopy shading and limitations on root and stem fluxes, and F_r, the effect of increased stomatal control, are adequately selected. The parameter F_r is used in the calculation of $K_{cb\,full}$ (Eqs. 9.11 and 9.12) and M_L is used in the computation of the density coefficient K_d (Equation 9.13). Both K_d and $K_{cb\,full}$ are used when computing K_{cb} for any crop or natural vegetation, with or without active ground cover (Equations 9.9 and 9.10). It has been shown that F_r plays a primary role in the estimation of K_{cb} and K_c, particularly for trees and vines that exercise substantial control over stomatal conductance, even under well-watered conditions.

To support the application of the A&P approach for most applications to compute $K_{cb\,A\&P}$, a selection of F_r and M_L values has been performed through a numerical search of parameter values that tend to cause $K_{cb\,A\&P}$ to match the tabulated K_{cb} values in Chapter 7 ($K_{cb\,TAB}$). The procedure consists of:

1. Selecting initial values for F_r [0.25 to 1.0] and M_L [1.00 to 2.00]. F_r is likely to be close to 1.0 when the crop has young and very green leaves, particularly annual crops, and close to 0.50 during the late season of deciduous woody fruit crops and annual crops that are harvested dry. M_L is close to 1.00 under dry crop conditions and close to 2.0 when the crop is not stressed; in addition, M_L is close to 2.0 when the canopy is less transparent to radiation. Alternatively, the parameters in Tables 9.1 to 9.4 are considered.
2. Computing crop $K_{cb\,A\&P}$ with observed/known values of $f_{c\,eff}$ and h and using the selected initial values for F_r and M_L, and then comparing the resulting $K_{cb\,A\&P}$ with the target value, which is commonly the standard tabulated value ($K_{cb\,target} = K_{cb\,TAB}$) for the same crop (Tables 7.1 to 7.3), or with a $K_{cb\,target} < K_{cb\,TAB}$ when the crop is known to be cultivated somewhat far from pristine conditions.
3. Repeating the computations using an iterative numerical search where the initial F_r values are progressively changed, but constrained by the limits of F_r, while comparing the successive values of $K_{cb\,A\&P}$ with $K_{cb\,target}$ values for the same crop, progressively decreasing the differences between both values until differences are negligible. The value for F_r needs to be progressively modified until the computed $K_{cb\,A\&P}$ effectively represents the crop under study.
4. When a F_r value is selected, the calculations are performed with the F_r value obtained in the previous calculation set, but now changing M_L within its selected interval aiming at minimizing the difference between $K_{cb\,A\&P}$ and $K_{cb\,target}$.
5. If increasing or decreasing M_L leads to increases in the difference between $K_{cb\,A\&P}$ and $K_{cb\,TAB}$, then the last value for M_L is retained, as well as the F_r used in the calculation; however, if the difference decreases, a new run must be performed until an M_L value is found that minimizes that difference. The value selected for F_r in Step 3 remains fixed throughout Steps 4 and 5, but it may be required to reapply the numerical search, focusing on F_r.
6. The procedure ends when the difference $K_{cb\,A\&P} - K_{cb\,target}$ is minimized and values for both M_L and F_r selected are within the respective intervals. These values can then be used with Eq. 9.9, 9.11, and 9.13 to calculate values for $K_{cb\,mid\,act}$ for any fraction of cover.

Tables 9.1 to 9.4 propose values for the parameters of the A&P approach. Tables 9.1 and 9.2 present M_L and F_r that are able to produce $K_{cb\,A\&P} = K_{cb\,TAB}$ for vegetable and field crops for mid-season and end-season, i.e., for most of the crop seasons. Table 9.3 presents M_L and F_r for $K_{cb\,A\&P}$ for fruit trees, shrubs and vines targeting $K_{cb\,TAB} \pm 5\%$. Table 9.4 refers to grasses. Since K_{cb} was not tabulated for grasses, the $K_{cb\,TAB}$ was estimated as $K_{cb} = K_c - 0.05$ as defined in Table 7.4 and with K_c values tabulated in Table 6.4.

TABLE 9.1

Parameters for the A&P approach aimed at estimating the standard K_{cb} for vegetable crops with presentation of the respective K_{cb} values

Crop Degree of ground cover or field system, training, and plant density	Crop stages	Observation parameters		Calibrated parameters		Standard K_{cb}
		$f_{c\ eff}$	h (m)	M_L	F_r	
a. Roots, tubers, bulbs, and stem vegetable crops						
a.1. Asparagus *(Asparagus officinalis)*	Mid	0.90	0.80	2.00	0.75	0.80
	End	0.20	0.25	2.00	0.50	0.20
a.2. Carrots *(Daucus carota)*	Mid	0.90	0.30	2.00	0.94	0.90
	End	0.90	0.30	2.00	0.78	0.75
a.3. Celeriac *(Apium graveolens)*	Mid	0.80	0.60	2.00	1.00	1.00
	End	0.80	0.60	2.00	0.95	0.95
a.4. Garlic *(Allium sativum)*	Mid	0.85	0.50	2.00	1.00	1.00
	End	0.80	0.45	2.00	0.55	0.50
a.5. Onions *(Allium cepa)*						
Dry	Mid	0.85	0.45	2.00	1.00	1.00
	End	0.75	0.40	2.00	0.65	0.60
Green	Mid	0.75	0.45	2.00	1.00	0.95
	End	0.75	0.40	2.00	1.00	0.80
Seed	Mid	0.75	0.65	2.00	1.00	0.95
	End	0.70	0.65	2.00	0.50	0.45
a.6. Parsnip *(Pastinaca sativa)*	Mid	0.85	0.40	2.00	1.00	1.00
	End	0.80	0.40	2.00	1.00	0.95
a.7. Potato *(Solanum tuberosum)*	Mid	0.90	0.60	2.00	1.00	1.05
Long season	End	0.50	0.45	2.00	0.45	0.30
Short season	End	0.55	0.55	2.00	0.70	0.50
a.8. Radish *(Raphanus sativus)*	Mid	0.80	0.30	2.00	1.00	0.90
	End	0.80	0.30	2.00	0.80	0.75
a.9. Rutabaga (*Brassica napobrassica*)	Mid	0.85	0.60	2.00	1.00	1.05
	End	0.85	0.60	1.80	0.90	0.90
a.10. Sweet potato (*Ipomoea batatas*)	Mid	0.95	0.50	2.00	1.00	1.05
	End	0.85	0.50	2.00	0.50	0.50
a.11. Table beets (*Beta vulgaris*)	Mid	0.85	0.40	2.00	1.00	1.00
	End	0.80	0.40	2.00	0.95	0.90
a.12. Turnip (*Brassica rapa* var. *rapa*)	Mid	0.85	0.60	2.00	1.00	1.00
	End	0.85	0.60	2.00	0.93	0.95
b. Leafy and flowers vegetable crops						
b.1. Artichoke *(Cynara scolymus)*	Mid	0.75	0.80	2.00	0.85	0.85
	End	0.75	0.80	2.00	0.80	0.80
b.2. Broccoli (*Brassica oleracea* var. *italica*)	Mid	0.90	0.60	2.00	0.96	1.00
	End	0.90	0.60	2.00	0.96	1.00
b.3. Brussel sprouts (*B. oleracea* v. *gemmifera*)	Mid	0.90	0.75	2.00	0.93	1.00
	End	0.90	0.75	2.00	0.93	1.00
b.4. Cabbage (*B. oleracea* var. *capitata*)	Mid	0.85	0.40	2.00	1.00	1.00
	End	0.85	0.40	2.00	0.94	0.90
b.5. Cauliflower (*B. oleracea* var. *botrytis*)	Mid	0.85	0.40	2.00	1.00	1.00
	End	0.85	0.40	2.00	0.96	0.95
b.6. Lettuce (*Lactuca sativa*)	Mid	0.85	0.35	2.00	1.00	1.00
	End	0.85	0.35	2.00	1.00	1.00
b.7. Spinach (*Spinacia oleracea*)	Mid	0.80	0.35	2.00	1.00	0.95
	End	0.80	0.35	2.00	1.00	0.95

TABLE 9.1 (CONT.)

Crop Degree of ground cover or field system, training, and plant density	Crop stages	Observation parameters		Calibrated parameters		Standard K_{cb}
		$f_{c\,eff}$	h (m)	M_L	F_r	
c. Fruit vegetable crops						
c.1. Bell pepper *(Capsicum annuum)*	Mid	0.90	0.70	2.00	1.00	1.05
	End	0.85	0.70	2.00	0.95	1.00
c.2. Cape gooseberry *(Physalis peruviana)*	Mid	0.15 – 0.30	0.5 – 1.0	1.5	1.00	0.40
1st year	End	0.10 – 0.25	0.5 – 1.0	1.5	1.00	0.30
Following years*	Mid	0.30 – 0.50	1.5 – 2.0	2.0	1.00	0.85
	End	0.25 – 0.45	1.5 – 2.0	2.0	0.65	0.50
c.3. Chili pepper (*Capsicum annuum*)	Mid	0.85	0.75	1.80	1.00	1.05
	End	0.80	0.70	1.80	0.70	0.70
c.4. Cucumber (*Cucumis sativus*)	Mid	0.90	0.40	2.00	0.96	0.95
For processing	End	0.85	0.40	2.00	0.84	0.80
Fresh market	End	0.90	0.40	2.00	0.70	0.70
c.5. Eggplant (*Solanum melongena*)	Mid	0.75	0.80	2.00	1.00	1.00
	End	0.70	0.75	2.00	0.93	0.90
c.6. Melon (*Cucumis melo*)	Mid	0.95	0.30	2.00	0.95	0.95
	End	0.85	0.30	2.00	0.80	0.75
c.7. Melon, cantaloupe (*C. melo* v. *cantalupensis*)	Mid	0.90	0.30	2.00	0.97	0.95
	End	0.85	0.30	2.00	0.75	0.70
c.8. Okra (*Abelmoschus esculentus*)	Mid	0.90	0.90	1.80	0.82	0.90
	End	0.85	0.90	1.70	0.70	0.75
c.9. Pumpkin, winter squash (*Cucurbita pepo*)	Mid	0.90	0.40	2.00	0.91	0.90
	End	0.85	0.40	1.70	0.58	0.55
c.10. Strawberries (*Fragaria* × *ananassa*)	Mid	0.75	0.20	2.00	0.90	0.75
	End	0.70	0.20	2.00	0.82	0.65
c.11. Tomato *(Solanum lycopersicum)*	Mid	0.90	0.70	2.00	1.00	1.05
For processing	End	0.80	0.65	2.00	0.75	0.75
Fresh market	End	0.80	0.65	2.00	0.90	0.90
c.12. Watermelon (*Citrullus lanatus*)	Mid	0.85	0.40	2.00	1.00	1.00
	End	0.80	0.40	2.00	0.65	0.60
c.13. Zucchini, Squash (*C. pepo*)	Mid	0.90	0.50	2.00	0.98	0.95
	End	0.85	0.50	2.00	0.70	0.65

* Cultivated as a perennial for two or more consecutive years to maximize productivity.

TABLE 9.2

Parameters for the A&P approach aimed at estimating the standard K_{cb} for field crops with presentation of the respective K_{cb} values

Crop Degree of ground cover or field system, training, and plant density	Crop stages	Observation parameters		Calibrated parameters		Standard K_{cb}
		$f_{c\,eff}$	h (m)	M_L	F_r	
a. Grain legumes						
a.1. Beans *(Phaseolus vulgaris)*	Mid	0.80	0.70	2.00	1.00	1.00
Green pods	End	0.80	0.70	2.00	0.85	0.85
Dry grain	End	0.55	0.55	2.00	0.38	0.30
a.2. Black and green gram (*Vigna mungo*)	Mid	0.95	0.90	2.00	0.90	1.05
Black gram (dry)	End	0.65	0.55	2.00	0.28	0.25
Green gram	End	0.95	0.85	2.00	0.28	0.30
a.3. Chickpea (*Cicer arietinum*)	Mid	0.90	0.70	2.00	0.95	1.00
	End	0.70	0.55	2.00	0.27	0.25
a.4. Cowpea (*Vigna unguiculata*)	Mid	0.80	0.80	2.00	0.98	1.00
Green pods	End	0.75	0.75	2.00	0.50	0.50
Dry grain	End	0.75	0.75	2.00	0.25	0.25
a.5. Faba bean (*Vicia faba*)	Mid	0.85	0.80	2.00	1.00	1.05
Fresh	End	0.85	0.80	2.00	0.90	0.95
Dry	End	0.60	0.70	2.00	0.35	0.30
a.6. Groundnut (peanut) (*Arachis hypogaea*)	Mid	0.90	0.50	2.00	0.98	1.00
	End	0.75	0.30	2.00	0.58	0.50
a.7. Lentil (*Lens culinaris*)	Mid	0.85	0.50	2.00	1.00	1.00
	End	0.65	0.40	2.00	0.23	0.20
a.8. Pea (*Pisum sativum*)	Mid	0.98	0.60	2.00	1.00	1.10
Fresh	End	0.98	0.55	2.00	0.97	1.05
Dry	End	0.80	0.50	2.00	0.22	0.20
a.9. Soybean (*Glycine max*)	Mid	0.98	0.80	2.00	0.93	1.05
	End	0.80	0.70	2.00	0.36	0.35
b. Fibre crops						
b.1 Cotton (*Gossypium hirsutum*)	Mid	0.90	1.20	2.00	1.00	1.05
	End	0.80	1.10	2.00	0.40	0.40
b.2. Flax (Linseed) (*Linum usitatissimum*)	Mid	0.85	1.00	2.00	1.00	1.00
	End	0.75	0.90	2.00	0.20	0.20
c. Oil crops						
c.1.Camelina (*Camelina sativa*)	Mid	0.95	0.80	2.00	1.00	1.05
	End	0.85	0.75	2.00	0.36	0.35
c.2. Canola (*Brassica napus*)	Mid	0.95	1.00	2.00	0.98	1.05
	End	0.90	0.90	2.00	0.25	0.25
c.3. Castorbean (*Ricinus communis*)	Mid	0.95	1.00	2.00	0.95	1.00
	End	0.85	0.85	2.00	0.31	0.30
c.4. Mustard (*Brassica juncea*)	Mid	0.95	2.00	2.00	0.90	1.05
	End	0.90	1.90	2.00	0.27	0.30
c.5. Safflower (*Carthamus tinctorius*)	Mid	0.90	1.10	2.00	0.96	1.00
	End	0.60	1.00	2.00	0.16	0.15
c.6. Sunflower (*Helianthus annuus*)	Mid	0.90	2.00	2.00	0.96	1.10
	End	0.80	1.95	2.00	0.19	0.20
d. Sugar crops						
d.1. Sugar beat *(Beta vulgaris)*	Mid	0.90	0.50	2.00	1.00	1.05
	End	0.85	0.50	2.00	0.65	0.65
d.2. Sugar cane (*Saccharum officinarum*)	Mid	0.95	4.00	2.00	0.97	1.15
	End	0.80	4.00	2.00	0.62	0.70

TABLE 9.2 (CONT.)

Crop Degree of ground cover or field system, training, and plant density	Crop stages	Observation parameters		Calibrated parameters		Standard K_{cb}
		$f_{c\ eff}$	h (m)	M_L	F_r	
e. Cereals and pseudocereals						
e.1. Barley (*Hordeum vulgare*)	Mid	0.90	0.90	2.00	1.00	1.05
	End	0.80	0.85	2.00	0.20	0.20
e.2. Maize (*Zea mays*)						
Grain	Mid	0.95	3.50	2.00	0.97	1.15
low grain moisture at harvest	End	0.75	3.30	2.00	0.21	0.25
high grain moisture at harvest	End	0.85	3.40	2.00	0.49	0.55
Silage	Mid	0.90	3.20	2.00	1.00	1.10
	End	0.85	3.10	2.00	0.75	0.80
Sweet	Mid	0.85	2.50	2.00	0.97	1.10
	End	0.85	2.50	2.00	0.90	1.00
e.3. Oats (*Avena sativa*)	Mid	0.90	1.10	2.00	0.96	1.00
	End	0.80	1.00	2.00	0.21	0.20
e.4 Pearl Millet (*Pennisetum glaucum*)	Mid	0.85	2.00	2.00	0.93	1.05
	End	0.70	1.85	2.00	0.19	0.20
e.5. Quinoa (*Chenopodium quinoa*)	Mid	0.80	1.20	2.00	1.00	1.05
	End	0.75	1.05	2.00	0.42	0.40
e.6. Rye (*Secale cereale*)	Mid	0.90	0.90	1.80	0.93	0.95
	End	0.80	0.80	1.50	0.21	0.20
e.7. Sorghum (*Sorghum bicolor*)						
Grain	Mid	0.85	2.00	2.00	0.89	1.00
	End	0.75	1.90	2.00	0.33	0.35
Silage	Mid	0.80	3.00	2.00	0.85	0.95
	End	0.75	3.00	2.00	0.77	0.85
Sweet	Mid	0.95	4.00	2.00	0.89	1.05
	End	0.85	4.00	2.00	0.77	0.90
e.8. Teff (*Eragrostis tef*)	Mid	0.80	1.10	2.00	1.00	1.00
	End	0.70	1.00	2.00	0.21	0.20
e.9. Wheat (*Triticum aestivum* and *T. durum*)	Mid	0.95	1.10	2.00	1.00	1.10
Winter						
low grain moisture at harvest	End	0.75	1.00	2.00	0.21	0.20
high grain moisture at harvest	End	0.85	1.00	2.00	0.45	0.45
Spring						
low grain moisture at harvest	End	0.75	1.00	2.00	0.21	0.20
high grain moisture at harvest	End	0.85	1.00	2.00	0.40	0.40
f. Rice (*Oryza sativa*)						
Flooded	Mid	0.95	1.00	2.00	1.00	1.15
Flooded, dry seeding	End	0.90	1.00	2.00	0.87	0.90
Flooded, anticipated cut-off	End	0.75	0.90	2.00	0.72	0.70
Intermittent	Mid	0.95	0.80	2.00	1.00	1.05
	End	0.90	0.70	2.00	0.85	0.85
Aerobic, irrigated	Mid	0.95	0.80	2.00	1.00	1.05
	End	0.95	0.75	2.00	0.69	0.70

TABLE 9.3

Parameters for the A&P approach aimed at estimating K_{cb} for fruit trees, shrubs and vines close to standard values in Table 7.3 (K_{cb} ± 5%)

Crop Degree of ground cover or field system, training, and plant density	Crop stages	Observation parameters		Calibrated parameters	
		$f_{c\ eff}$	h (m)	M_L	F_r
a. Mediterranean and warm temperate climates fruit trees and vines					
a.1. Olive *(Olea europea)*					
Young (<7 years traditional, < 4 years intensive)	Initial	0.10 – 0.25	1.5 – 2.0	1.0	1.00
	Mid	0.15 – 0.30		1.0	1.00
	End	0.15 – 0.30		1.0	1.00
Traditional, low, non-irrigated, vase, 100-200 pl ha^{-1}	Initial	0.10 – 0.20	2.3 – 3.0	1.5	0.70
	Mid	0.15 – 0.30	2.5 – 3.5	1.5	0.65
	End	0.15 – 0.30	2.5 – 3.5	1.5	0.65
Traditional, non-irrigated, medium, 200-300 pl ha^{-1}	Initial	0.15 – 0.25	2.5 – 4.0	1.5	0.70
	Mid	0.20 – 0.40	3.0 – 4.5	1.5	0.65
	End	0.20 – 0.40	3.0 – 4.5	1.5	0.65
Intensive, hedge prune, 300-800 pl ha^{-1}	Initial	0.25 – 0.35	2.8 – 3.3	1.5	0.70
	Mid	0.30 – 0.40	3.0 – 3.5	1.5	0.75
	End	0.30 – 0.40	3.0 – 3.5	1.4	0.70
Super-Intensive, low density, hedgerow, 800-1500 pl ha^{-1}	Initial	0.30 – 0.40	2.7 – 3.0	1.5	0.70
	Mid	0.35 – 0.45	3.0 – 3.5	1.6	0.75
	End	0.35 – 0.45	3.0 – 3.5	1.5	0.70
Super-intensive high density, hedgerow, 1 500-2 000 pl ha^{-1}	Initial	0.40 – 0.50	2.8 – 3.5	1.3	0.70
	Mid	0.45 – 0.55	3.0 – 4.0	1.4	0.75
	End	0.45 – 0.55	3.0 – 4.0	1.4	0.70
a.2. Table grapes *(Vitis vinifera)*					
Low (Young); diverse trellis and trainings	Initial	0.10 – 0.15	1.1 – 1.3	1.1	1.00
	Mid	0.25 – 0.40	1.3 – 1.5	1.4	1.00
	End	0.15 – 0.35	1.3 – 1.5	1.4	1.00
Medium; T-trellis, Y-trellis and VSP; 1 200-1 700 pl ha^{-1}	Initial	0.10 – 0.15	1.3 – 2.0	1.5	0.90
	Mid	0.40 – 0.60	1.5 – 2.2	1.5	0.95
	End	0.35 – 0.55	1.5 – 2.2	1.5	0.70
High; Y-trellis and overhead system; 1 200-1 700 pl ha^{-1}	Initial	0.15 – 0.20	1.8 – 2.4	1.5	0.90
	Mid	0.60 – 0.95	2.0 – 2.5	1.5	0.95
	End	0.55 – 0.90	2.0 – 2.5	1.5	0.70
a.3. Wine grapes *(Vitis vinifera)*					
Very low (Young); diverse trellis and trainings; 2 000-3 300 pl ha^{-1}	Initial	0.05 – 0.10	1.0 – 1.1	1.1	1.00
	Mid	0.10 – 0.15	1.2 – 1.5	1.1	1.00
	End	0.08 – 0.10	1.2 – 1.5	1.1	1.00
Low; diverse trellis and trainings; 2 000-3 300 pl ha	Initial	0.05 – 0.10	1.3 – 1.9	1.5	0.95
	Mid	0.15 – 0.35	1.5 – 2.0	1.5	0.90
	End	0.10 – 0.30	1.5 – 2.0	1.5	0.70
Medium; VSP, single & double Guyot, GDC, single & bilateral cordon, Lyre, Y-trellis; 2 000-3 300 pl ha^{-1}	Initial	0.10 – 0.15	1.3 – 1.9	1.5	0.95
	Mid	0.35 – 0.50	1.5 – 2.0	1.5	0.90
	End	0.30 – 0.40	1.5 – 2.0	1.5	0.70
High; VSP, GDC, Lyre, Y-trellis, T-trellis, QCT, Pergola; 2 000-3 300 pl ha^{-1}	Initial	0.13 – 0.15	1.3 – 2.3	1.5	0.95
	Mid	0.50 – 0.60	1.5 – 2.5	1.5	0.90
	End	0.45 – 0.55	1.5 – 2.5	1.5	0.70
Very high; Y-trellis, GDC and overhead system; 2 000-3 300 pl ha^{-1}	Initial	0.15 – 0.20	1.6 – 2.2	1.7	0.95
	Mid	0.60 – 0.70	1.8 – 2.5	1.5	0.90
	End	0.45 – 0.55	1.8 – 2.5	1.5	0.70

TABLE 9.3 (CONT.)

Crop Degree of ground cover or field system, training, and plant density	Crop stages	Observation parameters		Calibrated parameters	
		$f_{c\ eff}$	h (m)	M_L	F_r
a.4. Citrus trees					
a.4.1. Clementine *(Citrus clementina),* **Mandarin** *(C. reticulata),* **Lime** *(C. aurantifolia)*					
Young (<5 years); vase	Initial	0.10 – 0.20	1.3 – 1.5	1.8	1.00
	Mid	0.10 – 0.20	1.3 – 1.5	1.8	1.00
	End	0.10 – 0.20	1.3 – 1.5	1.8	1.00
Low; vase; <400 pl ha^{-1}	Initial	0.20 – 0.35	1.5 – 2.5	1.8	0.65
	Mid	0.20 – 0.35	1.5 – 2.5	1.8	0.75
	End	0.20 – 0.35	1.5 – 2.5	1.8	0.75
Medium; vase; 400-550 pl ha^{-1}	Initial	0.35 – 0.60	2.5 – 3.0	1.8	0.65
	Mid	0.35 – 0.60	2.5 – 3.0	1.8	0.70
	End	0.35 – 0.60	2.5 – 3.0	1.8	0.70
High; vase; >550 pl ha^{-1}	Initial	0.60 – 0.70	3.0 – 4.0	2.0	0.65
	Mid	0.60 – 0.70	3.0 – 4.0	2.0	0.70
	End	0.60 – 0.70	3.0 – 4.0	2.0	0.70
a.4.2. Lemon *(C. limon)*					
Young (<5 years) / vase	Initial	0.10 – 0.20	1.3 – 1.5	1.6	1.00
	Mid	0.10 – 0.20	1.3 – 1.5	1.8	1.00
	End	0.10 – 0.20	1.3 – 1.5	1.8	1.00
Low density; vase; <200 pl ha^{-1}	Initial	0.20 – 0.50	1.5 – 2.5	1.5	0.65
	Mid	0.20 – 0.50	1.5 – 2.5	1.9	0.70
	End	0.20 – 0.50	1.5 – 2.5	1.9	0.70
Medium density; base; 200-400 pl ha^{-1}	Initial	0.50 – 0.70	2.5 – 3.0	1.5	0.65
	Mid	0.50 – 0.70	2.5 – 3.0	1.9	0.70
	End	0.50 – 0.70	2.5 – 3.0	1.9	0.70
High density; vase; >400pl ha^{-1}	Initial	0.70 – 0.80	3.0 – 4.0	2.0	0.65
	Mid	0.70 – 0.80	3.0 – 4.0	2.0	0.70
	End	0.70 – 0.80	3.0 – 4.0	2.0	0.70
Hedgerow (Industry); > 1 000 pl ha^{-1}	Initial	0.60 – 0.75	2.5 – 3.5	2.0	0.65
	Mid	0.60 – 0.75	2.5 – 3.5	2.0	0.70
	End	0.60 – 0.75	2.5 – 3.5	2.0	0.70
a.4.3. Orange *(C. sinensis),* **Grapefruit** *(C. paradisi),* **Tangelo** *(C. tangelo)*					
Young (<5 years); vase	Initial	0.10 – 0.20	1.5 – 2.0	1.7	1.00
	Mid	0.10 – 0.20	1.5 – 2.0	1.7	1.00
	End	0.10 – 0.20	1.5 – 2.0	1.7	1.00
Low, vase; <400 pl ha^{-1}	Initial	0.20 – 0.40	2.0 – 3.0	1.8	0.60
	Mid	0.20 – 0.40	2.0 – 3.0	1.9	0.65
	End	0.20 – 0.40	2.0 – 3.0	1.8	0.65
Medium; vase; 400-600 pl ha^{-1}	Initial	0.40 – 0.70	3.0 – 4.0	1.8	0.60
	Mid	0.40 – 0.70	3.0 – 4.0	1.8	0.65
	End	0.40 – 0.70	3.0 – 4.0	1.5	0.65
High; vase; 600-950 pl ha^{-1}	Initial	0.70 – 0.80	4.0 – 4.5	2.0	0.60
	Mid	0.70 – 0.80	4.0 – 4.5	2.0	0.65
	End	0.70 – 0.80	4.0 – 4.5	2.0	0.65
Hedgerow (Industry); >1250 pl ha^{-1}	Initial	0.60 – 0.65	2.5 – 3.5	2.0	0.60
	Mid	0.60 – 0.65	2.5 – 3.5	2.0	0.65
	End	0.60 – 0.65	2.5 – 3.5	2.0	0.65

TABLE 9.3 (CONT.)

Crop Degree of ground cover or field system, training, and plant density	Crop stages	Observation parameters		Calibrated parameters	
		$f_{c\ eff}$	h (m)	M_L	F_r
a.5. Avocado *(Persea americana)*					
Young (<3 years), vase	Initial	0.10 – 0.20	1.5 – 2.0	1.5	1.00
	Mid	0.10 – 0.20	1.5 – 2.0	1.7	1.00
	End	0.10 – 0.20	1.5 – 2.0	1.7	1.00
Low ground cover; vase; <300 pl ha^{-1}	Initial	0.20 – 0.35	2.0 – 3.0	1.5	0.75
	Mid	0.20 – 0.35	2.0 – 3.0	2.0	0.80
	End	0.20 – 0.35	2.0 – 3.0	2.0	0.70
Medium to high; vase; 300-400 pl ha^{-1}	Initial	0.35 – 0.60	3.0 – 4.0	1.5	0.75
	Mid	0.35 – 0.60	3.0 – 4.0	2.0	0.80
	End	0.35 – 0.60	3.0 – 4.0	2.0	0.70
Very high; vase; >400 pl ha^{-1}	Initial	0.60 – 0.70	4.0 – 5.0	1.5	0.75
	Mid	0.60 – 0.70	4.0 – 5.0	2.0	0.80
	End	0.60 – 0.70	4.0 – 5.0	2.0	0.70
Hedgerow; medium density; 300-400 pl ha^{-1}	Initial	0.35 – 0.60	2.5 – 4.0	1.8	0.75
	Mid	0.35 – 0.60	2.5 – 4.0	1.8	0.80
	End	0.35 – 0.60	2.5 – 4.0	1.8	0.70
Hedgerow; high density; >400 pl ha^{-1}	Initial	0.60 – 0.85	3.0 – 4.0	2.0	0.75
	Mid	0.60 – 0.85	3.0 – 4.0	2.0	0.80
	End	0.60 – 0.85	3.0 – 4.0	2.0	0.70
a.6. Fig tree *(Ficus carica)*					
Young (<5 years)	Initial	0.05 – 0.10	1.0 – 1.1	1.5	1.00
	Mid	0.10 – 0.15	1.2 – 1.5	1.6	1.00
	End	0.05 – 0.10	1.2 – 1.5	1.5	1.00
Low, <400 pl ha^{-1}	Initial	0.10 – 0.15	1.1 – 1.5	1.6	0.65
	Mid	0.20 – 0.30	1.5 – 2.0	1.9	0.85
	End	0.10 – 0.20	1.5 – 2.0	1.6	0.60
Medium to high, 800-1 000 pl ha^{-1}	Initial	0.15 – 0.25	1.5 – 2.0	1.6	0.65
	Mid	0.30 – 0.50	2.0 – 2.5	2.0	0.85
	End	0.20 – 0.40	2.0 – 2.5	1.6	0.60
a.7. Loquat *(Eriobotrya japonica)*					
Young (4 <years); vase	Initial	0.15 – 0.25	1.5 – 2.5	1.1	1.00
	Mid	0.15 – 0.25	1.5 – 2.5	1.3	1.00
	End	0.15 – 0.25	1.5 – 2.5	1.3	1.00
Low to medium; vase; 450-600 pl ha^{-1}	Initial	0.25 – 0.50	2.5 – 3.5	1.5	0.55
	Mid	0.25 – 0.50	2.5 – 3.5	2.0	0.75
	End	0.25 – 0.50	2.5 – 3.5	2.0	0.65
High; vase; > 600 pl ha^{-1}	Initial	0.50 – 0.60	3.5 – 4.5	1.4	0.55
	Mid	0.50 – 0.60	3.5 – 4.5	2.0	0.75
	End	0.50 – 0.60	3.5 – 4.5	2.0	0.65
a.8. Persimmon *(Diospyros kaki)*					
Young (<5 years)	Initial	0.05 – 0.10	1.0 – 2.0	1.5	1.00
	Mid	0.10 – 0.15	1.5 – 2.0	1.6	1.00
	End	0.05 – 0.10	1.5 – 2.0	1.1	1.00
Low to medium, 300-500 pl ha^{-1}	Initial	0.10 – 0.15	1.5 – 2.0	1.5	0.65
	Mid	0.15 – 0.45	2.0 – 2.5	2.0	0.80
	End	0.10 – 0.20	2.0 – 2.5	2.0	0.50
High, 500-800 pl/ha^{-1}	Initial	0.13 – 0.15	2.0 – 2.5	1.5	0.65
	Mid	0.40 – 0.60	2.5 – 3.0	2.0	0.80
	End	0.35 – 0.55	2.5 – 3.0	2.0	0.50

TABLE 9.3 (CONT.)

Crop Degree of ground cover or field system, training, and plant density	Crop stages	Observation parameters		Calibrated parameters	
		$f_{c\ eff}$	h (m)	M_L	F_r
a.9. Pomegranate *(Punica granatum)*					
Young (<5 years)	Initial	0.10 – 0.15	1.0 – 1.5	1.1	1.00
	Mid	0.15 – 0.25	1.5 – 2.0	1.3	1.00
	End	0.10 – 0.15	1.5 – 2.0	1.3	1.00
Low; vase; 500-800 pl ha^{-1}	Initial	0.10 – 0.15	1.5 – 2.0	1.4	0.85
	Mid	0.25 – 0.40	2.0 – 2.5	1.5	0.75
	End	0.15 – 0.35	2.0 – 2.5	1.5	0.55
Medium; vase; 500-800 pl ha^{-1}	Initial	0.15 – 0.20	2.0 – 2.5	1.4	0.85
	Mid	0.40 – 0.60	2.5 – 3.0	1.5	0.75
	End	0.30 – 0.45	2.5 – 3.0	1.5	0.55
High; vase; >800 pl ha^{-1}	Initial	0.20 – 0.25	2.5 – 3.0	1.4	0.85
	Mid	0.60 – 0.75	3.0 – 3.5	1.8	0.70
	End	0.45 – 0.60	3.0 – 3.5	1.8	0.55
a.10. Tea *(Camellia sinensis)*					
Young (<2 years); low hedgerow; 7 000-13 500 pl ha^{-1}	Initial	0.40 – 0.50	0.6 – 0.7	1.2	1.00
	Mid	0.40 – 0.50	0.6 – 0.7	2.0	1.00
	End	0.40 – 0.50	0.6 – 0.7	2.0	1.00
Mature; hedgerow; 7 000-13 500 pl ha^{-1}	Initial	0.50 – 0.90	0.7 – 1.0	2.0	1.00
	Mid	0.50 – 0.90	0.7 – 1.0	2.0	1.00
	End	0.50 – 0.90	0.7 – 1.0	2.0	1.00
b. Temperate climate fruit trees, vines and shrubs					
b.1. Pome trees					
b.1.1. Apple *(Malus domestica)*					
Young (<4 years); central leader	Initial	0.05 – 0.10	1.5 – 2.5	1.4	1.00
	Mid	0.15 – 0.25	2.0 – 3.0	1.6	1.00
	End	0.10 – 0.15	2.0 – 3.0	1.3	1.00
Low; central leader; <1500 pl ha^{-1}	Initial	0.10 – 0.15	2.0 – 3.0	1.4	0.95
	Mid	0.25 – 0.40	3.0 – 4.0	1.6	0.90
	End	0.15 – 0.30	3.0 – 4.0	1.3	0.65
Medium; central leader; 1 500-3 000 pl ha^{-1}	Initial	0.15 – 0.20	2.5 – 3.5	1.5	0.95
	Mid	0.40 – 0.50	3.0 – 4.0	1.6	0.90
	End	0.30 – 0.40	3.0 – 4.0	1.3	0.65
High, central leader; 1 500-3 000 pl ha^{-1}	Initial	0.20 – 0.25	2.5 – 4.0	1.5	0.95
	Mid	0.50 – 0.70	3.0 – 4.5	1.6	0.90
	End	0.40 – 0.60	3.0 – 4.5	1.3	0.65
Very high, central leader; >3 000 pl ha^{-1}	Initial	0.20 – 0.25	2.5 – 3.0	1.5	0.95
	Mid	0.70 – 0.75	3.0 – 3.5	1.6	0.90
	End	0.60 – 0.65	3.0 – 3.5	1.3	0.65
b.1.2. Pear *(Pyrus communis)*					
Young (<5 years); all training systems	Initial	0.10 – 0.15	1.5 – 2.0	1.2	1.00
	Mid	0.25 – 0.35	2.0 – 2.5	1.3	1.00
	End	0.20 – 0.30	2.0 – 2.5	1.2	1.00
Low; vase;<1 000 pl ha^{-1}	Initial	0.10 – 0.15	1.5 – 2.0	1.6	0.95
	Mid	0.35 – 0.40	2.0 – 2.5	1.7	0.90
	End	0.30 – 0.35	2.0 – 2.5	1.5	0.65
Medium; central leader; 1 000-1 600 pl ha^{-1}	Initial	0.15 – 0.30	2.0 – 3.5	1.6	0.95
	Mid	0.35 – 0.60	2.5 – 4.0	2.0	0.90
	End	0.30 – 0.55	2.5 – 4.0	1.6	0.65
High, central leader, other trellis systems; >1 500 pl ha^{-1}	Initial	0.20 – 0.30	3.0 – 3.5	1.7	0.95
	Mid	0.60 – 0.75	3.5 – 4.0	2.0	0.90
	End	0.55 – 0.65	3.5 – 4.0	1.3	0.65

TABLE 9.3 (CONT.)

Crop Degree of ground cover or field system, training, and plant density	Crop stages	Observation parameters		Calibrated parameters	
		$f_{c\ eff}$	h (m)	M_L	F_r
b.2. Stone fruits					
b.2.1. Apricot *(Prunus armeniaca),* **plum** *(P. domestica* and *P. salicina)*					
Young, vase; (<5 years)	Initial	0.05 – 0.15	1.3 – 1.5	1.2	1.00
	Mid	0.20 – 0.40	2.0 – 2.5	1.4	1.00
	End	0.10 – 0.30	2.0 – 2.5	1.1	1.00
Low; vase; < 200 pl ha^{-1}	Initial	0.15 – 0.20	2.0 – 2.5	1.4	1.00
	Mid	0.40 – 0.65	2.5 – 3.0	1.5	0.85
	End	0.30 – 0.55	2.5 – 3.0	1.1	0.55
Medium; vase; 200-400 pl ha^{-1}	Initial	0.20 – 0.25	2.5 – 3.5	1.4	1.00
	Mid	0.65 – 0.70	3.0 – 4.5	1.6	0.85
	End	0.55 – 0.60	3.0 – 4.5	1.1	0.55
High; vase; > 400 pl ha^{-1}	Initial	0.20 – 0.25	2.5 – 4.5	1.4	1.00
	Mid	0.70 – 0.80	3.0 – 5.5	1.8	0.85
	End	0.60 – 0.70	3.0 – 5.5	1.2	0.55
Very high; vase; > 400 pl ha^{-1}	Initial	0.25 – 0.30	2.5 – 4.5	1.4	1.00
	Mid	0.80 – 0.90	3.0 – 5.5	1.8	0.85
	End	0.70 – 0.80	3.0 – 5.5	1.6	0.55
b.2.2. Cherry *(Prunus avium)*					
Young (<5 years)	Initial	0.10 – 0.15	2.0 – 2.5	1.2	1.00
	Mid	0.20 – 0.30	2.5 – 3.0	1.5	1.00
	End	0.15 – 0.25	2.5 – 3.0	1.3	1.00
Low; vase; <800 pl ha^{-1}	Initial	0.10 – 0.15	2.5 – 3.5	1.5	1.00
	Mid	0.25 – 0.40	3.0 – 4.0	1.6	0.90
	End	0.20 – 0.35	3.0 – 4.0	1.5	0.70
Medium to High (vigorous rootstock); vase; <800 pl ha^{-1}	Initial	0.15 – 0.20	3.0 – 4.0	1.3	1.00
	Mid	0.40 – 0.70	4.0 – 5.0	1.6	0.90
	End	0.35 – 0.65	4.0 – 5.0	1.4	0.70
Low to Medium (semi vigorous rootstock); central leader; 800-1250 pl ha^{-1}	Initial	0.15 – 0.20	3.0 – 4.0	1.3	1.00
	Mid	0.40 – 0.50	4.0 – 5.0	1.6	0.90
	End	0.35 – 0.45	4.0 – 5.0	1.4	0.70
Medium to High (semi-dwarf rootstock); central leader; >1 200 pl ha^{-1}	Initial	0.20 – 0.25	2.0 – 2.5	1.3	1.00
	Mid	0.40 – 0.70	2.5 – 3.5	1.7	0.90
	End	0.35 – 0.65	2.5 – 3.5	1.5	0.70
High (dwarf rootstock), central leader >1 200 pl ha^{-1}	Initial	0.25 – 0.30	1.5 – 2.5	1.3	1.00
	Mid	0.70 – 0.90	2.0 – 3.5	1.7	0.90
	End	0.65 – 0.80	2.0 – 3.5	1.5	0.70

TABLE 9.3 (CONT.)

Crop Degree of ground cover or field system, training, and plant density	Crop stages	Observation parameters		Calibrated parameters	
		$f_{c\,eff}$	h (m)	M_L	F_r
b.2.3. Peach *(Prunus persica)*					
Young (< 4 years)	Initial	0.10 – 0.15	1.0 – 1.5	1.2	1.00
	Mid	0.25 – 0.30	1.5 – 2.0	1.4	1.00
	End	0.15 – 0.20	1.5 – 2.0	1.2	1.00
Low (high vigour), vase; 400-600 pl ha^{-1}	Initial	0.10 – 0.20	1.5 – 3.0	1.3	1.00
	Mid	0.30 – 0.50	2.0 – 3.5	1.5	0.90
	End	0.20 – 0.40	2.0 – 3.5	1.5	0.60
Medium (medium vigour); vase; 400-700 pl ha^{-1}	Initial	0.20 – 0.25	3.0 – 4.0	1.2	1.00
	Mid	0.50 – 0.65	3.5 – 4.5	1.6	0.90
	End	0.40 – 0.55	3.5 – 4.5	1.5	0.60
High (low vigour), vase; >700 pl ha^{-1}	Initial	0.20 – 0.25	3.5 – 4.5	1.3	1.00
	Mid	0.65 – 0.80	4.0 – 5.0	1.7	0.90
	End	0.55 – 0.70	4.0 – 5.0	1.5	0.60
Medium (medium vigour), trellis; >1 100 pl ha^{-1}	Initial	0.10 – 0.20	2.5 – 3.5	1.4	1.00
	Mid	0.30 – 0.60	3.5 – 4.5	1.7	0.90
	End	0.25 – 0.55	3.5 – 4.5	1.5	0.60
High (low vigour), trellis >1 500 pl ha^{-1}	Initial	0.15 - 0.20	2.5 – 3.0	1.4	1.00
	Mid	0.60 – 0.75	3.5 – 4.0	1.7	0.90
	End	0.50 – 0.65	3.5 – 4.0	1.5	0.60
Very high (low vigour), trellis >2 000 pl ha^{-1}	Initial	0.20 – 0.25	3.0 – 3.5	1.4	1.00
	Mid	0.75 – 0.80	3.5 – 4.0	1.8	0.90
	End	0.65 – 0.70	3.5 – 4.0	1.5	0.60
b.3. Nut trees					
b.3.1. Almond *(Prunus dulcis)*					
Young (<4 years)	Initial	0.05 – 0.10	1.5 – 2.5	1.2	1.00
	Mid	0.20 – 0.30	2.5 – 3.0	1.3	1.00
	End	0.15 – 0.25	2.5 – 3.0	1.2	1.00
Low (high vigour), vase; <400 pl ha^{-1}	Initial	0.10 – 0.15	1.5 – 3.5	1.3	1.00
	Mid	0.20 – 0.40	2.0 – 4.0	1.6	0.90
	End	0.15 – 0.30	2.0 – 4.0	1.4	0.70
Medium (medium vigour), vase; <400 pl ha^{-1}	Initial	0.15 – 0.20	3.5 – 4.0	1.3	1.00
	Mid	0.40 – 0.55	4.0 – 4.5	1.6	0.90
	End	0.30 – 0.50	4.0 – 4.5	1.4	0.70
High (low vigour), vase; >200 pl ha^{-1}	Initial	0.15 – 0.20	4.0 – 4.5	1.3	1.00
	Mid	0.55 – 0.65	4.5 – 5.0	1.7	0.90
	End	0.50 – 0.60	4.5 – 5.0	1.4	0.70
Very high (low vigour), vase; >200 pl ha^{-1}	Initial	0.20 – 0.25	3.5 – 5.0	1.5	1.00
	Mid	0.65 – 0.80	4.0 – 5.5	1.7	0.90
	End	0.55 – 0.70	4.0 – 5.5	1.4	0.70
Hedgerow/Super-intensive (dwarf rootstock), >2500 pl ha^{-1}	Initial	0.10 – 0.15	1.5 – 2.0	1.3	1.00
	Mid	0.30 – 0.45	2.0 – 2.5	1.9	0.90
	End	0.25 – 0.40	2.0 – 2.5	1.4	0.70

TABLE 9.3 (CONT.)

Crop Degree of ground cover or field system, training, and plant density	Crop stages	Observation parameters		Calibrated parameters	
		$f_{c\ eff}$	h (m)	M_L	F_r
b.3.2. Hazelnut *(Corylus avellana)*					
Young (<6 years)	Initial	0.10 – 0.15	1.0 – 1.5	1.2	1.00
	Mid	0.20 – 0.35	1.5 – 2.0	1.5	1.00
	End	0.15 – 0.30	1.5 – 2.0	1.3	1.00
Low to medium; multiple stem-shrub system; 500 pl ha^{-1}	Initial	0.15 – 0.20	1.5 – 3.0	1.2	1.00
	Mid	0.35 – 0.55	2.0 – 3.5	1.6	0.80
	End	0.25 – 0.45	2.0 – 3.5	1.5	0.55
Medium to high; multiple stem-shrub system; 500-700 pl ha^{-1}	Initial	0.15 – 0.20	3.0 – 4.0	1.3	1.00
	Mid	0.55 – 0.70	3.5 – 4.5	1.6	0.80
	End	0.45 – 0.60	3.5 – 4.5	1.4	0.55
Very high; multiple stem-shrub system >700 pl ha^{-1}	Initial	0.20 – 0.25	4.0 – 4.5	1.3	1.00
	Mid	0.70 – 0.80	4.5 – 5.0	1.7	0.80
	End	0.60 – 0.70	4.5 – 5.0	1.3	0.55
Hedgerow/Super–intensive, > 1 000 pl ha^{-1}	Initial	0.20 – 0.25	2.0 – 3.5	1.3	1.00
	Mid	0.55 – 0.80	2.5 – 4.0	1.7	0.80
	End	0.45 – 0.70	2.5 – 4.0	1.3	0.55
b.3.3. Pecans *(Carya illinoinensis)*					
Young (<8 years)	Initial	0.10 – 0.15	6.0 – 7.0	1.2	1.00
	Mid	0.30 – 0.40	7.0 – 8.0	1.4	1.00
	End	0.20 – 0.30	7.0 – 8.0	1.3	1.00
Low; vase; >100 pl ha^{-1}	Initial	0.15 – 0.20	7.0 – 9.0	1.3	1.00
	Mid	0.40 – 0.60	8.0 – 10.0	1.6	0.70
	End	0.30 – 0.50	8.0 – 10.0	1.5	0.55
Medium to high; vase; 100-250 pl ha^{-1}	Initial	0.15 – 0.20	8.0 – 11.0	1.5	1.00
	Mid	0.60 – 0.80	10.0 – 13.0	1.6	0.70
	End	0.50 – 0.70	10.0 – 13.0	1.4	0.55
Very high; vase; 250-300 pl ha^{-1}	Initial	0.20 – 0.25	11.0 – 12.0	1.3	1.00
	Mid	0.80 – 0.90	13.0 – 14.0	1.8	0.70
	End	0.70 – 0.80	13.0 – 14.0	1.4	0.55
b.3.4. Pistachio *(Pistacia vera)*					
Young (<10 years)	Initial	0.05 – 0.10	2.0 – 2.5	1.2	1.00
	Mid	0.25 – 0.35	2.5 – 3.0	1.2	1.00
	End	0.15 – 0.20	2.5 – 3.0	1.2	1.00
Low; vase; >100 pl ha^{-1}	Initial	0.10 – 0.15	2.5 – 3.5	1.2	1.00
	Mid	0.35 – 0.50	3.0 – 4.0	1.5	0.85
	End	0.25 – 0.40	3.0 – 4.0	1.3	0.55
Medium to high; vase; 150-350 pl ha^{-1}	Initial	0.15 – 0.20	3.5 – 4.0	1.3	1.00
	Mid	0.50 – 0.70	4.0 – 4.5	1.5	0.85
	End	0.40 – 0.60	4.0 – 4.5	1.4	0.55
Very high; vase; >300 pl ha^{-1}	Initial	0.20 – 0.25	4.0 – 4.5	1.3	1.00
	Mid	0.70 – 0.85	4.5 – 5.0	1.5	0.85
	End	0.60 – 0.70	4.5 – 5.0	1.4	0.55
Hedgerow/Super-intensive; >1 000 pl ha^{-1}	Initial	0.15 – 0.20	2.0 – 2.5	1.3	1.00
	Mid	0.55 – 0.70	2.5 – 3.0	1.5	0.85
	End	0.40 – 0.55	2.5 – 3.0	1.4	0.55

TABLE 9.3 (CONT.)

Crop Degree of ground cover or field system, training, and plant density	Crop stages	Observation parameters		Calibrated parameters	
		$f_{c\ eff}$	h (m)	M_L	F_r
b.3.5. Walnut *(Juglans regia)*					
Young, vase (< 7 years)	Initial	0.10 – 0.25	4.0 – 5.0	1.1	1.00
	Mid	0.30 – 0.55	5.0 – 6.0	1.4	1.00
	End	0.20 – 0.45	5.0 – 6.0	1.1	1.00
Low; vase; <200 pl ha^{-1}	Initial	0.20 – 0.25	5.5 – 9.0	1.2	0.90
	Mid	0.55 – 0.75	6.0 – 10.0	1.5	0.90
	End	0.45 – 0.65	6.0 – 10.0	1.3	0.55
Medium to very high; vase; 200-550 pl ha^{-1}	Initial	0.25 – 0.30	9.0 – 10.0	1.3	0.90
	Mid	0.75 – 0.95	10.0 – 12.0	1.7	0.90
	End	0.65 – 0.85	10.0 – 12.0	1.5	0.55
b.4. Other temperate vines and shrubs					
b.4.1. Kiwifruit *(Actinidia deliciosa)*					
Young (<6 years); trellis	Initial	0.10 – 0.20	1.0 – 1.5	1.2	1.00
	Mid	0.35 – 0.60	1.5 – 2.0	1.2	1.00
	End	0.30 – 0.50	1.5 – 2.0	1.1	1.00
Medium density; T-bar trellis; <800 pl ha^{-1}	Initial	0.15 – 0.20	1.5 – 1.7	1.3	0.70
	Mid	0.60 – 0.80	2.0 – 2.2	2.0	0.80
	End	0.50 – 0.70	2.0 – 2.2	2.0	0.70
High density; pergola trellis; >800 pl ha^{-1}	Initial	0.20 – 0.25	1.5 – 2.0	1.3	0.70
	Mid	0.60 – 0.95	2.0 – 2.5	2.0	0.80
	End	0.55 – 0.90	2.0 – 2.5	2.0	0.70
b.4.2. Highbush blueberries *(Vaccinium corymbosum)*					
Young (< 3 years)	Initial	0.05 – 0.15	1.0 – 1.3	1.1	1.00
	Mid	0.25 – 0.50	1.3 – 1.5	1.3	1.00
	End	0.20 – 0.45	1.3 – 1.5	1.0	1.00
Common density, 2 500-5 000 pl ha^{-1}	Initial	0.20 – 0.25	1.3 – 1.7	1.6	1.00
	Mid	0.50 – 0.80	1.5 – 2.0	2.0	0.85
	End	0.45 – 0.75	1.5 – 2.0	2.0	0.65
b.4.3. Lowbush blueberries *(Vaccinium angustifolium)*					
Young (< 2 years)	Initial	0.10 – 0.20	0.3 – 0.4	1.1	1.00
	Mid	0.35 – 0.50	0.5 – 0.6	1.2	1.00
	End	0.30 – 0.45	0.5 – 0.6	1.0	1.00
Common density, > 5 000 pl ha^{-1}	Initial	0.15 – 0.20	0.45 – 0.75	1.6	1.00
	Mid	0.50 – 0.80	0.6 – 1.0	2.0	0.85
	End	0.45 – 0.75	0.6 – 1.0	2.0	0.70
b.4.4. Rabbiteye blueberry *(Vaccinium ashei)*					
Young (<3 years)	Initial	0.10 – 0.20	0.3 – 0.4	1.1	1.00
	Mid	0.35 – 0.50	0.5 – 0.6	1.2	1.00
	End	0.30 – 0.45	0.5 – 0.6	1.0	1.00
Common density, 2 500-5 000 pl ha^{-1}	Initial	0.15 – 0.20	0.45 – 0.75	1.6	1.00
	Mid	0.50 – 0.80	0.6 – 1.0	2.0	0.85
	End	0.45 – 0.75	0.6 – 1.0	2.0	0.70
b.4.5. Blackberry *(Rubus L. subgenus Rubus)*					
Young (< 3 years)	Initial	0.10 – 0.20	0.30 – 0.85	1.1	1.00
	Mid	0.15 – 0.50	0.40 – 1.00	1.2	1.00
	End	0.10 – 0.40	0.40 – 1.00	1.0	1.00
Common density, 2 000 – 5 000 pl ha^{-1}	Initial	0.15 – 0.20	0.85 – 1.5	1.5	0.80
	Mid	0.50 – 0.85	1.00 – 2.00	2.0	0.90
	End	0.45 – 0.75	1.00 – 2.00	2.0	0.50

TABLE 9.3 (CONT.)

Crop Degree of ground cover or field system, training, and plant density	Crop stages	Observation parameters		Calibrated parameters	
		$f_{c\ eff}$	h (m)	M_L	F_r
c. Sub–tropical and tropical orchards and plantations					
c.1. Sub–tropical and tropical small fruit trees and shrub					
c.1.1. Acerola *(Malpighia emarginata)*					
Young (<5 years)	Initial	0.10 – 0.45	0.7 – 1.5	1.1	1.00
	Mid	0.15 – 0.50	0.8 – 2.0	1.2	1.00
	End	0.10 – 0.48	0.7 – 2.0	1.2	1.00
Medium-High, 250-700 pl ha^{-1}	Initial	0.45 – 0.65	1.9 – 3.7	1.4	0.60
	Mid	0.50 – 0.70	2.0 – 4.0	1.5	0.65
	End	0.50 – 0.70	2.0 – 4.0	1.5	0.65
c.1.2. Carambola *(Averrhoa carambola)*					
Young (< 4 years)	Initial	0.15 – 0.35	0.7 – 1.8	1.8	1.00
	Mid	0.20 – 0.40	1.0 – 2.0	1.9	1.00
	End	0.20 – 0.40	1.0 – 2.0	1.9	1.00
Common density, 150-300 pl ha^{-1}	Initial	0.40 – 0.55	2.0 – 3.5	1.8	0.90
	Mid	0.45–0.60	3.0 – 4.0	2.0	1.00
	End	0.45–0.60	3.0 – 4.0	2.0	0.98
c.1.3. Cherimoya *(Annona cherimola)*					
Young (<3 years)	Initial	0.10 – 0.15	0.4 – 1.5	1.0	1.00
	Mid	0.10 – 0.15	0.4 – 1.5	1.4	1.00
	End	0.10 – 0.15	0.4 – 1.5	1.4	1.00
Low to medium, 125-400 pl ha^{-1}	Initial	0.10 – 0.30	1.4 – 3.3	1.6	0.70
	Mid	0.15 – 0.35	1.5 – 3.5	1.7	0.85
	End	0.15 – 0.35	1.5 – 3.5	1.7	0.75
Medium to high, 500-1250 pl ha^{-1}	Initial	0.25 – 0.55	3.0 – 3.2	1.8	0.70
	Mid	0.35 – 0.60	3.3 – 3.5	1.9	0.85
	End	0.35 – 0.60	3.3 – 3.5	1.9	0.75
c.1.4. Cocoa *(Theobroma cacao)*					
Young (<4 years), shading	Initial	0.10 – 0.45	0.7 – 1.8	1.3	1.00
	Mid	0.15 – 0.50	1.2 – 2.0	1.8	1.00
	End	0.15 – 0.50	1.2 – 2.0	1.6	1.00
Medium shading, <1 500 pl ha^{-1}	Initial	0.50 – 0.65	2.0 – 2.3	1.5	0.80
	Mid	0.55 – 0.70	2.2 – 2.5	2.0	0.83
	End	0.55 – 0.70	2.2 – 2.5	2.0	0.83
High shading, <1 500 pl ha^{-1}	Initial	0.70 – 0.85	2.0 – 3.5	1.8	0.80
	Mid	0.75 – 0.90	2.5 – 4.0	2.0	0.83
	End	0.75 – 0.90	2.5 – 4.0	1.8	0.83
c.1.5. Coffee *(Coffea arabica)*					
Young (<4 years)	Initial	0.10 – 0.15	1.0 – 1.5	1.6	1.00
	Mid	0.15 – 0.20	1.5 – 2.0	1.6	1.00
	End	0.15 – 0.20	1.5 – 2.0	1.6	1.00
Low to Medium, <6 000 pl ha^{-1}	Initial	0.20 – 0.45	1.5 – 2.0	1.8	0.85
	Mid	0.25 – 0.50	2.5 – 3.0	1.8	0.85
	End	0.25 – 0.50	2.5 – 3.0	1.8	0.85
High (including hedgerow), >6 000 pl ha^{-1}	Initial	0.50 – 0.75	1.5 – 2.0	1.8	0.85
	Mid	0.55 – 0.80	2.5 – 3.0	1.8	0.85
	End	0.55 – 0.80	2.5 – 3.0	1.8	0.85

TABLE 9.3 (CONT.)

Crop Degree of ground cover or field system, training, and plant density	Crop stages	Observation parameters		Calibrated parameters	
		$f_{c\ eff}$	h (m)	M_L	F_r
c.1.6. Dragon fruit *(Hylocereus undatus)*					
Young (<3 years)	Initial	0.10 – 0.15	0.3 – 1.8	1.1	1.00
	Mid	0.15 – 0.20	0.5 – 2.0	1.2	1.00
	End	0.10 – 0.20	0.4 – 2.0	1.2	1.00
Medium, <1 800 pl ha^{-1}	Initial	0.15 – 0.25	1.8 – 2.3	1.1	0.80
	Mid	0.20 – 0.30	2.0 – 2.5	1.5	1.00
	End	0.20 – 0.30	2.0 – 2.5	1.5	0.80
High, >1 800 pl ha^{-1}	Initial	0.25 – 0.65	1.9 – 2.3	1.3	0.80
	Mid	0.30 – 0.60	2.0 – 2.5	1.6	1.00
	End	0.30 – 0.55	2.0 – 2.5	1.6	0.80
High (hedgerow), >2 000 pl ha^{-1}	Initial	0.15 – 0.55	1.7 – 2.3	1.3	0.80
	Mid	0.25 – 0.60	2.0 – 2.5	1.5	1.00
	End	0.20 – 0.50	2.0 – 2.5	1.5	0.80
c.1.7. Guava *(Psidium guajava)*					
Young (<8 years)	Initial	0.10 – 0.15	0.5 – 1.5	1.2	1.00
	Mid	0.15 – 0.20	0.8 – 1.5	1.4	1.00
	End	0.10 – 0.15	0.8 – 1.5	1.4	1.00
Low to medium, <300 pl ha^{-1}	Initial	0.15 – 0.35	1.3 – 1.8	1.8	0.75
	Mid	0.20 – 0.40	1.5 – 2.0	2.0	0.85
	End	0.20 – 0.40	1.5 – 2.0	2.0	0.80
High, <350 pl ha^{-1}	Initial	0.35 – 0.55	1.8 – 2.3	1.8	0.75
	Mid	0.40 – 0.60	2.0 – 2.5	2.0	0.85
	End	0.40 – 0.60	2.0 – 2.5	2.0	0.80
Very high, >300 pl ha^{-1}	Initial	0.55 – 0.75	2.3 – 2.5	1.8	0.75
	Mid	0.60 – 0.80	2.5 – 3.0	2.0	0.85
	End	0.60 – 0.80	2.5 – 3.0	2.0	0.80
c.1.8. Jujube *(Ziziphus jujuba)*					
Young (<3 years)	Initial	0.05 – 0.10	1.0 – 1.5	1.2	1.00
	Mid	0.10 – 0.25	1.5 – 2.0	1.3	1.00
	End	0.05 – 0.15	1.5 – 2.0	1.2	1.00
Low to Medium; <1 100 pl ha^{-1}	Initial	0.10 – 0.15	1.5 – 2.5	1.4	0.70
	Mid	0.25 – 0.40	2.0 – 2.5	1.7	0.70
	End	0.20 – 0.35	2.0 – 2.5	1.6	0.60
High; >1 100 pl ha^{-1}	Initial	0.15 – 0.25	2.5 – 3.5	1.4	0.70
	Mid	0.40 – 0.70	3.0 – 4.0	1.8	0.70
	End	0.35 – 0.65	3.0 – 4.0	1.7	0.60
c.1.9. Papaya *(Carica papaya)*					
Young (<1 year)	Initial	0.15 – 0.35	0.85 – 1.50	1.2	1.00
	Mid	0.20 – 0.40	1.00 – 1.50	1.6	1.00
	End	0.20 – 0.40	1.00 – 1.50	1.6	1.00
Medium (single row), 3 000-4 500 pl ha^{-1}	Initial	0.40 – 0.65	1.00 – 2.50	1.3	0.70
	Mid	0.45 – 0.70	1.50 – 3.00	1.5	0.90
	End	0.45 – 0.70	1.50 – 3.00	1.5	0.70
High (single or double row), <7 000 pl ha^{-1}	Initial	0.70 – 0.80	2.50 – 3.50	1.5	0.70
	Mid	0.75 – 0.85	3.00 – 4.00	1.7	0.90
	End	0.75 – 0.85	3.00 – 4.00	1.7	0.70

TABLE 9.3 (CONT.)

Crop Degree of ground cover or field system, training, and plant density	Crop stages	Observation parameters		Calibrated parameters	
		$f_{c\ eff}$	h (m)	M_L	F_r
c.1.10. Passionfruit *(Passiflora edulis)*					
Young (< 0.5 years), Vertical shoot position (VSP)	Initial	0.10 – 0.15	1.3 – 1.8	1.5	1.00
	Mid	0.10 – 0.15	1.5 – 2.0	1.7	1.00
	End	0.10 – 0.13	1.5 – 2.0	1.7	1.00
Medium; VSP; 500 pl ha^{-1}	Initial	0.15 – 0.35	1.3 – 1.8	1.5	0.95
	Mid	0.15 – 0.35	1.5 – 2.0	2.0	1.00
	End	0.10 – 0.30	1.4 – 1.9	2.0	0.95
High; VSP; 1 000 pl ha^{-1}	Initial	0.35 – 0.45	1.4 – 1.8	1.5	0.95
	Mid	0.35 – 0.45	1.5 – 2.0	2.0	1.00
	End	0.30 – 0.40	1.4 – 1.8	2.0	0.95
Young, hedgerow	Initial	0.20 – 0.35	1.5 – 2.0	1.5	1.00
	Mid	0.20 – 0.35	1.5 – 2.0	1.7	1.00
	End	0.18 – 0.33	1.5 – 2.0	1.7	1.00
High; hedgerow; 1 000 pl ha^{-1}	Initial	0.35 – 0.70	1.5 – 2.0	1.5	0.95
	Mid	0.35 – 0.70	1.5 – 2.0	2.0	1.00
	End	0.33 – 0.60	1.3 – 1.8	2.0	0.95
Young, overhead trellis	Initial	0.18 – 0.60	1.8 – 2.2	1.4	1.00
	Mid	0.25 – 0.60	1.8 – 2.2	1.6	1.00
	End	0.18 – 0.55	1.8 – 2.2	1.6	1.00
Very high; overhead trellis; 2 000 pl ha^{-1}	Initial	0.60 – 0.75	1.8 – 2.2	1.5	0.95
	Mid	0.60 – 0.75	1.8 – 2.2	2.0	1.00
	End	0.55 – 0.70	1.8 – 2.2	2.0	0.95
c.2. Tropical fruit trees					
c.2.1. Cashew *(Anacardium occidentale),* **dwarf**					
Young (< 5 years)	Initial	0.10 – 0.45	0.8 – 2.8	1.0	1.00
	Mid	0.15 – 0.50	1.0 – 3.0	1.3	1.00
	End	0.10 – 0.45	1.0 – 3.0	1.3	1.00
Common density, 150-330 pl ha^{-1}	Initial	0.45 – 0.60	2.0 – 2.8	1.3	0.70
	Mid	0.50 – 0.65	3.0 – 5.0	1.5	0.60
	End	0.45 – 0.65	3.0 – 5.0	1.5	0.60
c.2.2. Jaboticaba *(Plinia peruviana)*					
Young (<5 years)	Initial	0.10 – 0.45	0.8 – 1.8	1.0	1.00
	Mid	0.15 – 0.50	1.0 – 2.0	1.1	1.00
	End	0.10 – 0.45	1.0 – 2.0	1.1	1.00
Common density, 250-500 pl ha^{-1}	Initial	0.40 – 0.80	2.5 – 4.0	1.3	0.60
	Mid	0.50 – 0.85	3.0 – 4.5	1.4	0.80
	End	0.45 – 0.80	3.0 – 4.5	1.4	0.70
c.2.3. Longan *(Dimocarpus longan)*					
Young (<5 years)	Initial	0.10 – 0.20	1.0 – 1.5	1.5	1.00
	Mid	0.15 – 0.30	1.5 – 2.0	1.6	1.00
	End	0.15 – 0.30	1.5 – 2.0	1.6	1.00
Common density, 250-300 pl ha^{-1}	Initial	0.30 – 0.60	2.5 – 3.5	1.9	0.80
	Mid	0.35 – 0.65	3.0 – 4.0	2.0	0.90
	End	0.35 – 0.65	3.0 – 4.0	2.0	0.90

TABLE 9.3 (CONT.)

Crop Degree of ground cover or field system, training, and plant density	Crop stages	Observation parameters		Calibrated parameters	
		$f_{c\,eff}$	h (m)	M_L	F_r
c.2.4. Lychee *(Litchi chinensis)*					
Young (<5 years); vase	Initial	0.10 – 0.15	1.0 – 2.0	1.5	1.00
	Mid	0.15 – 0.20	1.5 – 2.5	1.6	1.00
	End	0.15 – 0.20	1.5 – 2.5	1.6	1.00
Low to medium; vase; 100-200 pl ha^{-1}	Initial	0.20 – 0.25	2.5 – 3.5	1.9	0.80
	Mid	0.25 – 0.30	3.0 – 4.0	2.0	0.90
	End	0.20 – 0.30	3.0 – 4.0	2.0	0.90
High; vase; 200-400 pl ha^{-1}	Initial	0.30 – 0.60	4.0 – 4.5	1.9	0.80
	Mid	0.35 – 0.65	4.5 – 5.0	2.0	0.90
	End	0.30 – 0.60	4.5 – 5.0	2.0	0.90
c.2.5. Macadamia *(Macadamia integrifolia)*					
Young (<10 years)	Initial	0.10 – 0.40	0.40 – 1.60	1.1	1.00
	Mid	0.15 – 0.50	0.70 – 2.00	1.2	1.00
	End	0.10 – 0.45	0.65 – 1.80	1.2	1.00
Medium-High; 300-500 pl ha^{-1}	Initial	0.50 – 0.80	2.00 – 4.30	1.5	0.55
	Mid	0.55 – 0.85	2.50 – 4.50	1.7	0.55
	End	0.55 – 0.85	2.45 – 4.50	1.7	0.50
c.2.6. Mango *(Mangifera indica)*					
Young (<4 years)	Initial	0.10 – 0.20	0.75 – 2.00	1.4	1.00
	Mid	0.15 – 0.25	1.00 – 2.00	1.6	1.00
	End	0.10 – 0.25	1.00 – 2.00	1.6	1.00
Low to medium,100-250 pl ha-1	Initial	0.20 – 0.35	1.80 – 2.80	1.7	0.75
	Mid	0.25 – 0.40	2.00 – 3.00	2.0	0.85
	End	0.25 – 0.40	2.00 – 3.00	2.0	0.80
High, 400-600 pl ha-1	Initial	0.35 – 0.45	2.00 – 3.50	1.7	0.75
	Mid	0.40 – 0.50	2.50 – 4.00	2.0	0.85
	End	0.40 – 0.50	2.50 – 4.00	2.0	0.80
c.3. Palm, fiber and rubber plantations					
c.3.1 Coconut *(Cocos nucifera)* **and Açaí palm** *(Euterpe oleracea)*					
Young (<15 years for coconut, 3 years for açaí palm)	Initial	0.05 – 0.25	1.5 – 2.0	1.3	1.00
	Mid	0.10 – 0.30	1.8 – 2.5	1.4	1.00
	End	0.10 – 0.30	1.8 – 2.5	1.4	1.00
Medium, 100-200 pl ha^{-1}	Initial	0.30 – 0.45	2.5 – 4.5	1.6	0.80
	Mid	0.35 – 0.50	3.0 – 5.0	1.6	0.80
	End	0.45 – 0.50	3.0 – 5.0	1.6	0.80
High, 200 pl ha^{-1}	Initial	0.50 – 0.70	5.0 – 6.5	1.8	0.80
	Mid	0.55 – 0.75	5.5 – 7.0	1.9	0.80
	End	0.55 – 0.75	5.5 – 7.0	1.9	0.80
c.3.2. Date palm *(Phoenix dactylifera)*					
Young (<10 years)	Initial	0.10 – 0.15	2.5 – 3.5	1.4	1.00
	Mid	0.15 – 0.20	3.0 – 4.0	1.5	1.00
	End	0.15 – 0.20	3.0 – 4.0	1.5	1.00
Medium, 100-130 pl ha^{-1}	Initial	0.20 – 0.45	3.5 – 9.5	1.7	0.75
	Mid	0.25 – 0.50	4.0 – 10.0	1.8	0.75
	End	0.25 – 0.50	4.0 – 10.0	1.8	0.75
Very high, >200 pl ha^{-1}	Initial	0.65 – 0.80	9.5 – 13.0	1.7	0.75
	Mid	0.70 – 0.85	10.0 – 14.0	1.8	0.75
	End	0.70 – 0.85	10.0 – 14.0	1.8	0.75
Very high, >200 pl ha^{-1}	Initial	0.65 – 0.80	9.5 – 13.0	1.7	0.75
	Mid	0.70 – 0.85	10.0 – 14.0	1.8	0.75
	End	0.70 – 0.85	10.0 – 14.0	1.8	0.75

TABLE 9.3 (CONT.)

Crop Degree of ground cover or field system, training, and plant density	Crop stages	Observation parameters		Calibrated parameters	
		$f_{c\ eff}$	h (m)	M_L	F_r
c.3.3. Jatropha *(Jatropha curcas)*					
Young (<2 years); vase	Initial	0.10 – 0.15	0.45 – 0.85	1.6	1.00
	Mid	0.10 – 0.15	0.50 – 1.00	1.7	1.00
	End	0.10 – 0.15	0.50 – 1.00	1.7	1.00
Low to medium; vase; 1 100-1 500 pl ha^{-1}	Initial	0.08 – 0.25	0.8 – 1.9	1.7	0.90
	Mid	0.15 – 0.35	1.0 – 2.0	1.8	0.90
	End	0.10 – 0.25	1.0 – 2.0	1.8	0.90
High; vase; >1 500 pl ha	Initial	0.25 – 0.35	1.9 – 2.0	1.7	0.90
	Mid	0.35 – 0.50	2.0 – 3.0	1.8	0.90
	End	0.30 – 0.45	2.0 – 3.0	1.8	0.90
c.3.4. Oil palm *(Elaeis guineensis)*					
Young (<3 years)	Initial	0.10 – 0.20	1.0 – 2.0	1.3	1.00
	Mid	0.15 – 0.30	1.5 – 2.5	1.3	1.00
	End	0.15 – 0.25	1.5 – 2.5	1.3	1.00
Medium, 100-150 pl ha^{-1}	Initial	0.25 – 0.45	1.0 – 2.0	1.5	0.95
	Mid	0.30 – 0.50	2.5 – 5.0	1.5	0.95
	End	0.30 – 0.50	2.5 – 5.0	1.5	0.95
High, >150 pl ha^{-1}	Initial	0.45 – 0.55	4.5 – 6.0	1.5	0.95
	Mid	0.50 – 0.60	5.0 – 7.0	1.5	0.95
	End	0.50 – 0.60	5.0 – 7.0	1.5	0.95
c.3.5. Peach palm *(Bactris gasipaes)*					
Young (≤2.5 years)	Initial	0.10 – 0.20	1.0 – 1.5	1.2	1.00
	Mid	0.15 – 0.30	1.5 – 2.0	1.7	1.00
	End	0.15 – 0.30	1.5 – 2.0	1.5	1.00
Common density, 3 000-5 000 pl ha^{-1} for production of heart of palm, or "palmito"	Initial	0.30 – 0.50	2.0 – 3.0	1.3	0.85
	Mid	0.35 – 0.55	2.5 – 3.5	1.8	0.85
	End	0.35 – 0.55	2.5 – 3.5	1.7	0.85
c.3.6. Ramie *(Boehmeria nivea)*					
Young (< 2 years)	Initial	0.10 – 0.15	0.10 – 0.15	1.0	0.85
	Mid	0.10 – 0.60	0.70–2.00	1.6	0.85
	End	0.05 – 0.55	0.70–2.00	1.5	0.85
Common density, 40 000-60 000 pl ha^{-1}	Initial	0.10 – 0.15	1.8 – 2.0	1.3	0.85
	Mid	0.80 – 0.95	2.0 – 2.5	1.8	0.85
	End	0.75 – 0.90	2.0 – 2.5	1.8	0.85
c.3.7. Rubber tree *(Hevea brasiliensis)*					
Young (<6 years)	Initial	0.05 – 0.60	0.5 – 7.5	1.0	1.00
	Mid	0.10 – 0.70	0.7 – 8.0	1.2	1.00
	End	0.05 – 0.65	0.7 – 8.0	1.1	1.00
Common density; 400-700 pl ha^{-1}	Initial	0.75 – 0.80	8.0 – 13.5	1.2	0.69
	Mid	0.85 – 0.90	9.0 – 15.0	1.5	0.75
	End	0.80 – 0.85	9.0 – 15.0	1.4	0.69

TABLE 9.4

Parameters for the A&P approach aimed at estimating K_{cb} for grasses for grazing, for hay and for seed close to standard values of K_c in Table 6.4 ($K_{cb} = K_c \pm 0.05$)

Crop Degree of ground cover or field system, training, and plant density	Crop stages	Observed parameters		Calibrated parameters		Standard K_{cb}
		$f_{c\,eff}$	h (m)	M_L	F_r	
Alfalfa for hay; typical multiple cutting cycles	Ini	0.20-0.35	0.10-0.20	2	1	0.45
	Mid	0.95-0.98	0.50-0.75	2	1	1.05
	End	0.95-0.98	0.50-0.75	2	1	1.05
Alfalfa for seed	Ini	0.10-0.20	0.15-0.25	2	1	0.35
	Mid	0.95-0.98	0.40-0.50	2	1	1.05
	End	0.75-0.80	0.35-0.45	2	0.65	0.60
Grasses for grazing, high height of grazing cuts	Ini	0.75-0.80	0.10-0.20	2	1	0.85
	Mid	0.85-0.90	0.90-1.20	2	1	1.05
	End	0.85-0.90	0.90-1.20	2	0.85	0.85
Grasses for grazing, short height of grazing cuts	Ini	0.75-0.80	0.05-0.10	2	1	0.80
	Mid	0.90-0.95	0.10-0.30	2	1	0.95
	End	0.90-0.95	0.10-0.30	2	0.85	0.80
Grasses for grazing with large cut cycles	Ini	0.75-0.80	0.10-0.20	2	1	0.85
	Mid	0.85-0.90	0.30-0.50	2	0.95	0.90
	End	0.85-0.90	0.30-0.50	2	0.95	0.90
Grasses for grazing with short cut cycles	Ini	0.75-0.80	0.05-0.10	2	1	0.80
	Mid	0.85-0.90	0.15-0.25	2	0.95	0.90
	End	0.85-0.90	0.15-0.25	2	0.95	0.85
Grasses for seed production	Ini	0.10-0.15	0.05-0.10	2	1	0.25
	Mid	0.85-0.90	0.40-0.50	2	0.90	0.85
	End	0.80-0.85	0.35-0.45	2	0.65	0.65
Grasses for hay; typical cut cycles	Ini	0.30-0.45	0.05-0.10	2	1	0.50
	Mid	0.95-0.98	0.40-0.60	2	1	1.05
	End	0.95-0.98	0.40-0.60	2	1	1.00

Chapter 10

$ET_{c\ act}$ under various crop management practices

This chapter discusses various types of factors that may cause the values for $K_{c\ act}$ and $ET_{c\ act}$ to deviate from the standard K_c and K_{cb} values described in Chapters 6 and 7. These factors refer to the effects of surface mulches, intercropping, small areas of vegetation and estimation of K_c or K_{cb} for periods under conditions of frozen soil as well under non-active cover.

This chapter is intended to serve as a resource for situations where cultivation practices are known to deviate from those assumed in Tables 6.1 to 6.4 and 7.1 to 7.3 and where estimates of $K_{c\ act}$ and $ET_{c\ act}$ are still necessary. This chapter is by no means exhaustive; its intent is to show some of the procedures that can be used to make adjustments to K_c to account for deviations from standard conditions.

10.1 EFFECTS OF SURFACE MULCHES

Mulches are frequently used in vegetable production to reduce evaporation losses from the soil surface, to accelerate crop development in cool climates by increasing soil temperature, to reduce erosion, or to assist in weed control. Mulches may be composed of organic plant materials, or they may consist of plastic sheets or films.

10.1.1 Plastic mulches

Plastic mulches generally consist of thin sheets of polyethylene, or a similar material placed over the ground surface, especially along the plant rows. Holes are cut into the plastic film at plant spacings to allow the plant vegetation to emerge. Plastic mulches can be transparent, white, or black. Colour influences albedo, mainly during the early stages of the crop. However, as the intention is to use a simple procedure for adjusting K_c for mulched crops, only generalized recommendations are given here. The published literature offers contradictory information on impacts of plastic mulches on K_c and ET. However, it appears likely that black plastic increases the soil temperature more than transparent plastic, so that transparent plastic may cause slightly smaller differences in K_c relative to not adopting any type of mulching.

Plastic mulches can substantially reduce the evaporation of water from the soil surface. The impact of plastic mulch is likely more evident under trickle irrigation systems. Associated with the reduction in evaporation, however, there may occur a general increase in transpiration from the vegetation caused by the transfer of both sensible and radiative heat from the surface of the plastic cover to adjacent vegetation. Even though the transpiration rates under mulch may increase by an average of 10-30 percent over the season as compared to no mulch, the K_c values may decrease by an average of 10-30 percent due to a 30-80 percent reduction in soil evaporation. Ranges in observed reductions in K_c are given in Table 10.1 for three groups of crops: field crops referring to cereals, cotton and sugar cane; vegetable crops comprised of various species, and fruit tree and vine crops, namely apricot and pear orchards, and grape vineyards. Table 10.2 presents the percentage of reduction K_c in relation to the type of mulch which complements the information provided in Table 10.1.

TABLE 10.1

Approximate reductions (%) in K_c for various field and horticultural crops under nearly full plastic mulch and organic mulch relative to using no mulch when irrigation is by trickle irrigation

Crop	Early vegetative stages (%)	Mid-season (%)	Late and end season (%)	Notes
Field crops (maize, sorghum, wheat, okra, peanut, etc.)	5-25	5-10	5-10	Plastic films
Cotton	15-35	5-10	5-10	Plastic and straw
Sugar cane	15-35	10-15	10-15	Organic mulch
Vegetables (banana, cucumber, curcuma, onion, tomato, etc.)	10-35	5-10	5-10	Plastic and straw
Fruit tree and vine crops (apricot and pear orchards, and grapes vineyards)	15-35	10-25	10-25	Both plastic and organic mulche

TABLE 10.2

Approximate reductions (%) of soil evaporation (E_s) and crop coefficients (K_c) of straw and plastic film mulch relative to no mulch conditions

Parameter reduction	Mulch material	Early stages (%)	Mid-season (%)	Late & end season (%)
E_s	Straw	30-42	26-36	19-36
	Plastic film	38-58	31-51	25-34
K_c	Straw	14-34	8-17	8-17
	Plastic film	15-35	9-20	5-19
	Black	23-43	12-19	12-20
	Transparent	15-24	9-22	3-16
	White	27-33	-*	22-25

* Not given due to limited data available.

Literature has reported differences between mulch and no-mulch K_c that vary widely for the same types of crops, even for the same crops. This variability mainly relates to density of planting and cultivar and types with those producing more shading causing smaller differences. Variability is also created by measurement errors and biases. Soil texture and management impact reductions in K_c since soil surface conditions may or not favour evaporation losses. The latter vary according to the stages of development and amount of shading of the soil surface. It has been possible to reduce the wide ranges in impacts by grouping crops by stage since it is always observed that reductions in K_c are greater during the initial crop stage and during the first part of the vegetative growth stage, or the pre-anthesis stages, i.e., when ground cover f_c is smaller. Less mulching effect on decreasing K_c occurs by the start of mid-season since it is during that time that ground cover and f_c are higher, thus allowing less solar energy to reach the soil surface. By the end of season and during the second half of the late season, differences in the K_c value slightly increase again, especially when the soil is less shaded during this period. Users are therefore advised to divide the crop season into three time-periods when estimating reductions in K_c due to mulching as indicated in Table 10.1 and 10.2.

The smaller values of K_c reduction often refer to black plastic mulch because it has a low albedo, thus providing more energy for soil evaporation under mulch, thus contributing to a convergence of the K_c for mulched and non-mulched surfaces. Contrarily, mulches with transparent films likely tend to create large differences in K_c. However, users will be informed with better detail when searching in bibliographies.

When estimating basal K_{cb} for mulched crops, less adjustment is normally needed to the K_{cb} curve, likely being half the order of reduction in overall K_c, as it is generally understood that the "basal" evaporation of water from the soil surface that is associated with the standard K_{cb} value is less with a plastic mulch, though the transpiration is likely increased. The effect on K_{cb} can be greater with low density crops, e.g. with tree orchards and vine crops. Local calibration of K_{cb} (and K_c) for use with mulch is encouraged, namely using soil water balance models and computing soil evaporation K_e.

Where calculating the soil evaporation coefficient K_e with plastic mulch (Example 10.1), the f_w should represent the relative equivalent fraction of the ground surface that can contribute to evaporation through the vent holes in the plastic cover and to the fraction of surface that is wetted and is not covered by the mulch. The effective area of vent holes is normally two to four times the physical area of the vents (or even higher) to account for vapour transfer from under the sheet.

EXAMPLE 10.1
Effects of surface mulch

A plastic mulch is placed over cucumbers under drip irrigation. The mulch is clear plastic covering the entire field surface, with small openings at each plant. Adjust both the single and basal K_c values for this crop to reflect the presence of the mulch.

From Table 6.1, $K_{c\ ini}$, $K_{c\ mid}$ and $K_{c\ end}$ for fresh market cucumbers have values equal to 0.60, 1.00 and 0.75.

As the plastic mulch is continuous with only small vents at each plant, the $K_{c\ ini}$ is assumed to be only 0.10 (this value should be adjusted upward if precipitation occurs).

The $K_{c\ mid}$ and $K_{c\ end}$ values are estimated as:

$$K_{c\ mid} = 0.85\ (1.0) = 0.85$$

$$K_{c\ end} = 0.85\ (0.75) = 0.64$$

where the 0.85 multipliers are derived from Table 10.1 and reflect an assumed 15 percent reduction in ET_c due to the mulch, assuming an approximately weekly irrigation frequency.

From Table 7.1, the values for $K_{cb\ ini}$, $K_{cb\ mid}$, and $K_{cb\ end}$ are 0.15, 0.95 and 0.70 for this same cucumber crop. The $K_{cb\ ini}$ is assumed to be similar to the $K_{c\ ini}$ for mulched cover and is therefore set equal to 0.10. The $K_{cb\ mid}$ and $K_{cb\ end}$ values are estimated to be reduced by near 10 percent so that:

$$K_{cb\ mid} = 0.90\ (0.95) = 0.85$$

$$K_{cb\ end} = 0.90\ (0.7) = 0.63$$

These basal values are similar to the adjusted values for K_c. This is expected as evaporation from the mulch covered surface can be ignored. Additional adjustment to these K_c values to account for climate is necessary using Eq. 6.16 and 7.1.

The values for mean K_c and K_{cb} are similar with values of 0.10 for the initial stage, 0.85 for the mid-season stage and 0.64 at the end of the late season stage.

10.1.2 Organic mulches

Effects of organic mulches are considered through the differences between mulched and non-mulched K_c as in Table 10.1 and 10.2. Organic mulches are often used with orchard production and with row crops under reduced tillage operations. Organic mulches may consist of unincorporated plant residues or foreign material imported to the field such as straw. The depth of the organic mulch and the fraction of the soil surface covered can vary widely. These two parameters will affect the amount of reduction in evaporation from the soil surface, with K_c differences increasing when larger amounts of mulch are used, namely straw mulch, as well as when the percentage of ground coverage is larger.

A general rule when applying K_c from Tables 6.1, 6.2 and 6.3, and when applying the dual $K_{cb} + K_e$ method, is to reduce the amount of soil water evaporation by about 5 percent for each 10 percent of soil surface that is effectively covered by an organic mulch. For example, if 50 percent of the soil surface were covered by an organic crop mulch of high density, then the soil evaporation would be reduced by about 25 percent. In the case of $K_{c\ ini}$, which represents mostly evaporation from soil, one would reduce $K_{c\ ini}$ by about 25 percent. For less dense mulches, this percentage should be reduced by 5 or 10 percent. In the cases of $K_{c\ mid}$ and $K_{c\ end}$, one may reduce these values by 25 percent of the difference between ($K_{c\ mid} - K_{cb\ mid}$) and ($K_{c\ end} - K_{cb\ end}$) from Tables 6.1, 6.2 and 6.3 and Tables 7.1, 7.2 and 7.3. Generally, the differences between values in these Tables are only 5 - 10 percent for mid- and late-season, so that the adjustment to $K_{c\ mid}$ and $K_{c\ end}$ to account for an organic mulch may not be very large. However, one may instead consider utilizing information in Table 10.1 and 10.2.

When applying the dual crop coefficient approach ($K_c = K_{cb} + K_e$) with a separate water balance of the surface soil layer, the magnitude of the evaporation component ($K_e\ ET_o$) should be reduced by about 5 percent for each 10 percent of soil surface covered by the organic mulch. K_{cb} is not changed.

These recommendations are only approximate and attempt to account for the effects of partial reflection of solar radiation from residue, micro advection of heat from residue into the soil, lateral movement of soil water from below residue to exposed soil, and the insulating effect of the organic cover. As these parameters can vary widely, local observations and measurements are required if more precise estimates are required.

10.2 INTERCROPPING

Intercropping refers to the situation where two different crops are grown together within one field. For the estimation of the crop coefficient, a distinction is made between (see Figure 10.1):

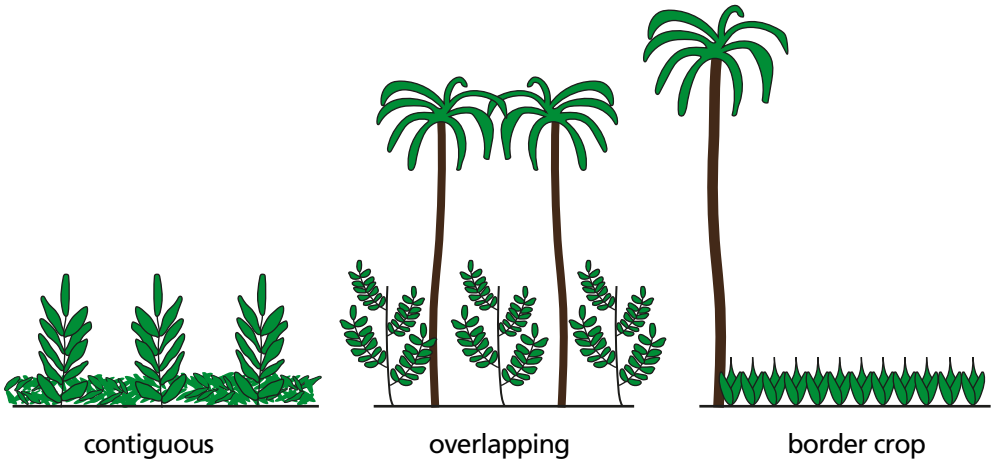
- contiguous vegetation, where the canopies of the two crops intermingle at some height (e.g. corn and beans intercropping);
- overlapping crops, where the canopy of one crop is well above that of the other so that the canopies cannot be considered to be contiguous (e.g. date trees overlapping pomegranate trees at an oasis); and
- border crops, where tall crops such as windbreaks border fields of shorter crops, or high trees border a field crop.

There is an upper limit to the energy available to evaporate water. This is represented by $K_{c\ max}$ (Equation 7.4 bis) for all crops in cultivated fields larger than 3 ha:

$$K_{c\ max} = \max\left(\left\{1.2 + [0.04\,(u_2 - 2) - 0.004\,(RH_{min} - 45)]\left(\frac{h}{3}\right)^{0.3}\right\}\{K_{cb} + 0.05\}\right) \quad (7.4\text{ bis})$$

where h is the height for the taller crop. Under all conditions when combining crop coefficients for multiple crops, K_c should be constrained by this upper bound ($K_c \leq K_{c\ max}$).

FIGURE 10.1
Different situations of intercropping: contiguous referring to two field crops, e.g. corn and beans; overlapping when tall trees and short trees are intercropped, e.g. date palm trees overlapping pomegranate trees at an oasis; and border crop tall tree crops are planted on the border of a field, e.g. date palm bordering alfalfa in a oasis, or when windbreaks protect horticultural crops



Source: Authors' own elaboration

10.2.1 Contiguous vegetation, single K_c

Where the taller crop has canopy foliage that extends down to the same elevation as that of the top of the shorter crop, the vegetation canopy can be considered to be contiguous. For example, in Africa and South America, maize and beans are frequently intercropped as contiguous vegetation, with one row of maize planted per one or more rows of beans. Another example is the cultivation of five to seven rows of wheat intercropped with three rows of maize in many areas of China.

Case of single K_c for similar fractions of ground covered by both crops

Where the leaf area or fraction of ground covered by the vegetation (f_c) is similar for each crop, the K_c in Tables 6.1 to 6.3 and Tables 7.1 to 7.3 for the taller crop (if this K_c is higher) can be taken to represent the entire field. The taller crop will act in some sense as a clothesline so that K_c (and ET_c) for the taller crop per unit of ground area is increased over that given in Tables 6.1 to 6.3 and Tables 7.1 to 7.3. However, the K_c (and ET_c) for the shorter crop will be reduced due to the windbreak effect by the taller crop. As a result, the K_c for the field as a whole may be similar to the weighted average of the K_c values for the two crops from Tables 6.1 to 6.3 and Tables 7.1 to 7.3, or, the total K_c may more closely follow the K_c predicted for a field sown entirely to the taller crop ($K_{c\ field} \approx K_{c\ taller\ crop}$). Yields for the shorter crop may be reduced relative to those for single cultivar production due to the effects of shading by the taller crop and the competition for soil water.

Case of single K_c for different fractions of ground covered by both crops

Where the fractions of ground covered by each crop are different, the K_c for an intercropped field can be estimated by weighting the K_c values for the individual crops according to the fraction of area covered by each crop and by the height of the crop:

$$K_{c\ field} = \frac{f_1 h_1 K_{c1} + f_2 h_2 K_{c2}}{f_1 h_1 + f_2 h_2} \quad (10.1)$$

where f_1 and f_2 are the fractions of the field surface planted to crops 1 and 2, h_1 and h_2 are the heights of crops 1 and 2, and K_{c1} and K_{c2} are the K_c values for crops 1 and 2.

EXAMPLE 10.2
Intercropped maize and beans

Determine the representative $K_{c\ mid}$ for a situation where a single 1 m wide row of maize is grown for each 2 m of beans, where $RH_{min} \approx 45\%$ and $u_2 \approx 2$ m/s.

From Table 6.2, the $K_{c\ mid}$ and h for maize is 1.20 and 2.50 m and the $K_{c\ mid}$ and h for beans is 1.05 and 0.60 m. No correction is needed for climate. The representative $K_{c\ mid}$ can be obtained by weighting the individual $K_{c\ mid}$ values according to the fraction of the field surface allocated to each crop ($f_1 \approx 0.30$ for maize and $f_2 \approx 0.70$ for beans) and according to the heights of the crops as (Eq. 10.1):

$$K_{c\ mid} = \frac{0.30\ (2.50)\ \ (1.20)\ +\ 0.70\ (0.60)\ \ (1.05)}{0.30\ (2.50)\ +\ 0.70\ (0.60)} = 1.15$$

Values can be obtained for daily K_c in a similar manner by constructing individual K_c curves and then weighting interpolated values from the individual K_c curves for any specific day using Eq. 10.1.

The crop coefficient for the mid-season and the entire field is 1.15.

10.2.2 When adopting the dual crop coefficient for relay intercropping

Relay intercropping describes a cropping pattern in which the lifecycle of one crop may only partially overlap that of the other crop. As a result, the timing of the midseason periods can be displaced between the crops, as shown in Figure 10.2, and the K_{cb} values may need to be calculated daily. The A&P approach (Chapter 9) can be used to estimate K_{cb} for relay intercropped crops by following the progression in the density of vegetation for each crop. The daily evolution of K_{cb} is estimated using the density coefficient, K_d, however modified from Equation 9.13. Since the intercropped crops will usually have different characteristics, it is necessary to know the fraction of the soil surface covered by each crop (Frac) and the height (h) of each crop. It is assumed that in the total crop cycle one crop is dominant (subscript dom) while the other is considered subordinate (subscript sub). The K_{cb} of the intercrop ($K_{cb\ inter}$) is estimated daily or weekly (Figure 10.2) as:

$$K_{cb\ inter} = \max\left[K_{cb\ sub} + K_{d\ dom}(K_{cb\ dom} - K_{cb\ sub});\ K_{cb\ dom} + K_{d\ sub}(K_{cb\ sub} - K_{cb\ dom})\right] \quad (10.2)$$

where $K_{cb\ dom}$ K_{cb} of the dominant crop when grown/cultivated alone [-],
$K_{cb\ sub}$ K_{cb} of the subordinated crop when grown/cultivated alone [-],

$K_{d\ dom}$ density coefficient for the dominant crop [-],
$K_{d\ sub}$ density coefficient for the subordinated crop [-].

K_d is estimated differently from Equation 9.13 and the rationale for changing the K_d calculation is to make K_d dependent on the height difference between the two associated crops since factors like the sunlight interception, aerodynamic canopy roughness and micro-advection of energy implied in K_d are then better accounted for. Thus, K_d is estimated for each crop as:

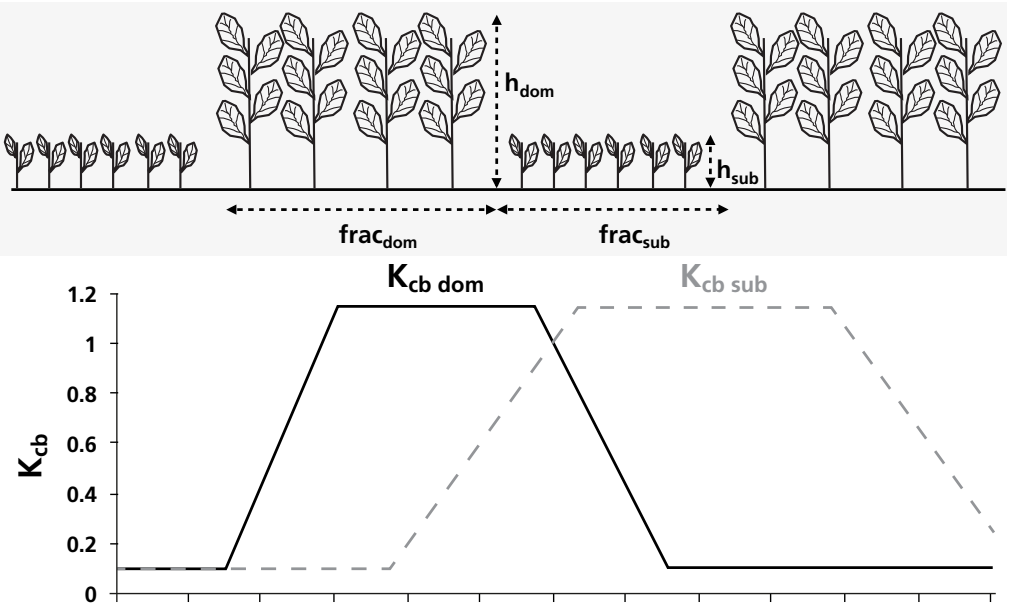
$$K_{d\ dom} = Frac_{dom}^{\left(\frac{1}{1+\max(h_{dom}-h_{sub};\,0)}\right)} \quad (10.3a)$$

$$K_{d\ sub} = Frac_{sub}^{\left(\frac{1}{1+\max(h_{sub}-h_{dom};\,0)}\right)} \quad (10.3b)$$

where $Frac_{dom}$ fraction of the soil surface cropped with the dominant crop [-],
$Frac_{sub}$ fraction of the soil surface cropped with the subordinated crops [-],
h_{dom} height of the dominant crop [m],
h_{sub} height of the subordinated crop [m].

The max function in Equations 10.3 considers that the conditions observed during earlier stages, where $K_{cb\ dom} > K_{cb\ sub}$ and $h_{dom} > h_{sub}$, may change with the development of the subordinated crop, thus with conditions changing to $K_{cb\ sub} > K_{cb\ dom}$ and $h_{sub} > h_{dom}$.

FIGURE 10.2
Contiguous or relay intercropping density and basal crop coefficient estimation (Eqs. 10.2 and 10.3)



Source: Authors' own elaboration

BOX 10.1

Estimation of the depletion fraction for no stress for relay intercropping

When performing the soil water balance for intercropped crops the estimation of the readily available water and of the stress coefficient needs an adjustment aiming to express the possible higher sensitivity to water deficits by one of the crops, particularly during mid-season stage. Thus, when both the dominant and the subordinated crops occupy an equal soil fraction, Frac, a higher weight is given to the depletion fraction for no stress value of the crop that has a largest demand for water, i.e. that has a higher K_{cb}.

Thus, the depletion fraction for no stress for the intercrop, p_{inter}, is performed as follows:

1) when there is only one component crop in the field, as may be the case during the earlier stages of the dominant crop and later stages of the subordinated crop, p_{inter} is that of the considered crop that is present;

2) during the co-growth period, p_{inter} is estimated as a weighted mean of the p values of both crops.

The respective weights consist of the fractions of the soil surface occupied by each crop (Frac) multiplied by the respective K_{cb} value. Thus, p_{inter} is computed as

$$p_{inter} = p_{dom} \qquad \text{If } Date_{act} < Date_{plant\ sub} \qquad (a)$$

$$p_{inter} = \frac{p_{dom} Frac_{dom} K_{cb\ dom} + p_{sub} Frac_{sub} K_{cb\ sub}}{Frac_{dom} K_{cb\ dom} + Frac_{sub} K_{cb\ sub}} \qquad \text{If } Date_{plant\ sub} \leq Date_{act} < Date_{plant\ dom} \qquad (b)$$

$$p_{inter} = p_{sub} \qquad \text{If } Date_{act} > Date_{harv\ dom} \qquad (c)$$

where

p_{dom}	is the value for depletion fraction for no stress for the dominant crop [-]
p_{sub}	is the value for depletion fraction for no stress for the subordinated crop [-]
p_{inter}	is the value for depletion fraction for no stress for the intercrop [-]
$Frac_{dom}$	is the fraction of cropped soil occupied by the dominant crop [-]
$Frac_{sub}$	is the fraction of cropped soil occupied by the subordinated crop [-]
$K_{cb\ dom}$	is the basal crop coefficient of the dominant crop when single cropped [-]
$K_{cb\ sub}$	is the basal crop coefficient of the subordinated crop when single cropped [-]
$Date_{act}$	is the current date relative to the intercrop estimations
$Date_{plant\ sub}$	is the sowing or planting date of the subordinated crop
$Date_{plant\ dom}$	is the sowing or planting date of the dominant crop

(a), (b) and (c) represent, respectively, the time periods between planting of the dominant crop and planting of the subordinated crop, the time period from planting of the subordinated crop and harvesting of the dominant crop, and the time period between harvesting of the dominant crop and harvesting of the subordinated crop.

BOX 10.2

Example of estimation of basal crop coefficient for relay intercropping wheat-sunflower

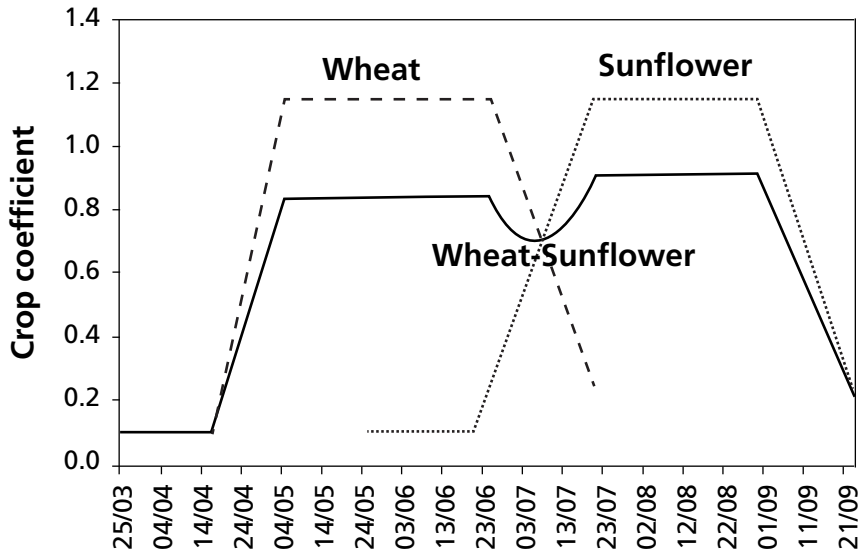
The results of applying the $K_{c\ inter}$, Equation 10.2, to a wheat-sunflower relay intercropping system are shown in the figure below. The $ET_{c\ act}$ and its partition were estimated using the SIMDualKc soil water balance model after careful calibration against observed soil water content data. The experimental studies included fields with separately grown crops and another with the crops intercropped. Therefore, the model was first calibrated for each individual crop and tested against the data collected in the field with the intercropped plants.

Wheat was sown by 25 March and harvested by 20 July while sunflower was sown by 25 May and harvested by 23 September. When single cropped, the maximum fraction of ground cover by wheat was 0.87 and by sunflower was 0.85. The maximum crop height was respectively 0.87 m and 1.95 m for wheat (dominant) and sunflower (subordinate). When intercropped, the fraction of the soil surface cropped with the dominant crop, wheat, was 0.54 while for the subordinated crop, sunflower, it was 0.46 (note that 0.54 + 0.46 = 1.0).

As represented in the Figure 10.3, the K_{cb} values of single cropped wheat and sunflower were 0.10, 1.15 and 0.25 for the initial, mid-season and end seasons, respectively. In the case of intercropping, $K_{cb\ inter}$ = 0.84 during the entire mid-season of wheat, because wheat only partially covered the ground. After the mid-season, the combined $K_{cb\ inter}$ decreases to 0.70 during wheat senescence when $K_{cb\ sunflower} < K_{cb\ wheat}$. The value of $K_{cb\ inter}$ increases smoothly when $K_{cb\ sunflower} > K_{cb\ wheat}$, and then reaches $K_{cb\ inter}$ = 0.91 when the wheat is harvested and the sunflower becomes the only crop and reaches mid-season (full growth). $K_{cb\ inter}$ value begins to decrease as sunflower starts to senesce and reaches the minimum value at harvest. The calculations for $K_{cb\ inter}$ are done on a daily basis.

FIGURE 10.3

Predicted daily intercropping basal crop curve (thick line), basal crop coefficient of the dominant crop (dash line) and basal crop coefficient for the subordinated crop (dot line) for wheat-sunflower relay intercropped at Hetao Irrigation District, Inner Mongolia, China



Source: Authors' own elaboration

10.3 OVERLAPPING VEGETATION

Where intercropping entails overlapping of another crop's spacing, the canopy of one crop is well above the other. This is the case, for example in southern California, where citrus trees are planted in tall date palm groves. Where a normal dense spacing is used for both the dates and for the citrus trees, the K_c may increase as the density of the combined vegetation increases, proportional to the increase in LAI (Example 10.3), with maximum K_c constrained by either $K_{c\ max}$ (Equation 7.4 or 9.8) or by $K_{cb\ full}$ (Equations 9.11) unless the total field area is small so that there is an additional clothesline or oasis effect.

EXAMPLE 10.3
Overlapping vegetation

A 20 ha date palm grove in Palm Desert, California, the United States has a tree spacing of 6 m. Interplanted among the rows of palms are small orange trees (50 percent canopy) on a 6 m spacing. The palm and citrus trees are 3 m from one another in the rows. Height of the palms is 10 m and height of the citrus is 3 m. The canopy foliage of the palms is well above that of the citrus so that the canopies cannot be considered as contiguous. Mean average RH_{min} during the mid-season is 20 percent and u_2 = 2 m s^{-1}. The $K_{c\ mid}$ from Table 6.3 for dates is 0.95 and when adjusted for humidity and wind using Eq. 6.16 is $K_{c\ mid}$ = 1.09. The $K_{c\ mid}$ from Table 6.3 for citrus having 50 percent canopy with no ground cover is 0.70 and when adjusted for humidity and wind using Eq. 6.16 is $K_{c\ mid}$ = 0.80.

The interplanting of citrus among the date palms has increased the total leaf area of the orchard so that ET_c for the combined planting (palms and citrus together) will be greater than for either planting alone. The estimated combined $K_{c\ mid}$ will be estimated as a function of the increase in total LAI. First the LAI values of the individual plantings are estimated by inverting Eq. 9.4 to solve for LAI:

$$LAI = -\ 1.4\ \ln\left[1 - \frac{K_{cb} - K_{c\ min}}{K_{cb\ full} - K_{c\ min}}\right]$$

where $K_{c\ min}$ is the minimum basal K_c for bare soil (≈ 0.15 to 0.20) and $K_{cb\ full}$ is the maximum mid-season K_c expected for the crop if there were complete ground cover, calculated using Eq. 9.4. Based on Eq. 9.4, with h = 10 m for the date palms and h = 3 m for the citrus, the $K_{cb\ full}$ values for the two crops, assuming complete ground cover for each, are $K_{cb\ full}$ = 1.34 for palms and $K_{cb\ full}$ = 1.30 for citrus (using RH_{min} = 20% and u_2 = 2 m s^{-1}). These estimates ignore effects of any unique stomatal control (F_r = 1.0). Therefore, using the above equation, the effective LAI values of the date palms and citrus are estimated to be approximately:

$$LAI_{palms} = -\ 1.4\ \ln[1-\ (1.09 - 0.15)/(1.34 - 0.15)] = 2.2$$

$$LAI_{citrus} = -\ 1.4\ \ln[1-\ (0.80 - 0.15)/(1.30 - 0.15)] = 1.2$$

Therefore, the effective LAI for the date palm-citrus combination is estimated to be approximately:

$$LAI_{combined} = LAI_{palms} + LAI_{citrus} = 2.2 + 1.2 = 3.4$$

The increase in $K_{c\ mid}$ for the date palm orchard resulting from the increase in LAI from the interplanting of citrus is estimated using a ratio of the LAI-based function introduced in Eq. 9.14, which results in the relationship:

$$K_{c\ mid} = K_{c\ mid,\ single\ crop}\left[\frac{1 - \exp\ (-0.7\ LAI_{combined})}{1 - \exp\ (-0.7\ LAI_{single\ crop})}\right]$$

EXAMPLE 10.3 (CONT.)

where $LAI_{combined}$ is the LAI for the two intercropped plantings combined and $LAI_{single\ crop}$ is the LAI for the taller, single crop. $K_{c\ mid}$ single crop is the mid-season K_c for the taller, single crop (in this case the date palms). In this application, the above equation is solved as:

$$K_{c\ mid} = 1.09 \left[\frac{1 - \exp(-0.7(3.4))}{1 - \exp(-0.7(2.2))} \right] = 1.26$$

Therefore, the $K_{c\ mid}$ estimated for the complex of date palms and citrus together is 1.26. This value is compared with the maximum expected $K_{c\ based}$ on energy limitations, represented by $K_{c\ max}$ of Eq. 7.4 which in this case for h=10 m is $K_{c\ max}$ = 1.34. Because $K_{c\ mid} < K_{c\ max}$ (i.e., 1.26 < 1.34), the $K_{c\ mid}$ = 1.26 is accepted as the approximate estimate of the $K_{c\ mid}$ for the intercropped field.

10.4 BORDER CROPS

Where tall crops such as windbreaks or date palms border fields of shorter crops, the upper storey of the taller crop can intercept extra sensible heat energy from the air stream. Under these conditions, the K_c is weighted according to the areas for each crop. However, prior to the weighting, the K_c for the border crop, if taller than the field (interior) crop, should be adjusted for a potential clothesline impact.

10.5 K_c FROM SMALL AREAS OF VEGETATION

Natural vegetation and some subsistence agriculture frequently occurs as small groups or stands of plants. The value for K_c for these small stands depends on the type and condition of other vegetation surrounding the small stand.

In the majority of cases for natural vegetation or for non-pristine agricultural vegetation, the value for K_c must adhere to upper limits for K_c of approximately 1.20-1.30, when the areal expanse of the vegetation is larger than about 2 000 m². This is required because ET from large areas of vegetation is governed by one-dimensional energy exchange principles and by the principle of conservation of energy. ET from small stands (< 2 000 m²) may be impacted by advected energy from beyond the stand and will adhere to these same principles and limits only where the vegetation height, leaf area, and soil water availability are similar to that of the surrounding vegetation.

10.6 K_c DURING NON-GROWING PERIODS

This section describes procedures for estimating ET_c and crop coefficients during non-growing periods. Non-growing periods are the periods during which the crop is dormant. Generally, the non-growing periods refer to periods of frost and continuously frozen conditions in case of annual crops and to the dormancy periods in the case of perennials, namely the evergreen trees and shrubs.

10.6.1 Annual winter crops

The type and condition of the ground surface during non-growing periods largely influences the range for ET_c. Where the surface is covered by nearly dead vegetation, or some type of organic mulch or crop residue, then the K_c will be similar to that for agriculture that uses a surface mulch. Where the surface is covered by weed growth or growth of "volunteer" plants, then the K_c will vary according to the leaf area or fraction of ground covered by the vegetation and by the availability of soil water. Where the surface is snow covered or frozen, then the K_c is difficult to predict and a constant value for K_c may have to be assumed (Box 10.4).

Where the ground is left mostly bare following harvest, then the K_c following harvest will be strongly influenced by the frequency and amount of precipitation or irrigation when a post-harvest leaching irrigation is applied. K_c for bare soil may be estimated as $K_c = K_{c\ ini}$ where $K_{c\ ini}$ is computed using the procedure proposed in Chapter 6. However, as for the wheat case in North China Plain (Box 10.4), if the post-harvest period is wet and K_c is then high, a $K_{c\ n\text{-}gro}$ of around 0.25-0.50 may be selected.

Where a daily soil water balance is applied, the user may elect to apply the dual K_{cb} approach (Chapter 7), preferably using a simulation computer model. If the topsoil layer dries to very low water contents during periods having no precipitation, the daily water balance calculation with K_{cb} not far from 0 will generally provide accurate estimates of ET_c during the non-growing periods.

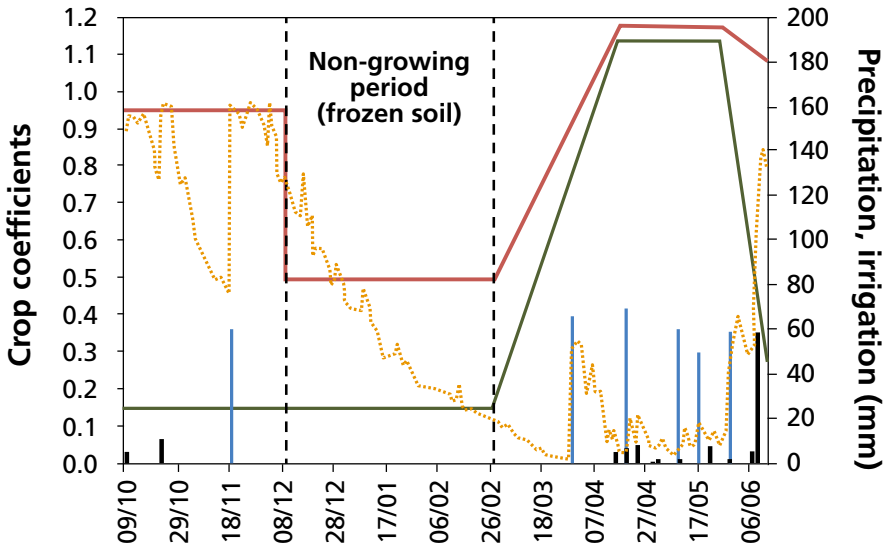
BOX 10.3

Case study of the estimation of basal and single crop coefficients for the non-growing period of a winter wheat crop under frozen soil conditions at Daxing, North China Plain South of Beijing

Results of applying the dual crop coefficient approach ($K_{cb} + K_e$) to a winter wheat crop cultivated at Daxing, North China Plain, are briefly analyzed. The observed $ET_{c\ act}$ data calculated with the soil water balance model SIMDualKc using soil water content measurements, and soil evaporation data measured with micro-lysimeters. Figure 10.4 shows the K_{cb} and K_e curves as well as the mean averaged $K_{c\ act}$ for the entire crop season. $K_{cb\ act}$ coincided with potential because the crop was not stressed.

Crop sowing was performed by early Fall, the 9th October, when the soil was wet by a previous large irrigation, and the non-growing period started when soil frozen initiated, by the 8th December. The development of the crop was quite limited due to weather conditions typical of a cold autumn. The soil kept frozen, with a few quite small snow events, and subject to freezing and thawing early December and late February. By 26 February, freezing in the root zone ended and the crop started to revive. The crop growth stage initiated then until the start of the mid-season in April. The K_{cb} curve resulting from the calibration of the model shows a constant value equal to $K_{cb\ ini}$ until reviving, i.e. a $K_{cb\ n\text{-}grow} = K_{cb\ ini}$. Differently, the actual single crop coefficient before the non-growing period, $K_{c\ act\ ini}$, was 0.95 due to stored soil water after the winter leaching irrigation. Resulting from averaging $K_{c\ n\text{-}grow}$ computed in various years of observations and from the simulate K_e curve, the value $K_{c\ n\text{-}grow} = 0.50$ was assumed. However, this value could be different if the $K_{c\ act}$ curve did not fall vertically but was an inclined straight line for the transition between the antecedent and the non-growing period.

FIGURE 10.4
Basal crop coefficient curve (full green line), evaporation coefficient curve (dotted brown line), and average actual single crop coefficient (thick full red line) for a basin irrigated winter wheat crop at Daxing, North China Plain



Source: Authors' own elaboration

Where the ground surface has a plant residue or other dead organic mulch cover, or where part of the unharvested crop remains on the surface in a dead or senesced condition, then the surface will respond similarly to a surface covered by mulch. In this case, K_c can be set equal to $K_{c\ ini}$ but the value for $K_{c\ ini}$ can be reduced by about 5 percent for each 10 percent of soil surface that is effectively covered by an organic mulch. Evaporation from dead, wet vegetation can be substantial for a few days following a precipitation event. Therefore, in the dual K_c approach, the value for f_c should be set close to zero to reflect the lack of green cover and f_w should be set equal to 1.0 to reflect the wetting of both soil and mulch cover by precipitation.

The dead mulch or vegetation will dry more quickly than would the underlying soil if it were exposed. The soil will be protected somewhat from evaporation by the dead mulch or vegetation cover. Therefore, total evaporation losses will be less than the predicted TEW. This can be accounted for by reducing the value for TEW by 5 percent for each 10 percent of soil surface that is effectively covered by an organic mulch or vegetation residues. The value for REW should be limited to REW ≤ TEW.

Where the ground surface is snow covered or frozen, any vegetation will be largely non-responsive and non-contributing to ET_c, and the amount of ET_c will be closely related to the availability of free water at the surface and to the albedo of the surface. The albedo of snow-covered surfaces can range from 0.40 for old, dirty snow cover, to 0.90 for fresh, dry snow. Therefore, the ET_c for snow-cover will be less than ET_o for grass, as 25-85 percent less shortwave energy is available. In addition, some energy must be used to melt the snow before evaporation.

Given the limited value of ET_o (or even ET_p) under snow-covered or frozen conditions, a single, average value may be best used to predict ET_c as for the example in Box 10.2 relative to winter wheat in North China Plain. The soil is deeply wet there because a large pre-sowing irrigation is applied for leaching of salts. Therefore, the period

antecedent to the non-growing period has K_c much larger than $K_{c\ n\text{-}gro}$. The user may select different K_c transitions between previous to and during the non-growing period, decreasing for a few days or abrupt like represented in Box 10.2.

For winter periods at Kimberly, Idaho, the United States, it was found that ET_c averaged 1 mm day^{-1} over nearly six month long winters (1 October to 30 March). Over the six-year study period, the ground was 50 percent covered by snow for 25 percent of the time. The ground, when exposed, was frozen about 50 percent of the time. The K_c averaged 0.25 during periods when the soil was not frozen but where frosts were occurring (October and early November). When the ground had 50 percent snow cover or greater, the ET_c averaged only 0.4 mm day^{-1}. Over the six-month non-growing period, total cumulative ET_c exceeded precipitation by about 50 mm. The ET_c/R_{so} ratio averaged only 0.17 over the six-month period, and averaged 0.11 from 1 Dec. – 10 Mar. The ET_c/R_s ratio averaged 0.23 over the six-month period, and averaged 0.15 from 1 Dec. – 10 Mar. A similar study conducted in Logan, Utah, the United States (latitude 41.6°N, elevation 1 350 m) over an eight-year period showed that ET_c varied widely with soil surface wetness and air temperature during the winter months. The "average" K_c from November to March was 0.5 for days having no snow cover. For days with snow cover, "ET_c" ranged from 0 to 1.5 mm day^{-1}.

10.6.2 Evergreen trees and shrubs

The period of dormancy of evergreen trees and shrubs generally has a K_c behaviour similar to annual crops where the soil is not frozen, or snow covered. The use of a soil water balance simulation model eases computations and should increase the accuracy of estimates when LAI and f_c inputs are adequate.

BOX 10.4

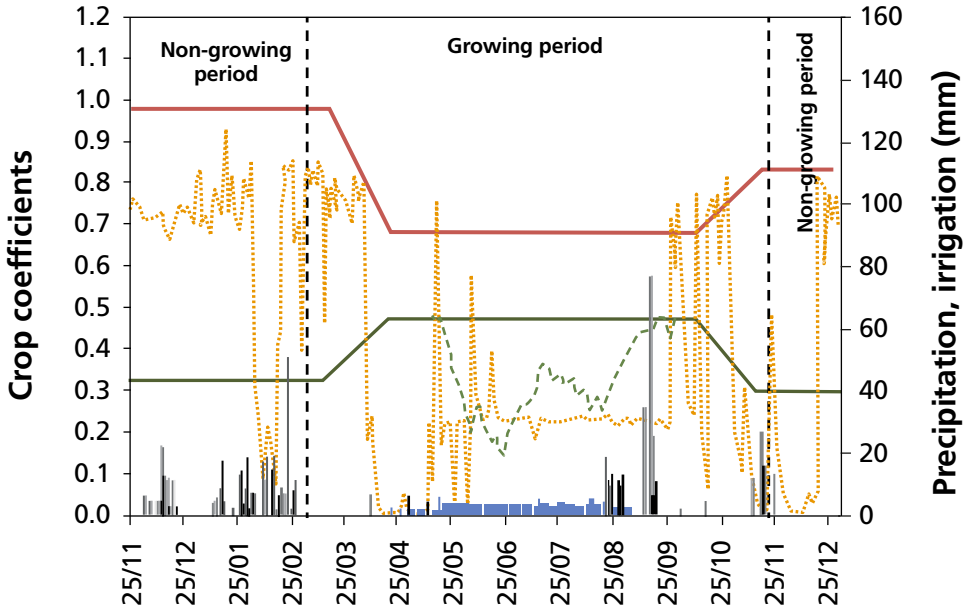
Estimation of crop coefficients for non-growing periods of an irrigated hedgerow olive orchard at Viana do Alentejo, Portugal

Crop coefficient curves resulting from applying the (K_{cb} + K_e) method to a super-high density hedgerow olive orchard are shown in the Figure 10.5. The observed $ET_{c\ act}$ data were measured using an eddy covariance system and actual T_c was measured with the sap flow method. This example uses the same data that were used in the case study of Box 8.4. The figure below shows the crop season variation of potential and actual K_{cb} and K_e curves as well as the variation of $K_{c\ act}$ along the full crop season. The respective curves were estimated from the calibration of the soil water balance model SIMDualKc using the ET_c and T_c measurements. Snow or soil freezing do not occur there but most of precipitation occurs during the non-growing period. Therefore, the K_c and K_{cb} curves show a behaviour resulting from the opposed rainfall distribution in the growing and non-growing periods. The olive orchard is mostly supplied by irrigation since rainfall is very short during the growing period.

The occurrence of precipitation and irrigation events is shown in the Figure 10.5, includes the out of season and dormant period. All irrigation wetting events occurred in the mid-season stage when precipitation and soil water storage were not sufficient to maintain the crop transpiration with the actual K_{cb} curve below the full line representing the potential K_{cb} curve. The soil evaporation K_e curve peaked after wetting events by rain or irrigation. It is important to note that even though values for K_c may be higher during the dormant period than during the growing period, the ET rates may typically be much lower during the dormant period as compared to during the growing period in mid- to high-latitudes because ET_o during the dormant periods is typically much lower than during midseason months.

FIGURE 10.5

Potential basal crop coefficient curve (full green line), actual K_{cb} curve (dashed green line), actual single averaged K_c (thick red full line) and soil evaporation coefficient curve (dotted line), for a drip irrigated hedgerow olive orchard at Viana do Alentejo, Portugal. Vertical full lines denote precipitation or irrigation events



Source: Authors' own elaboration

It is important that the K_c for vegetation during the non-growing period be limited according to the amount of soil water available to supply evapotranspiration. Otherwise, the law of conservation of mass will be violated. Under all conditions, the integration of K_c ET_o over the course of the non-growing period cannot exceed the sum of the precipitation occurring during the period plus any residual soil water in the root zone following harvest that can be depleted by the subsequent vegetation. The root zone in this case is the root zone for the weed or volunteer crops. A daily soil water balance may provide for the best estimate of soil water induced stress and associated reduction in K_c and ET_c namely when a simulation model is used. The soil water balances will automatically adhere to the law of conservation of mass, so that total ET_c from the weed or volunteer vegetation will not be overestimated.

During non-growing periods, weeds may germinate and grow. This vegetation is supplied with water from storage in the soil profile and from rainfall as it is the case of the olive orchard in a Mediterranean climate in Box 10.5. The amount of ground surface covered by vegetation will vary with soil management, e.g. the frequency and extent of soil tillage, and the availability of soil water and rain. If the single crop coefficient approach has to used in water management, the value for K_{cb} determined using procedures in Chapter 9 can be converted into an equivalent K_c by adding 0.05 to 0.15 according to the frequency of rainfall and surface wetting.

10.7 STRATEGIES, METHODS AND TECHNOLOGIES TO ACHIEVE WATER SAVINGS

In recent decades, research has produced a variety of strategies, methods, and technologies to reduce non-beneficial water use on farms for adaptation to the rising occurrence of water scarcity, agricultural droughts, and multi-sectoral rivalry for water, all which are increased by climate change. Adopting consumptive water-saving practices (WS_{ET}), i.e. that refer to $ET_{c\,act}$ requires accurate determination of crop water requirements using the FAO56 methods (K_c ET_o) under various water availability and management techniques, as described in the previous chapters. However, the non-consumptive water-saving practices (WS_{DP+RO}), that refer to controlling deep percolation (DP) and runoff (RO) at farm and off-farm, generally originate WS_{DP+RO} > WS_{ET}. Both DP and RO refer to excess irrigation water applied relative to ET_c and to limitations of the irrigation technologies and of the irrigators. In fact, water-saving practices must base not only on well-known net crop irrigation requirements but also on the availability of appropriate irrigation technologies and management.

This section, therefore, outlines ideas and methods for reducing non-beneficial consumptive and non-consumptive water use, mainly referring to ET, DP and RO ($NBWU_{ET}$ and $NBWU_{DP+RO}$) while maintaining high crop water productivity on farms. These achievements tipically require maintaining high levels of water use for transpiration and managing the irrigation systems targeting ET_c. In particular, this section describes water-saving routines linked to irrigation scheduling and management for irrigation systems, crop and soil management, mulching, remote sensing techniques, plant conditioners, and micro-climate adaptation. Controlling non-beneficial off-farm water use and non-consumptive farm water use imply different management and technological issues not dealt with in FAO56 but that are closely associated with the on-farm water management techniques and practices previously referred to and described hereafter. Off-farm reductions in water consumption include the adoption of modernized irrigation water collection, transport and distribution to farms and fields, and the use of automation for improving those related functions not described herein. All of these practices require an assessment of the local and regional hydrology to determine their overall impacts on reductions in water use.

It is important to distinguish between water "use" and water "consumption" when assessing impacts of irrigation on the fresh water resource. Water consumption is the conversion of liquid water to vapour via the ET process, while the term water use describes the total extraction of water from the environment and used as irrigation water. Water use includes both the consumed part of the extracted water and the non-consumed part. In other words, water use consists of the consumptive use and the non-consumed part, that remains on the land surface but does not infiltrate into the irrigated field, and that stored below the surface. The first is runoff and the second is deep percolation as referred to before. If the non-consumed component flows back to a surface stream or to an underlying aquifer, then it remains as part of the freshwater system and is available for later reuse. However, if the non-consumed component flows into a saline sink such as the ocean or brackish ground water, then it is no longer part of the freshwater system and is considered to be consumed. What is commonly considered to be "water savings' is the difference between the common total water use, beneficial or not, and the water use when the irrigator manages water to avoid excess ET, or DP or RO.

10.7.1 Water-saving of consumptive water use (WS_{ET})

Irrigation scheduling and management

The main aim of irrigation management and scheduling may differ from an area to another or from a farm to another depending on the WS purpose. The aim of the farmer is to maximize the net economic benefit by increasing yield and/or reducing costs. This may often be achieved by maximizing the beneficial consumption water use ($ET_{c\ act}$) and actual yield (Y_a). As there is a strong relation between crop transpiration (T_c) and CO_2 assimilation (gas exchange through stomata), higher Y_a is achieved at the expense of more water transpired, mainly when aiming at little or no water stress. Chapter 8 develops the bases for irrigation scheduling and describes yield-water computations. Practical issues for determining crop coefficients under field practices and management are dealt with in Chapter 9.

Under conditions of water scarcity, the objective of irrigation management may be to optimize crop water productivity (WP_{ET}), i.e. the marketable yield per unit of water consumed if such a solution is economically appropriate. This involves knowing the economic water productivity ratio (EWPR), which is the ratio of the value of the marketable yield to the unit cost of water applied. In some instances, the goal of maximizing the economics of production may be achieved by inducing controlled crop water stress to reduce vegetative growth and to optimize the quality of marketable yields, including eustress as previously defined ("good stress"), especially in some fruit tree crops and vineyards.

To reduce water consumption and optimize global WP and farm returns, an irrigation schedule must be planned, where such a plan may or not include deficit irrigation, depending upon the crop and water availability. For annual crops, optimizing WP must include selection of the best planting date according to the local climate and/or the foreseen weather conditions ahead of planting. That selection requires knowledge of the relationships between evapotranspiration or transpiration deficit and yield decrease (Equations 8.18 and 8.19, and Chapters 8 and 9). Using irrigation scheduling modelling, irrigation events may be planned using short and medium range weather forecasts, which are produced by various international organizations, e.g. ECMWF, the European Centre for Medium-Range Weather Forecasts, WAMIS, the World AgroMeteorological Information Service, or the Weather Prediction Center, USA National Oceanic and Atmospheric Administration. Irrigation scheduling with the FAO56 methodologies requires appropriate selection of K_c and K_{cb} values tabulated in Chapters 6 and 7, and revised in their application depending on various specific conditions including planting density and water and salinity stressors as discussed in Chapters 8 and 9, as well as in the previous sections of this current chapter.

Assuming that the tabulated K_c and K_{cb} are likely representing the upper maximum or near-maximum values since they refer to pristine conditions, those values may be difficult or unreasonable to achieve in common practice where it is essential to achieve water savings. The same upper limits apply to the field attainment of crop coefficients: Tabulated K_c and K_{cb} should represent reference values relative to those observed from field data and can be useful to verify data quality and data handling approaches, and thus to question results when locally observed values are well above or well below the tabulated values.

Limited deficit irrigation (DI) is specifically introduced into the irrigation practice to allow some degree of crop water stress and Y_a reduction, with a goal to reduce soil evaporation (E_s) and water and chemical losses, whilst maintaining transpiration at levels that produce economically viable yields. Deficit irrigation can be considered to

be a water-saving strategy where soil water depletion in the root zone goes below the optimum yield threshold and therefore transpiration decreases. Irrigation is applied up to a certain level below field capacity so that occasional rainfall can replenish the soil water storage capacity. Less frequent irrigation can reduce evaporation from the soil surface. However, the uneven distribution of rainfall and variable weather conditions pose a risk that may lead to Y_a losses larger than desired or planned. To be a sustainable practice, reduction of water consumption in deficit irrigation should be proportionally higher than the reduction in Y_a, so that WP_{ET} (kg m^{-3}) is optimized, and EWPR is always positive (above the economic feasibility threshold, EWPR>1.0).

Published research has provided inconsistent evidence on performance of deficit irrigation as a water-saving and economic strategy. In general, during years of water shortage, farmers may choose to adopt deficit irrigation schedules for their less profitable and more water-stress tolerant crops. To be economically viable, the cost of reduction of water use should be higher than the cost of reduction of the market value of Y_a. This may be difficult to achieve or manage, depending on the degree of deficits. Therefore, deficit irrigation is not always the preferred option by farmers due to the increased risk of both Y_a and monetary losses. Reducing the irrigated area may be a preferred solution, e.g. decreasing the irrigable area having annual crops in favour of watering of perennial crops. However, exceptions include the adoption of eustress in perennial crops where lower Y_a results in better quality fruit and higher profits, as for grapevine where beneficial effects such as decreased berry size or increased sugar content may lead to great benefit since wine quality may be improved.

Deficit irrigation can be implemented by applying a fixed water reduction strategy throughout the cropping season or targeted to some non-critical phenological stages in relation to water stress. Two main types of DI may be considered:

- *regulated deficit irrigation* (RDI) that targets certain phenological stages during which plants are less sensitive to water stress;
- *partial root-zone drying* (PRD), mostly used in fruit trees and vine crops, which targets the production of the plant hormone abscisic acid (ABA), which reduces leaf expansion and stomatal conductance by stressing one portion of the roots, whilst the other portion of the roots sustains transpiration.

Nevertheless, published research findings on RDI and PRD are not always consistent. Some results suggest that deficit irrigation and associated practices such as RDI and PRD may only be acceptable in reducing capital and operating costs of irrigation and to improve yield quality when carried out with using precision irrigation (timing, amount and placement) based on the knowledge of crop responses to water stress, which vary with crop and, often, with cultivar. Consequently, water-yield models, such as that described in Chapter 8, are often crop and crop cultivar specific.

Deficit irrigation is a highly complex practice that involves many variables. Its application must focus on overcoming the uncertainty of outcomes. In general, the applicability of deficit irrigation depends on site-specific conditions, crop types and related cultivars, and climatic variables (e.g., rainfall distribution and reference ET levels). Given the large variety of conditions and factors affecting crop water demand, yields and economic return, optimization modelling and decision-support tools are useful aids to design the most appropriate strategies for specific sites and for specific crops. Among such models are those used to derive various K_c and K_{cb} values tabulated in Chapters 6 and 7, e.g. CROPWAT, SIMDualKc and AquaCrop. These tools allow one to consider diverse criteria, primarily economic (maximizing profits), and select the irrigation scheduling and management issues that better lead to water-savings and to control environmental

impacts (minimizing leaching of nitrates and chemicals, controlling groundwater depletion and quality degradation). The FAO56 approach is used in such modelling applications and may be able to determine the effects of deficit irrigation on crop yield, non-beneficial crop water use, WP_{ET} and EWPR.

Crop and soil management

The planting date of annual crops likely impacts the amounts of beneficial and non-beneficial crop water use, namely the consumptive use, due to differences in responses to climatic factors, mainly the timing and distribution of rainfall. Anticipating or delaying planting dates is commonly done in relation to soil conditions for seeding and planting machines to move on the soil more easily and favouring the emergence of the new plants. But when farmers have predictive information on weather conditions, mainly rainfall, that adjustment of dates refer to the increased use of rainfall during the first periods of crop growth. This adjustment of the planting date is generally done using any kind of simulation tools that are able to predict the soil water content and $ET_{c\,act}$, therefore the gain in water availability. Adjusting that planting date requires that rainfall predictions are available to support farmers decisions. The availability of adequate K_c is also paramount.

Research has confirmed that the response of crops to water stress depends on critical periods in their phenological stages. Under conditions of water scarcity, irrigation during sensitive phenological stages (e.g. reproductive stage in wheat and maize) is favoured ahead of irrigation during less sensitive stages (e.g. vegetative stage). Again, this is favoured when SWB models are used based upon weather predictions. Crop and soil monitoring are also required.

Optimal crop growth and T_c may be reduced under conditions of limited supply of nutrients, fertilization and/or water, or under salinity conditions. This effect on $ET_{c\,act}$ can be accounted for in the FAO56 approach either by reducing K_c and K_{cb} coefficients empirically, or by adding the water stress coefficient (K_s) as described in previous chapters. Nutrient or other shortage effects manifest through a reduction in plant growth and canopy cover, and therefore a field can experience an increase in K_e due to more soil exposure. When irrigation is insufficient, fertilization should be reduced to avoid decreasing the osmotic potential.

Plant density and canopy size of perennials affect the utilization of resources. Reducing stand density makes more resources (e.g. water, nutrients) available to individual plants, in particular under rainfed agriculture. However, reduced plant density and canopy size are likely to increase E_s, thus increasing non-beneficial water use, i.e. E_s increases in detriment of T_c. In general, drier conditions are more suited to lower planting densities for perennials. Canopy cover is used in FAO56 guidelines to estimate the split of ET into E_s and T_c, because it approximates the ratio of radiation intercepted by the canopy and incoming solar radiation. Thus, Chapter 9 provides a guideline for the estimation of K_{cb} from the effective fraction of ground cover, $f_{c\,eff}$, and crop height, i.e. the A&P approach. This approach has been successfully applied to vegetable, field and tree crops and respective indicative parameters are tabulated in Chapter 9 to ease its application. The approach is particularly applicable to tree and shrub crops planted in rows given the large number of combinations of canopy sizes and densities (pruning), plant densities, row distances and orientations, wetted areas of micro-irrigation, etc. This type of modelling may be applied at the stage of planning and design of the cropping and irrigation system in order to provide for minimizing non-beneficial water consumption to assess different combinations of canopy management (pruning) and irrigation systems (e.g. drip vs micro-sprayers). However, the A&P approach, coupled with the dual K_c procedure, is mainly useful when applied to established orchards and vines to support planning and management of irrigation scheduling.

Inter-cropping (described in an earlier section of this chapter) and agro-forestry systems are popular practices to optimize the use of resources (water, nutrients and radiation energy). A higher WP_{ET} can be achieved in agro-forestry systems compared to mono-cropping because soil water between tree rows is consumed by inter-crop T_c rather than by E_s.

Soil management practices that may modify the volume of water infiltrated and stored in the soil, thereby affecting the ratio of E_s and T_c, are usually related to tillage (e.g. conservation or no-tillage), land preparation (land levelling or tilling ridge and furrows), or in-field rainwater harvesting. Soil additives and conditioners are materials used to increase the soil water and nutrient retention capacity or to reduce E_s. Many of these practices affect particularly the topsoil, which is the most dynamic soil layer in terms of mass and energy exchanges and may therefore impact K_c by decreasing K_e.

Less frequent irrigations reduce E_s compared to frequent irrigations, but they may increase the risk of crop water stress and favour Y_a reduction when ET_o is high. The frequency of irrigation wetting depends on various factors, namely the storage capacity of the soil, the type of irrigation system and the ET rate. Soil evaporation is site-specific depending upon climate through ET_o, the frequency of soil wetting, the use of mulches, and the fraction of ground shadowed by the canopy, which depends upon the crop density and the crop growth stage. E_s can be quantified through the partitioning of ET for the considered crop, namely using the dual K_c approach. E_s is limited by the shading of the crop canopy, which corresponds to limitations imposed to the available energy for evaporation at the soil surface. This approach is valid for all irrigation methods that wet the soil surface and is in proportion to the wetted fraction, as occurs for furrow irrigation, particularly, when alternate furrows are adopted.

As previously described in the current chapter, mulching is an old practice of covering the soil surface to reduce soil evaporation and surface runoff, regulate soil temperature, suppress weeds, improve soil health, and therefore improve crop growth. Different materials can be used for mulching, such as bark chips, straw, grass or other plant material, stones or plastic sheets, and humic substances. A wide variety of environmental and management factors have been reported in literature relative to the complex crop growth response to various types of mulch. Effects of mulches are greater at earlier crop stages, when the soil is yet bare or only partially covered by the crop canopy. As a result, those are periods when crop coefficients greatly can vary with the frequency of wettings and soil wetness. K_c can be estimated according to the previous mulching section of this chapter.

Only used in high-tech farming, plant conditioners include anti-transpirants, bio-stimulants and plant growth regulators. They are mainly applied to regulate harvesting dates of market produces, and to mitigate effects of environmental stresses. Although plant conditioners do not directly reduce non-beneficial water use, $NBWU_{ET}$, they affect T_c or the T_c/ET_c ratio and, therefore, their effects need to be considered when estimating total water consumption and in implementing irrigation scheduling. The use of plant conditioners does not change the standard K_c and K_{cb} coefficients but may act similarly to the stress coefficients in defining their actual values ($K_{c\ act}$ and $K_{cb\ act}$). They may affect soil evaporation when plant conditioners decrease (or increase) the crop cover fraction due to their effects on plants. Anti-transpirants may act by reducing stomatal conductance and then the reduced K_{cb} may be assessed using the F_r factor described in the A&P approach (Chapter 9). These types of conditioners may become more and more prevalent in the future as research develops means to reduce transpiration from plants while sustaining inflow of carbon dioxide.

Various practices may be used to condition micro-climate to reduce water losses and improve WS_{ET} through T_c. A typical case is sheltered or protected cultivation. However, as it happens with tunnels, the FAO PM-ET_o does not apply since the aerodynamic conditions are changed and do not respect the Penman-Monteith assumptions. Windbreaks, such as trees, may mitigate extreme winds and affect micro-climatic variables such as net radiation (by increased shading), temperature, humidity, heat and evaporation fluxes resulting in changes in evaporative demands and water vapour fluxes. The effects of tree windbreaks depend on surface roughness, height, orientation and porosity. Warmer, more humid and less windy conditions usually occur in the protected crop areas. Lower wind speed and more humid conditions reduce vapor pressure deficit and crop ET, mostly without decreasing dry matter assimilation. Reduction of sensible heat fluxes and surface resistances, on the other hand, would most likely favor transpiration and dry matter assimilation. Higher Y_{act} can therefore be expected in sheltered areas except when crops compete for resources with the windbreak plants (e.g. competition for water and nutrients). Generally, the trees in windbreaks are more tolerant of water shortage than agricultural crops and therefore can be irrigated less or not at all. Partial stomatal closure of the windbreak trees will cause a larger proportion of total available energy to be transformed into sensible heat rather than into evapotranspiration and total water consumption. As a result, total ET over large area of windbreaks and agriculture can be reduced.

10.7.2 Water-saving of non-consumptive water use (WS_{DP+RO})

Farm irrigation systems

Irrigation methods can be broadly classified as surface, sprinkler and micro-irrigation. This section focuses on water-saving practices that are typical for each irrigation method and applicable on farms. It should be born in mind that costs and benefits play a major role when assessing respective water-saving issues because economic returns of irrigation tend to decrease when water-saving practices are adopted. Water consumption and use under different irrigation methods show sometimes contrasting outcomes, depending on the type of crop, canopy cover, soil properties (mainly relative to infiltration), land slope, irrigation wetted area, and irrigation management.

Surface irrigation systems can have high performance when precise laser levelling is adopted, inflow rates are adequate and non-erosive, and discharge applications are well controlled. These rules apply to either basin, border or furrow irrigation. Precise land levelling for basin, furrow and border irrigation is usually fundamental for efficient application with surface irrigation methods. When applied water creates ponds on the land surface before infiltrating into the soil, it not only leads to increased evaporation losses but decreases distribution uniformity (DU) of the infiltrated water. As a result, there will be parts of the field receiving less and parts receiving excessive water. If only the average depletion is replaced, then field-average ET will be decreased due to water stress on the under-irrigated portion of the field and field-average yields will decrease.

To compensate for the less irrigated zones where the crop experiences stress, the farmer may tend to apply water in excess, thus promoting high deep percolation. Several authors reported that adopting precision land levelling in basin irrigation with controlled inflow rates and respective application time duration, reduced water application by nearly 40 percent relative to using less cared applications. However, the overall water consumption (evaporation and transpiration) may be similar but the cumulative soil evaporation may be less because the land surface is then usually wetted less frequently and transpiration increases because the canopy is increased and E_s is reduced.

Equipment for precise control of applied flow rates and volumes is rare where surface irrigation is traditional. The use of hand held equipment is not precise in control and largely depends upon the ability of the irrigant, which is another cause of non-uniformity of infiltrated volumes and, consequently, another reason why farmers may tend to apply excess water. In fact, the lack of precision control equipment makes it difficult to adopt precise scheduling in terms of managing discharge rates and volumes. For example, findings report a reduction in water application of up to 45% when surge-flow on alternate furrows is used, namely due to runoff control due to using automated gated pipes. Adopting alternate furrow irrigation may additionally reduce consumptive water use through reduced water application and less soil evaporation with little reduction in yield and increased WP_{ET} as compared to conventional furrow irrigation.

Achieving appropriate inflow control requires mechanization of water distribution to basins, borders or furrows, preferably automation and remote control. Thus, changing from traditional surface irrigation to modern practices requires investments that may be economically unsatisfactory. However, research has been declining and automated surface irrigation systems are not available in countries where traditional irrigation is predominant. In many regions, the trend has been to replace surface irrigation with sprinkler or micro-irrigation, which equipment is supported by strong markets as compared to those for surface irrigation equipment.

Rice is traditionally cultivated in flooded paddy fields where the basins' water depth is around 5 to 10 cm. Basins are surrounded by soil bunds that avoid RO. This water plays a large role: providing for crop ET, controlling impacts of temperature on the crop and controlling weed growth. When land levelling is improved, the flooding depth may be reduced to about 5 cm, which may decrease deep percolation (DP). When the climate is favourable, intermittent (nearly daily) flooding may be practiced with the soil water kept near saturation, thereby further decreasing DP. When infrequent irrigation by surface, sprinkler or micro-irrigation is practised with aerobic rice production, both DP and E_s may decrease, however depending upon soil and land surface conditions.

It is important to note that water-savings primarily refer to a reduction in DP and likely to little reduction to evaporation and the non-beneficial water consumption fraction. In other words, water use is reduced, but water consumption may be only slightly reduced. However, the standard K_c values change with the adopted management of the rice crop as tabulated in Chapter 6. As stated, net crop water requirements decrease from conventional flooding to flooding with dry seeding, flooding with anticipated cutoff and, finally, to aerobic rice. The adoption of different, specific K_c values is a condition for better achieving water saving.

Sprinkler irrigation systems may be stationary or continuous moving. The former includes portable and solid set systems and side roll. Continuous moving wheeled laterals are automated and include center-pivots, linear-move systems, travelling gun sprinklers and side-move systems. A primary objective of sprinkler systems is high water distribution uniformity throughout the irrigated field. Thus, potential water-saving practices aim at maintaining high DU when adopting sprinkler flow rates that favour water infiltration, spacing among sprinklers that provide for a good distribution of water throughout the field, low variability of pressure and discharge if the field is sloping, system maintenance and design for controlling wind effects.

If irrigation scheduling is appropriate in terms of the crop and soil water contents dynamics, these characteristics lead to high ratios of requirement to water applied, i.e., water application efficiencies. Sprinkler systems perform best when the soil infiltration is high, and wind speed is low. Sprinkler system design, mainly referring to the

selection of sprinklers sizes and spacing, pressure, flow rates, water application depths and diameters of piping, is fundamental to ensure uniform distribution of pressure and flow rates, and to reduce evaporative losses.

Practices and design to improve water infiltration and thereby reduce runoff require appropriate sprinkler selection. Because water applied by large sprinkler systems wets the entire ground surface, non-beneficial consumption of water can be substantial under high wind and high evaporative demand of the atmosphere. Wind-induced evaporation of spray may be highest when drops are small, and when the canopy cover is low and risers are high above the ground surface. Evaporation of spray can be reduced by placing nozzles closer to or even inside the plant canopy, using larger drop sizes and attempting to irrigate when wind speeds are low. Modern sprinkler systems, such as center-pivot and linear moving laterals, are often designed to apply small irrigation depths to avoid runoff. As a result, they must irrigate more frequently, thereby increasing total evaporation losses from soil. The more frequent irrigation reduces the risk of crop stress, but at the expense of increased E_s. The dual K_c procedure can be used to estimate the increases in K_e and E_s under frequent irrigation. Figures 6.1 and 6.2 can be used to estimate the increase in K_c under low vegetation cover.

Micro-irrigation systems include high-tech methods having large capital and maintenance investment. Systems are low-pressure and include surface and subsurface drip irrigation for row crops and orchards, micro-sprinklers and micro-sprayers for horticulture and under-tree irrigation, and bubblers for orchards. These irrigation systems wet a limited portion of the ground, targeting the crop root zone, mainly in case of orchards and vines. The limited wetting of the surface reduces the soil evaporation impact due to high frequency irrigation, thus providing for high WS_{ET} by reducing $NBWU_{ET}$. Achieving good performance with micro-irrigation systems requires good design, which complexity may be overcome using decision support and design models. WS practices resulting from good design aim at improving water application control, the emission uniformity (EU), and the management of irrigation, as well as automation and control.

The benefits of drip irrigation compared to surface and sprinkler irrigation include reduced water application, controlled operational water losses, less labour, low or no runoff and reduced energy consumption and pumping costs per unit area, as well as improved management of water, fertilization and pesticides for a range of high-profit crops. Micro-sprinklers and micro-sprayers have similar benefits, but the wetting pattern is wider and may be affected by wind. However, for under-tree irrigation, wind drift and E_s may be less than for field crops. Despite advantages, managing drip or micro-spray irrigation systems for peak performance is very demanding, including when automation is adopted.

High-tech micro-irrigation methods may not be economically feasible under all conditions since, for field crops, economic returns tend to decrease when more precision water-saving techniques are adopted. Benefits are generally greater for irrigating tree and vine crops having a better developed root system compared to annual crops. Water can be easily applied near the root zone of tree and vine crops, and it may be easier to apply deficit irrigation, in particular partial root-zone drying. Soil evaporation is controlled because the soil is only partially wetted near the crops, and often under the shadow of the crop canopy. When subsurface drip is applied, ground surface wetting occurs only by capillarity from the buried emitters. Drip irrigation, and border and basin irrigation, have advantages over sprinkler systems in saline environments by avoiding the wetting of the plant canopy. Drip irrigation can have reduced accumulation of salts in the root zone due to near continuous irrigation and therefore better suitability for salt sensitive crops.

In summary, the impact of the irrigation system type on consumptive water losses through evaporation and transpiration is controlled by the irrigation frequency, the fraction of soil surface wetted, the characteristics of spray drops and their evaporation in the case of sprinkler irrigation, and the wetting of the crop canopy in the case of sprinkler irrigation. These impacts can be generally quantified using the dual K_c approach. Differently, the processes of deep percolation and runoff, mainly depend upon the irrigation design and management, thus only indirectly from ET and K_c. However, the appropriate selection of K_c values permits the use of equipment and management tools that contribute to control DP and RO.

Off-farm irrigation systems

Water collection, transportation and distribution to farms may be a source of water losses in case of open channel conduits, much less when pipe systems are used. The latter respond quickly to management, and farmers may often receive water on demand, thus producing only minimal losses and causing adequate implementation and adoption of irrigation water distribution on demand. Gravity canal systems have a demanding operation and regulation to control discharges to low order canals or to distributors. Then, since it is difficult to operate on-demand, thus often producing supplies with discharges and pressure heads that vary and cause difficulties to the irrigation scheduling by the farmers and consequent losses. In addition, gravity systems respond slowly to regulation and management operative orders. Modern canal automation and permanent hydraulic monitoring, as well as distant command of gates, associated with using simulation of aggregated $ET_{c\ act}$ demands, allow an important water saving and appropriate delivery of water to the farms, thus producing WS at farm and off-farm.

10.7.3 Remote sensing techniques for irrigation crops monitoring

Remote sensing (RS) data and methodologies have been increasingly used for estimating parameters related to crop water requirements, namely K_c and K_{cb} as for many cases collected to build their tabulated values. By providing the spatial distribution of vegetation and ET, RS provides for a better understanding of within- and among fields variability, which is important to support irrigation management ranging from local farmers up to off-farm irrigation systems' managers. The increased number of satellite missions, improved sensor technology, enhanced spatial, temporal and spectral resolutions of data, and easier data access have all encouraged the use of RS for irrigation management. A description of RS-based techniques is provided in Chapter 1. RS estimates are valuable for assessing the occurrence of water shortage or poor distribution or other impacts caused by poor nutrient availability or salinity. The vegetation-based and thermal-based RS methods work well together as compliments in estimating both potential (standard) K_{cb} and actual $K_{c\ act}$.

Overall, the accurate estimation of K_{cb}, $K_{c\ act}$ and actual crop ET can be assisted by RS Improved estimation improves irrigation management and contributes both to the convergence of actual K_c and K_{cb} values towards better knowledge of standard values and to their distribution in space. Improved knowledge of K_c aids in achieving larger WP_{ET} and water savings, and minimizing non-beneficial water uses associated with over-irrigation. For successful application of RS, the availability of cloud-free satellite, UAV and/or manned aircraft images with an adequate temporal resolution of at least one clear image every 15 days is needed throughout the irrigation season. An important example of application of RS for irrigation management is SIMS, Satellite Irrigation Management Support. SIMS adopts the A&P approach to compute K_c and K_{cb}, using the fraction of ground cover estimated from RS. The model is used to support irrigation scheduling of perennials and annuals in California. The model is currently integrated with other RS irrigation management models in OpenET, which is aimed at RS advising on irrigation scheduling in the West of USA and assessing performance of RS viewed fields. Literature refers to several RS models used for monitoring and irrigators advising such as the SAMIR tool (Satellite Monitoring of Irrigation) used in Morocco, the AMSR2 used in Iran, the web-GIS SPIDER in Spain, or SEBAL for Jordan.

Annex 1
Nomenclature, symbols and units

NOMENCLATURE

List of principal symbols and acronyms (based upon the list adopted in FAO56)

AI	Aridity Index describing the aridity of the general region
ASW	available soil water, i.e. depth of water between the wilting point and field capacity [mm]
$A_{stage\ 1}$	fraction of the time step (day or hour) that resides in stage 1 evaporation [-]
A_{cm}	factor depending on the actual vegetation development [-]
a_1 to a_4	optimization parameters of the capillary rise parametric equation [various units]
a_d	empirical temperature adjustment coefficient for T_{dew} estimation from T_{mean} [°C]
a_D	optimization parameter of the deep percolation parametric equation [mm]
a_{psy}	coefficient relative to the ventilation of the psychrometer wet bulb [$°C^{-1}$]
a_s	fraction of extraterrestrial radiation reaching the earth on an overcast day [-]
a_s+b_s	fraction of extraterrestrial radiation reaching the earth on a clear sky day [-]
a_T	empirical temperature adjustment coefficient for T_{dew} estimation from T_{min} [°C]
b	rate of yield reduction per unit increase in EC_e [%/(dS m^{-1})]
b_1 to b_4	optimization adjustment parameters of the capillary rise parametric equation [-]
b_D	optimization parameter of the deep percolation parametric equation [-]
CEI_{TH}	mean annual Thornthwaite climatic evaporation index [mm]
CN	curve number [-]
CR	capillary rise [mm]
CR_{max}	potential or maximum capillary rise [mm d^{-1}]
CWR	crop water requirements [mm]
c	solute concentration of the liquid phase (kg m^{-3})
c_p	specific heat [MJ kg^{-1} $°C^{-1}$]
c_s	soil heat capacity [MJ m^{-3} $°C^{-1}$]
D	solute dispersion coefficient (m^2 s^{-1})

D_e	depth of cumulative evaporation (depletion) from the soil surface layer [mm]
D_r	depth of cumulative evapotranspiration (depletion) from the root zone [mm]
D_{REW}	cumulative depletion depth of the REW of the soil skin layer [mm]
DP	deep percolation [mm]
DP_e	deep percolation from the evaporation layer [mm]
DT	temperature difference [°C]
d	zero plane displacement height [m]
d_r	inverse relative distance Earth-Sun [-]
E	evaporation [mm d^{-1} or mm h^{-1}]
E_{pan}	pan evaporation [mm d^{-1} or mm h^{-1}]
E_s	evaporation from the soil [mm d^{-1} or mm h^{-1}]
EC_e	electrical conductivity of the saturation extract of the soil [dS m^{-1}]
$EC_{e\ threshold}$	EC_e of the soil above which the yield begins to decrease [dS m^{-1}]
$EC_{e\ 50}$	EC_e value at which the crop yield is reduced by 50% [dS m^{-1}]
EC_{iw}	electrical conductivity of the irrigation water [dS m^{-1}]
ET	evapotranspiration [mm d^{-1} or mm h^{-1} or any time fraction or season]
$ET_{a\ eb}$	actual crop evapotranspiration calculated via energy balance [mm d^{-1}]
ET_c	crop evapotranspiration under standard conditions [mm d^{-1} or mm h^{-1}]
$ET_{c\ act}$	actual crop evapotranspiration, i.e. under non-standard conditions [mm d^{-1} or mm h^{-1}]
ET_o	grass reference crop evapotranspiration [mm d^{-1} or mm h^{-1}]
ET_r	alfalfa reference crop evapotranspiration [mm d^{-1} or mm h^{-1}]
ET_{ref}	reference crop evapotranspiration [mm d^{-1} or mm h^{-1}]
ET_oF	fraction of ET to the grass reference crop ET (= K_c), as used in remote sensing [-]
ET_rF	fraction of ET to the alfalfa reference crop ET (= K_r), as used in remote sensing [-]
EWPR	economic water productivity ratio [-]
e°(T)	saturation vapour pressure at air temperature T [kPa]
e_s	saturation vapour pressure at a given time or averaged for a given time period [kPa]
e_a	actual vapour pressure at a given time or averaged for a given time period [kPa]
$e_s - e_a$	vapour pressure deficit [kPa]
F_r	crop canopy resistance correction factor [-]
$Frac_{dom}$	fraction of the soil surface under intercrop with the dominant crop [-]
$Frac_{sub}$	fraction of the soil surface cropped with the subordinated crop [-]
f_c	fraction of soil surface covered by vegetation (as observed from overhead) [-]

$1-f_c$	exposed soil fraction [-]
$f_{c\ eff}$	effective fraction of soil surface covered by vegetation [-]
f_1, f_2	fraction of soil surface cropped in an intercrop system for crop species 1 or 2 [-]
f_{ew}	fraction of soil surface both exposed and wetted (from where most evaporation occurs) [-]
f_w	fraction of soil surface wetted by rain or irrigation [-]
G	soil heat flux [MJ m^{-2} d^{-1} or MJ m^{-2} h^{-1} or W m^{-2}]
G_d	soil heat flux for 24-hour and ten-day periods [MJ m^{-2} d^{-1}]
GDD	growing degree days [°C]
GDD_{ini}	cumulative growing degree days for the initial growth stage [°C]
GDD_{dev}	cumulative growing degree days for the development growth stage [°C]
GDD_{mid}	cumulative growing degree days for the midseason growth stage [°C]
GDD_{late}	cumulative growing degree days for the late season growth stage [°C]
G_{hr}	soil heat flux for hourly or shorter periods [MJ m^{-2} h^{-1} or W m^{-2}]
G_{month}	soil heat flux for monthly periods [MJ m^{-2} d^{-1}]
G_{sc}	solar constant [0.0820 MJ m^{-2} min^{-1}]
GWC	groundwater contribution to evapotranspiration [mm]
g	gravitational acceleration [9.807 m s^{-2}]
H	sensible heat flux [MJ m^{-2} d^{-1} or MJ m^{-2} h^{-1} or W m^{-2}]
h or h_c	crop height [m]
h_{dom}	height of the dominant crop [m]
h_{sub}	height of the subordinated crop [m]
h	soil matric potential (or pressure head) [m]
h_ϕ	osmotic potential (or pressure head) [m]
I	irrigation depth [mm]
I_n	net irrigation depth [mm]
I_w	irrigation depth for the part of the surface wetted [mm]
J	number of day in the year [d] (=DOY)
K	soil hydraulic conductivity [m d^{-1} or mm h^{-1}]
K_B	clearness index for direct beam radiation [-]
K_c	standard crop coefficient [-]
$K_{c\ act}$	actual, non-standard, crop coefficient [-]
$K_{c\ ini}$	crop coefficient during the initial growth stage [-]
$K_{c\ mid}$	crop coefficient during the mid-season growth stage [-]
$K_{c\ end}$	crop coefficient at end of the late season growth stage [-]
$K_{c\ max}$	maximum value of crop coefficient (following rain or irrigation) [-]
$K_{c\ min}$	minimum value of crop coefficient (dry soil with no ground cover) [-]

$K_{c\ n\text{-}gro}$	crop coefficient during the non-growing crop stage [-]
K_{cb}	standard basal crop coefficient [-]
$K_{cb\ act}$	actual, non-standard, basal crop coefficient [-]
$K_{cb\ ini}$	basal crop coefficient during the initial growth stage [-]
$K_{cb\ mid}$	basal crop coefficient during the mid-season growth stage [-]
$K_{cb\ end}$	basal crop coefficient at end of the late season growth stage [-]
$K_{cb\ n\text{-}gro}$	basal crop coefficient during the non-growing crop stage [-]
$K_{cb\ dom}$	basal crop coefficient of the dominant crop [-]
$K_{cb\ sub}$	basal crop coefficient of the subordinated crop [-]
$K_{cb\ A\&P}$	basal crop coefficient estimated with the A&P approach [-]
$K_{cb\ cover}$	basal crop coefficient of the ground cover in the absence of tree foliage [-]
$K_{cb\ full}$	basal crop coefficient during mid-season for vegetation with full ground cover [-]
$K_{cb,\ h}$	K_{cb} for full cover vegetation under sub-humid and calm wind conditions [-]
K_D	index for diffuse radiation [-]
K_d	crop density coefficient [-]
$K_{d\ dom}$	density coefficient of the dominant crop [-]
$K_{d\ sub}$	density coefficient of the subordinated crop [-]
K_e	soil evaporation coefficient [-]
$K_{e\ max}$	maximum value of K_e coefficient (following rain or irrigation) [-]
K_p	pan coefficient [-]
K_r	soil evaporation reduction coefficient [-]
K_s	water stress coefficient [-]
K_t	clearness coefficient that is affected by turbidity and haze ($0 < K_t \leq 1.0$)
K_y	yield response factor [-]
k	von Karman's constant [0.41] [-]
k_h	multiplicative factor of crop height in $K_{cb\ full}$ equation [-]
k_{Rs}	adjustment coefficient for the Hargreaves' ET and radiation equation [$°C^{-0.5}$]
L_{ini}	length of initial growth stage [d]
L_{dev}	length of crop development growth stage [d]
L_{mid}	length of mid-season growth stage [d]
L_{late}	length of late season growth stage [d]
L_z	longitude of center of local time zone [degrees West of Greenwich]
L_m	longitude [degrees West of Greenwich]
LAI	leaf area index [m^2 (leaf area) m^{-2} (soil surface)]
LAI_{act}	field observed actual leaf area index [$m^2\ m^{-2}$]
LAI_{active}	active (sunlit) leaf area index [-]

LAI_{dense}	leaf area index expected for a crop under normal dense conditions [$m^2 m^{-2}$]
LF	leaching fraction [-]
LR	leaching requirement [-]
MAD	management allowed depletion [-]
M_L	multiplier on f_c describing the effect of canopy density [-]
N	maximum possible sunshine duration in a day, daylight hours [h]
$NBWU_{ET}$	non-beneficial consumptive water use referring to evapotranspiration [mm]
$NBWU_{DP+RO}$	non-beneficial non-consumptive water use referring to deep percolation and runoff [mm]
NDVI	normalized difference vegetation index [-]
NIR	net irrigation requirement [mm]
n	actual duration of sunshine in a day [h]
n/N	relative sunshine duration [-]
P	precipitation [mm]
P_e	effective precipitation [mm]
P	atmospheric pressure [kPa]
p	soil water depletion fraction for no stress [-]
p_{inter}	depletion fraction for relay intercropping [-]
p_{dom}	depletion fraction for no stress for the dominant crop [-]
p_{sub}	depletion fraction for no stress for the subordinated crop [-]
p_{salt}	soil water depletion fraction for no stress corrected for salinity [-]
q	flow rate [L s^{-1} or $m^3 s^{-1}$]
q	Darcy flux [m s^{-1}]
R_a	extraterrestrial radiation [MJ $m^{-2} d^{-1}$ or MJ $m^{-2} h^{-1}$ or W m^{-2}]
R_l	longwave radiation [MJ $m^{-2} d^{-1}$ or MJ $m^{-2} h^{-1}$ or W m^{-2}]
R_n	net radiation [MJ $m^{-2} d^{-1}$ or MJ $m^{-2} h^{-1}$ or W m^{-2}]
R_{nl}	net outgoing longwave radiation [MJ $m^{-2} d^{-1}$ or MJ $m^{-2} h^{-1}$ or W m^{-2}]
R_{ns}	net incoming solar shortwave radiation [MJ $m^{-2} d^{-1}$ or MJ $m^{-2} h^{-1}$ or W m^{-2}]
R_s	incoming solar shortwave radiation [MJ $m^{-2} d^{-1}$ or MJ $m^{-2} h^{-1}$ or W m^{-2}]
R_{so}	clear-sky incoming solar shortwave radiation [MJ $m^{-2} d^{-1}$ or MJ $m^{-2} h^{-1}$ or W m^{-2}]
R_s/R_{so}	relative incoming solar shortwave radiation [-]
r_a	aerodynamic resistance [s m^{-1}]
r_l	(bulk) stomatal resistance of a well-illuminated leaf [s m^{-1}]
r_{typ}	typical leaf resistance representing of well-watered agricultural vegetation [s m^{-1}]
r_s	(bulk) surface or canopy resistance [s m^{-1}]
RAW	root zone readily available soil water [mm]

RAW_{salt}	root zone readily available soil water adjusted for effects of soil salinity [mm]
REW	readily evaporable water from the soil surface layer [mm]
RH	relative humidity [%]
RH_{hr}	average hourly relative humidity [%]
RH_{max}	daily maximum relative humidity [%]
RH_{mean}	daily mean relative humidity [%]
RH_{min}	daily minimum relative humidity [%]
RO	surface runoff [mm]
S	sink term accounting for water uptake by plant roots ($m^3\ m^{-3}\ d^{-1}$ or $m^3\ m^{-3}\ h^{-1}$)
S_c	seasonal correction factor for solar time [h]
SAVI	soil adjusted vegetation index [-]
SF	subsurface flow [mm]
SWC	(volumetric) soil water content [$m^3\ m^{-3}$]
T	air temperature [°C]
T_{base}	base temperature, lower temperature threshold for the estimation of GDD [ºC]
T_c	crop transpiration [mm d^{-1} or mm h^{-1}]
$T_{c\ act}$	actual crop transpiration [mm d^{-1} or mm h^{-1}]
T_{dew}	dew point temperature [°C]
T_{dry}	dry bulb temperature [°C]
T_{ew}	depth of transpiration from the exposed and wetted fraction of the soil surface layer [mm]
T_K	absolute air temperature [K]
T_{Kv}	absolute virtual air temperature [K]
T_{max}	maximum air temperature [°C]
$T_{max,K}$	absolute maximum air temperature [K]
T_{mean}	mean air temperature [°C]
T_{min}	minimum air temperature [°C]
$T_{min,K}$	absolute minimum air temperature [K]
T_{upper}	upper temperature threshold for the estimation of GDD [ºC]
T_{wet}	wet bulb temperature [°C]
TAW	total available soil water of the root zone [mm]
TAW_{salt}	total available soil water adjusted for effects of soil salinity [mm]
TEW	total evaporable water from the soil surface layer [mm]
t	time [h or d]
t_w	time between irrigation events [d]
u_2	wind speed observed at, or adjusted to 2 m above ground surface [m s^{-1}]
u_z	wind speed observed at z m above ground surface [m s^{-1}]

VI	vegetation index [-]
W	precipitable water in the atmosphere [mm]
WP_c	crop water productivity [kg m^{-3}]
W_a	actual soil water storage [mm]
W_c	critical soil water storage [mm]
W_{FC}	soil water storage at field capacity of root zone [mm]
W_s	steady soil water storage [mm]
W_{WP}	soil water storage at the wilting point of root zone [mm]
WTD	water table depth [m]
WTD_a	actual water table depth [m]
WTD_c	critical water table depth [m].
Y_{act}	actual crop yield [kg ha^{-1}]
Y_{max}	maximum crop yield in absence of water and environment stresses [kg ha^{-1}]
Y_r	relative yield [-]
Z_e	depth of the surface soil layer from where soil evaporation occurs [m]
Z_r	rooting depth [m]
z	elevation, height above sea level [m],
z	soil depth [m]
z_h	height of humidity measurements [m]
z_m	height of wind speed measurements [m]
z_{oh}	roughness length governing heat and vapour transfer [m]
z_{om}	roughness length governing momentum transfer [m]
α	albedo [-]
β	Bowen ratio [-]
γ	psychrometric constant [kPa $°C^{-1}$]
γ_{psy}	psychrometric constant of instrument [kPa $°C^{-1}$]
Δ	slope of saturation vapour pressure curve [kPa $°C^{-1}$]
Δt	length of time interval [d]
ΔT	temperature difference [°C]
ΔT_{paired}	upward bias in $T_{min} - T_{dew}$ differences created by non-reference station aridity [°C]
Δz	effective soil depth [m]
δ	solar declination [rad]
ε	ratio molecular weight of water vapour/dry air (= 0.622)
η	mean angle of the sun above the horizon [rad]
θ	soil water content [m^3 m^{-3}]
θ_{FC}	soil water content at field capacity [m^3 m^{-3}]

θ_p	soil water content corresponding to the depletion fraction for no stress p [m^3 m^{-3}]
θ_{sub}	water content of the subsoil [m^3 m^{-3}]
θ_t	SWC threshold below which transpiration is reduced due to water stress [m^3 m^{-3}]
θ_{WP}	soil water content at the permanent wilting point [m^3 m^{-3}]
$\theta_{WP\ salt}$	soil water content at wilting point adjusted for effects of salinity [m^3 m^{-3}]
λ	latent heat of vaporization [MJ kg^{-1}]
λET	latent heat flux [MJ m^{-2} d^{-1} or MJ m^{-2} h^{-1} or W m^{-2}]
ρ_a	mean air density [kg m^{-3}]
ρ_w	density of water [kg m^{-3}]
σ	Stefan-Boltzmann constant [4.903 10^{-9} MJ K^{-4} m^{-2} d^{-1}]
φ	latitude [rad]
ω	solar time angle at midpoint of hourly or shorter period [rad]
ω_1	solar time angle at beginning of hourly or shorter period [rad]
ω_2	solar time angle at end of hourly or shorter period [rad]
ω_s	sunset hour angle [rad]

TERMINOLOGY

FAO56	Food and Agriculture Organization Irrigation and Drainage Paper 56 (1998)
FAO56 PM	FAO56 standardized Penman-Monteith equation
PM-ET_o	grass reference ET_o calculated using the FAO56 standardized Penman-Monteith equation
PM-ET_r	alfalfa reference ET_r calculated using an extension of the FAO56 Penman-Monteith equation.

ACRONYMS

A&P	Allen and Pereira approach
BREB	Bowen ratio energy balance
CIMIS	California Irrigation Management Information System
DI	Deficit irrigation
DOY	Number of day in the year
DU	Distribution uniformity
EBL	Equilibrium boundary layer
EC	Eddy covariance
ECMWF	European Center for Medium-Range Weather Forecasts
HWR	Height to width ratio

ISAREG	Soil water balance model
LDAS	Land data assimilation system
LSA-SAF	Satellite Applications Facility on Land Surface Analysis
METRIC	Mapping Evapotranspiration with Internalized Calibration model
NARR	North American Regional Reanalysis
PRD	Partial root-zone drying
QAQC	Quality assurance and quality control
RDI	Regulated deficit irrigation
RS	Remote sensing
RSSEB	Remote sensing surface energy balance
SEB	Surface energy balance
SEBAL	Surface Energy Balance Algorithm for Land remote sensing model
SEBS	Surface Energy Balance System remote sensing model
SIMDualKc	Soil water balance model adopting the dual crop coefficient approach
SIMS	Satellite Irrigation Management Support
SWB	Soil water balance
UNEP	United Nations Environment Programme
WAMIS	World AgroMeteorological Information Service, Weather Prediction Center
WMO	World Meteorological Organization
WS_{ET}	Water-saving practices focusing on crop evapotranspiration
WS_{DP+RO}	Water-saving practices focusing on controlling deep percolation and runoff

The statistical notation is given on last page of Annex 3.

PREFIXES

Units can be used as such or in multiples with the following prefixes:

Tera	(T) and which is	10^{12}	Centi	(c)	10^{-2}
Giga	(G)	10^{9}	Milli	(m)	10^{-3}
Mega	(M)	10^{6}	Micro	(μ)	10^{-6}
Kilo	(k)	10^{3}	Nano	(n)	10^{-9}
Hecto	(h)	10^{2}	Pico	(p)	10^{-12}
Deca	(da)	10^{1}	Femto	(f)	10^{-15}
Deci	(d)	10^{-1}	Atto	(a)	10^{-18}

TEMPERATURE

Standard unit: degree Celsius (°C)

degree Fahrenheit (°F)	(°C) = (°F – 32) 5/9
Kelvin (K)	1 K = (°C) + 273.15

PRESSURE (AIR PRESSURE, VAPOUR PRESSURE)

Standard unit: kilopascal (kPa)

millibar (mbar)	1 mbar = 0.1 kPa
bar	1 bar = 100 kPa
centimetre of water (cm)	1 cm of water = 0.09807 kPa
millimetre of mercury (mmHg)	1 mmHg = 0.1333 kPa
atmospheres (atm)	1 atm = 101.325 kPa
pound per square inch (psi)	1 psi = 6.896 kPa

WIND SPEED

Standard unit: metre per second (m s^{-1})

kilometre per day (km day^{-1})	1 km day^{-1} = 0.01157 m s^{-1}
nautical mile/hour (knot)	1 knot = 0.5144 m s^{-1}
foot per second (ft s^{-1})	1 ft/s = 0.3048 m s^{-1}

RADIATION

Standard unit: megajoule per square metre and per day (MJ m^{-2} day^{-1}) or as equivalent evaporation in mm per day (mm day^{-1})

equivalent evaporation (mm/day)	1 mm day^{-1} = 2.45 MJ m^{-2} day^{-1}
joule per cm^2 per day (J cm^{-2} day^{-1})	1 J cm^{-2} day^{-1} = 0.01 MJ m^{-2} day^{-1}
calorie per cm^2 per day (cal cm^{-2} day^{-1})	1 cal = 4.1868 J = 4.1868 10^{-6} MJ
	1 cal cm^{-2} day^{-1} = 4.1868 10^{-2} MJ m^{-2} day^{-1}
watt per m^2 (W m^{-2})	1 W = 1 J s^{-1}
	1 W m^{-2} = 0.0864 MJ m^{-2} day^{-1}

EVAPOTRANSPIRATION

Standard unit: millimetre per day (mm day^{-1}) or per hour (mm h^{-1})

m^3 per hectare per day (m^3 ha^{-1} day^{-1})	1 m^3 ha^{-1} day^{-1} = 0.1 mm day^{-1}
litre per second per hectare (l s^{-1} ha^{-1})	1 l s^{-1} ha^{-1} = 8.640 mm day^{-1}
equivalent radiation in megajoules per square metre per day (MJ m^{-2} day^{-1})	1 MJ m^{-2} day^{-1} = 0.408 mm day^{-1}

Annex 2
Meteorological tables

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TABLE A2.1
Atmospheric pressure (P) for different altitudes (z)

z (m)	p (kPa)	z (m)	p (kPa)	z (m)	p (kPa)	z (m)	p (kPa)
0	101.3	1 000	90.0	2 000	79.8	3 000	70.5
50	100.7	1 050	89.5	2 050	79.3	3 050	70.1
100	100.1	1 100	89.0	2 100	78.8	3 100	69.6
150	99.5	1 150	88.4	2 150	78.3	3 150	69.2
200	99.0	1 200	87.9	2 200	77.9	3 200	68.8
250	98.4	1 250	87.4	2 250	77.4	3 250	68.3
300	97.8	1 300	86.8	2 300	76.9	3 300	67.9
350	97.2	1 350	86.3	2 350	76.4	3 350	67.5
400	96.7	1 400	85.8	2 400	76.0	3 400	67.1
450	96.1	1 450	85.3	2 450	75.5	3 450	66.6
500	95.5	1 500	84.8	2 500	75.0	3 500	66.2
550	95.0	1 550	84.3	2 550	74.6	3 550	65.8
600	94.4	1 600	83.8	2 600	74.1	3 600	65.4
650	93.8	1 650	83.3	2 650	73.7	3 650	65.0
700	93.3	1 700	82.8	2 700	73.2	3 700	64.6
750	92.7	1 750	82.3	2 750	72.7	3 750	64.1
800	92.2	1 800	81.8	2 800	72.3	3 800	63.7
850	91.6	1 850	81.3	2 850	71.8	3 850	63.3
900	91.1	1 900	80.8	2 900	71.4	3 900	62.9
950	90.6	1 950	80.3	2 950	71.0	3 950	62.5
1 000	90.0	2 000	79.8	3 000	70.5	4 000	62.1

TABLE A2.2
Psychrometric constant (γ) for different altitudes (z)

$$\gamma = \frac{c_p P}{\varepsilon \lambda} = 0.665 \times 10^{-3} P \quad \text{(Eq. 3.3)}$$

Z (m)	γ kPa/°C	z (m)	γ kPa/°C	z (m)	γ kPa/°C	z (m)	γ kPa/°C
0	0.067	1 000	0.060	2 000	0.053	3 000	0.047
100	0.067	1 100	0.059	2 100	0.052	3 100	0.046
200	0.066	1 200	0.058	2 200	0.052	3 200	0.046
300	0.065	1 300	0.058	2 300	0.051	3 300	0.045
400	0.064	1 400	0.057	2 400	0.051	3 400	0.045
500	0.064	1 500	0.056	2 500	0.050	3 500	0.044
600	0.063	1 600	0.056	2 600	0.049	3 600	0.043
700	0.062	1 700	0.055	2 700	0.049	3 700	0.043
800	0.061	1 800	0.054	2 800	0.048	3 800	0.042
900	0.061	1 900	0.054	2 900	0.047	3 900	0.042
1 000	0.060	2 000	0.053	3 000	0.047	4 000	0.041

Based on $\lambda = 2.45$ MJ kg^{-1} at 20°C.

TABLE A2.3
Saturation vapour pressure (e^o (T)) for different temperatures (T)

$$e^o(T) = 0.6108 \exp\left[\frac{17.27\,T}{T+237.3}\right] \quad \text{(Eq. 3.6)}$$

T °C	e_s kPa	T °C	e°(T) kPa	e°(T) kPa	e°(T) kPa	T °C	e_s kPa
1.0	0.657	13.0	1.498	25.0	3.168	37.0	6.275
1.5	0.681	13.5	1.547	25.5	3.263	37.5	6.448
2.0	0.706	14.0	1.599	26.0	3.361	38.0	6.625
2.5	0.731	14.5	1.651	26.5	3.462	38.5	6.806
3.0	0.758	15.0	1.705	27.0	3.565	39.0	6.991
3.5	0.785	15.5	1.761	27.5	3.671	39.5	7.181
4.0	0.813	16.0	1.818	28.0	3.780	40.0	7.376
4.5	0.842	16.5	1.877	28.5	3.891	40.5	7.574
5.0	0.872	17.0	1.938	29.0	4.006	41.0	7.778
5.5	0.903	17.5	2.000	29.5	4.123	41.5	7.986
6.0	0.935	18.0	2.064	30.0	4.243	42.0	8.199
6.5	0.968	18.5	2.130	30.5	4.366	42.5	8.417
7.0	1.002	19.0	2.197	31.0	4.493	43.0	8.640
7.5	1.037	19.5	2.267	31.5	4.622	43.5	8.867
8.0	1.073	20.0	2.338	32.0	4.755	44.0	9.101
8.5	1.110	20.5	2.412	32.5	4.891	44.5	9.339
9.0	1.148	21.0	2.487	33.0	5.030	45.0	9.582
9.5	1.187	21.5	2.564	33.5	5.173	45.5	9.832
10.0	1.228	22.0	2.644	34.0	5.319	46.0	10.086
10.5	1.270	22.5	2.726	34.5	5.469	46.5	10.347
11.0	1.313	23.0	2.809	35.0	5.623	47.0	10.613
11.5	1.357	23.5	2.896	35.5	5.780	47.5	10.885
12.0	1.403	24.0	2.984	36.0	5.941	48.0	11.163
12.5	1.449	24.5	3.075	36.5	6.106	48.5	11.447

TABLE A2.4

Slope of vapour pressure curve (Δ) for different temperatures (T)

$$\Delta = \frac{4\,098\left[0.6108\exp\left(\frac{17.27\,T}{T+237.3}\right)\right]}{(T+237.3)^2} \qquad \text{(Eq. 3.8)}$$

T °C	Δ (kPa/°C)	T °C	Δ (kPa/°C)	T (°C)	Δ (kPa/°C)	T (°C)	Δ (kPa/°C)
1.0	0.047	13.0	0.098	25.0	0.189	37.0	0.342
1.5	0.049	13.5	0.101	25.5	0.194	37.5	0.350
2.0	0.050	14.0	0.104	26.0	0.199	38.0	0.358
2.5	0.052	14.5	0.107	26.5	0.204	38.5	0.367
3.0	0.054	15.0	0.110	27.0	0.209	39.0	0.375
3.5	0.055	15.5	0.113	27.5	0.215	39.5	0.384
4.0	0.057	16.0	0.116	28.0	0.220	40.0	0.393
4.5	0.059	16.5	0.119	28.5	0.226	40.5	0.402
5.0	0.061	17.0	0.123	29.0	0.231	41.0	0.412
5.5	0.063	17.5	0.126	29.5	0.237	41.5	0.421
6.0	0.065	18.0	0.130	30.0	0.243	42.0	0.431
6.5	0.067	18.5	0.133	30.5	0.249	42.5	0.441
7.0	0.069	19.0	0.137	31.0	0.256	43.0	0.451
7.5	0.071	19.5	0.141	31.5	0.262	43.5	0.461
8.0	0.073	20.0	0.145	32.0	0.269	44.0	0.471
8.5	0.075	20.5	0.149	32.5	0.275	44.5	0.482
9.0	0.078	21.0	0.153	33.0	0.282	45.0	0.493
9.5	0.080	21.5	0.157	33.5	0.289	45.5	0.504
10.0	0.082	22.0	0.161	34.0	0.296	46.0	0.515
10.5	0.085	22.5	0.165	34.5	0.303	46.5	0.526
11.0	0.087	23.0	0.170	35.0	0.311	47.0	0.538
11.5	0.090	23.5	0.174	35.5	0.318	47.5	0.550
12.0	0.092	24.0	0.179	36.0	0.326	48.0	0.562
12.5	0.095	24.5	0.184	36.5	0.334	48.5	0.574

TABLE A2.5
Number of the day in the year (J)

Day	January	February	March*	April*	May*	June*
1	1	32	60	91	121	152
2	2	33	61	92	122	153
3	3	34	62	93	123	154
4	4	35	63	94	124	155
5	5	36	64	95	125	156
6	6	37	65	96	126	157
7	7	38	66	97	127	158
8	8	39	67	98	128	159
9	9	40	68	99	129	160
10	10	41	69	100	130	161
11	11	42	70	101	131	162
12	12	43	71	102	132	163
13	13	44	72	103	133	164
14	14	45	73	104	134	165
15	15	46	74	105	135	166
16	16	47	75	106	136	167
17	17	48	76	107	137	168
18	18	49	77	108	138	169
19	19	50	78	109	139	170
20	20	51	79	110	140	171
21	21	52	80	111	141	172
22	22	53	81	112	142	173
23	23	54	82	113	143	174
24	24	55	83	114	144	175
25	25	56	84	115	145	176
26	26	57	85	116	146	177
27	27	58	86	117	147	178
28	28	59	87	118	148	179
29	29	(60)	88	119	149	180
30	30	-	89	120	150	181
31	31	-	90	-	151	-

*add 1 if leap year

J can be determined for each day (D) of month (M) by

J = **INTEGER**(275 M/9 – 30 + D) – 2
IF (M < 3) **THEN** J = J + 2
IF (leap year and (M > 2)) **THEN** J = J + 1

For ten-day calculations, compute J for day D = 5, 15 and 25

For monthly calculations, J at the middle of the month is approximately given by

J = **INTEGER**(30.4 M – 15)

TABLE A2.5 (CONT.)
Number of the day in the year (J)

Day	July*	August*	September*	October*	November*	December*
1	182	213	244	274	305	335
2	183	214	245	275	306	336
3	184	215	246	276	307	337
4	185	216	247	277	308	338
5	186	217	248	278	309	339
6	187	218	249	279	310	340
7	188	219	250	280	311	341
8	189	220	251	281	312	342
9	190	221	252	282	313	343
10	191	222	253	283	314	344
11	192	223	254	284	315	345
12	193	224	255	285	316	346
13	194	225	256	286	317	347
14	195	226	257	287	318	348
15	196	227	258	288	319	349
16	197	228	259	289	320	350
17	198	229	260	290	321	351
18	199	230	261	291	322	352
19	200	231	262	292	323	353
20	201	232	263	293	324	354
21	202	233	264	294	325	355
22	203	234	265	295	326	356
23	204	235	266	296	327	357
24	205	236	267	297	328	358
25	206	237	268	298	329	359
26	207	238	269	299	330	360
27	208	239	270	300	331	361
28	209	240	271	301	332	362
29	210	241	272	302	333	363
30	211	242	273	303	334	364
31	212	243	-	304	-	365

*add 1 if leap year

TABLE A2.6

Daily extraterrestrial radiation (R_a) for different latitudes for the 15th day of the month [1]

$$R_a = \frac{24(60)}{\pi} G_{sc} d_r [\omega_s \sin(\phi) \sin(\delta) + \cos(\phi) \cos(\delta) \sin(\omega_s)] \quad \text{(Eq. 3.18)}$$

Northern Hemisphere												Lat deg.	Southern Hemisphere											
Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
0.0	2.6	10.4	23.0	35.2	42.5	39.4	28.0	14.9	4.9	0.1	0.0	70	41.4	28.6	15.8	4.9	0.2	0.0	0.0	2.2	10.7	23.5	37.3	45.3
0.1	3.7	11.7	23.9	35.3	42.0	38.9	28.6	16.1	6.0	0.7	0.0	68	41.0	29.3	16.9	6.0	0.8	0.0	0.0	3.2	11.9	24.4	37.4	44.7
0.6	4.8	12.9	24.8	35.6	41.4	38.8	29.3	17.3	7.2	1.5	0.1	66	40.9	30.0	18.1	7.2	1.5	0.1	0.5	4.2	13.1	25.4	37.6	44.1
1.4	5.9	14.1	25.8	35.9	41.2	38.8	30.0	18.4	8.5	2.4	0.6	64	41.0	30.8	19.3	8.4	2.4	0.6	1.2	5.3	14.4	26.3	38.0	43.9
2.3	7.1	15.4	26.6	36.3	41.2	39.0	30.6	19.5	9.7	3.4	1.3	62	41.2	31.5	20.4	9.6	3.4	1.2	2.0	6.4	15.5	27.2	38.3	43.9
3.3	8.3	16.6	27.5	36.6	41.2	39.2	31.3	20.6	10.9	4.4	2.2	60	41.5	32.3	21.5	10.8	4.4	2.0	2.9	7.6	16.7	28.1	38.7	43.9
4.3	9.6	17.7	28.4	37.0	41.3	39.4	32.0	21.7	12.1	5.5	3.1	58	41.7	33.0	22.6	12.0	5.5	2.9	3.9	8.7	17.9	28.9	39.1	44.0
5.4	10.8	18.9	29.2	37.4	41.4	39.6	32.6	22.7	13.3	6.7	4.2	56	42.0	33.7	23.6	13.2	6.6	3.9	4.9	9.9	19.0	29.8	39.5	44.1
6.5	12.0	20.0	30.0	37.8	41.5	39.8	33.2	23.7	14.5	7.8	5.2	54	42.2	34.3	24.6	14.4	7.7	4.9	6.0	11.1	20.1	30.6	39.9	44.3
7.7	13.2	21.1	30.8	38.2	41.6	40.1	33.8	24.7	15.7	9.0	6.4	52	42.5	35.0	25.6	15.6	8.8	6.0	7.1	12.2	21.2	31.4	40.2	44.4
8.9	14.4	22.2	31.5	38.5	41.7	40.2	34.4	25.7	16.9	10.2	7.5	50	42.7	35.6	26.6	16.7	10	7.1	8.2	13.4	22.2	32.1	40.6	44.5
10.1	15.7	23.3	32.2	38.8	41.8	40.4	34.9	26.6	18.1	11.4	8.7	48	42.9	36.2	27.5	17.9	11.1	8.2	9.3	14.6	23.3	32.8	40.9	44.5
11.3	16.9	24.3	32.9	39.1	41.9	40.6	35.4	27.5	19.2	12.6	9.9	46	43.0	36.7	28.4	19.0	12.3	9.3	10.4	15.7	24.3	33.5	41.1	44.6
12.5	18.0	25.3	33.5	39.3	41.9	40.7	35.9	28.4	20.3	13.9	11.1	44	43.2	37.2	29.3	20.1	13.5	10.5	11.6	16.8	25.2	34.1	41.4	44.6
13.8	19.2	26.3	34.1	39.5	41.9	40.8	36.3	29.2	21.4	15.1	12.4	42	43.3	37.7	30.1	21.2	14.6	11.6	12.8	18.0	26.2	34.7	41.6	44.6
15.0	20.4	27.2	34.7	39.7	41.9	40.8	36.7	30.0	22.5	16.3	13.6	40	43.4	38.1	30.9	22.3	15.8	12.8	13.9	19.1	27.1	35.3	41.8	44.6
16.2	21.5	28.1	35.2	39.9	41.8	40.8	37	30.7	23.6	17.5	14.8	38	43.4	38.5	31.7	23.3	16.9	13.9	15.1	20.2	28.0	35.8	41.9	44.5
17.5	22.6	29.0	35.7	40.0	41.7	40.8	37.4	31.5	24.6	18.7	16.1	36	43.4	38.9	32.4	24.3	18.1	15.1	16.2	21.2	28.8	36.3	42.0	44.4
18.7	23.7	29.9	36.1	40.0	41.6	40.8	37.6	32.1	25.6	19.9	17.3	34	43.4	39.2	33.0	25.3	19.2	16.2	17.4	22.3	29.6	36.7	42.0	44.3
19.9	24.8	30.7	36.5	40.0	41.4	40.7	37.9	32.8	26.6	21.1	18.5	32	43.3	39.4	33.7	26.3	20.3	17.4	18.5	23.3	30.4	37.1	42.0	44.1

TABLE A2.6 (CONT.)

Northern Hemisphere												Lat deg.	Southern Hemisphere											
Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
21.1	25.8	31.4	36.8	40.0	41.2	40.6	38.0	33.4	27.6	22.2	19.8	30	43.1	39.6	34.3	27.2	21.4	18.5	19.6	24.3	31.1	37.5	42.0	43.9
22.3	26.8	32.2	37.1	40.0	40.9	40.4	38.2	33.9	28.5	23.3	21.0	28	43.0	39.8	34.8	28.1	22.5	19.7	20.7	25.3	31.8	37.8	41.9	43.6
23.4	27.8	32.8	37.4	39.9	40.6	40.2	38.3	34.5	29.3	24.5	22.2	26	42.8	39.9	35.3	29.0	23.5	20.8	21.8	26.3	32.5	38.0	41.8	43.3
24.6	28.8	33.5	37.6	39.7	40.3	39.9	38.3	34.9	30.2	25.5	23.3	24	42.5	40.0	35.8	29.8	24.6	21.9	22.9	27.2	33.1	38.3	41.7	43.0
25.7	29.7	34.1	37.8	39.5	40.0	39.6	38.4	35.4	31.0	26.6	24.5	22	42.2	40.1	36.2	30.6	25.6	23.0	24.0	28.1	33.7	38.4	41.4	42.6
26.8	30.6	34.7	37.9	39.3	39.5	39.3	38.3	35.8	31.8	27.7	25.6	20	41.9	40.0	36.6	31.3	26.6	24.1	25.0	28.9	34.2	38.6	41.2	42.1
27.9	31.5	35.2	38.0	39.0	39.1	38.9	38.2	36.1	32.5	28.7	26.8	18	41.5	40.0	37.0	32.1	27.5	25.1	26.0	29.8	34.7	38.7	40.9	41.7
28.9	32.3	35.7	38.1	38.7	38.6	38.5	38.1	36.4	33.2	29.6	27.9	16	41.1	39.9	37.2	32.8	28.5	26.2	27.0	30.6	35.2	38.7	40.6	41.2
29.9	33.1	36.1	38.1	38.4	38.1	38.1	38.0	36.7	33.9	30.6	28.9	14	40.6	39.7	37.5	33.4	29.4	27.2	27.9	31.3	35.6	38.7	40.2	40.6
30.9	33.8	36.5	38.0	38.0	37.6	37.6	37.8	36.9	34.5	31.5	30.0	12	40.1	39.6	37.7	34.0	30.2	28.1	28.9	32.1	36.0	38.6	39.8	40.0
31.9	34.5	36.9	37.9	37.6	37	37.1	37.5	37.1	35.1	32.4	31.0	10	39.5	39.3	37.8	34.6	31.1	29.1	29.8	32.8	36.3	38.5	39.3	39.4
32.8	35.2	37.2	37.8	37.1	36.3	36.5	37.2	37.2	35.6	33.3	32.0	8	38.9	39.0	37.9	35.1	31.9	30.0	30.7	33.4	36.6	38.4	38.8	38.7
33.7	35.8	37.4	37.6	36.6	35.7	35.9	36.9	37.3	36.1	34.1	32.9	6	38.3	38.7	38.0	35.6	32.7	30.9	31.5	34.0	36.8	38.2	38.2	38.0
34.6	36.4	37.6	37.4	36.0	35	35.3	36.5	37.3	36.6	34.9	33.9	4	37.6	38.3	38.0	36.0	33.4	31.8	32.3	34.6	37.0	38.0	37.6	37.2
35.4	37.0	37.8	37.1	35.4	34.2	34.6	36.1	37.3	37.0	35.6	34.8	2	36.9	37.9	38.0	36.4	34.1	32.6	33.1	35.2	37.1	37.7	37.0	36.4
36.2	37.5	37.9	36.8	34.8	33.4	33.9	35.7	37.2	37.4	36.3	35.6	0	36.2	37.5	37.9	36.8	34.8	33.4	33.9	35.7	37.2	37.4	36.3	35.6

(values in MJ m^{-2} day^{-1})[2]

[1] Values for R_a on the 15th day of the month provide a good estimate (error < 1 %) of R_a averaged over all days within the month. Only for high latitudes greater than 55° (N or S) during winter months deviations may be more than 1percent.

[2] Values can be converted to equivalent values in mm/day by dividing by Lambda = 2.45.

TABLE A2.7

Mean daylight hours (N) for different latitudes for the 15th of the month[1]

$$N = \frac{24}{\pi}\omega_s \qquad \text{(Eq. 3.31)}$$

Northern Hemisphere												Lat deg.	Southern Hemisphere											
Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
0.0	6.6	11.0	15.6	21.3	24.0	24.0	17.6	12.8	8.3	2.3	0.0	70	24.0	17.4	13.0	8.4	2.7	0.0	0.0	6.4	11.2	15.7	21.7	24.0
2.1	7.3	11.1	15.3	19.7	24.0	22.3	17.0	12.7	8.7	4.1	0.0	68	21.9	16.7	12.9	8.7	4.3	0.0	1.7	7.0	11.3	15.3	19.9	24.0
3.9	7.8	11.2	14.9	18.7	22.0	20.3	16.4	12.7	9.0	5.2	1.9	66	20.1	16.2	12.8	9.1	5.3	2.0	3.7	7.6	11.3	15.0	18.8	22.1
5.0	8.2	11.2	14.7	17.9	20.3	19.2	16.0	12.6	9.3	6	3.7	64	19.0	15.8	12.8	9.3	6.1	3.7	4.8	8.0	11.4	14.7	18.0	20.3
5.7	8.5	11.3	14.4	17.3	19.2	18.4	15.7	12.6	9.5	6.6	4.8	62	18.3	15.5	12.7	9.6	6.7	4.8	5.6	8.3	11.4	14.5	17.4	19.2
6.4	8.8	11.4	14.2	16.8	18.4	17.7	15.3	12.5	9.7	7.1	5.6	60	17.6	15.2	12.6	9.8	7.2	5.6	6.3	8.7	11.5	14.3	16.9	18.4
6.9	9.1	11.4	14.1	16.4	17.8	17.2	15.1	12.5	9.9	7.5	6.2	58	17.1	14.9	12.6	9.9	7.6	6.2	6.8	8.9	11.5	14.1	16.5	17.8
7.3	9.3	11.5	13.9	16.0	17.3	16.8	14.8	12.4	10.1	7.9	6.7	56	16.7	14.7	12.5	10.1	8.0	6.7	7.2	9.2	11.6	13.9	16.1	17.3
7.7	9.5	11.5	13.8	15.7	16.8	16.4	14.6	12.4	10.2	8.2	7.1	54	16.3	14.5	12.5	10.2	8.3	7.2	7.6	9.4	11.6	13.8	15.8	16.9
8.0	9.7	11.5	13.6	15.4	16.5	16.0	14.4	12.4	10.3	8.5	7.5	52	16.0	14.3	12.5	10.4	8.6	7.5	8.0	9.6	11.6	13.7	15.5	16.5
8.3	9.8	11.6	13.5	15.2	16.1	15.7	14.3	12.3	10.4	8.7	7.9	50	15.7	14.2	12.4	10.5	8.8	7.9	8.3	9.7	11.7	13.6	15.3	16.1
8.6	10.0	11.6	13.4	15.0	15.8	15.5	14.1	12.3	10.6	9.0	8.2	48	15.4	14.0	12.4	10.6	9.0	8.2	8.5	9.9	11.7	13.4	15.0	15.8
8.8	10.1	11.6	13.3	14.8	15.5	15.2	14.0	12.3	10.7	9.2	8.5	46	15.2	13.9	12.4	10.7	9.2	8.5	8.8	10.0	11.7	13.3	14.8	15.5
9.1	10.3	11.6	13.2	14.6	15.3	15.0	13.8	12.3	10.7	9.4	8.7	44	14.9	13.7	12.4	10.8	9.4	8.7	9.0	10.2	11.7	13.3	14.6	15.3
9.3	10.4	11.7	13.2	14.4	15.0	14.8	13.7	12.3	10.8	9.6	9.0	42	14.7	13.6	12.3	10.8	9.6	9	9.2	10.3	11.7	13.2	14.4	15.0
9.5	10.5	11.7	13.1	14.2	14.8	14.6	13.6	12.2	10.9	9.7	9.2	40	14.5	13.5	12.3	10.9	9.8	9.2	9.4	10.4	11.8	13.1	14.3	14.8
9.6	10.6	11.7	13.0	14.1	14.6	14.4	13.5	12.2	11.0	9.9	9.4	38	14.4	13.4	12.3	11.0	9.9	9.4	9.6	10.5	11.8	13.0	14.1	14.6
9.8	10.7	11.7	12.9	13.9	14.4	14.2	13.4	12.2	11.1	10.1	9.6	36	14.2	13.3	12.3	11.1	10.1	9.6	9.8	10.6	11.8	12.9	13.9	14.4
10.0	10.8	11.8	12.9	13.8	14.3	14.1	13.3	12.2	11.1	10.2	9.7	34	14.0	13.2	12.2	11.1	10.2	9.7	9.9	10.7	11.8	12.9	13.8	14.3

TABLE A2.7 (CONT.)

Northern Hemisphere												Lat deg.	Southern Hemisphere											
Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
10.1	10.9	11.8	12.8	13.6	14.1	13.9	13.2	12.2	11.2	10.3	9.9	32	13.9	13.1	12.2	11.2	10.4	9.9	10.1	10.8	11.8	12.8	13.7	14.1
10.3	11.0	11.8	12.7	13.5	13.9	13.8	13.1	12.2	11.3	10.5	10.1	30	13.7	13.0	12.2	11.3	10.5	10.1	10.2	10.9	11.8	12.7	13.5	13.9
10.4	11.0	11.8	12.7	13.4	13.8	13.6	13.0	12.2	11.3	10.6	10.2	28	13.6	13.0	12.2	11.3	10.6	10.2	10.4	11.0	11.8	12.7	13.4	13.8
10.5	11.1	11.8	12.6	13.3	13.6	13.5	12.9	12.1	11.4	10.7	10.4	26	13.5	12.9	12.2	11.4	10.7	10.4	10.5	11.1	11.9	12.6	13.3	13.6
10.7	11.2	11.8	12.6	13.2	13.5	13.3	12.8	12.1	11.4	10.8	10.5	24	13.3	12.8	12.2	11.4	10.8	10.5	10.7	11.2	11.9	12.6	13.2	13.5
10.8	11.3	11.9	12.5	13.1	13.3	13.2	12.8	12.1	11.5	10.9	10.7	22	13.2	12.7	12.1	11.5	10.9	10.7	10.8	11.2	11.9	12.5	13.1	13.3
10.9	11.3	11.9	12.5	12.9	13.2	13.1	12.7	12.1	11.5	11.0	10.8	20	13.1	12.7	12.1	11.5	11.1	10.8	10.9	11.3	11.9	12.5	13.0	13.2
11.0	11.4	11.9	12.4	12.8	13.1	13.0	12.6	12.1	11.6	11.1	10.9	18	13.0	12.6	12.1	11.6	11.2	10.9	11.0	11.4	11.9	12.4	12.9	13.1
11.1	11.5	11.9	12.4	12.7	12.9	12.9	12.5	12.1	11.6	11.2	11.1	16	12.9	12.5	12.1	11.6	11.3	11.1	11.1	11.5	11.9	12.4	12.8	12.9
11.3	11.6	11.9	12.3	12.6	12.8	12.8	12.5	12.1	11.7	11.3	11.2	14	12.7	12.4	12.1	11.7	11.4	11.2	11.2	11.5	11.9	12.3	12.7	12.8
11.4	11.6	11.9	12.3	12.6	12.7	12.6	12.4	12.1	11.7	11.4	11.3	12	12.6	12.4	12.1	11.7	11.4	11.3	11.4	11.6	11.9	12.3	12.6	12.7
11.5	11.7	11.9	12.2	12.5	12.6	12.5	12.3	12.1	11.8	11.5	11.4	10	12.5	12.3	12.1	11.8	11.5	11.4	11.5	11.7	11.9	12.2	12.5	12.6
11.6	11.7	11.9	12.2	12.4	12.5	12.4	12.3	12.0	11.8	11.6	11.5	8	12.4	12.3	12.1	11.8	11.6	11.5	11.6	11.7	12.0	12.2	12.4	12.5
11.7	11.8	12.0	12.1	12.3	12.3	12.3	12.2	12.0	11.9	11.7	11.7	6	12.3	12.2	12.0	11.9	11.7	11.7	11.7	11.8	12.0	12.1	12.3	12.3
11.8	11.9	12.0	12.1	12.2	12.2	12.2	12.1	12.0	11.9	11.8	11.8	4	12.2	12.1	12.0	11.9	11.8	11.8	11.8	11.9	12.0	12.1	12.2	12.2
11.9	11.9	12.0	12.0	12.1	12.1	12.1	12.1	12.0	12.0	11.9	11.9	2	12.1	12.1	12.0	12.0	11.9	11.9	11.9	11.9	12.0	12.0	12.1	12.1
12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0

[1] Values for N on the 15th day of the month provide a good estimate (error < 1 %) of N averaged over all days within the month. Only for high latitudes greater than 55° (N or S) during winter months deviations may be more than 1 percent.

TABLE A2.8
σT_K^4 (Stefan-Boltzmann law) at different temperatures (T)

With σ = 4.903 10^{-9} MJ K^{-4} m^{-2} day^{-1} and T_K = T[°C] + 273.15					
T (°C)	σT_K^4 (MJ m^{-2} d^{-1})	T (°C)	σT_K^4 (MJ m^{-2} d^{-1})	T (°C)	σT_K^4 (MJ m^{-2} d^{-1})
1.0	27.70	17.0	34.75	33.0	43.08
1.5	27.90	17.5	34.99	33.5	43.36
2.0	28.11	18.0	35.24	34.0	43.64
2.5	28.31	18.5	35.48	34.5	43.93
3.0	28.52	19.0	35.72	35.0	44.21
3.5	28.72	19.5	35.97	35.5	44.50
4.0	28.93	20.0	36.21	36.0	44.79
4.5	29.14	20.5	36.46	36.5	45.08
5.0	29.35	21.0	36.71	37.0	45.37
5.5	29.56	21.5	36.96	37.5	45.67
6.0	29.78	22.0	37.21	38.0	45.96
6.5	29.99	22.5	37.47	38.5	46.26
7.0	30.21	23.0	37.72	39.0	46.56
7.5	30.42	23.5	37.98	39.5	46.85
8.0	30.64	24.0	38.23	40.0	47.15
8.5	30.86	24.5	38.49	40.5	47.46
9.0	31.08	25.0	38.75	41.0	47.76
9.5	31.30	25.5	39.01	41.5	48.06
10.0	31.52	26.0	39.27	42.0	48.37
10.5	31.74	26.5	39.53	42.5	48.68
11.0	31.97	27.0	39.80	43.0	48.99
11.5	32.19	27.5	40.06	43.5	49.30
12.0	32.42	28.0	40.33	44.0	49.61
12.5	32.65	28.5	40.60	44.5	49.92
13.0	32.88	29.0	40.87	45.0	50.24
13.5	33.11	29.5	41.14	45.5	50.56
14.0	33.34	30.0	41.41	46.0	50.87
14.5	33.57	30.5	41.69	46.5	51.19
15.0	33.81	31.0	41.96	47.0	51.51
15.5	34.04	31.5	42.24	47.5	51.84
16.0	34.28	32.0	42.52	48.0	52.16
16.5	34.52	32.5	42.80	48.5	52.49

TABLE A2.9
Conversion factors to convert wind speed measured at given height (over grass) to wind speed measured at standard height of 2 m above ground surface

$$\text{conversion factor} = \frac{4.87}{\ln(67.8\ z - 5.42)} \qquad \text{(Eq. 3.50)}$$

z height (m)	conversion factor	z height (m)	conversion factor	z height (m)	conversion factor	z height (m)	conversion factor
-	-	2.2	0.980	4.2	0.865	6.0	0.812
-	-	2.4	0.963	4.4	0.857	6.5	0.802
-	-	2.6	0.947	4.6	0.851	7.0	0.792
-	-	2.8	0.933	4.8	0.844	7.5	0.783
1.0	1.178	3.0	0.921	5.0	0.838	8.0	0.775
1.2	1.125	3.2	0.910	5.2	0.833	8.5	0.767
1.4	1.084	3.4	0.899	5.4	0.827	9.0	0.760
1.6	1.051	3.6	0.889	5.6	0.822	9.5	0.754
1.8	1.023	3.8	0.881	5.8	0.817	10.0	0.748
2.0	1.000	4.0	0.872	6.0	0.812	10.5	0.742

Annex 3
Statistical analysis and correction of weather data series

A3.1 COMPLETING A DATA SET

Quite often data sets containing a weather variable Y_i observed at a given station are incomplete due to short interruptions in observations. Interruptions can be due to a large number of causes, the most frequent being the breakage or malfunction of instruments during a certain time period. When data are missing, it may be appropriate to complete these data sets from observations X_i from another nearby and reliable station.

However, to use portions of data set X_i to insert in data set Y_i, both data sets X_i and Y_i must be homogeneous. In other words, they need to represent the same conditions throughout time. The procedure for completing data sets must be applied after a test for homogeneity and any needed correction for non-homogeneity have been performed. The data completion procedure proposed herein consists of using an appropriate regression analysis.

The procedure for inserting data of a nearby station into an incomplete data set can be summarized as follows:

1. Select a nearby weather station for which the data set length covers all periods for which data are missing.
2. Characterize the data sets from the nearby station, X_i, and of the station having missing data, Y_i, by computing the mean and the standard deviation sx for the nearby weather station data set (X_i):

$$\bar{x} = \sum_{i=1}^{n} \frac{x_i}{n} \tag{A3.1}$$

$$s_x = \left(\sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{(n-1)} \right)^{1/2} \tag{A3.2}$$

and the mean y and standard deviation sy for the weather station having an incomplete data set (Y_i):

$$\bar{y} = \sum_{i=1}^{n} \frac{y_i}{n} \tag{A3.3}$$

$$s_y = \left(\sum_{i=1}^{n} \frac{(y_i - \bar{y})^2}{(n-1)} \right)^{1/2} \tag{A3.4}$$

for the periods when the data in both data sets are present, where x_i and y_i are individual observations from data sets X and Y, and n is the number of observations in each set.

3. Perform a regression of y_i on x_i for the periods when the data in both data sets are present and compute the regression parameters a and b:

$$\hat{y}_i = a + bx_i \tag{A3.5}$$

with

$$b = \frac{cov_{xy}}{s_x^2} = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} \tag{A3.6}$$

$$a = \bar{y} = b\bar{x} \tag{A3.7}$$

where cov_{xy} is the covariance between X_i and Y_i. Plotting all points x_i and y_i and the regression line b for the range of observed values, it can be observed if their deviations from the regression line increase or not when y increases. If deviations increase, the insertion of x_i values for completing the y_i series should not be performed because the two sites have a different behaviour relative to the particular weather variable under consideration and non-homogeneity may occur. Another nearby station should be selected.

4. When similitude is observed and homogeneity is accepted (see next section), continue computations calculating the correlation coefficient r:

$$r = \frac{cov_{xy}}{s_x s_y} = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\left(\sum_{i=1}^{n} (x_i - \bar{x})^2 \sum_{i=1}^{n} (y_i - \bar{y})^2\right)^{1/2}} \tag{A3.8}$$

Both a high r^2 ($r^2 \geq 0.7$) and a value for b that is within the range ($0.7 \leq b \leq 1.3$) indicate favourable conditions for replacing missing data in the incomplete data series. The parameters r^2 and b can be used as criteria for selecting the best nearby station.

5. For the missing data periods k = n+1, n+2..., m, using the regression equation characterized by the parameters a and b (Eq. A3.6 and A3.7), compute the missing data:

$$\hat{y}_k = a + b\, x_k \tag{A3.9}$$

6. Once the computed y_k data are inserted into Y_i the complete data set with dimension m will now be

$$Y_j = y_i \qquad (j = i = 1,\ldots,n)$$

and (A3.10)

$$Y_j = \hat{y}_k \qquad (j = k = n+1, n+2,\ldots,m)$$

The estimates $Y_i = \hat{y}_k$ taken from the regression equations are useful for predicting evapotranspiration but they cannot be treated as random variables.

A3.2 ANALYSIS OF THE HOMOGENEITY OF DATA SERIES

Weather data collected at a given weather station during a period of several years may be non-homogeneous when the data presents a sudden change in its mean and variance in relation to the original values. This phenomenon may occur due to several causes, some of which are related to changes in instrumentation and observation practices, and others which relate to modification of the environmental conditions of the site, such as rapid urbanization or, on the contrary, perhaps development of irrigation in the area.

Changes relative to data collection may be caused by:

- change in grass vegetation cover, its watering and management;
- change in location of the weather station, or in type of shelters;
- significant change in type of vegetation surrounding the weather station;
- change in type of sensor or instrument;
- change of observer and/or the timing of observations; and
- deterioration of sensors, or poor management of instruments leading to the malfunctioning of mechanical parts of instruments.

These changes cause observations made prior and after the change to belong to statistically different populations. Quite often, the occurrence of such changes and the dates when they occurred are not known, making it necessary to access whether a given data set can be considered homogeneous and, if not, to identify the sub-data series and perform the required corrections.

Procedures indicated hereafter are simple, well proven in practice and rely upon the statistical comparison of two data sets, one considered homogeneous and consisting of observations X_i performed in a nearby weather station, the other being the one under analysis and consisting of the observations Y_i of the same weather variable (T_{max}, T_{min}, u_2, RH_{max}, R_s, etc). Both sets X_i and Y_i should be collected at two stations that are in the same climatic region, have the same time length of observations and presenting the same trends in time despite the space variability when short time scales (daily, or 10-day) are utilized.

A3.2.1 Method of Cumulative Residuals

When relating two weather data sets from two weather stations, where the first is homogeneous, the data set of the second station can be considered homogeneous if their cumulative residuals from its linear regression with the first data set are not biased. The bias hypothesis can be tested for a selected probability p by verifying whether the residuals are contained within an ellipse which axis and depend on the probability p used to test the hypothesis. The magnitudes of and also depend on the size of the data set and on the standard deviation of the sample being tested.

This test uses the residuals of the linear regression of Y on X as described herein before. The residuals should follow a normal distribution with mean zero and standard deviation $s_{y,x}$, i.e. the error $\varepsilon_i \in N(0,s_{y,x})$. Thus, the residuals ε_i must be independent random variables, i.e., they should exhibit homoscedasticity.

The procedure for analysing the homogeneity of a weather data set Y_i of a given weather station can be summarized as follows:

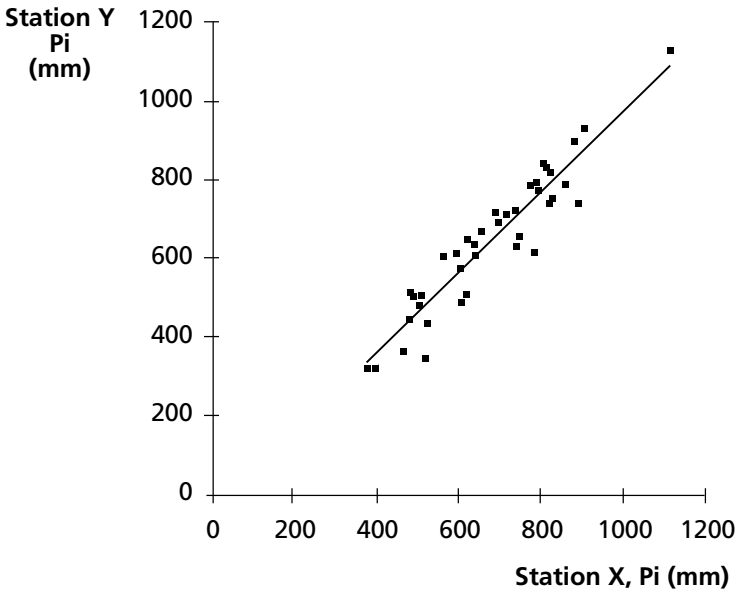
1. Select a reference weather station inside the same climatic region that is known to have a homogeneous data set X_i of the same weather variable. As an alternative, construct a "regional" homogeneous data set by averaging the observations at several weather stations in the same region.
2. Organize both data sets with their values x_i and y_i (i = 1, 2,..., n) in chronological order, where the initial timing and the time increment are identical for both data sets.
3. For both data sets, compute the mean and standard deviation (Eq. A3.1 to A3.4) for the homogeneous variable (x_i) and for the variable to be tested (y_i).

4. Calculate the regression line between the two variables y_i and x_i and the associated correlation coefficient (Eq. A3.5 to A3.8). The regression equation among the full data sets (subscript f) is expressed as:

$$\hat{y}_i = a_f + b_f x_i \qquad (i = 1, 2, ..., n) \qquad (A3.11)$$

Whenever possible, plot x_i, y_i and the regression line to visually verify whether the homoscedasticity hypothesis can be accepted (see Figure A3.1), i.e., when the deviations of y_i to the regression estimates $\hat{y}_i$ are within the same range for all x_i, thus when those deviations are not increasing (or decreasing) when the values of x_i are increasing.

FIGURE A3.1
Regression between two sets of weather data, with the X data set being homogeneous. The example shows that the homoscedasticity condition was satisfied



Data in this example were provided by J.L. Teixeira (personal communication, 1995)

5. Compute the residuals of the observed y_i values to the regression line (ε_i) (Eq. A3.5), the standard deviation $s_{y,x}$ of the residuals and the corresponding cumulative residual E_i:

$$\varepsilon_i = y_i - \hat{y}_i \qquad (A3.12)$$

$$s_{y,x} = s_y(1 - r^2)^{½} \qquad (A3.13)$$

$$E_i = \varepsilon_i + \sum_{j=1}^{i-1} \varepsilon_j \qquad (j = 1,....., i-1) \qquad (A3.14)$$

6. Select a probability p for accepting the hypothesis of homogeneity. The probability p = 80% for non-exceedance is commonly used. Then compute the ellipse equation having axes.

$$\alpha = n/2 \qquad (A3.15)$$

$$\beta = \frac{n}{(n-1)^{1/2}} z_p s_{y,x} \qquad \text{(A3.16)}$$

where n is the size of the sample under analysis,
z_p is standard normal variate for the probability p (usually p = 80%),
$s_{y,x}$ is the standard deviation of the residuals of y (Eq. A3.13).

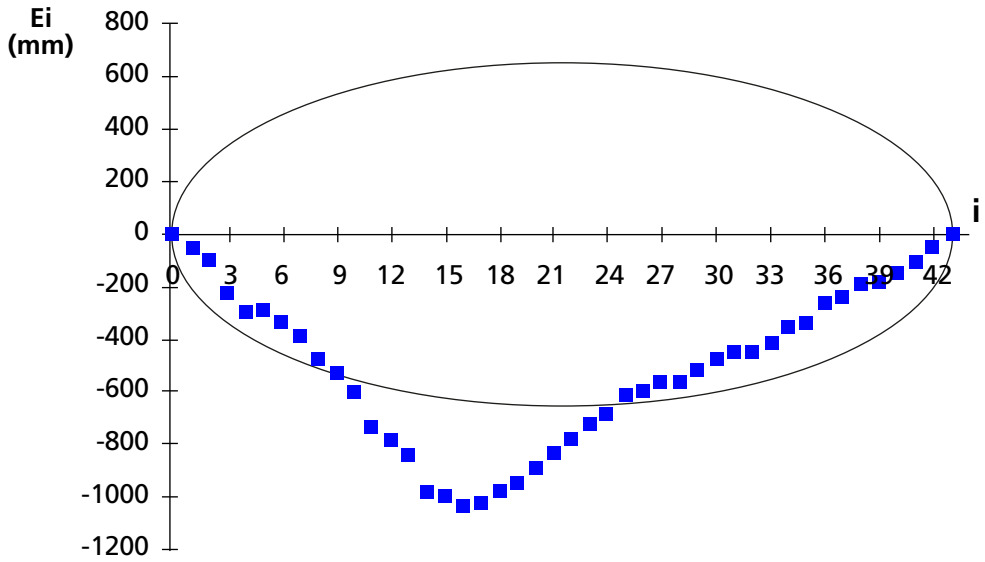
The parametric equation of the ellipse is then:

$$X = \alpha \cos(\theta)$$
$$Y = \beta \sin(\theta) \qquad \text{(A3.17)}$$

with θ [rad] varying from 0 to 2 π. When the computed values for E_i fall inside the ellipse (Eq. A3.17) it can be concluded, at the level of probability p, that there is no bias in the distribution of residuals, i.e., the data set y_i may be assumed as homogeneous.

7. Plot the cumulative residuals E_i against time using the time scale (interval) of the variable under analysis (Figure A3.2).
8. Draw the ellipse on the same plot and verify whether all E_i lie inside the ellipse. If they do, then the hypothesis of homogeneity is accepted at the p level of confidence.
9. If the hypothesis of homogeneity cannot be accepted (this is the case in Figure A3.2), then one can select the break point where it appears that E_i ceases to increase (or to decrease) and begins to decrease (or to increase), for example at i = 16 in Figure A3.2. This break point is termed k = i.

FIGURE A3.2
Plot of cumulative residuals against time and associated ellipse for the probability p = 80%, with results indicating that data set Y is not homogeneous (relative to data set X)



Source: FAO, 1998

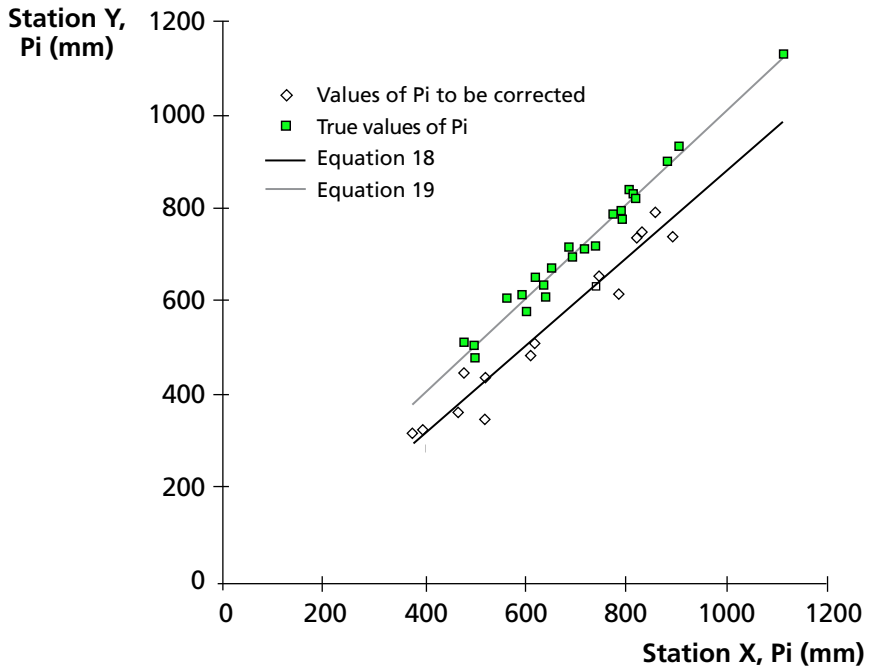
10. The data set is now divided into two subsets, the first from 1 to k and the second from k + 1 to n. Then, new regression equations are computed between Y and X for both subsets. If we presume that the second subset is homogeneous but that the first is not, then we have:

$$\hat{y}_i = a_{nh} + b_{nh} x_i \qquad (i = 1, 2, ...,k) \tag{A3.18}$$

$$\hat{y}_i = a_h + b_h x_i \qquad (i = k + 1, k + 2, ...,n) \tag{A3.19}$$

where the subscripts h and nh identify the regression coefficients of the homogeneous and the non-homogeneous subsets, respectively (see Figure A3.3).

FIGURE A3.3
The regression lines for the two subsets obtained from the data sets of Figures A3.1 and A3.2. Selection was made after definition of the break point in Figure A3.2



Source: FAO, 1998

11. Compute the differences between the two regression lines:

$$\Delta\hat{y}_i = (a_h + b_h x_i) - (a_{nh} + b_{nh} x_i) \tag{A3.20}$$

for the non-homogeneous set (i = 1, 2, ..., k)

12. Correct the non-homogeneous subset portion of data set Y:

$$y_{c,i} = y_i + \Delta\hat{y}_i \qquad (i = 1, 2, ...,k) \tag{A3.21}$$

where the subscript c identifies the corrected values. Thus, the corrected, homogeneous full set for weather variable Y consists of:

$$Y_i = y_{c,i} \qquad \text{for } i = 1, 2, \ldots, k$$

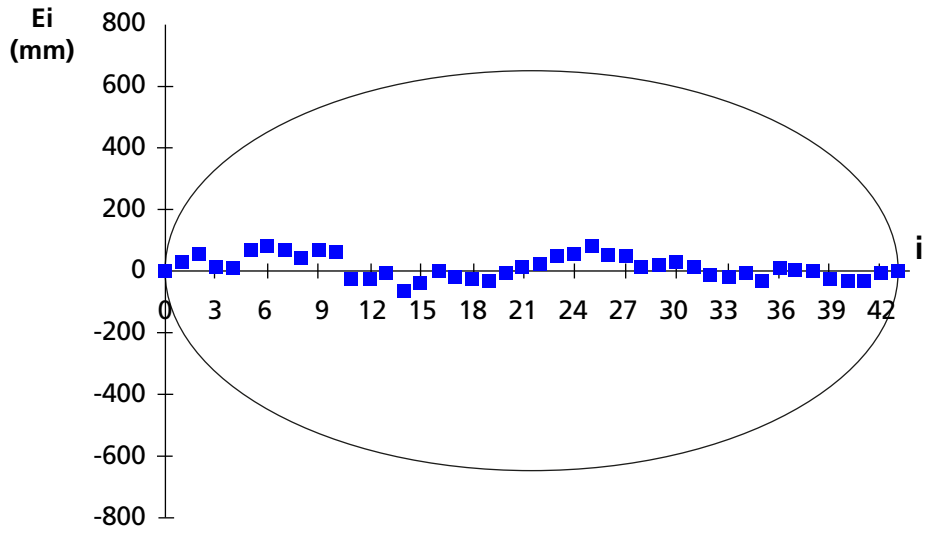
and (A3.22)

$$Y_i = y_i \qquad \text{for } i = k + 1, k + 2, \ldots, n$$

A similar procedure would be utilized if it was presumed that the second sub-set requires correction, rather than the first one.

Note that the variables Y_i are still considered to be random variables despite that the mean and the variance have been modified due to the correction introduced. To confirm the results of the correction of data set Y for homogeneity, the homogeneity test methodology can be applied again to the corrected variable Y to provide evidence of homogeneity in the graph of residuals. This has been done in Figure A3.4.

FIGURE A3.4
Plot of cumulative residuals against time and the associated ellipsis for p=80% after correction of variable y



Source: FAO, 1998

In this example, it was presumed that the latter sub-set (k to i) was the correct (representative) data set, or the data set displaying the desired attributes. It was therefore presumed that prior to time k, the readings were biased by instrument calibration, different location of the station or the instrument within the station, change in type or manufacturer of the instrument, or change in general environment of the station. It appears in Figure A3.3 that the data prior to i = k were biased downward by approximately 100 mm of annual precipitation.

A3.2.2 Double-Mass technique

The double-mass technique is also useful for assessing homogeneity in a weather data set. As with the method of cumulative residuals discussed above, the double-mass technique requires data sets from two weather stations, where X_i (i = 1, 2,...,n) is a

chronologic data set for a given weather variable observed for a certain time length at a "reference" station which is considered to be homogeneous, and where Y_i is a data set of the same variable, with the same time length, observed at another station and for which homogeneity needs to be analysed.

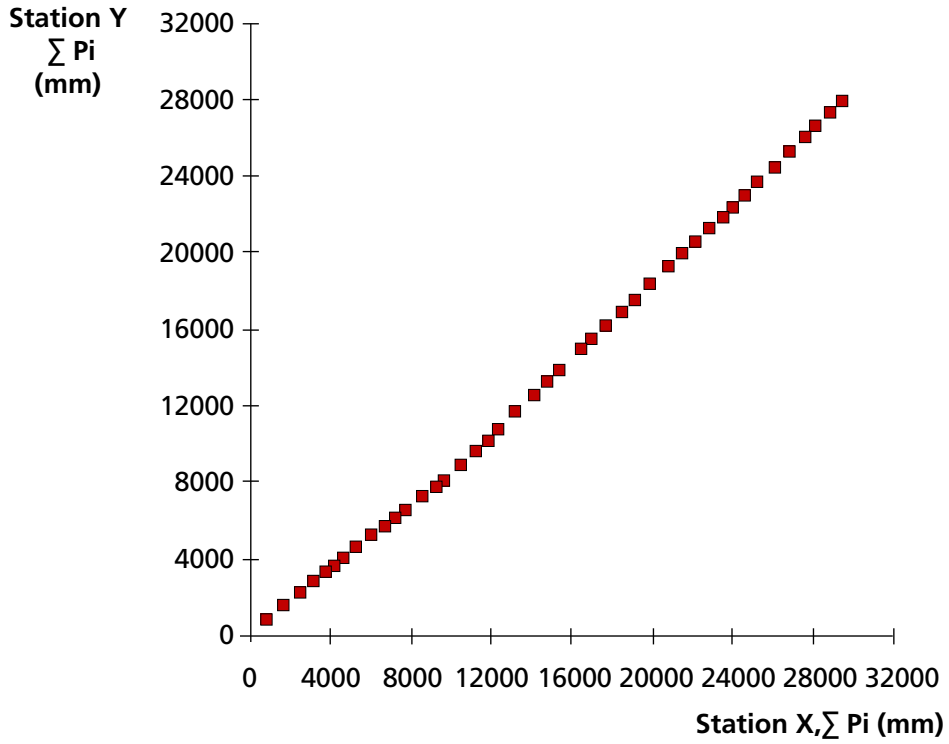
In the double-mass technique, starting with the first observed pair of values X_1 and Y_1, cumulative data sets are created by progressively summing values of X_i and Y_i to verify whether the long-term trends in variation of X_i and Y_i are the same. Thus, the following cumulative variables are obtained:

$$x_i = X_i \sum_{j=1}^{i-1} X_j \quad \text{(A3.23)}$$

$$y_i = Y_i \sum_{j=1}^{i-1} Y_j \quad \text{(A3.24)}$$

with i = 1, ..., n and j = 1,..., i – 1.

FIGURE A3.5
Double mass analysis applied to two series of precipitation when data from station Y are not homogeneous



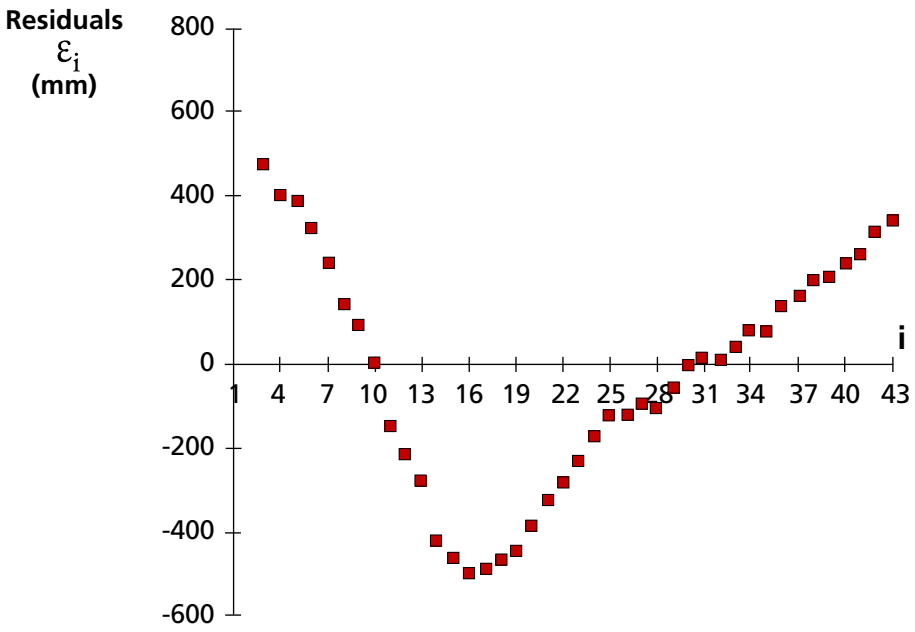
Source: FAO, 1998

These variables x_i and y_i are still considered to be random variables and are characterized by the mean and the standard deviation (Eq. A3.1 to A3.4). The y_i and x_i variables can be related through linear regression (Eq. A3.5 to A3.8). However, the double mass technique is typically applied as a graphical procedure.

The graphical application of the double-mass analysis is done by plotting all coordinate points x_i and y_i (Figure A3.5). The plot is then visually analysed to determine whether successive points of x_i and y_i follow a unique straight line, indicating the homogeneity of the data set Y_i relative to data set X_i. If there appears to be a break (or more than one break) in the plot of y_i to x_i, then there is a visual indication that the data series Y_i (or perhaps X_i) is not homogeneous (Figure A3.5). The break at coordinates x_k and y_k can be used to separate two subsets (i = 1, 2, ..., k) and (k + 1, k + 2, ..., n). One of the subsets is to be corrected. The appropriate one to correct needs to be identified by consulting the records of the weather station, when available.

FIGURE A3.6

Residuals of double mass to the straight line (Equation A3.26) indicating the non-homogeneity of the residuals of the series of precipitation of station Y



Source: FAO, 1998

Often, visual interpretation of the double-mass balance is difficult. Thus, the following numerical regression procedure is recommended.

1. Compute the regression line through the origin for the full set of data x_i and y_i

$$\hat{y}_i = b\,x_i \qquad (i = 1, 2, ..., n) \tag{A3.25}$$

with

$$b = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} \tag{A3.25a}$$

2. Compute the residuals to the regression line:

$$\varepsilon_i = y_i - bx_i \tag{A3.26}$$

3. Analyse the distribution of residuals: If the residuals plot as independent, random variables, then the set can be considered to be homogeneous. However, if the distribution of residuals is biased over i = k, then the homogeneity hypothesis

is rejected. The bias can be visually assessed by plotting (ε_i, i). The example in Figure A3.6 shows that residuals follow a trend of decreasing ε_i until i = k (= 16). Following that, the trend is to increase. This plot demonstrates a bias indicating that the data set Y is not homogeneous.

4. The break point at i = k defines two subsets (i = 1, 2, ..., k) and (i = k + 1, k + 2, ..., n). Using local information on data collection, the user must decide which subset requires correction.

5. When the first subset is homogeneous:

   a. compute the two regression lines, the first through the origin

$$\hat{y}_i = b_h x_i \qquad (i = 1, 2, ..., k) \tag{A3.27}$$

$$\hat{y}_{nh,i} = a_{nh} + b_{nh} x_i \qquad (i = k + 1, k + 2, ..., n) \tag{A3.28}$$

   where subscripts h and nh identify respectively the homogenous and non homogeneous subsets.

   b. compute the differences between both regression lines for i = k + 1, k + 2, ..., n

$$\Delta\hat{y}_i = b_h x_i - (a_{nh} + b_{nh} x_i) \tag{A3.29}$$

6. When the second subset is homogeneous

   a. compute the regression line for the homogeneous subset (i = k +1, k + 2, ..., n) after correcting the coordinates (x_i, y_i) using the coordinates of the break point (x_k, y_k), i.e. moving the origin of coordinates from (0, 0) to (x_k, y_k).

   This regression is therefore:

$$y_i - y_k = b_h (x_i - x_k) \tag{A3.30}$$

   thus

$$\hat{y}_i = (y_k - b_h x_k) + b_n x_i \qquad (i = k + 1, k + 2, ..., n) \tag{A3.31}$$

   b. compute the regression line for the non-homogeneous (nh) subset forced to the origin:

$$\hat{y}_i = b_{nh} x_i \qquad (i = 1, 2, ..., k) \tag{A3.32}$$

   c. compute the differences between the regression lines (A3.31) and (A3.32):

$$\Delta\hat{y}_i = [(y_k - b_h x_k) + b_h x_i] - b_{nh} x_i \tag{A3.33}$$

7. For both cases, correct the variables yi corresponding to the non-homogeneous subset as:

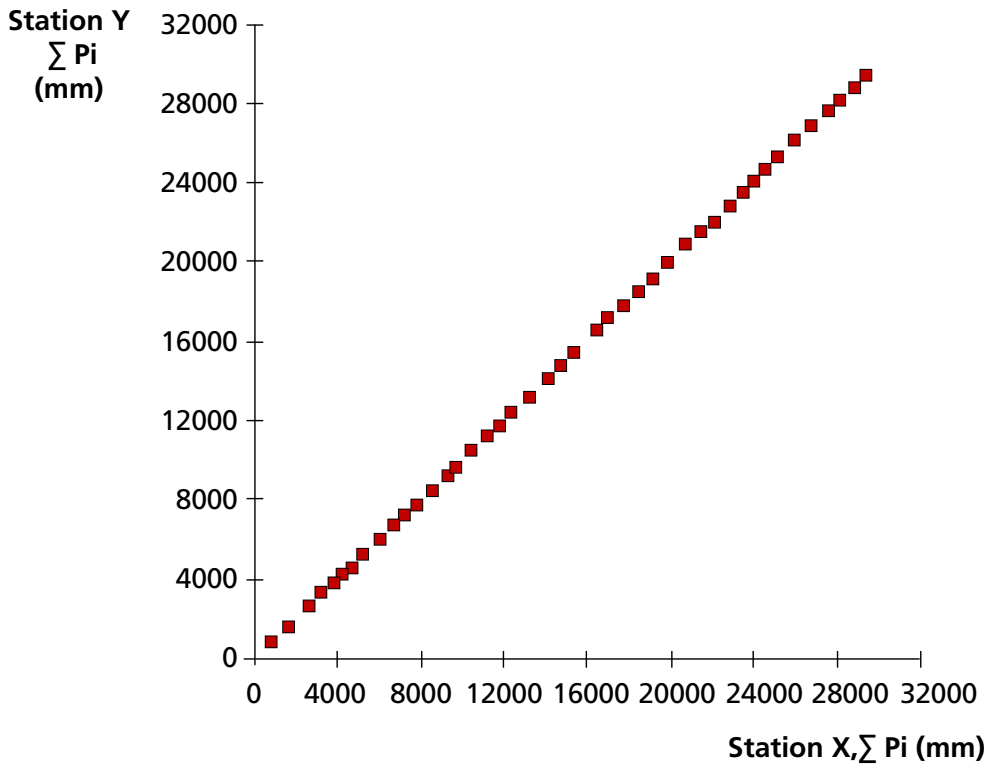
$$y_{c,i} = y_i + \Delta\hat{y}_i \tag{A3.34}$$

   with $\Delta\hat{y}_i$ given by Eq. A3.29 or A3.33.

8. Compute the corrected estimates of the weather variables Y_i by solving Eq. A3.24 for Y_i.

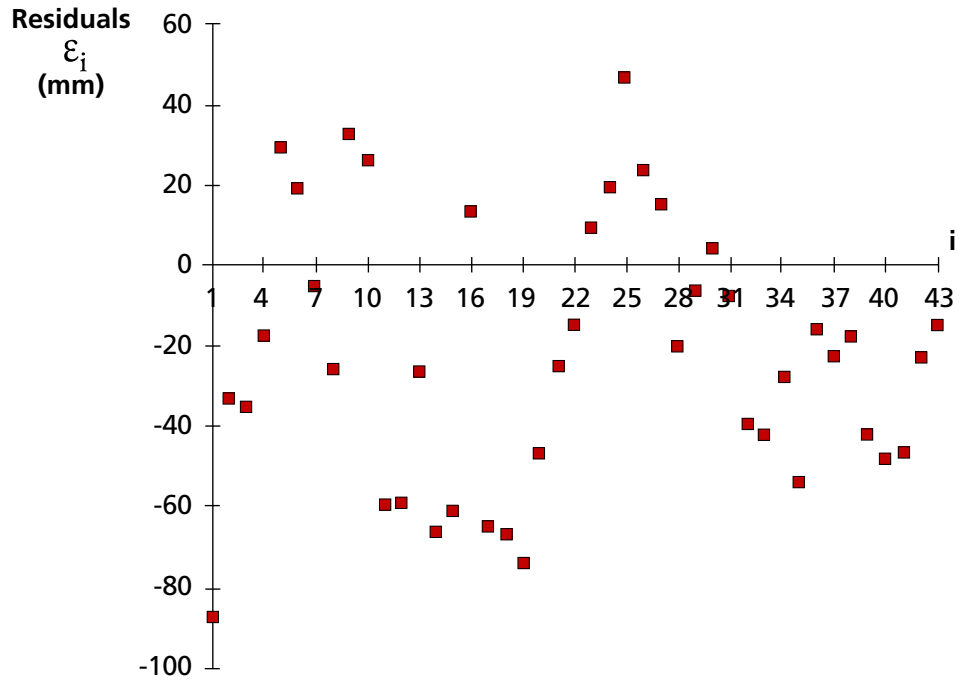
Figure A3.7 illustrates the double mass after correction of subset Y in Figure A3.3, where the cumulative sums now follow a straight line. Figure A3.8 is a plot of the corresponding residuals, which now follow a normal distribution. Similar verification can be easily made by the user.

FIGURE A3.7
Double mass after correction of data set Y (case of Figure A3.3)



Source: FAO, 1998

FIGURE A3.8
Residuals of the double mass after correction of data set Y (compare to Figure A3.4)



Source: FAO, 1998

A3.3 NOTATION IN STATISTICAL ANALYSIS

a	intercept (in a linear regression)
b	regression coefficient, slope
cov_{xy}	covariance of variables x and y
i	number of order of variable xi in the sample
j, k	number of a variable in a subset
n	size of the sample
p	probability
p(x)	probability distribution density function
r	regression coefficient
r^2	coefficient of determination
s_x	estimate of the standard deviation of the variable x
s_x^2	estimate of the variance of the variable x
s_y	estimate of the standard deviation of the variable y
s_y^2	estimate of the variance of the variable y
$s_{y,x}$	standard deviation of the residuals of y estimated from the regression
X	random variable
X_i	value of a variable in a data set
x_i	random variable
x_p	estimated value for the variable x with probability of non-exceedance p
$\bar{x}$	estimate of the mean, or mean of a sample of the random variable x_i
Y	transformed variable from X
Y_i	value of a variable in a data set
y_i	random variable
$\hat{y}_i$	value of y_i estimated from the regression
$\bar{y}$	estimate of the mean, or mean of a sample of the random variable y_i
Z, z	standard normal variable
z_p	value of the standard normal variable for the probability p
ε_i	residuals of y estimated from the regression
E_i	cumulative residuals
μ	mean of a population
σ	standard deviation of a population

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CHAPTER 1 - INTRODUCTION TO EVAPOTRANSPIRATION

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CHAPTERS 6 AND 7 – STANDARD ET_c: SINGLE AND BASAL CROP COEFFICIENTS (K_c AND K_{cb})

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Rev.1
FAO IRRIGATION AND DRAINAGE PAPERS

Crop evapotranspiration

Guidelines for computing crop water requirements

Second edition, revised 2025

As global agricultural systems face increasing pressure from climate variability, water scarcity, and the growing demand for food, there is a critical need for updated, science-based tools that support efficient and sustainable water use in agriculture. The original FAO Irrigation and Drainage Paper No. 56, published in 1998, set a global standard for computing crop evapotranspiration and estimating crop water requirements. Its methodologies have become integral to irrigation planning, agricultural water management, and policy development worldwide.

However, advancements in data science, sensor technology, and climate modeling—coupled with emerging challenges such as prolonged droughts and changing weather patterns—have made it essential to revisit and strengthen the technical foundations of this cornerstone publication. The revised FAO 56 responds directly to this need, integrating modern tools such as remote sensing, IoT applications, and enhanced energy balance methods to improve the precision, relevance, and accessibility of crop water estimation under current and future climate conditions.

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